Matter Application Cluster Specification

Version 1.6.1

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Abstract:	The Matter Application Clusters specified in this document are generic interfaces that are sufficiently general to be of use across a wide range of application domains.
Keywords:	Referenced in Chapter 1.

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Matter Application Clusters

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Introduction

The Matter Application Cluster specification defines generic interfaces that are sufficiently general to be of use across a wide range of application domains.

Scope and Purpose

This document specifies the Matter Application Cluster Library (MACL). The MACL is a repository for cluster functionality that is developed by the Connectivity Standards Alliance, and is a working library with regular updates as new functionality is added. A developer constructing a new application should use the MACL to find relevant cluster functionality that can be incorporated into the new application. Correspondingly, new clusters that are defined for applications should be considered for inclusion in the MACL.

The MACL consists of a number of sets of clusters. Clusters that are generally useful across many application domains are included in the General set. Clusters that are intended for use mainly in specific application domains are grouped together in domain oriented sets.

References

The following standards and specifications contain provisions, which through reference in this document constitute provisions of this specification. All the standards and specifications listed are normative references. At the time of publication, the editions indicated were valid. All standards and specifications are subject to revision, and parties to agreements based on this specification are encouraged to investigate the possibility of applying the most recent editions of the standards and specifications indicated below.

Connectivity Standards Alliance Reference Documents

Reference	Reference Location/URL	Description
[Aliro]	TBD	Aliro Specification - Under development
[Alliance-PNP]	https://groups.csa-iot.org/wg/members/document/21624	Organizational Processes and Procedures, 13-0625, revision 8, November 2021
[MatterCore]	https://groups.csa-iot.org/wg/members-all/document/27349	Matter Core Specification
[MatterDevLib]	https://groups.csa-iot.org/wg/members-all/document/27351	Matter Device Library Specification
[StandardNamespaces]	https://groups.csa-iot.org/wg/members-all/document/31936	Standard Namespaces

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External Reference Documents

Reference	Reference Location/URL	Description
[CENC]	https://www.iso.org/standard/84637.html	MPEG systems technologies Part 7: Common encryption in ISO base media file format files
[CENCISOBMFF]	https://www.iso.org/standard/70175.html	MPEG systems technologies Part 9: Common encryption of MPEG-2 transport streams
[CIE 1931]	https://en.wikipedia.org/wiki/CIE_1931_color_space	Commission Internationale de l'Éclairage CIE 1931 Color Space
[DASHIFINGEST]	https://dashif.org/Ingest/	DASH-IF Live Media Ingest Protocol
[DIALRegistry]	http://www.dial-multiscreen.org/dial-registry/namespace-database	DIAL Registry
[EU Code of Conduct for ESAs]	https://ses.jrc.ec.europa.eu/development-of-policy-proposals-for-energy-smart-appliances	EU Code of Conduct for ESAs
[HDMI]	https://hdmiforum.org/hdmi-forum-releases-version-2-1-hdmi-specification/	HDMI CEC specification
[IEEE 2030.5]	https://standards.ieee.org/ieee/2030.5/11216/	IEEE 2030.5
[ISO 4217]	https://www.iso.org/iso-4217-currency-codes.html	Currency Codes
[ISO 8601]	https://www.iso.org/iso-8601-date-and-time-format.html	Date and time format
[MPEGCMAF]	https://www.iso.org/standard/79106.html	Multimedia application format (MPEG-A) Part 19: Common media application format (CMAF) for segmented media
[ISOBMFF]	https://www.iso.org/standard/83102.html	Coding of audio-visual objects Part 12: ISO base media file format
[OpenADR]	https://www.openadr.org/	OpenADR Alliance
[PAS1878]	https://www.bsigroup.com/globalassets/localfiles/en-th/about-bsi/energy-smart-appliances-programme/bsi-pas-1878-energy-smart-appliances-system-functionality-and-architecture-th.pdf	PAS1878

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Reference	Reference Location/URL	Description
[RFC1738]	https://www.rfc-editor.org/rfc/rfc1738	Uniform Resource Locators (URL)
[RFC2045]	https://www.rfc-editor.org/rfc/rfc2045	MIME Part One: Format of Internet Message Bodies
[RFC2046]	https://www.rfc-editor.org/rfc/rfc2046	MIME Part Two: Media Types
[RFC3264]	https://www.rfc-editor.org/rfc/rfc3264	An Offer/Answer Model with the Session Description Protocol (SDP)
[RFC4122]	https://www.rfc-editor.org/rfc/rfc4122	A Universally Unique IDentifier (UUID) URN Namespace
[RFC5646]	https://tools.ietf.org/html/rfc5646	Tags for Identifying Languages
[RFC6236]	https://www.rfc-editor.org/rfc/rfc6236	Negotiation of Generic Image Attributes in the Session Description Protocol (SDP)
[RFC7728]	https://www.rfc-editor.org/rfc/rfc7728	RTP Stream Pause and Resume
[RFC8141]	https://www.rfc-editor.org/rfc/rfc8141	Uniform Resource Names (URNs)
[RFC8216bis]	https://datatracker.ietf.org/doc/html/draft-pantos-hls-rfc8216bis	HTTP Live Streaming 2nd Edition (HLS)
[RFC8445]	https://www.rfc-editor.org/rfc/rfc8445	Interactive Connectivity Establishment (ICE): A Protocol for Network Address Translator (NAT) Traversal
[RFC8489]	https://www.rfc-editor.org/rfc/rfc8489	Session Traversal Utilities for NAT (STUN)
[RFC8656]	https://www.rfc-editor.org/rfc/rfc8656	Traversal Using Relays around NAT (TURN)
[RFC8825]	https://www.rfc-editor.org/rfc/rfc8825	Overview: Real-Time Protocols for Browser-Based Applications
[RFC8834]	https://www.rfc-editor.org/rfc/rfc8834	Media Transport and Use of RTP in WebRTC
[RFC8839]	https://www.rfc-editor.org/rfc/rfc8839	Session Description Protocol (SDP) Offer/Answer Procedures for Interactive Connectivity Establishment (ICE)
[RFC8851]	https://www.rfc-editor.org/rfc/rfc8851	RTP Payload Format Restrictions
[RFC8866]	https://www.rfc-editor.org/rfc/rfc8866	SDP: Session Description Protocol

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Reference	Reference Location/URL	Description
[RFC9113]	https://www.rfc-editor.org/rfc/rfc9113	HTTP/2
[RFC9429]	https://www.rfc-editor.org/rfc/rfc9429	JavaScript Session Establishment Protocol (JSEP)
[SAE J2847/3_202311]	https://www.sae.org/standards/content/j2847/3_202311/	Communication for Plug-in Vehicles as a Distributed Energy Source
[SAREF4ENER]	https://saref.etsi.org/saref4ener/v1.1.2/	SAREF ontology for energy (ETSI TS 103 410-1)
[SEC 1]	https://www.secg.org/sec1-v2.pdf	SEC 1: Elliptic Curve Cryptography, Version 2.0, Certicom Research, May 2009
[Thread]	https://www.threadgroup.org	Thread 1.3.0 Specification
[RFC3986]	https://www.rfc-editor.org/rfc/rfc3986	Uniform Resource Identifier (URI)
[WakeOnLAN]	https://www.amd.com/system/files/TechDocs/20213.pdf	Wake on LAN Magic Packet specification

Provisional

Per [Alliance-PNP], when a specification is completed there may be sections of specification text (or smaller pieces of a section) that are not certifiable at this stage. These sections (or smaller pieces of a section) are marked as provisional prior to publishing the specification. This specification uses well-defined notation to mark Provisional Conformance (see [MatterCore], Section 7.3) or notes a section of text with the term "provisional".

List of Provisional Items

The following is a list of provisional items:

- Support for the Frequency feature of the Level Control Cluster is provisional.
- Support for InflowError, DrainError, TempTooLow, TempTooHigh, and WaterLevelError alarms in Dishwasher Alarm Cluster is provisional.
- Support for Content Control Cluster is provisional.
- Support for PowerInWatts feature of the Microwave Oven Control Cluster is provisional
- Support for the Ambient Context Sensing Cluster is provisional.
- Support for the Temperature Alarm Cluster is provisional.

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Chapter 1. General

The Cluster Library is made of individual chapters such as this one. See Document Control in the Cluster Library for a list of all chapters and documents. References between chapters are made using a *X.Y* notation where *X* is the chapter and *Y* is the sub-section within that chapter. References to external documents are contained in Chapter 1 and are made using [*Rn*] notation.

1.1. General Description

1.1.1. Introduction

The clusters specified in this document are generic interfaces that are sufficiently general to be of use across a wide range of application domains.

1.1.2. Cluster List

This section lists the general clusters as specified in this chapter.

Table 1. Overview of the General Clusters

ID	Cluster Name	Description
0x0003	Identify	Attributes and commands for putting a device into Identification mode (e.g., flashing a light)
0x0004	Groups	Cluster to manage the associated endpoint's membership into one or more groups to support groupcast interactions.
0x0062	Scenes Management	Attributes and commands for setting up and recalling a number of scenes for a device. Each scene corresponds to a set of stored values of specified device attributes.
0x0006	On/Off	Attributes and commands for switching devices between On and Off states.
0x0008	Level Control	Attributes and commands for controlling a characteristic of devices that can be set to a level between fully On and fully Off.
0x0045	Boolean State	Attribute and event for a boolean state variable

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ID	Cluster Name	Description
0x0080	Boolean State Configuration	Attributes and commands for configuration related to boolean state
0x0050	Mode Select	Allows a user to choose one mode option from several pre-defined values
n/a	Mode Base	Allows a user to choose one mode option from several pre-defined values
0x0508	Low Power	This cluster provides an interface for managing low power mode on a device.
0x0503	Wake On LAN	This cluster provides an interface for managing low power mode on a device that supports the Wake On LAN protocol.
0x003B	Switch	Attributes and events for various types of switch devices.
0x0060	Operational State	Commands and attributes for defining a device's operational state
n/a	Alarm Base	Base alarm cluster from which all alarms are derived
0x0097	Messages	Commands and attributes for sending messages to devices
0x0150	Service Area	Commands and attributes for controlling and querying the operating service areas of devices

1.2. Identify Cluster

This cluster supports an endpoint identification state (e.g., flashing a light), that indicates to an observer (e.g., an installer) which of several nodes and/or endpoints it is. It also supports a multicast request that any endpoint that is identifying itself to respond to the initiator.

The state of this cluster MAY be shared on more than one endpoint on a node.

> For Example: Two endpoints on a single node, one a temperature sensor, and one a humidity sensor, may both share the same cluster instance and therefore identification state (e.g. single LED on the node).

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1.2.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added
2	CCB 2808
3	All Hubs changes
4	New data model format and notation; add IdentifyType
5	Removed Query feature
6	Added Q quality for IdentifyTime attribute

1.2.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Utility	Endpoint	I

1.2.3. Cluster ID

ID	Name
0x0003	Identify

1.2.4. Data Types

1.2.4.1. IdentifyTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0x00	**None**	No presentation.	M
0x01	**LightOutput**	Light output of a lighting product.	M
0x02	**VisibleIndicator**	Typically a small LED.	M
0x03	**AudibleBeep**		M
0x04	**Display**	Presentation will be visible on display screen.	M

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Value	Name	Summary	Conformance
0x05	**Actuator**	Presentation will be conveyed by actuator functionality such as through a window blind operation or in-wall relay.	M

1.2.4.2. EffectIdentifierEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0x00	**Blink**	e.g., Light is turned on/off once.	M
0x01	**Breathe**	e.g., Light is turned on/off over 1 second and repeated 15 times.	M
0x02	**Okay**	e.g., Colored light turns green for 1 second; non-colored light flashes twice.	M
0x0B	**ChannelChange**	e.g., Colored light turns orange for 8 seconds; non-colored light switches to the maximum brightness for 0.5s and then minimum brightness for 7.5s.	M
0xFE	**FinishEffect**	Complete the current effect sequence before terminating. e.g., if in the middle of a breathe effect (as above), first complete the current 1s breathe effect and then terminate the effect.	M
0xFF	**StopEffect**	Terminate the effect as soon as possible.	M

1.2.4.3. EffectVariantEnum Type

This data type is derived from enum8.

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Value	Name	Summary	Conformance
0x00	**Default**	Indicates the default effect is used	M

1.2.5. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**IdentifyTime**	uint16	all	Q		RW VO	M
0x0001	**IdentifyType**	IdentifyTypeEnum	desc			R V	M

1.2.5.1. IdentifyTime Attribute

This attribute SHALL represent the remaining length of time, in seconds, that the endpoint will continue to identify itself.

If this attribute is set to a value other than 0 then the device SHALL enter its identification state, in order to indicate to an observer which of several nodes and/or endpoints it is. It is RECOMMENDED that this state consists of flashing a light with a period of 0.5 seconds. The IdentifyTime attribute SHALL be decremented every second while in this state.

If this attribute reaches or is set to the value 0 then the device SHALL terminate its identification state.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- When it changes from 0 to any other value and vice versa, or
- When it is written by a client, or
- When the value is set by an Identify command.

Since this attribute is not being reported during a regular countdown, clients SHOULD NOT rely on the reporting of this attribute in order to keep track of the remaining duration.

1.2.5.2. IdentifyType Attribute

This attribute SHALL indicate how the identification state is presented to the user.

This attribute SHALL contain one of the values defined in IdentifyTypeEnum. The value None SHALL NOT be used if the device is capable of presenting its identification state using one of the other methods defined in IdentifyTypeEnum.

1.2.6. Commands

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ID	Name	Direction	Response	Access	Conformance
0x00	**Identify**	client ⇒ server	Y	M	M
0x40	**TriggerEffect**	client ⇒ server	Y	M	O

1.2.6.1. Identify Command

This command starts or stops the receiving device identifying itself.

This command SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**IdentifyTime**	uint16	all			M

1.2.6.1.1. Effect on Receipt

On receipt of this command, the device SHALL set the IdentifyTime attribute to the value of the IdentifyTime field. This then starts, continues, or stops the device's identification state as detailed in IdentifyTime.

1.2.6.2. TriggerEffect Command

This command allows the support of feedback to the user, such as a certain light effect. It is used to allow an implementation to provide visual feedback to the user under certain circumstances such as a color light turning green when it has successfully connected to a network. The use of this command and the effects themselves are entirely up to the implementer to use whenever a visual feedback is useful but it is not the same as and does not replace the identify mechanism used during commissioning.

This command SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**EffectIdentifier**	EffectIdentifierEnum	desc			M
1	**EffectVariant**	EffectVariantEnum	desc			M

1.2.6.2.1. EffectIdentifier Field

This field SHALL indicate the identify effect to use and SHALL contain one of the non-reserved values in EffectIdentifierEnum.

All values of the EffectIdentifierEnum SHALL be supported. Implementors MAY deviate from the example light effects in EffectIdentifierEnum, but they SHOULD indicate during testing how they handle each effect.

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1.2.6.2.2. EffectVariant Field

This field SHALL indicate which variant of the effect, indicated in the EffectIdentifier field, SHOULD be triggered. If a device does not support the given variant, it SHALL use the default variant. This field SHALL contain one of the values in EffectVariantEnum.

1.2.6.2.3. Effect on Receipt

On receipt of this command, the device SHALL execute the trigger effect indicated in the EffectIdentifier and EffectVariant fields. If the EffectVariant field specifies a variant that is not supported on the device, it SHALL execute the default variant.

1.3. Groups Cluster

NOTE

Starting with the release of Matter when the Groupcast Cluster becomes certifiable, that cluster supersedes this Groups Cluster. Groups created on devices prior to that Matter release remain operational via the PerGroup feature in the Groupcast Cluster. All new group management operations SHOULD be performed through the Groupcast Cluster. Device types with existing mandatory conformance to the Groups Cluster SHALL retain Groups server cluster support for backward compatibility.

The Groups cluster manages, per endpoint, the content of the node-wide Group Table that is part of the underlying interaction layer.

In a network supporting fabrics, group IDs referenced by attributes or other elements of this cluster are scoped to the accessing fabric.

The Groups cluster is scoped to the endpoint. Groups cluster commands support discovering the endpoint membership in a group, adding the endpoint to a group, removing the endpoint from a group, removing endpoint membership from all groups. All commands defined in this cluster SHALL only affect groups scoped to the accessing fabric.

When group names are supported, the server stores a name string, which is set by the client for each assigned group and indicated in response to a client request.

Note that configuration of group addresses for outgoing commands is achieved using the Message Layer mechanisms where the Group Table is not involved. Hence this cluster does not play a part in that.

1.3.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added; CCB 1745 2100

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Revision	Description
2	CCB 2289
3	CCB 2310 2704
4	New data model format and notation

1.3.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Utility	Endpoint	G

1.3.3. Cluster ID

ID	Name
0x0004	Groups

1.3.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**GN**	GroupNames	O	The ability to store a name for a group

1.3.4.1. GroupNames Feature

The Group Names feature indicates the ability to store a name for a group when a group is added.

1.3.5. Data Types

1.3.5.1. NameSupportBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
7	**GroupNames**	The ability to store a name for a group.	M

1.3.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**NameSupport**	NameSupportBitmap	desc	F		R V	M

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1.3.6.1. NameSupport Attribute

This attribute provides legacy, read-only access to whether the Group Names feature is supported. The most significant bit, bit 7 (GroupNames), SHALL be equal to bit 0 of the FeatureMap attribute (GN Feature). All other bits SHALL be 0.

1.3.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**AddGroup**	client ⇒ server	AddGroupResponse	M F	M
0x01	**ViewGroup**	client ⇒ server	ViewGroupResponse	O F	M
0x02	**GetGroupMembership**	client ⇒ server	GetGroupMembershipResponse	O F	M
0x03	**RemoveGroup**	client ⇒ server	RemoveGroupResponse	M F	M
0x04	**RemoveAllGroups**	client ⇒ server	Y	M F	M
0x05	**AddGroupIfIdentifying**	client ⇒ server	Y	M F	M
0x00	**AddGroupResponse**	client ⇐ server	N		M
0x01	**ViewGroupResponse**	client ⇐ server	N		M
0x02	**GetGroupMembershipResponse**	client ⇐ server	N		M
0x03	**RemoveGroupResponse**	client ⇐ server	N		M

1.3.7.1. AddGroup Command

The AddGroup command allows a client to add group membership in a particular group for the server endpoint.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**GroupID**	group-id	min 1			M
1	**GroupName**	string	max 16			M

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1.3.7.1.1. GroupID Field

This field SHALL be used to identify the group and any associated key material to which the server endpoint is to be added.

1.3.7.1.2. GroupName Field

This field MAY be set to a human-readable name for the group. If the client has no name for the group, the GroupName field SHALL be set to the empty string.

Support of group names is optional and is indicated by the FeatureMap and NameSupport attribute.

1.3.7.1.3. Effect on Receipt

If the server does not support group names, the GroupName field SHALL be ignored.

On receipt of the AddGroup command, the server SHALL perform the following procedure:

1. If the command fields are not within constraints, the status SHALL be CONSTRAINT_ERROR, and the server continues from step 6.
2. If the receiving node requires security material to support the group ID and that material does not exist for this group ID, the status SHALL be UNSUPPORTED_ACCESS and the server continues from step 6.
3. If the server endpoint is a member of the group indicated by the GroupID, the group name SHALL be updated (if supported) to GroupName, the status SHALL be SUCCESS, and the server continues from step 6.
4. If there are no available resources to add the membership for the server endpoint, the status SHALL be RESOURCE_EXHAUSTED, and the server continues from step 6.
5. The server SHALL add the server endpoint as a member of the group indicated by the GroupID, the group name SHALL be updated (if supported) to GroupName, and the status SHALL be SUCCESS.
    a. If the GroupID had already been added to the Group Table because of a previous AddGroup or AddGroupIfIdentifying command and a GroupName is provided and the server supports GroupName storage, then the GroupName associated with the GroupID in the Group Table SHALL be updated to reflect the new GroupName provided for the Group, such that subsequent ViewGroup commands yield the same name for all endpoints which have a group association to the given GroupID.
6. If the AddGroup command was received as a unicast, the server SHALL generate an AddGroupResponse command with the Status field set to the evaluated status. If the AddGroup command was received as a groupcast, the server SHALL NOT generate an AddGroupResponse command.

See AddGroupResponse for a description of the response command.

1.3.7.2. ViewGroup Command

The ViewGroup command allows a client to request that the server responds with a ViewGroupResponse command containing the name string for a particular group.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**GroupID**	group-id	min 1			M

1.3.7.2.1. Effect on Receipt

On receipt of the ViewGroup command, the server SHALL perform the following procedure:

1. If the command fields are not within constraints, the status SHALL be CONSTRAINT_ERROR and the server continues from step 4.
2. If the server endpoint is a member of the group indicated by the GroupID, the status SHALL be SUCCESS, and the server continues from step 4.
3. Else the status SHALL be NOT_FOUND.
4. If the ViewGroup command was received as a unicast, the server SHALL generate a ViewGroupResponse command for the group, and the Status field set to the evaluated status. If the ViewGroup command was received as a groupcast, the server SHALL NOT generate a ViewGroupResponse command.

See ViewGroupResponse for a description of the response command.

1.3.7.3. GetGroupMembership Command

The GetGroupMembership command allows a client to inquire about the group membership of the server endpoint, in a number of ways.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**GroupList**	list[group-id]	all[min 1]			M

1.3.7.3.1. Effect on Receipt

On receipt of the GetGroupMembership command, the server SHALL respond with group membership information using the GetGroupMembershipResponse command as follows:

If the GroupList field is empty, the server SHALL respond with all group IDs indicating the groups of which the server endpoint is a member.

If the GroupList field contains at least one group ID indicating a group of which the server endpoint is a member, the server SHALL respond with each group ID indicating a group of which the server endpoint is a member that matches a group in the GroupList field.

If the GroupList field contains one or more group IDs but does not contain any group ID indicating a group of which the server endpoint is a member, the server SHALL only respond if the command is unicast. The response SHALL return with an empty GroupList field.

1.3.7.4. RemoveGroup Command

The RemoveGroup command allows a client to request that the server removes the membership for

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the server endpoint, if any, in a particular group.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**GroupID**	group-id	min 1			M

1.3.7.4.1. Effect on Receipt

On receipt of the RemoveGroup command, the server SHALL perform the following procedure:

1. If the command fields are not within constraints, the status SHALL be CONSTRAINT_ERROR and the server continues from step 4.
2. If the server endpoint is a member of the group indicated by the GroupID, the server SHALL remove the server endpoint membership in the group, the status SHALL be SUCCESS, and the server continues from step 4.
3. Else the status SHALL be NOT_FOUND.
4. If the RemoveGroup command was received as a unicast, the server SHALL generate a RemoveGroupResponse command with the Status field set to the evaluated status. If the RemoveGroup command was received as a groupcast, the server SHALL NOT generate a RemoveGroupResponse command.

See RemoveGroupResponse for a description of the response command.

Additionally, if the Scenes Management cluster is supported on the same endpoint, scenes associated with the indicated group SHALL be removed on that endpoint.

1.3.7.5. RemoveAllGroups Command

The RemoveAllGroups command allows a client to direct the server to remove all group associations for the server endpoint.

1.3.7.5.1. Effect on Receipt

On receipt of this command, the server SHALL remove all group memberships for the server endpoint from the Group Table. If the RemoveAllGroups command was received as unicast and a response is not suppressed, the server SHALL generate a response with the Status field set to SUCCESS.

Additionally, if the Scenes Management cluster is supported on the same endpoint, all scenes, except for scenes associated with group ID 0, SHALL be removed on that endpoint.

1.3.7.6. AddGroupIfIdentifying Command

The AddGroupIfIdentifying command allows a client to add group membership in a particular group for the server endpoint, on condition that the endpoint is identifying itself. Identifying functionality is controlled using the Identify cluster, (see Identify Cluster).

For correct operation of the AddGroupIfIdentifying command, any endpoint that supports the Groups server cluster SHALL also support the Identify server cluster.

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This command might be used to assist configuring group membership in the absence of a commissioning tool.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**GroupID**	group-id	min 1			M
1	**GroupName**	string	max 16			M

1.3.7.6.1. GroupID Field

This field SHALL be used to identify the group and any associated key material to which the server endpoint is to be added.

1.3.7.6.2. GroupName Field

This field MAY be set to a human-readable name for the group. If the client has no name for the group, the GroupName field SHALL be set to the empty string.

Support of group names is optional and is indicated by the FeatureMap and NameSupport attribute.

1.3.7.6.3. Effect on Receipt

If the server does not support group names, the GroupName field SHALL be ignored.

On receipt of the AddGroupIfIdentifying command, the server SHALL perform the following procedure:

1. The server verifies that it is currently identifying itself. If the server it not currently identifying itself, the status SHALL be SUCCESS, and the server continues from step 7.
2. If the command fields are not within constraints, the status SHALL be CONSTRAINT_ERROR and the server continues from step 7.
3. If the receiving node requires security material to support the group ID, and that material does not exist for this group ID, the status SHALL be UNSUPPORTED_ACCESS and the server continues from step 7.
4. If the server endpoint is a member of the group indicated by the GroupID, the status SHALL be SUCCESS and the server continues from step 7.
5. If there are no available resources to add the membership for the server endpoint, the status SHALL be RESOURCE_EXHAUSTED and the server continues from step 7.
6. The server SHALL add the server endpoint as a member of the group indicated by the GroupID, the group name SHALL be updated (if supported) to GroupName, and the status SHALL be SUCCESS.
    a. If the GroupID had already been added to the Group Table because of a previous AddGroup or AddGroupIfIdentifying command and a GroupName is provided and the server supports GroupName storage, then the GroupName associated with the GroupID in the Group Table SHALL be updated to reflect the new GroupName provided for the Group, such that subsequent ViewGroup commands yield the same name for all endpoints which have a group association to the given GroupID.

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7. If the AddGroupIfIdentifying command was received as unicast and the evaluated status is not SUCCESS, or if the AddGroupIfIdentifying command was received as unicast and the evaluated status is SUCCESS and a response is not suppressed, the server SHALL generate a response with the Status field set to the evaluated status.

1.3.7.7. AddGroupResponse Command

The AddGroupResponse is sent by the Groups cluster server in response to an AddGroup command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	enum8	desc			M
1	**GroupID**	group-id	min 1			M

1.3.7.7.1. Status Field

This field is set according to the Effect on Receipt section of the AddGroup command.

1.3.7.7.2. GroupID Field

This field is set to the GroupID field of the received AddGroup command.

1.3.7.8. ViewGroupResponse Command

The ViewGroupResponse command is sent by the Groups cluster server in response to a ViewGroup command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	enum8	desc			M
1	**GroupID**	group-id	min 1			M
2	**GroupName**	string	max 16			M

1.3.7.8.1. Status Field

This field is according to the Effect on Receipt section of the ViewGroup command.

1.3.7.8.2. GroupID Field

This field is set to the GroupID field of the received ViewGroup command.

1.3.7.8.3. GroupName Field

If the status is SUCCESS, and group names are supported, this field is set to the group name associated with that group in the Group Table; otherwise it is set to the empty string.

1.3.7.9. GetGroupMembershipResponse Command

The GetGroupMembershipResponse command is sent by the Groups cluster server in response to a

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GetGroupMembership command.

The GetGroupMembershipResponse command SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Capacity**	uint8	all	X		M
1	**GroupList**	list[group-id]	all[min 1]			M

1.3.7.9.1. Capacity Field

This field SHALL contain the remaining capacity of the Group Table of the node. The following values apply:

- 0 - No further groups MAY be added.
- 0 < Capacity < 0xFE - Capacity holds the number of groups that MAY be added.
- 0xFE - At least 1 further group MAY be added (exact number is unknown).
- null - It is unknown if any further groups MAY be added.

1.3.7.9.2. GroupList Field

The GroupList field SHALL contain either the group IDs of all the groups in the Group Table for which the server endpoint is a member of the group (in the case where the GroupList field of the received GetGroupMembership command was empty), or the group IDs of all the groups in the Group Table for which the server endpoint is a member of the group *and* for which the group ID was included in the the GroupList field of the received GetGroupMembership command (in the case where the GroupList field of the received GetGroupMembership command was not empty).
Zigbee: If the total number of groups will cause the maximum payload length of a frame to be exceeded, then the GroupList field SHALL contain only as many groups as will fit.

1.3.7.10. RemoveGroupResponse Command

The RemoveGroupResponse command is generated by the server in response to the receipt of a RemoveGroup command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	enum8	desc			M
1	**GroupID**	group-id	min 1			M

1.3.7.10.1. Status Field

This field is according to the Effect on Receipt section of the RemoveGroup command.

1.3.7.10.2. GroupID Field

This field is set to the GroupID field of the received RemoveGroup command.

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1.4. Scenes Management Cluster

The Scenes Management cluster provides attributes and commands for setting up and recalling scenes. Each scene corresponds to a set of stored values of specified attributes for one or more clusters on the same end point as the Scenes Management cluster.

In most cases scenes are associated with a particular group identifier. Scenes MAY also exist without a group, in which case the value 0 replaces the group identifier. Note that extra care is required in these cases to avoid a scene identifier collision, and that commands related to scenes without a group MAY only be unicast, i.e., they SHALL NOT be multicast or broadcast.

1.4.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Based on the ZCL Scenes Cluster, Updated the Cluster ID to 0x0062, the name to Scenes Management, removed the provisional status; Removed attributes SceneCount, CurrentScene, CurrentGroup, SceneValid, and NameSupport; Removed Explicit, TableSize and FabricScenes features; Removed EnhancedAddScene, EnhancedAddSceneResponse, EnhancedViewScene, EnhancedViewSceneResponse; TransitionTime field changed milliseconds in AddScene, ViewSceneResponse and RecallScene

1.4.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	S

1.4.3. Cluster ID

ID	Name
0x0062	Scenes Management

1.4.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

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Bit	Code	Feature	Conformance	Summary
0	**SN**	SceneNames	O	The ability to store a name for a scene

The following sections describe each feature in some detail. Further details are found within the specification.

1.4.4.1. SceneNames Feature

This feature indicates the ability to store a name for a scene when a scene is added.

1.4.5. Dependencies

Any endpoint that implements the Scenes Management server cluster SHALL also implement the Groups server cluster.

Note that the RemoveGroup command and the RemoveAllGroups command of the Groups cluster also remove scenes.

1.4.6. Handling of fabric-scoping

Attributes and commands for this cluster are scoped to the accessing fabric and SHALL only affect or reflect data related to the accessing fabric, with the exception of the SceneValid Field of the FabricSceneInfo attribute.

The following constraints apply in addition to any other stated requirements in individual data model elements:

- Any attribute read, attribute write or command invoked on the server when no accessing fabric is available SHALL fail with a status code of UNSUPPORTED_ACCESS returned to the client.
- When accessing scene information, implementations SHALL ensure that scenes with identical Group ID and Scene ID across fabrics will only access the data for the accessing fabric, so that the same identifier values used by different accessing fabrics do not cause mixing or overwriting of another fabric's scenes.
- Upon leaving a fabric with the RemoveFabric command of the Operational Credentials Cluster, all scenes data for the associated fabric SHALL be removed from the Scene Table.
- The Scene Table capacity for a given fabric SHALL be less than half (rounded down towards 0) of the Scene Table entries (as indicated in the SceneTableSize attribute), with a maximum of 253 entries (to allow expressing it in the GetSceneMembershipResponse command). If the Scene Table capacity is about to be exceeded by adding or storing a scene, then the resource exhaustion behavior of the associated command SHALL apply.

1.4.7. Data Types

1.4.7.1. CopyModeBitmap Type

This data type is derived from map8.

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Bit	Name	Summary	Conformance
0	**CopyAllScenes**	Copy all scenes in the scene table	M

1.4.7.2. SceneInfoStruct Type

Access Modifier: Fabric Scoped							
ID	**Name**	**Type**	**Constraint**	**Quality**	**Fallback**	**Access**	**Conformance**
0	**SceneCount**	uint8	all				M
1	**CurrentScene**	uint8	desc			S	M
2	**CurrentGroup**	group-id	all			S	M
3	**SceneValid**	bool	all			S	M
4	**RemainingCapacity**	uint8	max 253				M

1.4.7.2.1. SceneCount Field

This field SHALL indicate the number of scenes currently used in the server's Scene Table on the endpoint where the Scenes Management cluster appears.

This only includes the count for the associated fabric.

1.4.7.2.2. CurrentScene Field

This field SHALL indicate the scene identifier of the scene last invoked on the associated fabric. If no scene has been invoked, the value of this field SHALL be 0xFF, the undefined scene identifier.

1.4.7.2.3. CurrentGroup Field

This field SHALL indicate the group identifier of the scene last invoked on the associated fabric, or 0 if the scene last invoked is not associated with a group.

1.4.7.2.4. SceneValid Field

This field SHALL indicate whether the state of the server corresponds to that associated with the CurrentScene and CurrentGroup fields of the SceneInfoStruct they belong to. TRUE indicates that these fields are valid, FALSE indicates that they are not valid.

This field SHALL be set to False for all other fabrics when an attribute with the Scenes ("S") designation in the Quality column of another cluster present on the same endpoint is modified or when the current scene is modified by a fabric through the RecallScene or StoreScene commands, regardless

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of the fabric-scoped access quality of the command.

In the event where the SceneValid field is set to False for a fabric, the CurrentScene and CurrentGroup fields SHALL be the last invoked scene and group for that fabric. In the event where no scene was previously invoked for that fabric, the CurrentScene and CurrentGroup fields SHALL be their default values.

1.4.7.2.5. RemainingCapacity Field

This field SHALL indicate the remaining capacity of the Scene Table on this endpoint for the accessing fabric. Note that this value may change between reads, even if no entries are added or deleted on the accessing fabric, due to other clients associated with other fabrics adding or deleting entries that impact the resource usage on the device.

1.4.7.3. AttributeValuePairStruct Type

This data type indicates a combination of an identifier and the value of an attribute.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**AttributeID**	attrib-id	all				M
1	**ValueUnsigned8**	uint8					O.a
2	**ValueSigned8**	int8					O.a
3	**ValueUnsigned16**	uint16					O.a
4	**ValueSigned16**	int16					O.a
5	**ValueUnsigned32**	uint32					O.a
6	**ValueSigned32**	int32					O.a
7	**ValueUnsigned64**	uint64					O.a
8	**ValueSigned64**	int64					O.a

1.4.7.3.1. AttributeID Field

This field SHALL be present for all instances in a given ExtensionFieldSetStruct.

Which Value* field is used SHALL be determined based on the data type of the attribute indicated by AttributeID, as described in the Value* Fields subsection.

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The AttributeID field SHALL NOT refer to an attribute without the Scenes ("S") designation in the Quality column of the cluster specification.

1.4.7.3.2. ValueUnsigned8, ValueSigned8, ValueUnsigned16, ValueSigned16, ValueUnsigned32, ValueSigned32, ValueUnsigned64, ValueSigned64 Fields

These fields SHALL indicate the attribute value as part of an extension field set, associated with a given AttributeID under an ExtensionFieldSetStruct's ClusterID. Which of the fields is used SHALL be determined by the type of the attribute indicated by AttributeID as follows:

- Data types bool, map8, and uint8 SHALL map to ValueUnsigned8.
- Data types int8 SHALL map to ValueSigned8.
- Data types map16 and uint16 SHALL map to ValueUnsigned16.
- Data types int16 SHALL map to ValueSigned16.
- Data types map32, uint24, and uint32 SHALL map to ValueUnsigned32.
- Data types int24 and int32 SHALL map to ValueSigned32.
- Data types map64, uint40, uint48, uint56 and uint64 SHALL map to ValueUnsigned64.
- Data types int40, int48, int56 and int64 SHALL map to ValueSigned64.
- For derived types, the mapping SHALL be based on the base type. For example, an attribute of type percent SHALL be treated as if it were of type uint8, whereas an attribute of type percent100ths SHALL be treated as if it were of type uint16.
- For boolean nullable attributes, any value that is not 0 or 1 SHALL be considered to have the null value.
- For boolean non-nullable attributes, any value that is not 0 or 1 SHALL be considered to have the value FALSE.
- For non-boolean nullable attributes, any value that is not a valid numeric value for the attribute's type after accounting for range reductions due to being nullable and constraints SHALL be considered to have the null value for the type.
- For non-boolean non-nullable attributes, any value that is not a valid numeric value for the attribute's type after accounting for constraints SHALL be considered to be the valid attribute value that is closest to the provided value.
  - In the event that an invalid provided value is of equal numerical distance to the two closest valid values, the lowest of those values SHALL be considered the closest valid attribute value.

If the used field does not match the data type of the attribute indicated by AttributeID, the AttributeValuePairStruct SHALL be considered invalid.

Examples of processing are:

- ColorControl cluster CurrentX (AttributeID 0x0003) has a type of uint16 and is not nullable.
  - ValueUnsigned16 of 0xAB12 would be used as-is, as it is in range.
  - ValueUnsigned16 of 0xFF80 is outside of the range allowed for attribute CurrentX, and would be saturated to the closest valid value, which is the maximum of the attribute's con-

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straint range: 0xFEFF.

- LevelControl cluster CurrentLevel (AttributeID 0x0000) has a type of uint8 and is nullable.
  - ValueUnsigned8 of 0xA1 would be used as-is, as it is in range.
  - ValueUnsigned8 of 0xFF is outside the range allowed for nullable attribute CurrentLevel, and would be considered as the null value.

1.4.7.4. ExtensionFieldSetStruct Type

This data type indicates for a given cluster a set of attributes and their values.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**ClusterID**	cluster-id	all				M
1	**AttributeValueList**	list[AttributeValuePairStruct]	desc				M

1.4.7.4.1. ClusterID Field

This field SHALL indicate the cluster-id of the cluster whose attributes are in the AttributeValueList field.

1.4.7.4.2. AttributeValueList Field

This field SHALL indicate a set of attributes and their values which are stored as part of a scene.

Attributes which do not have the Scenes ("S") designation in the Quality column of their cluster specification SHALL NOT be used in the AttributeValueList field.

1.4.7.4.3. Form of ExtensionFieldSetStruct

The AttributeValuePairStructs in the AttributeValueList MAY be in any order.

The AttributeValueList SHOULD contain all the attributes with the Scenes ("S") quality as the specification of the cluster identified by ClusterID describes, but AttributeValuePairStruct MAY be omitted.

An example using the Color Control cluster (with features HS, XY, CT and EHUE but not CL):

- Attribute 0x0001, CurrentSaturation, S quality, optional based on feature, implemented
- Attribute 0x0003, CurrentX, S quality, optional based on feature, implemented
- Attribute 0x0004, CurrentY, S quality, optional based on feature, implemented
- Attribute 0x0007, ColorTemperatureMireds, S quality, optional based on feature, implemented
- Attribute 0x4000, EnhancedCurrentHue, S quality, optional based on feature, implemented
- Attribute 0x4001, EnhancedColorMode, S quality, mandatory, implemented
- Attribute 0x4002, ColorLoopActive, S quality, optional based on feature, NOT implemented
- Attribute 0x4003, ColorLoopDirection, S quality, optional based on feature, NOT implemented

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- Attribute 0x4004, ColorLoopTime, S quality, optional based on feature, NOT implemented

An ExtensionFieldSetStruct containing an invalid AttributeValuePairStruct SHALL be considered invalid.

1.4.7.5. Logical Scene Table

The Scene Table is used to store information for each scene capable of being invoked on the server. Each scene is defined for a particular group. The Scene Table is defined here as a conceptual illustration to assist in understanding the underlying data to be stored when scenes are defined. Though the Scene Table is defined here using the data model architecture rules and format, the design is not normative.

The Scene table is logically a list of fabric-scoped structs. The logical fields of each Scene Table entry struct are illustrated below. An ExtensionFieldSetStruct MAY be present for each Scenes-supporting cluster implemented on the same endpoint.

Quality: Fabric Scoped							
ID	**Name**	**Type**	**Constraint**	**Quality**	**Fallback**	**Access**	**Conformance**
0	**SceneGroupID**	group-id	all				M
1	**SceneID**	uint8	max 254				M
2	**SceneName**	string	max 16				SN
3	**SceneTransitionTime**	uint32	max 60000000		0		M
4	**ExtensionFields**	list[ExtensionFieldSetStruct]			empty		M

1.4.7.5.1. SceneGroupID Field

This field is the group identifier for which this scene applies, or 0 if the scene is not associated with a group.

1.4.7.5.2. SceneID Field

This field is unique within this group, which is used to identify this scene.

1.4.7.5.3. SceneName Field

The field is the name of the scene.

If scene names are not supported, any commands that write a scene name SHALL simply discard the name, and any command that returns a scene name SHALL return an empty string.

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1.4.7.5.4. SceneTransitionTime Field

This field is the amount of time, in milliseconds, it will take for a cluster to change from its current state to the requested state.

1.4.7.5.5. ExtensionFields Field

See the Scene Table Extensions subsections of individual clusters. A Scene Table Extension SHALL only use attributes with the Scene quality. Each ExtensionFieldSetStruct holds a set of values of these attributes for a cluster implemented on the same endpoint where the Scene ("S") designation appears in the quality column. A scene is the aggregate of all such fields across all clusters on the endpoint.

1.4.8. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**DoNotUse**					R V	X
0x0001	**SceneTableSize**	uint16	desc	F		R V	M
0x0002	**FabricSceneInfo**	list[SceneInfoStruct]	desc			R V F	M

1.4.8.1. SceneTableSize Attribute

This attribute SHALL indicate the number of entries in the Scene Table on this endpoint. This is the total across all fabrics; note that a single fabric cannot use all those entries (see Handling of fabric-scoping). The minimum size of this table, (i.e., the minimum number of scenes to support across all fabrics per endpoint) SHALL be 16, unless a device type in which this cluster is used, defines a larger value in the device type definition.

1.4.8.2. FabricSceneInfo Attribute

This attribute SHALL indicate a list of fabric scoped information about scenes on this endpoint.

The number of list entries for this attribute SHALL NOT exceed the number of supported fabrics by the device.

1.4.9. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**AddScene**	client ⇒ server	AddSceneResponse	M F	M
0x00	**AddSceneResponse**	client ⇐ server	N		M
0x01	**ViewScene**	client ⇒ server	ViewSceneResponse	O F	M

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ID	Name	Direction	Response	Access	Conformance
0x01	**ViewSceneResponse**	client ⇐ server	N		M
0x02	**RemoveScene**	client ⇒ server	RemoveSceneResponse	M F	M
0x02	**RemoveSceneResponse**	client ⇐ server	N		M
0x03	**RemoveAllScenes**	client ⇒ server	RemoveAllScenesResponse	M F	M
0x03	**RemoveAllScenesResponse**	client ⇐ server	N		M
0x04	**StoreScene**	client ⇒ server	StoreSceneResponse	M F	M
0x04	**StoreSceneResponse**	client ⇐ server	N		M
0x05	**RecallScene**	client ⇒ server	Y	O F	M
0x06	**GetSceneMembership**	client ⇒ server	GetSceneMembershipResponse	O F	M
0x06	**GetSceneMembershipResponse**	client ⇐ server	N		M
0x40	**CopyScene**	client ⇒ server	CopySceneResponse	M F	O
0x40	**CopySceneResponse**	client ⇐ server	N		CopyScene

1.4.9.1. Generic Usage Notes

The scene identifier 0, when used with group identifier 0, is reserved for the global scene used by the On/Off cluster.

1.4.9.2. AddScene Command

This command is used to add a new scene.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**GroupID**	group-id	all			M
1	**SceneID**	uint8	max 254			M
2	**TransitionTime**	uint32	max 60000000			M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
3	**SceneName**	string	max 16			M
4	**ExtensionFieldSetStructs**	list[ExtensionFieldSetStruct]	desc			M

It is not mandatory for an extension field set to be included in the command for every cluster on that endpoint that has a defined extension field set. Extension field sets MAY be omitted, including the case of no extension field sets at all.

1.4.9.2.1. GroupID Field

This field SHALL indicate the group identifier in the Group Table.

1.4.9.2.2. SceneID Field

This field SHALL indicate the scene identifier in the Scene Table.

1.4.9.2.3. TransitionTime Field

This field SHALL indicate the transition time of the scene, measured in milliseconds.

1.4.9.2.4. SceneName Field

This field SHALL indicate the name of the scene.

1.4.9.2.5. ExtensionFieldSetStructs Field

This field SHALL contains the list of extension fields.

1.4.9.2.6. Effect on Receipt

On receipt of this command, the server SHALL perform the following procedure:

1. If the value of the GroupID field is non-zero, the server verifies that the endpoint has an entry for that GroupID in the Group Table. If there is no such entry in the Group Table, the status SHALL be INVALID_COMMAND and the server continues from step 6.
2. If the ExtensionFieldSetStructs list is formatted in a way deemed invalid according to the ExtensionFieldSetStruct structure definition (see ExtensionFieldSetStruct), then the status SHALL be INVALID_COMMAND in the AddSceneResponse, and the server continues from step 6.
3. For each cluster referenced in the ExtensionFieldSetStructs and which is supported on the endpoint, the data related to that cluster in ExtensionFieldSetStructs is checked for internal consistency. The related checks are documented in the corresponding cluster specification. If one or more of these checks fails, then the status SHALL be INVALID_COMMAND in the AddSceneResponse, and the server continues from step 6.
4. If a scene already exists under the same group/scene identifier pair, in step 5 the already existing scene entry SHALL be replaced with the new scene being added. Otherwise, if a scene cannot be created due to the lack of Scene Table capacity for the given fabric, the status SHALL be

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RESOURCE_EXHAUSTED and the server continues from step 6.

5. The server adds the scene entry into its Scene Table with fields copied from the AddScene command data fields and the status SHALL be SUCCESS.
   a. Any ExtensionFieldSetStruct referencing a ClusterID that is not implemented on the endpoint MAY be omitted during processing.
   b. Any AttributeValuePairStruct referencing an AttributeID from the referenced cluster that is not implemented on the endpoint MAY be omitted during processing.
   c. If the ExtensionFieldSetStructs list has multiple entries listing the same ClusterID, the last one within the list SHALL be the one recorded.
   d. Within a single entry of the ExtensionFieldSetStructs list, if an ExtensionFieldSet contains the same AttributeID more than once, the last one within the ExtensionFieldSet SHALL be the one recorded.
6. If the AddScene command was received as a unicast, the server SHALL then generate an AddSceneResponse command with the Status field set to the evaluated status.

The SceneTransitionTime field of the Scene Table SHALL be updated with the value of the TransitionTime.

1.4.9.3. AddSceneResponse Command

This command is sent in response to the AddScene command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	status	desc			M
1	**GroupID**	group-id	all			M
2	**SceneID**	uint8	max 254			M

1.4.9.3.1. Status Field

This field SHALL be set according to the Effect on Receipt section for AddScene command.

1.4.9.3.2. GroupID Field

The GroupID field SHALL be set to the corresponding field of the received AddScene command.

1.4.9.3.3. SceneID Field

The SceneID field SHALL be set to the corresponding field of the received AddScene command.

1.4.9.3.4. When Generated

This command is generated in response to a received AddScene command.

1.4.9.4. ViewScene Command

This command returns the details for the requested scene.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**GroupID**	group-id	all			M
1	**SceneID**	uint8	max 254			M

1.4.9.4.1. GroupID Field

This field SHALL indicate the group identifier in the Group Table.

1.4.9.4.2. SceneID Field

This field SHALL indicate the scene identifier in the Scene Table.

1.4.9.4.3. Effect on Receipt

On receipt of this command, the server SHALL perform the following procedure:

1. If the value of the GroupID field is non-zero, the server verifies that the endpoint has an entry for that GroupID in the Group Table. If there is no such entry in the Group Table, the status SHALL be INVALID_COMMAND and the server continues from step 4.
2. The server verifies that the scene entry corresponding to the GroupID and SceneID fields exists in its Scene Table. If there is no such entry in its Scene Table, the status SHALL be NOT_FOUND and the server continues from step 4.
3. The server retrieves the requested scene entry from its Scene Table and the status SHALL be SUCCESS.
4. If the ViewScene command was received as a unicast, the server SHALL then generate a ViewSceneResponse command with the retrieved scene entry and the Status field set to the evaluated status.

NOTE	The order of attributes within the ExtensionFieldSetStruct MAY differ in implementation-defined ways from any potential order given in a prior AddScene.

1.4.9.5. ViewSceneResponse Command

This command is sent in response to the ViewScene command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	status	desc			M
1	**GroupID**	group-id	all			M
2	**SceneID**	uint8	max 254			M
3	**TransitionTime**	uint32	max 60000000			Status == SUCCESS
4	**SceneName**	string	max 16			Status == SUCCESS

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
5	**ExtensionFieldSetStructs**	list[ExtensionFieldSetStruct]				Status == SUCCESS

1.4.9.5.1. Status Field

This field SHALL be set according to the Effect on Receipt section for ViewScene command.

1.4.9.5.2. GroupID Field

The GroupID field SHALL be set to the corresponding field of the received ViewScene command.

1.4.9.5.3. SceneID Field

The SceneID field SHALL be set to the corresponding field of the received ViewScene command.

1.4.9.5.4. TransitionTime Field

This field SHALL be set to the corresponding field in the Scene Table entry.

1.4.9.5.5. SceneName Field

This field SHALL be set to the corresponding field in the Scene Table entry.

1.4.9.5.6. ExtensionFieldSetStructs Field

This field SHALL be set to the corresponding field in the Scene Table entry.

1.4.9.5.7. When Generated

This command is generated in response to a received ViewScene command.

1.4.9.6. RemoveScene Command

This command removes the specified scene.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**GroupID**	group-id	all			M
1	**SceneID**	uint8	max 254			M

1.4.9.6.1. GroupID Field

This field SHALL indicate the group identifier in the Group Table.

1.4.9.6.2. SceneID Field

This field SHALL indicate the scene identifier in the Scene Table.

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1.4.9.6.3. Effect on Receipt

On receipt of this command, the server SHALL perform the following procedure:

1. If the value of the GroupID field is non-zero, the server verifies that the endpoint has an entry for that GroupID in the Group Table. If there is no such entry in the Group Table, the status SHALL be INVALID_COMMAND and the server continues from step 4.
2. The server verifies that the scene entry corresponding to the GroupID and SceneID fields exists in its Scene Table. If there is no such entry in its Scene Table, the status SHALL be NOT_FOUND and the server continues from step 4.
3. The server removes the requested scene entry from its Scene Table and the status SHALL be SUCCESS.
4. If the RemoveScene command was received as a unicast, the server SHALL then generate a RemoveSceneResponse command with the Status field set to the evaluated status.

1.4.9.7. RemoveSceneResponse Command

This command is sent in response to the RemoveScene command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	status	desc			M
1	**GroupID**	group-id	all			M
2	**SceneID**	uint8	max 254			M

1.4.9.7.1. Status Field

This field SHALL be set according to the Effect on Receipt section for RemoveScene command.

1.4.9.7.2. GroupID Field

The GroupID field SHALL be set to the corresponding field of the received RemoveScene command.

1.4.9.7.3. SceneID Field

The SceneID field SHALL be set to the corresponding field of the received RemoveScene command.

1.4.9.7.4. When Generated

This command is generated in response to a received RemoveScene command.

1.4.9.8. RemoveAllScenes Command

This command removes all scenes for the specified GroupID.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**GroupID**	group-id	all			M

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1.4.9.8.1. GroupID Field

This field SHALL indicate the group identifier in the Group Table.

1.4.9.8.2. Effect on Receipt

On receipt of this command, the server SHALL perform the following procedure:

1. If the value of the GroupID field is non-zero, the server verifies that the endpoint has an entry for that GroupID in the Group Table. If there is no such entry in the Group Table, the status SHALL be INVALID_COMMAND and the server continues from step 3.
2. The server SHALL remove all scenes, corresponding to the value of the GroupID field, from its Scene Table and the status SHALL be SUCCESS.
3. If the RemoveAllScenes command was received as a unicast, the server SHALL then generate a RemoveAllScenesResponse command with the Status field set to the evaluated status.

1.4.9.9. RemoveAllScenesResponse Command

This command is sent in response to the RemoveAllScenes command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	status	desc			M
1	**GroupID**	group-id	all			M

1.4.9.9.1. Status Field

This field SHALL be set according to the Effect on Receipt section for RemoveAllScenes command.

1.4.9.9.2. GroupID Field

The GroupID field SHALL be set to the corresponding field of the received RemoveAllScenes command.

1.4.9.9.3. When Generated

This command is generated in response to a received RemoveAllScenes command.

1.4.9.10. StoreScene Command

This command will create a new entry for the requested scene.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**GroupID**	group-id	all			M
1	**SceneID**	uint8	max 254			M

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1.4.9.10.1. GroupID Field

This field SHALL indicate the group identifier in the Group Table.

1.4.9.10.2. SceneID Field

This field SHALL indicate the scene identifier in the Scene Table.

1.4.9.10.3. Effect on Receipt

On receipt of this command, the server SHALL perform the following procedure:

1. If the value of the GroupID field is non-zero, the server verifies that the endpoint has an entry for that GroupID in the Group Table. If there is no such entry in the Group Table, the status SHALL be INVALID_COMMAND and the server continues from step 4.
2. If a scene already exists under the same group/scene identifier pair, in step 3 the already existing scene entry SHALL be used to store the current state.
   If a scene cannot be created due to the lack of Scene Table capacity for the given fabric, the status SHALL be RESOURCE_EXHAUSTED and the server continues from step 4.
3. If a scene already exists under the same group/scene identifier pair, the ExtensionFieldSets of the stored scene SHALL be replaced with the ExtensionFieldSets corresponding to the current state of other clusters on the same endpoint and the other fields of the scene table entry SHALL remain unchanged.
   Otherwise, a new entry SHALL be added to the scene table, using the provided GroupID and SceneID, with SceneTransitionTime set to 0, with SceneName set to the empty string, and with ExtensionFieldSets corresponding to the current state of other clusters on the same endpoint.
   The status SHALL be SUCCESS.
4. If the StoreScene command was received as a unicast, the server SHALL then generate a StoreSceneResponse command with the Status field set to the evaluated status.

Note that if a scene to be stored requires a TransitionTime field and/or a SceneName field, these must be set up by a prior AddScene command, e.g., with no scene extension field sets.

1.4.9.11. StoreSceneResponse Command

This command is sent in response to the StoreScene command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	status	desc			M
1	**GroupID**	group-id	all			M
2	**SceneID**	uint8	max 254			M

1.4.9.11.1. Status Field

This field SHALL be set according to the Effect on Receipt section for StoreScene command.

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1.4.9.11.2. GroupID Field

The GroupID field SHALL be set to the corresponding field of the received StoreScene command.

1.4.9.11.3. SceneID Field

The SceneID field SHALL be set to the corresponding field of the received StoreScene command.

1.4.9.11.4. When Generated

This command is generated in response to a received StoreScene command.

1.4.9.12. RecallScene Command

This command will cause the specified scene to be recalled.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**GroupID**	group-id	all			M
1	**SceneID**	uint8	max 254			M
2	**TransitionTime**	uint32	max 60000000	X		O

1.4.9.12.1. GroupID Field

This field SHALL indicate the group identifier in the Group Table.

1.4.9.12.2. SceneID Field

This field SHALL indicate the scene identifier in the Scene Table.

1.4.9.12.3. TransitionTime Field

This field SHALL indicate the transition time of the scene, measured in milliseconds.

1.4.9.12.4. Effect on Receipt

On receipt of this command, the server SHALL perform the following procedure:

1. If the value of the GroupID field is non-zero, the server verifies that the endpoint has an entry for that GroupID in the Group Table. If there is no such entry in the Group Table, the status SHALL be INVALID_COMMAND and the server continues from step 4.
2. The server verifies that the scene entry corresponding to the GroupID and SceneID fields exists in its Scene Table. If there is no such entry in its Scene Table, the status SHALL be NOT_FOUND and the server continues from step 4.
3. The server retrieves the requested scene entry from its Scene Table. For each other cluster implemented on the endpoint, it SHALL retrieve any corresponding extension field sets from the Scene Table and set the attributes and corresponding state of the cluster accordingly. If there is no extension field set for a cluster, the state of that cluster SHALL remain unchanged. If

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an extension field set omits the values of any attributes, the values of these attributes SHALL remain unchanged. If an extension field set would cause an unknown or missing attribute to be set for any reason, that attribute SHALL be skipped. The status SHALL be SUCCESS.

4. If the RecallScene command was received as a unicast, the server SHALL then generate a response with the Status field set to the evaluated status.

If the TransitionTime data field is present in the command and its value is not equal to null, this field SHALL indicate the transition time in milliseconds. In all other cases (command data field not present or value equal to null), the SceneTransitionTime field of the Scene Table entry SHALL indicate the transition time. The transition time determines how long the transition takes from the old cluster state to the new cluster state. It is recommended that, where possible (e.g., it is not possible for attributes with Boolean data type), a gradual transition SHOULD take place from the old to the new state over this time. However, the exact transition algorithm is implementation-defined.

1.4.9.13. GetSceneMembership Command

This command can be used to get the used scene identifiers within a certain group, for the endpoint that implements this cluster.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**GroupID**	group-id	all			M

1.4.9.13.1. GroupID Field

This field SHALL indicate the group identifier in the Group Table.

1.4.9.13.2. Effect on Receipt

On receipt of this command, the server SHALL perform the following procedure:

1. If the value of the GroupID field is non-zero, the server verifies that the endpoint has an entry for that GroupID in the Group Table. If there is no such entry in the Group Table, the status SHALL be INVALID_COMMAND and the server continues from step 3.
2. The status SHALL be SUCCESS.
3. If the GetSceneMembership command was received as a unicast, the server SHALL then generate a GetSceneMembershipResponse command with the Status field set to the evaluated status.

1.4.9.14. GetSceneMembershipResponse Command

This command is sent in response to the GetSceneMembership command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	status	desc			M
1	**Capacity**	uint8	all	X		M
2	**GroupID**	group-id	all			M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
3	**SceneList**	list[uint8]				Status == Success

1.4.9.14.1. Status Field

This field SHALL be set according to the Effect on Receipt section for GetSceneMembership command.

1.4.9.14.2. Capacity Field

This field SHALL contain the remaining capacity of the Scene Table of the server (for all groups for the accessing fabric). The following values apply:

- 0 - No further scenes MAY be added.
- 0 < Capacity < 0xFE - Capacity holds the number of scenes that MAY be added.
- 0xFE - At least 1 further scene MAY be added (exact number is unknown).
- null - It is unknown if any further scenes MAY be added.

1.4.9.14.3. GroupID Field

This field SHALL be set to the corresponding field of the received GetSceneMembership command.

1.4.9.14.4. SceneList Field

If the status is not SUCCESS then this field SHALL be omitted, else this field SHALL contain the identifiers of all the scenes in the Scene Table with the corresponding Group ID.

1.4.9.14.5. When Generated

This command is generated in response to a received GetSceneMembership command.

1.4.9.15. CopyScene Command

This command allows a client to efficiently copy scenes from one group/scene identifier pair to another group/scene identifier pair.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Mode**	CopyModeBitmap	desc			M
1	**GroupIdentifierFrom**	group-id	all			M
2	**SceneIdentifierFrom**	uint8	max 254			M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
3	**GroupIdentifierTo**	group-id	all			M
4	**SceneIdentifierTo**	uint8	max 254			M

1.4.9.15.1. Mode Field

This field SHALL contain the information of how the scene copy is to proceed.

The CopyAllScenes bit of the Mode indicates whether all scenes are to be copied. If this value is set to 1, all scenes are to be copied and the SceneIdentifierFrom and SceneIdentifierTo fields SHALL be ignored. Otherwise this bit is set to 0.

1.4.9.15.2. GroupIdentifierFrom Field

This field SHALL indicate the identifier of the group from which the scene is to be copied. Together with the SceneIdentifierFrom field, this field uniquely identifies the scene to copy from the Scene Table.

1.4.9.15.3. SceneIdentifierFrom Field

This field SHALL indicate the identifier of the scene from which the scene is to be copied. Together with the GroupIdentifierFrom field, this field uniquely identifies the scene to copy from the Scene Table.

1.4.9.15.4. GroupIdentifierTo Field

This field SHALL indicate the identifier of the group to which the scene is to be copied. Together with the SceneIdentifierTo field, this field uniquely identifies the scene to copy to the Scene Table.

1.4.9.15.5. SceneIdentifierTo Field

This field SHALL indicate the identifier of the scene to which the scene is to be copied. Together with the GroupIdentifierTo field, this field uniquely identifies the scene to copy to the Scene Table.

1.4.9.15.6. Effect on Receipt

On receipt of this command, the server SHALL perform the following procedure:

1. If the value of either the GroupIdentifierFrom field or the Group Identifier To field is non-zero, the server verifies that the endpoint has an entry for these non-zero group identifiers in the Group Table. If there are no such entries in the Group Table, the status SHALL be INVALID_COMMAND and the server continues from step 5.
2. If the CopyAllScenes sub-field of the Mode field is set to 0, the server verifies that the scene entry corresponding to the GroupIdentifierFrom and SceneIdentifierFrom fields exists in its Scene Table. If there is no such entry in its Scene Table, the status SHALL be NOT_FOUND and the server continues from step 5.

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3. If the CopyAllScenes sub-field of the Mode field is set to 1, the server SHALL copy all its available scenes with group identifier equal to the GroupIdentifierFrom field under the group identifier specified in the GroupIdentifierTo field, leaving the scene identifiers the same. In this case, the SceneIdentifierFrom and SceneIdentifierTo fields SHALL be ignored.
   If the CopyAllScenes sub-field of the Mode field is set to 0, the server SHALL copy the Scene Table entry corresponding to the GroupIdentifierFrom and SceneIdentifierFrom fields to the Scene Table entry corresponding to the GroupIdentifierTo and SceneIdentifierTo fields.
   Regardless of the value of the CopyAllScenes subfield, if a scene already exists under the same group/scene identifier pair, it SHALL be replaced with the scene being copied.
   Regardless of the value of the CopyAllScenes subfield, if a scene cannot be copied due to the lack of Scene Table capacity for the given fabric, the status SHALL be RESOURCE_EXHAUSTED and the server continues from step 5. In this case, scenes already copied SHALL be kept.
4. The status SHALL be SUCCESS.
5. If the CopyScene command was received as a unicast, the server SHALL then generate a CopySceneResponse command with the Status field set to the evaluated status.

1.4.9.16. CopySceneResponse Command

This command is sent in response to the CopyScene command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	status	desc			M
1	**GroupIdentifierFrom**	group-id	all			M
2	**SceneIdentifierFrom**	uint8	max 254			M

1.4.9.16.1. Status Field

This field SHALL be set according to the Effect on Receipt section for the CopyScene command.

1.4.9.16.2. GroupIdentifierFrom Field

This field SHALL be set to the same values as in the corresponding fields of the received CopyScene command.

1.4.9.16.3. SceneIdentifierFrom Field

This field SHALL be set to the same values as in the corresponding fields of the received CopyScene command.

1.4.9.16.4. When Generated

The CopySceneResponse command is generated in response to a received CopyScene command.

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1.5. On/Off Cluster

Attributes and commands for turning devices on and off.

1.5.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added; CCB 1555
2	ZLO 1.0: StartUpOnOff
3	FeatureMap global attribute support with Level Control and Lighting feature
4	New data model format and notation
5	Addition of Dead Front behavior and associated FeatureMap entry
6	Addition of OffOnly feature and associated FeatureMap entry

1.5.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	OO

1.5.3. Cluster ID

ID	Name
0x0006	On/Off

1.5.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**LT**	Lighting	[!OFFONLY]	Behavior that supports lighting applications
1	**DF**	DeadFrontBehavior	[!OFFONLY]	Device has Dead Front behavior

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Bit	Code	Feature	Conformance	Summary
2	**OFFONLY**	OffOnly	[!(LT | DF)]	Device supports the OffOnly feature

1.5.4.1. Lighting Feature

This cluster is used for a lighting application.

On receipt of a Level Control cluster command that causes the OnOff attribute to be set to FALSE, the OnTime attribute SHALL be set to 0.

On receipt of a Level Control cluster command that causes the OnOff attribute to be set to TRUE, if the value of the OnTime attribute is equal to 0, the server SHALL set the OffWaitTime attribute to 0.

1.5.4.2. DeadFrontBehavior Feature

When this feature is supported, the device exposing this server cluster exhibits "dead front" behavior when the "OnOff" attribute is FALSE (Off). This "dead front" behavior includes:

- clusters other than this cluster that are also exposed MAY respond with failures to Invoke and Write interactions. Such failure responses when in a "dead front" SHALL be with an INVALID_IN_STATE status code.
- clusters other than this cluster MAY change the values of their attributes to best-effort values, due to the actual values not being defined or available in this state. Device type specifications that require support for the DF feature SHOULD define what these best-effort values are.
- Report Transactions SHALL continue to be generated. Such transactions MAY include best-effort values as noted above.
- Event generation logic for clusters other than this cluster is unchanged (noting possible use of best-effort attribute values as in the preceding bullets).

When this feature is supported and the OnOff attribute changes from TRUE to FALSE (e.g. when receiving an Off Command, or due to a manual interaction on the device), it SHALL start executing this "dead front" behavior.

When this feature is supported and the OnOff attribute changes from FALSE to TRUE (e.g. when receiving an On Command, or due to a manual interaction on the device), it SHALL stop executing this "dead front" behavior.

When this feature is supported, and any change of the "dead front" state leads to changes in attributes of other clusters due to the "dead front" feature, these attribute changes SHALL NOT be skipped or omitted from the usual processing associated with attribute changes. For example, if an attribute changes from value 4 to null on "dead front" behavior due to an Off command being received, this change SHALL be processed for reporting and subscriptions.

1.5.4.3. OffOnly Feature

When this feature is supported, the Off command SHALL be supported and the On and Toggle com-

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mands SHALL NOT be supported.

This feature is useful for devices which can be turned off via the Off command received by an instance of this cluster but cannot be turned on via commands received by an instance of this cluster due to regulatory requirements.

1.5.5. Data Types

1.5.5.1. OnOffControlBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**AcceptOnlyWhenOn**	Indicates a command is only accepted when in On state.	M

1.5.5.2. StartUpOnOffEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Off**	Set the OnOff attribute to FALSE	M
1	**On**	Set the OnOff attribute to TRUE	M
2	**Toggle**	If the previous value of the OnOff attribute is equal to FALSE, set the OnOff attribute to TRUE. If the previous value of the OnOff attribute is equal to TRUE, set the OnOff attribute to FALSE (toggle).	M

1.5.5.3. EffectIdentifierEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0x00	**DelayedAllOff**	Delayed All Off	M
0x01	**DyingLight**	Dying Light	M

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1.5.5.4. DelayedAllOffEffectVariantEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0x00	**DelayedOffFastFade**	Fade to off in 0.8 seconds	M
0x01	**NoFade**	No fade	M
0x02	**DelayedOffSlowFade**	50% dim down in 0.8 seconds then fade to off in 12 seconds	M

1.5.5.5. DyingLightEffectVariantEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0x00	**DyingLightFadeOff**	20% dim up in 0.5s then fade to off in 1 second	M

1.5.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**OnOff**	bool	all	N S		R V	M
0x4000	**GlobalSceneControl**	bool	all			R V	LT
0x4001	**OnTime**	uint16	all			RW VO	LT
0x4002	**OffWaitTime**	uint16	all			RW VO	LT
0x4003	**StartUpOnOff**	StartUpOnOffEnum	desc	N X		RW VM	LT

1.5.6.1. Scene Table Extensions

If the Scenes Management server cluster is implemented on the same endpoint, the following attribute SHALL be part of the ExtensionFieldSetStruct of the Scene Table.

- OnOff

1.5.6.2. OnOff Attribute

This attribute indicates whether the device type implemented on the endpoint is turned off or

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turned on, in these cases the value of the OnOff attribute equals FALSE, or TRUE respectively.

1.5.6.3. GlobalSceneControl Attribute

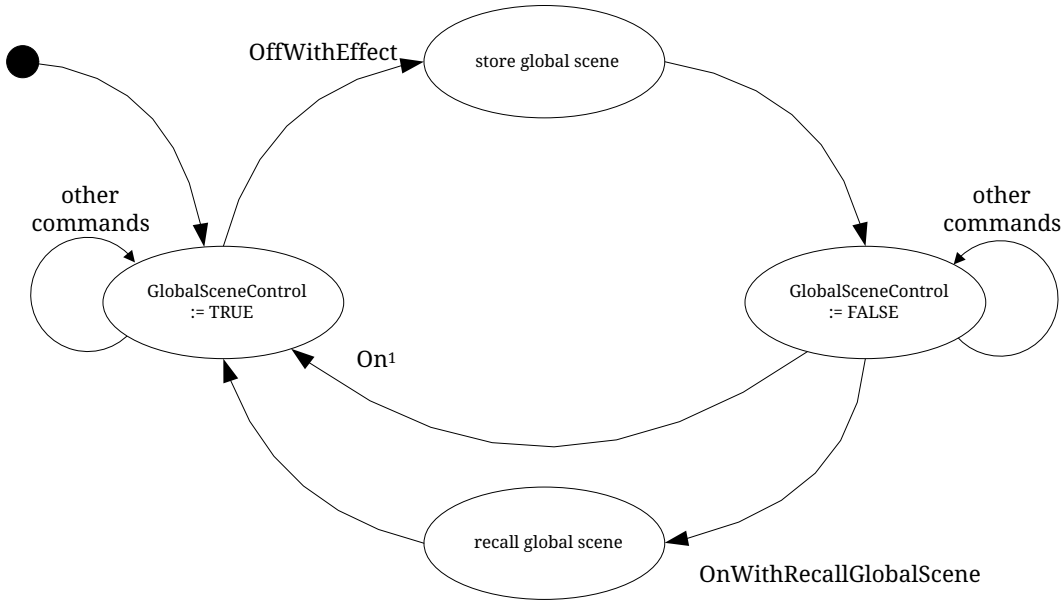
In order to support the use case where the user gets back the last setting of a set of devices (e.g. level settings for lights), a global scene is introduced which is stored when the devices are turned off and recalled when the devices are turned on. The global scene is defined as the scene that is stored with group identifier 0 and scene identifier 0.

This attribute is defined in order to prevent a second Off command storing the all-devices-off situation as a global scene, and to prevent a second On command destroying the current settings by going back to the global scene.

This attribute SHALL be set to TRUE after the reception of a command which causes the OnOff attribute to be set to TRUE, such as a standard On command, a MoveToLevel(WithOnOff) command, a RecallScene command or a OnWithRecallGlobalScene command.

This attribute is set to FALSE after reception of an OffWithEffect command.

These concepts are illustrated in Explanation of the Behavior of Store and Recall Global Scene functionality using a State Diagram.



Note 1: Any command which causes the OnOff attribute to be set to TRUE except OnWithRecallGlobalScene, e.g. On or Toggle.

Figure 1. Explanation of the Behavior of Store and Recall Global Scene functionality using a State Diagram

1.5.6.4. OnTime Attribute

This attribute specifies the length of time (in 1/10ths second) that the On state SHALL be maintained before automatically transitioning to the Off state when using the OnWithTimedOff command.

This attribute can be written at any time, but writing a value only has an effect when in the Timed On state.

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While in the Timed On state, this attribute SHALL be decremented every 1/10th second until it reaches 0. See OnWithTimedOff for more details.

1.5.6.5. OffWaitTime Attribute

This attribute specifies the length of time (in 1/10ths second) that the Off state SHALL be guarded to prevent another OnWithTimedOff command turning the server back to its On state (e.g., when leaving a room, the lights are turned off but an occupancy sensor detects the leaving person and attempts to turn the lights back on).

This attribute can be written at any time, but writing a value only has an effect when in the Timed On state followed by a transition to the Delayed Off state, or in the Delayed Off state.

While in the Off state after a Timed On state, this attribute SHALL be decremented every 1/10th second until it reaches 0. See OnWithTimedOff for more details.

1.5.6.6. StartUpOnOff Attribute

This attribute SHALL define the desired startup behavior of a device when it is supplied with power and this state SHALL be reflected in the OnOff attribute. If the value is null, the OnOff attribute is set to its previous value. Otherwise, the behavior is defined in the table defining StartUpOnOffEnum.

This behavior does not apply to reboots associated with OTA. After an OTA restart, the OnOff attribute SHALL return to its value prior to the restart.

1.5.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**Off**	client ⇒ server	Y	O	M
0x01	**On**	client ⇒ server	Y	O	!OFFONLY
0x02	**Toggle**	client ⇒ server	Y	O	!OFFONLY
0x40	**OffWithEffect**	client ⇒ server	Y	O	LT
0x41	**OnWithRecallGlobalScene**	client ⇒ server	Y	O	LT
0x42	**OnWithTimedOff**	client ⇒ server	Y	O	LT

1.5.7.1. Off Command

This command allows devices to be turned off.

1.5.7.1.1. Effect on Receipt

On receipt of the Off command, a server SHALL set the OnOff attribute to FALSE.

Additionally, when the OnTime attribute is supported, the server SHALL set the OnTime attribute to 0.

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1.5.7.2. On Command

This command allows devices to be turned on.

1.5.7.2.1. Effect on Receipt

If the OffOnly feature is supported, on receipt of the On command, an UNSUPPORTED_COMMAND failure status response SHALL be sent. Otherwise, on receipt of the On command, a server SHALL set the OnOff attribute to TRUE.

Additionally, when the OnTime and OffWaitTime attributes are both supported, if the value of the OnTime attribute is equal to 0, the server SHALL set the OffWaitTime attribute to 0.

1.5.7.3. Toggle Command

This command allows devices to be toggled.

1.5.7.3.1. Effect on Receipt

If the OffOnly feature is supported, on receipt of the Toggle command, an UNSUPPORTED_COMMAND failure status response SHALL be sent. Otherwise, on receipt of the Toggle command, if the value of the OnOff attribute is equal to FALSE, the server SHALL set the OnOff attribute to TRUE, otherwise, the server SHALL set the OnOff attribute to FALSE.

Additionally, when the OnTime and OffWaitTime attributes are both supported, if the value of the OnOff attribute is equal to FALSE and if the value of the OnTime attribute is equal to 0, the server SHALL set the OffWaitTime attribute to 0. If the value of the OnOff attribute is equal to TRUE, the server SHALL set the OnTime attribute to 0.

1.5.7.4. OffWithEffect Command

The OffWithEffect command allows devices to be turned off using enhanced ways of fading.

The OffWithEffect command SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**EffectIdentifier**	EffectIdentifierEnum	desc			M
1	**EffectVariant**	enum8	desc			M

1.5.7.4.1. EffectIdentifier Field

This field specifies the fading effect to use when turning the device off. This field SHALL contain one of the non-reserved values listed in EffectIdentifierEnum.

1.5.7.4.2. EffectVariant Field

This field is used to indicate which variant of the effect, indicated in the EffectIdentifier field,

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SHOULD be triggered. If the server does not support the given variant, it SHALL use the default variant. This field is dependent on the value of the EffectIdentifier field and SHALL contain one of the non-reserved values listed in either DelayedAllOffEffectVariantEnum or DyingLightEffectVariantEnum.

1.5.7.4.3. Effect on Receipt

On receipt of the OffWithEffect command the server SHALL check the value of the GlobalSceneControl attribute.

If the GlobalSceneControl attribute is equal to TRUE, the server SHALL store its settings in its global scene then set the GlobalSceneControl attribute to FALSE, then set the OnOff attribute to FALSE and if the OnTime attribute is supported set the OnTime attribute to 0.

If the GlobalSceneControl attribute is equal to FALSE, the server SHALL only set the OnOff attribute to FALSE.

1.5.7.5. OnWithRecallGlobalScene Command

This command allows the recall of the settings when the device was turned off.

1.5.7.5.1. Effect on Receipt

On receipt of the OnWithRecallGlobalScene command, if the GlobalSceneControl attribute is equal to TRUE, the server SHALL discard the command.

If the GlobalSceneControl attribute is equal to FALSE, the Scenes Management cluster server on the same endpoint SHALL recall its global scene, updating the OnOff attribute accordingly. The OnOff server SHALL then set the GlobalSceneControl attribute to TRUE.

Additionally, when the OnTime and OffWaitTime attributes are both supported, if the value of the OnTime attribute is equal to 0, the server SHALL set the OffWaitTime attribute to 0.

1.5.7.6. OnWithTimedOff Command

This command allows devices to be turned on for a specific duration with a guarded off duration so that SHOULD the device be subsequently turned off, further OnWithTimedOff commands, received during this time, are prevented from turning the devices back on. Further OnWithTimedOff commands received while the server is turned on, will update the period that the device is turned on.

The OnWithTimedOff command SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**OnOffControl**	OnOffControlBitmap	0 to 1			M
1	**OnTime**	uint16	max 0xFFFE			M
2	**OffWaitTime**	uint16	max 0xFFFE			M

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1.5.7.6.1. OnOffControl Field

This field contains information on how the server is to be operated.

1.5.7.6.1.1. AcceptOnlyWhenOn Bit

This bit specifies whether the OnWithTimedOff command is to be processed unconditionally or only when the OnOff attribute is equal to TRUE. If this sub-field is set to 1, the OnWithTimedOff command SHALL only be accepted if the OnOff attribute is equal to TRUE. If this sub-field is set to 0, the OnWithTimedOff command SHALL be processed unconditionally.

1.5.7.6.2. OnTime Field

This field is used to adjust the value of the OnTime attribute.

1.5.7.6.3. OffWaitTime Field

This field is used to adjust the value of the OffWaitTime attribute.

1.5.7.6.4. Effect on Receipt

On receipt of this command, if the AcceptOnlyWhenOn sub-field of the OnOffControl field is set to 1, and the value of the OnOff attribute is equal to FALSE, the command SHALL be discarded.

If the value of the OffWaitTime attribute is greater than zero and the value of the OnOff attribute is equal to FALSE, then the server SHALL set the OffWaitTime attribute to the minimum of the OffWaitTime attribute and the value specified in the OffWaitTime field.

In all other cases, the server SHALL set the OnTime attribute to the maximum of the OnTime attribute and the value specified in the OnTime field, set the OffWaitTime attribute to the value specified in the OffWaitTime field and set the OnOff attribute to TRUE.

If the values of the OnTime and OffWaitTime attributes are both not equal to 0xFFFF, the server SHALL then update these attributes every 1/10th second until both the OnTime and OffWaitTime attributes are equal to 0, as follows:

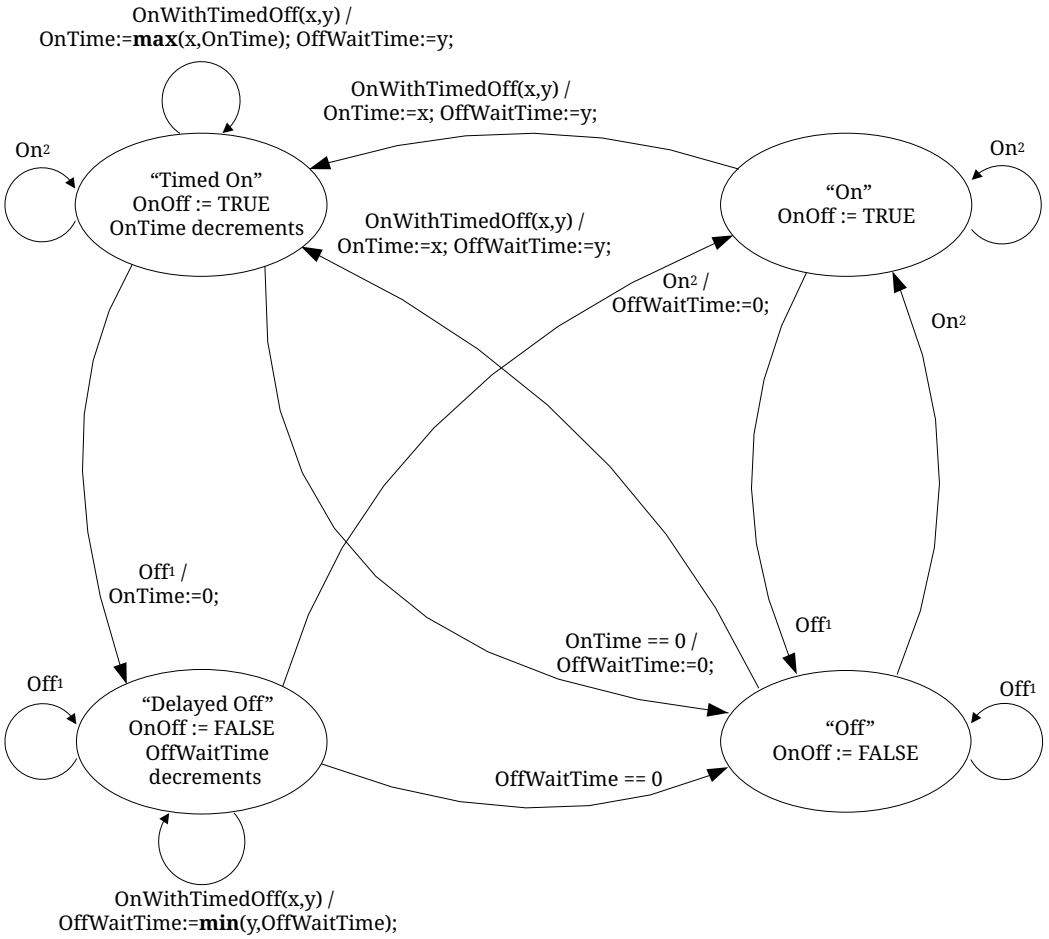
- If the value of the OnOff attribute is equal to TRUE and the value of the OnTime attribute is greater than zero, the server SHALL decrement the value of the OnTime attribute. If the value of the OnTime attribute reaches 0, the server SHALL set the OffWaitTime and OnOff attributes to 0 and FALSE, respectively.
- If the value of the OnOff attribute is equal to FALSE and the value of the OffWaitTime attribute is greater than zero, the server SHALL decrement the value of the OffWaitTime attribute. If the value of the OffWaitTime attribute reaches 0, the server SHALL terminate the update.

1.5.8. State Description

The operation of the On/Off cluster with respect to the On, Off, and OnWithTimedOff commands is illustrated in On/Off Cluster Operation State Machine. In this diagram, the values X and Y correspond to the OnTime and OffWaitTime fields, respectively, of the OnWithTimedOff command. In the Timed On state, the OnTime attribute is decremented every 1/10th second, unless its value equals 0xFFFF. Similarly, in the Delayed Off state, the OffWaitTime attribute is decremented every 1/10th

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second, unless its value equals 0xFFFF.



Off1 : Any command which causes the OnOff attribute to be set to FALSE, e.g. Off, Toggle or OffWithEffect.
On2 : Any command which causes the OnOff attribute to be set to TRUE, e.g. On, Toggle or OnWithRecallGlobalScene.

Figure 2. On/Off Cluster Operation State Machine

1.6. Level Control Cluster

This cluster provides an interface for controlling a characteristic of a device that can be set to a level, for example the brightness of a light or lamp, a pump's flow rate setpoint, etc.

1.6.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added

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Revision	Description
2	Added Options attribute, state change table; ZLO 1.0; Base cluster (no change) CCB 2085 1775 2281 2147
3	CCB 2574 2616 2659 2702 2814 2818 2819 2898
4	FeatureMap support with On/Off, Lighting and Frequency features
5	New data model format and notation
6	Added Q quality for CurrentLevel, CurrentFrequency and RemainingTime attributes, added clarifications for behavior related to Step value and adjusted constraint of DefaultMoveRate
7	Simplify constraint, added F quality, changed the conformance to mandatory for MinLevel and MaxLevel, and removed P quality

1.6.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	LVL

1.6.3. Cluster ID

Derived cluster specifications are defined elsewhere. This base cluster specification MAY be used for generic level control; however, it is recommended to derive another cluster to better define the application and domain requirements. If one of more derived cluster identifiers and the base identifier exists on a device endpoint, then they SHALL all represent a single instance of the device level control.

ID	Name
0x0008	Level Control

1.6.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary	Fallback
0	**OO**	OnOff	O	Dependency with the On/Off cluster	1

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Bit	Code	Feature	Conformance	Summary	Fallback
1	**LT**	Lighting	O	Behavior that supports lighting applications	0
2	**FQ**	Frequency	P	Supports frequency attributes and behavior.	0

The following sections describe each feature in some detail. Further details are found within the specification.

1.6.4.1. On/Off Feature

For many applications, a close relationship between this cluster and the On/Off cluster is needed. This section describes the dependencies that are required when an endpoint that implements this server cluster and also implements the On/Off server cluster. Before the On/Off feature bit in the FeatureMap existed, there was a dependency between this cluster and the On/Off cluster.

The OnOff attribute of the On/Off cluster and the CurrentLevel attribute of the Level Control cluster are intrinsically independent variables, as they are on different clusters. However, when both clusters are implemented on the same endpoint, dependencies MAY be introduced between them. Facilities are provided to introduce dependencies if required.

1.6.4.1.1. Effect of On/Off Commands on the CurrentLevel attribute

The OnLevel attribute determines whether commands of the On/Off cluster have a permanent effect on the CurrentLevel attribute or not. If this attribute is defined (i.e., implemented and not equal to null) they do have a permanent effect, otherwise they do not. There is always a temporary effect, due to fading up / down.

The effect on the Level Control cluster on receipt of the various commands of the On/Off cluster are as detailed in the Effect of On/Off Commands when associated with Level Control table. In this table, and throughout this cluster specification, 'level' means the value of the CurrentLevel attribute.

Table 2. Effect of On/Off Commands when associated with Level Control

Command	Action On Receipt
On	Temporarily store CurrentLevel. Set CurrentLevel to the minimum level allowed for the device. Change CurrentLevel to OnLevel, or to the stored level if OnLevel is not defined, over the time period OnOffTransitionTime.

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Command	Action On Receipt
Off	Temporarily store CurrentLevel. Change CurrentLevel to the minimum level allowed for the device over the time period OnOffTransitionTime. If OnLevel is not defined, set the CurrentLevel to the stored level.
Toggle	If the OnOff attribute has the value FALSE, proceed as for the On command. Otherwise proceed as for the Off command.

Intention of the actions described in the table above is that CurrentLevel, which was in effect before any of the On, Off or Toggle commands were issued, SHALL be restored, after the transition is completed. If another of these commands is received, before the transition is completed, the originally stored CurrentLevel SHALL be preserved and restored.

1.6.4.1.2. Effect of Level Control Commands on the OnOff Attribute

There are two sets of commands provided in the Level Control cluster. These are identical, except that the first set (MoveToLevel, Move and Step commands) SHALL NOT affect the OnOff attribute, whereas the second set ('with On/Off' variants) SHALL.

The first set is used to maintain independence between the CurrentLevel and OnOff attributes, so changing CurrentLevel has no effect on the OnOff attribute. As examples, this represents the behavior of a volume control with a separate mute button, or a 'turn to set level and press to turn on/off' light dimmer.

The second set is used to link the CurrentLevel and OnOff attributes. When the level is reduced to its minimum the OnOff attribute is automatically turned to FALSE, and when the level is increased above its minimum the OnOff attribute is automatically turned to TRUE. As an example, this represents the behavior of a light dimmer with no independent on/off switch.

For the Stop command, the StopWithOnOff is included solely for symmetry, to allow easy choice of one or other set of commands – both Stop commands are identical, because the dependency on On/Off is determined by the original command that is being stopped.

1.6.4.1.3. Effect of Level Control Commands Depends on OnOff

Before the Options attribute was introduced, all commands except those postfixed with ‘with On/Off’, had no effect if the OnOff attribute of the On/Off cluster, on the same endpoint, was FALSE. Even if the On/Off (OO) feature set bit is set to zero, this is still true. To allow such commands to function, please see the Options attribute below.

1.6.4.1.4. GlobalSceneControl and Commands with On/Off

If a MoveToLevel(WithOnOff), Move(WithOnOff) or Step(WithOnOff) command is received that causes a change to the value of the OnOff attribute of the On/Off cluster, the value of the GlobalSceneControl attribute of the On/Off cluster SHALL be updated according to section GlobalSceneControl.

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1.6.4.2. Lighting Feature

This feature supports an interface for controlling the level of a light source.

For the CurrentLevel attribute:

A value of 0x00 SHALL NOT be used.

A value of 0x01 SHALL indicate the minimum level that can be attained on a device.

A value of 0xFE SHALL indicate the maximum level that can be attained on a device.

A value of null SHALL represent an undefined value.

All other values are application specific gradations from the minimum to the maximum level.

1.6.4.3. Frequency Feature

NOTE	The Frequency feature is provisional.

1.6.5. Data Types

1.6.5.1. OptionsBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**ExecuteIfOff**	Dependency on On/Off cluster	LT | OO
1	**CoupleColorTemp-ToLevel**	Dependency on Color Control cluster	LT

1.6.5.1.1. ExecuteIfOff Bit

This bit indicates if this cluster has a dependency with the On/Off cluster.

1.6.5.1.2. CoupleColorTempToLevel Bit

This bit indicates if this cluster has a dependency with the Color Control cluster.

1.6.5.2. MoveModeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Up**	Increase the level	M
1	**Down**	Decrease the level	M

1.6.5.3. StepModeEnum Type

This data type is derived from enum8.

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Value	Name	Summary	Conformance
0	**Up**	Step upwards	M
1	**Down**	Step downwards	M

1.6.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**CurrentLevel**	uint8	MinLevel to MaxLevel	N Q S X		R V	M
0x0001	**RemainingTime**	uint16	all	Q		R V	LT
0x0002	**MinLevel**	uint8	max 254	F	1	R V	Rev >= v7, O
0x0003	**MaxLevel**	uint8	MinLevel to 254	F	254	R V	Rev >= v7, O
0x0004	**CurrentFrequency**	uint16	MinFrequency to MaxFrequency	Q S		R V	FQ
0x0005	**MinFrequency**	uint16	all			R V	FQ
0x0006	**MaxFrequency**	uint16	min MinFrequency			R V	FQ
0x0010	**OnOffTransitionTime**	uint16	all		0	RW VO	O
0x0011	**OnLevel**	uint8	MinLevel to MaxLevel	X		RW VO	M
0x0012	**OnTransitionTime**	uint16	all	X	null	RW VO	O
0x0013	**OffTransitionTime**	uint16	all	X	null	RW VO	O
0x0014	**DefaultMoveRate**	uint8	min 1	X	MS	RW VO	O
0x000F	**Options**	Options-Bitmap	desc			RW VO	M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x4000	**StartUpCurrentLevel**	uint8	desc	N X		RW VM	LT

1.6.6.1. Scene Table Extensions

If the Scenes Management server cluster is implemented on the same endpoint, the following attributes SHALL be part of the ExtensionFieldSetStruct of the Scene Table.

1. CurrentLevel
2. CurrentFrequency

Attributes in the scene table that are not supported by the device (according to the FeatureMap attribute) SHALL be present in the scene table but ignored. An extension field set value outside of the accepted values of the corresponding MoveTo<Target> command, e.g. a null value for the CurrentLevel attribute, corresponding to the MoveToLevel command, SHALL be ignored.

1.6.6.2. CurrentLevel Attribute

This attribute SHALL indicate the current level of this device. The meaning of 'level' is device dependent.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- At most once per second, or
- At the end of the movement/transition, or
- When it changes from null to any other value and vice versa.

1.6.6.3. RemainingTime Attribute

This attribute SHALL indicate the time remaining until the current command is complete - it is specified in 1/10ths of a second.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- When it changes from 0 to any value higher than 10, or
- When it changes, with a delta larger than 10, caused by the invoke of a command, or
- When it changes to 0.

For commands with a transition time or changes to the transition time less than 1 second, changes to this attribute SHALL NOT be reported.

As this attribute is not being reported during a regular countdown, clients SHOULD NOT rely on the reporting of this attribute in order to keep track of the remaining duration.

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1.6.6.4. MinLevel Attribute

This attribute SHALL indicate the minimum value of CurrentLevel that is capable of being assigned.

NOTE	This value is constrained by all lighting device types to 1, and its Conformance is Mandatory. As such, when the Lighting feature is supported this value SHALL be 1.

1.6.6.5. MaxLevel Attribute

This attribute SHALL indicate the maximum value of CurrentLevel that is capable of being assigned.

NOTE	This value is constrained by all lighting device types to 254, and its Conformance is Mandatory. As such, when the Lighting feature is supported this value SHALL be 254.

1.6.6.6. CurrentFrequency Attribute

This attribute SHALL indicate the frequency at which the device is at CurrentLevel. A CurrentFrequency of 0 is unknown.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- At most once per second, or
- At the start of the movement/transition, or
- At the end of the movement/transition.

1.6.6.7. MinFrequency Attribute

This attribute SHALL indicate the minimum value of CurrentFrequency that is capable of being assigned. MinFrequency SHALL be less than or equal to MaxFrequency. A value of 0 indicates undefined.

1.6.6.8. MaxFrequency Attribute

This attribute SHALL indicate the maximum value of CurrentFrequency that is capable of being assigned. MaxFrequency SHALL be greater than or equal to MinFrequency. A value of 0 indicates undefined.

1.6.6.9. Options Attribute

This attribute SHALL indicate the selected options of the device.

The Options attribute is a bitmap that determines the default behavior of some cluster commands. Each command that is dependent on the Options attribute SHALL first construct a temporary Options bitmap that is in effect during the command processing. The temporary Options bitmap has the same format and meaning as the Options attribute, but includes any bits that may be overridden by command fields.

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This attribute is meant to be changed only during commissioning.

Command execution SHALL NOT continue beyond the Options processing if all of these criteria are true:

- The command is one of the ‘without On/Off’ commands: Move, Move to Level, Step, or Stop.
- The On/Off cluster exists on the same endpoint as this cluster.
- The OnOff attribute of the On/Off cluster, on this endpoint, is FALSE.
- The value of the ExecuteIfOff bit is 0.

1.6.6.9.1. ExecuteIfOff Bit

If this bit is set, commands in this cluster are executed and potentially change the CurrentLevel attribute when the OnOff attribute of the On/Off cluster is FALSE.

1.6.6.9.2. CoupleColorTempToLevel Bit

If this bit is set, changes to the CurrentLevel attribute SHALL be coupled with the color temperature set in the Color Control cluster.

When not supporting the Lighting feature, this bit SHALL be zero and ignored.

1.6.6.10. OnOffTransitionTime Attribute

This attribute SHALL indicate the time taken to move to or from the target level when On or Off commands are received by an On/Off cluster on the same endpoint. It is specified in 1/10ths of a second.

The actual time taken SHOULD be as close to OnOffTransitionTime as the device is able. Please note that if the device is not able to move at a variable rate, the OnOffTransitionTime attribute SHOULD NOT be implemented.

1.6.6.11. OnLevel Attribute

This attribute SHALL indicate the value that the CurrentLevel attribute is set to when the OnOff attribute of an On/Off cluster on the same endpoint is set to TRUE, as a result of processing an On/Off cluster command. If the OnLevel attribute is not implemented, or is set to the null value, it has no effect. For more details see Effect of On/Off Commands on the CurrentLevel attribute.

OnLevel represents a mandatory field that was previously not present or optional. Implementers should be aware that older devices may not implement it.

1.6.6.12. OnTransitionTime Attribute

This attribute SHALL indicate the time taken to move the current level from the minimum level to the maximum level when an On command is received by an On/Off cluster on the same endpoint. It is specified in 1/10ths of a second. If this attribute is not implemented, or contains a null value, the OnOffTransitionTime SHALL be used instead.

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1.6.6.13. OffTransitionTime Attribute

This attribute SHALL indicate the time taken to move the current level from the maximum level to the minimum level when an Off command is received by an On/Off cluster on the same endpoint. It is specified in 1/10ths of a second. If this attribute is not implemented, or contains a null value, the OnOffTransitionTime SHALL be used instead.

1.6.6.14. DefaultMoveRate Attribute

This attribute SHALL indicate the movement rate, in units per second, when a Move command is received with a null value Rate parameter.

1.6.6.15. StartUpCurrentLevel Attribute

This attribute SHALL indicate the desired startup level for a device when it is supplied with power and this level SHALL be reflected in the CurrentLevel attribute. The values of the StartUpCurrentLevel attribute are listed below:

Value	Action on power up
0	Set the CurrentLevel attribute to the minimum value permitted on the device
null	Set the CurrentLevel attribute to its previous value
other values	Set the CurrentLevel attribute to this value

This behavior does not apply to reboots associated with OTA. After an OTA restart, the CurrentLevel attribute SHALL return to its value prior to the restart.

1.6.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**MoveToLevel**	client ⇒ server	Y	O	M
0x01	**Move**	client ⇒ server	Y	O	M
0x02	**Step**	client ⇒ server	Y	O	M
0x03	**Stop**	client ⇒ server	Y	O	M
0x04	**MoveToLevelWithOnOff**	client ⇒ server	Y	O	M
0x05	**MoveWithOnOff**	client ⇒ server	Y	O	M
0x06	**StepWithOnOff**	client ⇒ server	Y	O	M
0x07	**StopWithOnOff**	client ⇒ server	Y	O	M

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ID	Name	Direction	Response	Access	Conformance
0x08	**MoveToClosestFrequency**	client ⇒ server	Y	O	FQ

1.6.7.1. MoveToLevel Command

This command will move the device to the specified level.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Level**	uint8	max 254			M
1	**TransitionTime**	uint16	all	X		M
2	**OptionsMask**	Options-Bitmap	desc			M
3	**OptionsOverride**	Options-Bitmap	desc			M

1.6.7.1.1. Effect on Receipt

The OptionsMask and OptionsOverride fields SHALL both be present. Default values are provided to interpret missing fields from legacy devices. A temporary Options bitmap SHALL be created from the Options attribute, using the OptionsMask and OptionsOverride fields. Each bit of the temporary Options bitmap SHALL be determined as follows:

Each bit in the Options attribute SHALL determine the corresponding bit in the temporary Options bitmap, unless the OptionsMask field is present and has the corresponding bit set to 1, in which case the corresponding bit in the OptionsOverride field SHALL determine the corresponding bit in the temporary Options bitmap.

The resulting temporary Options bitmap SHALL then be processed as defined in the Options attribute.

On receipt of this command, a device SHALL move from its current level to the value given in the Level field. The meaning of 'level' is device dependent – e.g., for a light it MAY mean brightness level.

If the value of the Level field is below the MinLevel or above the MaxLevel for the device, the value SHALL be clipped to the applicable boundary value.

The movement SHALL be as continuous as technically practical, i.e., not a step function, and the time taken to move to the new level SHALL be equal to the value of the TransitionTime field, in tenths of a second, or as close to this as the device is able.

If the TransitionTime field takes the value null then the time taken to move to the new level SHALL instead be determined by the OnOffTransitionTime attribute. If OnOffTransitionTime, which is an optional attribute, is not present, the device SHALL move to its new level as fast as it is able.

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If the device is not able to move at a variable rate, the TransitionTime field MAY be disregarded.

1.6.7.2. Move Command

This command will move the device using the specified values.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**MoveMode**	MoveModeEnum	desc			M
1	**Rate**	uint8	all	X		M
2	**OptionsMask**	OptionsBitmap	desc			M
3	**OptionsOverride**	OptionsBitmap	desc			M

1.6.7.2.1. MoveMode Field

This field SHALL be one of the non-reserved values in MoveModeEnum.

1.6.7.2.2. Rate Field

This field SHALL indicate the rate of movement in units per second. The actual rate of movement SHOULD be as close to this rate as the device is able. If the Rate field is null, then the value of the DefaultMoveRate attribute SHALL be used if that attribute is supported and its value is not null. If the Rate field is null and the DefaultMoveRate attribute is either not supported or set to null, then the device SHOULD move as fast as it is able. If the device is not able to move at a variable rate, this field MAY be disregarded.

1.6.7.2.3. Effect on Receipt

On receipt of this command, if the Rate field has a value of zero, the command has no effect and a response SHALL be returned with the status code set to INVALID_COMMAND.

Otherwise, a device SHALL first create and process a temporary Options bitmap as described in the MoveToLevel command.

On receipt of this command, a device SHALL move from its current level in an up or down direction in a continuous fashion, as detailed below.

MoveMode	Action on Receipt
Up	Increase the device's level at the rate given in the Rate field. If the level reaches the maximum allowed for the device, stop.
Down	Decrease the device's level at the rate given in the Rate field. If the level reaches the minimum allowed for the device, stop.

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1.6.7.3. Step Command

This command will do a relative step change of the device using the specified values.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**StepMode**	StepModeEnum	desc			M
1	**StepSize**	uint8	all			M
2	**TransitionTime**	uint16	all	X		M
3	**OptionsMask**	OptionsBitmap	desc			M
4	**OptionsOverride**	OptionsBitmap	desc			M

1.6.7.3.1. StepMode Field

This field SHALL be one of the non-reserved values in StepModeEnum.

1.6.7.3.2. StepSize Field

This field SHALL indicate the change to CurrentLevel.

1.6.7.3.3. TransitionTime Field

This field SHALL indicate the time that SHALL be taken to perform the step, in tenths of a second. A step is a change in the CurrentLevel of StepSize units. The actual time taken SHOULD be as close to this as the device is able. If the TransitionTime field is equal to null, the device SHOULD move as fast as it is able.

If the device is not able to move at a variable rate, the TransitionTime field MAY be disregarded.

1.6.7.3.4. Effect on Receipt

On receipt of this command, if the StepSize field has a value of zero, the command has no effect and a response SHALL be returned with the status code set to INVALID_COMMAND.

Otherwise, a device SHALL first create and process a temporary Options bitmap as described in the MoveToLevel command.

On receipt of this command, a device SHALL move from its current level in an up or down direction as detailed below.

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StepMode	Action on Receipt
Up	Increase CurrentLevel by StepSize units, or until it reaches the maximum level allowed for the device if this reached in the process. In the latter case, the transition time SHALL be proportionally reduced.
Down	Decrease CurrentLevel by StepSize units, or until it reaches the minimum level allowed for the device if this reached in the process. In the latter case, the transition time SHALL be proportionally reduced.

1.6.7.4. Stop Command

This command will stop the actions of various other commands that are still in progress.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**OptionsMask**	Options-Bitmap	desc			M
1	**OptionsOverride**	Options-Bitmap	desc			M

1.6.7.4.1. Effect of Receipt

On receipt of this command, a device SHALL first create and process a temporary Options bitmap as described in the MoveToLevel command.

Upon receipt of this command, any MoveToLevel, Move or Step command (and their 'with On/Off' variants) currently in process SHALL be terminated. The value of CurrentLevel SHALL be left at its value upon receipt of the Stop command, and RemainingTime SHALL be set to zero.

This command has two entries in the Commands list, one for the MoveToLevel, Move and Step commands, and one for their 'with On/Off' counterparts. This is solely for symmetry, to allow easy choice of one or other set of commands – the Stop commands are identical, because the dependency on On/Off is determined by the original command that is being stopped.

1.6.7.5. MoveToClosestFrequency Command

This command will cause the device to change the current frequency to the requested value.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Frequency**	uint16	all			M

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1.6.7.5.1. Effect of Receipt

Upon receipt of this command, the device SHALL change its current frequency to the requested frequency, or to the closest frequency that it can generate. If the device cannot approximate the frequency, then it SHALL return a default response with an error code of CONSTRAINT_ERROR. Determining if a requested frequency can be approximated by a supported frequency is a manufacturer-specific decision.

1.6.7.6. 'With On/Off' Commands

The MoveToLevelWithOnOff, MoveWithOnOff and StepWithOnOff commands have identical data fields compared to the MoveToLevel, Move and Step commands respectively. They also have the same effects, except for the following additions.

Before commencing any command that has the effect of setting the CurrentLevel attribute above the minimum level allowed by the device, the OnOff attribute of the On/Off cluster on the same endpoint, if implemented, SHALL be set to TRUE (‘On’).

If any command that has the effect of setting the CurrentLevel attribute to the minimum level allowed by the device, the OnOff attribute of the On/Off cluster on the same endpoint, if implemented, SHALL be set to FALSE (‘Off’).

The StopWithOnOff command has identical data fields compared to the Stop command. Both Stop commands are identical, because the dependency on On/Off is determined by the original command that is being stopped.

1.6.8. State Change Table for Lighting

Below is a table of examples of state changes when Level Control and On/Off clusters are on the same endpoint and the Lighting feature is set.

EiO - ExecuteIfOff field in the Options attribute

OnOff – Attribute value of On/Off cluster: FALSE=‘Off’, TRUE=‘On’

MIN - MinLevel

MAX - MaxLevel

MID – Midpoint between MinLevel and MaxLevel

1.6.8.1. Lighting Device State Change examples

CurrentLevel	**EiO**	**OnOff**	**Physical Device**	**Command Before After**	**CurrentLevel**	**OnOff**	**Physical Device**	**Device Output Result**
any	0	FALSE	Off	MoveToLevel(l=*MID*, t=2 sec)	*same*	FALSE	Off	Stays off

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CurrentLevel	EiO	OnOff	Physical Device	Command Before After	CurrentLevel	OnOff	Physical Device	Device Output Result
any	0	FALSE	Off	MoveToLevelWithOnOff(l=*MID*, t=2 sec	*MID*	TRUE	On (midpoint brightness)	Turns on and output level adjusts or stays at half
any	1	FALSE	Off	MoveToLevel(l=*MID*, t=2 sec)	*MID*	FALSE	Off	Stays off
any	1	FALSE	Off	MoveToLevelWithOnOff(l=*MID*, t=2 sec)	*MID*	TRUE	On	Turns on and output level adjusts to or stays at half
any	1	FALSE	Off	Move(up, rate=64/s)	*MAX*	FALSE	Off	Stays off
any	1	FALSE	Off	MoveWithOnOff(up, rate=64/s)	*MAX*	TRUE	On	Turn on and output level adjusts to or stays at full
any	1	FALSE	Off	MoveWithOnOff(down, rate=64/s)	*MIN*	FALSE	Off	Stays off
any	*any*	TRUE	On (any brightness)	MoveToLevelWithOnOff(l=*MID*, t=2 sec)	*MID*	TRUE	On (midpoint brightness)	Output level adjusts to or stays at half
any	*any*	TRUE	On (any brightness)	MoveWithOnOff(up, rate=64/s)	*MAX*	TRUE	On (full brightness)	Output level adjusts to or stays at full

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Cur-rentLevel	EiO	OnOff	Physical Device	Com-mand Before After	Cur-rentLevel	OnOff	Physical Device	Device Output Result
any	*any*	TRUE	On (any bright-ness)	Move(down, rate=64/s)	*MIN*	TRUE	On (at minimum bright-ness)	Output level adjusts to minimum
any	*any*	TRUE	On (any bright-ness)	Move-With-OnOff(down, rate=64/s)	*MIN*	FALSE	Off	Output level adjusts to off

1.7. Boolean State Cluster

This cluster provides an interface to a boolean state.

1.7.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Removed P quality
3	Introduced ChangeEvent feature

1.7.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	BOOL

1.7.3. Cluster ID

ID	Name
0x0045	Boolean State

1.7.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

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Bit	Code	Feature	Conformance	Summary
0	**CHGEVENT**	ChangeEvent	[Rev >= v3]	Supports reporting change events

1.7.4.1. ChangeEvent Feature

This feature SHALL indicate that the StateChange event is supported and will be generated every time the state changes.

1.7.5. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**StateValue**	bool				R V	M

1.7.5.1. StateValue Attribute

This represents a boolean state.

The semantics of this boolean state are defined by the device type using this cluster.
For example, in a Contact Sensor device type, FALSE=open or no contact, TRUE=closed or contact.

1.7.6. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**StateChange**	INFO		V	CHGEVENT, O

1.7.6.1. StateChange Event

If this event is supported, it SHALL be generated when the StateValue attribute changes.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**StateValue**	bool				M

1.7.6.1.1. StateValue Field

This field SHALL indicate the new value of the StateValue attribute.

1.8. Boolean State Configuration Cluster

This cluster is used to configure a boolean sensor, including optional state change alarm features and configuration of the sensitivity level associated with the sensor.

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1.8.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Added FaultEvents feature

1.8.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	BOOLCFG

1.8.3. Cluster ID

ID	Name
0x0080	Boolean State Configuration

1.8.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**VIS**	Visual	O	Supports visual alarms
1	**AUD**	Audible	O	Supports audible alarms
2	**SPRS**	AlarmSuppress	[VIS | AUD]	Supports ability to suppress or acknowledge alarms
3	**SENSLVL**	SensitivityLevel	O	Supports ability to set sensor sensitivity
4	**FAULTEV**	FaultEvents	[Rev >= v2]	Supports reporting fault events

1.8.4.1. AlarmSuppress Feature

This feature SHALL indicate that the device is able to suppress the supported alarm modes, when the user acknowledges the alarm. This is intended to stop visual and/or audible alarms, when the user has become aware that the sensor is triggered, but it is no longer desired to have the alarm modes active on the device, e.g.:

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- The triggering cause have been resolved by the user, but the sensor has not yet stopped detecting the triggering cause.
- The user is not able to address the triggering cause, but is aware of the alarm and suppress/acknowledge it be addressed at a later point.

Acknowledge of alarms will for the remainder of this cluster be referred to as suppress.

A suppressed alarm is still considered active and will remain so unless it is actively disabled or the triggering condition is not longer present. The action of suppressing an alarm mode is only applicable to and is intended to stop the physical alarming, e.g. emitting a sound or blinking a light; it does not impact alarm reporting in AlarmsActive.

1.8.4.2. FaultEvents

This feature SHALL indicate that the device supports reporting sensor faults. When this feature is supported, there SHALL be at least one cause of internal failure that will generate the SensorFault.

1.8.5. Data Types

1.8.5.1. AlarmModeBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**Visual**	Visual alarming	VIS
1	**Audible**	Audible alarming	AUD

1.8.5.2. SensorFaultBitmap Type

This data type is derived from map16.

Bit	Name	Summary	Conformance
0	**GeneralFault**	Unspecified fault detected	M

1.8.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**CurrentSensitivityLevel**	uint8	max (SupportedSensitivityLevels - 1)	N		RW VO	SENSLVL

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0001	**SupportedSensitivityLevels**	uint8	2 to 10	F		R V	SENSLVL
0x0002	**DefaultSensitivityLevel**	uint8	max (SupportedSensitivityLevels - 1)	F	MS	R V	[SENSLVL]
0x0003	**AlarmsActive**	AlarmModeBitmap	all			R V	VIS | AUD
0x0004	**AlarmsSuppressed**	AlarmModeBitmap	all			R V	SPRS
0x0005	**AlarmsEnabled**	AlarmModeBitmap	all	N	MS	R V	[VIS | AUD]
0x0006	**AlarmsSupported**	AlarmModeBitmap	all	F		R V	VIS | AUD
0x0007	**SensorFault**	SensorFaultBitmap	all		0	R V	O

1.8.6.1. CurrentSensitivityLevel Attribute

This attribute SHALL indicate the currently selected sensitivity level.

If a write interaction to this attribute contains an unsupported sensitivity value, a CONSTRAINT_ERROR status SHALL be returned.

1.8.6.2. SupportedSensitivityLevels Attribute

This attribute SHALL indicate the number of supported sensitivity levels by the device.

These supported sensitivity levels SHALL be ordered by sensitivity, where a value of 0 SHALL be considered the lowest sensitivity level (least sensitive) and the highest supported value SHALL be considered the highest sensitivity level.
The number of supported sensitivity levels SHOULD represent unique sensitivity levels supported by the device.

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1.8.6.3. DefaultSensitivityLevel Attribute

This attribute SHALL indicate the default sensitivity level selected by the manufacturer.

1.8.6.4. AlarmsActive Attribute

This attribute SHALL indicate which specific alarm modes on the server are currently active. When the sensor is no longer triggered, this attribute SHALL be set to the inactive state, by setting the bit to 0, for all supported alarm modes.

If an alarm mode is not supported, the bit indicating this alarm mode SHALL always be 0.

A bit SHALL indicate whether the alarm mode inactive or not:

- 0 = Inactive
- 1 = Active

1.8.6.5. AlarmsSuppressed Attribute

This attribute SHALL indicate which specific alarm modes on the server are currently suppressed. When the sensor is no longer triggered, this attribute SHALL be set to the unsuppressed state, by setting the bit to 0, for all supported alarm modes.

If an alarm mode is not supported, the bit indicating this alarm mode SHALL always be 0.

A bit SHALL indicate whether the alarm mode is suppressed or not:

- 0 = Not suppressed
- 1 = Suppressed

1.8.6.6. AlarmsEnabled Attribute

This attribute SHALL indicate the alarm modes that will be emitted if the sensor is triggered.

If an alarm mode is not supported, the bit indicating this alarm mode SHALL always be 0.

A bit SHALL indicate whether the alarm mode is enabled or disabled:

- 0 = Disabled
- 1 = Enabled

1.8.6.7. AlarmsSupported Attribute

This attribute SHALL indicate the alarms supported by the sensor.

A bit SHALL indicate whether the alarm mode is supported:

- 0 = Not supported
- 1 = Supported

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1.8.6.8. SensorFault Attribute

This attribute SHALL indicate any faults registered by the device.

1.8.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**SuppressAlarm**	client ⇒ server	Y	O	SPRS
0x01	**EnableDisableAlarm**	client ⇒ server	Y	O	VIS | AUD

1.8.7.1. SuppressAlarm Command

This command will suppress the alarm modes specified.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**AlarmsToSuppress**	AlarmModeBitmap	all			M

1.8.7.1.1. AlarmsToSuppress Field

This field SHALL indicate the alarm modes to suppress.

1.8.7.1.2. Effect on Receipt

If any of the requested alarm modes are not supported this command SHALL be ignored and the server SHALL return an CONSTRAINT_ERROR status code.

If any of the requested alarm modes are not active, in case the sensor is not triggered or the alarm mode is disabled/not supported, this command SHALL be ignored and the server SHALL return an INVALID_IN_STATE status code.

In the case an alarm is already suppressed and the specific bit is set in the AlarmsSuppressed attribute, the bit SHALL be ignored and the remaining bits SHALL be evaluated.

For the valid bits in the AlarmsToSuppress field, the device SHALL suppress the specified alarm modes as requested in the AlarmsToSuppress field and set the AlarmsSuppressed attribute accordingly.

1.8.7.2. EnableDisableAlarm Command

This command allows setting the enable or disable value for the specified alarm modes.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**AlarmsToEnableDisable**	AlarmModeBitmap	all			M

1.8.7.2.1. AlarmsToEnableDisable Field

This field SHALL indicate the alarm modes to either enable or disable depending on the bit status, as specified for the AlarmsEnabled attribute.

1.8.7.2.2. Effect on Receipt

If any of the requested alarm modes are not supported this command SHALL be ignored and the server SHALL return an CONSTRAINT_ERROR status code.

If all the bits are valid, the value of the AlarmsEnabled attribute SHALL be set to the value of the AlarmsToEnableDisable field.

If an alarm mode is being enabled and the trigger condition is met, the device SHALL immediately activate the alarm mode and set the associated bit in the AlarmsActive attribute.

If an alarm mode is being disabled, any alarm mode which is either in the active or suppressed state SHALL be cleared and the alarm mode SHALL be considered not active.

1.8.8. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**AlarmsStateChanged**	INFO		V	VIS | AUD
0x01	**SensorFault**	INFO		V	FAULTEV, O

1.8.8.1. AlarmsStateChanged Event

This event SHALL be generated after any bits in the AlarmsActive and/or AlarmsSuppressed attributes change. This MAY occur in situations such as when internal processing by the server determines that an alarm mode becomes active or inactive, or when the SuppressAlarm or EnableDisableAlarm commands are processed in a way that some alarm modes becomes suppressed, active or inactive.

If several alarm modes change state at the same time, a single event combining multiple changes MAY be emitted instead of multiple events each representing a single change.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**AlarmsActive**	AlarmModeBitmap	all			M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
1	**AlarmsSuppressed**	AlarmModeBitmap	all			SPRS

1.8.8.1.1. AlarmsActive Field

This field SHALL indicate the state of active alarm modes, as indicated by the AlarmsActive attribute, at the time the event was generated.

1.8.8.1.2. AlarmsSuppressed Field

This field SHALL indicate the state of suppressed alarm modes, as indicated by the AlarmsSuppressed attribute, at the time the event was generated.

1.8.8.2. SensorFault Event

This event SHALL be generated when the device registers or clears a fault.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**SensorFault**	SensorFaultBitmap	all			M

1.8.8.2.1. SensorFault Field

This field SHALL indicate the value of the SensorFault attribute, at the time this event is generated.

1.9. Mode Select Cluster

This cluster provides an interface for controlling a characteristic of a device that can be set to one of several predefined values. For example, the light pattern of a disco ball, the mode of a massage chair, or the wash cycle of a laundry machine.

The server allows the client to set a mode on the server. A mode is one of a list of options that may be presented by a client for a user choice, or understood by the client, via the semantic tags on the mode.

A semantic tag is either a standard tag within a standard category namespace, or a manufacturer specific tag, within the namespace of the vendor ID of the manufacturer. If there is no semantic tag, the mode is anonymous, and the selection is made by the user solely based on the Label string.

Each cluster ID that indicates this specification SHALL define a distinct purpose for the cluster instance. For example: A LightBlinking cluster ID supports blinking modes for a light (and is described that way).

An anonymous mode SHALL support the derived cluster purpose. A manufacturer specific semantic tag SHALL support the derived cluster purpose. An anonymous mode SHALL NOT replace the meaning of a standard semantic tag, when one exists, for the cluster purpose.

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1.9.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	The MfgCode field was marked non-nullable, updated the related text and reordered sections

1.9.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	MOD

1.9.3. Cluster ID

ID	Name
0x0050	Mode Select

1.9.4. Features

This cluster SHALL support the FeatureMap global attribute:

Bit	Code	Feature	Conformance	Summary
0	**DEPONOFF**	OnOff	O	Dependency with the OnOff cluster

1.9.4.1. OnOff Feature

This feature creates a dependency between an OnOff cluster instance and this cluster instance on the same endpoint. See Section 1.9.6.6, “OnMode Attribute” for more information.

1.9.5. Data Types

1.9.5.1. SemanticTagStruct Type

A Semantic Tag is meant to be interpreted by the client for the purpose the cluster serves.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**MfgCode**	vendor-id	desc	F			M
1	**Value**	enum16	all	F			M

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1.9.5.1.1. Value Field

This field SHALL indicate the semantic tag within a semantic tag namespace which is either manufacturer specific or standard. For semantic tags in a standard namespace, see Standard Namespace.

1.9.5.1.2. MfgCode Field

This field SHALL indicate a manufacturer code (Vendor ID), and the Value field SHALL indicate a semantic tag defined by the manufacturer. Each manufacturer code supports a single namespace of values. The same manufacturer code and semantic tag value in separate cluster instances are part of the same namespace and have the same meaning. For example: a manufacturer tag meaning "pinch", has the same meaning in a cluster whose purpose is to choose the amount of sugar, or amount of salt.

1.9.5.2. ModeOptionStruct Type

This is a struct representing a possible mode of the server.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Label**	string	max 64	F	MS		M
1	**Mode**	uint8		F	MS		M
2	**SemanticTags**	list[SemanticTagStruct]	max 64	F	MS		M

1.9.5.2.1. Label Field

This field is readable text that describes the mode option that can be used by a client to indicate to the user what this option means. This field is meant to be readable and understandable by the user.

1.9.5.2.2. Mode Field

The Mode field is used to identify the mode option. The value SHALL be unique for every item in the SupportedModes attribute.

1.9.5.2.3. SemanticTags Field

This field is a list of semantic tags that map to the mode option. This MAY be used by clients to determine the meaning of the mode option as defined in a standard or manufacturer specific namespace. Semantic tags can help clients look for options that meet certain criteria. A semantic tag SHALL be either a standard tag or manufacturer specific tag as defined in each SemanticTagStruct list entry.

A mode option MAY have more than one semantic tag. A mode option MAY be mapped to a mixture of standard and manufacturer specific semantic tags.

All standard semantic tags are from a single namespace indicated by the StandardNamespace attribute.

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For example: A mode labeled "100%" can have both the `HIGH` (MS) and `MAX` (standard) semantic tag. Clients seeking the option for either `HIGH` or `MAX` will find the same option in this case.

1.9.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**Description**	**string**	max 64	F	MS	R V	M
0x0001	**StandardNamespace**	**enum16**	desc	F X	null	R V	M
0x0002	**SupportedModes**	**list[ModeOptionStruct]**	max 255	F	MS	R V	M
0x0003	**CurrentMode**	**uint8**	desc	N	MS	R V	M
0x0004	**StartUpMode**	**uint8**	desc	N X	MS	RW VO	O
0x0005	**OnMode**	**uint8**	desc	N X	null	RW VO	DEPONOFF

1.9.6.1. Description Attribute

This attribute describes the purpose of the server, in readable text.

For example, a coffee machine may have a Mode Select cluster for the amount of milk to add, and another Mode Select cluster for the amount of sugar to add. In this case, the first instance can have the description `Milk` and the second instance can have the description `Sugar`. This allows the user to tell the purpose of each of the instances.

1.9.6.2. StandardNamespace Attribute

This attribute, when not null, SHALL indicate a single standard namespace for any standard semantic tag value supported in this or any other cluster instance with the same value of this attribute. A null value indicates no standard namespace, and therefore, no standard semantic tags are provided in this cluster instance. Each standard namespace and corresponding values and value meanings SHALL be defined in another document.

1.9.6.3. SupportedModes Attribute

This attribute is the list of supported modes that may be selected for the CurrentMode attribute. Each item in this list represents a unique mode as indicated by the Mode field of the ModeOptionStruct. Each entry in this list SHALL have a unique value for the Mode field.

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1.9.6.4. CurrentMode Attribute

This attribute represents the current mode of the server.

The value of this field must match the Mode field of one of the entries in the `SupportedModes` attribute.

1.9.6.5. StartUpMode Attribute

The StartUpMode attribute value indicates the desired startup mode for the server when it is supplied with power.

If this attribute is not null, the CurrentMode attribute SHALL be set to the StartUpMode value, when the server is powered up, except in the case when the OnMode attribute overrides the StartUpMode attribute (see Section 1.9.6.6.1, “OnMode with Power Up”).

This behavior does not apply to reboots associated with OTA. After an OTA restart, the CurrentMode attribute SHALL return to its value prior to the restart.

The value of this field SHALL match the Mode field of one of the entries in the `SupportedModes` attribute.

If this attribute is not implemented, or is set to the null value, it SHALL have no effect.

1.9.6.6. OnMode Attribute

This attribute SHALL indicate the value of CurrentMode that depends on the state of the On/Off cluster on the same endpoint. If this attribute is not present or is set to null, it SHALL NOT have an effect, otherwise the CurrentMode attribute SHALL depend on the OnOff attribute of the On/Off cluster as described in the table below:

OnOff Change	CurrentMode Change
ON → OFF	No change
OFF → ON	Change CurrentMode to OnMode
OFF → OFF	No change
ON → ON	No change

The value of this field SHALL match the Mode field of one of the entries in the SupportedModes attribute.

1.9.6.6.1. OnMode with Power Up

If the On/Off feature is supported and the On/Off cluster attribute StartUpOnOff is present, with a value of On (turn on at power up), then the CurrentMode attribute SHALL be set to the OnMode attribute value when the server is supplied with power, except if the OnMode attribute is null.

1.9.7. Commands

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ID	Name	Direction	Response	Access	Conformance
0x00	**ChangeToMode**	client ⇒ server	Y	O	M

1.9.7.1. ChangeToMode Command

On receipt of this command, if the `NewMode` field indicates a valid mode transition within the supported list, the server SHALL set the CurrentMode attribute to the `NewMode` value, otherwise, the server SHALL respond with an `INVALID_COMMAND` status response.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**NewMode**	uint8	desc			M

1.10. Mode Base Cluster

This cluster provides an interface for controlling a characteristic of a device that can be set to one of several predefined values. For example, the light pattern of a disco ball, the mode of a massage chair, or the wash cycle of a laundry machine.

The server allows the client to set a mode on the server. A mode is one of a list of options that may be presented by a client for a user choice, or understood by the client, via the mode's tags.

A mode tag is either a standard tag within a standard category namespace, or a manufacturer specific tag, within the namespace of the vendor ID of the manufacturer.

Any derived cluster specification based on this cluster SHALL support the standard mode tag value definitions and command status definitions defined in this cluster and MAY define additional standard mode tag values and standard command status values that are supported in the respective derived cluster instances.

Each cluster ID that indicates this specification SHALL define a distinct purpose for the cluster instance. For example: A LightBlinking cluster ID supports blinking modes for a light (and is described that way).

An anonymous mode SHALL NOT replace the meaning of a standard mode tag, when one exists, for the cluster purpose.

1.10.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

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Revision	Description
2	ChangeToModeResponse command: StatusText must be provided for InvalidInMode status. Require at least one standard mode tag. Define reserved ranges for base/derived clusters.

1.10.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	MODB

1.10.3. Cluster ID

Cluster ID	Cluster Name
n/a	Mode Base

1.10.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Summary	Conformance
0	**DEPONOFF**	OnOff	Dependency with the OnOff cluster	O

1.10.4.1. OnOff Feature

This feature creates a dependency between an OnOff cluster instance and this cluster instance on the same endpoint. See OnMode for more information.

1.10.5. Data Types

1.10.5.1. ModeTagStruct Type

A Mode Tag is meant to be interpreted by the client for the purpose the cluster serves.

ID	Name	Type	Constraint	Quality	Fallback	Access	Confor-mance
0	**MfgCode**	vendor-id	desc				O
1	**Value**	enum16	all				M

1.10.5.1.1. MfgCode Field

If the MfgCode field exists, the Value field SHALL be in the manufacturer-specific value range (see Section 1.10.8, “Mode Namespace”).

This field SHALL indicate the manufacturer’s VendorID and it SHALL determine the meaning of the

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Value field.

The same manufacturer code and mode tag value in separate cluster instances are part of the same namespace and have the same meaning. For example: a manufacturer tag meaning "pinch" can be used both in a cluster whose purpose is to choose the amount of sugar, or in a cluster whose purpose is to choose the amount of salt.

1.10.5.1.2. Value Field

This field SHALL indicate the mode tag within a mode tag namespace which is either manufacturer specific or standard.

1.10.5.2. ModeOptionStruct Type

This is a struct representing a possible mode of the server.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Label**	string	max 64	F			M
1	**Mode**	uint8		F			M
2	**ModeTags**	list[ModeTagStruct]	max 8	F			M

1.10.5.2.1. Label Field

This field SHALL indicate readable text that describes the mode option, so that a client can provide it to the user to indicate what this option means. This field is meant to be readable and understandable by the user.

1.10.5.2.2. Mode Field

This field is used to identify the mode option.

1.10.5.2.3. ModeTags Field

This field SHALL contain a list of tags that are associated with the mode option. This can be used by clients to determine the full or the partial semantics of a certain mode, depending on which tags they understand, using standard definitions and/or manufacturer specific namespace definitions.

The standard mode tags are defined in this cluster specification. For the derived cluster instances, if the specification of the derived cluster defines a namespace, the set of standard mode tags also includes the mode tag values from that namespace.

Mode tags can help clients look for options that meet certain criteria, render the user interface, use the mode in an automation, or to craft help text their voice-driven interfaces. A mode tag SHALL be either a standard tag or a manufacturer specific tag, as defined in each ModeTagStruct list entry.

A mode option MAY have more than one mode tag. A mode option MAY be associated with a mixture of standard and manufacturer specific mode tags. A mode option SHALL be associated with at

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least one standard mode tag.

Each mode tag in this field SHALL be distinct from other mode tags in this field. For example, a simplified list containing [Auto, Auto] would not be allowed.

A few examples are provided below.

- A mode named "100%" can have both the High (manufacturer specific) and Max (standard) mode tag. Clients seeking the mode for either High or Max will find the same mode in this case.
- A mode that includes a LowEnergy tag can be displayed by the client using a widget icon that shows a green leaf.
- A mode that includes a LowNoise tag may be used by the client when the user wishes for a lower level of audible sound, less likely to disturb the household's activities.
- A mode that includes a LowEnergy tag (standard, defined in this cluster specification) and also a Delicate tag (standard, defined in the namespace of a Laundry Mode derived cluster).
- A mode that includes both a generic Quick tag (defined here), and Vacuum and Mop tags, (defined in the RVC Clean cluster that is a derivation of this cluster).

1.10.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SupportedModes**	list[ModeOptionStruct]	2 to 255	F		R V	M
0x0001	**CurrentMode**	uint8	desc	N		R V	M
0x0002	**StartUpMode**	uint8	desc	N X	MS	RW VO	O
0x0003	**OnMode**	uint8	desc	N X		RW VO	DEPONOFF

1.10.6.1. SupportedModes Attribute

This attribute SHALL contain the list of supported modes that may be selected for the CurrentMode attribute. Each item in this list represents a unique mode as indicated by the Mode field of the ModeOptionStruct.

Each entry in this list SHALL have a unique value for the Mode field.

Each entry in this list SHALL have a unique value for the Label field.

The set of ModeTags listed in each entry in this list SHALL be distinct from the sets of ModeTags listed in the other entries. This comparison SHALL NOT depend on the order of the ModeTags in the lists. Two sets SHALL be considered distinct if one of them contains an element that the other one does not. Note that the two sets could have a non-empty intersection, or one could be a subset of the other, and still be distinct.

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Simplified examples of allowed ModeTags lists:

SupportedModes Index	Simplified mode tag list	Reason
0	Cool	Subset is ok
1	Cool, MfrTag1	Overlap is ok, the set was made unique by adding a manufacturer specific tag
2	Cool, MfrTag2	Overlap is ok, list is unique from index 1

Simplified examples of disallowed ModeTags lists:

SupportedModes Index	Simplified mode tag list	Reason
0	Cool, MfrTag1	Tag set is duplicated from index 1
1	MfrTag1, Cool	Tag set is duplicated from index 0

1.10.6.2. CurrentMode Attribute

This attribute SHALL indicate the current mode of the server.

The value of this field SHALL match the Mode field of one of the entries in the SupportedModes attribute.

The value of this attribute may change at any time via an out-of-band interaction outside of the server, such as interactions with a user interface, via internal mode changes due to autonomously progressing through a sequence of operations, on system time-outs or idle delays, or via interactions coming from a fabric other than the one which last executed a ChangeToMode.

1.10.6.3. StartUpMode Attribute

This attribute SHALL indicate the desired startup mode for the server when it is supplied with power.

If this attribute is not null, the CurrentMode attribute SHALL be set to the StartUpMode value, when the server is powered up, except in the case when the OnMode attribute overrides the StartUpMode attribute (see Section 1.10.6.4.1, “OnMode with Power Up”).

This behavior does not apply to reboots associated with OTA. After an OTA restart, the CurrentMode attribute SHALL return to its value prior to the restart.

The value of this field SHALL match the Mode field of one of the entries in the SupportedModes attribute.

If this attribute is not implemented, or is set to the null value, it SHALL have no effect.

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1.10.6.4. OnMode Attribute

This attribute SHALL indicate whether the value of CurrentMode depends on the state of the On/Off cluster on the same endpoint. If this attribute is not present or is set to null, there is no dependency, otherwise the CurrentMode attribute SHALL depend on the OnOff attribute in the On/Off cluster as described in the table below:

OnOff Change	CurrentMode Change
ON → OFF	No change
OFF → ON	Change CurrentMode to OnMode
OFF → OFF	No change
ON → ON	No change

The value of this field SHALL match the Mode field of one of the entries in the SupportedModes attribute.

1.10.6.4.1. OnMode with Power Up

If the On/Off feature is supported and the On/Off cluster attribute StartUpOnOff is present, with a value of On (turn on at power up), then the CurrentMode attribute SHALL be set to the OnMode attribute value when the server is supplied with power, except if the OnMode attribute is null.

1.10.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**ChangeToMode**	client ⇒ server	ChangeToModeResponse	O	M
0x01	**ChangeToModeResponse**	client ⇐ server	N		M

1.10.7.1. ChangeToMode Command

This command is used to change device modes.

On receipt of this command the device SHALL respond with a ChangeToModeResponse command.

This command SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**NewMode**	uint8	desc			M

1.10.7.1.1. NewMode Field

If the NewMode field doesn't match the Mode field of any entry of the SupportedModes list, the ChangeToModeResponse command's Status field SHALL indicate UnsupportedMode and the Status-

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Text field SHALL be included and MAY be used to indicate the issue, with a human readable string, or include an empty string.

If the NewMode field matches the Mode field of one entry of the SupportedModes list, but the device is not able to transition as requested, the ChangeToModeResponse command SHALL:

- Have the Status set to a product-specific Status value representing the error, or GenericFailure if a more specific error cannot be provided. See Status field for details.
- Provide a human readable string in the StatusText field.

If the NewMode field matches the Mode field of one entry of the SupportedModes list and the device is able to transition as requested, the server SHALL transition into the mode associated with NewMode, the ChangeToModeResponse command SHALL have the Status field set to Success, the StatusText field MAY be supplied with a human readable string or include an empty string and the CurrentMode field SHALL be set to the value of the NewMode field.

If the NewMode field is the same as the value of the CurrentMode attribute the ChangeToModeResponse command SHALL have the Status field set to Success and the StatusText field MAY be supplied with a human readable string or include an empty string.

1.10.7.2. ChangeToModeResponse Command

This command is sent by the device on receipt of the ChangeToMode command. This command SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	enum8	desc			M
1	**StatusText**	string	max 64			[Status == Success], M

1.10.7.2.1. Status Field

1.10.7.2.1.1. Mode Base Status Code Ranges

The following table defines the enumeration ranges for the ChangeToModeResponse command's Status field values.

Status Code Range	Range Name	Summary
0x00 to 0x3F	**CommonCodes**	Common standard values defined in the generic Mode Base cluster specification.
0x40 to 0x7F	**DerivedClusterCodes**	Derived cluster specific standard values defined in the derived Mode Base cluster specification.

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Status Code Range	Range Name	Summary
0x80 to 0xBF	**MfgCodes**	Manufacturer specific values. For the derived Mode Base cluster instances, these are manufacturer specific under the derived cluster.

The set of Status field values defined in each of the generic or derived Mode Base cluster specifications is called StatusCode.

1.10.7.2.1.2. Mode Base Status CommonCodes Range

The following table defines the common StatusCode values.

Status Code	Name	Summary
0x00	**Success**	Switching to the mode indicated by the NewMode field is allowed and possible. The CurrentMode attribute is set to the value of the NewMode field.
0x01	**UnsupportedMode**	The value of the NewMode field doesn't match any entries in the SupportedModes attribute.
0x02	**GenericFailure**	Generic failure code, indicating that switching to the mode indicated by the NewMode field is not allowed or not possible.
0x03	**InvalidInMode**	The received request cannot be handled due to the current mode of the device

The derived cluster code definitions SHALL NOT duplicate the common code definitions. For example, a derived cluster specification SHALL NOT define a status code with the same semantic as the common code of 0x01 (UnsupportedMode).

If the Status field is set to Success, the StatusText field is optional.

If the Status field is set to UnsupportedMode, the StatusText field SHALL be an empty string.

If the Status field is not set to Success or UnsupportedMode, the StatusText field SHALL include a vendor-defined error description which can be used to explain the error to the user. For example, if the Status field is set to InvalidInMode, the StatusText field SHOULD indicate why the transition to the requested mode is not allowed, given the current mode of the device, which may involve other clusters.

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1.10.8. Mode Namespace

This section provides the definitions of the mode tag ranges and of the common standard mode tag values available in the instances of this cluster.

The following table defines the enumeration ranges for the ModeTagStruct Value field values.

Mode Tag Value Range	Range Name	Summary
0x0000 to 0x3FFF	**CommonTags**	Common standard values defined in this cluster specification.
0x4000 to 0x7FFF	**DerivedClusterTags**	Derived cluster specific standard values defined in the derived cluster specification.
0x8000 to 0xBFFF	**MfgTags**	Manufacturer-specific values. For the derived cluster instances, these are manufacturer specific under the derived cluster.

The derived cluster specific standard value definitions SHALL not duplicate the common standard value definitions. For example, a derived cluster specification can't define a mode tag value with the same mode as the common standard tag value of 0x0001 (Quick).

The set of ModeTagStruct Value field values defined in each of the generic or derived Mode Base cluster specifications is called ModeTag.

The following table defines the common ModeTag values.

Mode Tag Value	Name
0x0000	**Auto**
0x0001	**Quick**
0x0002	**Quiet**
0x0003	**LowNoise**
0x0004	**LowEnergy**
0x0005	**Vacation**
0x0006	**Min**
0x0007	**Max**
0x0008	**Night**
0x0009	**Day**

1.10.8.1. Auto Tag

The device decides which options, features and setting values to use.

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1.10.8.2. Quick Tag

The mode of the device is optimizing for faster completion.

1.10.8.3. Quiet Tag

The device is silent or barely audible while in this mode.

1.10.8.4. LowNoise Tag

Either the mode is inherently low noise or the device optimizes for that.

1.10.8.5. LowEnergy Tag

The device is optimizing for lower energy usage in this mode. Sometimes called "Eco mode".

1.10.8.6. Vacation Tag

A mode suitable for use during vacations or other extended absences.

1.10.8.7. Min Tag

The mode uses the lowest available setting value.

1.10.8.8. Max Tag

The mode uses the highest available setting value.

1.10.8.9. Night Tag

The mode is recommended or suitable for use during night time.

1.10.8.10. Day Tag

The mode is recommended or suitable for use during day time.

1.11. Low Power Cluster

This cluster provides an interface for managing low power mode on a device.

This cluster would be supported on an endpoint that represents a physical device with a low power mode. This cluster provides a sleep() command to allow clients to manually put the device into low power mode. There is no command here to wake up a sleeping device because that operation often involves other protocols such as Wake On LAN. Most devices automatically enter low power mode based upon inactivity.

The cluster server for Low Power is implemented by a device that supports a low power mode, such as a TV, Set-top box, or Smart Speaker.

NOTE	We have considered a “DisableLowPowerMode” command but have not added it due to suspected issues with energy consumption regulations. This can be added in

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the future.

1.11.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

1.11.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	LOWPOWER

1.11.3. Cluster ID

ID	Name
0x0508	Low Power

1.11.4. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**Sleep**	client ⇒ server	Y	O	M

1.11.4.1. Sleep Command

This command SHALL put the device into low power mode.

1.12. Wake On LAN Cluster

This cluster provides an interface for managing low power mode on a device that supports the Wake On LAN or Wake On Wireless LAN (WLAN) protocol (see [Wake On LAN]).

This cluster would be supported on IP devices that have a low power mode AND support the ability to be woken up using the Wake on LAN or Wake on WLAN protocol. This cluster provides the device MAC address which is a required input to the Wake on LAN protocol. Besides the MAC address, this cluster provides an optional link-local IPv6 address which is useful to support "Wake on Direct Packet" used by some Ethernet and Wi-Fi devices.

Acting on the MAC address or link-local IPv6 address information does require the caller to be in the same broadcast domain as the destination. To wake the destination up, the caller sends a multi-cast-based magic UDP packet that contains destination’s MAC address in the UDP payload to FF02::1, the IPv6 all-nodes link-local multicast group address. If the optional link-local address is provided by the destination through this cluster, the caller also sends the magic UDP packet in unicast to that link-local address. This unicast-based method is particularly useful for Wi-Fi devices, since due to

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lack of MAC layer retransmission mechanism, multicast over Wi-Fi is not as reliable as unicast. If a device provides the link-local address in this cluster, its Ethernet controller or Wi-Fi radio SHALL respond to the IPv6 neighbor solicitation message for the link-local address without the need to wake host CPU up. In order to receive the magic or neighbor solicitation packets in multicast, the Wi-Fi devices must support Group Temporal Key (GTK) rekey operation in low power mode.

Most devices automatically enter low power mode based upon inactivity.

The cluster server for Wake on LAN or Wake on WLAN is implemented by a device that supports the Wake on LAN/WLAN protocol, such as a TV, Set-top Box, or Smart Speaker.

1.12.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

1.12.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	WAKEONLAN

1.12.3. Cluster ID

ID	Name
0x0503	Wake on LAN

1.12.4. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**MACAddress**	string	max 12	F		R V	O
0x0001	**LinkLocalAddress**	ipv6adr	desc	F		R V	O

1.12.4.1. MACAddress Attribute

This attribute SHALL indicate the current MAC address of the device. Only 48-bit MAC Addresses SHALL be used for this attribute as required by the Wake on LAN protocol.

Format of this attribute SHALL be an upper-case hex-encoded string representing the hex address, like `12345678ABCD`.

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1.12.4.2. LinkLocalAddress Attribute

This attribute SHALL indicate the current link-local address of the device. Only 128-bit IPv6 link-local addresses SHALL be used for this attribute.

NOTE	Some companies may consider MAC Address to be protected data subject to PII handling considerations and will therefore choose not to include it or read it. The MAC Address can often be determined using ARP in IPv4 or NDP in IPv6.

1.13. Switch Cluster

This cluster exposes interactions with a switch device, for the purpose of using those interactions by other devices.
Two types of switch devices are supported: latching switch (e.g. rocker switch) and momentary switch (e.g. push button), distinguished with their feature flags.
Interactions with the switch device are exposed as attributes (for the latching switch) and as events (for both types of switches).
An interested client MAY subscribe to these attributes/events and thus be informed of the interactions, and can perform actions based on this, for example by sending commands to perform an action such as controlling a light or a window shade.

1.13.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Behavior clarifications for ambiguous combinations of events; Introduction of TotalNumberOfPressesCounted == 0 when MultiPressMax is exceeded; Introduction of ActionSwitch feature flag.

1.13.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	SWTCH

1.13.3. Cluster ID

ID	Name
0x003B	Switch

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1.13.4. Features

These feature flags SHALL be used to indicate the physics as well as the detection/reporting capabilities of the device represented by the cluster.

Bit	Code	Feature	Conformance	Summary
0	**LS**	LatchingSwitch	O.a	Switch is latching
1	**MS**	MomentarySwitch	O.a	Switch is momentary
2	**MSR**	MomentarySwitchRelease	[MS & !AS]	Switch supports release events generation
3	**MSL**	MomentarySwitchLongPress	[MS & (MSR | AS)]	Switch supports long press detection
4	**MSM**	MomentarySwitchMultiPress	AS, [MS & MSR]	Switch supports multi-press detection
5	**AS**	ActionSwitch	[MS]	Switch is momentary, targeted at specific user actions (focus on multi-press and optionally long press) with a reduced event generation scheme

1.13.4.1. LatchingSwitch Feature

This feature flag is for a switch that maintains its position after being pressed (or turned).

1.13.4.2. MomentarySwitch Feature

This feature flag is for a switch that does not maintain its position after being pressed (or turned). After releasing, it goes back to its idle position.

1.13.4.3. MomentarySwitchRelease Feature

This feature flag is for a momentary switch that can distinguish and report release events.

1.13.4.4. MomentarySwitchLongPress Feature

This feature flag is for a momentary switch that can distinguish and report long presses from short presses.

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1.13.4.5. MomentarySwitchMultiPress Feature

This feature flag is for a momentary switch that can distinguish and report double press and potentially multiple presses with more events, such as triple press, etc.

1.13.4.6. ActionSwitch Feature

This feature flag indicates simplified handling of events for multi-press-capable switches. See Section 1.13.8, “Sequence of events for MultiPress”.

1.13.5. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**NumberOfPositions**	uint8	min 2	F		R V	M
0x0001	**CurrentPosition**	uint8	max (NumberOfPositions - 1)	N		R V	M
0x0002	**MultiPressMax**	uint8	min 2	F		R V	MSM

1.13.5.1. NumberOfPositions Attribute

This attribute SHALL indicate the maximum number of positions the switch has. Any kind of switch has a minimum of 2 positions. Also see Section 1.13.10, “Multi Position Details” for the case NumberOfPositions>2.

1.13.5.2. CurrentPosition Attribute

This attribute SHALL indicate the position of the switch.

The valid range is zero to NumberOfPositions - 1.

CurrentPosition value 0 SHALL be assigned to the default position of the switch: for example the "open" state of a rocker switch, or the "idle" state of a push button switch.

1.13.5.3. MultiPressMax Attribute

This attribute SHALL indicate how many consecutive presses can be detected and reported by a momentary switch which supports multi-press (MSM feature flag set).

For example, a momentary switch supporting single press, double press and triple press, but not quad press and beyond, would return the value 3.

When more than MultiPressMax presses are detected within a multi-press sequence:

- The server for cluster revision < 2 SHOULD generate a MultiPressComplete event with the Total-

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NumberOfPressesCounted field set to the value of the MultiPressMax attribute, and avoid generating any further InitialPress and MultiPressOngoing events until the switch has become fully idle (i.e. no longer in the process of counting presses within the multipress).

- The server for cluster revision >= 2 SHALL generate a MultiPressComplete event with the TotalNumberOfPressesCounted field set to zero (indicating an aborted sequence), and SHALL NOT generate any further InitialPress and MultiPressOngoing events until the switch has become fully idle (i.e. no longer in the process of counting presses within the multipress).

This approach avoids unintentionally causing intermediate actions where there is a very long sequence of presses beyond MultiPressMax that MAY be taken in account specially by switches (e.g. to trigger special behavior such as factory reset for which generating events towards the client is not appropriate).

1.13.6. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**SwitchLatched**	INFO		V	LS
0x01	**InitialPress**	INFO		V	MS
0x02	**LongPress**	INFO		V	MSL
0x03	**ShortRelease**	INFO		V	MSR
0x04	**LongRelease**	INFO		V	MSL
0x05	**MultiPressOngoing**	INFO		V	MSM & !AS
0x06	**MultiPressComplete**	INFO		V	MSM

1.13.6.1. SwitchLatched Event

This event SHALL be generated, when the latching switch is moved to a new position.
It MAY have been delayed by debouncing within the switch.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**NewPosition**	uint8	0 to (NumberOfPositions - 1)			M

1.13.6.1.1. NewPosition Field

This field SHALL indicate the new value of the CurrentPosition attribute, i.e. after the move.

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1.13.6.2. InitialPress Event

This event SHALL be generated, when the momentary switch starts to be pressed (after debouncing).

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**NewPosition**	uint8	0 to (NumberOfPositions - 1)			M

1.13.6.2.1. NewPosition Field

This field SHALL indicate the new value of the CurrentPosition attribute, i.e. while pressed.

1.13.6.3. LongPress Event

This event SHALL be generated when the momentary switch has been pressed for a "long" time. The time interval constituting a "long" time is manufacturer-determined, since it depends on the switch physics.

- When the AS feature flag is set, this event:
  - SHALL NOT be generated during a multi-press sequence (since a long press is a separate cycle from any multi-press cycles);
  - SHALL only be generated after the first InitialPress following a MultiPressComplete when a long press is detected after the idle time.
- Else, when the MSM feature flag is set, this event:
  - SHALL NOT be generated during a multi-press sequence (since a long press is a separate cycle from any multi-press cycles);
  - SHALL only be generated after the first InitialPress following a MultiPressComplete when a long press is detected after the idle time;
  - SHALL NOT be generated after a MultiPressOngoing event without an intervening MultiPressComplete event.

The above constraints imply that for a given activity detection cycle of a switch having MSM and/or MSL feature flags set, the entire activity is either a single long press detection cycle of (InitialPress, LongPress, LongRelease), or a single multi-press detection cycle (ending in MultiPressComplete), where presses that would otherwise be reported as long presses are instead reported as a counted press in the MultiPressComplete event, and as InitialPress/ShortRelease pairs otherwise (where applicable).

The rationale for this constraint is the ambiguity of interpretation of events when mixing long presses and multi-press events.

The data of this event SHALL contain the following information:

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**NewPosition**	uint8	0 to (NumberOfPositions - 1)			M

1.13.6.3.1. NewPosition Field

This field SHALL indicate the new value of the CurrentPosition attribute, i.e. while pressed.

1.13.6.4. ShortRelease Event

If the server has the Action Switch (AS) feature flag set, this event SHALL NOT be generated at all, since setting the Action Switch feature flag forbids the Momentary Switch ShortRelease (MSR) feature flag from being set. Otherwise, the following paragraphs describe the situations where this event is generated.

This event SHALL be generated, when the momentary switch has been released (after debouncing).

- If the server has the Momentary Switch LongPress (MSL) feature flag set, then this event SHALL be generated when the switch is released if ***no*** LongPress event had been generated since the previous InitialPress event.
- If the server does not have the Momentary Switch LongPress (MSL) feature flag set, this event SHALL be generated when the switch is released - even when the switch was pressed for a long time.
- Also see Section 1.13.7, “Sequence of generated events”.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PreviousPosition**	uint8	0 to (NumberOfPositions - 1)			M

1.13.6.4.1. PreviousPosition Field

This field SHALL indicate the previous value of the CurrentPosition attribute, i.e. just prior to release.

1.13.6.5. LongRelease Event

This event SHALL be generated, when the momentary switch has been released (after debouncing) and after having been pressed for a long time, i.e. this event SHALL be generated when the switch is released if a LongPress event has been generated since the previous InitialPress event. Also see Section 1.13.7, “Sequence of generated events”.

The data of this event SHALL contain the following information:

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PreviousPosition**	uint8	0 to (NumberOfPositions - 1)			M

1.13.6.5.1. PreviousPosition Field

This field SHALL indicate the previous value of the CurrentPosition attribute, i.e. just prior to release.

1.13.6.6. MultiPressOngoing Event

If the server has the Action Switch (AS) feature flag set, this event SHALL NOT be generated at all. Otherwise, the following paragraphs describe the situations where this event is generated.

This event SHALL be generated to indicate how many times the momentary switch has been pressed in a multi-press sequence, *during* that sequence. See Section 1.13.8, “Sequence of events for MultiPress”.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**NewPosition**	uint8	0 to (NumberOfPositions - 1)			M
1	**CurrentNumberOfPressesCounted**	uint8	2 to MultiPressMax			M

1.13.6.6.1. NewPosition Field

This field SHALL indicate the new value of the CurrentPosition attribute, i.e. while pressed.

1.13.6.6.2. CurrentNumberOfPressesCounted Field

This field SHALL contain:

- a value of 2 when the second press of a multi-press sequence has been detected,
- a value of 3 when the third press of a multi-press sequence has been detected,
- a value of N when the Nth press of a multi-press sequence has been detected.

1.13.6.7. MultiPressComplete Event

This event SHALL be generated to indicate how many times the momentary switch has been pressed in a multi-press sequence, after it has been detected that the sequence has *ended*. See Sec-

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tion 1.13.8, “Sequence of events for MultiPress”.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PreviousPosition**	uint8	0 to (NumberOfPositions - 1)			M
1	**TotalNumberOfPressesCounted**	uint8	max MultiPressMax			M

The PreviousPosition field SHALL indicate the previous value of the CurrentPosition attribute, i.e. just prior to release.

The TotalNumberOfPressesCounted field SHALL contain:

- a value of 0 when there was an aborted multi-press sequence, where the number of presses goes beyond MultiPressMax presses,
- a value of 1 when there was exactly one press in a multi-press sequence (and the sequence has ended), i.e. there was no double press (or more),
- a value of 2 when there were exactly two presses in a multi-press sequence (and the sequence has ended),
- a value of 3 when there were exactly three presses in a multi-press sequence (and the sequence has ended),
- a value of N when there were exactly N presses in a multi-press sequence (and the sequence has ended).

NOTE

The introduction of TotalNumberOfPressesCounted supporting the value 0 MAY impact clients of switches using cluster revision 1 since such servers would not use this value of TotalNumberOfPressesCounted to indicate an aborted sequence. Clients SHOULD always act using the TotalNumberOfPressesCounted field taken into account since for values from 1 to MultiPressMax, the user action that led to the event was different depending on the count.

1.13.7. Sequence of generated events

This section describes the sequence of events that will be generated by three types of momentary switches (distinguished by their feature flags). For each switch type, we will define the sequence of generated events for these three interactions:

1. Sequence for a switch which is pressed briefly.
2. Sequence for a switch pressed for a long time.
3. Sequence for a switch pressed for a very long time.

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In the three interactions described in the subsections below, if the NumberOfPositions attribute is equal to 2, the InitialPress and LongPress events have the NewPosition field set to 1 and the ShortRelease and LongRelease events have the PreviousPosition field set to 1. For larger values of the NumberOfPositions attribute, see Section 1.13.10, “Multi Position Details”.

1.13.7.1. Switches with long-press detection support (MSL feature flag set)

This switch (with feature flags MS & MSL both set) SHALL generate either a sequence of two or three (depending on how long the switch is pressed) events for one interaction cycle, in the order given below and illustrated in the figure below.

- Upon a short press:
  - Generate InitialPress
  - If the MSR feature flag is set and the AS feature flag is NOT set, generate ShortRelease after the previous event, when release is detected.
  - If the MSM feature flag is set, generate MultiPressComplete(1) after the previous event(s).
- Upon a long press (or very long press):
  - Generate InitialPress
  - Generate LongPress after the previous event, when long press is detected
  - Generate LongRelease (and NOT MultiPressComplete(1)) after the previous event, when switch release is detected.

A LongPress event SHALL be generated exactly once for this case and SHALL NOT be repeated, irrespective how long the switch is pressed.

The image shows a time representation of the state of the switch (low=not pressed, high=pressed) with the colored dots indicating the various events generated at that moment in time.

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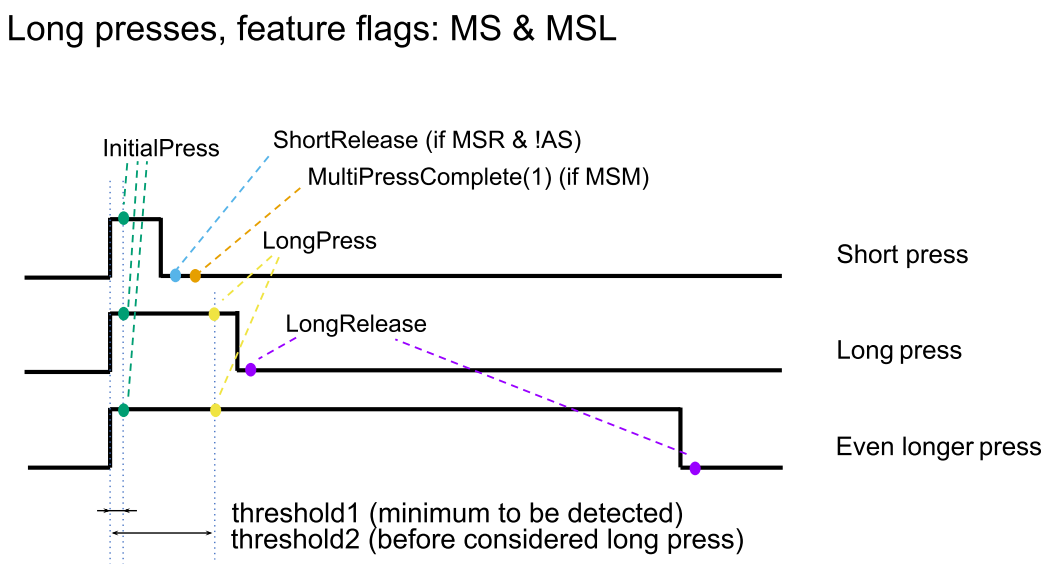


Figure 3. Switch device with MSL feature flag set (single press case)

1.13.7.2. Switches without long-press detection support (MSL feature flag not set)

This switch (with feature flags states MS & !MSL) SHALL NOT generate LongPress and LongRelease events and therefore it SHALL generate a sequence of two events for one interaction cycle, irrespective of how long the switch is pressed, in the order given and illustrated below, based on whether the MSM, AS and MSR flags are set.

- Any press length:
  1. Generate InitialPress
  2. If the MSR feature flag is set and the AS feature flag is NOT set, generate ShortRelease after the previous event, when the release is detected.
  3. If the MSM feature flag is set, generate MultiPressComplete(1) after the previous event(s).

NOTE	Even after a "long" period being pressed, the release event is never LongRelease. A device without the MSL feature flag SHALL NOT generate the LongPress and LongRelease events.

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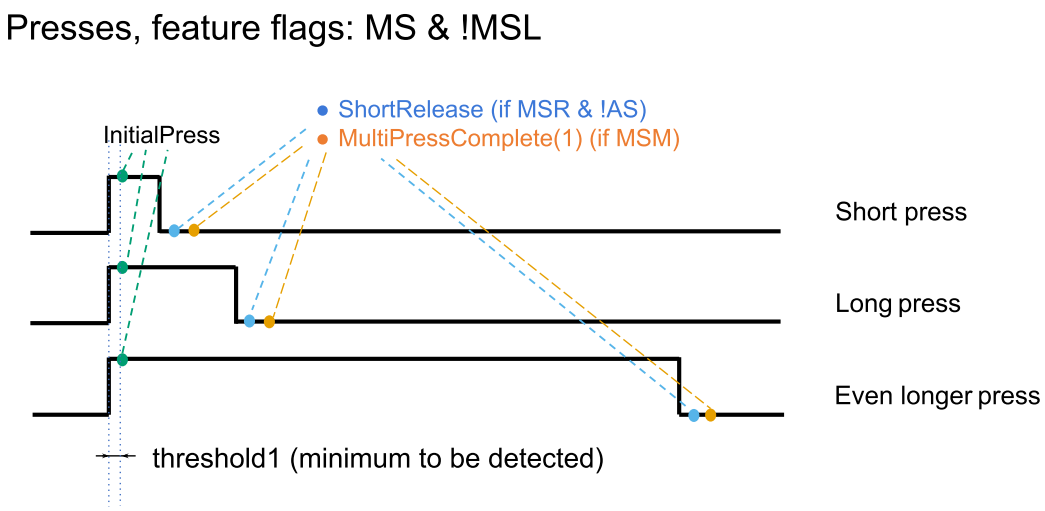


Figure 4. Switch device without MSL feature flag not set (single press case)

1.13.7.3. Switches with most basic MS-only feature flag set, not (MSR | MSL | MSM | AS)

This switch (with feature flags states MS & !MSR & !MSL & !MSM & !AS) SHALL generate a single InitialPress event for one interaction cycle, irrespective of how long the switch is pressed, as illustrated in the figure below.

A device with this set of feature flags SHALL NOT generate any of the ShortRelease, LongPress and LongRelease events.

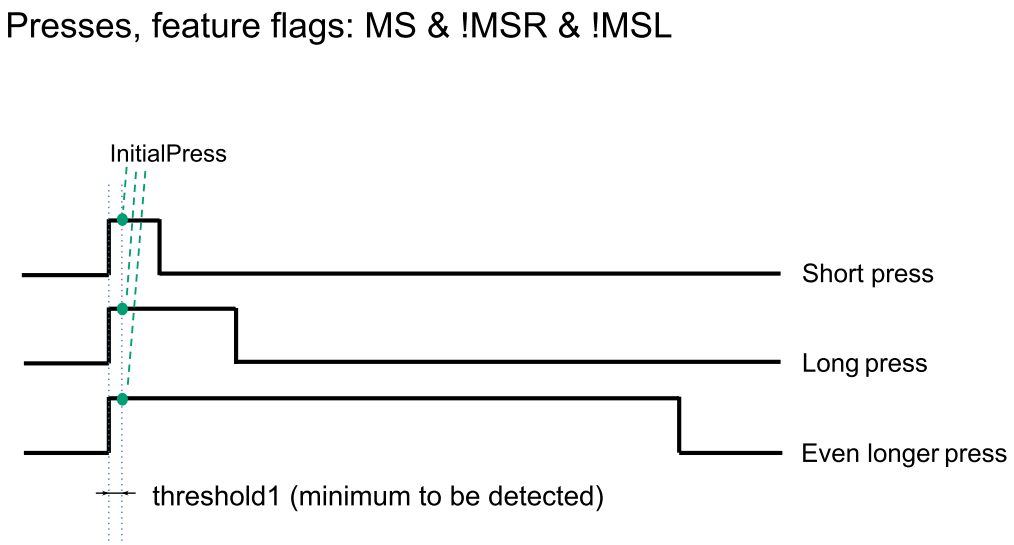


Figure 5. Switch device delivering only 'InitialPress' events (single press case)

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1.13.8. Sequence of events for MultiPress

Multi-press detection is a feature of momentary switches (indicated with feature flag MSM being set, with optional AS feature flag set) that they can count and report sequences of press-release cycles within a certain time frame, for example to indicate that the user has pressed the switch once, twice or three (or even more) times in succession. The definition of the time window for this detection is manufacturer-specific since it depends on the switch physics. The maximum number of presses which can be detected and reported SHALL be indicated in attribute MultiPressMax. The minimum value and default value for MultiPressMax are both 2.

A switch supporting MultiPress SHALL set feature flags MS & MSM, and MAY optionally, according to conformance rules of each flag, set flags MSL, MSR, and AS.

The ActionSwitch (AS) feature flag is a behavior selector which, when set, reduces the number of events generated by the switch.

A press cycle when MSM is set is **either** a long-press cycle (if MSL set), **or** a multi-press cycle.

- If the MSL feature flag is set and the cycle is deemed "long" during the first press cycle after InitialPress, then the switch SHALL react as described in Section 1.13.7.1, “Switches with long-press detection support (MSL feature flag set)”, and SHALL NOT consider this to be the start of a multi-press sequence (see penultimate case illustrated in Figure 7, “Switch device with MSM and AS feature flags (multi-press case)”).
- If the MSL feature flag is not set or if the first button press cycle is short, then the switch SHALL count presses until the switch becomes sufficiently idle, and SHALL NOT generate LongPress and LongRelease events for intermediate presses within the multi-press cycle that are "long".

1.13.8.1. Multi-press behavior when ActionsSwitch (AS) feature flag is not set

A switch with the MSM feature flag set but the AS flag unset:

- SHALL generate an InitialPress event at the start of every button press within a button multi-press cycle;
- SHALL generate a ShortRelease event at the end of every button press within a button multi-press cycle if MSR is set;
- SHALL generate a MultiPressOngoing event after the InitialPress event of the second and subsequent presses of a button multi-press cycle;
- SHALL generate a MultiPressComplete at the end of a multi-press cycle, including single presses;
- SHALL NOT (for cluster revision >=2) or SHOULD NOT (for cluster revision <2) generate any LongPress and LongRelease events within the multi-press cycle.

The MultiPressOngoing event is provided for parties interested in keeping track of the actual presses during the multi-press sequence. The MultiPressComplete event is provided for parties interested in the outcome of the whole sequence: after the multi-press sequence has ended, they will receive the MultiPressComplete event indicating how many times the switch was pressed.

In the figure below, several sequences of user interaction are indicated:

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- Single press sequence: after the press and release moments, the InitialPress SHALL be generated, and ShortRelease events SHALL be generated if the MSR feature flag is set. After some further time, when the switch has detected that there is no second press, it SHALL generate MultiPressComplete(1) since it has detected that the sequence consisted of one press. No MultiPressOngoing event SHALL be generated for this case.
- Double press sequence: after each of the press and release moments, the InitialPress SHALL be generated, and ShortRelease events SHALL be generated if the MSR feature flag is set. Additionally, when the switch is pressed for the second time, the MultiPressOngoing(2) event SHALL be generated, as the switch has detected the second press. Note that this event coincides with the second InitialPress event; both SHALL be generated. After some further time, when the switch has detected that there is no third press, it SHALL generate MultiPressComplete(2) since it has detected that the sequence consisted of two presses.
- Triple press sequence: after each press and release moments, the InitialPress SHALL be generated, and ShortRelease events SHALL be generated if the MSR feature flag is set. Additionally, when the switch is pressed for the second time, the MultiPressOngoing(2) event SHALL be generated, as the switch has detected the second press. Note that this event coincides with the second InitialPress event; both SHALL be generated. Additionally, when the switch is pressed for the third time, the MultiPressOngoing(3) event SHALL be generated, as the switch has detected the third press. Note that this event coincides with the third InitialPress event; both SHALL be generated. After some further time, when the switch has detected that there is no fourth press, it SHALL generate MultiPressComplete(3) since it has detected that the sequence consisted of three presses.
  - This case specifically does NOT generate LongPress and LongRelease events within the sequence (even though the figure below illustrates a longer press duration for the second press in one of the cases), since multi-press sequences SHOULD/SHALL NOT generate LongPress and LongRelease events within the sequence (see LongPress for more details).
- N presses starting with a long press (if MSL feature flag is set): this starts with a long press sequence wish SHALL be considered according to Section 1.13.7.1, “Switches with long-press detection support (MSL feature flag set)”. Then it is a sequence similar to triple press, with N presses instead of 3.

For the above cases where multiple events need to be generated at the same time, the MultiPressOngoing event SHALL be generated directly after the InitialPress event.

NOTE	The numbers in parentheses in the bulleted text above and in the figure below indicate the value of the CurrentNumberOfPressesCounted resp. TotalNumberOfPressesCounted field in the event data.
NOTE	As with the other figures, sufficient debounce time needs to be take into account for the detection of press and release events. This is included in the figure, and has been left out of the description above for readability, but SHALL be applied.
NOTE	Several of the events require very low congestion and latency in the final system to be usable for real-time control and use cases. Care must be taken during design not to rely on very low latency and real-time event timestamps if the low-latency

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assumptions do not hold.

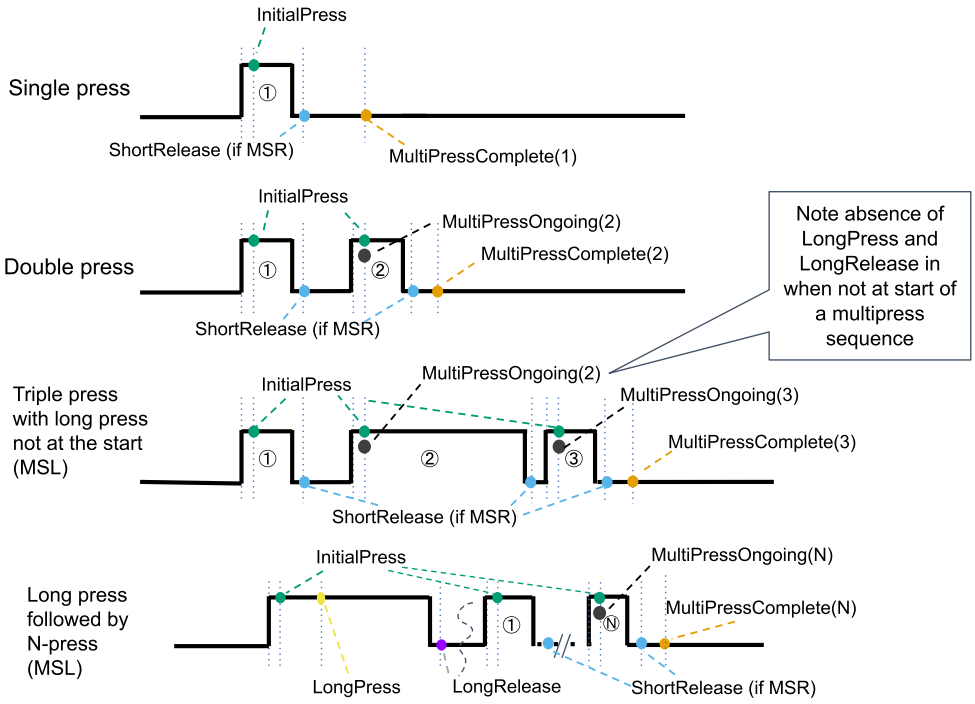


Figure 6. Switch device with MSM and !AS feature flags (multi-press case)

1.13.8.2. Multi-press behavior when ActionsSwitch (AS) feature flag is set

When the ActionSwitch feature flag is set, the behavior for all presses:

- The switch SHALL only generate an InitialPress event at the start of every button press cycle (whether long or not);
- The switch SHALL NOT generate ShortRelease events;
- The switch SHALL NOT generate MultiPressOngoing events.

Because of these simplifications, clients can decide to act purely on the following to determine user action:

- LongRelease (if MSL feature flag is set) to determine the end of a long press.
  - InitialPress and LongPress (if MSL feature flag is set), if needed, to detect the start of the press and long-press periods so that actions such as "do action X while long pressed and stop on release" can be supported.
- MultiPressComplete to determine the end of a series of presses (0 .. MultiPressMax)
  - Only the final count of presses detected by the switch is provided.
  - The value of 0 indicates an aborted press (e.g. maybe too many presses in a row, button held down for too long, etc).
  - The value of 1 indicates a single press, 2 a double-press, etc.

Because of the omission of ShortRelease, MultiPressOngoing and the "inner" InitialPress events, less network traffic is generated, while not missing the overall "action" intent from the end-user.

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In the figure below, several sequences of user interaction are indicated:

- Single press sequence: after the initial press detection, the InitialPress event is generated. After some further time, when the switch has detected that there is no second press, and the press is not a long press (if MSL feature flag is set), it generates the MultiPressComplete(1) event since it has detected that the sequence consisted of one press.
- Double press sequence: after the initial press detection, the InitialPress event is generated. After some further time, when the switch has detected that there is no third press, it generates the MultiPressComplete(2) event since it has detected that the sequence consisted of two presses.
- N-press sequence: after the initial press detection, the InitialPress event is generated. Additionally, when the switch is pressed for the second time, we note that the press is long, but there is not a LongPress or LongRelease event, since long presses inside (not at the start) of a multi-press sequence are treated as short presses. After some further time, when the switch has detected that there are no further presses beyond N presses, it generates a MultiPressComplete(N) event since it has detected that the sequence consisted of N presses.
- Long press sequence followed by N-press sequence:
  - This case contains to two separate sequences, since a long press sequence is always an independent sequence, and does not get reported as "long" within multi-presses (see previous example case).
  - The first sequence is a Long Press sequence of: InitialPress, LongPress, and finally, LongRelease.
  - The second sequence, whether immediate or not, is a N-press sequence of: InitialPress, and finally, MultiPressComplete(N).
- N-press sequence with N > MultiPressMax: after the initial press detection, the InitialPress event is generated. After some further time, when the switch has detected that there are no further presses beyond N presses, it generates a MultiPressComplete(0) event since it has detected that the multi-press sequence is done, but since there were more presses than MultiPressMax, the sequence uses 0 for TotalNumberOfPressesCounted to indicate invalid complete sequence.

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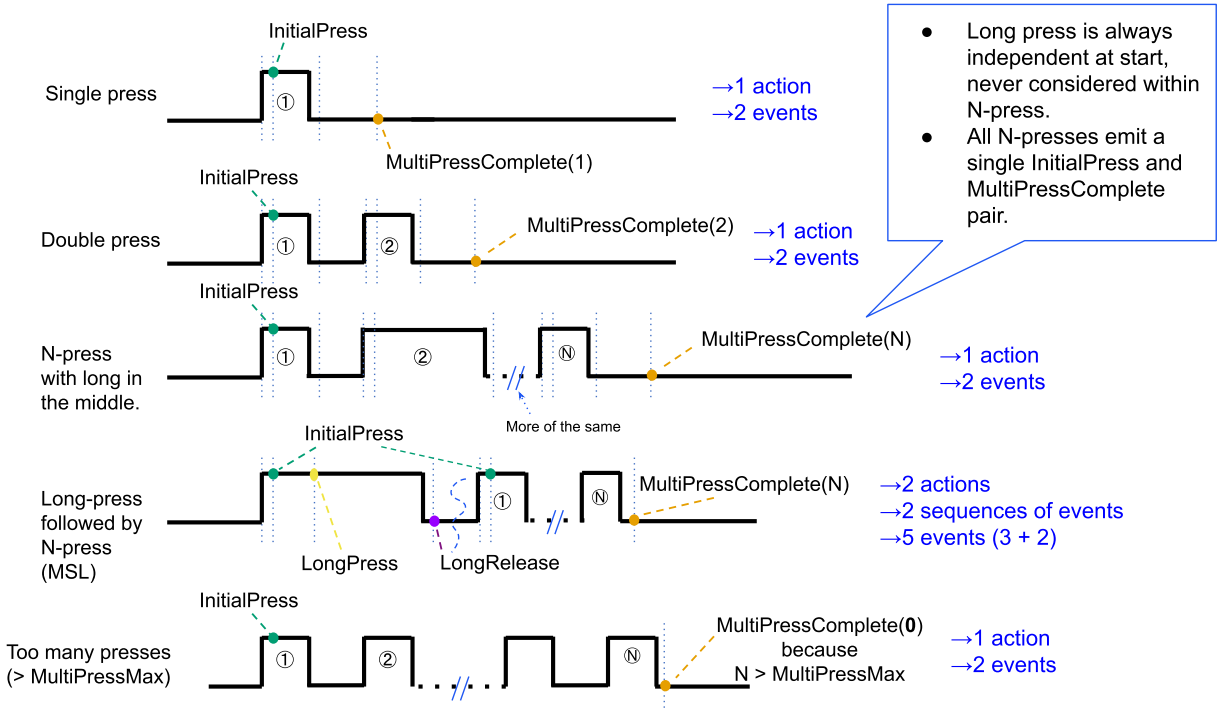


Figure 7. Switch device with MSM and AS feature flags (multi-press case)

1.13.9. Summary of cases for MS feature flag

Given the variety of situations that may arise due to the different switch capabilities, the following describes the edges that would unambiguously convey user action intents:

- MS & !MSR & !MSM: Every action cycle is only a single InitialPress event.
- MS & !MSM: Every action cycle ends on ShortRelease or LongRelease (if MSL feature flag is set), depending on long or short press and whether MSR/MSL feature flags are set.
- MS & MSM: Every action cycle ends on either LongRelease (if MSL feature flag is set and first press was long) or MultiPressComplete.
  - The AS feature flag does not change the outcome of which events can be used for action detection.
  - The AS feature flag is backwards-compatible with previous revisions of this cluster.
  - The AS feature flag primarily offers a reduction of events generated.

The above list is relevant for the case of a client which does not use the events indicating progress during the action cycle (i.e. ShortRelease, MultiPressOngoing, repeated InitialPress) for its operation.

1.13.10. Multi Position Details

This section will discuss some archetypes of switch devices with more than 2 positions and how they SHALL set attribute values and generate events matching their characteristics.

For the SwitchLatched, InitialPress, LongPress and MultiPressOngoing events, the field NewPosition SHALL be set to the value corresponding to the new position to which the switch was moved. For the ShortRelease, LongRelease and MultiPressComplete events, the field PreviousPosition SHALL be

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set to the value corresponding to the position of the switch just preceding the (latest) release.

1.13.10.1. Latching Switch with N stable positions (N>2) with fixed sequence

With such a device, the user can move the switch from a position M to positions M-1 and M+1 (either with a wraparound between the end positions, or fixed stop at the end positions).
On each interaction with the switch device, it SHALL generate a SwitchLatched event with the New-Position field set to the value associated with the new position.
Due to the physical constraints, such an event will have a NewPosition field which is equal to the previous NewPosition field plus or minus 1 (modulo NumPositions if the switch does not have end stops).

In a first example, a switch has 3 positions, associated with values 0, 1 and 2. In this case, wrap-around is not possible: from position 0 it can only be moved to position 1.

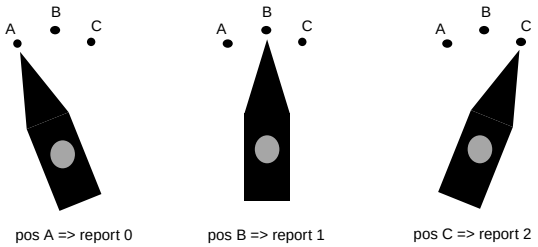


Figure 8. Rotary switch device with 3 positions

In another example, a switch has 8 positions, associated with values 0 through 7. In this case, the physics of the switch allow wraparound: from position 0 it can be moved to position 1 or to position 7.

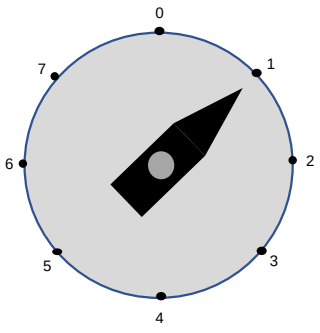


Figure 9. Rotary switch device with 8 positions and wraparound

1.13.10.2. Latching Switch with N stable positions (N>2) with random sequence (example: radio buttons)

With such a device, the user can press any of the available buttons, so this switch does not show the incrementing or decrementing behavior of NewPosition which we discussed for the latching switch

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with fixed sequence. In the example in the figure below, the 5 buttons are labeled "A" through "E" for the user and are associated with values 0 through 4. The user first presses the "A" button, and the switch device generates a SwitchLatched event with NewPosition set to 0. Then the user first presses the "D" button, and the switch device generates a SwitchLatched event with NewPosition set to 4.

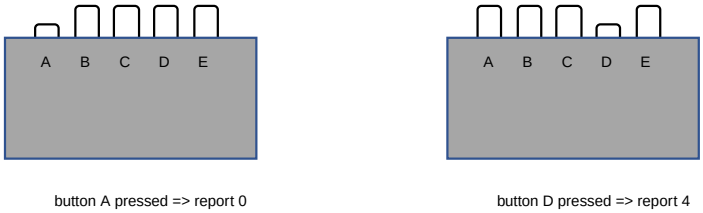


Figure 10. Switch device with radio buttons

1.13.10.3. Momentary Switch with 2 or more non-stable positions

For a Momentary Switch with more than 1 stable position, it depends on the physics of the switch device which sequence of events will be generated.

NOTE

In this section we will mention only the InitialPress and ShortRelease events. The switch device could also generate the other events defined above for a momentary switch, depending on the capabilities of the switch device and the interaction with the switch device.

The first variant (figure below, example: up/down control for window blinds) shows a switch in neutral position (left figure) which corresponds to CurrentPosition=0. The user can press the top side (position value 1) or the bottom side (position value 2). It is not possible to go directly from position 1 to position 2 or vice versa - the switch will always need to go through the neutral position 0.

So when the user presses the top side of the switch, the InitialPress (NewPosition=1) event will be generated. When they release the top side, the ShortRelease (PreviousPosition=1) event will be generated. The user continues to press the bottom side, and the event InitialPress (NewPosition=2) is generated. Upon release of the bottom side, the event ShortRelease (PreviousPosition=2) is generated.

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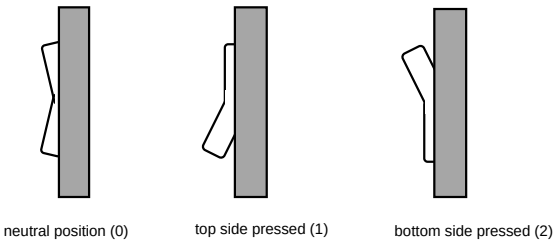


Figure 11. Up/down control switch device

Another variant (figure below, example: joystick) has a control handle with a neutral position in the middle (left figure) which corresponds to CurrentPosition=0. The handle can be moved along the dotted lines.

In the middle figure, the user moves the handle to the East position and then releases it (which makes it return to the neutral middle position). This generates this sequence of events:

```
InitialPress (NewPosition=3)       // move to East
ShortRelease (PreviousPosition=3)  // back to middle (from East)
```

In the righthand figure, the user moves the handle to the SouthWest position, then to the South position and then releases it (which makes it return to the neutral middle position). This generates this sequence of events:

```
InitialPress (NewPosition=6)       // move to SouthWest
InitialPress (NewPosition=5)       // move to South
ShortRelease (PreviousPosition=5)  // back to middle (from South)
```

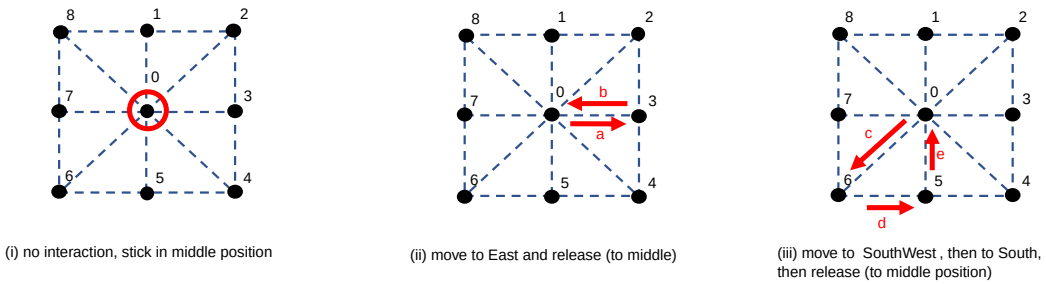


Figure 12. Switch device with joystick

Therefore, in the "joystick" variation, there could be a succession of InitialPress events without an intermediate ShortRelease events - unlike the "window blinds control" variation which will always have such an intermediate ShortRelease event.

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1.14. Operational State Cluster

This cluster supports remotely monitoring and, where supported, changing the operational state of any device where a state machine is a part of the operation.

This cluster defines common states, scoped to this cluster (e.g. Stopped, Running, Paused, Error). A derived cluster specification may define more states scoped to the derivation. Manufacturer specific states are supported in this cluster and any derived clusters thereof. When defined in a derived instance, such states are scoped to the derivation.

Actual state transitions are dependent on both the implementation, and the requirements that may additionally be imposed by a derived cluster.

An implementation that supports remotely starting its operation can make use of this cluster's Start command to do so. A device that supports remote pause or stop of its currently selected operation can similarly make use of this cluster's Pause and Stop commands to do so. The ability to remotely pause or stop is independent of how the operation was started (for example, an operation started by using a manual button press can be stopped by using a Stop command if the device supports remotely stopping the operation).

Additionally, this cluster provides events for monitoring the operational state of the device.

1.14.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	The Pause and Resume commands are usable in all compatible states and defined reserved ranges for base/derived clusters
3	Changed CountdownTime attribute to Q quality

1.14.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	OPSTATE

1.14.3. Cluster ID

ID	Name
0x0060	Operational State

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1.14.4. Data Types

1.14.4.1. OperationalStateEnum Type

This type defines the set of known operational state values, and is derived from enum8. The following table defines the applicable ranges for values that are defined within this type. All values that are undefined SHALL be treated as reserved. As shown by the table, states that may be specific to a certain Device Type or other modality SHALL be defined in a derived cluster of this cluster.

Value	Name	Summary
0x00 to 0x3F	**GeneralStates**	Generally applicable values for state, defined herein
0x40 to 0x7F	**DerivedClusterStates**	Derived Cluster defined states
0x80 to 0xBF	**ManufacturerStates**	Vendor specific states

The derived cluster-specific state definitions SHALL NOT duplicate any general state definitions. That is, a derived cluster specification of this cluster cannot define states with the same semantics as the general states defined below.

A manufacturer-specific state definition SHALL NOT duplicate the general state definitions or derived cluster state definitions. That is, a manufacturer-defined state defined for this cluster or a derived cluster thereof cannot define a state with the same semantics as the general states defined below or states defined in a derived cluster. Such manufacturer-specific state definitions SHALL be scoped in the context of the Vendor ID present in the Basic Information cluster.

The following table defines the generally applicable states.

Value	Name	Summary	Conformance
0x00	**Stopped**	The device is stopped	M
0x01	**Running**	The device is operating	M
0x02	**Paused**	The device is paused during an operation	M
0x03	**Error**	The device is in an error state	M

1.14.4.2. OperationalStateStruct Type

The OperationalStateStruct is used to indicate a possible state of the device.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**OperationalStateID**	OperationalStateEnum	all		0	M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
1	**OperationalStateLabel**	string	max 64			OperationalStateID >= 0x80 & OperationalStateID <= 0xBF

1.14.4.2.1. OperationalStateID Field

This SHALL be populated with a value from the OperationalStateEnum.

1.14.4.2.2. OperationalStateLabel Field

This field is present when the OperationalStateID is from the set reserved for Manufacturer Specific States. If present, this SHALL contain a human-readable description of the operational state.

1.14.4.3. ErrorStateEnum Type

This type defines the set of known operational error values, and is derived from enum8. The following table defines the applicable ranges for values that are defined within this type. All values that are undefined SHALL be treated as reserved. As shown by the table, errors that may be specific to a certain Device Type or other modality SHALL be defined in a derived cluster of this cluster.

Value	Name	Summary
0x00 to 0x3F	**GeneralErrors**	Generally applicable values for error, defined herein
0x40 to 0x7F	**DerivedClusterErrors**	Derived Cluster defined errors
0x80 to 0xBF	**ManufacturerError**	Vendor specific errors

The derived cluster-specific error definitions SHALL NOT duplicate the general error definitions. That is, a derived cluster specification of this cluster cannot define errors with the same semantics as the general errors defined below.

The manufacturer-specific error definitions SHALL NOT duplicate the general error definitions or derived cluster-specific error definitions. That is, a manufacturer-defined error defined for this cluster or a derived cluster thereof cannot define errors with the same semantics as the general errors defined below or errors defined in a derived cluster. Such manufacturer-specific error definitions SHALL be scoped in the context of the Vendor ID present in the Basic Information cluster.

The set of ErrorStateID field values defined in each of the generic or derived Operational State cluster specifications is called ErrorState.

1.14.4.3.1. ErrorStateEnum GeneralErrors Range

The following table defines the generally applicable ErrorState values.

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Value	Name	Summary	Conformance
0x00	**NoError**	The device is not in an error state	M
0x01	**UnableToStartOrResume**	The device is unable to start or resume operation	M
0x02	**UnableToCompleteOperation**	The device was unable to complete the current operation	M
0x03	**CommandInvalidInState**	The device cannot process the command in its current state	M

1.14.4.4. ErrorStateStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ErrorStateID**	ErrorStateEnum	all		0	M
1	**ErrorStateLabel**	string	max 64		empty	ErrorStateID >= 0x80 & ErrorStateID <= 0xBF
2	**ErrorStateDetails**	string	max 64		empty	O

1.14.4.4.1. ErrorStateID Field

This SHALL be populated with a value from the ErrorStateEnum.

1.14.4.4.2. ErrorStateLabel Field

This field is present when the ErrorStateID is from the set reserved for Manufacturer Specific errors. If present, this SHALL contain a human-readable description of the error state.

1.14.4.4.3. ErrorStateDetails Field

This SHALL be a human-readable string that provides details about the error condition. As an example, if the ErrorStateID indicates that the device is a Robotic Vacuum that is stuck, the ErrorStateDetails contains "left wheel blocked".

1.14.5. Attributes

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**PhaseList**	list[string]	max 32[max 64]	X	MS	R V	M
0x0001	**CurrentPhase**	uint8	desc	X	MS	R V	M
0x0002	**CountdownTime**	elapsed-s	max 259200	Q X	null	R V	O
0x0003	**OperationalStateList**	list[OperationalStateStruct]	desc		MS	R V	M
0x0004	**OperationalState**	OperationalStateEnum	all			R V	M
0x0005	**OperationalError**	ErrorStateStruct	desc			R V	M

1.14.5.1. PhaseList Attribute

This attribute SHALL indicate a list of names of different phases that the device can go through for the selected function or mode. The list may not be in sequence order. For example in a washing machine this could include items such as "pre-soak", "rinse", and "spin". These phases are manufacturer specific and may change when a different function or mode is selected.

A null value indicates that the device does not present phases during its operation. When this attribute's value is null, the CurrentPhase attribute SHALL also be set to null.

1.14.5.2. CurrentPhase Attribute

This attribute represents the current phase of operation being performed by the server. This SHALL be the positional index representing the value from the set provided in the PhaseList Attribute, where the first item in that list is an index of 0. Thus, this attribute SHALL have a maximum value that is "length(PhaseList) - 1".

This attribute SHALL be null if the PhaseList attribute is null or if the PhaseList attribute is an empty list.

1.14.5.3. CountdownTime Attribute

This attribute SHALL represent the estimated time left before the operation is completed, in seconds.

A value of 0 (zero) means that the operation has completed.

A value of null represents that there is no time currently defined until operation completion. This MAY happen, for example, because no operation is in progress or because the completion time is

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unknown.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- If it has changed due to a change in the CurrentPhase or OperationalState attributes, or
- When it changes from 0 to any other value and vice versa, or
- When it changes from null to any other value and vice versa, or
- When it increases, or
- When there is any increase or decrease in the estimated time remaining that was due to progressing insight of the server's control logic, or
- When it changes at a rate significantly different from one unit per second.

Changes to this attribute merely due to the normal passage of time with no other dynamic change of device state SHALL NOT be reported.

As this attribute is not being reported during a regular countdown, clients SHOULD NOT rely on the reporting of this attribute in order to keep track of the remaining duration.

1.14.5.4. OperationalStateList Attribute

This attribute describes the set of possible operational states that the device exposes. An operational state is a fundamental device state such as Running or Error. Details of the phase of a device when, for example, in a state of Running are provided by the CurrentPhase attribute.

All devices SHALL, at a minimum, expose the set of states matching the commands that are also supported by the cluster instance, in addition to Error. The set of possible device states are defined in the OperationalStateEnum. A device type requiring implementation of this cluster SHALL define the set of states that are applicable to that specific device type.

1.14.5.5. OperationalState Attribute

This attribute specifies the current operational state of a device. This SHALL be populated with a valid OperationalStateID from the set of values in the OperationalStateList Attribute.

1.14.5.6. OperationalError Attribute

This attribute SHALL specify the details of any current error condition being experienced on the device when the OperationalState attribute is populated with Error. See Section 1.14.4.4, "ErrorStateStruct Type" for general requirements on the population of this attribute.

When there is no error detected, this SHALL have an ErrorStateID of NoError.

1.14.6. Commands

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ID	Name	Direction	Response	Access	Conformance
0x00	**Pause**	client ⇒ server	Operational-Comman-dResponse	O	Resume, O
0x01	**Stop**	client ⇒ server	Operational-Comman-dResponse	O	Start, O
0x02	**Start**	client ⇒ server	Operational-Comman-dResponse	O	O
0x03	**Resume**	client ⇒ server	Operational-Comman-dResponse	O	Pause, O
0x04	**Operational-Comman-dResponse**	client ⇐ server	N		Pause | Stop | Start | Resume

Note that it is entirely possible due to regulatory or other reasons for an instance of this cluster to expose no possible commands. When that occurs, this cluster does not provide any ability to actuate the device, instead it provides readable (and by extension, can be subscribed to) information as to the state of the device only. The commands that are supported SHALL be exposed by the device in the AcceptedCommandList global attribute.

1.14.6.1. Pause Command

This command SHALL be supported if the device supports remotely pausing the operation. If this command is supported, the Resume command SHALL also be supported.

On receipt of this command, the device SHALL pause its operation if it is possible based on the current function of the server. For example, if it is at a point where it is safe to do so and/or permitted, but can be restarted from the point at which pause occurred.

If this command is received when already in the Paused state the device SHALL respond with an OperationalCommandResponse command with an ErrorStateID of NoError but take no further action.

A device that receives this command in any state which is not Pause-compatible SHALL respond with an OperationalCommandResponse command with an ErrorStateID of CommandInvalidInState and SHALL take no further action.

States are defined as Pause-compatible as follows:

- For states defined in this cluster specification, in Table 3, “Pause Compatibility”.
- For states defined by derived cluster specifications, in the corresponding specifications.
- For manufacturer-specific states, by the manufacturer.

A device that is unable to honor the Pause command for whatever reason SHALL respond with an

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OperationalCommandResponse command with an ErrorStateID of CommandInvalidInState but take no further action.

Otherwise, on success:

- The OperationalState attribute SHALL be set to Paused.
- The device SHALL respond with an OperationalCommandResponse command with an ErrorStateID of NoError.

The following table defines the compatibility of this cluster's states with the Pause command.

Table 3. Pause Compatibility

State Value	State Name	Pause-Compatible
0x00	**Stopped**	N
0x01	**Running**	Y
0x02	**Paused**	Y
0x03	**Error**	N

1.14.6.2. Stop Command

This command SHALL be supported if the device supports remotely stopping the operation.

On receipt of this command, the device SHALL stop its operation if it is at a position where it is safe to do so and/or permitted. Restart of the device following the receipt of the Stop command SHALL require attended operation unless remote start is allowed by the device type and any jurisdiction governing remote operation of the device.

If this command is received when already in the Stopped state the device SHALL respond with an OperationalCommandResponse command with an ErrorStateID of NoError but take no further action.

A device that is unable to honor the Stop command for whatever reason SHALL respond with an OperationalCommandResponse command with an ErrorStateID of CommandInvalidInState but take no further action.

Otherwise, on success:

- The OperationalState attribute SHALL be set to Stopped.
- The device SHALL respond with an OperationalCommandResponse command with an ErrorStateID of NoError.

1.14.6.3. Start Command

This command SHALL be supported if the device supports remotely starting the operation. If this command is supported, the 'Stop command SHALL also be supported.

On receipt of this command, the device SHALL start its operation if it is safe to do so and the device is in an operational state from which it can be started. There may be either regulatory or manufac-

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turer-imposed safety and security requirements that first necessitate some specific action at the device before a Start command can be honored. In such instances, a device SHALL respond with a status code of CommandInvalidInState if a Start command is received prior to the required on-device action.

If this command is received when already in the Running state the device SHALL respond with an OperationalCommandResponse command with an ErrorStateID of NoError but take no further action.

A device that is unable to honor the Start command for whatever reason SHALL respond with an OperationalCommandResponse command with an ErrorStateID of UnableToStartOrResume but take no further action.

Otherwise, on success:

- The OperationalState attribute SHALL be set to Running.
- The device SHALL respond with an OperationalCommandResponse command with an ErrorStateID of NoError.

1.14.6.4. Resume Command

This command SHALL be supported if the device supports remotely resuming the operation. If this command is supported, the Pause command SHALL also be supported.

On receipt of this command, the device SHALL resume its operation from the point it was at when it received the Pause command, or from the point when it was paused by means outside of this cluster (for example by manual button press).

If this command is received when already in the Running state the device SHALL respond with an OperationalCommandResponse command with an ErrorStateID of NoError but take no further action.

A device that receives this command in any state which is not Resume-compatible SHALL respond with an OperationalCommandResponse command with an ErrorStateID of CommandInvalidInState and SHALL take no further action.

States are defined as Resume-compatible as follows:

- For states defined in this cluster specification, in Table 4, “Resume Compatibility”.
- For states defined by derived cluster specifications, in the corresponding specifications.
- For manufacturer-specific states, by the manufacturer.

The following table defines the compatibility of this cluster’s states with the Resume command.

Table 4. Resume Compatibility

State Value	State Name	Resume-Compatible
0x00	**Stopped**	N
0x01	**Running**	Y

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State Value	State Name	Resume-Compatible
0x02	**Paused**	Y
0x03	**Error**	N

A device that is unable to honor the Resume command for any other reason SHALL respond with an OperationalCommandResponse command with an ErrorStateID of UnableToStartOrResume but take no further action.

Otherwise, on success:

- The OperationalState attribute SHALL be set to the most recent non-Error operational state prior to entering the Paused state.
- The device SHALL respond with an OperationalCommandResponse command with an ErrorStateID of NoError.

1.14.6.5. OperationalCommandResponse Command

This command SHALL be supported by an implementation if any of the other commands defined by this cluster are supported (i.e. listed in the AcceptedCommandList global attribute). This command SHALL also be supported by an implementation of a derived cluster as a response to any commands that MAY be additionally defined therein.

This command SHALL be generated in response to any of the Start, Stop, Pause, or Resume commands. The data for this command SHALL be as follows:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0x00	**CommandResponseState**	ErrorStateStruct	all			M

1.14.6.5.1. CommandResponseState Field

This SHALL indicate the success or otherwise of the attempted command invocation. On a successful invocation of the attempted command, the ErrorStateID SHALL be populated with NoError. See the individual command sections for additional specific requirements on population.

1.14.7. Events

ID	Name	Priority	Access	Conformance
0x00	**OperationalError**	CRITICAL	V	M
0x01	**OperationCompletion**	INFO	V	O

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1.14.7.1. OperationalError Event

This event is generated when a reportable error condition is detected. A device that generates this event SHALL also set the OperationalState attribute to Error, indicating an error condition.

This event SHALL contain the following fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ErrorState**	ErrorStateStruct	all			M

1.14.7.2. OperationCompletion Event

This event SHOULD be generated when the overall operation ends, successfully or otherwise. For example, the completion of a cleaning operation in a Robot Vacuum Cleaner, or the completion of a wash cycle in a Washing Machine.

It is highly RECOMMENDED that appliances device types employing the Operational State cluster support this event, even if it is optional. This assists clients in executing automations or issuing notifications at critical points in the device operation cycles.

This event SHALL contain the following fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**CompletionErrorCode**	enum8	all			M
1	**TotalOperationalTime**	elapsed-s	all	X		O
2	**PausedTime**	elapsed-s	all	X		O

1.14.7.2.1. CompletionErrorCode Field

This field provides an indication of the state at the end of the operation. This field SHALL have a value from the ErrorStateEnum set. A value of NoError indicates success, that is, no error has been detected.

1.14.7.2.2. TotalOperationalTime Field

The total operational time, in seconds, from when the operation was started via an initial Start command or autonomous/manual starting action, until the operation completed. This includes any time spent while paused. There may be cases whereby the total operational time exceeds the maximum value that can be conveyed by this attribute, in such instances, this attribute SHALL be populated with null.

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1.14.7.2.3. PausedTime Field

The total time spent in the paused state, in seconds. There may be cases whereby the total paused time exceeds the maximum value that can be conveyed by this attribute, in such instances, this attribute SHALL be populated with null.

1.15. Alarm Base Cluster

This cluster is a base cluster from which clusters for particular alarms for a device type can be derived. Each derivation SHALL define the values for the AlarmBitmap data type used in this cluster. Each derivation SHALL define which alarms are latched.

1.15.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Changed event field ID’s to match SDK

1.15.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	ALARM

1.15.3. Cluster ID

ID	Name
n/a	Alarm Base

1.15.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**RESET**	Reset	O	Supports the ability to reset alarms

1.15.4.1. Reset Feature

This feature indicates that alarms can be reset via the Reset command.

1.15.5. Data Types

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1.15.5.1. AlarmBitmap Type

This data type SHALL be a map32 with values defined by the derived cluster. The meaning of each bit position SHALL be consistent for all attributes in a derived cluster. That is, if bit 0 is defined for an alarm, the Latch, State, and Supported information for that alarm are also bit 0.

1.15.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**Mask**	Alarm-Bitmap	all			R V	M
0x0001	**Latch**	Alarm-Bitmap	all	F		R V	RESET
0x0002	**State**	Alarm-Bitmap	all			R V	M
0x0003	**Supported**	Alarm-Bitmap	all	F		R V	M

1.15.6.1. Mask Attribute

This attribute SHALL indicate a bitmap where each bit set in the Mask attribute corresponds to an alarm that SHALL be enabled.

1.15.6.2. Latch Attribute

This attribute SHALL indicate a bitmap where each bit set in the Latch attribute SHALL indicate that the corresponding alarm will be latched when set, and will not reset to inactive when the underlying condition which caused the alarm is no longer present, and so requires an explicit reset using the Reset command.

1.15.6.3. State Attribute

This attribute SHALL indicate a bitmap where each bit SHALL represent the state of an alarm. The value of true means the alarm is active, otherwise the alarm is inactive.

1.15.6.4. Supported Attribute

This attribute SHALL indicate a bitmap where each bit SHALL represent whether or not an alarm is supported. The value of true means the alarm is supported, otherwise the alarm is not supported.

If an alarm is not supported, the corresponding bit in Mask, Latch, and State SHALL be false.

1.15.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**Reset**	client ⇒ server	Y	O	RESET

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ID	Name	Direction	Response	Access	Conformance
0x01	**ModifyEnabledAlarms**	client ⇒ server	Y	O	O

1.15.7.1. Reset Command

This command resets active and latched alarms (if possible). Any generated Notify event SHALL contain fields that represent the state of the server after the command has been processed.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Alarms**	AlarmBitmap	all			M

1.15.7.1.1. Alarms Field

This field SHALL indicate a bitmap where each bit set in this field corresponds to an alarm that SHALL be reset to inactive in the State attribute unless the alarm definition requires manual intervention. If the alarms indicated are successfully reset, the response status code SHALL be SUCCESS, otherwise, the response status code SHALL be FAILURE.

1.15.7.2. ModifyEnabledAlarms Command

This command allows a client to request that an alarm be enabled or suppressed at the server.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Mask**	AlarmBitmap	all			M

1.15.7.2.1. Mask Field

This field SHALL indicate a bitmap where each bit set in the this field corresponds to an alarm that SHOULD be enabled or suppressed. A value of 1 SHALL indicate that the alarm SHOULD be enabled while a value of 0 SHALL indicate that the alarm SHOULD be suppressed.

A server that receives this command with a Mask that includes bits that are set for unknown alarms SHALL respond with a status code of INVALID_COMMAND.

A server that receives this command with a Mask that includes bits that are set for alarms which are not supported, as indicated in the Supported attribute, SHALL respond with a status code of INVALID_COMMAND.

A server that is unable to enable a currently suppressed alarm, or is unable to suppress a currently enabled alarm SHALL respond with a status code of FAILURE; otherwise the server SHALL respond with a status code of SUCCESS.

On a SUCCESS case, the server SHALL also change the value of the Mask attribute to the value of the

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Mask field from this command. After that the server SHALL also update the value of its State attribute to reflect the status of the new alarm set as indicated by the new value of the Mask attribute.

1.15.8. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**Notify**	INFO		V	M

1.15.8.1. Notify Event

This event SHALL be generated when one or more alarms change state.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Active**	Alarm-Bitmap	all			M
1	**Inactive**	Alarm-Bitmap	all			M
2	**State**	Alarm-Bitmap	all			M
3	**Mask**	Alarm-Bitmap	all			M

1.15.8.1.1. Active Field

This field SHALL indicate those alarms that have become active.

1.15.8.1.2. Inactive Field

This field SHALL indicate those alarms that have become inactive.

1.15.8.1.3. Mask Field

This field SHALL be a copy of the Mask attribute when this event was generated.

1.15.8.1.4. State Field

This field SHALL be a copy of the new State attribute value that resulted in the event being generated. That is, this field SHALL have all the bits in Active set and SHALL NOT have any of the bits in Inactive set.

1.16. Messages Cluster

This cluster provides an interface for passing messages to be presented by a device.

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1.16.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added
2	Updated from SE1.4 version; CCB 1819
3	Initial Matter release; renamed from EnergyMessaging to Messages

1.16.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	MESS

1.16.3. Cluster ID

ID	Name
0x0097	Messages

1.16.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**CONF**	ReceivedConfirmation	O	
1	**RESP**	ConfirmationResponse	[CONF]	
2	**RPLY**	ConfirmationReply	[CONF]	
3	**PROT**	ProtectedMessages	O	

1.16.4.1. ReceivedConfirmation Feature

This feature SHALL indicate that the device can get confirmation from a user that the message was received.

1.16.4.2. ConfirmationResponse Feature

This feature SHALL indicate that the device is capable of presenting a list of responses to the user and recording the user's choice of response.

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1.16.4.3. ConfirmationReply Feature

This feature SHALL indicate that the device is capable of collecting a free-form text response to a message.

1.16.4.4. ProtectedMessages Feature

This feature SHALL indicate that the device is capable of requiring the user to authenticate before viewing a message; e.g. entering a PIN or password before viewing a message with billing information.

1.16.5. Data Types

1.16.5.1. MessageID Type

This data type is an octstr of fixed length 16, containing the binary encoding of a UUID as specified in [RFC4122].

1.16.5.2. MessageControlBitmap Type

This data type is derived from map16, and indicates control information related to a message.

Bit	Name	Summary	Conformance
0	**ConfirmationRequired**	Message requires confirmation from user	CONF
1	**ResponseRequired**	Message requires response from user	RESP
2	**ReplyMessage**	Message supports reply message from user	RPLY
3	**MessageConfirmed**	Message has already been confirmed	CONF
4	**MessageProtected**	Message required PIN/password protection	PROT

1.16.5.2.1. ConfirmationRequired Bit

This bit SHALL indicate that the message originator requests a confirmation of receipt by the user. If confirmation is required, the device SHOULD present the message until it is either confirmed by the user selecting a confirmation option, or the message expires.

1.16.5.2.2. ResponseRequired Bit

This bit SHALL indicate that a MessagePresented event SHOULD be generated based on the response of the user to the message.

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1.16.5.2.3. ReplyMessage Bit

This bit SHALL indicate that a free-form user reply is to be included in the confirmation of receipt.

1.16.5.2.4. MessageConfirmed Bit

This bit SHALL indicate the current confirmation state of a message, which is useful in the event that there are multiple Messages cluster client devices on a network.

1.16.5.2.5. MessageProtected Bit

This bit SHALL indicate that user authentication (e.g. by password or PIN) is required before viewing a message.

1.16.5.3. FutureMessagePreferenceEnum Type

This data type is derived from enum8.

A display device MAY include this preference in the MessageComplete event as a hint to clients about how to handle future similar messages.

Value	Name	Summary	Conformance
0	**Allowed**	Similar messages are allowed	M
1	**Increased**	Similar messages should be sent more often	M
2	**Reduced**	Similar messages should be sent less often	M
3	**Disallowed**	Similar messages should not be sent	M
4	**Banned**	No further messages should be sent	M

1.16.5.4. MessagePriorityEnum Type

This data type is derived from enum8.

Priority SHOULD be used to decide which messages to show when the number of eligible messages is larger than the device's capacity to present them.

Value	Name	Summary	Conformance
0	**Low**	Message to be transferred with a low level of importance	M

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Value	Name	Summary	Conformance
1	**Medium**	Message to be transferred with a medium level of importance	M
2	**High**	Message to be transferred with a high level of importance	M
3	**Critical**	Message to be transferred with a critical level of importance	M

1.16.5.5. MessageStruct Type

This represents a single message.

Access Modifier: Fabric Scoped							
ID	**Name**	**Type**	**Constraint**	**Quality**	**Fallback**	**Access**	**Conformance**
0	**MessageID**	MessageID	all			S	M
1	**Priority**	MessagePriorityEnum				S	M
2	**MessageControl**	MessageControlBitmap				S	M
3	**StartTime**	epoch-s		X		S	M
4	**Duration**	uint64	all	X		S	M
5	**MessageText**	string	max 256			S	M
6	**Responses**	list[MessageResponseOptionStruct]	max 4			S	RESP

1.16.5.5.1. MessageID Field

This field SHALL indicate a globally unique ID for this message.

1.16.5.5.2. Priority Field

This field SHALL indicate the priority level for this message.

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1.16.5.5.3. MessageControl Field

This field SHALL indicate control information related to the message.

1.16.5.5.4. StartTime Field

This field SHALL indicate the time in UTC at which the message becomes available to be presented. A null value SHALL indicate "now."

1.16.5.5.5. Duration Field

This field SHALL indicate the amount of time, in milliseconds, after the StartTime during which the message is available to be presented. A null value SHALL indicate "until changed".

1.16.5.5.6. MessageText Field

This field SHALL indicate a string containing the message to be presented.

1.16.5.5.7. Responses Field

This field SHALL indicate a list of potential responses to the message. The entries in this list SHALL have unique values of MessageResponseID.

If the ResponseRequired bit is set on the message but this list is empty, the device SHALL provide a generic acknowledgement button, e.g. "OK".

If the ResponseRequired bit is not set on the message, this list SHALL be ignored.

1.16.5.6. MessageResponseOptionStruct Type

This represents a possible response to a message.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**MessageResponseID**	uint32	min 1				M
1	**Label**	string	max 32				M

1.16.5.6.1. MessageResponseID Field

This field SHALL indicate a unique unsigned 32-bit number identifier for this message response option.

1.16.5.6.2. Label Field

This field SHALL indicate the text for this option; e.g. "Yes", "No", etc.

1.16.6. Attributes

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**Messages**	list[MessageStruct]	max 8			R V F	M
0x0001	**ActiveMessageIDs**	list[MessageID]	max 8			R V	M

1.16.6.1. Messages Attribute

This attribute SHALL indicate a list of queued messages.

In addition to filtering based upon fabric, to preserve user privacy, the server MAY further limit the set of messages returned in a read request. At minimum, the server SHALL return to a client those messages that the client itself created/submitted.

1.16.6.2. ActiveMessageIDs Attribute

This attribute SHALL indicate a list of the MessageIDs of the Messages currently being presented. If this list is empty, no messages are currently being presented.

This list SHALL NOT be fabric-scoped; it SHALL contain MessageIDs for all Messages being presented, no matter what fabric the client that queued them is on.

1.16.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**PresentMessagesRequest**	client ⇒ server	Y	O F	M
0x01	**CancelMessagesRequest**	client ⇒ server	Y	O F	M

1.16.7.1. PresentMessagesRequest Command

Upon receipt, this SHALL cause the message in the passed fields to be appended to the Messages attribute.

If appending the message would cause the number of messages to be greater than the capacity of the list, the device SHALL NOT append any message to Messages, and SHALL return a status code of RESOURCE_EXHAUSTED.

When displaying a message in response to this command, an indication (ex. visual) of the origin node of the command SHALL be provided. This could be in the form of a friendly name label which uniquely identifies the node to the user. This friendly name label is typically assigned by the Matter Admin at the time of commissioning and, when it's a device, is often editable by the user. It might be a combination of a company name and friendly name, for example, "Acme" or "Acme Streaming Service on Alice's Phone".

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NOTE	It is currently not specified where the friendly name label can be found on the node, meaning that clients SHOULD NOT rely on a certain method they happen to observe in a particular server instance, since other instances could employ a different method.

The device SHOULD make it possible for the user to view which nodes have access to this cluster and to individually remove privileges for each node.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**MessageID**	MessageID	all			M
1	**Priority**	MessagePriorityEnum				M
2	**MessageControl**	MessageControlBitmap				M
3	**StartTime**	epoch-s		X		M
4	**Duration**	uint64	all	X		M
5	**MessageText**	string	max 256			M
6	**Responses**	list[MessageResponseOptionStruct]	max 4			RESP

1.16.7.1.1. MessageID Field

This field SHALL indicate a globally unique ID for this message. See MessageID.

1.16.7.1.2. Priority Field

This field SHALL indicate the priority level for this message. See Priority.

1.16.7.1.3. MessageControl Field

This field SHALL indicate control information related to the message. See MessageControl.

1.16.7.1.4. StartTime Field

This field SHALL indicate the time in UTC at which the message becomes available to be presented. A null value SHALL indicate "now." See StartTime.

1.16.7.1.5. Duration Field

This field SHALL indicate the amount of time, in milliseconds, after the StartTime during which the message is available to be presented. A null value SHALL indicate "until changed". See Duration.

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1.16.7.1.6. MessageText Field

This field SHALL indicate a string containing the message to be presented. See MessageText.

1.16.7.1.7. Responses Field

This field SHALL indicate a list of potential responses to the message. The entries in this list SHALL have unique values of MessageResponseID.

If the ResponseRequired bit is set on the message but this list is empty, the device SHALL provide a generic acknowledgement button, e.g. "OK".

If the ResponseRequired bit is not set on the message, this list SHALL be ignored.

See Responses.

1.16.7.2. CancelMessagesRequest Command

This command will cancel the message IDs specified.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**MessageIDs**	list[MessageID]	max 8			M

1.16.7.2.1. MessageIDs Field

This field SHALL indicate the MessageIDs for the messages being cancelled.

Cancelling a message SHALL cause it to be removed from Messages, cause its MessageID to be removed from ActiveMessageIDs and cause any active presentation of the message to cease.

Message IDs in this command that indicate messages that do not exist in Messages, or that are not scoped to the fabric of the sender, SHALL be ignored.

1.16.8. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**MessageQueued**	INFO		V	M
0x01	**MessagePresented**	INFO		V	M
0x02	**MessageComplete**	INFO		V	M

1.16.8.1. MessageQueued Event

This event SHALL be generated when a message is added to the messages attribute.

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Access Modifier: Fabric Sensitive						
ID	**Name**	**Type**	**Constraint**	**Quality**	**Fallback**	**Conformance**
0	**MessageID**	MessageID				M

1.16.8.1.1. MessageID Field

This field SHALL indicate the MessageID for newly added message.

1.16.8.2. MessagePresented Event

This event SHALL be generated when the message is presented to the user.

Access Modifier: Fabric Sensitive						
ID	**Name**	**Type**	**Constraint**	**Quality**	**Fallback**	**Conformance**
0	**MessageID**	MessageID				M

1.16.8.2.1. MessageID Field

This field SHALL indicate the MessageID for the message being presented.

1.16.8.3. MessageComplete Event

This event SHALL be generated when the message is confirmed by the user, or when the Duration field of the message has elapsed without confirmation.

Access Modifier: Fabric Sensitive						
ID	**Name**	**Type**	**Constraint**	**Quality**	**Fallback**	**Conformance**
0	**MessageID**	MessageID				M
1	**ResponseID**	uint32	all	X		RESP
2	**Reply**	string	max 256	X		RPLY
3	**FutureMessagesPreference**	FutureMessagePreferenceEnum	all	X		M

1.16.8.3.1. MessageID Field

This field SHALL indicate the MessageID for the message being confirmed.

1.16.8.3.2. ResponseID Field

This field SHALL indicate the MessageResponseID selected by the user. If there was no response before the Duration field of the message has elapsed, this field SHALL be null.

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1.16.8.3.3. Reply Field

This field SHALL indicate a user-provided reply to the message. If there was no reply, or the message did not have the ReplyRequired bit set, this field SHALL be null.

1.16.8.3.4. FutureMessagesPref Field

This field SHALL indicate a user-provided preference for the delivery of similar messages in the future. A null value SHALL indicate no change in preference.

1.17. Service Area Cluster

This cluster provides an interface for controlling the areas where a device should operate, for reporting the status at each area, and for querying the current area.

The device MAY operate at one area at a time, as in the case of a mobile device, such as a robot. Other devices MAY operate at (service) multiple areas simultaneously, as in the case of a sensor that can monitor multiple areas. This cluster specification uses the term "operate" to describe both the operating and servicing actions, regardless of the device type.

The cluster allows the client to select one or more areas on the server, to indicate where the device SHOULD attempt to operate. An area is one of a list of options that MAY be presented by a client for a user choice, or understood by the client, via the semantic data of the area.

The area semantic data is a combination of semantic tags, indicating one or more of the following: the building floor, area type, landmark, and relative position.

1.17.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Rename InitialTimeEstimate to EstimatedTime

1.17.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	SEAR

1.17.3. Cluster ID

ID	Name
0x0150	Service Area

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1.17.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**SELRUN**	SelectWhileRunning	O	The device allows changing the selected areas while running
1	**PROG**	ProgressReporting	O	The device implements the progress reporting feature
2	**MAPS**	Maps	O	The device has map support

1.17.4.1. SelectWhileRunning Feature

This feature indicates whether this device allows changing the selected areas, by using the SelectAreas command, while operating.

1.17.5. Data Types

1.17.5.1. LandmarkInfoStruct Type

The data from this structure indicates a landmark and position relative to the landmark.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**LandmarkTag**	tag	all				M
1	**RelativePositionTag**	tag	all	X			M

1.17.5.1.1. LandmarkTag Field

This field SHALL indicate that the area is associated with a landmark.

This field SHALL be the ID of a landmark semantic tag, located within the Common Landmark Namespace. For example, this tag MAY indicate that the area refers to an area next to a table.

1.17.5.1.2. RelativePositionTag Field

This field SHALL identify the position of the area relative to a landmark. This is a static description of a zone known to the server, and this field never reflects the device's own proximity or position relative to the landmark, but that of the zone.

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This field SHALL be the ID of a relative position semantic tag, located within the Common Relative Position Namespace.

If the RelativePositionTag field is null, this field indicates proximity to the landmark. Otherwise, the RelativePositionTag field indicates the position of the area relative to the landmark indicated by the LandmarkTag field. For example, this tag, in conjunction with the LandmarkTag field, MAY indicate that the area refers to a zone under a table.

1.17.5.2. AreaInfoStruct Type

The data from this structure indicates the name and/or semantic data describing an area, as detailed below.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**LocationInfo**	location-desc	all	X			M
1	**LandmarkInfo**	LandmarkInfoStruct	all	X			M

This data type includes the LocationInfo field, with the following fields: LocationName, FloorNumber, AreaType. Additional semantic data MAY be available in the LandmarkInfo field.

For an area description to be meaningful, it SHALL have at least one of the following:

- a non-empty name (LocationInfo's LocationName field)

OR

- some semantic data (one or more of these: FloorNumber, AreaType or LandmarkTag)

The normative text from the remainder of this section describes these constraints.

If the LocationInfo field is null, the LandmarkInfo field SHALL NOT be null.

If the LandmarkInfo field is null, the LocationInfo field SHALL NOT be null.

If LocationInfo is not null, and its LocationName field is an empty string, at least one of the following SHALL NOT be null:

- LocationInfo's FloorNumber field
- LocationInfo's AreaType field
- LandmarkInfo field

If all three of the following are null, LocationInfo's LocationName field SHALL NOT be an empty string:

- LocationInfo's FloorNumber field
- LocationInfo's AreaType field

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- LandmarkInfo field

1.17.5.2.1. LocationInfo Field

This field SHALL indicate the name of the area, floor number and/or area type.

A few examples are provided below.

- An area can have LocationInfo's LocationName field set to "blue room", and the AreaType field set to the ID of a "Living Room" semantic tag. Clients wishing to direct the device to operate in (or service) the living room can use this area.
- An area can have LocationInfo set to null, the LandmarkInfo's LandmarkTag field set to the ID of the "Table" landmark semantic tag, and the RelativePositionTag field set to the ID of the "Under" position semantic tag. With such an area indication, the client can request the device to operate in (or service) the area located under the table.

1.17.5.2.2. LandmarkInfo Field

This field SHALL indicate an association with a landmark. A value of null indicates that the information is not available or known. For example, this may indicate that the area refers to a zone next to a table.

If this field is not null, that indicates that the area is restricted to the zone where the landmark is located, as indicated by the LandmarkTag and, if not null, by the RelativePositionTag fields, rather than to the entire room or floor where the landmark is located, if those are indicated by the LocationInfo field.

1.17.5.3. MapStruct Type

This is a struct representing a map.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**MapID**	uint32			MS		M
1	**Name**	string	max 64		MS		M

1.17.5.3.1. MapID Field

This field SHALL represent the map's identifier.

1.17.5.3.2. Name Field

This field SHALL represent a human understandable map description.

For example: "Main Floor", or "Second Level".

1.17.5.4. AreaStruct Type

This is a struct representing an area known to the server.

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**AreaID**	uint32			MS		M
1	**MapID**	uint32	desc	X	MS		M
2	**AreaInfo**	AreaInfoStruct			MS		M

1.17.5.4.1. AreaID Field

This field SHALL represent the identifier of the area.

1.17.5.4.2. MapID Field

This field SHALL indicate the map identifier which the area is associated with. A value of null indicates that the area is not associated with a map.

If the SupportedMaps attribute is not empty, this field SHALL match the MapID field of an entry from the SupportedMaps attribute's list. If the SupportedMaps attribute is empty, this field SHALL be null.

1.17.5.4.3. AreaInfo Field

This field SHALL contain data describing the area.

This SHOULD be used by clients to determine the name and/or the full, or the partial, semantics of a certain area.

NOTE	If any entries on the SupportedAreas attribute's list have the AreaInfo field missing the semantic data, the client MAY remind the user to assign the respective data.

1.17.5.5. ProgressStruct Type

This is a struct indicating the progress.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**AreaID**	uint32			MS		M
1	**Status**	OperationalStatusEnum			MS		M
2	**TotalOperationalTime**	elapsed-s		X	MS		O
3	**EstimatedTime**	elapsed-s		X	MS		O

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1.17.5.5.1. AreaID Field

This field SHALL indicate the identifier of the area, and the identifier SHALL be an entry in the SupportedAreas attribute's list.

1.17.5.5.2. Status Field

This field SHALL indicate the operational status of the device regarding the area indicated by the AreaID field.

1.17.5.5.3. TotalOperationalTime Field

This field SHALL indicate the total operational time, in seconds, from when the device started to operate at the area indicated by the AreaID field, until the operation finished, due to completion or due to skipping, including any time spent while paused.

A value of null indicates that the total operational time is unknown.

There MAY be cases where the total operational time exceeds the maximum value that can be conveyed by this attribute, and in such instances this attribute SHALL be populated with null.

This attribute SHALL be null if the Status field is not set to Completed or Skipped.

1.17.5.5.4. EstimatedTime Field

This field SHALL indicate the estimated time for the operation, in seconds, from when the device will start operating at the area indicated by the AreaID field, until the operation completes, excluding any time spent while not operating in the area.

A value of null indicates that the estimated time is unknown. If the estimated time is unknown, or if it exceeds the maximum value that can be conveyed by this attribute, this attribute SHALL be null.

After initializing the ProgressStruct instance, the server SHOULD NOT change the value of this field, except when repopulating the entire instance, to avoid excessive reporting of the Progress attribute changes.

1.17.5.6. OperationalStatusEnum Type

This data type is derived from enum8.

The following table defines the status values.

Value	Name	Summary	Conformance
0x00	**Pending**	The device has not yet started operating at the given area, or has not finished operating at that area but it is not currently operating at the area	M

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Value	Name	Summary	Conformance
0x01	**Operating**	The device is currently operating at the given area	M
0x02	**Skipped**	The device has skipped the given area, before or during operating at it, due to a SkipArea command, due an out of band command (e.g. from the vendor's application), due to a vendor specific reason, such as a time limit used by the device, or due the device ending operating unsuccessfully	M
0x03	**Completed**	The device has completed operating at the given area	M

1.17.5.6.1. SelectAreasStatus Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0x00	**Success**	Attempting to operate in the areas identified by the entries of the NewAreas field is allowed and possible, the SelectedAreas attribute is set to the value of the NewAreas field	M
0x01	**UnsupportedArea**	The value of at least one of the entries of the NewAreas field doesn't match any entries in the SupportedAreas attribute	M

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Value	Name	Summary	Conformance
0x02	**InvalidInMode**	The received request cannot be handled due to the current mode of the device	M
0x03	**InvalidSet**	The set of values is invalid. For example, areas on different floors, that a robot knows it can't reach on its own	M

1.17.5.6.2. SkipAreaStatus Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0x00	**Success**	Skipping the area is allowed and possible, or the device was operating at the last available area and has stopped	M
0x01	**InvalidAreaList**	The SelectedAreas attribute is empty.	M
0x02	**InvalidInMode**	The received request cannot be handled due to the current mode of the device. For example, the CurrentArea attribute is null or the device is not operating	M
0x03	**InvalidSkippedArea**	The SkippedArea field doesn't match an entry in the SupportedAreas list	M

1.17.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SupportedAreas**	list[AreaStruct]	max 255		MS	R V	M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0001	**SupportedMaps**	list[MapStruct]	max 255		MS	R V	MAPS
0x0002	**SelectedAreas**	list[uint32]	desc		empty	R V	M
0x0003	**CurrentArea**	uint32	desc	X	null	R V	desc
0x0004	**EstimatedEndTime**	epoch-s		Q X	null	R V	[CurrentArea]
0x0005	**Progress**	list[ProgressStruct]	max 255		empty	R V	PROG

1.17.6.1. SupportedAreas Attribute

This attribute SHALL contain the list of areas that can be included in the SelectedAreas attribute's list. Each item in this list represents a unique area, as indicated by the AreaID field of AreaStruct.

Each entry in this list SHALL have a unique value for the AreaID field.

If the SupportedMaps attribute is not empty, each entry in this list SHALL have a unique value for the combination of the MapID and AreaInfo fields.

If the SupportedMaps attribute is empty, each entry in this list SHALL have a unique value for the AreaInfo field and SHALL have the MapID field set to null.

An empty value indicates that the device is currently unable to provide the list of supported areas.

NOTE due to the maximum size of this list and to the fact that the entries MAY include strings (see LocationName), care must be taken by implementers to avoid creating a data structure that is overly large, which can result in significant latency in accessing this attribute.

The value of this attribute MAY change at any time via an out-of-band interaction outside of the server, such as interactions with a user interface, or due to internal device changes.

When removing entries in the SupportedAreas attribute list the server SHALL adjust the values of the SelectedAreas, CurrentArea, and Progress attributes such that they only reference valid entries in the updated SupportedAreas attribute list. These changes to the SelectedAreas, CurrentArea, and Progress attributes MAY result in the server setting some or all of them to empty (for SelectedAreas and Progress) or null (for CurrentArea), or updating them with data that matches the constraints from the description of the respective attributes. These actions are required to ensure having a consistent representation of the maps and locations available to the clients.

The SupportedAreas attribute list changes mentioned above SHOULD NOT be allowed while the device is operating, to reduce the impact on the clients, and the potential confusion for the users.

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A few examples are provided below.

Valid list of areas:

- AreaID=0, LocationName="yellow bedroom", MapID=null
- AreaID=1, LocationName="orange bedroom", MapID=null

Valid list of areas:

- AreaID=5, LocationName="hallway", MapID=1
- AreaID=3, LocationName="hallway", MapID=2

1.17.6.2. SupportedMaps Attribute

This attribute SHALL contain the list of supported maps.

A map is a full or a partial representation of a home, known to the device. For example:

- a single level home MAY be represented using a single map
- a two level home MAY be represented using two maps, one for each level
- a single level home MAY be represented using two maps, each including a different set of rooms, such as "map of living room and kitchen" and "map of bedrooms and hallway"
- a single level home MAY be represented using one map for the indoor areas (living room, bedrooms etc.) and one for the outdoor areas (garden, swimming pool etc.)

Each map includes one or more areas - see the SupportedAreas attribute. In the context of this cluster specification, a map is effectively a group label for a set of areas, rather than a graphical representation that the clients can display to the users. The clients that present the list of available areas for user selection (see the SelectAreas command) MAY choose to filter the SupportedAreas list based on the associated map. For example, the clients MAY allow the user to indicate that the device is to operate on the first floor, and allow the user to choose only from the areas situated on that level.

If empty, that indicates that the device is currently unable to provide this information.

Each entry in this list SHALL have a unique value for the MapID field.

Each entry in this list SHALL have a unique value for the Name field.

NOTE	due to the maximum size of this list and to the fact that the entries MAY include strings (see the Name field of the MapStruct data type), care must be taken by implementers to avoid creating a data structure that is overly large, which can result in significant latency in accessing this attribute.

The value of this attribute MAY change at any time via an out-of-band interaction outside of the server, such as interactions with a user interface.

When updating the SupportedMaps attribute list by deleting entries, or by setting the attribute to an empty list, the SupportedAreas attribute SHALL be updated such that all entries in that list meet the constraints indicated in the description of the SupportedAreas attribute. This MAY result in the

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server removing entries from the SupportedAreas attribute list. See the SupportedAreas attribute description for the implications of changing that attribute.

The SupportedMaps attribute list changes mentioned above SHOULD NOT be allowed while the device is operating, to reduce the impact on the clients, and the potential confusion for the users.

1.17.6.3. SelectedAreas Attribute

This attribute SHALL indicate the set of areas where the device SHOULD attempt to operate.

The mobile devices MAY travel without operating across any areas while attempting to reach the areas indicated by the SelectedAreas attribute. For example, a robotic vacuum cleaner MAY drive without cleaning when traveling without operating.

If this attribute is empty, the device is not constrained to operate in any specific areas.

If this attribute is not empty:

- each item in this list SHALL match the AreaID field of an entry in the SupportedAreas attribute's list
- each entry in this list SHALL have a unique value

1.17.6.4. CurrentArea Attribute

If the device is mobile, this attribute SHALL indicate the area where the device is currently located, regardless of whether it is operating or not, such as while traveling between areas.

If the device is not mobile and can operate at multiple areas sequentially, this attribute SHALL indicate the area which is currently being serviced, or the area which is currently traversed by the device. For example, a camera device MAY use this attribute to indicate which area it currently takes video of (serviced area) or which area it currently has in view but not taking video of (e.g. an area which is traversed while panning).

NOTE	A device MAY traverse an area regardless of the status of the area (pending, skipped, or completed).

If a device can simultaneously operate at multiple areas, such as in the case of a sensor that can monitor multiple areas at the same time, the CurrentArea attribute SHALL NOT be implemented, since it doesn't apply. Else this attribute SHALL be optionally implemented.

A null value indicates that the device is currently unable to provide this information. For example, the device is traversing an unknown area, or the SupportedAreas attribute was updated and the area where the device is located was removed from that list.

If not null, the value of this attribute SHALL match the AreaID field of an entry on the SupportedAreas attribute's list.

1.17.6.5. EstimatedEndTime Attribute

This attribute SHALL represent the estimated Epoch time for completing operating at the area indi-

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cated by the CurrentArea attribute, in seconds.

A value of 0 means that the operation has completed.

When this attribute is null, that represents that there is no time currently defined until operation completion. This MAY happen, for example, because no operation is in progress or because the completion time is unknown.

This attribute SHALL be null if the CurrentArea attribute is null.

If the Progress attribute is available, and it contains an entry matching CurrentArea, the server MAY use the time estimate provided in the EstimatedTime field of that entry to compute the EstimatedEndTime attribute.

The value of this attribute SHALL only be reported in the following cases:

- when it changes to or from 0
- when it decreases
- when it changes to or from null

NOTE	If the device is capable of pausing its operation, this attribute MAY be set to null, to indicate that completion time is unknown, or increment the value while being in the paused state.

1.17.6.6. Progress Attribute

This attribute SHALL indicate the operating status at one or more areas.

Each entry in this list SHALL have a unique value for the AreaID field.

For each entry in this list, the AreaID field SHALL match an entry on the SupportedAreas attribute's list.

When this attribute is empty, that represents that no progress information is currently available.

If the SelectedAreas attribute is empty, indicating the device is not constrained to operate in any specific areas, the Progress attribute list MAY change while the device operates, due to the device adding new entries dynamically, when it determines which ones it can attempt to operate at.

If the SelectedAreas attribute is not empty, and the device starts operating:

- the Progress attribute list SHALL be updated so each entry of SelectedAreas has a matching Progress list entry, based on the AreaID field
- the length of the Progress and SelectedAreas list SHALL be the same
- the entries in the Progress list SHALL be initialized by the server, by having their status set to Pending or Operating, and the TotalOperationalTime field set to null

When the device ends operation unexpectedly, such as due to an error, the server SHALL update all Progress list entries with the Status field set to Operating or Pending to Skipped.

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When the device finishes operating, successfully or not, it SHALL NOT change the Progress attribute, except in the case of an unexpected end of operation as described above, or due to changes to the SupportedMaps or SupportedAreas attributes, so the clients can retrieve the progress information at that time.

NOTE	if the device implements the Operational Status cluster, or a derivation of it, in case the device fails to service any locations in the SelectedAreas list before ending the operation, it SHOULD use the Operational Status cluster to indicate that the device was unable to complete the operation (see the UnableToCompleteOperation error from that cluster specification). The clients SHOULD then read the Progress attribute, and indicate which areas have been successfully serviced (marked as completed).

1.17.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**SelectAreas**	client ⇒ server	SelectAreasResponse	O	M
0x01	**SelectAreasResponse**	client ⇐ server	N		M
0x02	**SkipArea**	client ⇒ server	SkipAreaResponse	O	[CurrentArea | Progress]
0x03	**SkipAreaResponse**	client ⇐ server	N		SkipArea

1.17.7.1. SelectAreas Command

This command is used to select a set of device areas, where the device is to operate.

On receipt of this command the device SHALL respond with a SelectAreasResponse command.

This command SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**NewAreas**	list[uint32]	desc			M

1.17.7.1.1. NewAreas Field

This field indicates which areas the device is to operate at.

If this field is empty, that indicates that the device is to operate without being constrained to any specific areas, and the operation will not allow skipping using the SkipArea Command, otherwise the field SHALL be a list of unique values that match the AreaID field of entries on the SupportedAreas list.

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1.17.7.1.2. Effect on Receipt

If the NewAreas field contains any duplicated entries, the device SHALL operate on only the first instance of an entry in the list.

For example, if the list `[1, 2, 2, 3, 2]` were received, the device would ignore the duplicated instances of `2` in the list and operate as if the list `[1, 2, 3]` had been received instead.

If the device determines that it can't operate at all areas from the list, the SelectAreasResponse command's Status field SHALL indicate InvalidSet.

If at least one entry on the NewAreas field doesn't match the AreaID field of any entry of the SupportedAreas list, the SelectAreasResponse command's Status field SHALL indicate UnsupportedArea.

If the current state of the device doesn't allow for the areas to be selected, the SelectAreasResponse command SHALL have the Status field set to InvalidInMode.

Unless the SelectWhileRunning feature indicates that selecting areas is allowed while the device is in a non-idle state, such as operating or paused, the SelectAreasResponse command SHALL have the Status field set to InvalidInMode if this restriction prevents changing the selected areas.

Otherwise, if the device accepts the request, the server will attempt to operate at the areas indicated by the entries of the NewArea field, the SelectAreasResponse command SHALL have the Status field set to Success, and the SelectedAreas attribute SHALL be set to the value of the NewAreas field.

NOTE	The device MAY start to operate immediately or wait until specifically requested to operate. The behavior of the device is manufacturer specific.

If the NewAreas field is the same as the value of the SelectedAreas attribute the SelectAreasResponse command SHALL have the Status field set to Success and the StatusText field MAY be supplied with a human readable string or include an empty string.

1.17.7.2. SelectAreasResponse Command

This command is sent by the device on receipt of the SelectAreas command. This command SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	SelectAreasStatus	all			M
1	**StatusText**	string	max 256			M

1.17.7.2.1. Status Field

If the Status field is set to Success or UnsupportedArea, the server MAY use a non-empty string for the StatusText field to provide additional information. For example, if Status is set to Unsupport-

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edArea, the server MAY use StatusText to indicate which areas are unsupported.

If the Status field is not set to Success, or UnsupportedArea, the StatusText field SHALL include a vendor-defined error description which can be used to explain the error to the user. For example, if the Status field is set to InvalidInMode, the StatusText field SHOULD indicate why the request is not allowed, given the current mode of the device, which MAY involve other clusters.

1.17.7.3. SkipArea Command

This command is used to skip the given area, and to attempt operating at other areas on the SupportedAreas attribute list.

On receipt of this command the device SHALL respond with a SkipAreaResponse command.

This command SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**SkippedArea**	uint32	desc			M

1.17.7.3.1. SkippedArea Field

The SkippedArea field indicates the area to be skipped.

The SkippedArea field SHALL match an entry in the SupportedAreas list.

1.17.7.3.2. Effect on Receipt

If the SelectedAreas attribute is empty, the SkipAreaResponse command's Status field SHALL indicate InvalidAreaList.

If the SkippedArea field does not match an entry in the SupportedAreas attribute, the SkipAreaResponse command's Status field SHALL indicate InvalidSkippedArea.

If the SkipArea command can not be handled by the device, e.g. due to the current state, the SkipAreaResponse command's Status field SHALL indicate InvalidInMode.

Otherwise, if the device accepts the request, the device SHALL return a SkipAreaResponse with the Status field set to Success and:

- If the device is currently operating at the area identified by SkippedArea, as indicated by either the CurrentArea or the Progress attributes, if implemented, the device SHALL stop operating at that area.
- If the Progress attribute is implemented, the entry corresponding to SkippedArea SHALL be updated to indicate that the area was skipped.
- The device SHALL attempt to operate only at the areas in the SelectedAreas attribute list where operating has not been skipped or completed, using a vendor defined order.
- If the device has either skipped or completed operating at all areas on the SelectedAreas

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attribute list, the server SHALL stop operating.

1.17.7.4. SkipAreaResponse Command

This command is sent by the device on receipt of the SkipArea command.

This command SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	SkipAreaStatus	all			M
1	**StatusText**	string	max 256			M

1.17.7.4.1. Status Field

If the Status field is set to Success or InvalidAreaList, the server MAY use a non-empty string for the StatusText field to provide additional information. For example, if Status is set to InvalidAreaList, the server MAY use StatusText to indicate why this list is invalid.

If the Status field is not set to Success or InvalidAreaList, the StatusText field SHALL include a vendor defined error description which can be used to explain the error to the user. For example, if the Status field is set to InvalidInMode, the StatusText field SHOULD indicate why the request is not allowed, given the current mode of the device, which MAY involve other clusters.

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Chapter 2. Measurement and Sensing

The Cluster Library is made of individual chapters such as this one. See Document Control in the Cluster Library for a list of all chapters and documents. References between chapters are made using a *X.Y* notation where *X* is the chapter and *Y* is the sub-section within that chapter. References to external documents are contained in Chapter 1 and are made using [*Rn*] notation.

2.1. General Description

2.1.1. Introduction

The clusters specified in this document are generic measurement and sensing interfaces that are sufficiently general to be of use across a wide range of application domains.

2.1.2. Cluster List

This section lists the measurement and sensing clusters as specified in this chapter.

Table 5. Overview of the Measurement and Sensing Clusters

Cluster ID	Cluster Name	Description
0x0400	Illuminance Measurement	Attributes and commands for configuring the measurement of illuminance, and reporting illuminance measurements
0x0402	Temperature Measurement	Attributes and commands for configuring the measurement of temperature, and reporting temperature measurements
0x0403	Pressure Measurement	Attributes and commands for configuring the measurement of pressure, and reporting pressure measurements
0x0404	Flow Measurement	Attributes and commands for configuring the measurement of flow, and reporting flow rates
0x0405	Relative Humidity Measurement	Supports configuring the measurement of relative humidity, and reporting relative humidity measurements of water in the air
0x0406	Occupancy Sensing	Attributes and commands for configuring occupancy sensing, and reporting occupancy status

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Cluster ID	Cluster Name	Description
various	Resource Monitoring	Attributes and commands for reporting conditions of various resources
0x005B	Air Quality Measurement	Attributes for reporting air quality classification
various	Concentration Measurement	Attributes and aliases for concentration measurements
0x005C	Smoke and CO Alarm	An interface to smoke and CO alarms
0x0091	Electrical Energy Measurement	Attributes and commands for measuring electrical energy
0x0090	Electrical Power Measurement	Attributes and commands for measuring electrical power
0x0550	Zone Management	Attributes and commands for configuration and management of sensing zones
0x0430	Soil Measurement	Attributes and commands for configuring the measurement of soil, and reporting soil measurements
0x0431	Ambient Context Sensing	Attributes and commands for configuring the measurement
0x0064	Temperature Alarm	Alarm definitions for temperature measurement devices

2.1.3. Data Types

2.1.3.1. MeasurementTypeEnum Type

This data type is derived from enum16.

Value	Name	Summary	Conformance
0	**Unspecified**		M
1	**Voltage**	Voltage in millivolts (mV)	M
2	**ActiveCurrent**	Active current in milliamps (mA)	M
3	**ReactiveCurrent**	Reactive current in milliamps (mA)	M
4	**ApparentCurrent**	Apparent current in milliamps (mA)	M

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Value	Name	Summary	Conformance
5	**ActivePower**	Active power in milli-watts (mW)	M
6	**ReactivePower**	Reactive power in milli-volt-amps reactive (mVAR)	M
7	**ApparentPower**	Apparent power in mil-livolt-amps (mVA)	M
8	**RMSVoltage**	Root mean squared voltage in millivolts (mV)	M
9	**RMSCurrent**	Root mean squared current in milliamps (mA)	M
10	**RMSPower**	Root mean squared power in milliwatts (mW)	M
11	**Frequency**	AC frequency in milli-hertz (mHz)	M
12	**PowerFactor**	Power Factor ratio in +/- 1/100ths of a per-cent.	M
13	**NeutralCurrent**	AC neutral current in milliamps (mA)	M
14	**ElectricalEnergy**	Electrical energy in mil-liwatt-hours (mWh)	M
15	**ReactiveEnergy**	Reactive power in milli-volt-amp-hours reac-tive (mVARh)	M
16	**ApparentEnergy**	Apparent power in mil-livolt-amp-hours (mVAh)	M
17	**SoilMoisture**	Soil moisture in per-cent	M

2.1.3.2. MeasurementAccuracyRangeStruct Type

This struct represents the accuracy of a measurement for a range of measurement values. Accuracy SHALL be expressed as a maximum +/- percentage of the true value, a maximum +/- fixed value of the true value, or both.

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**RangeMin**	int64	-2^{62} to 2^{62}				M
1	**RangeMax**	int64	-2^{62} to 2^{62}				M
2	**PercentMax**	percent100ths	all				O.a+
3	**PercentMin**	percent100ths	max PercentTypical				[PercentMax]
4	**PercentTypical**	percent100ths	PercentMin to PercentMax				[PercentMin]
5	**FixedMax**	uint64	max (2^{62} - 1)				O.a+
6	**FixedMin**	uint64	max FixedMax				[FixedMax]
7	**FixedTypical**	uint64	FixedMin to FixedMax				[FixedMin]

- If both PercentMax and FixedMax are indicated, then for a given true value in the range between RangeMin and RangeMax,
  - the reported value SHALL be less than or equal to the sum of the true value, FixedMax and PercentMax percent of the true value.
  - the reported value SHALL be greater than or equal to the true value minus the sum of FixedMax and PercentMax percent of the true value.
- If only PercentMax is indicated, then for a given true value in the range between RangeMin and RangeMax,
  - the reported value SHALL be less than or equal to the sum of the true value and PercentMax percent of the true value.
  - the reported value SHALL be greater than or equal to the true value minus PercentMax percent of the true value.
- If only FixedMax is indicated, then for a given true value in the range between RangeMin and RangeMax,
  - the reported value SHALL be less than or equal to the sum of the true value and FixedMax.
  - the reported value SHALL be greater than or equal to the true value minus FixedMax.

2.1.3.2.1. RangeMin Field

This field SHALL indicate the minimum measurement value for the specified level of accuracy.

The value of this field SHALL be greater than or equal to the value of the MinMeasuredValue field

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on the encompassing MeasurementAccuracyStruct.

The value of this field SHALL be less than or equal to the value of the MaxMeasuredValue field on the encompassing MeasurementAccuracyStruct.

2.1.3.2.2. RangeMax Field

This field SHALL indicate the maximum measurement value for the specified level of accuracy.

The value of this field SHALL be greater than the value of the RangeMin field.

The value of this field SHALL be greater than or equal to the value of the MinMeasuredValue field on the encompassing MeasurementAccuracyStruct.

The value of this field SHALL be less than or equal to the value of the MaxMeasuredValue field on the encompassing MeasurementAccuracyStruct.

2.1.3.2.3. PercentMax Field

This field SHALL indicate the maximum +/- percentage accuracy for the associated measurement.

2.1.3.2.4. PercentMin Field

This field SHALL indicate the minimum +/- percentage accuracy for the associated measurement.

2.1.3.2.5. PercentTypical Field

This field SHALL indicate the typical +/- percentage accuracy for the associated measurement.

2.1.3.2.6. FixedMax Field

This field SHALL indicate the maximum +/- fixed accuracy for the associated measurement, in the unit indicated by MeasurementType.

2.1.3.2.7. FixedMin Field

This field SHALL indicate the minimum +/- fixed accuracy for the associated measurement, in the unit indicated by MeasurementType.

2.1.3.2.8. FixedTypical Field

This field SHALL indicate the typical +/- fixed accuracy for the associated measurement, in the unit indicated by MeasurementType.

2.1.3.3. MeasurementAccuracyStruct Type

This struct represents the set of accuracy ranges for a given measurement, the maximum and minimum values for the measurement, and whether the measurement is directly measured or just estimated from other information.

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**MeasurementType**	MeasurementTypeEnum	all				M
1	**Measured**	bool	all				M
2	**MinMeasuredValue**	int64	-2^{62} to 2^{62}				M
3	**MaxMeasuredValue**	int64	-2^{62} to 2^{62}				M
4	**AccuracyRanges**	list[MeasurementAccuracyRangeStruct]	min 1				M

2.1.3.3.1. MeasurementType Field

This field SHALL indicate the type of measurement for the accuracy provided.

2.1.3.3.2. Measured Field

This field SHALL indicate whether the associated measurement was directly measured. If this field is not set to true, then the associated measurement was estimated.

2.1.3.3.3. MinMeasuredValue Attribute

This field SHALL indicate the minimum value that can be measured.

2.1.3.3.4. MaxMeasuredValue Attribute

This field SHALL indicate the maximum value that can be measured.

2.1.3.3.5. AccuracyRanges Field

This field SHALL indicate a list of measurement ranges and their associated accuracies.

The value of the RangeMin field on the first MeasurementAccuracyRangeStruct in this list SHALL be equal to the value of the MinMeasuredValue field.

The value of the RangeMax field on the last MeasurementAccuracyRangeStruct in this list SHALL be less than or equal to the value of the MaxMeasuredValue field.

The value of the RangeMin field on each MeasurementAccuracyRangeStruct in this list other than the first SHALL be one more the value of the RangeMax field on the previous MeasurementAccuracyRangeStruct in this list (i.e. there SHALL be no gaps in the accuracy ranges, and the ranges SHALL

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be in increasing order).

2.1.4. Measured Value

This section provides requirements on the attributes MeasuredValue, MinMeasuredValue, MaxMeasuredValue. Accuracy of MeasuredValue is discussed in the following section.

2.1.4.1. Constraint

Where MinMeasuredValue or MaxMeasuredValue attributes are mandatory the null value MAY be used to indicate that a limit is unknown.

For any measurement cluster with MeasuredValue, MinMeasuredValue and MaxMeasuredValue attributes, the following SHALL be always be true:

- If MinMeasuredValue and MaxMeasuredValue are both known, then MaxMeasuredValue SHALL be greater than MinMeasuredValue.
- If MaxMeasuredValue is known, then MeasuredValue SHALL be less than or equal to MaxMeasuredValue.
- If MinMeasuredValue is known, then MeasuredValue SHALL be greater than or equal to MinMeasuredValue.

2.1.5. Measurement Accuracy

Measurement clusters MAY express the accuracy of their measurements with a Tolerance attribute expressing a simple magnitude of error, or with a MeasurementAccuracyStruct expressing magnitude or percentage ranges of error for different ranges of measured values.

2.1.5.1. Tolerance Attribute

For any measurement cluster with a MeasuredValue and Tolerance attribute, when Tolerance is implemented the following SHALL always be true:

- The Tolerance attribute SHALL indicate the magnitude of the possible error that is associated with MeasuredValue attribute, using the same units and resolution. The true value SHALL be in the range (MeasuredValue – Tolerance) to (MeasuredValue + Tolerance).
- If known, the true value SHALL never be outside the possible physical range. Some examples:
  - a temperature SHALL NOT be below absolute zero
  - a concentration SHALL NOT be negative

2.2. Illuminance Measurement Cluster

The Illuminance Measurement cluster provides an interface to illuminance measurement functionality, including configuration and provision of notifications of illuminance measurements.

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2.2.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added; CCB 2048 2049 2050
2	CCB 2167
3	New data model format and notation
4	Removed P quality
5	Added F quality to MinMeasuredValue, MaxMeasuredValue, Tolerance and LightSensorType attributes

2.2.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	ILL

2.2.3. Cluster ID

ID	Name
0x0400	Illuminance Measurement

2.2.4. Data Types

2.2.4.1. LightSensorTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Photodiode**	Indicates photodiode sensor type	M
1	**CMOS**	Indicates CMOS sensor type	M
64 to 254	**MS**	Reserved for manufacturer specific light sensor types	O

2.2.5. Attributes

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**MeasuredValue**	uint16	0, MinMeasuredValue to MaxMeasuredValue	X		R V	M
0x0001	**MinMeasuredValue**	uint16	1 to 65533	F X		R V	M
0x0002	**MaxMeasuredValue**	uint16	min (MinMeasuredValue + 1)	F X		R V	M
0x0003	**Tolerance**	uint16	max 2048	F		R V	O
0x0004	**LightSensorType**	LightSensorType-Enum	all	F X	null	R V	O

2.2.5.1. MeasuredValue Attribute

This attribute SHALL indicate the illuminance in Lux (symbol lx) as follows:

- MeasuredValue = 10,000 x $\log_{10}$(illuminance) + 1,

where 1 lx <= illuminance <= 3.576 Mlx, corresponding to a MeasuredValue in the range 1 to 0xFFFE.

The MeasuredValue attribute can take the following values:

- 0 indicates a value of illuminance that is too low to be measured,
- MinMeasuredValue <= MeasuredValue <= MaxMeasuredValue under normal circumstances,
- null indicates that the illuminance measurement is invalid.

The MeasuredValue attribute is updated continuously as new measurements are made.

2.2.5.2. MinMeasuredValue Attribute

This attribute SHALL indicate the minimum value of MeasuredValue that can be measured. A value of null indicates that this attribute is not defined. See Measured Value for more details.

2.2.5.3. MaxMeasuredValue Attribute

This attribute SHALL indicate the maximum value of MeasuredValue that can be measured. A value of null indicates that this attribute is not defined. See Measured Value for more details.

2.2.5.4. Tolerance Attribute

See Measured Value.

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2.2.5.5. LightSensorType Attribute

This attribute SHALL indicate the electronic type of the light sensor. This attribute SHALL be set to one of the non-reserved values listed in LightSensorTypeEnum or null in case the sensor type is unknown.

2.3. Temperature Measurement Cluster

This cluster provides an interface to temperature measurement functionality, including configuration and provision of notifications of temperature measurements.

2.3.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added
2	CCB 2241 2370
3	CCB 2823
4	New data model format and notation
5	Removed P quality
6	Added F quality to MinMeasuredValue, MaxMeasuredValue and Tolerance attributes

2.3.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	TMP

2.3.3. Cluster ID

ID	Name
0x0402	Temperature Measurement

2.3.4. Attributes

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**MeasuredValue**	temperature	MinMeasuredValue to MaxMeasuredValue	X		R V	M
0x0001	**MinMeasuredValue**	temperature	-27315 to 32766	F X		R V	M
0x0002	**MaxMeasuredValue**	temperature	min (MinMeasuredValue + 1)	F X		R V	M
0x0003	**Tolerance**	uint16	max 2048	F	0	R V	O

2.3.4.1. MeasuredValue Attribute

This attribute SHALL indicate the measured temperature.

The null value indicates that the temperature is unknown.

2.3.4.2. MinMeasuredValue Attribute

This attribute SHALL indicate the minimum value of MeasuredValue that is capable of being measured. See Measured Value for more details.

The null value indicates that the value is not available.

2.3.4.3. MaxMeasuredValue Attribute

This attribute indicates the maximum value of MeasuredValue that is capable of being measured. See Measured Value for more details.

The null value indicates that the value is not available.

2.3.4.4. Tolerance Attribute

See Measured Value.

2.4. Pressure Measurement Cluster

This cluster provides an interface to pressure measurement functionality, including configuration and provision of notifications of pressure measurements.

2.4.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

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Revision	Description
1	Mandatory global ClusterRevision attribute added
2	CCB 2241 2370
3	New data model format and notation
4	Removed P quality
5	Added F quality to MinMeasuredValue, MaxMeasuredValue, Tolerance and Scaled attributes

2.4.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	PRS

2.4.3. Cluster ID

ID	Name
0x0403	Pressure Measurement

2.4.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**EXT**	Extended	O	Extended range and resolution

2.4.5. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**MeasuredValue**	int16	MinMeasuredValue to MaxMeasuredValue	X		R V	M
0x0001	**MinMeasuredValue**	int16	max 32766	F X		R V	M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0002	**MaxMeasuredValue**	int16	(MinMeasuredValue + 1) to 32767	F X		R V	M
0x0003	**Tolerance**	uint16	max 2048	F	0	R V	O
0x0010	**ScaledValue**	int16	MinScaledValue to MaxScaledValue	X		R V	EXT
0x0011	**MinScaledValue**	int16	max 32766	F X		R V	EXT
0x0012	**MaxScaledValue**	int16	(MinScaledValue + 1) to 32767	F X		R V	EXT
0x0013	**ScaledTolerance**	uint16	max 2048	F	0	R V	[EXT]
0x0014	**Scale**	int8	min -127	F		R V	EXT

2.4.5.1. MeasuredValue Attribute

This attribute SHALL represent the pressure in kPa as follows:

MeasuredValue = 10 x Pressure [kPa]

The null value indicates that the value is not available.

2.4.5.2. MinMeasuredValue Attribute

This attribute SHALL indicate the minimum value of MeasuredValue that can be measured. See Measured Value for more details.

The null value indicates that the value is not available.

2.4.5.3. MaxMeasuredValue Attribute

This attribute SHALL indicate the maximum value of MeasuredValue that can be measured. See Measured Value for more details.

The null value indicates that the value is not available.

2.4.5.4. Tolerance Attribute

See Measured Value.

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2.4.5.5. ScaledValue Attribute

This attribute SHALL represent the pressure in Pascals as follows:

ScaledValue = 10^{Scale} x Pressure [Pa]

The null value indicates that the value is not available.

2.4.5.6. MinScaledValue Attribute

This attribute SHALL indicate the minimum value of ScaledValue that can be measured.

The null value indicates that the value is not available.

2.4.5.7. MaxScaledValue Attribute

This attribute SHALL indicate the maximum value of ScaledValue that can be measured.

The null value indicates that the value is not available.

2.4.5.8. ScaledTolerance Attribute

This attribute SHALL indicate the magnitude of the possible error that is associated with ScaledValue. The true value is located in the range

(ScaledValue – ScaledTolerance) to (ScaledValue + ScaledTolerance).

2.4.5.9. Scale Attribute

This attribute SHALL indicate the base 10 exponent used to obtain ScaledValue (see ScaledValue).

2.5. Flow Measurement Cluster

This cluster provides an interface to flow measurement functionality, including configuration and provision of notifications of flow measurements.

2.5.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added
2	CCB 2241 2370
3	New data model format and notation
4	Removed P quality

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Revision	Description
5	Added F quality to MinMeasuredValue, MaxMeasuredValue and Tolerance attributes

2.5.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	FLW

2.5.3. Cluster ID

ID	Name
0x0404	Flow Measurement

2.5.4. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**MeasuredValue**	uint16	MinMeasuredValue to MaxMeasuredValue	X		R V	M
0x0001	**MinMeasuredValue**	uint16	max 65533	F X		R V	M
0x0002	**MaxMeasuredValue**	uint16	min (MinMeasuredValue + 1)	F X		R V	M
0x0003	**Tolerance**	uint16	max 2048	F	0	R V	O

2.5.4.1. MeasuredValue Attribute

This attribute SHALL indicate the flow in m^3/h as follows:

MeasuredValue = 10 x Flow

The null value indicates that the flow measurement is unknown, otherwise the range SHALL be as described in Measured Value.

2.5.4.2. MinMeasuredValue Attribute

This attribute SHALL indicate the minimum value of MeasuredValue that can be measured. See Measured Value for more details.

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The null value indicates that the value is not available.

2.5.4.3. MaxMeasuredValue Attribute

This attribute SHALL indicate the maximum value of MeasuredValue that can be measured. See Measured Value for more details.

The null value indicates that the value is not available.

2.5.4.4. Tolerance Attribute

See Measured Value.

2.6. Water Content Measurement Clusters

This is a base cluster. The server cluster provides an interface to water content measurement functionality. The measurement is reportable and may be configured for reporting. Water content measurements currently is, but are not limited to relative humidity.

2.6.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added
2	CCB 2241
3	New data model format and notation
4	Removed P quality
5	Added F quality to MinMeasuredValue, MaxMeasuredValue and Tolerance attributes

2.6.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	RH

2.6.3. Cluster ID

ID	Name
0x0405	Relative Humidity Measurement

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2.6.4. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**MeasuredValue**	uint16	MinMeasuredValue to MaxMeasuredValue	X		R V	M
0x0001	**MinMeasuredValue**	uint16	max 9999	F X		R V	M
0x0002	**MaxMeasuredValue**	uint16	(MinMeasuredValue + 1) to 10000	F X		R V	M
0x0003	**Tolerance**	uint16	max 2048	F		R V	O

2.6.4.1. MeasuredValue Attribute

MeasuredValue represents the water content in % as follows:

MeasuredValue = 100 x water content

Where 0% < = water content < = 100%, corresponding to a MeasuredValue in the range 0 to 10000.

The maximum resolution this format allows is 0.01%.

MinMeasuredValue and MaxMeasuredValue define the range of the sensor.

The null value indicates that the measurement is unknown, otherwise the range SHALL be as described in Measured Value.

MeasuredValue is updated continuously as new measurements are made.

2.6.4.2. MinMeasuredValue Attribute

The MinMeasuredValue attribute indicates the minimum value of MeasuredValue that can be measured. The null value means this attribute is not defined. See Measured Value for more details.

2.6.4.3. MaxMeasuredValue Attribute

The MaxMeasuredValue attribute indicates the maximum value of MeasuredValue that can be measured. The null value means this attribute is not defined. See Measured Value for more details.

2.6.4.4. Tolerance Attribute

See Measured Value.

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2.7. Occupancy Sensing Cluster

The server cluster provides an interface to occupancy sensing functionality based on one or more sensing modalities, including configuration and provision of notifications of occupancy status.

2.7.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added
2	Physical Contact Occupancy feature with mandatory OccupancySensorTypeBitmap
3	New data model format and notation
4	Remove nullable from PhysicalContact delay attributes and ability to not report transitions
5	Rework to add modalities: radar-, ambient-sensing- and vision-based sensors; add sensitivity setting (via co-located BooleanStateConfiguration cluster on same endpoint); add new attribute for tuning the sensor's reporting; add event OccupancyChanged; add N quality for HoldTime and other timing attributes; describe how to accommodate legacy clients
6	Removed P quality
7	Added OccupancyEvent feature

2.7.1.1. Backward compatibility

With the introduction of new sensor modalities in cluster revision 5, the indication for those modalities SHALL be done using feature bits. Indication of those new modalities is not feasible via the legacy attributes OccupancySensorType and OccupancySensorTypeBitmap. Servers SHALL set these attributes for backward compatibility with clients implementing a cluster revision <= 4 as described in OccupancySensorType and OccupancySensorTypeBitmap Attributes. Clients implementing a cluster revision >= 5 SHOULD use the server's ClusterRevision attribute, and SHOULD ignore these legacy attributes.

With the same revision 5, the attribute HoldTime (along with the HoldTimeLimits) is used to determine the reporting when occupancy is no longer detected. This replaces the 9 legacy attributes PIROccupiedToUnoccupiedDelay through PhysicalContactUnoccupiedToOccupiedThreshold. Servers MAY use the applicable attributes from this set of legacy attributes if and only if HoldTime is supported (see Attribute table and the corresponding explanation for details on conformance).

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For the case of a server which supports none of the modalities supported in ClusterRevision <= 4 (i.e. none of PIR, US and PHY), specific mappings for the legacy attributes OccupancySensorType and OccupancySensorTypeBitmap are provided below, representing this sensor as a PIR sensor for best-effort backward compatibility. For this case, the legacy timing attributes linked to PIR (i.e. PIROccupiedToUnoccupiedDelay through PIRUnoccupiedToOccupiedThreshold) apply.

2.7.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	OCC

2.7.3. Cluster ID

Cluster ID	Cluster Name
0x0406	Occupancy Sensing

2.7.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below. For sensor devices implementing multiple modalities, please see the guidance provided in the Occupancy Sensor device type description.

Bit	Code	Feature	Conformance	Summary
0	**OTHER**	Other	O.a+	Supports sensing using a modality not listed in the other bits
1	**PIR**	PassiveInfrared	O.a+	Supports sensing using PIR (Passive InfraRed)
2	**US**	Ultrasonic	O.a+	Supports sensing using UltraSound
3	**PHY**	PhysicalContact	O.a+	Supports sensing using a physical contact
4	**AIR**	ActiveInfrared	O.a+	Supports sensing using Active InfraRed measurement (e.g. time-of-flight or transflective/reflective IR sensing)

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Bit	Code	Feature	Conformance	Summary
5	**RAD**	Radar	O.a+	Supports sensing using radar waves (microwave)
6	**RFS**	RFSensing	O.a+	Supports sensing using analysis of radio signals, e.g.: RSSI, CSI and/or any other metric from the signal
7	**VIS**	Vision	O.a+	Supports sensing based on analyzing images
9	**OCCEVENT**	OccupancyEvent	[Rev >= v7]	Supports generating OccupancyChanged events

2.7.5. Data Types

2.7.5.1. OccupancyBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**Occupied**	Indicates the sensed occupancy state	M

2.7.5.1.1. Occupied Bit

If this bit is set, it SHALL indicate the occupied state else if the bit if not set, it SHALL indicate the unoccupied state.

2.7.5.2. OccupancySensorTypeBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**PIR**	Indicates a passive infrared sensor.	M
1	**Ultrasonic**	Indicates a ultrasonic sensor.	M
2	**PhysicalContact**	Indicates a physical contact sensor.	M

NOTE This enum is as defined in ClusterRevision 4 and its definition SHALL NOT be

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extended; the feature flags provide the sensor modality (or modalities) for later cluster revisions. See Backward Compatibility section.

2.7.5.3. OccupancySensorTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**PIR**	Indicates a passive infrared sensor.	M
1	**Ultrasonic**	Indicates a ultrasonic sensor.	M
2	**PIRAndUltrasonic**	Indicates a passive infrared and ultrasonic sensor.	M
3	**PhysicalContact**	Indicates a physical contact sensor.	M

NOTE This enum is as defined in ClusterRevision 4 and its definition SHALL NOT be extended; the feature flags provide the sensor modality (or modalities) for later cluster revisions. See Backward Compatibility section.

2.7.5.4. HoldTimeLimitsStruct Type

This structure provides information on the server's supported values for the HoldTime attribute.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**HoldTimeMin**	uint16	min 1				M
1	**HoldTimeMax**	uint16	min maxOf(HoldTimeMin, 10)				M
2	**HoldTimeDefault**	uint16	HoldTimeMin to HoldTimeMax				M

2.7.5.4.1. HoldTimeMin Field

This field SHALL specify the minimum value of the server's supported value for the HoldTime attribute, in seconds.

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2.7.5.4.2. HoldTimeMax Field

This field SHALL specify the maximum value of the server's supported value for the HoldTime attribute, in seconds.

2.7.5.4.3. HoldTimeDefault Field

This field SHALL specify the (manufacturer-determined) default value of the server's HoldTime attribute, in seconds. This is the value that a client who wants to reset the settings to a valid default SHOULD use.

2.7.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**Occupancy**	OccupancyBitmap	0 to 1			R V	M
0x0001	**OccupancySensorType**	OccupancySensorTypeEnum	desc	F		R V	M, D
0x0002	**OccupancySensorTypeBitmap**	OccupancySensorTypeBitmap	0 to 7	F		R V	M, D
0x0003	**HoldTime**	uint16	HoldTimeLimits.HoldTimeMin to HoldTimeLimits.HoldTimeMax	N		RW VM	O
0x0004	**HoldTimeLimits**	HoldTimeLimitsStruct		F		R V	HoldTime
0x0010	**PIROccupiedToUnoccupiedDelay**	uint16	all	N	0	RW VM	[HoldTime & (PIR | (!PIR & !US & !PHY))], D

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0011	**PIRUnoccupiedToOccupiedDelay**	uint16	all	N	0	RW VM	(HoldTime & (PIR | (!PIR & !US & !PHY))) & PIRUnoccupiedToOccupiedThreshold, [HoldTime & (PIR | (!PIR & !US & !PHY))], D
0x0012	**PIRUnoccupiedToOccupiedThreshold**	uint8	1 to 254	N	1	RW VM	(HoldTime & (PIR | (!PIR & !US & !PHY))) & PIRUnoccupiedToOccupiedDelay, [HoldTime & (PIR | (!PIR & !US & !PHY))], D
0x0020	**UltrasonicOccupiedToUnoccupiedDelay**	uint16	all	N	0	RW VM	[HoldTime & US], D
0x0021	**UltrasonicUnoccupiedToOccupiedDelay**	uint16	all	N	0	RW VM	HoldTime & US & UltrasonicUnoccupiedToOccupiedThreshold, [HoldTime & US], D

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0022	**UltrasonicUnoccupiedToOccupiedThreshold**	uint8	1 to 254	N	1	RW VM	HoldTime & US & UltrasonicUnoccupiedToOccupiedDelay, [HoldTime & US], D
0x0030	**PhysicalContactOccupiedToUnoccupiedDelay**	uint16	all	N	0	RW VM	[HoldTime & PHY], D
0x0031	**PhysicalContactUnoccupiedToOccupiedDelay**	uint16	all	N	0	RW VM	HoldTime & PHY & PhysicalContactUnoccupiedToOccupiedThreshold, [HoldTime & PHY], D
0x0032	**PhysicalContactUnoccupiedToOccupiedThreshold**	uint8	1 to 254	N	1	RW VM	HoldTime & PHY & PhysicalContactUnoccupiedToOccupiedDelay, [HoldTime & PHY], D

2.7.6.1. Occupancy Attribute

This attribute SHALL indicate the sensed (processed) status of occupancy. For compatibility reasons this is expressed as a bitmap where the status is indicated in bit 0: a value of 1 means occupied, and 0 means unoccupied, with the other bits set to 0; this can be considered equivalent to a boolean.

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2.7.6.2. OccupancySensorType and OccupancySensorTypeBitmap Attributes

These attributes SHALL indicate the type of sensor. In ClusterRevision <= 4, these were used to specify the type of the occupancy sensor using a bitmap. For ClusterRevision >= 5, the server SHALL indicate the type of sensor using the feature flags, and the server SHALL provide these attributes for the backward compatibility with clients implementing a cluster revision <= 4, using the mapping according to the following table. Also see Backward Compatibility section.

Feature Flag Value			Value of OccupancySensorTypeBitmap	Value of OccupancySensorType
PIR	**US**	**PHY**		
0	0	0	PIR *	PIR *
1	0	0	PIR	PIR
0	1	0	Ultrasonic	Ultrasonic
1	1	0	PIR + Ultrasonic	PIRAndUltrasonic
0	0	1	PhysicalContact	PhysicalContact
1	0	1	PhysicalContact + PIR	PIR **
0	1	1	PhysicalContact + Ultrasonic	Ultrasonic **
1	1	1	PhysicalContact + PIR + Ultrasonic	PIRAndUltrasonic **

* In case the feature flags PIR, US and PHY are all set to 0, these attributes are set to mimic a PIR sensor since this likely will give the best compatibility with clients implementing a cluster revision <= 4, which are not aware of the feature flags and the added modalities. In this case, timing parameters (when HoldTime is supported) are expressed using the PIROccupiedToUnoccupiedDelay, PIRUnoccupiedToOccupiedDelay and PIRUnoccupiedToOccupiedThreshold attributes.

** These combinations of sensor type were not defined for OccupancySensorType in ClusterRevision <= 4, so they are mapped to a combination which *was* defined in that version.

2.7.6.3. HoldTime Attribute

This attribute SHALL specify the time delay, in seconds, before the sensor changes to its unoccupied state after the last detection of occupancy in the sensed area. This is equivalent to the legacy *OccupiedToUnoccupiedDelay attributes.

Low values of HoldTime SHOULD be avoided since they could lead to many reporting messages. A value 0 for HoldTime SHALL NOT be used.

The figure below illustrates this with an example of how this attribute is used for a PIR sensor. It uses threshold detection to generate an "internal detection" signal, which needs post-processing to become usable for transmission (traffic shaping). The bit in the Occupancy attribute will be set to 1 when the internal detection signal goes high, and will stay at 1 for HoldTime after the (last) instance

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where the internal detection signal goes low.

The top half of the figure shows the case of a single trigger: the bit in the Occupancy attribute will be 1 for the duration of the PIR signal exceeding the threshold plus HoldTime. The bottom half of the figure shows the case of multiple triggers: the second trigger starts before the HoldTime of the first trigger has expired; this results in a single period of the bit in the Occupancy attribute being 1. The bit in the Occupancy attribute will be set to 1 from the start of the first period where the PIR signal exceeds the threshold until HoldTime after the last moment where the PIR exceeded the threshold.

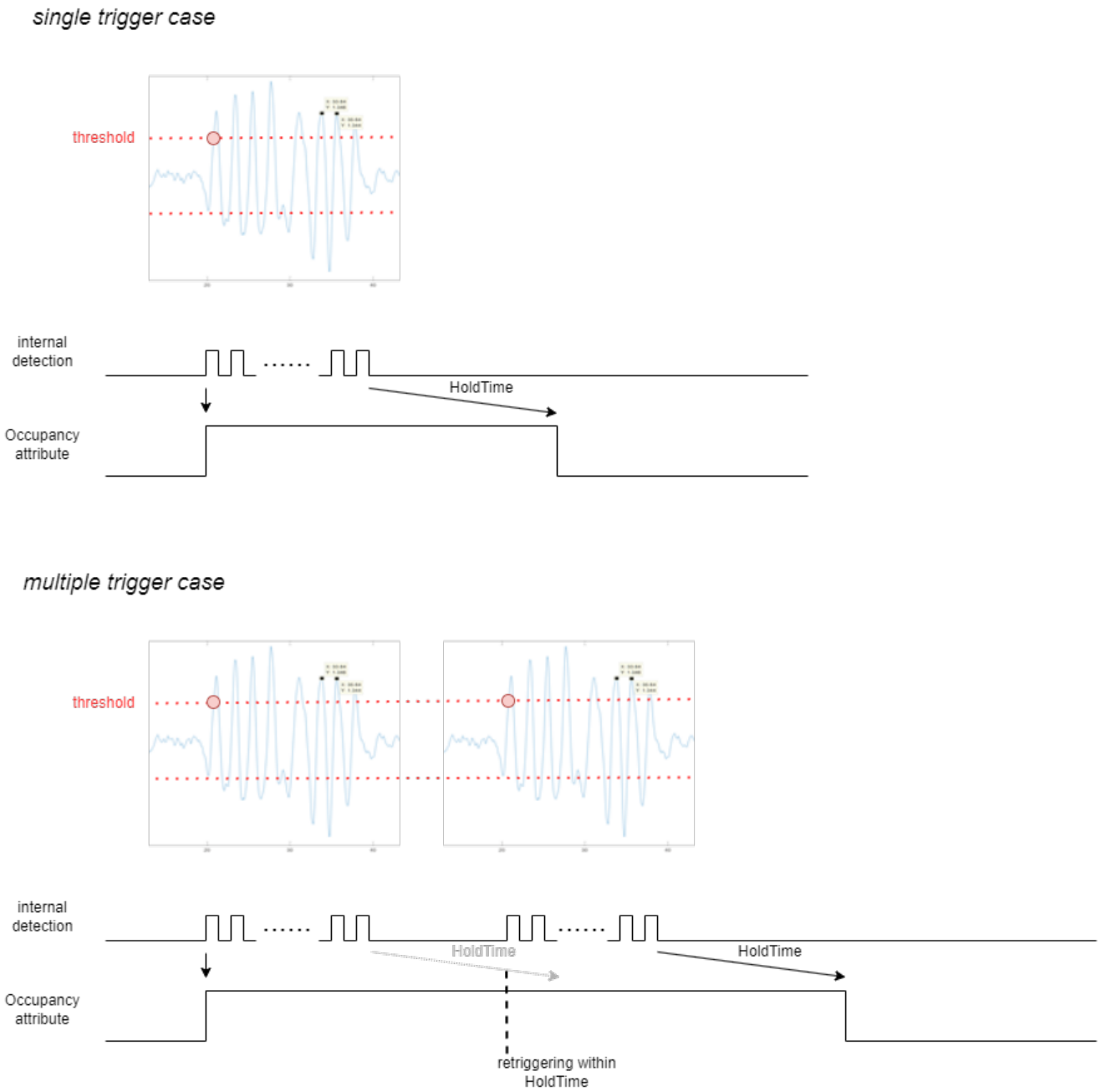


Figure 13. Processing of PIR signal towards Occupancy attribute using HoldTime

2.7.6.4. HoldTimeLimits Attribute

This attribute SHALL indicate the server's limits, and default value, for the HoldTime attribute.

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2.7.6.5. Attributes to support legacy implementations

The following 3 sets of 3 attributes each were used in ClusterRevision <= 4 to indicate timings and conditions to tune the reporting by the sensor - separately for each of the three sensor modalities defined in that revision. In ClusterRevision >= 5, these are replaced by the HoldTime attribute (shared for all modalities). A server with ClusterRevision >= 5 MAY (under the conformance conditions stated in the attributes table above) provide these attributes for backward compatibility with clients implementing a cluster revision <= 4. When doing so, the provided values for these legacy *OccupiedToUnoccupied attributes SHALL follow the values in the HoldTime attribute when updated, and vice-versa, such that any present *OccupiedToUnoccupiedDelay attribute maintains equality to HoldTime.

Also see Backward Compatibility section.

Some, but not all, of the implications of the conformance rules for these attributes are:

- if attribute HoldTime is not supported, these 9 legacy attributes SHALL NOT be supported
- if modality PIR is supported, while both PHY and US modalities are not supported,
  - the 3 legacy attributes related to PIR MAY be supported (if allowed by the other rules)
  - the 6 legacy attributes related to Ultrasonic and PhysicalContact SHALL NOT be supported
- if modality US is supported while both PIR and PHY modalities are not supported:
  - the 3 legacy attributes related to UltraSonic MAY be supported (if allowed by the other rules)
  - the 6 legacy attributes related to PIR and PhysicalContact SHALL NOT be supported
- if modality PHY is supported while both PIR and US modalities are not supported:
  - the 3 legacy attributes related to PhysicalContact MAY be supported (if allowed by the other rules)
  - the 6 legacy attributes related to PIR and Ultrasonic SHALL NOT be supported
- the legacy attributes related to UnoccupiedToOccupiedDelay and UnoccupiedToOccupiedThreshold for a certain modality SHALL be both supported or both not supported
- if none of the modalities PIR, US, and PHY is supported (this case is presented as a PIR sensor for compatibility):
  - the 3 legacy attributes related to PIR MAY be supported (if allowed by the other rules)
  - the 6 legacy attributes related to Ultrasonic and PhysicalContact SHALL NOT be supported

For cases where more than one of the PIR, PHY and US modalities is supported *and* the timing parameters are provided for more than one modality, it is manufacturer specific how to combine those timing parameters.

2.7.6.6. PIROccupiedToUnoccupiedDelay Attribute

This attribute SHALL specify the time delay, in seconds, before the PIR sensor changes to its unoccupied state after the last detection of occupancy in the sensed area.

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2.7.6.7. PIRUnoccupiedToOccupiedDelay Attribute

This attribute SHALL specify the time delay, in seconds, before the PIR sensor changes to its occupied state after the first detection of occupancy in the sensed area.

2.7.6.8. PIRUnoccupiedToOccupiedThreshold Attribute

This attribute SHALL specify the number of occupancy detection events that must occur in the period PIRUnoccupiedToOccupiedDelay, before the PIR sensor changes to its occupied state.

2.7.6.9. UltrasonicOccupiedToUnoccupiedDelay Attribute

This attribute SHALL specify the time delay, in seconds, before the Ultrasonic sensor changes to its unoccupied state after the last detection of occupancy in the sensed area.

2.7.6.10. UltrasonicUnoccupiedToOccupiedDelay Attribute

This attribute SHALL specify the time delay, in seconds, before the Ultrasonic sensor changes to its occupied state after the first detection of occupancy in the sensed area.

2.7.6.11. UltrasonicUnoccupiedToOccupiedThreshold Attribute

This attribute SHALL specify the number of occupancy detection events that must occur in the period UltrasonicUnoccupiedToOccupiedDelay, before the Ultrasonic sensor changes to its occupied state.

2.7.6.12. PhysicalContactOccupiedToUnoccupiedDelay Attribute

This attribute SHALL specify the time delay, in seconds, before the physical contact occupancy sensor changes to its unoccupied state after detecting the unoccupied event.

2.7.6.13. PhysicalContactUnoccupiedToOccupiedDelay Attribute

This attribute SHALL specify the time delay, in seconds, before the physical contact sensor changes to its occupied state after the first detection of the occupied event.

2.7.6.14. PhysicalContactUnoccupiedToOccupiedThreshold Attribute

This attribute SHALL specify the number of occupancy detection events that must occur in the period PhysicalContactUnoccupiedToOccupiedDelay, before the PhysicalContact sensor changes to its occupied state.

2.7.7. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**Occupancy-Changed**	INFO		V	OCCEVENT, O

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2.7.7.1. OccupancyChanged Event

If this event is supported, it SHALL be generated when the Occupancy attribute changes.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Occupancy**	Occupancy-Bitmap				M

2.7.7.1.1. Occupancy Field

This field SHALL indicate the new value of the Occupancy attribute.

2.8. Resource Monitoring Clusters

This generic cluster provides an interface to the current condition of a resource. A resource is a component of a device that is designed to be replaced, refilled, or emptied when exhausted or full. Examples of resources include filters, cartridges, and water tanks. While batteries fit this definition they are not intended to be used with this cluster. Use the power source cluster for batteries instead.

NOTE	This cluster is not meant to be used for monitoring of the system resources, such as processing, memory utilization, networking properties, etc.

This cluster SHALL be used via an alias to a specific resource type (see Cluster IDs).

2.8.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

2.8.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	REPM

2.8.3. Cluster IDs

The table below is a list of aliased Cluster IDs which represent different resource types and conform to this cluster definition.

ID	Name	PICS
0x0071	HEPA Filter Monitoring	HEPAFREMON

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ID	Name	PICS
0x0072	Activated Carbon Filter Monitoring	ACFREMON
0x0079	Water Tank Level Monitoring	WTLREPMON

2.8.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**CON**	Condition	O	Supports monitoring the condition of the resource in percentage
1	**WRN**	Warning	O	Supports warning indication
2	**REP**	ReplacementProductList	O	Supports specifying the list of replacement products

2.8.5. Data Types

2.8.5.1. DegradationDirectionEnum Type

This data type is derived from enum8.

Indicates the direction in which the condition of the resource changes over time.

Value	Name	Summary	Conformance
0	**Up**	The degradation of the resource is indicated by an upwards moving/increasing value	M
1	**Down**	The degradation of the resource is indicated by a downwards moving/decreasing value	M

2.8.5.2. ChangeIndicationEnum Type

This data type is derived from enum8.

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Value	Name	Summary	Conformance
0	**OK**	Resource is in good condition, no intervention required	M
1	**Warning**	Resource will be exhausted soon, intervention will shortly be required	WRN
2	**Critical**	Resource is exhausted, immediate intervention is required	M

2.8.5.3. ProductIdentifierTypeEnum Type

This data type is derived from enum8.

Indicate the type of identifier used to describe the product. Devices SHOULD use globally-recognized IDs over OEM specific ones.

Value	Name	Summary	Conformance
0	**UPC**	12-digit Universal Product Code	M
1	**GTIN-8**	8-digit Global Trade Item Number	M
2	**EAN**	13-digit European Article Number	M
3	**GTIN-14**	14-digit Global Trade Item Number	M
4	**OEM**	Original Equipment Manufacturer part number	M

2.8.5.4. ReplacementProductStruct Type

Indicates the product identifier that can be used as a replacement for the resource.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**ProductIdentifierType**	ProductIdentifierTypeEnum	desc				M
1	**ProductIdentifierValue**	string	max 20				M

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2.8.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**Condition**	percent				R V	CON
0x0001	**DegradationDirection**	DegradationDirectionEnum	desc	F		R V	CON
0x0002	**ChangeIndication**	ChangeIndicationEnum				R V	M
0x0003	**InPlaceIndicator**	bool				R V	O
0x0004	**LastChangedTime**	epoch-s	all	N X	null	RW VO	O
0x0005	**ReplacementProductList**	list[ReplacementProductStruct]	max 5	F		R V	REP

2.8.6.1. Condition Attribute

This attribute SHALL indicate the current condition of the resource in percent.

2.8.6.2. DegradationDirection Attribute

This attribute SHALL indicate the direction of change for the condition of the resource over time, which helps to determine whether a higher or lower condition value is considered optimal.

2.8.6.3. ChangeIndication Attribute

This attribute SHALL be populated with a value from ChangeIndicationEnum that is indicative of the current requirement to change the resource.

2.8.6.4. InPlaceIndicator Attribute

This attribute SHALL indicate whether a resource is currently installed. A value of true SHALL indicate that a resource is installed. A value of false SHALL indicate that a resource is not installed.

2.8.6.5. LastChangedTime Attribute

This attribute MAY indicates the time at which the resource has been changed, if supported by the server. The attribute SHALL be null if it was never set or is unknown.

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2.8.6.6. ReplacementProductList Attribute

This attribute SHALL indicate the list of supported products that may be used as replacements for the current resource. Each item in this list represents a unique ReplacementProductStruct.

2.8.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**ResetCondition**	client ⇒ server	Y	O	O

2.8.7.1. ResetCondition Command

Upon receipt, the device SHALL reset the Condition and ChangeIndicator attributes, indicating full resource availability and readiness for use, as initially configured. Invocation of this command MAY cause the LastChangedTime to be updated automatically based on the clock of the server, if the server supports setting the attribute.

2.9. Air Quality Cluster

This cluster provides an interface to air quality classification using distinct levels with human-readable labels.

2.9.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

2.9.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	AIRQUAL

2.9.3. Cluster ID

ID	Name
0x005B	Air Quality

2.9.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

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Bit	Code	Feature	Conformance	Summary
0	**FAIR**	Fair	O	Cluster supports the Fair air quality level
1	**MOD**	Moderate	O	Cluster supports the Moderate air quality level
2	**VPOOR**	VeryPoor	O	Cluster supports the Very poor air quality level
3	**XPOOR**	ExtremelyPoor	O	Cluster supports the Extremely poor air quality level

2.9.5. Data Types

2.9.5.1. AirQualityEnum Type

This data type is derived from enum8.

The AirQualityEnum provides a representation of the quality of the analyzed air. It is up to the device manufacturer to determine the mapping between the measured values and their corresponding enumeration values.

Value	Name	Summary	Conformance
0	**Unknown**	The air quality is unknown.	M
1	**Good**	The air quality is good.	M
2	**Fair**	The air quality is fair.	FAIR
3	**Moderate**	The air quality is moderate.	MOD
4	**Poor**	The air quality is poor.	M
5	**VeryPoor**	The air quality is very poor.	VPOOR
6	**ExtremelyPoor**	The air quality is extremely poor.	XPOOR

2.9.6. Attributes

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**AirQuality**	AirQualityEnum	desc			R V	M

2.9.6.1. AirQuality Attribute

This attribute SHALL indicate a value from AirQualityEnum that is indicative of the currently measured air quality.

2.10. Concentration Measurement Clusters

The server cluster provides an interface to concentration measurement functionality.

This cluster SHALL to be used via an alias to a specific substance (see Cluster IDs).

2.10.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added
2	CCB 2882
3	Cluster redesigned to add support for Level Indication, Peak/Average Measurement, Medium/Unit of Measurement and Uncertainty
4	Removed P quality
5	Added F quality to MinMeasuredValue, MaxMeasuredValue and Uncertainty attributes

2.10.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	CONC

2.10.3. Cluster IDs

The table below is a list of Cluster IDs that conform to this specification.

ID	Name	PICS
0x040C	Carbon Monoxide Concentration Measurement	CMOCONC

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ID	Name	PICS
0x040D	Carbon Dioxide Concentration Measurement	CDOCONC
0x0413	Nitrogen Dioxide Concentration Measurement	NDOCONC
0x0415	Ozone Concentration Measurement	OZCONC
0x042A	PM2.5 Concentration Measurement	PMICONC
0x042B	Formaldehyde Concentration Measurement	FLDCONC
0x042C	PM1 Concentration Measurement	PMHCONC
0x042D	PM10 Concentration Measurement	PMKCONC
0x042E	Total Volatile Organic Compounds Concentration Measurement	TVOCCONC
0x042F	Radon Concentration Measurement	RNCONC

2.10.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**MEA**	NumericMeasurement	O.a+	Cluster supports numeric measurement of substance
1	**LEV**	LevelIndication	O.a+	Cluster supports basic level indication for substance using the ConcentrationLevel enum
2	**MED**	MediumLevel	[LEV]	Cluster supports the Medium Concentration Level
3	**CRI**	CriticalLevel	[LEV]	Cluster supports the Critical Concentration Level

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Bit	Code	Feature	Conformance	Summary
4	**PEA**	PeakMeasurement	[MEA]	Cluster supports peak numeric measurement of substance
5	**AVG**	AverageMeasure-ment	[MEA]	Cluster supports average numeric measurement of substance

2.10.5. Data Types

2.10.5.1. MeasurementUnitEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**PPM**	Parts per Million (10^6)	MEA
1	**PPB**	Parts per Billion (10^9)	MEA
2	**PPT**	Parts per Trillion (10^{12})	MEA
3	**MGM3**	Milligram per m^3	MEA
4	**UGM3**	Microgram per m^3	MEA
5	**NGM3**	Nanogram per m^3	MEA
6	**PM3**	Particles per m^3	MEA
7	**BQM3**	Becquerel per m^3	MEA

Where mentioned, Billion refers to 10^9, Trillion refers to 10^{12} (short scale).

2.10.5.2. MeasurementMediumEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Air**	The measurement is being made in Air	M
1	**Water**	The measurement is being made in Water	M
2	**Soil**	The measurement is being made in Soil	M

2.10.5.3. LevelValueEnum Type

This data type is derived from enum8.

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Value	Name	Summary	Conformance
0	**Unknown**	The level is Unknown	M
1	**Low**	The level is considered Low	M
2	**Medium**	The level is considered Medium	MED
3	**High**	The level is considered High	M
4	**Critical**	The level is considered Critical	CRI

2.10.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**MeasuredValue**	single	MinMeasuredValue to MaxMeasuredValue	X		R V	MEA
0x0001	**MinMeasuredValue**	single	all	F X		R V	MEA
0x0002	**MaxMeasuredValue**	single	min MinMeasuredValue	F X		R V	MEA
0x0003	**PeakMeasuredValue**	single	MinMeasuredValue to MaxMeasuredValue	X		R V	PEA
0x0004	**PeakMeasuredValueWindow**	elapsed-s	max 604800			R V	PEA
0x0005	**AverageMeasuredValue**	single	MinMeasuredValue to MaxMeasuredValue	X		R V	AVG

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0006	**AverageMeasuredValueWindow**	elapsed-s	max 604800			R V	AVG
0x0007	**Uncertainty**	single	all	F	MS	R V	[MEA]
0x0008	**MeasurementUnit**	MeasurementUnitEnum	all	F		R V	MEA
0x0009	**MeasurementMedium**	MeasurementMediumEnum	all	F		R V	M
0x000A	**LevelValue**	LevelValueEnum	all			R V	LEV

2.10.6.1. MeasuredValue Attribute

This attribute SHALL indicate the most recent measurement as a single-precision floating-point number. MeasuredValue's unit is represented by MeasurementUnit.

A value of null indicates that the measurement is unknown or outside the valid range.

MinMeasuredValue and MaxMeasuredValue define the valid range for MeasuredValue.

2.10.6.2. MinMeasuredValue Attribute

This attribute SHALL indicate the minimum value of MeasuredValue that is capable of being measured. A MinMeasuredValue of null indicates that the MinMeasuredValue is not defined.

2.10.6.3. MaxMeasuredValue Attribute

This attribute SHALL indicate the maximum value of MeasuredValue that is capable of being measured. A MaxMeasuredValue of null indicates that the MaxMeasuredValue is not defined.

2.10.6.4. PeakMeasuredValue Attribute

This attribute SHALL indicate the maximum value of MeasuredValue that has been measured during the PeakMeasuredValueWindow. If this attribute is provided, the PeakMeasuredValueWindow attribute SHALL also be provided.

2.10.6.5. PeakMeasuredValueWindow Attribute

This attribute SHALL indicate the window of time used for determining the PeakMeasuredValue. The value is in seconds.

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2.10.6.6. AverageMeasuredValue Attribute

This attribute SHALL indicate the average value of MeasuredValue that has been measured during the AverageMeasuredValueWindow. If this attribute is provided, the AverageMeasuredValueWindow attribute SHALL also be provided.

2.10.6.7. AverageMeasuredValueWindow Attribute

This attribute SHALL indicate the window of time used for determining the AverageMeasuredValue. The value is in seconds.

2.10.6.8. Uncertainty Attribute

This attribute SHALL indicate the range of error or deviation that can be found in MeasuredValue and PeakMeasuredValue. This is considered a +/- value and should be considered to be in MeasurementUnit.

2.10.6.9. MeasurementUnit Attribute

This attribute SHALL indicate the unit of MeasuredValue. See MeasurementUnitEnum.

2.10.6.10. MeasurementMedium Attribute

This attribute SHALL indicate the medium in which MeasuredValue or LevelValue is being measured. See MeasurementMediumEnum.

2.10.6.11. LevelValue Attribute

This attribute SHALL indicate the level of the substance detected. See LevelValueEnum.

2.11. Smoke CO Alarm Cluster

This cluster provides an interface for observing and managing the state of smoke and CO alarms.

2.11.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Added Inoperative to ExpressedStateEnum and added Unmounted attribute

2.11.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	SMOKECO

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2.11.3. Cluster ID

ID	Name
0x005C	Smoke CO Alarm

2.11.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**SMOKE**	SmokeAlarm	O.a+	Supports Smoke alarm
1	**CO**	COAlarm	O.a+	Supports CO alarm

2.11.5. Data Types

2.11.5.1. AlarmStateEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Normal**	Nominal state, the device is not alarming	M
1	**Warning**	Warning state	O
2	**Critical**	Critical state	M

2.11.5.1.1. Normal Value

This value SHALL indicate that this alarm is not alarming.

2.11.5.1.2. Warning Value

This value SHALL indicate that this alarm is in a warning state. Alarms in this state SHOULD be subject to being muted via physical interaction.

2.11.5.1.3. Critical Value

This value SHALL indicate that this alarm is in a critical state. Alarms in this state SHALL NOT be subject to being muted via physical interaction.

2.11.5.2. SensitivityEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**High**	High sensitivity	O

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Value	Name	Summary	Conformance
1	**Standard**	Standard Sensitivity	M
2	**Low**	Low sensitivity	O

2.11.5.3. ExpressedStateEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Normal**	Nominal state, the device is not alarming	M
1	**SmokeAlarm**	Smoke Alarm state	SMOKE
2	**COAlarm**	CO Alarm state	CO
3	**BatteryAlert**	Battery Alert State	M
4	**Testing**	Test in Progress	M
5	**HardwareFault**	Hardware Fault Alert State	M
6	**EndOfService**	End of Service Alert State	M
7	**InterconnectSmoke**	Interconnected Smoke Alarm State	O
8	**InterconnectCO**	Interconnected CO Alarm State	O
9	**Inoperative**	Hardware is not able to detect Smoke or CO	[Rev >= v2]

2.11.5.3.1. Normal Value

This value SHALL indicate that this alarm is not alarming.

2.11.5.3.2. SmokeAlarm Value

This value SHALL indicate that this alarm is currently expressing visual indication of Smoke Alarm. This value SHALL indicate that the alarm is currently expressing audible indication of Smoke Alarm unless the DeviceMuted attribute is supported and set to Muted.

2.11.5.3.3. COAlarm Value

This value SHALL indicate that this alarm is currently expressing visual indication of CO Alarm. This value SHALL indicate that the alarm is currently expressing audible indication of CO Alarm unless the DeviceMuted attribute is supported and set to Muted.

2.11.5.3.4. BatteryAlert Value

This value SHALL indicate that this alarm is currently expressing visual indication of Critical Low

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Battery. This value SHALL indicate that the alarm is currently expressing audible indication of Critical Low Battery unless the DeviceMuted attribute is supported and set to Muted.

2.11.5.3.5. Testing Value

This value SHALL indicate that this alarm is currently expressing visual and audible indication of SelfTest.

2.11.5.3.6. HardwareFault Value

This value SHALL indicate that this alarm is currently expressing visual indication of Hardware Fault. This value SHALL indicate that the alarm is currently expressing audible indication of Hardware Fault unless the DeviceMuted attribute is supported and set to Muted.

2.11.5.3.7. EndOfService Value

This value SHALL indicate that this alarm is currently expressing visual indication of End Of Service. This value SHALL indicate that the alarm is currently expressing audible indication of End of Service unless the DeviceMuted attribute is supported and set to Muted.

2.11.5.3.8. InterconnectSmoke Value

This value SHALL indicate that this alarm is currently expressing visual indication of Smoke Alarm caused by Interconnect. This value SHALL indicate that the alarm is currently expressing audible indication of Smoke Alarm caused by Interconnect unless the DeviceMuted attribute is supported and set to Muted.

2.11.5.3.9. InterconnectCO Value

This value SHALL indicate that this alarm is currently expressing visual indication of CO Alarm caused by Interconnect. This value SHALL indicate that the alarm is currently expressing audible indication of CO Alarm caused by Interconnect unless the DeviceMuted attribute is supported and set to Muted.

2.11.5.3.10. Inoperative Value

This value SHALL indicate that the hardware is currently unable to detect any Smoke or Carbon Monoxide (CO). The value SHALL only be set when the inoperability is caused either by

- the battery is too low (BatteryAlert has the value "critical"),
- hardware is defective (HardwareFaultAlert is true) or
- the device being unmounted or physically disabled (Unmounted is true).

2.11.5.4. MuteStateEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**NotMuted**	Not Muted	M

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Value	Name	Summary	Conformance
1	**Muted**	Muted	M

2.11.5.4.1. NotMuted Value

This value SHALL indicate that the device is not muted.

2.11.5.4.2. Muted Value

This value SHALL indicate that the device is muted.

2.11.5.5. EndOfServiceEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Normal**	Device has not expired	M
1	**Expired**	Device has reached its end of service	M

2.11.5.5.1. Expired Value

This value SHALL indicate that the device has reached its end of service, and needs to be replaced.

2.11.5.5.2. Normal Value

This value SHALL indicate that the device has not yet reached its end of service, and does not need to be imminently replaced.

2.11.5.6. ContaminationStateEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Normal**	Nominal state, the sensor is not contaminated	M
1	**Low**	Low contamination	O
2	**Warning**	Warning state	O
3	**Critical**	Critical state, will cause nuisance alarms	M

2.11.5.6.1. Normal Value

This value SHALL indicate that the smoke sensor has nominal contamination levels, no customer action is required.

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2.11.5.6.2. Low Value

This value SHALL indicate that the smoke sensor has detectable contamination levels, but the contamination is too low to cause a visible or audible alarm.

2.11.5.6.3. Warning Value

This value SHALL indicate that the smoke sensor has contamination levels in a warning state. At this level, the contamination may cause a visible or audible alarm. User intervention is suggested.

2.11.5.6.4. Critical Value

This value SHALL indicate that the smoke sensor has contamination levels in a critical state. At this level, the contamination should cause a visible or audible alarm. User intervention is required. Critical contamination of the sensor SHALL also be reflected as a HardwareFault.

2.11.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**ExpressedState**	ExpressedStateEnum	all	N		R V	M
0x0001	**SmokeState**	AlarmStateEnum	all	N		R V	SMOKE
0x0002	**COState**	AlarmStateEnum	all	N		R V	CO
0x0003	**BatteryAlert**	AlarmStateEnum	all	N		R V	M
0x0004	**DeviceMuted**	MuteStateEnum	all	N		R V	O
0x0005	**TestInProgress**	bool	all			R V	M
0x0006	**HardwareFaultAlert**	bool	all	N		R V	M
0x0007	**EndOfServiceAlert**	EndOfServiceEnum	all	N		R V	M
0x0008	**InterconnectSmokeAlarm**	AlarmStateEnum	all			R V	O
0x0009	**InterconnectCOAlarm**	AlarmStateEnum	all			R V	O
0x000A	**ContaminationState**	ContaminationStateEnum	all			R V	[SMOKE]

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x000B	**SmokeSensitivityLevel**	SensitivityEnum	all			RW VM	[SMOKE]
0x000C	**ExpiryDate**	epoch-s	all	F		R V	O
0x000D	**Unmounted**	bool	all		false	R V	[Rev >= v2]

2.11.6.1. ExpressedState Attribute

This attribute SHALL indicate the visibly- and audibly-expressed state of the alarm. When multiple alarm conditions are being reflected in the server, this attribute SHALL indicate the condition with the highest priority. Priority order of conditions is determined by the manufacturer and SHALL be supplied as a part of certification procedure. If the value of ExpressedState is not Normal, the attribute corresponding to the value SHALL NOT be Normal. For example, if the ExpressedState is set to SmokeAlarm, the value of the SmokeState will indicate the severity of the alarm (Warning or Critical). Clients SHOULD also read the other attributes to be aware of further alarm conditions beyond the one indicated in ExpressedState.

Visible expression is typically a LED light pattern. Audible expression is a horn or speaker pattern. Audible expression SHALL BE suppressed if the DeviceMuted attribute is supported and set to Muted.

2.11.6.2. SmokeState Attribute

This attribute SHALL indicate whether the device's smoke sensor is currently triggering a smoke alarm.

2.11.6.3. COState Attribute

This attribute SHALL indicate whether the device's CO sensor is currently triggering a CO alarm.

2.11.6.4. BatteryAlert Attribute

This attribute SHALL indicate whether the power resource fault detection mechanism is currently triggered at the device. If the detection mechanism is triggered, this attribute SHALL be set to Warning or Critical, otherwise it SHALL be set to Normal. The battery state SHALL also be reflected in the Power Source cluster representing the device's battery using the appropriate supported attributes and events.

If the sensor cannot operate in the set state, the ExpressedState attribute SHALL be set to Inoperative.

2.11.6.5. DeviceMuted Attribute

This attribute SHALL indicate the whether the audible expression of the device is currently muted.

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Audible expression is typically a horn or speaker pattern.

2.11.6.6. TestInProgress Attribute

This attribute SHALL indicate whether the device self-test is currently activated. If the device self-test is activated, this attribute SHALL be set to True, otherwise it SHALL be set to False.

2.11.6.7. HardwareFaultAlert Attribute

This attribute SHALL indicate whether the hardware fault detection mechanism is currently triggered. If the detection mechanism is triggered, this attribute SHALL be set to True, otherwise it SHALL be set to False.

When this attribute is set to True, and the sensor cannot operate in this state, the ExpressedState attribute SHALL be set to Inoperative.

2.11.6.8. EndOfServiceAlert Attribute

This attribute SHALL indicate whether the end-of-service has been triggered at the device. This attribute SHALL be set to Expired when the device reaches the end-of-service.

2.11.6.9. InterconnectSmokeAlarm Attribute

This attribute SHALL indicate whether the interconnected smoke alarm is currently triggering by branching devices. When the interconnected smoke alarm is being triggered, this attribute SHALL be set to Warning or Critical, otherwise it SHALL be set to Normal.

2.11.6.10. InterconnectCOAlarm Attribute

This attribute SHALL indicate whether the interconnected CO alarm is currently triggering by branching devices. When the interconnected CO alarm is being triggered, this attribute SHALL be set to Warning or Critical, otherwise it SHALL be set to Normal.

2.11.6.11. ContaminationState Attribute

This attribute SHALL indicate the contamination level of the smoke sensor.

2.11.6.12. SmokeSensitivityLevel Attribute

This attribute SHALL indicate the sensitivity level of the smoke sensor configured on the device.

2.11.6.13. ExpiryDate Attribute

This attribute SHALL indicate the date when the device reaches its stated expiry date. After the ExpiryDate has been reached, the EndOfServiceAlert SHALL start to be triggered. To account for better customer experience across time zones, the EndOfServiceAlert MAY be delayed by up to 24 hours after the ExpiryDate. Similarly, clients MAY delay any actions based on the ExpiryDate by up to 24 hours to best align with the local time zone.

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2.11.6.13.1. Unmounted Attribute

This attribute SHALL indicate that the hardware is currently not mechanically or electrically mounted in its designated installation position.

When this attribute is set to True, and the sensor cannot operate in this state, the ExpressedState attribute SHALL be set to Inoperative.

2.11.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**SelfTestRequest**	client ⇒ server	Y	O	O

2.11.7.1. SelfTestRequest Command

This command SHALL initiate a device self-test. The return status SHALL indicate whether the test was successfully initiated. Only one SelfTestRequest may be processed at a time. When the value of the ExpressedState attribute is any of SmokeAlarm, COAlarm, Testing, InterconnectSmoke, InterconnectCO, the device SHALL NOT execute the self-test, and SHALL return status code BUSY.

Upon successful acceptance of SelfTestRequest, the TestInProgress attribute SHALL be set to True and ExpressedState attribute SHALL be set to Testing. Any faults identified during the test SHALL be reflected in the appropriate attributes and events. Upon completion of the self test procedure, the SelfTestComplete event SHALL be generated, the TestInProgress attribute SHALL be set to False and ExpressedState attribute SHALL be updated to reflect the current state of the server.

2.11.8. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**SmokeAlarm**	CRITICAL		V	SMOKE
0x01	**COAlarm**	CRITICAL		V	CO
0x02	**LowBattery**	INFO		V	M
0x03	**HardwareFault**	INFO		V	M
0x04	**EndOfService**	INFO		V	M
0x05	**SelfTestComplete**	INFO		V	M
0x06	**AlarmMuted**	INFO		V	O
0x07	**MuteEnded**	INFO		V	O
0x08	**InterconnectSmokeAlarm**	CRITICAL		V	[SMOKE]

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ID	Name	Priority	Quality	Access	Conformance
0x09	**Interconnect-COAlarm**	CRITICAL		V	[CO]
0x0A	**AllClear**	INFO		V	M

2.11.8.1. SmokeAlarm Event

This event SHALL be generated when SmokeState attribute changes to either Warning or Critical state.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**AlarmSeverityLevel**	AlarmStateEnum	all			M

2.11.8.1.1. AlarmSeverityLevel Field

This field SHALL indicate the current value of the SmokeState attribute.

2.11.8.2. COAlarm Event

This event SHALL be generated when COState attribute changes to either Warning or Critical state.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**AlarmSeverityLevel**	AlarmStateEnum	all			M

2.11.8.2.1. AlarmSeverityLevel Field

This field SHALL indicate the current value of the COState attribute.

2.11.8.3. LowBattery Event

This event SHALL be generated when BatteryAlert attribute changes to either Warning or Critical state.

The data of this event SHALL contain the following information:

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**AlarmSeverityLevel**	AlarmStateEnum	all			M

2.11.8.3.1. AlarmSeverityLevel Field

This field SHALL indicate the current value of the BatteryAlert attribute.

2.11.8.4. HardwareFault Event

This event SHALL be generated when the device detects a hardware fault that leads to setting HardwareFaultAlert to True.

2.11.8.5. EndOfService Event

This event SHALL be generated when the EndOfServiceAlert is set to Expired.

2.11.8.6. SelfTestComplete Event

This event SHALL be generated when the SelfTest completes, and the attribute TestInProgress changes to False.

2.11.8.7. AlarmMuted Event

This event SHALL be generated when the DeviceMuted attribute changes to Muted.

2.11.8.8. MuteEnded Event

This event SHALL be generated when DeviceMuted attribute changes to NotMuted.

2.11.8.9. InterconnectSmokeAlarm Event

This event SHALL be generated when the device hosting the server receives a smoke alarm from an interconnected sensor.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**AlarmSeverityLevel**	AlarmStateEnum	all			M

2.11.8.9.1. AlarmSeverityLevel Field

This field SHALL indicate the current value of the InterconnectSmokeAlarm attribute.

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2.11.8.10. InterconnectCOAlarm Event

This event SHALL be generated when the device hosting the server receives a CO alarm from an interconnected sensor.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**AlarmSeverityLevel**	AlarmStateEnum	all			M

2.11.8.10.1. AlarmSeverityLevel Field

This field SHALL indicate the current value of the InterconnectCOAlarm attribute.

2.11.8.11. AllClear Event

This event SHALL be generated when ExpressedState attribute returns to Normal state.

2.12. Electrical Energy Measurement Cluster

This cluster provides a mechanism for querying data about the electrical energy imported or provided by the server.

2.12.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Added support for apparent and reactive energy measurement

2.12.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	EEM

2.12.3. Cluster ID

ID	Name
0x0091	Electrical Energy Measurement

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2.12.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**IMPE**	ImportedEnergy	O.a+	Measurement of energy imported by the server
1	**EXPE**	ExportedEnergy	O.a+	Measurement of energy provided by the server
2	**CUME**	CumulativeEnergy	O.b+	Measurements are cumulative
3	**PERE**	PeriodicEnergy	O.b+	Measurements are periodic
4	**APPE**	ApparentEnergy	P, O	Measurements report apparent energy
5	**REAE**	ReactiveEnergy	P, O	Measurements report reactive energy

2.12.4.1. ImportedEnergy Feature

The feature indicates the server is capable of measuring how much energy is imported by the server.

2.12.4.2. ExportedEnergy Feature

The feature indicates the server is capable of measuring how much energy is exported by the server.

2.12.4.3. CumulativeEnergy Feature

The feature indicates the server is capable of measuring how much energy has been imported or exported by the server over the device's lifetime. This measurement MAY start from when a device's firmware is updated to include this feature, when a device's firmware is updated to correct measurement errors, or when a device is factory reset.

2.12.4.4. PeriodicEnergy Feature

The feature indicates the server is capable of measuring how much energy has been imported or exported by the server during a certain period of time. The start and end times for measurement periods SHALL be determined by the server, and MAY represent overlapping periods.

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2.12.5. Data Types

2.12.5.1. MeasurementTypeEnum Type

This data type is derived from enum16.

Value	Name	Summary	Conformance
0	**Unspecified**		M
1	**Voltage**	Voltage in millivolts (mV)	M
2	**ActiveCurrent**	Active current in milliamps (mA)	M
3	**ReactiveCurrent**	Reactive current in milliamps (mA)	M
4	**ApparentCurrent**	Apparent current in milliamps (mA)	M
5	**ActivePower**	Active power in milliwatts (mW)	M
6	**ReactivePower**	Reactive power in millivolt-amps reactive (mVAR)	M
7	**ApparentPower**	Apparent power in millivolt-amps (mVA)	M
8	**RMSVoltage**	Root mean squared voltage in millivolts (mV)	M
9	**RMSCurrent**	Root mean squared current in milliamps (mA)	M
10	**RMSPower**	Root mean squared power in milliwatts (mW)	M
11	**Frequency**	AC frequency in millihertz (mHz)	M
12	**PowerFactor**	Power Factor ratio in +/- 1/100ths of a percent.	M
13	**NeutralCurrent**	AC neutral current in milliamps (mA)	M
14	**ElectricalEnergy**	Electrical energy in milliwatt-hours (mWh)	M

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Value	Name	Summary	Conformance
15	**ReactiveEnergy**	Reactive power in milli-volt-amp-hours reactive (mVARh)	M
16	**ApparentEnergy**	Apparent power in millivolt-amp-hours (mVAh)	M

2.12.5.1.1. MeasurementAccuracyRangeStruct Type

This struct represents the accuracy of a measurement for a range of measurement values. Accuracy SHALL be expressed as a maximum +/- percentage of the true value, a maximum +/- fixed value of the true value, or both.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**RangeMin**	int64	-2^{62} to 2^{62}				M
1	**RangeMax**	int64	-2^{62} to 2^{62}				M
2	**PercentMax**	percent100ths	all				O.a+
3	**PercentMin**	percent100ths	max PercentTypical				[PercentMax]
4	**PercentTypical**	percent100ths	PercentMin to PercentMax				[PercentMin]
5	**FixedMax**	uint64	max $2^{62} - 1$				O.a+
6	**FixedMin**	uint64	max FixedMax				[FixedMax]
7	**FixedTypical**	uint64	FixedMin to FixedMax				[FixedMin]

- If both PercentMax and FixedMax are indicated, then for a given true value in the range between RangeMin and RangeMax,
  - the reported value SHALL be less than or equal to the sum of the true value, FixedMax and PercentMax percent of the true value.
  - the reported value SHALL be greater than or equal to the true value minus the sum of FixedMax and PercentMax percent of the true value.
- If only PercentMax is indicated, then for a given true value in the range between RangeMin and RangeMax,
  - the reported value SHALL be less than or equal to the sum of the true value and PercentMax

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percent of the true value.

- the reported value SHALL be greater than or equal to the true value minus PercentMax percent of the true value.

- If only FixedMax is indicated, then for a given true value in the range between RangeMin and RangeMax,
  - the reported value SHALL be less than or equal to the sum of the true value and FixedMax.
  - the reported value SHALL be greater than or equal to the true value minus FixedMax.

2.12.5.1.1.1. RangeMin Field

This field SHALL indicate the minimum measurement value for the specified level of accuracy.

The value of this field SHALL be greater than or equal to the value of the MinMeasuredValue field on the encompassing MeasurementAccuracyStruct.

The value of this field SHALL be less than or equal to the value of the MaxMeasuredValue field on the encompassing MeasurementAccuracyStruct.

2.12.5.1.1.2. RangeMax Field

This field SHALL indicate the maximum measurement value for the specified level of accuracy.

The value of this field SHALL be greater than the value of the RangeMin field.

The value of this field SHALL be greater than or equal to the value of the MinMeasuredValue field on the encompassing MeasurementAccuracyStruct.

The value of this field SHALL be less than or equal to the value of the MaxMeasuredValue field on the encompassing MeasurementAccuracyStruct.

2.12.5.1.1.3. PercentMax Field

This field SHALL indicate the maximum +/- percentage accuracy for the associated measurement.

2.12.5.1.1.4. PercentMin Field

This field SHALL indicate the minimum +/- percentage accuracy for the associated measurement.

2.12.5.1.1.5. PercentTypical Field

This field SHALL indicate the typical +/- percentage accuracy for the associated measurement.

2.12.5.1.1.6. FixedMax Field

This field SHALL indicate the maximum +/- fixed accuracy for the associated measurement, in the unit indicated by MeasurementType.

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2.12.5.1.1.7. FixedMin Field

This field SHALL indicate the minimum +/- fixed accuracy for the associated measurement, in the unit indicated by MeasurementType.

2.12.5.1.1.8. FixedTypical Field

This field SHALL indicate the typical +/- fixed accuracy for the associated measurement, in the unit indicated by MeasurementType.

2.12.5.1.2. MeasurementAccuracyStruct Type

This struct represents the set of accuracy ranges for a given measurement, the maximum and minimum values for the measurement, and whether the measurement is directly measured or just estimated from other information.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**MeasurementType**	MeasurementTypeEnum	all				M
1	**Measured**	bool	all				M
2	**MinMeasuredValue**	int64	-2^{62} to 2^{62}				M
3	**MaxMeasuredValue**	int64	-2^{62} to 2^{62}				M
4	**AccuracyRanges**	list[MeasurementAccuracyRangeStruct]	min 1				M

2.12.5.1.2.1. MeasurementType Field

This field SHALL indicate the type of measurement for the accuracy provided.

2.12.5.1.2.2. Measured Field

This field SHALL indicate whether the associated measurement was directly measured. If this field is not set to true, then the associated measurement was estimated.

2.12.5.1.2.3. MinMeasuredValue Attribute

This field SHALL indicate the minimum value that can be measured.

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2.12.5.1.2.4. MaxMeasuredValue Attribute

This field SHALL indicate the maximum value that can be measured.

2.12.5.1.2.5. AccuracyRanges Field

This field SHALL indicate a list of measurement ranges and their associated accuracies.

The value of the RangeMin field on the first MeasurementAccuracyRangeStruct in this list SHALL be equal to the value of the MinMeasuredValue field.

The value of the RangeMax field on the last MeasurementAccuracyRangeStruct in this list SHALL be less than or equal to the value of the MaxMeasuredValue field.

The value of the RangeMin field on each MeasurementAccuracyRangeStruct in this list other than the first SHALL be one more the value of the RangeMax field on the previous MeasurementAccuracyRangeStruct in this list (i.e. there SHALL be no gaps in the accuracy ranges, and the ranges SHALL be in increasing order).

2.12.5.2. EnergyMeasurementStruct Type

This struct SHALL indicate the amount of energy measured during a given measurement period.

A server which does not have the ability to determine the time in UTC, or has not yet done so, SHALL use the system time fields to specify the measurement period and observation times.

A server which has determined the time in UTC SHALL use the timestamp fields to specify the measurement period. Such a server MAY also include the systime fields to indicate how many seconds had passed since boot for a given timestamp; this allows for client-side resolution of UTC time for previous reports that only included systime.

Elements using this data type SHALL indicate whether it represents cumulative or periodic energy, e.g. in the name or in the element description.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Energy**	energy-mWh	0 to 2^{62}				M
1	**StartTimestamp**	epoch-s					desc
2	**EndTimestamp**	epoch-s	min (StartTimestamp + 1)				desc
3	**StartSystime**	systime-ms					desc
4	**EndSystime**	systime-ms	min (StartSystime + 1)				desc

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
5	**ApparentEnergy**	energy-mVAh	0 to 2^{62}				P, APPE
6	**ReactiveEnergy**	energy-mVARh	0 to 2^{62}				P, REAE

2.12.5.2.1. Energy Field

This field SHALL be the reported energy.

If the EnergyMeasurementStruct represents cumulative energy, then this SHALL represent the cumulative energy recorded at either the value of the EndTimestamp field or the value of the EndSystime field, or both.

If the EnergyMeasurementStruct represents periodic energy, then this SHALL represent the energy recorded during the period specified by either the StartTimestamp and EndTimestamp fields, the period specified by the StartSystime and EndSystime fields, or both.

2.12.5.2.2. StartTimestamp Field

This field SHALL indicate the timestamp in UTC of the beginning of the period during which the value of the Energy field was measured.

If this EnergyMeasurementStruct represents cumulative energy, this field SHALL be omitted.

Otherwise, if the server had determined the time in UTC at or before the beginning of the measurement period, this field SHALL be indicated.

Otherwise, if the server had not yet determined the time in UTC at or before the beginning of the measurement period, or does not have the capability of determining the time in UTC, this field SHALL be omitted.

2.12.5.2.3. EndTimestamp Field

This field SHALL indicate the timestamp in UTC of the end of the period during which the value of the Energy field was measured.

If the server had determined the time in UTC by the end of the measurement period, this field SHALL be indicated.

Otherwise, if the server had not yet determined the time in UTC by the end of the measurement period, or does not have the capability of determining the time in UTC, this field SHALL be omitted.

2.12.5.2.4. StartSystime Field

This field SHALL indicate the time elapsed since boot at the beginning of the period during which the value of the Energy field was measured.

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If this EnergyMeasurementStruct represents cumulative energy, this field SHALL be omitted.

Otherwise, if the server had not yet determined the time in UTC at the start of the measurement period, or does not have the capability of determining the time in UTC, this field SHALL be indicated.

Otherwise, if the server had determined the time in UTC at or before the beginning of the measurement period, this field MAY be omitted; if it is indicated, its value SHALL be the time elapsed since boot at the UTC time indicated in StartTimestamp.

2.12.5.2.5. EndSystime Field

This field SHALL indicate the time elapsed since boot at the end of the period during which the value of the Energy field was measured.

If the server had not yet determined the time in UTC by the end of the measurement period, or does not have the capability of determining the time in UTC, this field SHALL be indicated.

Otherwise, if the server had determined the time in UTC by the end of the measurement period, this field MAY be omitted; if it is indicated, its value SHALL be the time elapsed since boot at the UTC time indicated in EndTimestamp.

2.12.5.2.6. ApparentEnergy Field

This field SHALL indicate the reported apparent energy.

If the EnergyMeasurementStruct represents cumulative energy, then this SHALL represent the cumulative apparent energy recorded at either the value of the EndTimestamp field or the value of the EndSystime field, or both.

If the EnergyMeasurementStruct represents periodic energy, then this SHALL represent the apparent energy recorded during the period specified by either the StartTimestamp and EndTimestamp fields, the period specified by the StartSystime and EndSystime fields, or both.

2.12.5.2.7. ReactiveEnergy Field

This field SHALL be the reported reactive energy.

If the EnergyMeasurementStruct represents cumulative energy, then this SHALL represent the cumulative reactive energy recorded at either the value of the EndTimestamp field or the value of the EndSystime field, or both.

If the EnergyMeasurementStruct represents periodic energy, then this SHALL represent the reactive energy recorded during the period specified by either the StartTimestamp and EndTimestamp fields, the period specified by the StartSystime and EndSystime fields, or both.

2.12.5.3. CumulativeEnergyResetStruct Type

This struct SHALL represent the times at which cumulative measurements were last zero, either due to initialization of the device, or an internal reset of the cumulative value.

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**ImportedResetTimestamp**	epoch-s		X	null		[IMPE]
1	**ExportedResetTimestamp**	epoch-s		X	null		[EXPE]
2	**ImportedResetSystime**	systime-ms		X	null		[IMPE]
3	**ExportedResetSystime**	systime-ms		X	null		[EXPE]

2.12.5.3.1. ImportedResetTimestamp Field

This field SHALL indicate the timestamp in UTC when the value of the Energy field on the CumulativeEnergyImported attribute was most recently zero.

If the server had determined the time in UTC when the value of the Energy field on the CumulativeEnergyImported attribute was most recently zero, this field SHALL be indicated.

Otherwise, if the server had not yet determined the time in UTC when the value of the Energy field on the CumulativeEnergyImported attribute was most recently zero, or does not have the capability of determining the time in UTC, this field SHALL be omitted.

If the timestamp in UTC when the value of the Energy field on the CumulativeEnergyImported attribute was most recently zero cannot currently be determined, a value of null SHALL be returned.

2.12.5.3.2. ExportedResetTimestamp Field

This field SHALL indicate the timestamp in UTC when the value of the Energy field on the CumulativeEnergyExported attribute was most recently zero.

If the server had determined the time in UTC when the value of the Energy field on the CumulativeEnergyExported attribute was most recently zero, this field SHALL be indicated.

Otherwise, if the server had not yet determined the time in UTC when the value of the Energy field on the CumulativeEnergyExported attribute was most recently zero, or does not have the capability of determining the time in UTC, this field SHALL be omitted.

If the timestamp in UTC when the value of the Energy field on the CumulativeEnergyExported attribute was most recently zero cannot currently be determined, a value of null SHALL be returned.

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2.12.5.3.3. ImportedResetSystime Field

This field SHALL indicate the time elapsed since boot when the value of the Energy field on the CumulativeEnergyImported attribute was most recently zero.

If the server had not yet determined the time in UTC when the value of the Energy field on the CumulativeEnergyImported attribute was most recently zero, or does not have the capability of determining the time in UTC, this field SHALL be indicated.

Otherwise, if the server had determined the time in UTC when the value of the Energy field on the CumulativeEnergyImported attribute was most recently zero, this field MAY be omitted; if it is indicated, its value SHALL be the time elapsed since boot at the UTC time indicated in ImportedResetTimestamp.

2.12.5.3.4. ExportedResetSystime Field

This field SHALL indicate the time elapsed since boot when the value of the Energy field on the CumulativeEnergyExported attribute was most recently zero.

If the server had not yet determined the time in UTC when the value of the Energy field on the CumulativeEnergyExported attribute was most recently zero, or does not have the capability of determining the time in UTC, this field SHALL be indicated.

Otherwise, if the server had determined the time in UTC when the value of the Energy field on the CumulativeEnergyExported attribute was most recently zero, this field MAY be omitted; if it is indicated, its value SHALL be the time elapsed since boot at the UTC time indicated in ImportedResetTimestamp.

2.12.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**Accuracy**	MeasurementAccuracyStruct		F		R V	M
0x0001	**CumulativeEnergyImported**	EnergyMeasurementStruct		Q X		R V	IMPE & CUME
0x0002	**CumulativeEnergyExported**	EnergyMeasurementStruct		Q X		R V	EXPE & CUME
0x0003	**PeriodicEnergyImported**	EnergyMeasurementStruct		Q X		R V	IMPE & PERE

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0004	**PeriodicEnergyExported**	EnergyMeasurementStruct		Q X		R V	EXPE & PERE
0x0005	**CumulativeEnergyReset**	CumulativeEnergyResetStruct		X	null	R V	[CUME]

2.12.6.1. Accuracy Attribute

This attribute SHALL indicate the accuracy of energy measurement by this server. The value of the MeasurementType field on this MeasurementAccuracyStruct SHALL be ElectricalEnergy.

2.12.6.2. CumulativeEnergyImported Attribute

This attribute SHALL indicate the most recent measurement of cumulative energy imported by the server over the lifetime of the device, and the timestamp of when the measurement was recorded.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

If the cumulative energy imported cannot currently be determined, a value of null SHALL be returned.

2.12.6.3. CumulativeEnergyExported Attribute

This attribute SHALL indicate the most recent measurement of cumulative energy exported by the server over the lifetime of the device, and the timestamp of when the measurement was recorded.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

If the cumulative energy exported cannot currently be determined, a value of null SHALL be returned.

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2.12.6.4. PeriodicEnergyImported Attribute

This attribute SHALL indicate the most recent measurement of energy imported by the server and the period during which it was measured.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

If the periodic energy imported cannot currently be determined, a value of null SHALL be returned.

2.12.6.5. PeriodicEnergyExported Attribute

This attribute SHALL indicate the most recent measurement of energy exported by the server and the period during which it was measured.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

If the periodic energy exported cannot currently be determined, a value of null SHALL be returned.

2.12.6.6. CumulativeEnergyReset Attribute

This attribute SHALL indicate when cumulative measurements were most recently zero.

2.12.7. Events

This cluster SHALL support these events:

ID	Name	Priority	Quality	Access	Conformance
0	**CumulativeEnergyMeasured**	INFO		V	CUME
1	**PeriodicEnergyMeasured**	INFO		V	PERE

2.12.7.1. CumulativeEnergyMeasured Event

This event SHALL be generated when the server takes a snapshot of the cumulative energy

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imported by the server, exported from the server, or both, but not more frequently than the rate mentioned in the description above of the related attribute.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**EnergyImported**	EnergyMeasurementStruct				CUME & IMPE
1	**EnergyExported**	EnergyMeasurementStruct				CUME & EXPE

2.12.7.1.1. EnergyImported Field

This field SHALL be the value of CumulativeEnergyImported attribute at the timestamp indicated in its EndTimestamp field, EndSystime field, or both.

2.12.7.1.2. EnergyExported Field

This field SHALL be the value of CumulativeEnergyExported attribute at the timestamp indicated in its EndTimestamp field, EndSystime field, or both.

2.12.7.2. PeriodicEnergyMeasured Event

This event SHALL be generated when the server reaches the end of a reporting period for imported energy, exported energy, or both.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**EnergyImported**	EnergyMeasurementStruct				PERE & IMPE
1	**EnergyExported**	EnergyMeasurementStruct				PERE & EXPE

2.12.7.2.1. EnergyImported Field

This field SHALL be the value of PeriodicEnergyImported attribute at the timestamp indicated in its EndTimestamp field, EndSystime field, or both.

2.12.7.2.2. EnergyExported Field

This field SHALL be the value of PeriodicEnergyExported attribute at the timestamp indicated in its

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EndTimestamp field, EndSystime field, or both.

2.13. Electrical Power Measurement Cluster

This cluster provides a mechanism for querying data about electrical power as measured by the server.

2.13.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Changed reactive and apparent power fields to use new data types
3	Changed range of apparent current field to allow negative apparent current

2.13.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	EPM

2.13.3. Cluster ID

ID	Name
0x0090	Electrical Power Measurement

2.13.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**DIRC**	DirectCurrent	O.a+	Supports measurement of direct current
1	**ALTC**	AlternatingCurrent	O.a+	Supports measurement of alternating current
2	**POLY**	PolyphasePower	[ALTC]	Supports polyphase measurements

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Bit	Code	Feature	Conformance	Summary
3	**HARM**	Harmonics	[ALTC]	Supports measurement of AC harmonics
4	**PWRQ**	PowerQuality	[ALTC]	Supports measurement of AC harmonic phases

2.13.4.1. DirectCurrent Feature

This feature indicates the cluster can measure a direct current.

2.13.4.2. AlternatingCurrent Feature

This feature indicates the cluster can measure an alternating current.

2.13.4.3. PolyphasePower Feature

This feature indicates the cluster represents the collective measurements for a Polyphase power supply.

2.13.4.4. Harmonics Feature

This feature indicates the cluster can measure the harmonics of an alternating current.

2.13.4.5. PowerQuality Feature

This feature indicates the cluster can measure the harmonic phases of an alternating current.

2.13.5. Data Types

2.13.5.1. PowerModeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Unknown**		M
1	**DC**	Direct current	M
2	**AC**	Alternating current, either single-phase or polyphase	M

2.13.5.2. MeasurementTypeEnum Type

This data type is derived from enum16.

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Value	Name	Summary	Conformance
0	**Unspecified**		M
1	**Voltage**	Voltage in millivolts (mV)	M
2	**ActiveCurrent**	Active current in milliamps (mA)	M
3	**ReactiveCurrent**	Reactive current in milliamps (mA)	M
4	**ApparentCurrent**	Apparent current in milliamps (mA)	M
5	**ActivePower**	Active power in milliwatts (mW)	M
6	**ReactivePower**	Reactive power in millivolt-amps reactive (mVAR)	M
7	**ApparentPower**	Apparent power in millivolt-amps (mVA)	M
8	**RMSVoltage**	Root mean squared voltage in millivolts (mV)	M
9	**RMSCurrent**	Root mean squared current in milliamps (mA)	M
10	**RMSPower**	Root mean squared power in milliwatts (mW)	M
11	**Frequency**	AC frequency in millihertz (mHz)	M
12	**PowerFactor**	Power Factor ratio in +/- 1/100ths of a percent.	M
13	**NeutralCurrent**	AC neutral current in milliamps (mA)	M
14	**ElectricalEnergy**	Electrical energy in milliwatt-hours (mWh)	M
15	**ReactiveEnergy**	Reactive power in millivolt-amp-hours reactive (mVARh)	M
16	**ApparentEnergy**	Apparent power in millivolt-amp-hours (mVAh)	M

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2.13.5.2.1. MeasurementAccuracyRangeStruct Type

This struct represents the accuracy of a measurement for a range of measurement values. Accuracy SHALL be expressed as a maximum +/- percentage of the true value, a maximum +/- fixed value of the true value, or both.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**RangeMin**	int64	-2^{62} to 2^{62}				M
1	**RangeMax**	int64	-2^{62} to 2^{62}				M
2	**PercentMax**	percent100ths	all				O.a+
3	**PercentMin**	percent100ths	max PercentTypical				[PercentMax]
4	**PercentTypical**	percent100ths	PercentMin to PercentMax				[PercentMin]
5	**FixedMax**	uint64	max 2^{62} - 1				O.a+
6	**FixedMin**	uint64	max FixedMax				[FixedMax]
7	**FixedTypical**	uint64	FixedMin to FixedMax				[FixedMin]

- If both PercentMax and FixedMax are indicated, then for a given true value in the range between RangeMin and RangeMax,
  - the reported value SHALL be less than or equal to the sum of the true value, FixedMax and PercentMax percent of the true value.
  - the reported value SHALL be greater than or equal to the true value minus the sum of FixedMax and PercentMax percent of the true value.
- If only PercentMax is indicated, then for a given true value in the range between RangeMin and RangeMax,
  - the reported value SHALL be less than or equal to the sum of the true value and PercentMax percent of the true value.
  - the reported value SHALL be greater than or equal to the true value minus PercentMax percent of the true value.
- If only FixedMax is indicated, then for a given true value in the range between RangeMin and RangeMax,
  - the reported value SHALL be less than or equal to the sum of the true value and FixedMax.
  - the reported value SHALL be greater than or equal to the true value minus FixedMax.

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2.13.5.2.1.1. RangeMin Field

This field SHALL indicate the minimum measurement value for the specified level of accuracy.

The value of this field SHALL be greater than or equal to the value of the MinMeasuredValue field on the encompassing MeasurementAccuracyStruct.

The value of this field SHALL be less than or equal to the value of the MaxMeasuredValue field on the encompassing MeasurementAccuracyStruct.

2.13.5.2.1.2. RangeMax Field

This field SHALL indicate the maximum measurement value for the specified level of accuracy.

The value of this field SHALL be greater than the value of the RangeMin field.

The value of this field SHALL be greater than or equal to the value of the MinMeasuredValue field on the encompassing MeasurementAccuracyStruct.

The value of this field SHALL be less than or equal to the value of the MaxMeasuredValue field on the encompassing MeasurementAccuracyStruct.

2.13.5.2.1.3. PercentMax Field

This field SHALL indicate the maximum +/- percentage accuracy for the associated measurement.

2.13.5.2.1.4. PercentMin Field

This field SHALL indicate the minimum +/- percentage accuracy for the associated measurement.

2.13.5.2.1.5. PercentTypical Field

This field SHALL indicate the typical +/- percentage accuracy for the associated measurement.

2.13.5.2.1.6. FixedMax Field

This field SHALL indicate the maximum +/- fixed accuracy for the associated measurement, in the unit indicated by MeasurementType.

2.13.5.2.1.7. FixedMin Field

This field SHALL indicate the minimum +/- fixed accuracy for the associated measurement, in the unit indicated by MeasurementType.

2.13.5.2.1.8. FixedTypical Field

This field SHALL indicate the typical +/- fixed accuracy for the associated measurement, in the unit indicated by MeasurementType.

2.13.5.2.2. MeasurementAccuracyStruct Type

This struct represents the set of accuracy ranges for a given measurement, the maximum and mini-

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mum values for the measurement, and whether the measurement is directly measured or just estimated from other information.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**MeasurementType**	MeasurementTypeEnum	all				M
1	**Measured**	bool	all				M
2	**MinMeasuredValue**	int64	-2^{62} to 2^{62}				M
3	**MaxMeasuredValue**	int64	-2^{62} to 2^{62}				M
4	**AccuracyRanges**	list[MeasurementAccuracyRangeStruct]	min 1				M

2.13.5.2.2.1. MeasurementType Field

This field SHALL indicate the type of measurement for the accuracy provided.

2.13.5.2.2.2. Measured Field

This field SHALL indicate whether the associated measurement was directly measured. If this field is not set to true, then the associated measurement was estimated.

2.13.5.2.2.3. MinMeasuredValue Attribute

This field SHALL indicate the minimum value that can be measured.

2.13.5.2.2.4. MaxMeasuredValue Attribute

This field SHALL indicate the maximum value that can be measured.

2.13.5.2.2.5. AccuracyRanges Field

This field SHALL indicate a list of measurement ranges and their associated accuracies.

The value of the RangeMin field on the first MeasurementAccuracyRangeStruct in this list SHALL be equal to the value of the MinMeasuredValue field.

The value of the RangeMax field on the last MeasurementAccuracyRangeStruct in this list SHALL be less than or equal to the value of the MaxMeasuredValue field.

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The value of the RangeMin field on each MeasurementAccuracyRangeStruct in this list other than the first SHALL be one more the value of the RangeMax field on the previous MeasurementAccuracyRangeStruct in this list (i.e. there SHALL be no gaps in the accuracy ranges, and the ranges SHALL be in increasing order).

2.13.5.3. MeasurementRangeStruct Type

This struct SHALL indicate the maximum and minimum values of a given measurement type during a measurement period, along with the observation times of these values.

A server which does not have the ability to determine the time in UTC, or has not yet done so, SHALL use the system time fields to specify the measurement period and observation times.

A server which has determined the time in UTC SHALL use the timestamp fields to specify the measurement period and observation times. Such a server MAY also include the systime fields to indicate how many seconds had passed since boot for a given timestamp; this allows for client-side resolution of UTC time for previous reports that only included systime.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**MeasurementType**	MeasurementTypeEnum					M
1	**Min**	int64	-2^{62} to 2^{62}				M
2	**Max**	int64	-2^{62} to 2^{62}				M
3	**StartTimestamp**	epoch-s					EndTimestamp
4	**EndTimestamp**	epoch-s	min (StartTimestamp + 1)				desc
5	**MinTimestamp**	epoch-s					EndTimestamp
6	**MaxTimestamp**	epoch-s	min (MinTimestamp + 1)				EndTimestamp
7	**StartSystime**	systime-ms					EndSystime
8	**EndSystime**	systime-ms	min (StartSystime + 1)				desc
9	**MinSystime**	systime-ms					EndSystime

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
10	**MaxSystime**	systime-ms	min (MinSystime + 1)				EndSystime

2.13.5.3.1. MeasurementType Field

This field SHALL be the type of measurement for the range provided.

2.13.5.3.2. Min Field

This field SHALL be the smallest measured value for the associated measurement over either the period between StartTimestamp and EndTimestamp, or the period between StartSystime and EndSystime, or both.

2.13.5.3.3. Max Field

This field SHALL be the largest measured value for the associated measurement over the period between either StartTimestamp and EndTimestamp or the period between StartSystime and EndSystime, or both.

2.13.5.3.4. StartTimestamp Field

This field SHALL be the timestamp in UTC of the beginning of the measurement period.

If the server had not yet determined the time in UTC at or before the beginning of the measurement period, or does not have the capability of determining the time in UTC, this field SHALL be omitted.

2.13.5.3.5. EndTimestamp Field

This field SHALL be the timestamp in UTC of the end of the measurement period.

If the server had not yet determined the time in UTC at or before the beginning of the measurement period, or does not have the capability of determining the time in UTC, this field SHALL be omitted.

2.13.5.3.6. MinTimestamp Field

This field SHALL be the most recent timestamp in UTC that the value in the Min field was measured.

This field SHALL be greater than or equal to the value of the StartTimestamp field.

This field SHALL be less than or equal to the value of the EndTimestamp field.

2.13.5.3.7. MaxTimestamp Field

This field SHALL be the most recent timestamp in UTC of the value in the Max field.

This field SHALL be greater than or equal to the value of the StartTimestamp field.

This field SHALL be less than or equal to the value of the EndTimestamp field.

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2.13.5.3.8. StartSystime Field

This field SHALL be the time since boot of the beginning of the measurement period.

If the server had determined the time in UTC at or before the start of the measurement period, this field MAY be omitted along with the EndSystime, MinSystime, and MaxSystime fields.

2.13.5.3.9. EndSystime Field

This field SHALL be the time since boot of the end of the measurement period.

If the server had determined the time in UTC at the end of the measurement period, this field MAY be omitted along with the StartSystime field, MinSystime, and MaxSystime fields.

2.13.5.3.10. MinSystime Field

This field SHALL be the measurement time since boot of the value in the Min field was measured.

This field SHALL be greater than or equal to the value of the StartSystime field.

This field SHALL be less than or equal to the value of the EndSystime field.

2.13.5.3.11. MaxSystime Field

This field SHALL be the measurement time since boot of the value in the Max field.

This field SHALL be greater than or equal to the value of the StartSystime field.

This field SHALL be less than or equal to the value of the EndSystime field.

2.13.5.4. HarmonicMeasurementStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Order**	uint8	min 1				M
1	**Measurement**	int64	-2^{62} to 2^{62}	X			M

2.13.5.4.1. Order Field

This field SHALL be the order of the harmonic being measured. Typically this is an odd number, but servers may choose to report even harmonics.

2.13.5.4.2. Measurement Field

This field SHALL be the measured value for the given harmonic order.

For the Harmonic Currents attribute, this value is the most recently measured harmonic current reading in milliamps (mA). A positive value indicates that the measured harmonic current is positive, and a negative value indicates that the measured harmonic current is negative.

For the Harmonic Phases attribute, this value is the most recent phase of the given harmonic order

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in millidegrees (mDeg). A positive value indicates that the measured phase is leading, and a negative value indicates that the measured phase is lagging.

If this measurement is not currently available, a value of null SHALL be returned.

2.13.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**PowerMode**	PowerModeEnum				R V	M
0x0001	**NumberOfMeasurementTypes**	uint8	max 32	F		R V	M
0x0002	**Accuracy**	list[MeasurementAccuracyStruct]	1 to NumberOfMeasurementTypes	F		R V	M
0x0003	**Ranges**	list[MeasurementRangeStruct]	0 to NumberOfMeasurementTypes	Q	empty	R V	O
0x0004	**Voltage**	voltage-mV	-2^{62} to 2^{62}	Q X	null	R V	O
0x0005	**ActiveCurrent**	amperage-mA	-2^{62} to 2^{62}	Q X	null	R V	O
0x0006	**ReactiveCurrent**	amperage-mA	-2^{62} to 2^{62}	Q X	null	R V	[ALTC]
0x0007	**ApparentCurrent**	amperage-mA	-2^{62} to 2^{62}	Q X	null	R V	[ALTC]
0x0008	**ActivePower**	power-mW	-2^{62} to 2^{62}	Q X		R V	M
0x0009	**ReactivePower**	power-mVAR	-2^{62} to 2^{62}	Q X	null	R V	[ALTC]
0x000A	**ApparentPower**	power-mVA	-2^{62} to 2^{62}	Q X	null	R V	[ALTC]
0x000B	**RMSVoltage**	voltage-mV	-2^{62} to 2^{62}	Q X	null	R V	[ALTC]
0x000C	**RMSCurrent**	amperage-mA	-2^{62} to 2^{62}	Q X	null	R V	[ALTC]

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x000D	**RMSPower**	power-mW	-2^{62} to 2^{62}	Q X	null	R V	[ALTC]
0x000E	**Frequency**	int64	0 to 1000000	Q X	null	R V	[ALTC]
0x000F	**HarmonicCurrents**	list[HarmonicMeasurementStruct]	max 25	Q X		R V	HARM
0x0010	**HarmonicPhases**	list[HarmonicMeasurementStruct]	max 25	Q X		R V	PWRQ
0x0011	**PowerFactor**	int64	-10000 to 10000	Q X	null	R V	[ALTC]
0x0012	**NeutralCurrent**	amperage-mA	-2^{62} to 2^{62}	Q X	null	R V	[POLY]

2.13.6.1. PowerMode Attribute

This SHALL indicate the current mode of the server. For some servers, such as an EV, this may change depending on the mode of charging or discharging.

2.13.6.2. NumberOfMeasurementTypes Attribute

This SHALL indicate the maximum number of measurement types the server is capable of reporting.

2.13.6.3. Accuracy Attribute

This SHALL indicate a list of accuracy specifications for the measurement types supported by the server. There SHALL be an entry for ActivePower, as well as any other measurement types implemented by this server.

2.13.6.4. Ranges Attribute

This SHALL indicate a list of measured ranges for different measurement types. Each measurement type SHALL have at most one entry in this list, representing the range of measurements in the most recent measurement period.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

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The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

2.13.6.5. Voltage Attribute

This SHALL indicate the most recent Voltage reading in millivolts (mV).

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

If the voltage cannot be measured, a value of null SHALL be returned.

2.13.6.6. ActiveCurrent Attribute

This SHALL indicate the most recent ActiveCurrent reading in milliamps (mA).

A positive value represents current flowing into the server, while a negative value represents current flowing out of the server.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

If the current cannot be measured, a value of null SHALL be returned.

2.13.6.7. ReactiveCurrent Attribute

This SHALL indicate the most recent ReactiveCurrent reading in milliamps (mA).

A positive value represents current flowing into the server, while a negative value represents current flowing out of the server.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

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If the current cannot be measured, a value of null SHALL be returned.

2.13.6.8. ApparentCurrent Attribute

This SHALL indicate the most recent ApparentCurrent (square root sum of the squares of active and reactive currents) reading in milliamps (mA).

A positive value represents current flowing into the server, while a negative value represents current flowing out of the server.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

If the active or reactive currents cannot be measured, a value of null SHALL be returned.

2.13.6.9. ActivePower Attribute

This SHALL indicate the most recent ActivePower reading in milliwatts (mW). If the power cannot be measured, a value of null SHALL be returned.

A positive value represents power imported, while a negative value represents power exported.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

If the Polyphase Power feature is set, this value represents the combined active power imported or exported.

2.13.6.10. ReactivePower Attribute

This SHALL indicate the most recent ReactivePower reading in millivolt-amps reactive (mVAR).

A positive value represents power imported, while a negative value represents power exported.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

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The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

If the reactive power cannot be measured, a value of null SHALL be returned.

If the Polyphase Power feature is supported, this value represents the combined reactive power imported or exported.

2.13.6.11. ApparentPower Attribute

This SHALL indicate the most recent ApparentPower reading in millivolt-amps (mVA).

A positive value represents power imported, while a negative value represents power exported.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

If the apparent power cannot be measured, a value of null SHALL be returned.

2.13.6.12. RMSVoltage Attribute

This SHALL indicate the most recent RMSVoltage reading in millivolts (mV).

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

If the RMS voltage cannot be measured, a value of null SHALL be returned.

2.13.6.13. RMSCurrent Attribute

This SHALL indicate the most recent RMSCurrent reading in milliamps (mA).

A positive value represents current flowing into the server, while a negative value represents current flowing out of the server.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

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The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

If the RMS current cannot be measured, a value of null SHALL be returned.

2.13.6.14. RMSPower Attribute

This SHALL indicate the most recent RMSPower reading in milliwatts (mW).

A positive value represents power imported, while a negative value represents power exported.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

If the RMS power cannot be measured, a value of null SHALL be returned.

2.13.6.15. Frequency Attribute

This SHALL indicate the most recent Frequency reading in millihertz (mHz).

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

If the frequency cannot be measured, a value of null SHALL be returned.

2.13.6.16. HarmonicCurrents Attribute

This SHALL indicate a list of HarmonicMeasurementStruct values, with each HarmonicMeasurementStruct representing the harmonic current reading for the harmonic order specified by Order.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

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2.13.6.17. HarmonicPhases Attribute

This SHALL indicate a list of HarmonicMeasurementStruct values, with each HarmonicMeasurementStruct representing the most recent phase of the harmonic current reading for the harmonic order specified by Order.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

2.13.6.18. PowerFactor Attribute

This SHALL indicate the Power Factor ratio in +/- 1/100ths of a percent.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

2.13.6.19. NeutralCurrent Attribute

This SHALL indicate the most recent NeutralCurrent reading in milliamps (mA). Typically this is a derived value, taking the magnitude of the vector sum of phase currents.

If the neutral current cannot be measured or derived, a value of null SHALL be returned.

A positive value represents an imbalance between the phase currents when power is imported.

A negative value represents an imbalance between the phase currents when power is exported.

The reporting interval of this attribute SHALL be manufacturer dependent. The server MAY choose to omit publication of deltas considered not meaningful.

The server SHALL NOT mark this attribute ready for report if the last time this was done was more recently than 1 second ago.

The server MAY delay marking this attribute ready for report for longer periods if needed, however the server SHALL NOT delay marking this attribute as ready for report for longer than 60 seconds.

2.13.7. Events

This cluster SHALL support the following event:

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ID	Name	Priority	Quality	Access	Conformance
0	**MeasurementPeriodRanges**	INFO		V	Ranges

2.13.7.1. MeasurementPeriodRanges Event

If supported, this event SHALL be generated at the end of a measurement period. The start and end times for measurement periods SHALL be determined by the server, and MAY represent overlapping periods.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Ranges**	list[MeasurementRangeStruct]	1 to NumberOfMeasurementTypes			M

2.13.7.1.1. Ranges Field

This SHALL indicate the value of the Ranges attribute at the time of event generation.

2.14. Zone Management Cluster

This cluster provides an interface to manage regions of interest, or Zones, which can be either manufacturer or user defined.

This cluster also defines a Trigger, which is a set of conditions and timing that apply to a Zone and allow for events to be generated or the triggering state to be used by other clusters such as Push AV Stream Transport Cluster.

2.14.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial version of the Zone Management cluster

2.14.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	ZONEMGMT

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2.14.3. Cluster ID

ID	Name
0x0550	Zone Management

2.14.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**TWODCART**	TwoDimensional-CartesianZone	O	Supports Two Dimensional Cartesian Zones
1	**PERZONESENS**	PerZoneSensitivity	[TWODCART]	Supports a sensitivity value per Zone
2	**USERDEFINED**	UserDefined	[TWODCART]	Supports user defined zones
3	**FOCUSZONES**	FocusZones	[USERDEFINED]	Supports user defined focus zones

2.14.4.1. TwoDimensionalCartesianZone Feature

When this feature is supported, Zones based on a 2 Dimensional Cartesian plane MAY be defined and SHALL be represented by a TwoDCartesianZoneStruct. Within a TwoDCartesianZoneStruct the bounding of the zone SHALL be a polygon defined by a list of vertices comprising X (horizontal) and Y (vertical) coordinates, with each vertex defining the point where adjacent edges meet and an implicit connection between the last and first vertices in the list.

The origin (0,0) SHALL be located at the top left of the Cartesian plane, with positive X and Y values moving right and down across the Cartesian plane respectively.

2.14.4.2. PerZoneSensitivity Feature

When this feature is supported, the ZoneTriggerControlStruct SHALL be used for specifying a zone specific value for the sensitivity of that zone to trigger events. If not supported, only the Sensitivity Attribute SHALL be used.

2.14.4.3. UserDefined Feature

When this feature is supported, the device allows for creating and managing user defined zones via commands.

2.14.4.4. FocusZones Feature

When this feature is supported, the device allows for creating and managing user defined Focus

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Value zones via commands.

2.14.5. Data Types

2.14.5.1. ZoneTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**TwoDCARTZone**	Indicates a Two Dimensional Cartesian Zone	M

2.14.5.2. ZoneUseEnum Type

This data type is derived from enum8, and is used to indicate intended Zone usage.

Value	Name	Summary	Conformance
0	**Motion**	Indicates Zone is intended to detect Motion	M
1	**Privacy**	Indicates Zone is intended to protect privacy	M
2	**Focus**	Indicates Zone provides a focus area	FOCUSZONES

2.14.5.2.1. Motion Value

This value indicates the Zone is intended to be used for motion detection

2.14.5.2.2. Privacy Value

This value indicates the Zone is intended to be used for privacy blocking. All pixels within Privacy Zones SHALL be replaced with black.

2.14.5.2.3. Focus Value

This value indicates the Zone is intended to be a focal point for quality or analysis. Implementations MAY increase encoding quality within this type of Zone at the expense of other areas.

2.14.5.3. ZoneSourceEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Mfg**	Indicates a Manufacturer defined Zone.	M

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Value	Name	Summary	Conformance
1	**User**	Indicates a User defined Zone.	M

2.14.5.3.1. Mfg Value

This value indicates the Zone is built-in and provided by the manufacturer of the device. Zones of this type can't be created or modified using commands in this cluster.

2.14.5.3.2. User Value

This value indicates the Zone was defined and created by a user. Zones of this type can be created, modified or deleted using commands in this cluster.

2.14.5.4. ZoneEventTriggeredReasonEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Motion**	Zone event triggered because motion is detected	M

2.14.5.5. ZoneEventStoppedReasonEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**ActionStopped**	Indicates that whatever triggered the Zone event has stopped being detected.	M
1	**Timeout**	Indicates that the max duration for detecting triggering activity has been reached.	M

2.14.5.6. TwoDCartesianVertexStruct Type

This struct is used to encode a point on the 2 Dimensional Cartesian Plane for the TwoDCartesian-Zone feature.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**X**	uint16	all				M
1	**Y**	uint16	all				M

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2.14.5.6.1. X Field

This field SHALL represent the position of the vertex along the horizontal (x) axis.

2.14.5.6.2. Y Field

This field SHALL represent the position of the vertex along the vertical (y) axis.

2.14.5.7. TwoDCartesianZoneStruct Type

This struct is used to encode all information needed to define a TwoDCartesianZone.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Name**	string	max 32				M
1	**Use**	ZoneUseEnum	all				M
2	**Vertices**	list[TwoDCartesianVertexStruct]	3 to 12				M
3	**Color**	string	7, 9				O

2.14.5.7.1. Name Field

The Name field SHALL be a string representing the name of the Zone. This is not guaranteed to be unique.

2.14.5.7.2. Use Field

The Use field SHALL be a Zone Use Enum representing the purpose of the Zone.

2.14.5.7.3. Vertices Field

The Vertices field SHALL be a list of vertices of type TwoDCartesianVertexStruct. These vertices define a simple polygon on the TwoD Cartesian plane, which represents the bounds of the TwoD Cartesian Zone with an implicit connection between the last and first list items.

2.14.5.7.4. Color Field

This field SHALL indicate the color, in RGB or RGBA, used for attaching a color to the Zone definition and is a purely informational value to help in uniformly presenting Zones in User Interfaces and MAY be ignored. The value SHALL conform to the 6-digit or 8-digit format defined for CSS sRGB hexadecimal color notation [https://www.w3.org/TR/css-color-4/#hex-notation]. If a 6-digit format is used, then the alpha component SHALL assume the value of 0 meaning fully transparent interior.

Examples:

- #00FFFF for R=0x00, G=0xFF, B=0xFF, A absent - For a light-blue zone with full transparency.

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- #00FFFF80 for R=0x00, G=0xFF, B=0xFF, A=0x80 - For a light-blue zone with partial interior transparency.
- #000000FF for R=0x00, G=0x00, B=0x00, A=0xFF - For a Privacy type zone that is black and fully opaque interior.

2.14.5.8. ZoneInformationStruct Type

This struct is used to encode basic information about a Zone without containing the specifics of how the zone is defined.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**ZoneID**	ZoneID	all				M
1	**ZoneType**	ZoneType-Enum	all				M
2	**ZoneSource**	ZoneSourceEnum	all				M
3	**TwoDCartesianZone**	TwoDCartesianZoneStruct	all				TWODCART & (ZoneType == TwoDCARTZone)

2.14.5.8.1. ZoneID Field

This field SHALL indicate the unique ZoneID of the Zone.

2.14.5.8.2. ZoneType Field

This field SHALL indicate the zone type which defines the Zone.

2.14.5.8.3. ZoneSource Field

This field SHALL indicate the source of the Zone.

2.14.5.8.4. TwoDCartesianZone Field

This field SHALL indicate the detailed information for the TwoDCartesianZone.

2.14.5.9. ZoneTriggerControlStruct Type

This struct is used to encode a set of values for controlling the generation of ZoneTriggered and ZoneStopped events from the Node.

Zone events can be triggered due to many underlying reasons, such as a motion sensor on the device, and this is intended to be manufacturer-specific. When a triggering activity is initially detected, the Node SHALL generate a ZoneTriggered event.

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This places the Node in a triggered state, at which point the Node SHALL internally track two values.

TimeSinceInitialTrigger

The time in seconds since the initial triggering activity.

TriggerDetectedDuration

Initially set to the InitialDuration value.

If the TriggerDetectedDuration value is exceeded by the TimeSinceInitialTrigger, the Node SHALL generate a ZoneStopped event with the reason parameter set to ActionStopped.

However, if additional triggering actions are detected during this period, the Node SHALL increase the TriggerDetectedDuration value by the AugmentationDuration value. This process can occur repeatedly but after the first increase of TriggerDetectedDuration the Node SHALL NOT increase the TriggerDetectedDuration value unless the previous TriggerDetectedDuration has been exceeded by the TimeSinceInitialTrigger.

If the TimeSinceInitialTrigger value exceeds the MaxDuration value, the Node SHALL generate a ZoneStopped with the reason parameter set to Timeout.

Once a ZoneStopped event has been generated, the Node SHALL stop detecting the triggering activity for the period of the BlindDuration value.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**ZoneID**	ZoneID	all				M
1	**InitialDuration**	elapsed-s	1 to 65535				M
2	**AugmentationDuration**	elapsed-s	max InitialDuration				M
3	**MaxDuration**	elapsed-s	min InitialDuration				M
4	**BlindDuration**	elapsed-s	all				M
5	**Sensitivity**	uint8	1 to SensitivityMax				PERZONESENS

2.14.5.9.1. ZoneID Field

This field SHALL indicate the unique ZoneID of the Zone this Trigger applies to.

2.14.5.9.2. InitialDuration Field

This field SHALL indicate the initial duration in seconds after triggering activity is first detected before the Node could generate a ZoneStopped event.

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2.14.5.9.3. AugmentationDuration Field

This field SHALL indicate the duration in seconds that the TriggerDetectedDuration value is to be extended by if the triggering activity is still detected during this period.

2.14.5.9.4. MaxDuration Field

This field SHALL indicate the maximum duration in seconds after the initial triggering activity detection that additional triggering activity will be detected.

2.14.5.9.5. BlindDuration Field

This field SHALL indicate the duration in seconds after a ZoneStopped event is generated that the Node SHALL NOT generate any ZoneTriggered events.

2.14.5.9.6. Sensitivity Field

This field SHALL indicate the per-zone sensitivity of the underlying zone triggering detection mechanism. The higher the value, the more sensitive the detection. The actual meaning of the values is implementation-specific.

2.14.5.10. ZoneID Type

This data type is derived from uint16.

It represents a defined Zone. This value starts at 0 and monotonically increases by 1 with each new Zone provisioned by the Node or permanently created by the device maker. A value incremented past 65534 SHALL wrap to 0. The Node SHALL verify that the incremented ID does not match any other ID. If such a match is found, the ID SHALL be incremented until a unique ID is found.

A Node SHALL only store a single Zone for a particular ZoneID and that ID SHALL NOT change over the lifetime of the Zone.

2.14.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**MaxUserDefinedZones**	uint8	min 5	F		R V	USERDEFINED
0x0001	**MaxZones**	uint8	min 1	F		R V	M
0x0002	**Zones**	list[ZoneInformationStruct]	0 to MaxZones	N		R V	M
0x0003	**Triggers**	list[ZoneTriggerControlStruct]	0 to MaxZones	N		R V	M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0004	**SensitivityMax**	uint8	2 to 10	F		R V	M
0x0005	**Sensitivity**	uint8	1 to SensitivityMax	N		RW VO	!PERZONESENS
0x0006	**TwoDCartesianMax**	TwoDCartesianVertexStruct		F		R V	TWODCART

2.14.6.1. MaxUserDefinedZones Attribute

This attribute SHALL specify the maximum number of user-defined zones that can be supported by the Node. This value is manufacturer-defined.

2.14.6.2. MaxZones Attribute

This attribute SHALL specify the maximum number of zones allowed to created. This value SHALL be the sum of the number of predefined Mfg Zones, and MaxUserDefinedZones, if supported. This value is manufacturer-defined.

2.14.6.3. Zones Attribute

This attribute SHALL specify all currently defined zones as a list of ZoneInformationStruct. Use the commands from this cluster to add, update or remove entries.

2.14.6.4. Triggers Attribute

This attribute SHALL specify all currently defined triggers controlling the generation of ZoneTriggered and ZoneStopped events and SHALL be a list of ZoneTriggerControlStruct. To add an entry use CreateOrUpdateTrigger. To remove an entry use RemoveTrigger.

2.14.6.5. SensitivityMax Attribute

This attribute SHALL specify the hardware specific value for the number of supported sensitivity levels. This value is manufacturer defined. If the PerZoneSensitivity feature is supported, the value of this attribute determines valid values for the Sensitivity field in ZoneTriggerControlStruct; if the PerZoneSensitivity feature is not supported, the value of this attribute determines valid values for the Sensitivity Attribute. Implementations require two to ten levels of sensitivity control in order to ensure that there is some user-level customization of the Trigger.

2.14.6.6. Sensitivity Attribute

This attribute SHALL specify the sensitivity of the underlying zone triggering detection mechanism if the PerZoneSensitivity features is not supported. The higher the value the more sensitive the detection. The actual meaning of the values is implementation specific.

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2.14.6.7. TwoDCartesianMax Attribute

This attribute SHALL specify the maximum X and Y points that are allowed for TwoD Cartesian Zones. If this cluster is on the same endpoint as Camera AV Stream Management Cluster, these values SHALL be equal to the value of `SensorWidth - 1` and `SensorHeight - 1` from the VideoSensorParams attribute.

2.14.7. Commands

ID	Name	Direction	Response	Access	Quality	Conformance
0x00	**CreateTwoDCartesianZone**	client ⇒ server	CreateTwoDCartesianZoneResponse	M		TWODCART & USERDEFINED
0x01	**CreateTwoDCartesianZoneResponse**	client ⇐ server	N			TWODCART & USERDEFINED
0x02	**UpdateTwoDCartesianZone**	client ⇒ server	Y	M		TWODCART & USERDEFINED
0x03	**RemoveZone**	client ⇒ server	Y	M		USERDEFINED
0x04	**CreateOrUpdateTrigger**	client ⇒ server	Y	M		M
0x05	**RemoveTrigger**	client ⇒ server	Y	M		M

2.14.7.1. CreateTwoDCartesianZone Command

This command SHALL create and store a TwoD Cartesian Zone.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Zone**	TwoDCartesianZoneStruct				M

2.14.7.1.1. Zone Field

The Zone field SHALL be a TwoDCartesianZoneStruct representing all information required to define the TwoD Cartesian Zone.

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2.14.7.1.2. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If the current number of UserDefined Zones within Zones equals MaxUserDefinedZones:
  - Fail the command with the status code RESOURCE_EXHAUSTED, and end processing with no other side effects.
- If the Use field of the passed in *Zone* is Focus Value and the device does not support FocusZones:
  - Fail the command with the status code CONSTRAINT_ERROR, and end processing with no other side effects.
- If any X or Y points, in the Vertices field, of the TwoDCartesianZone field, of the passed in *Zone*, are greater than the corresponding point values in TwoDCartesianMax:
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- If the Vertices and Use fields, of the TwoDCartesianZone field, of the passed in *Zone*, exactly match the corresponding fields of any entry in the Zones attribute:
  - Fail the command with the status code ALREADY_EXISTS, and end processing with no other side effects.
- If the resulting shape of the Vertices field, of the TwoDCartesianZone field, of the passed in *Zone*, results in any self-intersecting lines:
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- Create a new ZoneID as described in ZoneID within Zones
- Create a new ZoneInformationStruct setting ZoneID to the new ZoneID, ZoneSource to User, the passed in *Zone* as TwoDCartesianZone, and ZoneType to TwoDCARTZone.
- Add that ZoneInformationStruct as a new entry in Zones.
- Return the new ZoneID in the corresponding CreateTwoDCartesianZoneResponse command.

2.14.7.2. CreateTwoDCartesianZoneResponse Command

This command SHALL be generated in response to a CreateTwoDCartesianZone command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ZoneID**	ZoneID				M

2.14.7.2.1. ZoneID Field

The ZoneID field SHALL be an unsigned 16 bit integer representing the unique ZoneID.

2.14.7.3. UpdateTwoDCartesianZone Command

The UpdateTwoDCartesianZone SHALL update a stored TwoD Cartesian Zone.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ZoneID**	ZoneID				M
1	**Zone**	TwoDCartesianZoneStruct				M

2.14.7.3.1. ZoneID Field

The ZoneID field SHALL be a ZoneID of the Zone to be updated.

2.14.7.3.2. Zone Field

The Zone field SHALL be a TwoDCartesianZoneStruct representing updated Zone information.

2.14.7.3.3. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If no entry can be found for the ZoneID in the Zones attribute:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- If the requested zone's ZoneSource is Mfg:
  - Fail the command with the status code CONSTRAINT_ERROR, and end processing with no other side effects.
- If the requested zone's ZoneType is not TwoDCARTZone:
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- If any requested X or Y points in the vertices are greater than the corresponding point values in TwoDCartesianMax:
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- If the requested zone's Vertices and Use fields exactly match any *other* existing Zone definition:
  - Fail the command with the status code ALREADY_EXISTS, and end processing with no other side effects.
- If the provided vertices result in a self-intersecting shape:
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- Update the Zone details for the ZoneID entry within Zones.

2.14.7.4. RemoveZone Command

This command SHALL remove the user-defined Zone indicated by ZoneID.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ZoneID**	ZoneID				M

2.14.7.4.1. ZoneID Field

The ZoneID field SHALL be a ZoneID of the Zone to be removed.

2.14.7.4.2. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If no entry can be found for the passed in *ZoneID* in the Zones attribute:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- If the requested zone's ZoneSource is Mfg:
  - Fail the command with the status code CONSTRAINT_ERROR, and end processing with no other side effects.
- If the passed in *ZoneID* is found in the ZoneID field of any entry in the Triggers list:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
- Remove the entry for the passed in *ZoneID* from the Zones list.

2.14.7.5. CreateOrUpdateTrigger Command

This command is used to create or update a Trigger for the specified motion Zone.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Trigger**	ZoneTriggerControlStruct				M

2.14.7.5.1. Trigger Field

This field SHALL be a ZoneTriggerControlStruct representing all information required to define the Trigger conditions.

2.14.7.5.2. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If no entry can be found for the ZoneID field value of the passed in *Trigger*, in the Zones attribute:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.

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- If the Zone use for that entry is not Motion Value:
  - Fail the command with the status code CONSTRAINT_ERROR, and end processing with no other side effects.
- Add or update the passed in *Trigger* entry in the Triggers attribute.

2.14.7.6. RemoveTrigger Command

This command SHALL remove the Trigger for the provided ZoneID.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ZoneID**	ZoneID				M

2.14.7.6.1. ZoneID Field

The ZoneID field SHALL be a ZoneID of the Zone Trigger to be removed.

2.14.7.6.2. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If no entry can be found for the passed in *ZoneID* in the Triggers attribute:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- Remove the entry associated with *ZoneID* from the Triggers attribute.

2.14.8. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**ZoneTriggered**	INFO		V	M
0x01	**ZoneStopped**	INFO		V	M

2.14.8.1. ZoneTriggered Event

This event SHALL be generated when a Zone is first triggered.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0x0000	**Zone**	ZoneID	all			M
0x0001	**Reason**	ZoneEventTriggeredReasonEnum	all			M

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2.14.8.1.1. Zone Field

This field SHALL contain the ZoneID of the Zone that triggered.

2.14.8.1.2. Reason Field

This field SHALL indicate why the zone was triggered.

2.14.8.2. ZoneStopped Event

This event SHALL be generated when either the TriggerDetectedDuration value is exceeded by the TimeSinceInitialTrigger value or the MaxDuration value is exceeded by the TimeSinceInitialTrigger value, as described in Section 2.14.5.9, “ZoneTriggerControlStruct”.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0x0000	**Zone**	ZoneID	all			M
0x0001	**Reason**	ZoneEventStoppedReasonEnum	all			M

2.14.8.2.1. Zone Field

This field SHALL contain the ZoneID of the Zone that stopped.

2.14.8.2.2. Reason Field

This field SHALL indicate why the zone stopped triggering.

2.15. Soil Measurement Cluster

This cluster provides an interface to soil measurement functionality, including configuration and provision of notifications of soil measurements.

2.15.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

2.15.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	SOIL

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2.15.3. Cluster ID

ID	Name
0x0430	Soil Measurement

2.15.4. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SoilMoistureMeasurementLimits**	MeasurementAccuracyStruct		F		R V	M
0x0001	**SoilMoistureMeasuredValue**	percent	SoilMoistureMeasurementLimits.MinMeasuredValue to SoilMoistureMeasurementLimits.MaxMeasuredValue	X		R V	M

2.15.4.1. SoilMoistureMeasurementLimits Attribute

This attribute SHALL indicate the limits for the SoilMoistureMeasuredValue attribute.

Given the measurements are in percentage, the MinMeasuredValue field in the SoilMoistureMeasurementLimits attribute SHALL NOT be less than 0 and SHALL NOT be greater than 99. The MaxMeasuredValue field in the SoilMoistureMeasurementLimits attribute SHALL NOT be less than (SoilMoistureMinMeasurableValue + 1) and SHALL NOT be greater than 100. The MeasurementType field value SHALL be set to SoilMoisture.

There SHALL only be a single entry in the AccuracyRanges list of the SoilMoistureMeasurementLimits attribute. The entry SHALL cover the full measurement range, meaning that the value of the RangeMin field SHALL be equal to the value of the MinMeasuredValue field and the value of the RangeMax field SHALL be equal to the value of the MaxMeasuredValue field. The entry SHALL only indicate a PercentMax value and the value SHALL NOT be greater than 10.00 percent.

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2.15.4.2. SoilMoistureMeasuredValue Attribute

This attribute SHALL indicate the water content of the soil in percentage.

The null value indicates that the measurement is unknown e.g. no measurement has been performed yet.

2.16. Ambient Context Sensing Cluster

This server cluster provides an interface to ambient context sensing functionality.

NOTE	Support for this cluster is provisional.

2.16.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

2.16.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	ACS

2.16.3. Cluster ID

ID	Name	Conformance
0x0431	Ambient Context Sensing	P

2.16.4. General Description

This cluster exposes ambient context sensing information, such as human activity, object identification, crowd detection, ambient audio context, and situational prediction. The ambient context sensing could be provided by various technologies. The sensing capabilities are indicated in the AmbientContextFeatureMap of this cluster.

The ambient context sensing information is exposed jointly with a boolean detection attribute and an ambient context type attribute where the detailed sensing context is conveyed through semantic tags from the StandardNamespaces such as Identified Human Activity, Identified Object, and Identified Sound namespaces. When multiple concurrent ambient context sensing events are detected by the underlying technology, the ambient context sensing information can be exposed either as independent occurrences or a single combined occurrence.

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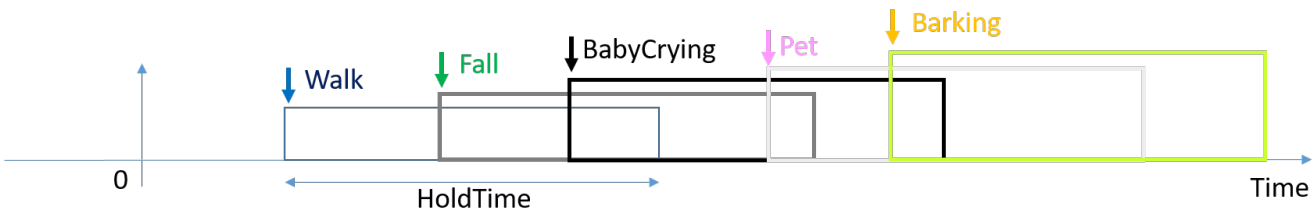
2.16.4.1. Multiple Ambient Context Detection

This cluster supports the concurrent detection of multiple ambient context sensing events through AmbientContextType attribute. The order of the list element SHALL reflect the event detection sequence such that a lower index element of the list indicates a more recently detected ambient context event. For example, index 0 represents the most recent detection, index 1 represents an older detection, and index 2 would be the oldest detection. Each ambient context element of the list remains in the AmbientContextType list according to the HoldTime requirements. Each event start time information can be accessed through the event data field of AmbientContextDetectStarted event function.

2.16.4.2. Internal Timing Tracking

The server is expected to track the event start time and maintain the triggered state properly for each ambient context sensing event exposed in the AmbientContextType attribute. When multiple ambient context sensing events occur, each sensing event time information is expected to be tracked independently according to sensor manufacturer's internal implementation. If the same ambient context type occurs within the HoldTime duration, the triggered state time SHALL be extended by adding HoldTime time in accordance with HoldTime functional implementation described in Occupancy Sensing Cluster. This time addition SHALL be allowed only once per each HoldTime period. This addition process can occur repeatedly until the aggregated triggered state duration does not exceed HoldTimeMax which SHALL be the maximum allowable continuous duration for a triggered event state.

2.16.4.3. Multiple Ambient Context Example



Time (Event Detection Sequence)	AmbientContextType List Value
Walking	{NULL,4B,0x03}
Fall	[{NULL,4B,0x01}, {NULL,4B,0x03}]
BabyCrying	[{NULL,4A,0x06}, {NULL,4B,0x01}, {NULL,4B,0x03}]
Pet	[{NULL,49,0x05}, {NULL,4A,0x06}, {NULL,4B,0x01}]
Barking	[{NULL,4A,0X04}, {NULL,49,0x05}, {NULL,4A,0x06}]

Figure 14. Multiple Ambient Context Event Detection

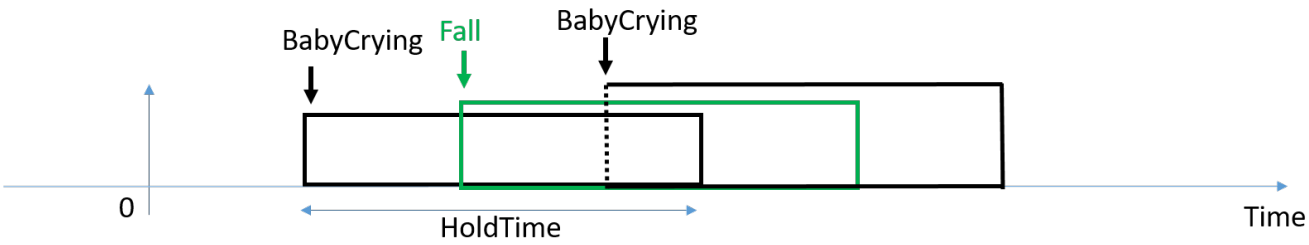
The above example provides a graphical representation and the corresponding AmbientContextType attribute value for five sequential ambient context events detected in the order of [Run, Fall, Crying, Pet, Barking]. The rectangular block illustrates the HoldTime triggered state of each ambient context event occurrence (also applicable to the Boolean attribute such as HumanActivityDetected). The table below details the AmbientContextType attribute value change according to the

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sequential sensing event occurrence.

Each specific ambient context sensing event is identified by the combination of Namespace ID and Tag ID from one of the following namespaces: Identified Human Activity Namespace, Identified Object Namespace, Identified Sound Namespace.

By the time the "Pet" object identification event occurs, the earlier "Walk" event is no longer presented in the AmbientContextType attribute value since the triggered state of “Walk” event has expired. As described in the InternalTimingTracking, the server is expected to track the event start and end time of each ambient context sensing event through manufacturer’s internal time-tracking implementation.



Time (Event Detection Sequence)	AmbientContextType List Value
BabyCrying	{NULL,4A,0x06}
Fall	[{NULL,4B,0x01}, {NULL,4A,0x06}]
BabyCrying	[{NULL,4A,0x06}, {NULL,4B,0x01}] ~~[{NULL,4A,0x06}, {NULL,4B,0x01}, {NULL,4A,0x06}]~~

Figure 15. Same Ambient Context Detection Repetition

If the same type of ambient context sensing event occurs consecutively within the HoldTime duration, then only a single instance of that ambient context type information SHALL be present in the AmbientContextType attribute as illustrated in the example above.

2.16.5. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**HA**	HumanActivity	P, O.a+	Supports various human actions and activities classification
1	**OC**	ObjectCounting	P, O.a+	Supports object counting
2	**OI**	ObjectIdentification	P, O.a+	Supports object identification

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Bit	Code	Feature	Conformance	Summary
3	**AUD**	SoundIdentification	P, O.a+	Supports sound identification
4	**PRED**	PredictedActivity	P, O	Supports predicting various human actions and activities

2.16.6. Data Types

2.16.6.1. HoldTimeLimitsStruct Type

This structure provides information on the server's supported values for the HoldTime attribute.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**HoldTimeMin**	uint16	min 1				P, M
1	**HoldTimeMax**	uint16	min maxOf(HoldTimeMin, 10)				P, M
2	**HoldTimeDefault**	uint16	HoldTimeMin to HoldTimeMax				P, M

2.16.6.1.1. HoldTimeMin Field

This field SHALL specify the minimum value supported by the server for the HoldTime attribute, in seconds.

2.16.6.1.2. HoldTimeMax Field

This field SHALL specify the maximum value supported by the server for the HoldTime attribute, in seconds. This field also specifies the maximum duration time that is allowed to be continuously in triggered detection state.

2.16.6.1.3. HoldTimeDefault Field

This field SHALL specify the (manufacturer-determined) default value of the server's HoldTime attribute, in seconds. This is the value that a client who wants to reset the settings to a valid default SHOULD use.

2.16.6.2. AmbientContextTypeStruct Type

This structure provides information on the server's supported values for the Ambient Context type

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attribute.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**AmbientContextSensed**	list[SemanticTagStruct]	max 2				P, M

2.16.6.2.1. AmbientContextSensed Field

This field specifies the detail ambient context information associated with the Boolean detection attributes, HumanActivityDetected, ObjectIdentified, and AudioContextDetected. The detail ambient context information SHALL be represented by the namespace ID and semantic tag ID of the SemanticTagStruct available from Identified Human Activity, Identified Object, and Identified Sound namespaces in the StandardNamespaces. When AmbientContextSensed field contains multiple data elements, it SHALL indicate a single combined ambient context event rather than unrelated independent ambient context events. For an example, if a joint event exposure of "Child Fall" is intended, then the AmbientContextType attribute can be exposed as follows:

```
{
  "AmbientContextSensed": "[
     {
       NULL, // MfgCode: null == standard
       0x4B, // IdentifiedObject namespace
       2     // "Child" tag in that namespace
     },
     {
       NULL, // MfgCode: null == standard
       0x49, // IdentifiedHumanActivity namespace
       1     // "Fall" tag in that namespace
     }
   ]",
}
```

where AmbientContextSensed field contains the SemanticTag data list of "Child" tag ID (=2) from IdentifiedObject namespace (=0x4B) and "Fall" tag ID (=1) from IdentifiedHumanActivity namespace (=0x49). However, if two independent events exposure is intended, then the AmbientContextType attribute can be exposed as follows:

```
{
  {
    "AmbientContextSensed": "{NULL,0x4B,1}",
  }
```

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```
  {
    "AmbientContextSensed": "{NULL,0x49,2}",
  }
}
```

where each AmbientContextSensed field contains only one individual ambient sensing context. In order to prevent confusion arising from excessive joint permutations, AmbientContextSensed field SHALL NOT include more than two ambient context events.

2.16.6.3. ObjectCountConfigStruct Type

This structure provides information on the server's supported values for the ObjectCountConfig attribute.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**CountingObject**	SemanticTagStruct					P, M
1	**ObjectCountThreshold**	uint16	min 1				P, M

2.16.6.3.1. CountingObject Field

This field SHALL indicate the object to be detected and counted. If the MfgCode field in CountingObject is NULL, the object SHALL be specified by ObjectIdentified namespace ID and its tag ID available from the AmbientContextTypeSupported attribute.

2.16.6.3.2. ObjectCountThreshold Field

This field SHALL indicate the minimum number of detected objects to render the true Boolean state of CountThresholdReached attribute.

2.16.6.4. PredictedActivityStruct Type

This data structure provides information on future predicted activities.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**StartTimestamp**	epoch-s	max (EndTimestamp - 1)				P, M
1	**EndTimestamp**	epoch-s	min (StartTimestamp + 1)				P, M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
2	**AmbientContextType**	list[SemanticTagStruct]	max 100				P, HA | OI | AUD
3	**CrowdDetected**	bool	all				P, OC
4	**CrowdCount**	uint8	1 to 254				P, [OC]
5	**Confidence**	percent	all				P, M

2.16.6.4.1. StartTimestamp Field

This field SHALL indicate the predicted start time for the predicted activity.

2.16.6.4.2. EndTimestamp Field

This field SHALL indicate the predicted end time for the predicted activity.

2.16.6.4.3. AmbientContextType Field

This field SHALL indicate the predicted state of the AmbientContextType attribute for the specified time period.

2.16.6.4.4. CrowdDetected Field

This field SHALL indicate the predicted state of the CrowdDetected attribute for the specified time period.

2.16.6.4.5. CrowdCount Field

This field SHALL indicate the predicted value of the CrowdCount attribute for the specified time period.

2.16.6.4.6. Confidence Field

This field SHALL indicate confidence level for the predicted activity state.

A value of 100% SHALL indicate a complete certainty of the predicted occupancy state, while a 0% value SHALL indicate no certainty. The algorithm to calculate the likelihood of a predicted occupancy state is not specified and is considered manufacturer specific.

2.16.7. Attributes

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**HumanActivityDetected**	bool	all			R V	P, HA
0x0001	**ObjectIdentified**	bool	all			R V	P, OI
0x0002	**AudioContextDetected**	bool	all			R V	P, AUD
0x0003	**AmbientContextType**	list[AmbientContextTypeStruct]	1 to SimultaneousDetectionLimit			R V	P, HA | OI | AUD
0x0004	**AmbientContextTypeSupported**	list[SemanticTagStruct]	max 50			R V	P, HA | OI | AUD
0x0005	**ObjectCountThresholdReached**	bool	all			R V	P, OC & OI
0x0006	**ObjectCountConfig**	ObjectCountConfigStruct		N		RW VM	P, OC & OI
0x0007	**ObjectCount**	uint16	min 1		0	R V	P, [OC & OI]
0x0008	**SimultaneousDetectionLimit**	uint8	1 to 10	F		R V	P, M
0x0009	**HoldTime**	uint16	HoldTimeLimits.HoldTimeMin to HoldTimeLimits.HoldTimeMax	N		RW VM	P, M
0x000A	**HoldTimeLimits**	HoldTimeLimitsStruct		F		R V	P, M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x000B	**PredictedActivity**	list[PredictedActivityStruct]	max 20			R V	P, PRED

2.16.7.1. HumanActivityDetected Attribute

This attribute SHALL indicate the human activity detection in Boolean data. The detected human activity type can be found from the AmbientContextType attribute.

2.16.7.2. ObjectIdentified Attribute

This attribute SHALL indicate the occurrence of object identification in Boolean data. The identified object information can be found from the AmbientContextType attribute.

2.16.7.3. AudioContextDetected Attribute

This attribute SHALL indicate the ambient audio context detection in Boolean data. The detected audio context type can be found from the AmbientContextType attribute.

2.16.7.4. AmbientContextType Attribute

This attribute SHALL indicate the details for the currently observed and detected ambient context. This attribute supports multiple simultaneous ambient context detections. The attribute expression rule is defined in the MultipleAmbientSensingDetection section. The total number of simultaneous ambient context detections is constrained by the SimultaneousDetectionLimit attribute.

2.16.7.5. AmbientContextTypeSupported Attribute

This attribute SHALL indicate the list of ambient context detection types supported by the server. Each supported ambient context detection type element SHALL correspond to a feature type supported in the AmbientContextFeatureMap and SHALL be represented by a SemanticTagStruct from one of the following StandardNamespaces: Identified Human Activity Namespace, Identified Object Namespace, or Identified Sound Namespace.

2.16.7.6. ObjectCountThresholdReached Attribute

This attribute SHALL indicate whether the number count of the specified object is greater or equal to the threshold specified by the ObjectCountThreshold. The object being counted SHALL be limited to one identified object type and SHALL be identified by the tag ID from the Identified Object namespace, as presented in the AmbientContextTypeSupported attribute.

2.16.7.7. ObjectCountConfig Attribute

This attribute SHALL specify the configuration parameters required to support an object counting feature, including the identification of the object to be detected and counted, as well as the counting threshold value for the ObjectCountThresholdReached detection.

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2.16.7.8. ObjectCount Attribute

This optional attribute SHALL indicate the number of objects detected in the area covered by the sensor. ObjectCount SHALL be exposed only when the ObjectCountThresholdReached attribute is true.

2.16.7.9. SimultaneousDetectionLimit Attribute

This attribute SHALL indicate the maximum number of simultaneous multiple ambient context detections supported by the server. If an additional detection event causes the total number of simultaneous detection events to exceed a SimultaneousDetectionLimit, the oldest ambient sensing detection event SHALL be removed and the latest detection SHALL be added. The consecutive recurrence of the same ambient context sensing event within the HoldTime duration SHALL not increase the total number of simultaneous detection events. If simultaneous detection is not supported, then the value SHALL be set to 1.

2.16.7.10. HoldTime Attribute

This attribute SHALL indicate the time duration (in seconds) of True state before transitioning its sensing detection state from True to False after the last detection. Low HoldTime value SHOULD be avoided to prevent excessive data reporting for subscription. This attribute is equivalent to the HoldTime attribute of the OccupancySensing cluster. For further information, refer to the HoldTime attribute description of the Occupancy Sensing Cluster. The HoldTime SHALL be applied individually to each ambient context detection occurrence individually. A more detailed HoldTime implementation example for multiple simultaneous ambient context detections can be found in theMultipleAmbientSensingDetection section.

2.16.7.11. HoldTimeLimits Attribute

This attribute SHALL indicate the server's limits, and default value, for the HoldTime attribute. This is equivalent to the HoldTimeLimits attribute of the Occupancy Sensing Cluster attribute. For further information, refer to the HoldTimeLimits attribute description of the Occupancy Sensing Cluster.

2.16.7.12. PredictedActivity Attribute

This attribute SHALL indicate the server's prediction of upcoming changes to the monitored area's ambient context.

The value of the StartTimestamp field on each PredictedActivityStruct in this list other than the first SHALL be greater than the value of the EndTimestamp field on the previous PredictedActivityStruct in this list.

2.16.8. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**AmbientContextDetectStarted**	INFO		V	P, HA | OI | AUD | (OC & OI)

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ID	Name	Priority	Quality	Access	Conformance
0x01	**AmbientContextDetectEnded**	INFO		V	P, HA | OI | AUD | (OC & OI)

2.16.8.1. AmbientContextDetectStarted Event

This event SHALL be generated when a new ambient context detection is added to the AmbientContextType attribute.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**AmbientContextDetected**	AmbientContextTypeStruct				P, HA | OI | OC | AUD
1	**ObjectCountThresholdReached**	bool	all			P, OC & OI
2	**ObjectCount**	uint16	min 1		NULL	P, [OC & OI]

2.16.8.1.1. AmbientContextDetected Field

This field SHALL indicate the detail ambient context information that triggers this event reporting. The detail ambient context information SHALL be represented by the namespace ID and semantic tag ID available from Identified Human Activity, Identified Object, and Identified Sound namespaces in the StandardNamespaces. For the object counting feature, the AmbientContextDetected field SHALL represent the object being counted.

2.16.8.1.2. ObjectCountThresholdReached Field

This field SHALL indicate an ObjectCountThresholdReached attribute value when the event reporting is triggered by the object counting threshold detection.

2.16.8.1.3. ObjectCount Field

This field SHALL indicate the number of objects detected in the area covered by the sensor when ObjectCountThresholdReached attribute is changed to True.

2.16.8.2. AmbientContextDetectEnded Event

This event SHALL be generated when the ambient context previously reported by an AmbientContextDetectStarted event is removed from the AmbientContextType attribute. This termination event doesn't necessarily reflect or coincide with the end of the actual event progression. For example, while both AmbientContextDetectStarted and AmbientContextDetectEnded events are used to inform the "sleeping" event occurrence, the AmbientContextDetectEnded event doesn't necessarily indicate the actual end of "sleeping" action. The AmbientContextDetectStarted event start time information is provided to facilitate the mapping of the matching the AmbientContextDetectStarted

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event and the AmbientContextDetectEnded event for event logging purpose. If a server is time-synchronized or capable of supporting both the POSIX time stamp and the system time stamp, a server SHALL provide the EventStartTimePos data field instead of the EventStartTimeSys data field.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**EventStartTimePos**	posix-ms	all			P, O.a
1	**EventStartTimeSys**	systime-ms	all			P, O.a

2.16.8.2.1. EventStartTimePos Field

This field SHALL indicate the POSIX time stamp when the corresponding AmbientContextDetectStarted Event was generated.

2.16.8.2.2. EventStartTimeSys Field

This field SHALL indicate the system time stamp when the corresponding AmbientContextDetectStarted Event was generated.

2.17. Temperature Alarm Cluster

This cluster is a derived cluster of Alarm Base cluster and provides the alarm definition related to temperature measurements.

NOTE	Support for this cluster is provisional.

2.17.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

2.17.2. Classification

Hierarchy	Role	Scope	PICS Code
Alarm Base	Application	Endpoint	TEMPALM

2.17.3. Cluster ID

Cluster ID	Cluster Name	Conformance
0x0064	Temperature Alarm	P

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2.17.4. Features

Bit	Code	Feature	Conformance	Summary
20	**OVER**	OverTemperature	O.a+	Supports activating alarms when a temperature measurement goes over a threshold
21	**UNDER**	UnderTemperature	O.a+	Supports activating alarms when a temperature measurement goes under a threshold
22	**MAJOR**	MajorThreshold	O	Supports the major threshold for alarms
23	**MINOR**	MinorThreshold	[MAJOR]	Supports the minor threshold for alarms
24	**OCRIADJ**	OverCriticalAdjustable	[OVER]	Supports the ability to adjust the over critical temperature threshold
25	**OMAJADJ**	OverMajorAdjustable	[OVER & MAJOR]	Supports the ability to adjust the over major temperature threshold
26	**OMINADJ**	OverMinorAdjustable	[OVER & MINOR]	Supports the ability to adjust the over minor temperature threshold
27	**UMINADJ**	UnderMinorAdjustable	[UNDER & MINOR]	Supports the ability to adjust the under minor temperature threshold
28	**UMAJADJ**	UnderMajorAdjustable	[UNDER & MAJOR]	Supports the ability to adjust the under major temperature threshold
29	**UCRIADJ**	UnderCriticalAdjustable	[UNDER]	Supports the ability to adjust the under critical temperature threshold

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2.17.5. Data Types

2.17.5.1. AlarmBitmap Type

This data type is derived from map32.

Bit	Name	Summary	Conformance
0	**CriticalOverTemperatureAlarm**	The measured temperature is above the critical threshold.	M
1	**MajorOverTemperatureAlarm**	The measured temperature is above the major threshold.	M
2	**MinorOverTemperatureAlarm**	The measured temperature is above the minor threshold.	M
3	**MinorUnderTemperatureAlarm**	The measured temperature is below the minor threshold.	M
4	**MajorUnderTemperatureAlarm**	The measured temperature is below the major threshold.	M
5	**CriticalUnderTemperatureAlarm**	The measured temperature is below the critical threshold.	M

2.17.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0080	**CriticalOverTemperatureThreshold**	temperature	min (maxOf(CriticalUnderTemperatureThreshold, MajorOverTemperatureThreshold, MinorOverTemperatureThreshold) + 1)			R V	OVER

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0081	**MajorOverTemperatureThreshold**	temperature	min (maxOf(MajorUnderTemperatureThreshold, MinorOverTemperatureThreshold) + 1)			R V	OVER & MAJOR
0x0082	**MinorOverTemperatureThreshold**	temperature	min (MinorUnderTemperatureThreshold + 1)			R V	OVER & MINOR
0x0083	**MinorUnderTemperatureThreshold**	temperature	max (MinorOverTemperatureThreshold - 1)			R V	UNDER & MINOR
0x0084	**MajorUnderTemperatureThreshold**	temperature	max (minOf(MajorOverTemperatureThreshold, MinorUnderTemperatureThreshold) - 1)			R V	UNDER & MAJOR

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0085	**CriticalUnderTemperatureThreshold**	temperature	max (minOf(CriticalOverTemperatureThreshold, MajorUnderTemperatureThreshold, MinorUnderTemperatureThreshold) - 1)			R V	UNDER

2.17.6.1. CriticalOverTemperatureThreshold Attribute

This attribute SHALL indicate the threshold for temperature measurement over which the CriticalOverTemperature alarm SHALL be active.

2.17.6.2. MajorOverTemperatureThreshold Attribute

This attribute SHALL indicate the threshold for temperature measurement over which the MajorOverTemperature alarm SHALL be active.

2.17.6.3. MinorOverTemperatureThreshold Attribute

This attribute SHALL indicate the threshold for temperature measurement over which the MinorOverTemperature alarm SHALL be active.

2.17.6.4. MinorUnderTemperatureThreshold Attribute

This attribute SHALL indicate the threshold for temperature measurement under which the MinorUnderTemperature alarm SHALL be active.

2.17.6.5. MajorUnderTemperatureThreshold Attribute

This attribute SHALL indicate the threshold for temperature measurement under which the MajorUnderTemperature alarm SHALL be active.

2.17.6.6. CriticalUnderTemperatureThreshold Attribute

This attribute SHALL indicate the threshold for temperature measurement under which the CriticalUnderTemperature alarm SHALL be active.

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2.17.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x80	**SetTemperatureAlarmThresholds**	client ⇒ server	Y	O	OCRIADJ | OMAJADJ | OMINADJ | UMINADJ | UMAJADJ | UCRIADJ

2.17.7.1. SetTemperatureAlarmThresholds Command

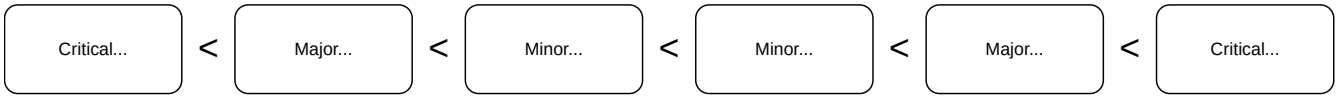
This command will set the alarm thresholds for the specified values.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**CriticalOverTemperatureThreshold**	temperature	min (maxOf(CriticalUnderTemperatureThreshold, MajorOverTemperatureThreshold, MinorOverTemperatureThreshold) + 1)			[OCRIADJ].b+
1	**MajorOverTemperatureThreshold**	temperature	min (maxOf(MajorUnderTemperatureThreshold, MinorOverTemperatureThreshold) + 1)			[OMAJADJ].b+
2	**MinorOverTemperatureThreshold**	temperature	min (MinorUnderTemperatureThreshold + 1)		32765	[OMINADJ].b+

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
3	**MinorUnderTemperatureThreshold**	temperature	max (MinorOverTemperatureThreshold - 1)		-27314	[UMINADJ].b+
4	**MajorUnderTemperatureThreshold**	temperature	max (minOf(MajorOverTemperatureThreshold, MinorUnderTemperatureThreshold) - 1)			[UMAJADJ].b+
5	**CriticalUnderTemperatureThreshold**	temperature	max (minOf(CriticalOverTemperatureThreshold, MajorUnderTemperatureThreshold, MinorUnderTemperatureThreshold) - 1)			[UCRIADJ].b+

NOTE The constraints related to the field values in the table above SHALL have the relationship as represented by the illustration below:



2.17.7.1.1. CriticalOverTemperatureThreshold Field

This field SHALL specify the new value of the CriticalOverTemperatureThreshold attribute.

2.17.7.1.2. MajorOverTemperatureThreshold Field

This field SHALL specify the new value of the MajorOverTemperatureThreshold attribute.

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2.17.7.1.3. MinorOverTemperatureThreshold Field

This field SHALL specify the new value of the MinorOverTemperatureThreshold attribute.

2.17.7.1.4. MinorUnderTemperatureThreshold Field

This field SHALL specify the new value of the MinorUnderTemperatureThreshold attribute.

2.17.7.1.5. MajorUnderTemperatureThreshold Field

This field SHALL specify the new value of the MajorUnderTemperatureThreshold attribute.

2.17.7.1.6. CriticalUnderTemperatureThreshold Field

This field SHALL specify the new value of the CriticalUnderTemperatureThreshold attribute.

2.17.7.1.7. Effect on Receipt

On receipt of this command, the device SHALL attempt to adjust the supplied temperature thresholds.

If all threshold changes are successful, the server SHALL reevaluate the alarm state based on the new threshold values.

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Chapter 3. Lighting

The Cluster Library is made of individual chapters such as this one. See Document Control in the Cluster Library for a list of all chapters and documents. References between chapters are made using a *X.Y* notation where *X* is the chapter and *Y* is the sub-section within that chapter. References to external documents are contained in Chapter 1 and are made using [*Rn*] notation.

3.1. General Description

3.1.1. Introduction

The clusters specified in this document are for use typically in lighting applications, but MAY be used in any application domain.

3.1.2. Terms

HSV: Hue, Saturation, Value. A color space, also known as HSB (Hue, Saturation, Brightness). This is a well-known transformation of the RGB (Red, Green, Blue) color space. For more information see e.g., http://en.wikipedia.org/wiki/HSV_color_space.

Illuminance: The density of incident luminous flux on a surface. Illuminance is the standard metric for lighting levels, and is measured in lux (lx).

3.1.3. The Dimming Light Curve

The dimming curve is recommended to be logarithmic, as defined by the following equation:

$$\%Light = 10^{\left(\frac{Level-1}{\left(\frac{253}{3}\right)}\right)-1}$$

Where: %Light is the percent light output of the ballast and Level is an 8-bit integer between 1 (0.1% light output) and 254 (100% output) that is adjusted for MinLevel and MaxLevel using the following equation:

Level = MinLevel + (MaxLevel - MinLevel) * (CurrentLevel - 1) / 253.

NOTE	Value 255 is not used.

The table below gives a couple of examples of the dimming light curve for different values of MinLevel, MaxLevel and CurrentLevel.

Table 6. Examples of The Dimming Light Curve

MinLevel	MaxLevel	CurrentLevel	Level	%Light
1	254	1	1	0.10%
1	254	10	10	0.13%

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MinLevel	MaxLevel	CurrentLevel	Level	%Light
1	254	100	100	1.49%
1	254	254	254	100%
170	254	1	170	10.1%
170	254	10	173	11.0%
170	254	100	203	24.8%
170	254	254	254	100%
170	230	1	170	10.1%
170	230	10	172	10.7%
170	230	100	193	19.2%
170	230	254	230	51.9%

3.1.4. Cluster List

This section lists the lighting specific clusters as specified in this chapter.

Table 7. Overview of the Lighting Clusters

Cluster ID	Cluster Name	Description
0x0300	Color Control	Attributes and commands for controlling the color of a color-capable light.

3.2. Color Control Cluster

This cluster provides an interface for changing the color of a light. Color is specified according to the CIE 1931 Color space. Color control is carried out in terms of x,y values, as defined by this specification.

Additionally, color MAY optionally be controlled in terms of color temperature, or as hue and saturation values based on optionally variable RGB and W color points. It is recommended that the hue and saturation are interpreted according to the HSV (a.k.a. HSB) color model.

Control over luminance is not included, as this is provided by means of the Level Control for Lighting cluster. It is recommended that the level provided by this cluster be interpreted as representing a proportion of the maximum intensity achievable at the current color.

3.2.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

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Revision	Description
1	Mandatory global ClusterRevision attribute added; CCB 2028
2	Added Options attribute, CCB 2085 2104 2124 2230; ZLO 1.0
3	CCB 2501 2814 2839 2840 2843 2861
4	All Hubs changes
5	New data model format and notation, FeatureMap support
6	Added clarifications to Scenes support for Matter
7	Added Q quality for CurrentHue, CurrentSaturation, CurrentX, CurrentY, ColorTemperatureMireds, EnhancedCurrentHue and RemainingTime attributes. Added clarifications for behavior related to Step values. Added constraints on minimum acceptable color temperature values in mireds.
8	Change WhitePoint* and ColorPoint* attributes from RW to R (CCB 2490, TCR 11893)
9	Removed P quality

3.2.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	CC

3.2.3. Cluster ID

ID	Name
0x0300	Color Control

3.2.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**HS**	HueSaturation	EHUE, O	Supports color specification via hue/saturation.
1	**EHUE**	EnhancedHue	CL, O	Enhanced hue is supported.

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Bit	Code	Feature	Conformance	Summary
2	**CL**	ColorLoop	O	Color loop is supported.
3	**XY**	XY	O	Supports color specification via XY.
4	**CT**	ColorTemperature	O	Supports specification of color temperature.

3.2.5. Dependencies

3.2.5.1. Coupling color temperature to Level Control

If the Level Control for Lighting cluster identifier 0x0008 is supported on the same endpoint as the Color Control cluster and color temperature is supported, it is possible to couple changes in the current level to the color temperature.

The CoupleColorTempToLevel bit of the Options attribute of the Level Control cluster indicates whether the color temperature is to be linked with the CurrentLevel attribute in the Level Control cluster.

If the CoupleColorTempToLevel bit of the Options attribute of the Level Control cluster is equal to 1 and the ColorMode or EnhancedColorMode attribute is set to 2 (ColorTemperatureMireds) then a change in the CurrentLevel attribute SHALL affect the ColorTemperatureMireds attribute. This relationship is manufacturer specific, with the qualification that the maximum value of the CurrentLevel attribute SHALL correspond to a ColorTemperatureMired attribute value equal to the CoupleColorTempToLevelMinMireds attribute. This relationship is one-way so a change to the ColorTemperatureMireds attribute SHALL NOT have any effect on the CurrentLevel attribute.

In order to simulate the behavior of an incandescent bulb, a low value of the CurrentLevel attribute SHALL be associated with a high value of the ColorTemperatureMireds attribute (i.e., a low value of color temperature in kelvins).

If the CoupleColorTempToLevel bit of the Options attribute of the Level Control cluster is equal to 0, there SHALL be no link between color temperature and current level.

3.2.5.2. Independent transition in hue and saturation

Various commands in this cluster can be used to start transitions in hue and/or saturation.

- When a transition in hue is in progress, and a command to change saturation (MoveSaturation (with MoveMode!=Stop), StepSaturation, MoveToSaturation) is received by the server, this latter command SHALL NOT interrupt the ongoing transition in hue.
- When a transition in saturation is in progress, and a command to change hue (MoveHue (with MoveMode!=Stop), EnhancedMoveHue (with MoveMode!=Stop) StepHue, EnhancedStepHue, MoveToHue, EnhancedMoveToHue) is received by the server, this latter command SHALL NOT

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interrupt the ongoing transition in saturation.

3.2.6. Data Types

3.2.6.1. ColorCapabilitiesBitmap Type

This data type is derived from map16.

Bit	Name	Summary	Conformance
0	**HueSaturation**	Supports color specification via hue/saturation.	HS
1	**EnhancedHue**	Enhanced hue is supported.	EHUE
2	**ColorLoop**	Color loop is supported.	CL
3	**XY**	Supports color specification via XY.	XY
4	**ColorTemperature**	Supports color specification via color temperature.	CT

3.2.6.2. OptionsBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**ExecuteIfOff**	Dependency on On/Off cluster	M

3.2.6.2.1. ExecuteIfOff Bit

This bit SHALL indicate if this cluster server instance has a dependency with the On/Off cluster.

3.2.6.3. UpdateFlagsBitmap Type

This data type is derived from map8 and is used in the ColorLoopSet command.

Bit	Name	Summary	Conformance
0	**UpdateAction**	Device adheres to the associated action field.	M
1	**UpdateDirection**	Device updates the associated direction attribute.	M

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Bit	Name	Summary	Conformance
2	**UpdateTime**	Device updates the associated time attribute.	M
3	**UpdateStartHue**	Device updates the associated start hue attribute.	M

3.2.6.3.1. UpdateAction Bit

This bit SHALL indicate whether the server adheres to the Action field in order to process the command.

- 0 = Device SHALL ignore the Action field.
- 1 = Device SHALL adhere to the Action field.

3.2.6.3.2. UpdateDirection Bit

This bit SHALL indicate whether the device updates the ColorLoopDirection attribute with the Direction field.

- 0 = Device SHALL ignore the Direction field.
- 1 = Device SHALL update the ColorLoopDirection attribute with the value of the Direction field.

3.2.6.3.3. UpdateTime Bit

This bit SHALL indicate whether the device updates the ColorLoopTime attribute with the Time field.

- 0 = Device SHALL ignore the Time field.
- 1 = Device SHALL update the value of the ColorLoopTime attribute with the value of the Time field.

3.2.6.3.4. UpdateStartHue Bit

This bit SHALL indicate whether the device updates the ColorLoopStartEnhancedHue attribute with the value of the StartHue field.

- 0 = Device SHALL ignore the StartHue field.
- 1 = Device SHALL update the value of the ColorLoopStartEnhancedHue attribute with the value of the StartHue field.

3.2.6.4. DriftCompensationEnum Type

This data type is derived from enum8.

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Value	Name	Summary	Conformance
0	**None**	There is no compensation.	M
1	**OtherOrUnknown**	The compensation is based on other or unknown mechanism.	M
2	**TemperatureMonitoring**	The compensation is based on temperature monitoring.	M
3	**OpticalLuminanceMonitoringAndFeedback**	The compensation is based on optical luminance monitoring and feedback.	M
4	**OpticalColorMonitoringAndFeedback**	The compensation is based on optical color monitoring and feedback.	M

3.2.6.5. ColorModeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**CurrentHueAndCurrentSaturation**	The current hue and saturation attributes determine the color.	M
1	**CurrentXAndCurrentY**	The current X and Y attributes determine the color.	M
2	**ColorTemperatureMireds**	The color temperature attribute determines the color.	M

3.2.6.6. EnhancedColorModeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**CurrentHueAndCurrentSaturation**	The current hue and saturation attributes determine the color.	M
1	**CurrentXAndCurrentY**	The current X and Y attributes determine the color.	M

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Value	Name	Summary	Conformance
2	**ColorTemperatureMireds**	The color temperature attribute determines the color.	M
3	**EnhancedCurrentHueAndCurrentSaturation**	The enhanced current hue and saturation attributes determine the color.	M

3.2.6.7. DirectionEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Shortest**	Shortest distance	M
1	**Longest**	Longest distance	M
2	**Up**	Up	M
3	**Down**	Down	M

3.2.6.8. MoveModeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Stop**	Stop the movement	M
1	**Up**	Move in an upwards direction	M
3	**Down**	Move in a downwards direction	M

3.2.6.9. StepModeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
1	**Up**	Step in an upwards direction	M
3	**Down**	Step in a downwards direction	M

3.2.6.10. ColorLoopActionEnum Type

This data type is derived from enum8.

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Value	Name	Summary	Conformance
0	**Deactivate**	De-activate the color loop.	M
1	**ActivateFromColorLoopStartEnhancedHue**	Activate the color loop from the value in the ColorLoopStartEnhancedHue field.	M
2	**ActivateFromEnhancedCurrentHue**	Activate the color loop from the value of the EnhancedCurrentHue attribute.	M

3.2.6.11. ColorLoopDirectionEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Decrement**	Decrement the hue in the color loop.	M
1	**Increment**	Increment the hue in the color loop.	M

3.2.7. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**CurrentHue**	uint8	max 254	N Q		R V	HS
0x0001	**CurrentSaturation**	uint8	max 254	S N Q		R V	HS
0x0002	**RemainingTime**	uint16	max 65534	Q	0	R V	O
0x0003	**CurrentX**	uint16	max 65279	S N Q		R V	XY
0x0004	**CurrentY**	uint16	max 65279	S N Q		R V	XY
0x0005	**DriftCompensation**	DriftCompensationEnum	all		-	R V	O
0x0006	**CompensationText**	string	max 254		-	R V	O

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0007	**ColorTemperatureMireds**	uint16	max 65279	S N Q		R V	CT
0x0008	**ColorMode**	ColorModeEnum	all	N		R V	M
0x000F	**Options**	OptionsBitmap	desc			RW VO	M
0x0010	**NumberOfPrimaries**	uint8	max 6	F X		R V	M
0x0011	**Primary1X**	uint16	max 65279	F		R V	NumberOfPrimaries > 0, O
0x0012	**Primary1Y**	uint16	max 65279	F		R V	NumberOfPrimaries > 0, O
0x0013	**Primary1Intensity**	uint8	all	F X		R V	NumberOfPrimaries > 0, O
0x0015	**Primary2X**	uint16	max 65279	F		R V	NumberOfPrimaries > 1, O
0x0016	**Primary2Y**	uint16	max 65279	F		R V	NumberOfPrimaries > 1, O
0x0017	**Primary2Intensity**	uint8	all	F X		R V	NumberOfPrimaries > 1, O
0x0019	**Primary3X**	uint16	max 65279	F		R V	NumberOfPrimaries > 2, O
0x001A	**Primary3Y**	uint16	max 65279	F		R V	NumberOfPrimaries > 2, O
0x001B	**Primary3Intensity**	uint8	all	F X		R V	NumberOfPrimaries > 2, O

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0020	**Primary4X**	uint16	max 65279	F		R V	NumberOfPrimaries > 3, O
0x0021	**Primary4Y**	uint16	max 65279	F		R V	NumberOfPrimaries > 3, O
0x0022	**Primary4Intensity**	uint8	all	F X		R V	NumberOfPrimaries > 3, O
0x0024	**Primary5X**	uint16	max 65279	F		R V	NumberOfPrimaries > 4, O
0x0025	**Primary5Y**	uint16	max 65279	F		R V	NumberOfPrimaries > 4, O
0x0026	**Primary5Intensity**	uint8	all	F X		R V	NumberOfPrimaries > 4, O
0x0028	**Primary6X**	uint16	max 65279	F		R V	NumberOfPrimaries > 5, O
0x0029	**Primary6Y**	uint16	max 65279	F		R V	NumberOfPrimaries > 5, O
0x002A	**Primary6Intensity**	uint8	all	F X		R V	NumberOfPrimaries > 5, O
0x0030	**WhitePointX**	uint16	max 65279			R V	O
0x0031	**WhitePointY**	uint16	max 65279			R V	O
0x0032	**ColorPointRX**	uint16	max 65279			R V	O
0x0033	**ColorPointRY**	uint16	max 65279			R V	O
0x0034	**ColorPointRIntensity**	uint8	all	X		R V	O

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0036	**ColorPointGX**	uint16	max 65279			R V	O
0x0037	**ColorPointGY**	uint16	max 65279			R V	O
0x0038	**ColorPointGIntensity**	uint8	all	X		R V	O
0x003A	**ColorPointBX**	uint16	max 65279			R V	O
0x003B	**ColorPointBY**	uint16	max 65279			R V	O
0x003C	**ColorPointBIntensity**	uint8	all	X		R V	O
0x4000	**EnhancedCurrentHue**	uint16	all	N Q S		R V	EHUE
0x4001	**EnhancedColorMode**	EnhancedColorModeEnum	all	N S		R V	M
0x4002	**ColorLoopActive**	uint8	max 1	N S		R V	CL
0x4003	**ColorLoopDirection**	ColorLoopDirectionEnum	all	N S		R V	CL
0x4004	**ColorLoopTime**	uint16	all	N S		R V	CL
0x4005	**ColorLoopStartEnhancedHue**	uint16	all			R V	CL
0x4006	**ColorLoopStoredEnhancedHue**	uint16	all			R V	CL

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x400A	**ColorCapabilities**	ColorCapabilities-Bitmap	max 0x1F			R V	M
0x400B	**ColorTempPhysicalMinMireds**	uint16	1 to 65279			R V	CT
0x400C	**ColorTempPhysicalMaxMireds**	uint16	max 65279			R V	CT
0x400D	**CoupleColorTempToLevelMinMireds**	uint16	ColorTempPhysicalMinMireds to ColorTemperatureMireds			R V	CT
0x4010	**StartUpColorTemperatureMireds**	uint16	1 to 65279	N X		RW VM	CT

3.2.7.1. Scene Table Extensions

If the Scenes Management server cluster is implemented on the same endpoint, the following attributes SHALL be part of the ExtensionFieldSetStruct of the Scene Table. If an attribute is not implemented, the value that will be used for it in the AttributeValuePairStruct is given in parentheses.

1. CurrentX (0)
2. CurrentY (0)
3. EnhancedCurrentHue (null)
4. CurrentSaturation (null)
5. ColorLoopActive (0)
6. ColorLoopDirection (Decrement)
7. ColorLoopTime (0)

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8. ColorTemperatureMireds (null)
9. EnhancedColorMode

Since there is a direct relation between ColorTemperatureMireds and XY, color temperature, if supported, is stored as XY in the scenes table.

Attributes in the scene table that are not supported by the device (according to the FeatureMap attribute) SHALL be present in the scene table but ignored.

If the Scenes Management cluster server is implemented on the same endpoint, and the cluster revision is 6 or above, then all extension fields in the list above, including up to EnhancedColorMode, SHALL be present in the extension fields set for a Scene to be considered valid. This disambiguates the color mode used within the Scene.

3.2.7.2. CurrentHue Attribute

The CurrentHue attribute contains the current hue value of the light. It is updated as fast as practical during commands that change the hue.

The hue in degrees SHALL be related to the CurrentHue attribute by the relationship:

```
Hue = "CurrentHue" * 360 / 254
```

where CurrentHue is in the range from 0 to 254 inclusive.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- At most once per second or
- At the end of the movement/transition.

3.2.7.3. CurrentSaturation Attribute

This attribute SHALL indicate the current saturation value of the light. It is updated as fast as practical during commands that change the saturation.

The saturation (on a scale from 0.0 to 1.0) SHALL be related to the CurrentSaturation attribute by the relationship:

```
Saturation = "CurrentSaturation" / 254
```

where CurrentSaturation is in the range from 0 to 254 inclusive.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- At most once per second or
- At the end of the movement/transition.

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3.2.7.4. RemainingTime Attribute

This attribute SHALL indicate the time remaining, in 1/10ths of a second, until transitions due to the currently active command will be complete.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- When it changes from 0 to any value higher than 10, or
- When it changes, with a delta larger than 10, caused by the invoke of a command, or
- When it changes to 0.

For commands with a transition time or changes to the transition time less than 1 second, changes to this attribute SHALL NOT be reported.

As this attribute is not being reported during a regular countdown, clients SHOULD NOT rely on the reporting of this attribute in order to keep track of the remaining duration.

3.2.7.5. CurrentX Attribute

This attribute SHALL indicate the current value of the normalized chromaticity value x, as defined in the CIE xyY Color Space. It is updated as fast as practical during commands that change the color.

The value of x SHALL be related to the CurrentX attribute by the relationship

```
x = "CurrentX" / 65536
```

where CurrentX is in the range from 0 to 65279 inclusive.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- At most once per second or
- At the end of the movement/transition.

3.2.7.6. CurrentY Attribute

This attribute SHALL indicate the current value of the normalized chromaticity value y, as defined in the CIE xyY Color Space. It is updated as fast as practical during commands that change the color.

The value of y SHALL be related to the CurrentY attribute by the relationship

```
y = "CurrentY" / 65536
```

where CurrentY is in the range from 0 to 65279 inclusive.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- At most once per second or
- At the end of the movement/transition.

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3.2.7.7. DriftCompensation Attribute

This attribute SHALL indicate what mechanism, if any, is in use for compensation for color/intensity drift over time.

3.2.7.8. CompensationText Attribute

This attribute SHALL contain a textual indication of what mechanism, if any, is in use to compensate for color/intensity drift over time.

3.2.7.9. ColorTemperatureMireds Attribute

This attribute SHALL indicate **a scaled inverse** of the current value of the color temperature. The unit of ColorTemperatureMireds is the mired (micro reciprocal degree), a.k.a. mirek (micro reciprocal kelvin). It is updated as fast as practical during commands that change the color.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- At most once per second or
- At the end of the movement/transition.

The color temperature value in kelvins SHALL be related to the ColorTemperatureMireds attribute in mired by the relationship

```
"Color temperature [K]" = "1,000,000" / "ColorTemperatureMireds"
```

where ColorTemperatureMireds is in the range from 1 to 65279 inclusive, giving a color temperature range from 1,000,000 K to 15.32 K.

If this attribute is implemented then the ColorMode attribute SHALL also be implemented.

3.2.7.10. ColorMode Attribute

This attribute SHALL indicate which attributes are currently determining the color of the device.

The value of the ColorMode attribute cannot be written directly - it is set upon reception of any command in section Commands to the appropriate mode for that command.

3.2.7.11. Options Attribute

This attribute SHALL indicate a bitmap that determines the default behavior of some cluster commands. Each command that is dependent on the Options attribute SHALL first construct a temporary Options bitmap that is in effect during the command processing. The temporary Options bitmap has the same format and meaning as the Options attribute, but includes any bits that may be overridden by command fields.

This attribute is meant to be changed only during commissioning.

Below is the format and description of the Options attribute and temporary Options bitmap and the effect on dependent commands.

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Command execution SHALL NOT continue beyond the Options processing if all of these criteria are true:

- The On/Off cluster exists on the same endpoint as this cluster.
- The OnOff attribute of the On/Off cluster, on this endpoint, is FALSE.
- The value of the ExecuteIfOff bit is 0.

3.2.7.11.1. ExecuteIfOff Bit

- 0 = Do not execute command if the OnOff attribute in the On/Off cluster is FALSE.
- 1 = Execute command if the OnOff attribute in the On/Off cluster is FALSE.

3.2.7.12. EnhancedCurrentHue Attribute

This attribute SHALL indicate the non-equidistant steps along the CIE 1931 color triangle, and it provides 16-bits precision.

The upper 8 bits of this attribute SHALL be used as an index in the implementation specific XY lookup table to provide the non-equidistant steps. The lower 8 bits SHALL be used to interpolate between these steps in a linear way in order to provide color zoom for the user.

To provide compatibility with clients not supporting EHUE, the CurrentHue attribute SHALL contain a hue value in the range 0 to 254, calculated from the EnhancedCurrentHue attribute.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- At most once per second or
- At the end of the movement/transition.

3.2.7.13. EnhancedColorMode Attribute

This attribute SHALL indicate which attributes are currently determining the color of the device.

To provide compatibility with clients not supporting EHUE, the ColorMode attribute SHALL indicate CurrentHueAndCurrentSaturation when the EnhancedCurrentHue attribute indicates EnhancedCurrentHueAndCurrentSaturation. For other values of the EnhancedCurrentHue attribute, the value of the ColorMode attribute SHALL be the same as the value of the EnhancedCurrentHue attribute. If the ColorMode attribute is changed, its new value SHALL be copied to the EnhancedColorMode attribute.

3.2.7.14. ColorLoopActive Attribute

This attribute SHALL indicate the current active status of the color loop. If this attribute has the value 0, the color loop SHALL NOT be active. If this attribute has the value 1, the color loop SHALL be active.

3.2.7.15. ColorLoopDirection Attribute

This attribute SHALL indicate the current direction of the color loop. If this attribute has the value

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Decrement, the EnhancedCurrentHue attribute SHALL be decremented. If this attribute has the value Increment, the EnhancedCurrentHue attribute SHALL be incremented.

3.2.7.16. ColorLoopTime Attribute

This attribute SHALL indicate the number of seconds it SHALL take to perform a full color loop, i.e., to cycle all values of the EnhancedCurrentHue attribute (between 0 and 65534).

3.2.7.17. ColorLoopStartEnhancedHue Attribute

This attribute SHALL indicate the value of the EnhancedCurrentHue attribute from which the color loop SHALL be started.

3.2.7.18. ColorLoopStoredEnhancedHue Attribute

This attribute SHALL indicate the value of the EnhancedCurrentHue attribute before the color loop was started. Once the color loop is complete, the EnhancedCurrentHue attribute SHALL be restored to this value.

3.2.7.19. ColorCapabilities Attribute

This attribute SHALL indicate the color control capabilities of the device.

Bits 0-4 of the ColorCapabilities attribute SHALL have the same values as the corresponding bits of the FeatureMap attribute. All other bits in ColorCapabilities SHALL be 0.

3.2.7.20. ColorTempPhysicalMinMireds Attribute

This attribute SHALL indicate the minimum mired value supported by the hardware. ColorTempPhysicalMinMireds corresponds to the maximum color temperature in kelvins supported by the hardware.
ColorTempPhysicalMinMireds <= ColorTemperatureMireds.

3.2.7.21. ColorTempPhysicalMaxMireds Attribute

This attribute SHALL indicate the maximum mired value supported by the hardware. ColorTempPhysicalMaxMireds corresponds to the minimum color temperature in kelvins supported by the hardware.
ColorTemperatureMireds <= ColorTempPhysicalMaxMireds.

3.2.7.22. CoupleColorTempToLevelMinMireds Attribute

This attribute SHALL indicate a lower bound on the value of the ColorTemperatureMireds attribute for the purposes of coupling the ColorTemperatureMireds attribute to the CurrentLevel attribute when the CoupleColorTempToLevel bit of the Options attribute of the Level Control cluster is equal to 1. When coupling the ColorTemperatureMireds attribute to the CurrentLevel attribute, this value SHALL correspond to a CurrentLevel value of 254 (100%).

This attribute SHALL be set such that the following relationship exists:
ColorTempPhysicalMinMireds <= CoupleColorTempToLevelMinMireds <= ColorTemperatureMireds

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Note that since this attribute is stored as a micro reciprocal degree (mired) value (i.e. color temperature in kelvins = 1,000,000 / CoupleColorTempToLevelMinMireds), the CoupleColorTempToLevelMinMireds attribute corresponds to an upper bound on the value of the color temperature in kelvins supported by the device.

3.2.7.23. StartUpColorTemperatureMireds Attribute

This attribute SHALL indicate the desired startup color temperature value the light SHALL use when it is supplied with power and this value SHALL be reflected in the ColorTemperatureMireds attribute. In addition, the ColorMode and EnhancedColorMode attributes SHALL be set to 2 (ColorTemperatureMireds). The values of the StartUpColorTemperatureMireds attribute are listed in the table below,

Value	Action on power up
1 to 65279	Set the ColorTemperatureMireds attribute to this value.
null	Set the ColorTemperatureMireds attribute to its previous value.

3.2.7.24. NumberOfPrimaries Attribute

This attribute SHALL indicate the number of color primaries implemented on this device. A value of null SHALL indicate that the number of primaries is unknown.

Where this attribute is implemented, the attributes below for indicating the “x” and “y” color values of the primaries SHALL also be implemented for each of the primaries from 1 to NumberOfPrimaries, without leaving gaps. Implementation of the Primary1Intensity attribute and subsequent intensity attributes is optional.

3.2.7.25. Primary1X Attribute

This attribute SHALL indicate the normalized chromaticity value x for this primary, as defined in the CIE xyY Color Space.

The value of x SHALL be related to the Primary1X attribute by the relationship

x = Primary1X / 65536 (Primary1X in the range 0 to 65279 inclusive)

3.2.7.26. Primary1Y Attribute

This attribute SHALL indicate the normalized chromaticity value y for this primary, as defined in the CIE xyY Color Space.

The value of y SHALL be related to the Primary1Y attribute by the relationship

y = Primary1Y / 65536 (Primary1Y in the range 0 to 65279 inclusive)

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3.2.7.27. Primary1Intensity Attribute

This attribute SHALL indicate a representation of the maximum intensity of this primary as defined in Section 3.1.3, “The Dimming Light Curve”, normalized such that the primary with the highest maximum intensity contains the value 254.

A value of null SHALL indicate that this primary is not available.

3.2.7.28. Primary2X, Primary2Y, Primary2Intensity, Primary3X, Primary3Y, Primary3Intensity, Primary4X, Primary4Y, Primary4Intensity, Primary5X, Primary5Y, Primary5Intensity, Primary6X, Primary6Y and Primary6Intensity Attributes

These attributes SHALL represent the capabilities of the 2nd, 3rd, 4th, 5th and 6th primaries, where present, in the same way as for the Primary1X, Primary1Y and Primary1Intensity attributes.

3.2.7.29. WhitePointX Attribute

This attribute SHALL indicate the normalized chromaticity value x, as defined in the CIE xyY Color Space, of the current white point of the device.

The value of x SHALL be related to the WhitePointX attribute by the relationship

x = WhitePointX / 65536 (WhitePointX in the range 0 to 65279 inclusive)

3.2.7.30. WhitePointY Attribute

This attribute SHALL indicate the normalized chromaticity value y, as defined in the CIE xyY Color Space, of the current white point of the device.

The value of y SHALL be related to the WhitePointY attribute by the relationship

y = WhitePointY / 65536 (WhitePointY in the range 0 to 65279 inclusive)

3.2.7.31. ColorPointRX Attribute

This attribute SHALL indicate the normalized chromaticity value x, as defined in the CIE xyY Color Space, of the red color point of the device.

The value of x SHALL be related to the ColorPointRX attribute by the relationship

x = ColorPointRX / 65536 (ColorPointRX in the range 0 to 65279 inclusive)

3.2.7.32. ColorPointRY Attribute

This attribute SHALL indicate the normalized chromaticity value y, as defined in the CIE xyY Color Space, of the red color point of the device.

The value of y SHALL be related to the ColorPointRY attribute by the relationship

y = ColorPointRY / 65536 (ColorPointRY in the range 0 to 65279 inclusive)

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3.2.7.33. ColorPointRIntensity Attribute

This attribute SHALL indicate a representation of the relative intensity of the red color point as defined in Section 3.1.3, “The Dimming Light Curve”, normalized such that the color point with the highest relative intensity contains the value 254.

A value of null SHALL indicate an invalid value.

3.2.7.34. ColorPointGX, ColorPointGY, ColorPointGIntensity, ColorPointBX, ColorPointBY and ColorPointBIntensity Attributes

These attributes SHALL represent the chromaticity values and intensities of the green and blue color points, in the same way as for the ColorPointRX, ColorPointRY and ColorPointRIntensity attributes.

If any one of these red, green or blue color point attributes is implemented then they SHALL all be implemented.

3.2.8. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**MoveToHue**	client ⇒ server	Y	O	HS
0x01	**MoveHue**	client ⇒ server	Y	O	HS
0x02	**StepHue**	client ⇒ server	Y	O	HS
0x03	**MoveToSaturation**	client ⇒ server	Y	O	HS
0x04	**MoveSaturation**	client ⇒ server	Y	O	HS
0x05	**StepSaturation**	client ⇒ server	Y	O	HS
0x06	**MoveToHueAndSaturation**	client ⇒ server	Y	O	HS
0x07	**MoveToColor**	client ⇒ server	Y	O	XY
0x08	**MoveColor**	client ⇒ server	Y	O	XY
0x09	**StepColor**	client ⇒ server	Y	O	XY
0x0A	**MoveToColorTemperature**	client ⇒ server	Y	O	CT
0x40	**EnhancedMoveToHue**	client ⇒ server	Y	O	EHUE
0x41	**EnhancedMoveHue**	client ⇒ server	Y	O	EHUE

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ID	Name	Direction	Response	Access	Conformance
0x42	**EnhancedStepHue**	client ⇒ server	Y	O	EHUE
0x43	**EnhancedMoveToHueAndSaturation**	client ⇒ server	Y	O	EHUE
0x44	**ColorLoopSet**	client ⇒ server	Y	O	CL
0x47	**StopMoveStep**	client ⇒ server	Y	O	HS | XY | CT
0x4B	**MoveColorTemperature**	client ⇒ server	Y	O	CT
0x4C	**StepColorTemperature**	client ⇒ server	Y	O	CT

3.2.8.1. Generic Usage Notes

When asked to change color via one of these commands, the implementation SHALL select a color, within the limits of the hardware of the device, which is as close as possible to that requested. The determination as to the true representations of color is out of the scope of this specification. However, as long as the color data fields of the received command are within the permitted range of this specification and no error condition applies, the resulting status code SHALL be SUCCESS.

For example the MoveToColorTemperature command: if the target color temperature is not achievable by the hardware then the color temperature SHALL be clipped at the physical minimum or maximum achievable (depending on the direction of the color temperature transition) when the device reaches that color temperature (which MAY be before the requested transition time).

If a color loop is active (i.e., the ColorLoopActive attribute is equal to 1), it SHALL only be stopped by sending a specific ColorLoopSet command frame with a request to deactivate the color loop (i.e., the color loop SHALL NOT be stopped on receipt of another command which has a 'stop' semantic such as the EnhancedMoveToHue command with MoveMode==Stop, or the StopMoveStep command). In addition, while a color loop is active, a manufacturer MAY choose to ignore incoming color commands which affect a change in hue.

3.2.8.2. Note on Change of ColorMode

The first action taken when any one of these commands is received is to change the ColorMode attribute to the appropriate value for the command (see individual commands). Note that, when moving from one color mode to another (e.g., CurrentX/CurrentY to CurrentHue/CurrentSaturation), the starting color for the command is formed by calculating the values of the new attributes (in this case CurrentHue, CurrentSaturation) from those of the old attributes (in this case CurrentX and CurrentY).

When moving from a mode to another mode that has a more restricted color range (e.g., Cur-

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rentX/CurrentY to CurrentHue/CurrentSaturation, or CurrentHue/CurrentSaturation to ColorTemperatureMireds) it is possible for the current color value to have no equivalent in the new mode. The behavior in such cases is manufacturer dependent, and therefore it is recommended to avoid color mode changes of this kind during use.

3.2.8.3. Use of the OptionsMask and OptionsOverride Fields

The OptionsMask and OptionsOverride Fields SHALL both be present. Default values are provided to interpret missing fields from legacy devices. A temporary Options bitmap SHALL be created from the Options attribute, using the OptionsMask and OptionsOverride Fields. Each bit of the temporary Options bitmap SHALL be determined as follows:

Each bit in the Options attribute SHALL determine the corresponding bit in the temporary Options bitmap, unless the OptionsMask field is present and has the corresponding bit set to 1, in which case the corresponding bit in the OptionsOverride field SHALL determine the corresponding bit in the temporary Options bitmap.

The resulting temporary Options bitmap SHALL then be processed as defined in section Options.

3.2.8.4. MoveToHue Command

This command will move the device to the requested hue using a transition.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Hue**	uint8	max 254			M
1	**Direction**	DirectionEnum	all			M
2	**TransitionTime**	uint16	max 65534			M
3	**OptionsMask**	OptionsBitmap	desc			M
4	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.4.1. Hue Field

This field SHALL indicate the hue to be moved to.

3.2.8.4.2. Direction Field

This field SHALL indicate the movement direction.

3.2.8.4.3. TransitionTime Field

This field SHALL indicate, in 1/10ths of a second, the time that SHALL be taken to move to the new hue.

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3.2.8.4.4. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.4.5. Effect on Receipt

On receipt of this command, a device SHALL also set the ColorMode attribute to the value 0 and then SHALL move from its current hue to the value given in the Hue field.

The movement SHALL be continuous, i.e., not a step function, and the time taken to move to the new hue SHALL be equal to the TransitionTime field.

As hue is effectively measured on a circle, the new hue MAY be moved to in either direction. The direction of hue change is given by the Direction field. If Direction is 'Shortest distance', the direction is taken that involves the shortest path round the circle. This case corresponds to expected normal usage. If Direction is 'Longest distance', the direction is taken that involves the longest path round the circle. This case can be used for 'rainbow effects'. In both cases, if both distances are the same, the Up direction SHALL be taken.

3.2.8.5. MoveHue Command

This command will change the hue of the device with a requested rate.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**MoveMode**	MoveModeEnum	all			M
1	**Rate**	uint8	all			M
2	**OptionsMask**	OptionsBitmap	desc			M
3	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.5.1. MoveMode Field

This field SHALL indicate the mode of movement.

3.2.8.5.2. Rate Field

This field SHALL indicate the rate of movement in steps per second. A step is a change in the device's hue of one unit.

3.2.8.5.3. OptionsMask and OptionsOverride field

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

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3.2.8.5.4. Effect on Receipt

On receipt of this command,

- If the MoveMode field is set to 1 (Up) or 3 (Down):
  - If the Rate field has a value of zero, the command has no effect and a response SHALL be returned with the status code set to INVALID_COMMAND.
  - Otherwise, a device SHALL set the ColorMode attribute to the value 0 (CurrentHueAndCurrentSaturation) and SHALL then move from its current hue in an up or down direction in a continuous fashion, as detailed in the table.
- If the MoveMode field is set to 0 (Stop), the Rate field SHALL be completely ignored.

MoveMode	Action on Receipt
Stop	If moving, stop, else ignore the command (i.e., the command is accepted but has no effect). This SHALL stop any ongoing hue and/or saturation transition.
Up	Increase the device's hue at the rate given in the Rate field. If the hue reaches the maximum allowed for the device, then wraparound and proceed from its minimum allowed value.
Down	Decrease the device's hue at the rate given in the Rate field. If the hue reaches the minimum allowed for the device, then wraparound and proceed from its maximum allowed value.

3.2.8.6. StepHue Command

This command will change the hue of the device using a step and transition.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**StepMode**	StepModeEnum	all			M
1	**StepSize**	uint8	all			M
2	**TransitionTime**	uint8	all			M
3	**OptionsMask**	OptionsBitmap	desc			M
4	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.6.1. StepMode Field

This field SHALL indicate the mode of the step to be performed.

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3.2.8.6.2. StepSize Field

This field SHALL indicate the change to be added to (or subtracted from) the current value of the device's hue.

3.2.8.6.3. TransitionTime Field

This field SHALL indicate, in 1/10ths of a second, the time that SHALL be taken to perform the step. A step is a change in the device's hue of Step size units.

NOTE	Here the TransitionTime data field is of data type uint8, where uint16 is more common for TransitionTime data fields in other clusters / commands.

3.2.8.6.4. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.6.5. Effect on Receipt

On receipt of this command, if the StepSize field has a value of zero, the command has no effect and a response SHALL be returned with the status code set to INVALID_COMMAND.

Otherwise, a device SHALL set the ColorMode attribute to the value 0 (CurrentHueAndCurrentSaturation) and SHALL then move from its current hue in an up or down direction by one step, as detailed in the table.

StepMode	Action on Receipt
Up	Increase the device's hue by one step, in a continuous fashion. If the hue value reaches the maximum value then wraparound and proceed from the minimum allowed value.
Down	Decrease the device's hue by one step, in a continuous fashion. If the hue value reaches the minimum value then wraparound and proceed from the maximum allowed value.

3.2.8.7. MoveToSaturation Command

This command will move the device to the requested saturation using a transition.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Saturation**	uint8	max 254			M
1	**TransitionTime**	uint16	max 65534			M
2	**OptionsMask**	Options-Bitmap	desc			M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
3	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.7.1. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.7.2. Effect on Receipt

On receipt of this command, a device SHALL set the ColorMode attribute to the value 0 (CurrentHueAndCurrentSaturation) and SHALL then move from its current saturation to the value given in the Saturation field.

The movement SHALL be continuous, i.e., not a step function, and the time taken to move to the new saturation SHALL be equal to the TransitionTime field, in 1/10ths of a second.

3.2.8.8. MoveSaturation Command

This command will change the saturation of the device with a requested rate.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**MoveMode**	MoveModeEnum	all			M
1	**Rate**	uint8	all			M
2	**OptionsMask**	OptionsBitmap	desc			M
3	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.8.1. MoveMode Field

This field SHALL indicate the mode of movement, as described in the MoveHue command.

3.2.8.8.2. Rate Field

This field SHALL indicate the rate of movement in steps per second. A step is a change in the device's saturation of one unit.

3.2.8.8.3. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

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3.2.8.8.4. Effect on Receipt

On receipt of this command,

- If the MoveMode field is set to 1 (Up) or 3 (Down)
  - If the Rate field has a value of zero, the command has no effect and a response SHALL be returned with the status code set to INVALID_COMMAND.
  - Otherwise, a device SHALL set the ColorMode attribute to the value 0 (CurrentHueAndCurrentSaturation) and SHALL then move from its current saturation in an up or down direction in a continuous fashion, as detailed in the table.
- If the MoveMode field is set to 0 (Stop), the Rate field SHALL be completely ignored.

MoveMode	Action on Receipt
Stop	If moving, stop, else ignore the command (i.e., the command is accepted but has no effect). This SHALL stop any ongoing hue and/or saturation transition.
Up	Increase the device's saturation at the rate given in the Rate field. If the saturation reaches the maximum allowed for the device, stop.
Down	Decrease the device's saturation at the rate given in the Rate field. If the saturation reaches the minimum allowed for the device, stop.

3.2.8.9. StepSaturation Command

This command will change the saturation of the device using a step and transition.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**StepMode**	StepModeEnum	all			M
1	**StepSize**	uint8	all			M
2	**TransitionTime**	uint8	all			M
3	**OptionsMask**	OptionsBitmap	desc			M
4	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.9.1. StepMode Field

This field SHALL indicate the mode of the step to be performed, as described in the StepHue command.

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3.2.8.9.2. StepSize Field

This field SHALL indicate the change to be added to (or subtracted from) the current value of the device's saturation.

3.2.8.9.3. TransitionTime Field

This field SHALL indicate, in 1/10ths of a second, the time that SHALL be taken to perform the step. A step is a change in the device's saturation of Step size units.

NOTE	Here the TransitionTime data field is of data type uint8, where uint16 is more common for TransitionTime data fields in other clusters / commands.

3.2.8.9.4. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.9.5. Effect on Receipt

On receipt of this command, if the StepSize field has a value of zero, the command has no effect and a response SHALL be returned with the status code set to INVALID_COMMAND.

Otherwise, a device SHALL set the ColorMode attribute to the value 0 (CurrentHueAndCurrentSaturation) and SHALL then move from its current saturation in an up or down direction by one step.

StepMode	Action on Receipt
Up	Increase the device's saturation by one step, in a continuous fashion. However, if the saturation value is already the maximum value then do nothing.
Down	Decrease the device's saturation by one step, in a continuous fashion. However, if the saturation value is already the minimum value then do nothing.

3.2.8.10. MoveToHueAndSaturation Command

This command will move the device to the requested hue and saturation using a transition.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Hue**	uint8	max 254			M
1	**Saturation**	uint8	max 254			M
2	**TransitionTime**	uint16	max 65534			M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
3	**OptionsMask**	OptionsBitmap	desc			M
4	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.10.1. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.10.2. Effect on Receipt

On receipt of this command, a device SHALL set the ColorMode attribute to the value 0 (CurrentHueAndCurrentSaturation) and SHALL then move from its current hue and saturation to the values given in the Hue and Saturation fields.

The movement SHALL be continuous, i.e., not a step function, and the time taken to move to the new color SHALL be equal to the TransitionTime field, in 1/10ths of a second.

The path through color space taken during the transition is not specified, but it is recommended that the shortest path is taken through color space, i.e., movement is 'in a straight line' across the CIE xyY Color Space.

3.2.8.11. MoveToColor Command

This command will move the device to the requested color using a transition.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ColorX**	uint16	max 65279			M
1	**ColorY**	uint16	max 65279			M
2	**TransitionTime**	uint16	max 65534			M
3	**OptionsMask**	OptionsBitmap	desc			M
4	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.11.1. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

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3.2.8.11.2. Effect on Receipt

On receipt of this command, a device SHALL set the value of the ColorMode attribute, where implemented, to 1 (CurrentXAndCurrentY), and SHALL then move from its current color to the color given in the ColorX and ColorY fields.

The movement SHALL be continuous, i.e., not a step function, and the time taken to move to the new color SHALL be equal to the TransitionTime field, in 1/10ths of a second.

The path through color space taken during the transition is not specified, but it is recommended that the shortest path is taken through color space, i.e., movement is 'in a straight line' across the CIE xyY Color Space.

3.2.8.12. MoveColor Command

This command will change the color of the device with a requested rate.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**RateX**	int16	all			M
1	**RateY**	int16	all			M
2	**OptionsMask**	Options-Bitmap	desc			M
3	**OptionsOverride**	Options-Bitmap	desc			M

3.2.8.12.1. RateX Field

This field SHALL indicate the rate of movement in steps per second. A step is a change in the device's CurrentX attribute of one unit.

3.2.8.12.2. RateY Field

This field SHALL indicate the rate of movement in steps per second. A step is a change in the device's CurrentY attribute of one unit.

3.2.8.12.3. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.12.4. Effect on Receipt

On receipt of this command, if both the RateX and RateY fields contain a value of zero, no movement SHALL be carried out, and the command execution SHALL have no effect other than stopping the operation of any previously received command of this cluster. While this command (when used with both the RateX and RateY fields set to zero) can be used to stop any operation of this cluster, it is recommended to instead use the StopMoveStep command to achieve this goal.

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Otherwise, a device SHALL set the value of the ColorMode attribute, where implemented, to 1 (CurrentXAndCurrentY), and SHALL then move from its current color in a continuous fashion according to the rates specified. This movement SHALL continue until the target color for the next step cannot be implemented on this device.

3.2.8.13. StepColor Command

This command will change the color of the device using a step and transition.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**StepX**	int16	all			M
1	**StepY**	int16	all			M
2	**TransitionTime**	uint16	max 65534			M
3	**OptionsMask**	OptionsBitmap	desc			M
4	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.13.1. StepX and StepY Fields

These fields SHALL indicate the change to be added to the device's CurrentX attribute and CurrentY attribute respectively.

3.2.8.13.2. TransitionTime Field

The field SHALL indicate, in 1/10ths of a second, the time that SHALL be taken to perform the color change.

3.2.8.13.3. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.13.4. Effect on Receipt

On receipt of this command, if both the StepX and StepY fields contain a value of zero, the command has no effect and a response SHALL be returned with the status code set to INVALID_COMMAND.

Otherwise, a device SHALL set the value of the ColorMode attribute, where implemented, to 1 (CurrentXAndCurrentY), and SHALL then move from its current color by the color step indicated.

The movement SHALL be continuous, i.e., not a step function, and the time taken to move to the new color SHALL be equal to the TransitionTime field, in 1/10ths of a second.

The path through color space taken during the transition is not specified, but it is recommended

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that the shortest path is taken through color space, i.e., movement is 'in a straight line' across the CIE xyY Color Space.

Note also that if the required step is larger than can be represented by signed 16-bit integers then more than one step command SHOULD be issued.

3.2.8.14. MoveToColorTemperature Command

This command will move the device to the requested color temperature using a transition.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ColorTemperatureMireds**	uint16	max 65279			M
1	**TransitionTime**	uint16	max 65534			M
2	**OptionsMask**	OptionsBitmap	desc			M
3	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.14.1. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.14.2. Effect on Receipt

On receipt of this command, a device SHALL set the value of the ColorMode attribute, where implemented, to 2 (ColorTemperatureMireds), and SHALL then move from its current color to the color given by the ColorTemperatureMired field.

The movement SHALL be continuous, i.e., not a step function, and the time taken to move to the new color SHALL be equal to the TransitionTime field, in 1/10ths of a second.

By definition of this color mode, the path through color space taken during the transition is along the 'Black Body Line'.

If the ColorTemperatureMireds field is less than the value of the ColorTempPhysicalMinMireds attribute, the value of the ColorTempPhysicalMinMireds attribute SHALL be used as the final target for the ColorTemperatureMireds attribute.

If the ColorTemperatureMireds field is greater than the value of the ColorTempPhysicalMaxMireds attribute, the value of the ColorTempPhysicalMaxMireds attribute SHALL be used as the final target for the ColorTemperatureMireds attribute.

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3.2.8.15. EnhancedMoveToHue Command

This command allows the light to be moved in a smooth continuous transition from their current hue to a target hue.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**EnhancedHue**	uint16	all			M
1	**Direction**	DirectionEnum	all			M
2	**TransitionTime**	uint16	max 65534			M
3	**OptionsMask**	OptionsBitmap	desc			M
4	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.15.1. EnhancedHue Field

This field SHALL indicate the target enhanced hue for the light.

3.2.8.15.2. Direction Field

This field SHALL indicate the movement direction.

3.2.8.15.3. TransitionTime Field

This field SHALL indicate the transition time, as described in the MoveToHue command.

3.2.8.15.4. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.15.5. Effect on Receipt

On receipt of this command, a device SHALL set the ColorMode attribute to 0 (CurrentHueAndCurrentSaturation) and set the EnhancedColorMode attribute to the value 3 (EnhancedCurrentHueAndCurrentSaturation). The device SHALL then move from its current enhanced hue to the value given in the EnhancedHue field.

The movement SHALL be continuous, i.e., not a step function, and the time taken to move to the new enhanced hue SHALL be equal to the TransitionTime field.

3.2.8.16. EnhancedMoveHue Command

This command allows the light to start a continuous transition starting from their current hue.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**MoveMode**	MoveModeEnum	desc			M
1	**Rate**	uint16	all			M
2	**OptionsMask**	OptionsBitmap	desc			M
3	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.16.1. MoveMode Field

This field SHALL indicate the mode of movement, as described in the MoveHue command.

3.2.8.16.2. Rate field

This field SHALL indicate the rate of movement in steps per second. A step is a change in the enhanced hue of a device by one unit.

3.2.8.16.3. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.16.4. Effect on receipt

On receipt of this command,

- If the MoveMode field is set to 1 (Up) or 3 (Down)
  - If the Rate field has a value of zero, the command has no effect and a response SHALL be returned with the status code set to INVALID_COMMAND.
  - Otherwise, a device SHALL set the ColorMode attribute to 0 (CurrentHueAndCurrentSaturation) and set the EnhancedColorMode attribute to the value 3 (EnhancedCurrentHueAndCurrentSaturation). The device SHALL then move from its current enhanced hue in an up or down direction in a continuous fashion, as detailed in the table.
- If the MoveMode field is set to 0 (Stop), the Rate field SHALL be completely ignored.

MoveMode	Action on Receipt
Stop	If moving, stop, else ignore the command (i.e., the command is accepted but has no effect). This SHALL stop any ongoing hue and/or saturation transition.

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MoveMode	Action on Receipt
Up	Increase the device's enhanced hue at the rate given in the Rate field. If the enhanced hue reaches the maximum allowed for the device, wraparound and proceed from its minimum allowed value.
Down	Decrease the device's enhanced hue at the rate given in the Rate field. If the hue reaches the minimum allowed for the device, wraparound and proceed from its maximum allowed value.

3.2.8.17. EnhancedStepHue Command

This command allows the device to be moved in a stepped transition from their current hue.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**StepMode**	StepModeEnum	desc			M
1	**StepSize**	uint16	all			M
2	**TransitionTime**	uint16	max 65534			M
3	**OptionsMask**	OptionsBitmap	desc			M
4	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.17.1. StepMode Field

This field SHALL indicate the mode of the step to be performed, as described in the StepHue command.

3.2.8.17.2. StepSize Field

This field SHALL indicate the change to be added to (or subtracted from) the current value of the device's enhanced hue.

3.2.8.17.3. TransitionTime Field

The field SHALL indicate, in units of 1/10ths of a second, the time that SHALL be taken to perform the step. A step is a change to the device's enhanced hue of a magnitude corresponding to the StepSize field.

NOTE	Here TransitionTime data field is of data type uint16, while the TransitionTime data field of the StepHue command is of data type uint8.

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3.2.8.17.4. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.17.5. Effect on Receipt

On receipt of this command, if the StepSize field has a value of zero, the command has no effect and a response SHALL be returned with the status code set to INVALID_COMMAND.

Otherwise, a device SHALL set the ColorMode attribute to 0 (CurrentHueAndCurrentSaturation) and the EnhancedColorMode attribute to the value 3 (EnhancedCurrentHueAndCurrentSaturation). The device SHALL then move from its current enhanced hue in an up or down direction by one step, as detailed in the table.

StepMode	Action on Receipt
Up	Increase the device's enhanced hue by one step. If the enhanced hue reaches the maximum allowed for the device, wraparound and proceed from its minimum allowed value.
Down	Decrease the device's enhanced hue by one step. If the hue reaches the minimum allowed for the device, wraparound and proceed from its maximum allowed value.

3.2.8.18. EnhancedMoveToHueAndSaturation Command

This command allows the light to be moved in a smooth continuous transition from their current hue to a target hue and from their current saturation to a target saturation.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**EnhancedHue**	uint16	all			M
1	**Saturation**	uint8	max 254			M
2	**TransitionTime**	uint16	max 65534			M
3	**OptionsMask**	OptionsBitmap	desc			M
4	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.18.1. EnhancedHue Field

This field SHALL indicate the target enhanced hue for the light.

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3.2.8.18.2. Saturation Field

This field SHALL indicate the saturation, as described in the MoveToHueAndSaturation command.

3.2.8.18.3. TransitionTime Field

This field SHALL indicate the transition time, as described in the MoveToHue command.

3.2.8.18.4. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.18.5. Effect on Receipt

On receipt of this command, a device SHALL set the ColorMode attribute to the value 0 (CurrentHueAndCurrentSaturation) and set the EnhancedColorMode attribute to the value 3 (EnhancedCurrentHueAndCurrentSaturation). The device SHALL then move from its current enhanced hue and saturation to the values given in the EnhancedHue and Saturation fields.

The movement SHALL be continuous, i.e., not a step function, and the time taken to move to the new color SHALL be equal to the TransitionTime field, in 1/10ths of a second.

The path through color space taken during the transition is not specified, but it is recommended that the shortest path is taken through color space, i.e., movement is 'in a straight line' across the CIE xyY Color Space.

3.2.8.19. ColorLoopSet Command

This command allows a color loop to be activated such that the color light cycles through its range of hues.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**UpdateFlags**	UpdateFlagsBitmap	all			M
1	**Action**	ColorLoopActionEnum	all			M
2	**Direction**	ColorLoopDirectionEnum	all			M
3	**Time**	uint16	all			M
4	**StartHue**	uint16	all			M
5	**OptionsMask**	OptionsBitmap	desc			M
6	**OptionsOverride**	OptionsBitmap	desc			M

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3.2.8.19.1. UpdateFlags Field

This field SHALL indicate which color loop attributes to update (from the values supplied in the other fields, see field descriptions below) before the color loop is started.

3.2.8.19.2. Action Field

This field SHALL indicate the action to take for the color loop.

3.2.8.19.3. Direction Field

This field SHALL indicate the direction for the color loop.

3.2.8.19.4. Time Field

This field SHALL indicate the number of seconds over which to perform a full color loop.

3.2.8.19.5. Start Hue Field

This field SHALL indicate the starting hue to use for the color loop.

3.2.8.19.6. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.19.7. Effect on Receipt

On receipt of this command, the device SHALL first update its color loop attributes according to the value of the UpdateFlags field, as follows.

- If the UpdateDirection sub-field is set to 1, the device SHALL set the ColorLoopDirection attribute to the value of the Direction field.
- If the UpdateTime sub-field is set to 1, the device SHALL set the ColorLoopTime attribute to the value of the Time field.
- If the UpdateStartHue sub-field is set to 1, the device SHALL set the ColorLoopStartEnhancedHue attribute to the value of the StartHue field.
- If the UpdateAction sub-field of the UpdateFlags field is set to 1, the device SHALL adhere to the action specified in the Action field, as follows.
  - If the value of the Action field is set to 0 and the color loop is active, i.e. the ColorLoopActive attribute is set to 1, the device SHALL de-active the color loop, set the ColorLoopActive attribute to 0 and set the EnhancedCurrentHue attribute to the value of the ColorLoopStoredEnhancedHue attribute.
  - If the value of the Action field is set to 0 and the color loop is inactive, i.e. the ColorLoopActive attribute is set to 0, the device SHALL ignore the action update component of the command, i.e. the EnhancedCurrentHue attribute SHALL NOT be updated (contrasting with the case described in the previous bullet).
  - If the value of the Action field is set to 1, the device SHALL set the ColorLoopStoredEnhancedHue attribute to the value of the EnhancedCurrentHue attribute, set the Color-

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LoopActive attribute to 1 and activate the color loop, starting from the value of the ColorLoopStartEnhancedHue attribute.

- If the value of the Action field is set to 2, the device SHALL set the ColorLoopStoredEnhancedHue attribute to the value of the EnhancedCurrentHue attribute, set the ColorLoopActive attribute to 1 and activate the color loop, starting from the value of the EnhancedCurrentHue attribute.

If the color loop is active, the device SHALL cycle over the complete range of values of the EnhancedCurrentHue attribute in the direction of the ColorLoopDirection attribute over the time specified in the ColorLoopTime attribute. The level of increments/decrements is application specific.

If the color loop is active (and stays active), the device SHALL immediately react on updates of the ColorLoopDirection and ColorLoopTime attributes.

3.2.8.20. StopMoveStep Command

This command is provided to allow MoveTo and Step commands to be stopped.

NOTE This automatically provides symmetry to the Level Control cluster.

NOTE The StopMoveStep command has no effect on an active color loop.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**OptionsMask**	OptionsBitmap	desc			M
1	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.20.1. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.20.2. Effect on Receipt

Upon receipt of this command, any MoveTo, Move or Step command currently in process SHALL be terminated. The values of the CurrentHue, EnhancedCurrentHue and CurrentSaturation attributes SHALL be left at their present value upon receipt of the StopMoveStep command, and the RemainingTime attribute SHALL be set to zero.

3.2.8.21. MoveColorTemperature Command

This command allows the color temperature of the light to be moved at a specified rate.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**MoveMode**	MoveModeEnum	all			M
1	**Rate**	uint16	all			M
2	**ColorTemperatureMinimumMireds**	uint16	max 65279			M
3	**ColorTemperatureMaximumMireds**	uint16	max 65279			M
4	**OptionsMask**	OptionsBitmap	desc			M
5	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.21.1. MoveMode Field

This field SHALL indicate the mode of movement, as described in the MoveHue command.

3.2.8.21.2. Rate Field

This field SHALL indicate the rate of movement in steps per second. A step is a change in the color temperature of a device by one unit.

3.2.8.21.3. ColorTemperatureMinimumMireds Field

This field SHALL indicate a lower bound on the ColorTemperatureMireds attribute (≡ an upper bound on the color temperature in kelvins) for the current move operation such that:

ColorTempPhysicalMinMireds <= ColorTemperatureMinimumMireds field <= ColorTemperatureMireds

As such if the move operation takes the ColorTemperatureMireds attribute towards the value of the ColorTemperatureMinimumMireds field it SHALL be clipped so that the above invariant is satisfied. If the ColorTemperatureMinimumMireds field is set to 0, ColorTempPhysicalMinMireds SHALL be used as the lower bound for the ColorTemperatureMireds attribute.

3.2.8.21.4. ColorTemperatureMaximumMireds Field

This field SHALL indicate an upper bound on the ColorTemperatureMireds attribute (≡ a lower bound on the color temperature in kelvins) for the current move operation such that:

ColorTemperatureMireds <= ColorTemperatureMaximumMireds field <= ColorTempPhysicalMaxMireds

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As such if the move operation takes the ColorTemperatureMireds attribute towards the value of the ColorTemperatureMaximumMireds field it SHALL be clipped so that the above invariant is satisfied. If the ColorTemperatureMaximumMireds field is set to 0, ColorTempPhysicalMaxMireds SHALL be used as the upper bound for the ColorTemperatureMireds attribute.

3.2.8.21.5. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.21.6. Effect on Receipt

On receipt of this command,

- If the MoveMode field is set to 1 (Up) or 3 (Down)
  - If the Rate field has a value of zero, the command has no effect and a response SHALL be returned with the status code set to INVALID_COMMAND.
  - Otherwise, a device SHALL set both the ColorMode and EnhancedColorMode attributes to 2 (ColorTemperatureMireds) The device SHALL then move from its current color temperature in an up or down direction in a continuous fashion, as detailed in the table.
- If the MoveMode field is set to 0 (Stop), the Rate field SHALL be completely ignored.

MoveMode	Action on Receipt
Stop	If moving, stop the operation, else ignore the command (i.e., the command is accepted but has no effect).
Up	Increase the ColorTemperatureMireds attribute (≡ decrease the color temperature in kelvins) at the rate given in the Rate field. If the ColorTemperatureMireds attribute reaches the maximum allowed for the device (via either the ColorTemperatureMaximumMireds field or the ColorTempPhysicalMaxMireds attribute), the move operation SHALL be stopped.
Down	Decrease the ColorTemperatureMireds attribute (≡ increase the color temperature in kelvins) at the rate given in the Rate field. If the ColorTemperatureMireds attribute reaches the minimum allowed for the device (via either the ColorTemperatureMinimumMireds field or the ColorTempPhysicalMinMireds attribute), the move operation SHALL be stopped.

3.2.8.22. StepColorTemperature Command

This command allows the color temperature of the light to be stepped with a specified step size.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**StepMode**	StepModeEnum	all			M
1	**StepSize**	uint16	all			M
2	**TransitionTime**	uint16	max 65534			M
3	**ColorTemperatureMinimumMireds**	uint16	max 65279			M
4	**ColorTemperatureMaximumMireds**	uint16	max 65279			M
5	**OptionsMask**	OptionsBitmap	desc			M
6	**OptionsOverride**	OptionsBitmap	desc			M

3.2.8.22.1. StepMode Field

This field SHALL indicate the mode of the step to be performed, as described in the StepHue command.

3.2.8.22.2. StepSize Field

This field SHALL indicate the change to be added to (or subtracted from) the current value of the device's color temperature.

3.2.8.22.3. TransitionTime Field

This field SHALL indicate, in units of 1/10ths of a second, the time that SHALL be taken to perform the step. A step is a change to the device's color temperature of a magnitude corresponding to the StepSize field.

3.2.8.22.4. ColorTemperatureMinimumMireds Field

This field SHALL indicate a lower bound on the ColorTemperatureMireds attribute (≡ an upper bound on the color temperature in kelvins) for the current step operation such that:

ColorTempPhysicalMinMireds <= ColorTemperatureMinimumMireds field <= ColorTemperatureMireds

As such if the step operation takes the ColorTemperatureMireds attribute towards the value of the ColorTemperatureMinimumMireds field it SHALL be clipped so that the above invariant is satisfied. If the ColorTemperatureMinimumMireds field is set to 0, ColorTempPhysicalMinMireds SHALL be

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used as the lower bound for the ColorTemperatureMireds attribute.

3.2.8.22.5. ColorTemperatureMaximumMireds Field

This field SHALL indicate an upper bound on the ColorTemperatureMireds attribute (≡ a lower bound on the color temperature in kelvins) for the current step operation such that:

ColorTemperatureMireds ≤ ColorTemperatureMaximumMireds field ≤ ColorTempPhysicalMaxMireds

As such if the step operation takes the ColorTemperatureMireds attribute towards the value of the ColorTemperatureMaximumMireds field it SHALL be clipped so that the above invariant is satisfied. If the ColorTemperatureMaximumMireds field is set to 0, ColorTempPhysicalMaxMireds SHALL be used as the upper bound for the ColorTemperatureMireds attribute.

3.2.8.22.6. OptionsMask and OptionsOverride Fields

These fields SHALL be processed according to section Use of the OptionsMask and OptionsOverride Fields.

3.2.8.22.7. Effect on Receipt

On receipt of this command, if the StepSize field has a value of zero, the command has no effect and a response SHALL be returned with the status code set to INVALID_COMMAND.

Otherwise, a device SHALL set both the ColorMode and EnhancedColorMode attributes to 2 (ColorTemperatureMireds). The device SHALL then move from its current color temperature in an up or down direction by one step, as detailed in the table.

StepMode	Action on Receipt
Up	Increase the ColorTemperatureMireds attribute (≡ decrease the color temperature in kelvins) by one step. If the ColorTemperatureMireds attribute reaches the maximum allowed for the device (via either the ColorTemperatureMaximumMireds field or the ColorTempPhysicalMaxMireds attribute), the step operation SHALL be stopped.
Down	Decrease the ColorTemperatureMireds attribute (≡ increase the color temperature in kelvins) by one step. If the ColorTemperatureMireds attribute reaches the minimum allowed for the device (via either the ColorTemperatureMinimumMireds field or the ColorTempPhysicalMinMireds attribute), the step operation SHALL be stopped.

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Chapter 4. HVAC

The Cluster Library is made of individual chapters such as this one. References between chapters are made using a *X.Y* notation where *X* is the chapter and *Y* is the sub-section within that chapter.

4.1. General Description

4.1.1. Introduction

The clusters specified in this document are for use typically in HVAC applications, but MAY be used in any application domain.

4.1.2. Terms

Precooling: Cooling a building in the early (cooler) part of the day, so that the thermal mass of the building decreases cooling needs in the later (hotter) part of the day.

4.1.3. Cluster List

This section lists the HVAC specific clusters as specified in this chapter.

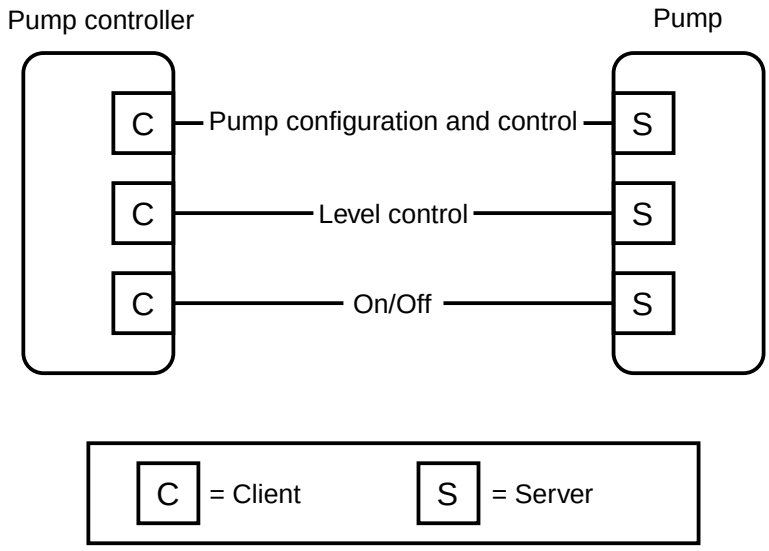
Table 8. Overview of the HVAC Clusters

Cluster ID	Cluster Name	Description
0x0200	Pump Configuration and Control	An interface for configuring and controlling pumps.
0x0201	Thermostat	An interface for configuring and controlling the functionality of a thermostat
0x0202	Fan Control	An interface for controlling a fan
0x0204	Thermostat User Interface Configuration	An interface for configuring the user interface of a thermostat (which MAY be remote from the thermostat)
0x0081	Valve Configuration and Control	An interface for configuring and controlling the functionality of a valve

4.2. Pump Configuration and Control Cluster

The Pump Configuration and Control cluster provides an interface for the setup and control of pump devices, and the automatic reporting of pump status information. Note that control of pump speed is not included – speed is controlled by the On/Off and Level Control clusters.

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Note: Device names are examples for illustration purposes only

Figure 16. Typical Usage of Pump Configuration and Control Cluster

4.2.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added
2	All Hubs changes
3	New data model format and notation, added additional events
4	Added feature map
5	Removed P quality

4.2.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	PCC

4.2.3. Cluster ID

ID	Name
0x0200	Pump Configuration and Control

4.2.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

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Bit	Code	Feature	Conformance	Summary
0	**PRSCONST**	ConstantPressure	O.a+	Supports operating in constant pressure mode
1	**PRSCOMP**	CompensatedPressure	O.a+	Supports operating in compensated pressure mode
2	**FLW**	ConstantFlow	O.a+	Supports operating in constant flow mode
3	**SPD**	ConstantSpeed	O.a+	Supports operating in constant speed mode
4	**TEMP**	ConstantTemperature	O.a+	Supports operating in constant temperature mode
5	**AUTO**	Automatic	O	Supports operating in automatic mode
6	**LOCAL**	LocalOperation	O	Supports operating using local settings

4.2.5. Dependencies

Where external pressure, flow, and temperature measurements are processed by this cluster (see ControlMode Attribute), these are provided by a Pressure Measurement cluster, a Flow Measurement cluster, and a Temperature Measurement client cluster, respectively. These 3 client clusters are used for connection to a remote sensor device. The pump is able to use the sensor measurement provided by a remote sensor for regulation of the pump speed.

Note that control of the pump setpoint is not included in this cluster – the On/Off and Level Control clusters (see Typical Usage of Pump Configuration and Control Cluster) MAY be used by a pump device to turn it on and off and control its setpoint. Note that the Pump Configuration and Control cluster MAY override on/off/setpoint settings for specific operation modes (see OperationMode Attribute for detailed description of the operation and control of the pump.).

4.2.6. Data Types

4.2.6.1. PumpStatusBitmap Type

This data type is derived from map16.

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Bit	Name	Summary	Conformance
0	**DeviceFault**	A fault related to the system or pump device is detected.	M
1	**SupplyFault**	A fault related to the supply to the pump is detected.	M
2	**SpeedLow**	Setpoint is too low to achieve.	M
3	**SpeedHigh**	Setpoint is too high to achieve.	M
4	**LocalOverride**	Device control is overridden by hardware, such as an external STOP button or via a local HMI.	M
5	**Running**	Pump is currently running	M
6	**RemotePressure**	A remote pressure sensor is used as the sensor for the regulation of the pump.	M
7	**RemoteFlow**	A remote flow sensor is used as the sensor for the regulation of the pump.	M
8	**RemoteTemperature**	A remote temperature sensor is used as the sensor for the regulation of the pump.	M

4.2.6.1.1. DeviceFault Bit

If this bit is set, it MAY correspond to an event in the range 2-16, see Events.

4.2.6.1.2. SupplyFault Bit

If this bit is set, it MAY correspond to an event in the range 0-1 or 13, see Events.

4.2.6.1.3. LocalOverride Bit

While this bit is set, the EffectiveOperationMode is adjusted to Local. Any request changing OperationMode SHALL generate a FAILURE error status until LocalOverride is cleared on the physical device. When LocalOverride is cleared, the device SHALL return to the operation mode set in OperationMode.

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4.2.6.1.4. RemotePressure Bit

If this bit is set, EffectiveControlMode is ConstantPressure and the setpoint for the pump is interpreted as a percentage of the range of the remote sensor ([MinMeasuredValue – MaxMeasuredValue]).

4.2.6.1.5. RemoteFlow Bit

If this bit is set, EffectiveControlMode is ConstantFlow, and the setpoint for the pump is interpreted as a percentage of the range of the remote sensor ([MinMeasuredValue – MaxMeasuredValue]).

4.2.6.1.6. RemoteTemperature Bit

If this bit is set, EffectiveControlMode is ConstantTemperature, and the setpoint for the pump is interpreted as a percentage of the range of the remote sensor ([MinMeasuredValue – MaxMeasuredValue])

4.2.6.2. OperationModeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Normal**	The pump is controlled by a setpoint, as defined by a connected remote sensor or by the ControlMode attribute.	M
1	**Minimum**	This value sets the pump to run at the minimum possible speed it can without being stopped.	SPD
2	**Maximum**	This value sets the pump to run at its maximum possible speed.	SPD
3	**Local**	This value sets the pump to run with the local settings of the pump, regardless of what these are.	LOCAL

4.2.6.2.1. Normal Value

If the pump is running in this operation mode the setpoint is an internal variable which MAY be controlled between 0% and 100%, e.g., by means of the Level Control cluster

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4.2.6.3. ControlModeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**ConstantSpeed**	The pump is running at a constant speed.	SPD
1	**ConstantPressure**	The pump will regulate its speed to maintain a constant differential pressure over its flanges.	PRSCONST
2	**ProportionalPressure**	The pump will regulate its speed to maintain a constant differential pressure over its flanges.	PRSCOMP
3	**ConstantFlow**	The pump will regulate its speed to maintain a constant flow through the pump.	FLW
5	**ConstantTemperature**	The pump will regulate its speed to maintain a constant temperature.	TEMP
7	**Automatic**	The operation of the pump is automatically optimized to provide the most suitable performance with respect to comfort and energy savings.	AUTO

4.2.6.3.1. ConstantSpeed Value

The setpoint is interpreted as a percentage of the range derived from the [MinConstSpeed – MaxConstSpeed] attributes.

4.2.6.3.2. ConstantPressure Value

The setpoint is interpreted as a percentage of the range of the sensor used for this control mode. In case of the internal pressure sensor, this will be the range derived from the [MinConstPressure – MaxConstPressure] attributes. In case of a remote pressure sensor, this will be the range derived from the [MinMeasuredValue – MaxMeasuredValue] attributes of the remote pressure sensor.

4.2.6.3.3. ProportionalPressure Value

The setpoint is interpreted as a percentage of the range derived of the [MinCompPressure – Max-

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CompPressure] attributes. The internal setpoint will be lowered (compensated) dependent on the flow in the pump (lower flow ⇒ lower internal setpoint).

4.2.6.3.4. ConstantFlow Value

The setpoint is interpreted as a percentage of the range of the sensor used for this control mode. In case of the internal flow sensor, this will be the range derived from the [MinConstFlow – MaxConstFlow] attributes. In case of a remote flow sensor, this will be the range derived from the [MinMeasuredValue – MaxMeasuredValue] attributes of the remote flow sensor.

4.2.6.3.5. ConstantTemperature Value

The setpoint is interpreted as a percentage of the range of the sensor used for this control mode. In case of the internal temperature sensor, this will be the range derived from the [MinConstTemp – MaxConstTemp] attributes. In case of a remote temperature sensor, this will be the range derived from the [MinMeasuredValue – MaxMeasuredValue] attributes of the remote temperature sensor.

4.2.6.3.6. Automatic Value

This behavior is manufacturer defined. The pump can be stopped by setting the setpoint of the level control cluster to 0, or by using the On/Off cluster. If the pump is started (at any setpoint), the speed of the pump is entirely determined by the pump.

4.2.7. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**MaxPressure**	int16	all	F X	null	R V	M
0x0001	**MaxSpeed**	uint16	all	F X	null	R V	M
0x0002	**MaxFlow**	uint16	all	F X	null	R V	M
0x0003	**MinConstPressure**	int16	all	F X	null	R V	PRSCONST, [AUTO]
0x0004	**MaxConstPressure**	int16	all	F X	null	R V	PRSCONST, [AUTO]
0x0005	**MinCompPressure**	int16	all	F X	null	R V	PRSCOMP, [AUTO]
0x0006	**MaxCompPressure**	int16	all	F X	null	R V	PRSCOMP, [AUTO]
0x0007	**MinConstSpeed**	uint16	all	F X	null	R V	SPD, [AUTO]
0x0008	**MaxConstSpeed**	uint16	all	F X	null	R V	SPD, [AUTO]
0x0009	**MinConstFlow**	uint16	all	F X	null	R V	FLW, [AUTO]

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x000A	**MaxConstFlow**	uint16	all	F X	null	R V	FLW, [AUTO]
0x000B	**MinConstTemp**	int16	min -27315	F X	null	R V	TEMP, [AUTO]
0x000C	**MaxConstTemp**	int16	min -27315	F X	null	R V	TEMP, [AUTO]
0x0010	**PumpStatus**	PumpStatusBitmap	desc		0	R V	O
0x0011	**EffectiveOperationMode**	OperationModeEnum	desc	N	desc	R V	M
0x0012	**EffectiveControlMode**	ControlModeEnum	desc	N	desc	R V	M
0x0013	**Capacity**	int16	all	X	null	R V	M
0x0014	**Speed**	uint16	all	X	null	R V	O
0x0015	**LifetimeRunningHours**	uint24	all	N X	0	RW VM	O
0x0016	**Power**	uint24	all	X	null	R V	O
0x0017	**LifetimeEnergyConsumed**	uint32	all	N X	0	RW VM	O
0x0020	**OperationMode**	OperationModeEnum	desc	N	0	RW VM	M
0x0021	**ControlMode**	ControlModeEnum	desc	N	0	RW VM	O
0x0022	**AlarmMask**						D

4.2.7.1. MaxPressure Attribute

This attribute specifies the maximum pressure the pump can achieve. It is a physical limit, and does not apply to any specific control mode or operation mode.

Valid range is -3,276.7 kPa to 3,276.7 kPa (steps of 0.1 kPa).
This attribute SHALL be null if the value is invalid.

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4.2.7.2. MaxSpeed Attribute

This attribute specifies the maximum speed the pump can achieve. It is a physical limit, and does not apply to any specific control mode or operation mode.

Valid range is 0 to 65,534 RPM (steps of 1 RPM).
This attribute SHALL be null if the value is invalid.

4.2.7.3. MaxFlow Attribute

This attribute specifies the maximum flow the pump can achieve. It is a physical limit, and does not apply to any specific control mode or operation mode.

Valid range is 0 m^3/h to 6,553.4 m^3/h (steps of 0.1 m^3/h).
This attribute SHALL be null if the value is invalid.

4.2.7.4. MinConstPressure Attribute

This attribute specifies the minimum pressure the pump can achieve when it is working with the ControlMode attribute set to ConstantPressure.

Valid range is –3,276.7 kPa to 3,276.7 kPa (steps of 0.1 kPa).
This attribute SHALL be null if the value is invalid.

4.2.7.5. MaxConstPressure Attribute

This attribute specifies the maximum pressure the pump can achieve when it is working with the ControlMode attribute set to ConstantPressure.

Valid range is –3,276.7 kPa to 3,276.7 kPa (steps of 0.1 kPa).
This attribute SHALL be null if the value is invalid.

4.2.7.6. MinCompPressure Attribute

This attribute specifies the minimum compensated pressure the pump can achieve when it is working with the ControlMode attribute set to ProportionalPressure.

Valid range is –3,276.7 kPa to 3,276.7 kPa (steps of 0.1 kPa).
This attribute SHALL be null if the value is invalid.

4.2.7.7. MaxCompPressure Attribute

This attribute specifies the maximum compensated pressure the pump can achieve when it is working with the ControlMode attribute set to ProportionalPressure.

Valid range is –3,276.7 kPa to 3,276.7 kPa (steps of 0.1 kPa).
This attribute SHALL be null if the value is invalid.

4.2.7.8. MinConstSpeed Attribute

This attribute specifies the minimum speed the pump can achieve when it is working with the Con-

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trolMode attribute set to ConstantSpeed.

Valid range is 0 to 65,534 RPM (steps of 1 RPM).
This attribute SHALL be null if the value is invalid.

4.2.7.9. MaxConstSpeed Attribute

This attribute specifies the maximum speed the pump can achieve when it is working with the ControlMode attribute set to ConstantSpeed.

Valid range is 0 to 65,534 RPM (steps of 1 RPM).
This attribute SHALL be null if the value is invalid.

4.2.7.10. MinConstFlow Attribute

This attribute specifies the minimum flow the pump can achieve when it is working with the ControlMode attribute set to ConstantFlow.

Valid range is 0 m^3/h to 6,553.4 m^3/h (steps of 0.1 m^3/h).
This attribute SHALL be null if the value is invalid.

4.2.7.11. MaxConstFlow Attribute

This attribute specifies the maximum flow the pump can achieve when it is working with the ControlMode attribute set to ConstantFlow.

Valid range is 0 m^3/h to 6,553.4 m^3/h (steps of 0.1 m^3/h).
This attribute SHALL be null if the value is invalid.

4.2.7.12. MinConstTemp Attribute

This attribute specifies the minimum temperature the pump can maintain in the system when it is working with the ControlMode attribute set to ConstantTemperature.

Valid range is –273.15 °C to 327.67 °C (steps of 0.01 °C).
This attribute SHALL be null if the value is invalid.

4.2.7.13. MaxConstTemp Attribute

This attribute specifies the maximum temperature the pump can maintain in the system when it is working with the ControlMode attribute set to ConstantTemperature.

MaxConstTemp SHALL be greater than or equal to MinConstTemp

Valid range is –273.15 °C to 327.67 °C (steps of 0.01 °C).
This attribute SHALL be null if the value is invalid.

4.2.7.14. PumpStatus Attribute

This attribute specifies the activity status of the pump functions as listed in PumpStatusBitmap. Where a pump controller function is active, the corresponding bit SHALL be set to 1. Where a pump

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controller function is not active, the corresponding bit SHALL be set to 0.

4.2.7.15. EffectiveOperationMode Attribute

This attribute specifies current effective operation mode of the pump as defined in OperationModeEnum.

The value of the EffectiveOperationMode attribute is the same as the OperationMode attribute, unless one of the following points are true:

- The pump is physically set to run with the local settings
- The LocalOverride bit in the PumpStatus attribute is set,

See OperationMode Attribute and ControlMode Attribute for a detailed description of the operation and control of the pump.

4.2.7.16. EffectiveControlMode Attribute

This attribute specifies the current effective control mode of the pump as defined in ControlModeEnum.

This attribute contains the control mode that currently applies to the pump. It will have the value of the ControlMode attribute, unless one of the following points are true:

- The ControlMode attribute is set to Automatic. In this case, the value of the EffectiveControlMode SHALL match the behavior of the pump.
- A remote sensor is used as the sensor for regulation of the pump. In this case, EffectiveControlMode will display ConstantPressure, ConstantFlow or ConstantTemperature if the remote sensor is a pressure sensor, a flow sensor or a temperature sensor respectively, regardless of the value of the ControlMode attribute.

In case the ControlMode attribute is not included on the device and no remote sensors are connected, the value of the EffectiveControlMode SHALL match the vendor-specific behavior of the pump.

See OperationMode Attribute and ControlMode Attribute for detailed a description of the operation and control of the pump.

4.2.7.17. Capacity Attribute

This attribute specifies the actual capacity of the pump as a percentage of the effective maximum setpoint value. It is updated dynamically as the speed of the pump changes.

If the value is not available (the measurement or estimation of the speed is done in the pump), this attribute will indicate the null value.

Valid range is 0 % to 163.835% (0.005 % granularity). Although this attribute is a signed value, values of capacity less than zero have no physical meaning.

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4.2.7.18. Speed Attribute

This attribute specifies the actual speed of the pump measured in RPM. It is updated dynamically as the speed of the pump changes.

If the value is not available (the measurement or estimation of the speed is done in the pump), this attribute will indicate the null value.

Valid range is 0 to 65,534 RPM.

4.2.7.19. LifetimeRunningHours Attribute

This attribute specifies the accumulated number of hours that the pump has been powered and the motor has been running. It is updated dynamically as it increases. It is preserved over power cycles of the pump. If LifeTimeRunningHours rises above maximum value it “rolls over” and starts at 0 (zero).

This attribute is writeable, in order to allow setting to an appropriate value after maintenance. If the value is not available, this attribute will indicate the null value.

Valid range is 0 to 16,777,214 hrs.

4.2.7.20. Power Attribute

This attribute specifies the actual power consumption of the pump in Watts. The value of this attribute is updated dynamically as the power consumption of the pump changes.

This attribute is read only. If the value is not available (the measurement of power consumption is not done in the pump), this attribute will indicate the null value.

Valid range is 0 to 16,777,214 Watts.

4.2.7.21. LifetimeEnergyConsumed Attribute

This attribute specifies the accumulated energy consumption of the pump through the entire lifetime of the pump in kWh. The value of the LifetimeEnergyConsumed attribute is updated dynamically as the energy consumption of the pump increases. If LifetimeEnergyConsumed rises above maximum value it “rolls over” and starts at 0 (zero).

This attribute is writeable, in order to allow setting to an appropriate value after maintenance.

Valid range is 0 kWh to 4,294,967,294 kWh.
This attribute SHALL be null if the value is unknown.

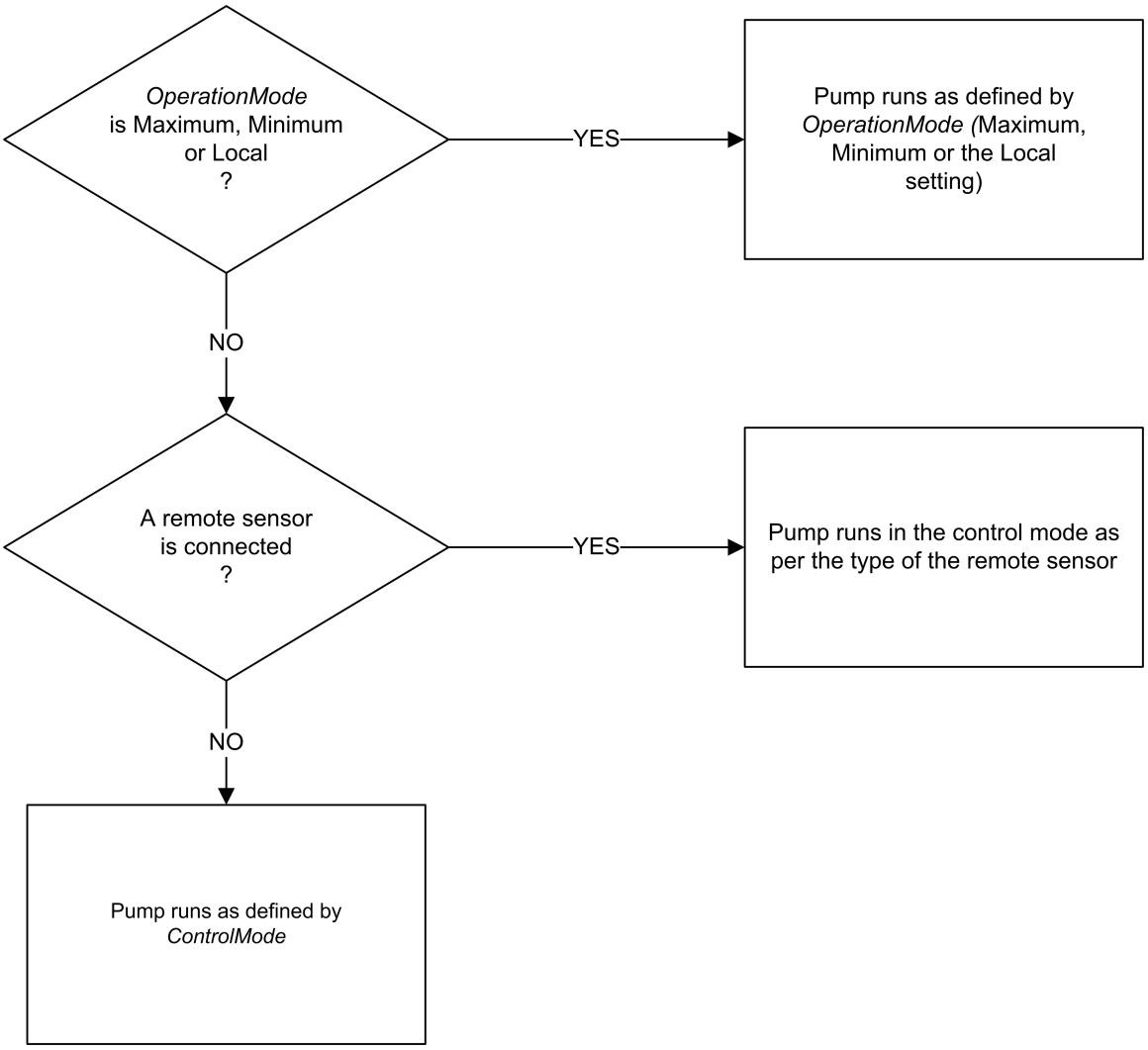
4.2.7.22. OperationMode Attribute

This attribute specifies the operation mode of the pump as defined in OperationModeEnum.

The actual operating mode of the pump is a result of the setting of the attributes OperationMode, ControlMode and the optional connection of a remote sensor. The operation and control is prioritized as shown in the scheme below:

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Priority Scheme of Pump Operation and Control



If this attribute is Maximum, Minimum or Local, the OperationMode attribute decides how the pump is operated.

If this attribute is Normal and a remote sensor is connected to the pump, the type of the remote sensor decides the control mode of the pump. A connected remote pressure sensor will make the pump run in control mode Constant pressure and vice versa for flow and temperature type sensors. This is regardless of the setting of the ControlMode attribute.

If this attribute is Normal and no remote sensor is connected, the control mode of the pump is decided by the ControlMode attribute.

OperationMode MAY be changed at any time, even when the pump is running. The behavior of the pump at the point of changing the value of this attribute is vendor-specific.

In the case a device does not support a specific operation mode, the write interaction to this attribute with an unsupported operation mode value SHALL be ignored and a response containing the status of CONSTRAINT_ERROR SHALL be returned.

4.2.7.23. ControlMode Attribute

This attribute specifies the control mode of the pump as defined in ControlModeEnum.

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See OperationMode Attribute for a detailed description of the operation and control of the pump.

ControlMode MAY be changed at any time, even when the pump is running. The behavior of the pump at the point of changing is vendor-specific.

In the case a device does not support a specific control mode, the write interaction to this attribute with an unsupported control mode value SHALL be ignored and a response containing the status of CONSTRAINT_ERROR SHALL be returned.

4.2.8. Events

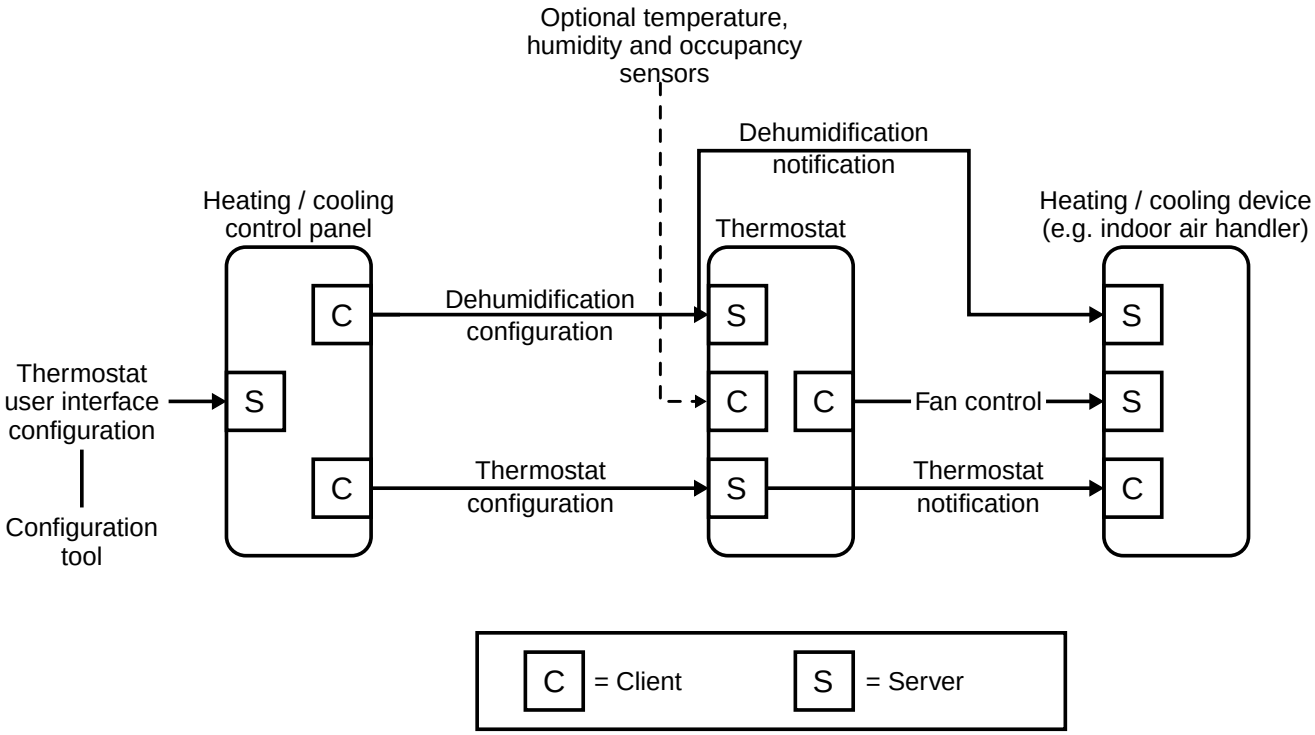
ID	Name	Priority	Quality	Access	Conformance
0x00	**SupplyVoltageLow**	INFO		V	O
0x01	**SupplyVoltageHigh**	INFO		V	O
0x02	**PowerMissingPhase**	INFO		V	O
0x03	**SystemPressureLow**	INFO		V	O
0x04	**SystemPressureHigh**	INFO		V	O
0x05	**DryRunning**	CRITICAL		V	O
0x06	**MotorTemperatureHigh**	INFO		V	O
0x07	**PumpMotorFatalFailure**	CRITICAL		V	O
0x08	**ElectronicTemperatureHigh**	INFO		V	O
0x09	**PumpBlocked**	CRITICAL		V	O
0x0A	**SensorFailure**	INFO		V	O
0x0B	**ElectronicNonFatalFailure**	INFO		V	O
0x0C	**ElectronicFatalFailure**	CRITICAL		V	O
0x0D	**GeneralFault**	INFO		V	O
0x0E	**Leakage**	INFO		V	O
0x0F	**AirDetection**	INFO		V	O

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ID	Name	Priority	Quality	Access	Conformance
0x10	**TurbineOperation**	INFO		V	O

4.3. Thermostat Cluster

This cluster provides an interface to the functionality of a thermostat.



Note: Device names are examples for illustration purposes only

Figure 17. Example Usage of the Thermostat and Related Clusters

4.3.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added; fixed some defaults; CCB 1823, 1480
2	CCB 1981 2186 2249 2250 2251; NFR Thermostat Setback
3	CCB 2477 2560 2773 2777 2815 2816 3029
4	All Hubs changes

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Revision	Description
5	New data model format and notation, added FeatureMap, collapsed attribute sets, clarified edge cases around limits, default value of xxxSetpointLimit now respects AbsxxxSetpointLimit
6	Introduced the LTNE feature and adapted text (spec issue #5778)
7	Update constraints on local temperature calibration and minimum setpoint deadband and introduced Presets and MatterScheduleConfiguration features
8	Added comment regarding writing to ControlSequenceOfOperation
9	Removed AlarmMask attribute and AlarmCodeBitmap Type
10	Removed Zigbee related elements and P quality, deprecated ThermostatProgrammingOperationMode and removed ProgrammingOperationModeBitmap. and deprecated Setback feature and related elements.
11	Added support for Thermostat suggestions. Added events.

4.3.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	TSTAT

4.3.3. Cluster ID

ID	Name
0x0201	Thermostat

4.3.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

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Bit	Code	Feature	Conformance	Summary
0	**HEAT**	Heating	AUTO, O.a+	Thermostat is capable of managing a heating device
1	**COOL**	Cooling	AUTO, O.a+	Thermostat is capable of managing a cooling device
2	**OCC**	Occupancy	O	Supports Occupied and Unoccupied setpoints
4	**SB**	Setback	D	
5	**AUTO**	AutoMode	O	Supports a System Mode of Auto
6	**LTNE**	LocalTemperatureNotExposed	O	Thermostat does not expose the LocalTemperature Value in the LocalTemperature attribute
7	**MSCH**	MatterScheduleConfiguration	O	Supports enhanced schedules
8	**PRES**	Presets	O	Thermostat supports setpoint presets
9	**TEVT**	Events	P, O	Thermostat supports events
10	**TSUGGEST**	ThermostatSuggestions	[PRES]	Thermostat supports suggestions

4.3.4.1. LocalTemperatureNotExposed Feature

This feature indicates that the Calculated Local Temperature used internally is unavailable to report externally, for example due to the temperature control being done by a separate subsystem which does not offer a view into the currently measured temperature, but allows setpoints to be provided.

4.3.4.2. MatterScheduleConfiguration Feature

This feature indicates that the thermostat is capable of schedules. If this feature is supported, the thermostat SHALL support a mechanism to do time synchronization.

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4.3.4.3. ThermostatSuggestions Feature

This feature indicates that the thermostat can process suggestions. If this feature is supported, the thermostat SHALL support a mechanism to do time synchronization.

4.3.5. Units of Temperature

Temperatures within this cluster are represented by types using units of degree Celsius.

The temperature data type used throughout this cluster is defined in the Derived Data Types section of the Data Model.

The following temperature-related data types are also defined in and used throughout this cluster:

- TemperatureDifference
- SignedTemperature
- UnsignedTemperature

While temperature values SHALL be transferred over the air using these types, this does not limit how Thermostats may display or store temperature values. Thermostats which display temperature values SHOULD follow the recommendations in Conversion of Temperature Values for Display.

CAUTION	*Calculations with temperature attributes* Where calculations or comparisons are performed, attribute values SHALL be converted to a common type. In many cases, it is not sufficient to simply use the integer representation as the scaling from °C to integer value differs.

4.3.6. Setpoint Limits

There are a number of attributes which impose limits on setpoint values. This imposes constraints which SHALL be maintained by any mechanism which modifies a limit or setpoint. Individual attribute descriptions detail the actions to be taken should a conflict arise while modifying the value.

User configurable limits SHALL be within device limits:

- AbsMinHeatSetpointLimit <= MinHeatSetpointLimit <= MaxHeatSetpointLimit <= AbsMaxHeatSetpointLimit
- AbsMinCoolSetpointLimit <= MinCoolSetpointLimit <= MaxCoolSetpointLimit <= AbsMaxCoolSetpointLimit

Setpoints SHALL be within user configurable limits:

- MinHeatSetpointLimit <= OccupiedHeatingSetpoint <= MaxHeatSetpointLimit
- MinCoolSetpointLimit <= OccupiedCoolingSetpoint <= MaxCoolSetpointLimit
- MinHeatSetpointLimit <= UnoccupiedHeatingSetpoint <= MaxHeatSetpointLimit
- MinCoolSetpointLimit <= UnoccupiedCoolingSetpoint <= MaxCoolSetpointLimit

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If, and only if, the AUTO feature is supported, a deadband SHALL be maintained between Heating and Cooling setpoints and limits:

- AbsMinHeatSetpointLimit <= (AbsMinCoolSetpointLimit - MinSetpointDeadBand)
- AbsMaxHeatSetpointLimit <= (AbsMaxCoolSetpointLimit - MinSetpointDeadBand)
- MinHeatSetpointLimit <= (MinCoolSetpointLimit - MinSetpointDeadBand)
- MaxHeatSetpointLimit <= (MaxCoolSetpointLimit - MinSetpointDeadBand)
- OccupiedHeatingSetpoint <= (OccupiedCoolingSetpoint - MinSetpointDeadBand)
- UnoccupiedHeatingSetpoint <= (UnoccupiedCoolingSetpoint - MinSetpointDeadBand)

4.3.7. Re-evaluation of Current Thermostat Suggestion

This section describes when the server SHALL re-evaluate the value of the CurrentThermostatSuggestion attribute and what the new value SHALL be set to.

The server SHALL re-evaluate the value of the CurrentThermostatSuggestion attribute when any of the following happen:

1. The value of the ThermostatSuggestions attribute is updated.
2. Any timers scheduled by a previous re-evaluation expire.
3. The state of the server changes in a way that would affect the results of the re-evaluation.

When re-evaluating the value of the CurrentThermostatSuggestion attribute:

1. If the server does not have its time synchronized, it SHALL set the CurrentThermostatSuggestion attribute to null.
2. Otherwise,
   a. Any timers scheduled by a previous re-evaluation SHALL be canceled.
   b. The server SHALL remove any entries in the ThermostatSuggestions attribute whose ExpirationTime is less than or equal to the current timestamp in UTC.
   c. If the value of the ThermostatSuggestions attribute has no entries with a value in the EffectiveTime field that is less than or equal to the current timestamp in UTC
      i. The CurrentThermostatSuggestion attribute SHALL be set to null.
   d. Otherwise,
      i. The server SHALL update the CurrentThermostatSuggestion attribute to the value of one of the entries whose EffectiveTime is less than or equal to the current timestamp in UTC. See Section 4.3.8, “Guidelines For Determination of Current Thermostat Suggestion” for which entry in the entries the server MAY choose to be the current thermostat suggestion.
   e. The server SHALL schedule a timer to expire at the earliest of the set of times that are greater than the current timestamp in UTC and are present in either the EffectiveTime or the ExpirationTime fields of the entries in the ThermostatSuggestions attribute. When the timer expires, the server SHALL re-evaluate the suggestions as described in Section 4.3.7,

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"Re-evaluation of Current Thermostat Suggestion" and set the CurrentThermostatSuggestion attribute accordingly.

4.3.8. Guidelines For Determination of Current Thermostat Suggestion

The following is the recommended procedure that the Thermostat SHOULD use to resolve conflicts when there are multiple entries in the ThermostatSuggestions attribute which may be applicable at the same time:

1. If the list of entries in the ThermostatSuggestions attribute has multiple suggestions with the same PresetHandle, the server SHOULD calculate the number of entries with the same PresetHandle. The server SHOULD choose the entry with the most recent EffectiveTime from the entries with the highest count of the same PresetHandle.
2. If there is only one entry in the ThermostatSuggestions attribute, whose PresetHandle matches the PresetHandle in an entry of the Presets attribute and has a PresetScenario that is one of Occupied, Sleep, Wake or GoingToSleep (i.e. indicating a user is home) the server SHOULD choose that entry.
3. If there are multiple entries in the ThermostatSuggestions attribute, whose PresetHandle matches the PresetHandle in an entry of the Presets attribute and has a PresetScenario that is one of Occupied, Sleep, Wake or GoingToSleep (i.e. indicating a user is home) the server SHOULD choose one of these entries based on the other rules listed, such as highest count or most recent EffectiveTime.
4. The server MAY also decide to choose an entry with the most recent EffectiveTime.
5. In addition to the above guidelines, a thermostat MAY implement some additional decision making policies to determine the CurrentThermostatSuggestion.
6. Once the server has determined an appropriate entry as the current thermostat suggestion, it SHALL set the CurrentThermostatSuggestion to the chosen entry.
7. Otherwise, if the above steps or any additional manufacturer specific decision making, does not result in the server being able to determine which entry to follow, the thermostat SHALL NOT follow any of the suggestions, set the CurrentThermostatSuggestion attribute to null and set the ThermostatSuggestionNotFollowingReason attribute to ConflictingSuggestions.

4.3.9. Dependencies

For remote temperature sensing, the Temperature Measurement client cluster MAY be included on the same endpoint.

For occupancy sensing, the Occupancy Sensing client cluster MAY be included on the same endpoint.

4.3.10. Data Types

4.3.10.1. ACErrorCodeBitmap Type

This data type is derived from map32.

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Bit	Name	Summary	Conformance
0	**CompressorFail**	Compressor Failure or Refrigerant Leakage	M
1	**RoomSensorFail**	Room Temperature Sensor Failure	M
2	**OutdoorSensorFail**	Outdoor Temperature Sensor Failure	M
3	**CoilSensorFail**	Indoor Coil Temperature Sensor Failure	M
4	**FanFail**	Fan Failure	M

4.3.10.1.1. CoolingStage Bits

These bits SHALL indicate what stage of cooling the HVAC system is using.

- 00 = Cool Stage 1
- 01 = Cool Stage 2
- 10 = Cool Stage 3
- 11 = Reserved

4.3.10.1.2. HeatingStage Bits

These bits SHALL indicate what stage of heating the HVAC system is using.

- 00 = Heat Stage 1
- 01 = Heat Stage 2
- 10 = Heat Stage 3
- 11 = Reserved

4.3.10.1.3. HeatingIsHeatPump Bit

This bit SHALL indicate whether the HVAC system is conventional or a heat pump.

- 0 = Conventional
- 1 = Heat Pump

4.3.10.1.4. HeatingUsesFuel Bit

This bit SHALL indicate whether the HVAC system uses fuel.

- 0 = Does not use fuel
- 1 = Uses fuel

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4.3.10.2. OccupancyBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**Occupied**	Indicates the occupancy state	M

4.3.10.2.1. Occupied Bit

If this bit is set, it SHALL indicate the occupied state else if the bit if not set, it SHALL indicate the unoccupied state.

4.3.10.3. PresetTypeFeaturesBitmap Type

This data type is derived from map16.

Bit	Name	Summary	Conformance
0	**Automatic**	Preset may be automatically activated by the thermostat	M
1	**SupportsNames**	Preset supports user-provided names	M

4.3.10.4. RelayStateBitmap Type

This data type is derived from map16.

Bit	Name	Summary	Conformance
0	**Heat**	Heat Stage On	M
1	**Cool**	Cool Stage On	M
2	**Fan**	Fan Stage On	M
3	**HeatStage2**	Heat 2nd Stage On	M
4	**CoolStage2**	Cool 2nd Stage On	M
5	**FanStage2**	Fan 2nd Stage On	M
6	**FanStage3**	Fan 3rd Stage On	M

4.3.10.5. RemoteSensingBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**LocalTemperature**	Calculated Local Temperature is derived from a remote node	M

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Bit	Name	Summary	Conformance
1	**OutdoorTemperature**	OutdoorTemperature is derived from a remote node	OutdoorTemperature
2	**Occupancy**	Occupancy is derived from a remote node	OCC

4.3.10.6. ScheduleTypeFeaturesBitmap Type

This data type is derived from map16.

Bit	Name	Summary	Conformance
0	**SupportsPresets**	Supports presets	[PRES].b+
1	**SupportsSetpoints**	Supports setpoints	O.b+
2	**SupportsNames**	Supports user-provided names	O
3	**SupportsOff**	Supports transitioning to SystemModeOff	O

4.3.10.6.1. SupportsPresets Bit

This bit SHALL indicate that any ScheduleStruct with a SystemMode field whose value matches the SystemMode field on the encompassing ScheduleTypeStruct supports specifying presets on ScheduleTransitionStructs contained in its Transitions field.

4.3.10.6.2. SupportsSetpoints Bit

This bit SHALL indicate that any ScheduleStruct with a SystemMode field whose value matches the SystemMode field on the encompassing ScheduleTypeStruct supports specifying setpoints on ScheduleTransitionStructs contained in its Transitions field.

4.3.10.6.3. SupportsNames Bit

This bit SHALL indicate that any ScheduleStruct with a SystemMode field whose value matches the SystemMode field on the encompassing ScheduleTypeStruct supports setting the value of the Name field.

4.3.10.6.4. SupportsOff Bit

This bit SHALL indicate that any ScheduleStruct with a SystemMode field whose value matches the SystemMode field on the encompassing ScheduleTypeStruct supports setting its SystemMode field to Off.

4.3.10.7. ScheduleDayOfWeekBitmap Type

This data type is derived from map8.

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Bit	Name	Summary	Conformance
0	**Sunday**	Sunday	M
1	**Monday**	Monday	M
2	**Tuesday**	Tuesday	M
3	**Wednesday**	Wednesday	M
4	**Thursday**	Thursday	M
5	**Friday**	Friday	M
6	**Saturday**	Saturday	M
7	**Away**	Away or Vacation	M

4.3.10.8. ScheduleModeBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**HeatSetpointPresent**	Adjust Heat Setpoint	M
1	**CoolSetpointPresent**	Adjust Cool Setpoint	M

4.3.10.9. ThermostatSuggestionNotFollowingReasonBitmap Type

This data type is derived from map16.

Bit	Name	Summary	Conformance
0	**DemandResponseEvent**	Thermostat is responding to a Demand Response event	M
1	**OngoingHold**	Thermostat has an ongoing setpoint hold	M
2	**Schedule**	Thermostat is following a schedule	M
3	**Occupancy**	Thermostat is following the occupancy signal	M
4	**VacationMode**	Thermostat is set to Vacation mode	M
5	**TimeOfUseCostSavings**	Thermostat is following a Time Of Use based cost savings plan	M
6	**PreCoolingOrPreHeating**	Thermostat is precooling or preheating based on an energy forecast signal	M

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Bit	Name	Summary	Conformance
7	**ConflictingSuggestions**	Thermostat has conflicting suggestions and is unable to choose one	M

4.3.10.10. ACCapacityFormatEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**BTUh**	British Thermal Unit per Hour	O

4.3.10.11. ACCompressorTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Unknown**	Unknown compressor type	O
1	**T1**	Max working ambient 43 °C	O
2	**T2**	Max working ambient 35 °C	O
3	**T3**	Max working ambient 52 °C	O

4.3.10.12. ACLouverPositionEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
1	**Closed**	Fully Closed	O
2	**Open**	Fully Open	O
3	**Quarter**	Quarter Open	O
4	**Half**	Half Open	O
5	**ThreeQuarters**	Three Quarters Open	O

4.3.10.13. ACRefrigerantTypeEnum Type

This data type is derived from enum8.

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Value	Name	Summary	Conformance
0	**Unknown**	Unknown Refrigerant Type	O
1	**R22**	R22 Refrigerant	O
2	**R410a**	R410a Refrigerant	O
3	**R407c**	R407c Refrigerant	O

4.3.10.14. ACTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Unknown**	Unknown AC Type	O
1	**CoolingFixed**	Cooling and Fixed Speed	O
2	**HeatPumpFixed**	Heat Pump and Fixed Speed	O
3	**CoolingInverter**	Cooling and Inverter	O
4	**HeatPumpInverter**	Heat Pump and Inverter	O

4.3.10.15. SetpointRaiseLowerModeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Heat**	Adjust Heat Setpoint	HEAT
1	**Cool**	Adjust Cool Setpoint	COOL
2	**Both**	Adjust Heat Setpoint and Cool Setpoint	HEAT | COOL

4.3.10.16. ControlSequenceOfOperationEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**CoolingOnly**	Heat and Emergency are not possible	[COOL]
1	**CoolingWithReheat**	Heat and Emergency are not possible	[COOL]

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Value	Name	Summary	Conformance
2	**HeatingOnly**	Cool and precooling (see Terms) are not possible	[HEAT]
3	**HeatingWithReheat**	Cool and precooling are not possible	[HEAT]
4	**CoolingAndHeating**	All modes are possible	[HEAT & COOL]
5	**CoolingAndHeating-WithReheat**	All modes are possible	[HEAT & COOL]

NOTE

CoolingAndHeating

A thermostat indicating it supports CoolingAndHeating (or CoolingAndHeatingWithReheat) SHOULD be able to request heating or cooling on demand and will usually support the Auto SystemMode.

Systems which support cooling **or** heating, requiring external intervention to change modes or where the whole building must be in the same mode, SHOULD report CoolingOnly or HeatingOnly based on the current capability.

4.3.10.17. PresetScenarioEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
1	**Occupied**	The thermostat-controlled area is occupied	M
2	**Unoccupied**	The thermostat-controlled area is unoccupied	M
3	**Sleep**	Users are likely to be sleeping	M
4	**Wake**	Users are likely to be waking up	M
5	**Vacation**	Users are on vacation	M
6	**GoingToSleep**	Users are likely to be going to sleep	M
254	**UserDefined**	Custom presets	M

4.3.10.17.1. Undefined Value

This value SHALL have no specified scenario.

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4.3.10.17.2. Occupied Value

This value SHALL indicate the preset for periods when the thermostat's temperature-controlled area is occupied. It is intended for thermostats that can automatically determine occupancy.

4.3.10.17.3. Unoccupied Value

This value SHALL indicate the preset for periods when the thermostat's temperature-controlled area is unoccupied. It is intended for thermostats that can automatically determine occupancy.

4.3.10.17.4. Sleep Value

This value SHALL indicate the preset for periods when users are likely to be asleep.

4.3.10.17.5. Wake Value

This value SHALL indicate the preset for periods when users are likely to be waking up.

4.3.10.17.6. Vacation Value

This value SHALL indicate the preset for periods when users are on vacation, or otherwise out-of-home for extended periods of time.

4.3.10.17.7. GoingToSleep Value

This value SHALL indicate the preset for periods when users are likely to be going to sleep.

4.3.10.17.8. UserDefined Value

This value SHALL indicate a free-form preset; when set, the Name field on PresetStruct SHALL NOT be null.

4.3.10.18. SetpointChangeSourceEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Manual**	Manual, user-initiated setpoint change via the thermostat	O
1	**Schedule**	Schedule/internal programming-initiated setpoint change	[MSCH]
2	**External**	Externally-initiated setpoint change (e.g., DRLC cluster command, attribute write)	O

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4.3.10.19. StartOfWeekEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Sunday**		M
1	**Monday**		M
2	**Tuesday**		M
3	**Wednesday**		M
4	**Thursday**		M
5	**Friday**		M
6	**Saturday**		M

4.3.10.20. SystemModeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Off**	The Thermostat does not generate demand for Cooling or Heating	O
1	**Auto**	Demand is generated for either Cooling or Heating, as required	AUTO
3	**Cool**	Demand is only generated for Cooling	[COOL]
4	**Heat**	Demand is only generated for Heating	[HEAT]
5	**EmergencyHeat**	2nd stage heating is in use to achieve desired temperature	[HEAT]
6	**Precooling**	(see Terms)	[COOL]
7	**FanOnly**		O
8	**Dry**		O
9	**Sleep**		O

4.3.10.21. ThermostatRunningModeEnum Type

This data type is derived from enum8.

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Value	Name	Summary	Conformance
0	**Off**	The Thermostat does not generate demand for Cooling or Heating	O
3	**Cool**	Demand is only generated for Cooling	[COOL]
4	**Heat**	Demand is only generated for Heating	[HEAT]

4.3.10.22. TemperatureSetpointHoldEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**SetpointHoldOff**	Follow scheduling program	M
1	**SetpointHoldOn**	Maintain current setpoint, regardless of schedule transitions	M

4.3.10.23. PresetStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**PresetHandle**	octstr	max 16	X			M
1	**PresetScenario**	PresetScenarioEnum	all				M
2	**Name**	string	max 64	X	null		O
3	**CoolingSetpoint**	temperature	desc		26°C		COOL
4	**HeatingSetpoint**	temperature	desc		20°C		HEAT
5	**BuiltIn**	bool	all	X	false		M

4.3.10.23.1. PresetHandle Field

This field SHALL indicate a device generated identifier for this preset. It SHALL be unique on the device, and SHALL NOT be reused after the associated preset has been deleted.

This field SHALL only be null when the encompassing PresetStruct is appended to the Presets attribute for the purpose of creating a new Preset. Refer to Presets for the creation of Preset han-

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dles.

4.3.10.23.2. PresetScenario Field

This field SHALL indicate the associated PresetScenarioEnum value for this preset.

4.3.10.23.3. Name Field

This field SHALL indicate a name provided by a user. The null value SHALL indicate no name.

Within each subset of presets sharing the same PresetScenario field value, there SHALL NOT be any presets with the same value, including null as a value, in the Name field.

4.3.10.23.4. CoolingSetpoint Field

This field SHALL indicate the cooling setpoint for the preset. Refer to Setpoint Limits for value constraints.

4.3.10.23.5. HeatingSetpoint Field

This field SHALL indicate the heating setpoint for the preset. Refer to Setpoint Limits for value constraints.

4.3.10.23.6. BuiltIn Field

This field SHALL indicate whether the preset is marked as "built-in", meaning that it can be modified, but it cannot be deleted.

4.3.10.24. PresetTypeStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**PresetScenario**	PresetScenarioEnum	all				M
1	**NumberOfPresets**	uint8	all		0		M
2	**PresetTypeFeatures**	PresetTypeFeaturesBitmap	all		0		M

4.3.10.24.1. PresetScenario Field

This field SHALL specify a PresetScenarioEnum value supported by this thermostat.

4.3.10.24.2. NumberOfPresets Field

This field SHALL specify a limit for the number of presets for this PresetScenarioEnum.

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4.3.10.24.3. PresetTypeFeatures Field

This field SHALL specify a bitmap of features for this PresetTypeStruct.

4.3.10.25. WeeklyScheduleTransitionStruct Type

This represents a single transition in a Thermostat schedule

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**TransitionTime**	uint16	max 1439				M
1	**HeatSetpoint**	temperature	all	X			M
2	**CoolSetpoint**	temperature	all	X			M

4.3.10.25.1. TransitionTime Field

This field SHALL represent the start time of the schedule transition during the associated day. The time will be represented by a 16 bits unsigned integer to designate the minutes since midnight. For example, 6am will be represented by 360 minutes since midnight and 11:30pm will be represented by 1410 minutes since midnight.

4.3.10.25.2. HeatSetpoint Field

This field SHALL represent the heat setpoint to be applied at this associated transition start time.

4.3.10.25.3. CoolSetpoint Field

This field SHALL represent the cool setpoint to be applied at this associated transition start time.

4.3.10.26. ScheduleStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**ScheduleHandle**	octstr	max 16	X			M
1	**SystemMode**	SystemModeEnum	desc				M
2	**Name**	string	max 64				O
3	**PresetHandle**	octstr	max 16				O

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
4	**Transitions**	list[ScheduleTransitionStruct]	1 to NumberOfScheduleTransitions		empty		M
5	**BuiltIn**	bool	all	X	false		M

4.3.10.26.1. ScheduleHandle Field

This field SHALL indicate a device generated identifier for this schedule. It SHALL be unique on the device, and SHALL NOT be reused after the associated schedule has been deleted.

This field SHALL only be null when the encompassing ScheduleStruct is appended to the Schedules attribute for the purpose of creating a new Schedule. Refer to Schedules for the creation of Schedule handles.

4.3.10.26.2. SystemMode Field

This field SHALL specify the default thermostat system mode for transitions in this schedule. The only valid values for this field SHALL be Auto, Heat, and Cool.

4.3.10.26.3. Name Field

This field SHALL specify a name for the ScheduleStruct.

4.3.10.26.4. PresetHandle Field

This field SHALL indicate the default PresetHandle value for transitions in this schedule.

4.3.10.26.5. Transitions Field

This field SHALL specify a list of transitions for the schedule.

This field SHALL NOT contain more than one ScheduleStruct with the same TransitionTime field and overlapping DayOfWeek fields; i.e. there SHALL be no duplicate transitions.

If the NumberOfScheduleTransitionPerDay attribute is not null, then for each bit in ScheduleDayOfWeekBitmap, the number of transitions with that bit set in DayOfWeek SHALL NOT be greater than the value of the NumberOfScheduleTransitionPerDay attribute.

For the purposes of determining which ScheduleStruct in this list is currently active, the current time SHALL be the number of minutes past midnight in the display value of the current time, not the actual number of minutes that have elapsed since midnight. On days which transition into or out of daylight saving time, certain values may repeat or not occur during the transition period.

A ScheduleTransitionStruct in this list SHALL be active if the current day of the week matches its DayOfWeek field and the current time is greater than or equal to the TransitionTime, but less than the TransitionTime on any other ScheduleTransitionStruct in the Transitions field whose DayOfWeek field also matches the current day of the week.

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If the current time is less than every ScheduleTransitionStruct whose DayOfWeek field also matches the current day of the week, the server SHALL attempt the same process to identify the active ScheduleTransitionStruct for the day preceding the previously attempted day of the week, repeating until an active ScheduleTransitionStruct is found or the attempted day is the current day of the week again. If no active ScheduleTransitionStruct is found, then the active ScheduleTransitionStruct SHALL be the ScheduleTransitionStruct with the largest TransitionTime field from the set of ScheduleTransitionStructs whose DayOfWeek field matches the current day of the week.

4.3.10.26.6. BuiltIn Field

This field SHALL indicate whether the schedule is marked as "built-in", meaning that it can be modified, but it cannot be deleted.

4.3.10.27. ScheduleTransitionStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**DayOfWeek**	ScheduleDayOfWeekBitmap	desc				M
1	**TransitionTime**	uint16	max 1439				M
2	**PresetHandle**	octstr	max 16				[PRES]
3	**SystemMode**	SystemModeEnum	desc				O
4	**CoolingSetpoint**	temperature	desc				[COOL]
5	**HeatingSetpoint**	temperature	desc				[HEAT]

This struct provides a time of day and a set of days of the week for a state transition within a schedule. The thermostat SHALL use the following order of precedence for determining a new setpoint at the time of transition:

1. If the PresetHandle field is provided, then the setpoint for the PresetStruct in the Presets attribute with that identifier SHALL be used
2. If either the HeatingSetpoint or CoolingSetpoint is provided, then it SHALL be used
   a. If the SystemMode field is provided, the HeatingSetpoint and CoolingSetpoint fields SHALL be interpreted using the SystemMode field
   b. If the SystemMode field is not provided, the HeatingSetpoint and CoolingSetpoint fields

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SHALL be interpreted using the SystemMode field on the parent ScheduleStruct

3. If neither the PresetHandle field or any Setpoint field is provided, then the PresetHandle field on the parent ScheduleStruct SHALL be used to determine the active PresetStruct
4. If the PresetHandle is not indicated and no setpoint is provided for the current SystemMode, the server SHALL use a default value for the current SystemMode.

If the setpoint was derived from a preset, then the ActivePresetHandle SHALL be set to the PresetHandle of that preset.

If a CoolingSetpoint was used to determine the cooling setpoint:

- If the server supports the OCC feature, and the Occupied bit is not set on the Occupancy attribute, then the UnoccupiedCoolingSetpoint attribute SHALL be set to the CoolingSetpoint
- Otherwise, the OccupiedCoolingSetpoint attribute SHALL be set to the CoolingSetpoint

If a HeatingSetpoint was used to determine the heating setpoint:

- If the server supports the OCC feature, and the Occupied bit is not set on the Occupancy attribute, then the UnoccupiedHeatingSetpoint attribute SHALL be set to the HeatingSetpoint
- Otherwise, the OccupiedHeatingSetpoint attribute SHALL be set to the HeatingSetpoint

The ScheduleTransitionStruct SHALL be invalid if all the following are true:

- The HeatingSetpoint field is not provided
- The PresetHandle field is not provided
- The PresetHandle field on the encompassing ScheduleStruct is not provided
- The SystemMode field is provided and has the value Heat or Auto, or the SystemMode field on the parent ScheduleStruct has the value Heat or Auto

The ScheduleTransitionStruct SHALL be invalid if all the following are true:

- The CoolingSetpoint field is not provided
- The PresetHandle field is not provided
- The PresetHandle field on the encompassing ScheduleStruct is not provided
- The SystemMode field is provided and has the value Cool or Auto, or the SystemMode field on the parent ScheduleStruct has the value Cool or Auto

4.3.10.27.1. DayOfWeek Field

This field SHALL specify a bitmask of days of the week that the transition applies to. The Vacation bit SHALL NOT be set; vacation schedules SHALL be set via the vacation preset.

4.3.10.27.2. TransitionTime Field

This SHALL specify the time of day at which the transition becomes active, in terms of minutes within the day representing the wall clock, where 0 is 00:00:00, 1 is 00:01:00 and 1439 is 23:59:00.

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Handling of transitions during the changeover of Daylight Saving Time is implementation-dependent.

4.3.10.27.3. PresetHandle Field

This field SHALL specify the preset used at the TransitionTime. If this field is provided, then the SystemMode, CoolingSetpoint and HeatingSetpoint fields SHALL NOT be provided.

4.3.10.27.4. SystemMode Field

This SHALL specify the default mode to which the thermostat will switch for this transition, overriding the default for the schedule. The only valid values for this field SHALL be Auto, Heat, Cool and Off. This field SHALL only be included when the required system mode differs from the schedule's default SystemMode.

4.3.10.27.5. CoolingSetpoint Field

This field SHALL specify the cooling setpoint for the transition. If PresetHandle is set, this field SHALL NOT be included. Refer to Setpoint Limits for value constraints.

4.3.10.27.6. HeatingSetpoint Field

This field SHALL specify the cooling setpoint for the transition. If PresetHandle is set, this field SHALL NOT be included. Refer to Setpoint Limits for value constraints.

4.3.10.28. ScheduleTypeStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**SystemMode**	SystemModeEnum	desc				M
1	**NumberOfSchedules**	uint8	max NumberOfSchedules		0		M
2	**ScheduleTypeFeatures**	ScheduleTypeFeaturesBitmap	desc		0		M

4.3.10.28.1. SystemMode Field

This field SHALL specify a SystemModeEnum supported by this thermostat for Schedules. The only valid values for this field SHALL be Auto, Heat, and Cool.

4.3.10.28.2. NumberOfSchedules Field

This field SHALL specify a limit for the number of Schedules for this SystemMode.

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4.3.10.28.3. ScheduleTypeFeatures Field

This field SHALL specify a bitmap of features for this schedule entry. At least one of SupportsPresets and SupportsSetpoints SHALL be set.

4.3.10.29. ThermostatSuggestionStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**UniqueID**	uint8	all				M
1	**PresetHandle**	octstr	max 16				M
2	**EffectiveTime**	epoch-s	all				M
3	**ExpirationTime**	epoch-s	all				M

4.3.10.29.1. UniqueID Field

This field SHALL have a generated identifier that identifies a distinct entry of type ThermostatSuggestionStruct.

4.3.10.29.2. PresetHandle Field

This field SHALL indicate the PresetHandle of the PresetStruct that represents the thermostat suggestion value.

4.3.10.29.3. EffectiveTime Field

This field SHALL indicate the UTC timestamp at which the suggestion SHALL take effect.

4.3.10.29.4. ExpirationTime Field

This field SHALL indicate the UTC timestamp at which the suggestion SHALL expire.

4.3.10.30. TemperatureDifference Type

This data type is derived from int16 and represents a temperature difference with a resolution of 0.01°C.

- *value* = (*temperature in °C*) x 100

- -4°C ⇒ -400
- 123.45°C ⇒ 12345

The full (non-null) range of -327.67°C to 327.67°C may be used.

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4.3.10.31. SignedTemperature Type

This data type is derived from int8 and represents a temperature from -12.7°C to 12.7°C with a resolution of 0.1°C.

- *value* = (*temperature in °C*) x 10

- -4°C ⇒ -40
- 12.3°C ⇒ 123

This type is employed where compactness of representation is important and where the resolution and range are still satisfactory.

4.3.10.32. UnsignedTemperature Type

This data type is derived from uint8 and represents a temperature from 0°C to 25.5°C with a resolution of 0.1°C.

- *value* = (*temperature in °C*) x 10

- 4°C ⇒ 40
- 12.3°C ⇒ 123

This type is employed where compactness of representation is important and where the resolution and range are still satisfactory.

4.3.11. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**LocalTemperature**	temperature	all	X		R V	M
0x0001	**OutdoorTemperature**	temperature	all	X	null	R V	O
0x0002	**Occupancy**	OccupancyBitmap	all			R V	OCC
0x0003	**AbsMinHeatSetpointLimit**	temperature	desc	F	7°C	R V	[HEAT]
0x0004	**AbsMaxHeatSetpointLimit**	temperature	desc	F	30°C	R V	[HEAT]

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0005	**AbsMinCoolSetpointLimit**	temperature	desc	F	16°C	R V	[COOL]
0x0006	**AbsMaxCoolSetpointLimit**	temperature	desc	F	32°C	R V	[COOL]
0x0007	**PICoolingDemand**						D
0x0008	**PIHeatingDemand**						D
0x0009	**HVACSystemTypeConfiguration**						D
0x0010	**LocalTemperatureCalibration**	SignedTemperature	all	N	0°C	RW VM	[!LTNE]
0x0011	**OccupiedCoolingSetpoint**	temperature	desc	N	26°C	RW VO	COOL
0x0012	**OccupiedHeatingSetpoint**	temperature	desc	N	20°C	RW VO	HEAT
0x0013	**UnoccupiedCoolingSetpoint**	temperature	desc	N	26°C	RW VO	COOL & OCC
0x0014	**UnoccupiedHeatingSetpoint**	temperature	desc	N	20°C	RW VO	HEAT & OCC
0x0015	**MinHeatSetpointLimit**	temperature	desc	N	AbsMinHeatSetpointLimit	RW VM	[HEAT]

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0016	**MaxHeatSetpointLimit**	temperature	desc	N	AbsMaxHeatSetpointLimit	RW VM	[HEAT]
0x0017	**MinCoolSetpointLimit**	temperature	desc	N	AbsMinCoolSetpointLimit	RW VM	[COOL]
0x0018	**MaxCoolSetpointLimit**	temperature	desc	N	AbsMaxCoolSetpointLimit	RW VM	[COOL]
0x0019	**MinSetpointDeadBand**	SignedTemperature	0 to 12.7°C	N	2°C	R[W] VM	AUTO
0x001A	**RemoteSensing**	RemoteSensingBitmap	all	N	0	RW VM	O
0x001B	**ControlSequenceOfOperation**	ControlSequenceOfOperationEnum	desc	N		RW VM	M
0x001C	**SystemMode**	SystemModeEnum	desc	N		RW VM	M
0x001E	**ThermostatRunningMode**	ThermostatRunningModeEnum	desc		0	R V	TEVT & AUTO, [AUTO]
0x0023	**TemperatureSetpointHold**	TemperatureSetpointHoldEnum	desc	N	0	RW VM	O
0x0024	**TemperatureSetpointHoldDuration**	uint16	max 1440	N X	null	RW VM	O
0x0025	**ThermostatProgrammingOperationMode**						D

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0029	**ThermostatRunningState**	RelayStateBitmap	desc		-	R V	O
0x0030	**SetpointChangeSource**	SetpointChangeSourceEnum	desc		0	R V	O
0x0031	**SetpointChangeAmount**	TemperatureDifference	all	X	null	R V	O
0x0032	**SetpointChangeSourceTimestamp**	epoch-s	all		MS	R V	O
0x0034	**OccupiedSetback**						D
0x0035	**OccupiedSetbackMin**						D
0x0036	**OccupiedSetbackMax**						D
0x0037	**UnoccupiedSetback**						D
0x0038	**UnoccupiedSetbackMin**						D
0x0039	**UnoccupiedSetbackMax**						D
0x003A	**EmergencyHeatDelta**	UnsignedTemperature	all	N	25.5°C	RW VM	O
0x0040	**ACType**	ACTypeEnum	desc	N	0	RW VM	O

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0041	**ACCapacity**	uint16	all	N	0	RW VM	O
0x0042	**ACRefrigerantType**	ACRefrigerantTypeEnum	desc	N	0	RW VM	O
0x0043	**ACCompressorType**	ACCompressorTypeEnum	desc	N	0	RW VM	O
0x0044	**ACErrorCode**	ACErrorCodeBitmap	all		0	RW VM	O
0x0045	**ACLouverPosition**	ACLouverPositionEnum	desc	N	0	RW VM	O
0x0046	**ACCoilTemperature**	temperature	all	X	null	R V	O
0x0047	**ACCapacityFormat**	ACCapacityFormatEnum	desc	N	0	RW VM	O
0x0048	**PresetTypes**	list[PresetTypeStruct]	1 to 7	F		R V	PRES
0x0049	**ScheduleTypes**	list[ScheduleTypeStruct]	1 to 3	F		R V	MSCH
0x004A	**NumberOfPresets**	uint8	min 1	F		R V	PRES
0x004B	**NumberOfSchedules**	uint8	min 1	F		R V	MSCH
0x004C	**NumberOfScheduleTransitions**	uint8	min 1	F		R V	MSCH

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x004D	**NumberOfScheduleTransitionPerDay**	uint8	min 1	F X		R V	MSCH
0x004E	**ActivePresetHandle**	octstr	max 16	N X		R V	PRES
0x004F	**ActiveScheduleHandle**	octstr	max 16	N X		R V	MSCH
0x0050	**Presets**	list[PresetStruct]	max NumberOfPresets	N T		RW VM	PRES
0x0051	**Schedules**	list[ScheduleStruct]	max NumberOfSchedules	N T		RW VM	MSCH
0x0052	**SetpointHoldExpiryTimestamp**	epoch-s	all	N X	null	R V	O
0x0053	**MaxThermostatSuggestions**	uint8	min 5	F		R V	TSUGGEST
0x0054	**ThermostatSuggestions**	list[ThermostatSuggestionStruct]	max MaxThermostatSuggestions	N		R V	TSUGGEST
0x0055	**CurrentThermostatSuggestion**	ThermostatSuggestionStruct	all	X		R V	TSUGGEST
0x0056	**ThermostatSuggestionNotFollowingReason**	ThermostatSuggestionNotFollowingReasonBitmap	all	X		R V	TSUGGEST

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4.3.11.1. Calculated Local Temperature

The local temperature SHALL be calculated from the *measured temperature*, including any adjustments applied by LocalTemperatureCalibration attribute (if any) as follows:

Calculated Local Temperature = (*measured temperature*) + LocalTemperatureCalibration

The *measured temperature* value may be local, or remote, depending on the value of the RemoteSensing attribute when supported.

All setpoint attributes in the Thermostat cluster SHALL be triggered based off this calculated value (i.e., measured temperature and any calibration offset).

If the LocalTemperatureNotExposed feature is present, the behavior of the thermostat SHALL be that the equipment's temperature control uses the calculated local temperature even if that value is not reported in the LocalTemperature attribute.

4.3.11.2. LocalTemperature Attribute

This attribute SHALL indicate the current Calculated Local Temperature, when available.

- If the LTNE feature is not supported:
  - If the LocalTemperatureCalibration is invalid or currently unavailable, the attribute SHALL report null.
  - If the LocalTemperatureCalibration is valid, the attribute SHALL report that value.
- Otherwise, if the LTNE feature is supported, there is no feedback externally available for the LocalTemperatureCalibration. In that case, the LocalTemperature attribute SHALL always report null.

4.3.11.3. OutdoorTemperature Attribute

This attribute SHALL indicate the outdoor temperature, as measured locally or remotely (over the network).

4.3.11.4. Occupancy Attribute

This attribute SHALL indicate whether the heated/cooled space is occupied or not, as measured locally or remotely (over the network).

4.3.11.5. AbsMinHeatSetpointLimit Attribute

This attribute SHALL indicate the absolute minimum level that the heating setpoint MAY be set to. This is a limitation imposed by the manufacturer.

Refer to Setpoint Limits for constraints

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4.3.11.6. AbsMaxHeatSetpointLimit Attribute

This attribute SHALL indicate the absolute maximum level that the heating setpoint MAY be set to. This is a limitation imposed by the manufacturer.

Refer to Setpoint Limits for constraints

4.3.11.7. AbsMinCoolSetpointLimit Attribute

This attribute SHALL indicate the absolute minimum level that the cooling setpoint MAY be set to. This is a limitation imposed by the manufacturer.

Refer to Setpoint Limits for constraints

4.3.11.8. AbsMaxCoolSetpointLimit Attribute

This attribute SHALL indicate the absolute maximum level that the cooling setpoint MAY be set to. This is a limitation imposed by the manufacturer.

Refer to Setpoint Limits for constraints

4.3.11.9. HVACSystemTypeConfiguration Attribute

This attribute SHALL indicate the HVAC system type controlled by the thermostat. If the thermostat uses physical DIP switches to set these parameters, this information SHALL be available read-only from the DIP switches. If these parameters are set via software, there SHALL be read/write access in order to provide remote programming capability.

4.3.11.10. LocalTemperatureCalibration Attribute

This attribute SHALL indicate the offset the Thermostat server SHALL make to the measured temperature (locally or remotely) to adjust the Calculated Local Temperature prior to using, displaying or reporting it.

The purpose of this attribute is to adjust the calibration of the Thermostat server per the user's preferences (e.g., to match if there are multiple servers displaying different values for the same HVAC area) or compensate for variability amongst temperature sensors.

If a Thermostat client attempts to write LocalTemperatureCalibration attribute to an unsupported value (e.g., out of the range supported by the Thermostat server), the Thermostat server SHALL respond with a status of SUCCESS and set the value of LocalTemperatureCalibration to the upper or lower limit reached.

NOTE	Prior to revision 8 of this cluster specification the value of this attribute was constrained to a range of -2.5°C to 2.5°C.

4.3.11.11. OccupiedCoolingSetpoint Attribute

This attribute SHALL indicate the cooling mode setpoint when the room is occupied.

Refer to Setpoint Limits for constraints.

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If an attempt is made to set this attribute to a value greater than MaxCoolSetpointLimit or less than MinCoolSetpointLimit, a response with the status code CONSTRAINT_ERROR SHALL be returned.

If this attribute is set to a value that is less than (OccupiedHeatingSetpoint + MinSetpointDeadBand), the value of OccupiedHeatingSetpoint SHALL be adjusted to (OccupiedCoolingSetpoint - MinSetpointDeadBand).

If the occupancy status of the room is unknown, this attribute SHALL be used as the cooling mode setpoint.

If a client changes the value of this attribute, the server supports the PRES feature, and the server either does not support the OCC feature or the Occupied bit is set on the Occupancy attribute, the value of the ActivePresetHandle attribute SHALL be set to null.

4.3.11.12. OccupiedHeatingSetpoint Attribute

This attribute SHALL indicate the heating mode setpoint when the room is occupied.

Refer to Setpoint Limits for constraints.

If an attempt is made to set this attribute to a value greater than MaxHeatSetpointLimit or less than MinHeatSetpointLimit, a response with the status code CONSTRAINT_ERROR SHALL be returned.

If this attribute is set to a value that is greater than (OccupiedCoolingSetpoint - MinSetpointDeadBand), the value of OccupiedCoolingSetpoint SHALL be adjusted to (OccupiedHeatingSetpoint + MinSetpointDeadBand).

If the occupancy status of the room is unknown, this attribute SHALL be used as the heating mode setpoint.

If a client changes the value of this attribute, the server supports the PRES feature, and the server either does not support the OCC feature or the Occupied bit is set on the Occupancy attribute, the value of the ActivePresetHandle attribute SHALL be set to null.

4.3.11.13. UnoccupiedCoolingSetpoint Attribute

This attribute SHALL indicate the cooling mode setpoint when the room is unoccupied.

Refer to Setpoint Limits for constraints.

If an attempt is made to set this attribute to a value greater than MaxCoolSetpointLimit or less than MinCoolSetpointLimit, a response with the status code CONSTRAINT_ERROR SHALL be returned.

If this attribute is set to a value that is less than (UnoccupiedHeatingSetpoint + MinSetpointDeadBand), the value of UnoccupiedHeatingSetpoint SHALL be adjusted to (UnoccupiedCoolingSetpoint - MinSetpointDeadBand).

If the occupancy status of the room is unknown, this attribute SHALL NOT be used.

If a client changes the value of this attribute, the server supports the PRES and OCC features, and the Occupied bit is not set on the Occupancy attribute, the value of the ActivePresetHandle attribute

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SHALL be set to null.

4.3.11.14. UnoccupiedHeatingSetpoint Attribute

This attribute SHALL indicate the heating mode setpoint when the room is unoccupied.

Refer to Setpoint Limits for constraints.

If an attempt is made to set this attribute to a value greater than MaxHeatSetpointLimit or less than MinHeatSetpointLimit, a response with the status code CONSTRAINT_ERROR SHALL be returned.

If this attribute is set to a value that is greater than (UnoccupiedCoolingSetpoint - MinSetpointDeadBand), the value of UnoccupiedCoolingSetpoint SHALL be adjusted to (UnoccupiedHeatingSetpoint + MinSetpointDeadBand).

If the occupancy status of the room is unknown, this attribute SHALL NOT be used.

If a client changes the value of this attribute, the server supports the PRES and OCC features, and the Occupied bit is not set on the Occupancy attribute, the value of the ActivePresetHandle attribute SHALL be set to null.

4.3.11.15. MinHeatSetpointLimit Attribute

This attribute SHALL indicate the minimum level that the heating setpoint MAY be set to.

This attribute, and the following three attributes, allow the user to define setpoint limits more constrictive than the manufacturer imposed ones. Limiting users (e.g., in a commercial building) to such setpoint limits can help conserve power.

Refer to Setpoint Limits for constraints. If an attempt is made to set this attribute to a value which conflicts with setpoint values then those setpoints SHALL be adjusted by the minimum amount to permit this attribute to be set to the desired value. If an attempt is made to set this attribute to a value which is not consistent with the constraints and cannot be resolved by modifying setpoints then a response with the status code CONSTRAINT_ERROR SHALL be returned.

4.3.11.16. MaxHeatSetpointLimit Attribute

This attribute SHALL indicate the maximum level that the heating setpoint MAY be set to.

Refer to Setpoint Limits for constraints. If an attempt is made to set this attribute to a value which conflicts with setpoint values then those setpoints SHALL be adjusted by the minimum amount to permit this attribute to be set to the desired value. If an attempt is made to set this attribute to a value which is not consistent with the constraints and cannot be resolved by modifying setpoints then a response with the status code CONSTRAINT_ERROR SHALL be returned.

4.3.11.17. MinCoolSetpointLimit Attribute

This attribute SHALL indicate the minimum level that the cooling setpoint MAY be set to.

Refer to Setpoint Limits for constraints. If an attempt is made to set this attribute to a value which conflicts with setpoint values then those setpoints SHALL be adjusted by the minimum amount to

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permit this attribute to be set to the desired value. If an attempt is made to set this attribute to a value which is not consistent with the constraints and cannot be resolved by modifying setpoints then a response with the status code CONSTRAINT_ERROR SHALL be returned.

4.3.11.18. MaxCoolSetpointLimit Attribute

This attribute SHALL indicate the maximum level that the cooling setpoint MAY be set to.

Refer to Setpoint Limits for constraints. If an attempt is made to set this attribute to a value which conflicts with setpoint values then those setpoints SHALL be adjusted by the minimum amount to permit this attribute to be set to the desired value. If an attempt is made to set this attribute to a value which is not consistent with the constraints and cannot be resolved by modifying setpoints then a response with the status code CONSTRAINT_ERROR SHALL be returned.

4.3.11.19. MinSetpointDeadBand Attribute

On devices which support the AUTO feature, this attribute SHALL indicate the minimum difference between the Heat Setpoint and the Cool Setpoint.

Refer to Setpoint Limits for constraints.

NOTE	Prior to revision 8 of this cluster specification the value of this attribute was constrained to a range of 0°C to 2.5°C.

NOTE	For backwards compatibility, this attribute is optionally writeable. However any writes to this attribute SHALL be silently ignored.

4.3.11.20. RemoteSensing Attribute

This attribute SHALL indicate when the local temperature, outdoor temperature and occupancy are being sensed by remote networked sensors, rather than internal sensors.

If the LTNE feature is present in the server, the LocalTemperature RemoteSensing bit value SHALL always report a value of 0.

If the LocalTemperature RemoteSensing bit is written with a value of 1 when the LTNE feature is present, the write SHALL fail and the server SHALL report a CONSTRAINT_ERROR.

4.3.11.21. ControlSequenceOfOperation Attribute

This attribute SHALL indicate the overall operating environment of the thermostat, and thus the possible system modes that the thermostat can operate in.

If an attempt is made to write to this attribute, the server SHALL silently ignore the write and the value of this attribute SHALL remain unchanged. This behavior is in place for backwards compatibility with existing thermostats.

4.3.11.22. SystemMode Attribute

This attribute SHALL indicate the current operating mode of the thermostat. Its value SHALL be

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limited by the ControlSequenceOfOperation attribute.

4.3.11.23. ThermostatRunningMode Attribute

This attribute SHALL indicate the running mode of the thermostat. This attribute uses the same values as SystemModeEnum but can only be Off, Cool or Heat. This attribute is intended to provide additional information when the thermostat's system mode is in auto mode.

4.3.11.24. StartOfWeek Attribute

This attribute SHALL indicate the day of the week that this thermostat considers to be the start of week for weekly setpoint scheduling.

This attribute MAY be able to be used as the base to determine if the device supports weekly scheduling by reading the attribute. Successful response means that the weekly scheduling is supported.

4.3.11.25. NumberOfWeeklyTransitions Attribute

This attribute SHALL indicate how many weekly schedule transitions the thermostat is capable of handling.

4.3.11.26. NumberOfDailyTransitions Attribute

This attribute SHALL indicate how many daily schedule transitions the thermostat is capable of handling.

4.3.11.27. TemperatureSetpointHold Attribute

This attribute SHALL indicate the temperature hold status on the thermostat. If hold status is on, the thermostat SHOULD maintain the temperature setpoint for the current mode until a system mode change. If hold status is off, the thermostat SHOULD follow the setpoint transitions specified by its internal scheduling program. If the thermostat supports setpoint hold for a specific duration, it SHOULD also implement the TemperatureSetpointHoldDuration attribute.

If the server supports a setpoint hold for a specific duration, it SHOULD also implement the SetpointHoldExpiryTimestamp attribute.

If this attribute is updated to SetpointHoldOn and the TemperatureSetpointHoldDuration has a non-null value and the SetpointHoldExpiryTimestamp is supported, the server SHALL update the SetpointHoldExpiryTimestamp with a value of current UTC timestamp, in seconds, plus the value in TemperatureSetpointHoldDuration multiplied by 60.

If this attribute is updated to SetpointHoldOff and the SetpointHoldExpiryTimestamp is supported, the server SHALL set the SetpointHoldExpiryTimestamp to null.

4.3.11.28. TemperatureSetpointHoldDuration Attribute

This attribute SHALL indicate the period in minutes for which a setpoint hold is active. Thermostats that support hold for a specified duration SHOULD implement this attribute. The null value indicates the field is unused. All other values are reserved.

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If this attribute is updated to a non-null value and the TemperatureSetpointHold is set to SetpointHoldOn and the SetpointHoldExpiryTimestamp is supported, the server SHALL update SetpointHoldExpiryTimestamp with a value of current UTC timestamp, in seconds, plus the new value of this attribute multiplied by 60.

If this attribute is set to null and the SetpointHoldExpiryTimestamp is supported, the server SHALL set the SetpointHoldExpiryTimestamp to null.

4.3.11.29. ThermostatRunningState Attribute

This attribute SHALL indicate the current relay state of the heat, cool, and fan relays.

Unimplemented outputs SHALL be treated as if they were Off.

4.3.11.30. SetpointChangeSource Attribute

This attribute SHALL indicate the source of the current active OccupiedCoolingSetpoint or OccupiedHeatingSetpoint (i.e., who or what determined the current setpoint).

This attribute enables service providers to determine whether changes to setpoints were initiated due to occupant comfort, scheduled programming or some other source (e.g., electric utility or other service provider). Because automation services MAY initiate frequent setpoint changes, this attribute clearly differentiates the source of setpoint changes made at the thermostat.

4.3.11.31. SetpointChangeAmount Attribute

This attribute SHALL indicate the delta between the current active OccupiedCoolingSetpoint or OccupiedHeatingSetpoint and the previous active setpoint. This attribute is meant to accompany the SetpointChangeSource attribute; devices implementing SetpointChangeAmount SHOULD also implement SetpointChangeSource.

The null value indicates that the previous setpoint was unknown.

4.3.11.32. SetpointChangeSourceTimestamp Attribute

This attribute SHALL indicate the time in UTC at which the SetpointChangeAmount attribute change was recorded.

4.3.11.33. EmergencyHeatDelta Attribute

This attribute SHALL indicate the delta between the Calculated Local Temperature and the OccupiedHeatingSetpoint or UnoccupiedHeatingSetpoint attributes at which the Thermostat server will operate in emergency heat mode.

If the difference between the Calculated Local Temperature and OccupiedCoolingSetpoint or UnoccupiedCoolingSetpoint is greater than or equal to the EmergencyHeatDelta and the Thermostat server's SystemMode attribute is in a heating-related mode, then the Thermostat server SHALL immediately switch to the SystemMode attribute value that provides the highest stage of heating (e.g., emergency heat) and continue operating in that running state until the OccupiedHeatingSetpoint value is reached. For example:

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- Calculated Local Temperature = 10.0°C
- OccupiedHeatingSetpoint = 16.0°C
- EmergencyHeatDelta = 2.0°C

⇒ OccupiedHeatingSetpoint - Calculated Local Temperature ≥? EmergencyHeatDelta

⇒ 16°C - 10°C ≥? 2°C

⇒ TRUE >>> Thermostat server changes its SystemMode to operate in 2nd stage or emergency heat mode

The purpose of this attribute is to provide Thermostat clients the ability to configure rapid heating when a setpoint is of a specified amount greater than the measured temperature. This allows the heated space to be quickly heated to the desired level set by the user.

4.3.11.34. ACType Attribute

This attribute SHALL indicate the type of Mini Split ACTypeEnum of Mini Split AC is defined depending on how Cooling and Heating condition is achieved by Mini Split AC.

4.3.11.35. ACCapacity Attribute

This attribute SHALL indicate capacity of Mini Split AC in terms of the format defined by the ACCapacityFormat attribute

4.3.11.36. ACRefrigerantType Attribute

This attribute SHALL indicate type of refrigerant used within the Mini Split AC.

4.3.11.37. ACCompressorType Attribute

This attribute SHALL indicate the type of compressor used within the Mini Split AC.

4.3.11.38. ACErrorCode Attribute

This attribute SHALL indicate the type of errors encountered within the Mini Split AC.

4.3.11.39. ACLouverPosition Attribute

This attribute SHALL indicate the position of Louver on the AC.

4.3.11.40. ACCoilTemperature Attribute

This attribute SHALL indicate the temperature of the AC coil, as measured locally or remotely (over the network).

4.3.11.41. ACCapacityFormat Attribute

This attribute SHALL indicate the format for the ACCapacity attribute.

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4.3.11.42. PresetTypes Attribute

This attribute SHALL indicate the supported PresetScenarioEnum values, limits on how many presets can be created for each PresetScenarioEnum, and whether or not a thermostat can transition automatically to a given scenario.

The list SHALL contain at least one entry.
The list SHALL NOT be larger than the number of supported PresetScenarioEnum values (maximum 7).
The list SHALL NOT contain any PresetTypeStruct entries with duplicate PresetScenarioEnum values.

4.3.11.43. ScheduleTypes Attribute

This attribute SHALL indicate the supported SystemMode values for Schedules, limits on how many schedules can be created for each SystemMode value, and whether or not a given SystemMode value supports transitions to Presets, target setpoints, or both.

The list SHALL contain at least one entry.
The list SHALL NOT be larger than the number of supported schedule SystemMode values (maximum 3, since the data type only allows Auto, Heat and Cool).
The list SHALL NOT contain any ScheduleTypeStruct entries with duplicate SystemModeEnum values.

4.3.11.44. NumberOfPresets Attribute

This attribute SHALL indicate the maximum number of entries supported by the Presets attribute.

4.3.11.45. NumberOfSchedules Attribute

This attribute SHALL indicate the maximum number of entries supported by the Schedules attribute.

4.3.11.46. NumberOfScheduleTransitions Attribute

This attribute SHALL indicate the maximum number of transitions per Schedules attribute entry.

4.3.11.47. NumberOfScheduleTransitionPerDay Attribute

This attribute SHALL indicate the maximum number of transitions per day of the week supported by each Schedules attribute entry. If this value is null, there is no limit on the number of transitions per day.

4.3.11.48. ActivePresetHandle Attribute

This attribute SHALL indicate the PresetHandle of the active preset. If this attribute is null, then there is no active preset.

4.3.11.49. ActiveScheduleHandle Attribute

This attribute SHALL indicate the ScheduleHandle of the active schedule. A null value in this

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attribute indicates that there is no active schedule.

4.3.11.50. Presets Attribute

This attribute SHALL contain the current list of configured presets.

On receipt of a write request:

1. If the PresetHandle field is null, the PresetStruct SHALL be treated as an added preset, and the device SHALL create a new unique value for the PresetHandle field.
   a. If the BuiltIn field is true, a response with the status code CONSTRAINT_ERROR SHALL be returned.
2. If the PresetHandle field is not null, the PresetStruct SHALL be treated as a modification of an existing preset.
   a. If the value of the PresetHandle field does not match any of the existing presets, a response with the status code NOT_FOUND SHALL be returned.
   b. If the value of the PresetHandle field is duplicated on multiple presets in the updated list, a response with the status code CONSTRAINT_ERROR SHALL be returned.
   c. If the BuiltIn field is true, and the PresetStruct in the current value with a matching PresetHandle field has a BuiltIn field set to false, a response with the status code CONSTRAINT_ERROR SHALL be returned.
   d. If the BuiltIn field is false, and the PresetStruct in the current value with a matching PresetHandle field has a BuiltIn field set to true, a response with the status code CONSTRAINT_ERROR SHALL be returned.
3. If the specified PresetScenarioEnum value does not exist in PresetTypes, a response with the status code CONSTRAINT_ERROR SHALL be returned.
4. If the Name is set, but the associated PresetTypeStruct does not have the SupportsNames bit set, a response with the status code CONSTRAINT_ERROR SHALL be returned.
5. If appending the received PresetStruct to the pending list of Presets would cause the total number of pending presets to exceed the value of the NumberOfPresets attribute, a response with the status code RESOURCE_EXHAUSTED SHALL be returned.
6. If appending the received PresetStruct to the pending list of Presets would cause the total number of pending presets whose PresetScenario field matches the appended preset's PresetScenario field to exceed the value of the NumberOfPresets field on the PresetTypeStruct whose PresetScenario matches the appended preset's PresetScenario field, a response with the status code RESOURCE_EXHAUSTED SHALL be returned.
7. Otherwise, the write SHALL be pended until receipt of a commit request, and the status code SUCCESS SHALL be returned.
   a. If the BuiltIn field is null:
      i. If there is a PresetStruct in the current value with a matching PresetHandle field, the BuiltIn field on the pending PresetStruct SHALL be set to the value of the BuiltIn on the matching PresetStruct.
      ii. Otherwise, the BuiltIn field on the pending PresetStruct SHALL be set to false.

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On an attempt to commit, the status of this attribute SHALL be determined as follows:

1. For all existing presets:
   a. If, after applying all pending changes, the updated value of the Presets attribute would not contain a PresetStruct with a matching PresetHandle field, indicating the removal of the PresetStruct, the server SHALL check for invalid removal of the PresetStruct:
      i. If the BuiltIn field is true on the removed PresetStruct, the attribute status SHALL be CONSTRAINT_ERROR.
      ii. If the MSCH feature is supported and the removed PresetHandle would be referenced by any PresetHandle on any ScheduleTransitionStruct on any ScheduleStruct in the updated value of the Schedules attribute, the attribute status SHALL be INVALID_IN_STATE.
      iii. If the removed PresetHandle is equal to the value of the ActivePresetHandle attribute, the attribute status SHALL be INVALID_IN_STATE.
2. If the attribute status has not yet been determined:
   a. The attribute status SHALL be SUCCESS.
   b. For all existing presets:
      i. If, after applying all pending changes, the updated value of the Presets attribute would not contain a PresetStruct with a matching PresetHandle field, indicating the removal of the PresetStruct, the server SHALL ensure that the preset being removed is unused, as follows:
         A. If the PresetHandle field of the removed preset is equal to the value of the PresetHandle field of the CurrentThermostatSuggestion attribute's value, the CurrentThermostatSuggestion attribute SHALL be set to null.
         B. If the PresetHandle field of the removed preset is equal to the value of the PresetHandle field of one or more of the entries in the ThermostatSuggestions attribute, the server SHALL delete any such entries from the ThermostatSuggestions attribute.

4.3.11.51. Schedules Attribute

This attribute SHALL contain a list of ScheduleStructs.

On receipt of a write request:

1. For all schedules in the write request:
   a. If the ScheduleHandle field is null, the ScheduleStruct SHALL be treated as an added schedule, and the device SHALL create a new unique value for the ScheduleHandle field.
      i. If the BuiltIn field is true, a response with the status code CONSTRAINT_ERROR SHALL be returned.
   b. Otherwise, if the ScheduleHandle field is not null, the ScheduleStruct SHALL be treated as a modification of an existing schedule.
      i. If the value of the ScheduleHandle field does not match any of the existing schedules, a response with the status code NOT_FOUND SHALL be returned.
      ii. If the BuiltIn field is true, and the ScheduleStruct in the current value with a matching

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ScheduleHandle field has a BuiltIn field set to false, a response with the status code CONSTRAINT_ERROR SHALL be returned.

iii. If the BuiltIn field is false, and the ScheduleStruct in the current value with a matching ScheduleHandle field has a BuiltIn field set to true, a response with the status code CONSTRAINT_ERROR SHALL be returned.

c. If the specified SystemMode does not exist in ScheduleTypes, a response with the status code CONSTRAINT_ERROR SHALL be returned.

d. If the number of transitions exceeds the NumberOfScheduleTransitions value, a response with the status code RESOURCE_EXHAUSTED SHALL be returned.

e. If the value of the NumberOfScheduleTransitionPerDay attribute is not null, and the number of transitions on any single day of the week exceeds the NumberOfScheduleTransitionPerDay value, a response with the status code RESOURCE_EXHAUSTED SHALL be returned.

f. If the PresetHandle field is present, but the associated ScheduleTypeStruct does not have the SupportsPresets bit set, a response with the status code CONSTRAINT_ERROR SHALL be returned.

g. If the PresetHandle field is present, but after applying all pending changes, the Presets attribute would not contain a PresetStruct whose PresetHandle field matches the value of the PresetHandle field, a response with the status code CONSTRAINT_ERROR SHALL be returned.

h. If the Name is set, but the associated ScheduleTypeStruct does not have the SupportsNames bit set, a response with the status code CONSTRAINT_ERROR SHALL be returned.

i. For all transitions in all schedules in the write request:

i. If the PresetHandle field is present, but the ScheduleTypeStruct matching the value of the SystemMode field on the encompassing ScheduleStruct does not have the SupportsPresets bit set, a response with the status code CONSTRAINT_ERROR SHALL be returned.

j. If the PresetHandle field is present, but after applying all pending changes, the Presets attribute would not contain a PresetStruct whose PresetHandle field matches the value of the PresetHandle field, a response with the status code CONSTRAINT_ERROR SHALL be returned.

i. If the SystemMode field is present, but the ScheduleTypeStruct matching the value of the SystemMode field on the encompassing ScheduleStruct does not have the SupportsSetpoints bit set, a response with the status code CONSTRAINT_ERROR SHALL be returned.

ii. If the SystemMode field is has a value of SystemModeOff, but the ScheduleTypeStruct matching the value of the SystemMode field on the encompassing ScheduleStruct does not have the SupportsOff bit set, a response with the status code CONSTRAINT_ERROR SHALL be returned.

k. If the HeatingSetpoint field is present, but the ScheduleTypeStruct matching the value of the SystemMode field on the encompassing ScheduleStruct does not have the SupportsSetpoints bit set, a response with the status code CONSTRAINT_ERROR SHALL be returned.

l. If the CoolingSetpoint field is present, but the ScheduleTypeStruct matching the value of the SystemMode field on the encompassing ScheduleStruct does not have the SupportsSetpoints bit set, a response with the status code CONSTRAINT_ERROR SHALL be returned.

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2. If appending the received ScheduleStruct to the pending list of Schedules would cause the total number of pending schedules to exceed the value of the NumberOfSchedules attribute, a response with the status code RESOURCE_EXHAUSTED SHALL be returned.
3. If appending the received ScheduleStruct to the pending list of Schedules would cause the total number of pending schedules whose SystemMode field matches the appended schedule's SystemMode field to exceed the value of the NumberOfSchedules field on the ScheduleTypeStruct whose SystemMode field matches the appended schedule's SystemMode field, a response with the status code RESOURCE_EXHAUSTED SHALL be returned.
4. Otherwise, the write SHALL be pended until receipt of a commit request, and the attribute status SHALL be SUCCESS.
    a. If the BuiltIn field is null:
        i. If there is a ScheduleStruct in the current value with a matching ScheduleHandle field, the BuiltIn field on the pending ScheduleStruct SHALL be set to the value of the BuiltIn on the matching ScheduleStruct.
        ii. Otherwise, the BuiltIn field on the pending ScheduleStruct SHALL be set to false.

On an attempt to commit, the status of this attribute SHALL be determined as follows:

1. For all existing schedules:
    a. If, after applying all pending changes, the updated value of the Schedules attribute would not contain a ScheduleStruct with a matching ScheduleHandle field, indicating the removal of the ScheduleStruct, the server SHALL check for invalid removal of the ScheduleStruct:
        i. If the BuiltIn field is true on the removed ScheduleStruct, the attribute status SHALL be CONSTRAINT_ERROR.
        ii. If the removed ScheduleHandle is equal to the value of the ActiveScheduleHandle attribute, the attribute status SHALL be INVALID_IN_STATE.
2. Otherwise, the attribute status SHALL be SUCCESS.

4.3.11.52. SetpointHoldExpiryTimestamp Attribute

If there is a known time when the TemperatureSetpointHold SHALL be cleared, this attribute SHALL contain the timestamp in UTC indicating when that will happen. If there is no such known time, this attribute SHALL be null.

If the TemperatureSetpointHold is set to SetpointHoldOn and the TemperatureSetpointHoldDuration is set to null, this attribute SHALL be set to null indicating there is a hold on the Thermostat without a duration.

If the TemperatureSetpointHold is set to SetpointHoldOff, this attribute SHALL be set to null indicating there is no hold on the Thermostat.

4.3.11.53. MaxThermostatSuggestions Attribute

This attribute SHALL indicate the maximum number of entries supported by the ThermostatSuggestions attribute.

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4.3.11.54. ThermostatSuggestions Attribute

This attribute SHALL indicate an unordered set of thermostat suggestions.

The suggested value shall be a PresetHandle that SHALL match the PresetHandle of one of the entries in the Presets attribute.

The Thermostat MAY use this information to ensure user comfort while also prioritizing other factors (e.g. energy savings, cost, and so on). Entries with effective times in the future MAY be used for pre-cool or pre-heat decisions.

See Section 4.3.7, “Re-evaluation of Current Thermostat Suggestion” for what to do if this attribute’s value changes.

4.3.11.55. CurrentThermostatSuggestion Attribute

This attribute SHALL indicate an entry in the entries, which identifies the current thermostat suggestion.

A value of null SHALL indicate that there is no current thermostat suggestion. When this attribute’s value changes to a null value, the server MAY decide to set the ActivePresetHandle attribute to a value of its choice, based on schedules, occupancy sensors, etc.

When the server is "following the suggestion" that means that the server SHALL ensure that the value of the ActivePresetHandle attribute matches the value of this attribute.

When there is a current thermostat suggestion and the server is unable to follow the suggestion or the server is unable to choose a current thermostat suggestion due to conflicting suggestions, it SHALL set the ThermostatSuggestionNotFollowingReason attribute to a non-null value as described in the definition of the ThermostatSuggestionNotFollowingReason attribute. Otherwise, the server SHALL follow the suggestion and set the ThermostatSuggestionNotFollowingReason attribute to null.

Whenever the state of the server changes such that it might need to start or stop following the suggestion, the server SHALL re-evaluate whether it is doing so and update the ActivePresetHandle and ThermostatSuggestionNotFollowingReason attributes as needed.

4.3.11.56. ThermostatSuggestionNotFollowingReason Attribute

This attribute SHALL indicate the reasons the Thermostat is unable to follow suggestions.

When the server is unable to follow the suggestion, it SHALL set the appropriate bits in the value of the ThermostatSuggestionNotFollowingReason attribute to indicate the reasons due to which the suggestion is not being followed. The value of the ThermostatSuggestionNotFollowingReason attribute SHALL be 0 (all bits cleared) if there are no bits defined in ThermostatSuggestionNotFollowingReasonBitmap that represent the reasons the suggestion is not being followed.

If the CurrentThermostatSuggestion attribute is null, this attribute SHALL be set to null.

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4.3.12. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**SetpointRaiseLower**	client ⇒ server	Y	O	M
0x05	**SetActiveScheduleRequest**	client ⇒ server	Y	O	MSCH
0x06	**SetActivePresetRequest**	client ⇒ server	Y	O	PRES
0x07	**AddThermostatSuggestion**	client ⇒ server	AddThermostatSuggestionResponse	M	TSUGGEST
0x02	**AddThermostatSuggestionResponse**	client ⇐ server	N		TSUGGEST
0x08	**RemoveThermostatSuggestion**	client ⇒ server	Y	M	TSUGGEST

4.3.12.1. SetpointRaiseLower Command

This command will raise or lower the setpoint based on the provided values.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Mode**	SetpointRaiseLowerModeEnum	desc			M
1	**Amount**	int8	all			M

4.3.12.1.1. Mode Field

The field SHALL specify which setpoints are to be adjusted.

4.3.12.1.1.1. Heat Value

If the server does not support the HEAT feature then it SHALL respond with INVALID_COMMAND. If the server supports the AUTO feature and the resulting setpoint would be invalid solely due to MinSetpointDeadBand then the Cooling setpoint SHALL be increased sufficiently to maintain the deadband.

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4.3.12.1.1.2. Cool Value

If the server does not support the COOL feature then it SHALL respond with INVALID_COMMAND. If the server supports the AUTO feature and the resulting setpoint would be invalid solely due to MinSetpointDeadBand then the Heating setpoint SHALL be decreased sufficiently to maintain the deadband.

4.3.12.1.1.3. Both Value

The client MAY indicate Both regardless of the server feature support. The server SHALL only adjust the setpoint that it supports and not respond with an error.

4.3.12.1.2. Amount Field

This field SHALL indicate the amount (possibly negative) that should be added to the setpoint(s), in steps of 0.1°C.

4.3.12.1.3. Effect on Receipt

Upon receipt, the attributes for the indicated setpoint(s) SHALL have the amount specified in the Amount field added to them. If the Occupancy feature is supported, the affected setpoints(s) SHALL be determined based on the current occupancy state, otherwise the occupied setpoints SHALL be adjusted. If the resulting value is outside the limits imposed by MinCoolSetpointLimit, MaxCoolSetpointLimit, MinHeatSetpointLimit and MaxHeatSetpointLimit, the value is clamped to those limits. This is not considered an error condition.

4.3.12.2. SetActiveScheduleRequest Command

This command will set the active schedule to the provided schedule handle.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ScheduleHandle**	octstr	max 16			M

4.3.12.2.1. ScheduleHandle Field

This field SHALL specify the value of the ScheduleHandle field on the ScheduleStruct to be made active.

4.3.12.2.2. Effect on Receipt

Upon receipt, if the Schedules attribute contains a ScheduleStruct whose ScheduleHandle field matches the value of the ScheduleHandle field, the server SHALL set the thermostat's ActiveScheduleHandle attribute to the value of the ScheduleHandle field.

Otherwise, the server SHALL return a status code of INVALID_COMMAND.

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4.3.12.3. SetActivePresetRequest Command

This command will set the active preset to the provided preset handle.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PresetHandle**	octstr	max 16	X		M

4.3.12.3.1. PresetHandle Field

This field SHALL specify the value of the PresetHandle field on the PresetStruct to be made active. If the field is set to null, that indicates there should be no active preset.

4.3.12.3.2. Effect on Receipt

Upon receipt of this command,

1. If the PresetHandle field is null, the server SHALL clear the ActivePresetHandle attribute by setting it to null.
2. Otherwise,
   a. If this command is received with a PresetHandle such that the Presets attribute does not contain a PresetStruct whose PresetHandle field matches the value of the PresetHandle, the server SHALL return a status code of INVALID_COMMAND.
   b. The server SHALL set the ActivePresetHandle attribute to the value of the PresetHandle field.
   c. The server SHALL adjust the thermostat's setpoints according to the preset identified by the PresetHandle field. If the Occupancy feature is supported, the server SHALL determine the setpoints to set based on the current occupancy state, otherwise the occupied setpoints SHALL be set.
3. The server SHALL return a status code of SUCCESS.

4.3.12.4. AddThermostatSuggestion Command

This command will add a new suggestion based on the specified values.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PresetHandle**	octstr	max 16			M
1	**EffectiveTime**	epoch-s	desc	X		M
2	**ExpirationInMinutes**	uint16	30 to 1440			M

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4.3.12.4.1. PresetHandle Field

This field SHALL specify the value of the thermostat suggestion.

4.3.12.4.2. EffectiveTime Field

This field SHALL indicate the timestamp in UTC at which the thermostat suggestion SHALL become available. If this field is set to null, that means the thermostat suggestion SHALL become available immediately.

4.3.12.4.3. ExpirationInMinutes Field

This field SHALL indicate the duration in minutes after which the thermostat suggestion provided by this command SHALL expire.

4.3.12.4.4. Effect on Receipt

1. If the server does not have its time synchronized, the server SHALL return INVALID_IN_STATE.
2. If this command is received with a PresetHandle field such that the Presets attribute does not contain a PresetStruct whose PresetHandle field matches the value of the PresetHandle field, the server SHALL return a status code of NOT_FOUND.
3. If the size of the ThermostatSuggestions attribute is equal to the value in MaxThermostatSuggestions, the server SHALL return a status code of RESOURCE_EXHAUSTED.
4. If the value of the EffectiveTime field is greater than the current time in UTC plus 24 hours, the server SHALL return a status code of INVALID_COMMAND.
5. If there are no errors then:
   a. The server SHALL update the ThermostatSuggestions attribute by adding an entry whose:
      i. PresetHandle field SHALL be set to the value in the PresetHandle field of this command.
      ii. EffectiveTime field SHALL be set as follows:
         A. If the EffectiveTime field of this command is null, the EffectiveTime field SHALL be set to a timestamp in UTC representing now.
         B. Otherwise, the EffectiveTime field SHALL be set to the value in the EffectiveTime field of this command.
      iii. ExpirationTime field SHALL be set to the value in the EffectiveTime field of the struct, plus the value in the ExpirationInMinutes field of this command, with the latter converted to seconds.
      iv. UniqueID field SHALL be set to a generated identifier that is distinct from all other entries in the ThermostatSuggestions attribute.
   b. The server SHALL return a AddThermostatSuggestionResponse.

4.3.12.5. AddThermostatSuggestionResponse Command

This command is sent in response to the AddThermostatSuggestion command.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**UniqueID**	uint8	all			M

4.3.12.5.1. UniqueID Field

This field SHALL contain the value in the UniqueID field of the entry that was added to the ThermostatSuggestions attribute via the AddThermostatSuggestion command for which this response SHALL be sent.

4.3.12.6. RemoveThermostatSuggestion Command

This command will remove the specified suggestion.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**UniqueID**	uint8	all			M

4.3.12.6.1. UniqueID Field

This field SHALL contain a unique identifier for an entry in the ThermostatSuggestions attribute that SHALL be removed.

4.3.12.6.2. Effect on Receipt

1. If there is a entry in the ThermostatSuggestions attribute, whose UniqueID field matches the UniqueID field passed in this command,
   a. The server SHALL remove the entry from the corresponding list.
   b. The server SHALL return a status code of SUCCESS.
2. Otherwise, the server SHALL return a status code of NOT_FOUND.

4.3.13. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**SystemModeChange**	INFO		V	P, TEVT
0x01	**LocalTemperatureChange**	INFO		V	P, TEVT & !LTNE
0x02	**OccupancyChange**	INFO		V	P, TEVT & OCC
0x03	**SetpointChange**	INFO		V	P, TEVT
0x04	**RunningStateChange**	INFO		V	P, TEVT

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ID	Name	Priority	Quality	Access	Conformance
0x05	**RunningModeChange**	INFO		V	P, TEVT & AUTO
0x06	**ActiveScheduleChange**	INFO		V	P, TEVT & MSCH
0x07	**ActivePresetChange**	INFO		V	P, TEVT & PRES

4.3.13.1. SystemModeChange Event

This event SHALL be generated when the SystemMode attribute changes.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PreviousSystemMode**	SystemModeEnum				O
1	**CurrentSystemMode**	SystemModeEnum				M

4.3.13.1.1. PreviousSystemMode Field

This field SHALL indicate the previous value of the SystemMode attribute. If the previous value is unavailable, this field SHALL be omitted.

4.3.13.1.2. CurrentSystemMode Field

This field SHALL indicate the current (after the change that caused the event to be generated) value of the SystemMode attribute.

4.3.13.2. LocalTemperatureChange Event

This event SHALL be generated when the LocalTemperature attribute changes significantly.

Significant changes are:

- Changes from null to not null, or from not null to null.
- Changes from one not-null value to another not-null value that are sufficiently large, as determined by the server.

LocalTemperatureChange events SHALL NOT be generated more often than once every 60 seconds.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**CurrentLocalTemperature**	temperature		X		M

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4.3.13.2.1. CurrentLocalTemperature Field

This field SHALL indicate the current value of the LocalTemperature attribute.

4.3.13.3. OccupancyChange Event

This event SHALL be generated when the Occupancy attribute changes.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PreviousOccupancy**	OccupancyBitmap				O
1	**CurrentOccupancy**	OccupancyBitmap				M

4.3.13.3.1. PreviousOccupancy Field

This field SHALL indicate the previous value of the Occupancy attribute. If the previous value is unavailable, this field SHALL be omitted.

4.3.13.3.2. CurrentOccupancy Field

This field SHALL indicate the current (after the change that caused the event to be generated) value of the Occupancy attribute.

4.3.13.4. SetpointChange Event

This event SHALL be generated when the value of any of the OccupiedHeatingSetpoint, UnoccupiedHeatingSetpoint, OccupiedCoolingSetpoint, or UnoccupiedCoolingSetpoint attributes is changed.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**SystemMode**	SystemModeEnum	Heat, Cool			M
1	**Occupancy**	OccupancyBitmap			1	OCC
2	**PreviousSetpoint**	temperature				O
3	**CurrentSetpoint**	temperature				M

4.3.13.4.1. SystemMode Field

This field SHALL indicate the system mode associated with the changed attribute. If the changed attribute is OccupiedHeatingSetpoint or UnoccupiedHeatingSetpoint, the value of this field SHALL be Heat. If the changed attribute is OccupiedCoolingSetpoint or UnoccupiedCoolingSetpoint, the value of this field SHALL be Cool.

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4.3.13.4.2. Occupancy Field

This field SHALL indicate the occupancy associated with the changed attribute. If the changed attribute is OccupiedHeatingSetpoint or OccupiedCoolingSetpoint, the value of this field SHALL be 1. If the changed attribute is UnoccupiedHeatingSetpoint or UnoccupiedCoolingSetpoint, the value of this field SHALL be 0.

4.3.13.4.3. PreviousSetpoint Field

This field SHALL indicate the previous value of the changed attribute. If the previous value is unavailable, this field SHALL be omitted.

4.3.13.4.4. CurrentSetpoint Field

This field SHALL indicate the current (after the change that caused the event to be generated) value of the changed attribute.

4.3.13.5. RunningModeChange Event

This event SHALL be generated when the ThermostatRunningMode attribute changes.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PreviousRunningMode**	ThermostatRunningModeEnum				O
1	**CurrentRunningMode**	ThermostatRunningModeEnum				M

4.3.13.5.1. PreviousRunningMode Field

This field SHALL indicate the previous value of the ThermostatRunningMode attribute. If the previous value is unavailable, this field SHALL be omitted.

4.3.13.5.2. CurrentRunningMode Field

This field SHALL indicate the current (after the change that caused the event to be generated) value of the ThermostatRunningMode attribute.

4.3.13.6. RunningStateChange Event

This event SHALL be generated when the ThermostatRunningState attribute changes.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PreviousRunningState**	RelayStateBitmap				O

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
1	**CurrentRunningState**	RelayStateBitmap				M

4.3.13.6.1. PreviousRunningState Field

This field SHALL indicate the previous value of the ThermostatRunningState attribute. If the previous value is unavailable, this field SHALL be omitted.

4.3.13.6.2. CurrentRunningState Field

This field SHALL indicate the current (after the change that caused the event to be generated) value of the ThermostatRunningState attribute.

4.3.13.7. ActivePresetChange Event

This event SHALL be generated when the ActivePresetHandle attribute changes.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PreviousPresetHandle**	octstr	max 16	X		O
1	**CurrentPresetHandle**	octstr	max 16	X		M

4.3.13.7.1. PreviousPresetHandle Field

This field SHALL indicate the previous value of the ActivePresetHandle attribute. If the previous value is unavailable, this field SHALL be omitted.

4.3.13.7.2. CurrentPresetHandle Field

This field SHALL indicate the current (after the change that caused the event to be generated) value of the ActivePresetHandle attribute.

4.3.13.8. ActiveScheduleChange Event

This event SHALL be generated when the ActiveScheduleHandle attribute changes.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PreviousScheduleHandle**	octstr	max 16	X		O

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
1	**CurrentScheduleHandle**	octstr	max 16	X		M

4.3.13.8.1. PreviousScheduleHandle Field

This field SHALL indicate the previous value of the ActiveScheduleHandle attribute. If the previous value is unavailable, this field SHALL be omitted.

4.3.13.8.2. CurrentScheduleHandle Field

This field SHALL indicate the current (after the change that caused the event to be generated) value of the ActiveScheduleHandle attribute.

4.4. Fan Control Cluster

This cluster specifies an interface to control the speed of a fan.

4.4.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added
2	New data model format and notation; Percent, speed and motion settings; General cleanup
3	Addition of AirflowDirection and Step command
4	Change conformance for FanModeSequenceEnum
5	Clarified attribute usage, added conformance column
6	Removed Zigbee related elements, removed P quality

4.4.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	FAN

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4.4.3. Cluster ID

ID	Name
0x0202	Fan Control

4.4.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**SPD**	MultiSpeed	O	0-SpeedMax Fan Speeds
1	**AUT**	Auto	O	Automatic mode supported for fan speed
2	**RCK**	Rocking	O	Rocking movement supported
3	**WND**	Wind	O	Wind emulation supported
4	**STEP**	Step	O	Step command supported
5	**DIR**	AirflowDirection	O	Airflow Direction attribute is supported

4.4.4.1. MultiSpeed Feature

Legacy Fan Control cluster revision 0-1 defined 3 speeds (low, medium and high) plus automatic speed control but left it up to the implementer to decide what was supported. Therefore, it is assumed that legacy client implementations are capable of determining, from the server, the number of speeds supported between 1, 2, or 3, and whether automatic speed control is supported.

The MultiSpeed feature includes attributes that support a running fan speed value from 0 to SpeedMax.

See Section 4.4.6.6.1, “Speed Rules” for more details.

4.4.5. Data Types

4.4.5.1. RockBitmap Type

This data type is derived from map8.

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Bit	Name	Summary	Conformance
0	**RockLeftRight**	Indicate rock left to right	M
1	**RockUpDown**	Indicate rock up and down	M
2	**RockRound**	Indicate rock around	M

4.4.5.2. WindBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**SleepWind**	Indicate sleep wind	M
1	**NaturalWind**	Indicate natural wind	M

4.4.5.2.1. SleepWind Bit

The fan speed, based on current settings, SHALL gradually slow down to a final minimum speed. For this process, the sequence, speeds and duration are MS.

4.4.5.2.2. NaturalWind Bit

The fan speed SHALL vary to emulate natural wind. For this setting, the sequence, speeds and duration are MS.

4.4.5.3. StepDirectionEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Increase**	Step moves in increasing direction	M
1	**Decrease**	Step moves in decreasing direction	M

4.4.5.4. AirflowDirectionEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Forward**	Airflow is in the forward direction	M
1	**Reverse**	Airflow is in the reverse direction	M

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4.4.5.5. FanModeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Off**	Fan is off	M
1	**Low**	Fan using low speed	O
2	**Medium**	Fan using medium speed	[Low]
3	**High**	Fan using high speed	M
4	**On**		D
5	**Auto**	Fan is using auto mode	AUT
6	**Smart**	Fan is using smart mode	D

4.4.5.6. FanModeSequenceEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**OffLowMedHigh**	Fan is capable of off, low, medium and high modes	[!AUT].a
1	**OffLowHigh**	Fan is capable of off, low and high modes	[!AUT].a
2	**OffLowMedHighAuto**	Fan is capable of off, low, medium, high and auto modes	[AUT].b
3	**OffLowHighAuto**	Fan is capable of off, low, high and auto modes	[AUT].b
4	**OffHighAuto**	Fan is capable of off, high and auto modes	[AUT].b
5	**OffHigh**	Fan is capable of off and high modes	[!AUT].a

4.4.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**FanMode**	FanModeEnum	desc	N		RW VO	M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0001	**FanModeSequence**	FanModeSequenceEnum	all	F		R V	M
0x0002	**PercentSetting**	percent	max 100	X		RW VO	M
0x0003	**PercentCurrent**	percent	max 100			R V	M
0x0004	**SpeedMax**	uint8	1 to 100	F		R V	SPD
0x0005	**SpeedSetting**	uint8	max SpeedMax	X		RW VO	SPD
0x0006	**SpeedCurrent**	uint8	max SpeedMax			R V	SPD
0x0007	**RockSupport**	RockBitmap	min 1	F		R V	RCK
0x0008	**RockSetting**	RockBitmap	desc			RW VO	RCK
0x0009	**WindSupport**	WindBitmap	min 1	F		R V	WND
0x000A	**WindSetting**	WindBitmap	desc			RW VO	WND
0x000B	**AirflowDirection**	AirflowDirectionEnum	all			RW VO	DIR

4.4.6.1. FanMode Attribute

This attribute SHALL indicate the current speed mode of the fan.

This attribute SHALL be set to one of the values in FanModeEnum supported by the server as indicated in the FanModeSequence attribute. The Low value SHALL be supported if and only if the FanModeSequence attribute value is less than 4. The Medium value SHALL be supported if and only if the FanModeSequence attribute value is 0 or 2.

This attribute MAY be written by a client to request a different fan mode. The server SHALL return INVALID_IN_STATE to indicate that the fan is not in a state where this attribute can be changed to the requested value.

The server MAY have values that this attribute can never be set to or that will be ignored by the server. For example, where this cluster appears on the same or another endpoint as other clusters

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with a system dependency, for example the Thermostat cluster, attempting to set this attribute to Off may not be allowed by the system.

If an attempt is made to set this attribute to a value not supported by the server as indicated in the FanModeSequence attribute, the server SHALL respond with CONSTRAINT_ERROR.

When this attribute is successfully written to, the PercentSetting and SpeedSetting (if present) attributes SHALL be set to appropriate values, as defined by Section 4.4.6.3.1, “Percent Rules” and Section 4.4.6.6.1, “Speed Rules” respectively, unless otherwise specified below.

When this attribute is set to any valid value, the PercentCurrent and SpeedCurrent (if present) attributes SHALL indicate the actual currently operating fan speed, unless otherwise specified below.

4.4.6.1.1. Off Value

Successfully setting the attribute value to Off SHALL set the values of these attributes to 0 (zero):

- PercentSetting
- PercentCurrent
- SpeedSetting (if present)
- SpeedCurrent (if present)

4.4.6.1.2. Auto Value

Successfully setting the attribute value to Auto SHALL set the values of these attributes to null:

- PercentSetting
- SpeedSetting (if present)

These attributes SHALL continue to indicate the current state of the fan while this attribute value is Auto:

- PercentCurrent
- SpeedCurrent (if present)

4.4.6.1.3. On Value

If a client attempts to write a value of On, the attribute SHALL be set to High.

4.4.6.1.4. Smart Value

If a client attempts to write a value of Smart, this attribute SHALL be set to Auto if the AUT feature is supported, otherwise the attribute SHALL be set to High.

4.4.6.2. FanModeSequence Attribute

This attribute indicates the fan speed ranges that SHALL be supported by the server.

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4.4.6.3. PercentSetting Attribute

This attribute SHALL indicate the speed setting for the fan with a value of 0 indicating that the fan is off and a value of 100 indicating that the fan is set to run at its maximum speed. If the FanMode attribute is set to Auto, the value of this attribute SHALL be set to null.

This attribute MAY be written to by a client to indicate a new fan speed. If a client writes null to this attribute, the attribute value SHALL NOT change. If the fan is in a state where this attribute cannot be changed to the requested value, the server SHALL return INVALID_IN_STATE.

When this attribute is successfully written, the server SHALL set the value of the FanMode and SpeedSetting (if present) attributes to values that abide by the mapping requirements listed below.

4.4.6.3.1. Percent Rules

It is up to the server implementation to map between ranges of the PercentSetting attribute, the SpeedSetting (if present) attribute, and the FanMode attribute enumerated values that are supported as indicated in the FanModeSequence attribute per the requirements below.

There SHALL be a mapping between the values of the FanMode attribute and the values of this attribute by dividing the domain of values of this attribute into ranges as follows:

- The ranges SHALL be non-overlapping, contiguous, and in ascending order.
- Every value from 0 to 100 SHALL be in exactly 1 range.
- Every value in a lower range shall be less than every value in any higher range.
- The value 0 (zero) in this attribute SHALL be in a range by itself. This range SHALL map to Off.
- If Auto is supported by the server, the null value SHALL be in a range by itself. This range SHALL map to Auto.
- There shall be a one-to-one mapping between each range and each value in FanMode.
- Every range SHALL map to a greater value in the FanMode attribute than all lower-valued ranges.

The following table illustrates an example of such a mapping. The actual mapping provided by the server is determined by the manufacturer subject to the rules mandated in this specification. Note that the values in this example are contiguous, non-overlapping and span from 0 to 100 with no missing values. Note also that each range appears in an ascending order, 0 appears by itself in a range that maps to Off, and null appears by itself in a range that maps to Auto.

Attribute	**Values**				
FanMode	Off	Low	Medium	High	Auto
PercentSetting	0	1 to 33	34 to 66	67 to 100	null

If a client successfully sets the FanMode attribute, the server SHALL set the PercentSetting attribute to some value within the range which maps to the FanMode setting. If no such value exists, the attribute value SHALL NOT change. If a client successfully sets the PercentSetting attribute, the server SHALL set the FanMode attribute to the value of the corresponding range, and the SpeedSetting attribute (if present) according to the equation below. If a client successfully sets the Per-

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centSetting attribute to 0, the server SHALL set the FanMode attribute to Off and the SpeedSetting (if present) attribute to 0.

If the MultiSpeed feature is supported, the value of the SpeedSetting and SpeedCurrent (speed) attributes SHALL be calculated from the PercentSetting or PercentCurrent (percent) attributes respectively using the following formula:

- speed = ceil(SpeedMax * (percent * 0.01))

> For example: If the SpeedMax attribute is 42 (42 speed fan) and PercentSetting is changed to 25, then SpeedSetting and SpeedCurrent become 11 (rounding up 10.5).

4.4.6.4. PercentCurrent Attribute

This attribute SHALL indicate the actual currently operating fan speed, or zero to indicate that the fan is off. There MAY be a temporary mismatch between the value of this attribute and the value of the PercentSetting attribute due to other system requirements or constraints that would not allow the fan to operate at the requested setting.

For example, if the value of this attribute is currently 50%, and the PercentSetting attribute is newly set to 25%, the value of this attribute MAY stay above 25% for a period necessary to dissipate internal heat, maintain product operational safety, etc.

When the value of the FanMode attribute is AUTO, the value of this attribute MAY vary across the range over time.

See Section 4.4.6.3.1, “Percent Rules” for more details.

4.4.6.5. SpeedMax Attribute

This attribute SHALL indicate the maximum value to which the SpeedSetting attribute can be set.

4.4.6.6. SpeedSetting Attribute

This attribute SHALL indicate the speed setting for the fan. This attribute MAY be written by a client to indicate a new fan speed. If the FanMode attribute is set to Auto, the value of this attribute SHALL be set to null.

The server SHALL support all values between 0 and SpeedMax.

If a client writes null to this attribute, the attribute value SHALL NOT change. If the fan is in a state where this attribute cannot be changed to the requested value, the server SHALL return INVALID_IN_STATE.

When this attribute is successfully written to, the server SHALL set the value of the FanMode and PercentSetting attributes to values that abide by the mapping requirements listed below.

4.4.6.6.1. Speed Rules

It is up to the server implementation to map between ranges of the SpeedSetting attribute, the Per-

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centSetting attribute, and the FanMode attribute enumerated values that are supported as indicated in the FanModeSequence attribute per the requirements below.

There SHALL be a mapping between the values of the FanMode attribute and the values of this attribute by dividing the domain of values of this attribute into ranges as follows:

- The ranges SHALL be non-overlapping, contiguous, and in ascending order.
- Every value from 0 to SpeedMax SHALL be in exactly 1 range.
- Every value in any lower range shall be less than every value in any higher range.
- The value 0 (zero) in this attribute SHALL be in a range by itself. This range SHALL map to Off.
- If Auto is supported by the server, the null value SHALL be in a range by itself. This range SHALL map to Auto.
- There shall be a one-to-one mapping between each range and each value in FanMode.
- Every range SHALL map to a greater value in the FanMode attribute than all lower-valued ranges.

If a client successfully sets the FanMode attribute, the server SHALL set the SpeedSetting attribute to some value within the range which maps to the FanMode setting, as well as the PercentSetting attribute according to the equation below. If no such value exists, the attribute value SHALL NOT change.

If a client successfully sets the SpeedSetting attribute to 0, the server SHALL set the FanMode attribute to Off and the PercentSetting attribute to 0.

The following table illustrates an example of such a mapping with a value of 6 for the SpeedMax attribute. The actual mapping provided by the server is determined by the manufacturer subject to the rules mandated in this specification. Note that the values in this example are contiguous, non-overlapping and span from 0 to 6 with no missing values. Note also that each range appears in an ascending order, 0 appears by itself in a range that maps to Off, and null appears by itself in a range that maps to Auto.

Attribute	**Values**				
FanMode	Off	Low	Medium	High	Auto
SpeedSetting	0	1 to 2	3 to 4	5 to 6	null

The value of the PercentSetting and PercentCurrent (percent) attributes SHALL be calculate from the SpeedSetting and SpeedCurrent (speed) attributes respectively using the following formula:

- percent = floor(speed/SpeedMax * 100)

> For example: If the SpeedMax attribute is 42 (42 speed fan) and SpeedSetting attribute is changed to 11, then PercentSetting and PercentCurrent become 26.

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4.4.6.7. SpeedCurrent Attribute

This attribute SHALL indicate the actual currently operating fan speed, or zero to indicate that the fan is off. There MAY be a temporary mismatch between the value of this attribute and the value of the SpeedSetting attribute due to other system requirements or constraints that would not allow the fan to operate at the requested setting.

For example, if the value of this attribute is currently 5, and the SpeedSetting attribute is newly set to 2, the value of this attribute MAY stay above 2 for a period necessary to dissipate internal heat, maintain product operational safety, etc.

When the value of the FanMode attribute is AUTO, the value of this attribute MAY vary across the range over time.

See Section 4.4.6.6.1, “Speed Rules” for more details.

4.4.6.8. RockSupport Attribute

This attribute is a bitmap that indicates the rocking motions that are supported by the server.

If this attribute is supported by the server, at least one bit SHALL be set in this attribute.

4.4.6.9. RockSetting Attribute

This attribute is a bitmap that indicates the currently active fan rocking motion setting. Each bit SHALL only be set to 1, if the corresponding bit in the RockSupport attribute is set to 1, otherwise a status code of CONSTRAINT_ERROR SHALL be returned.

If a combination of supported bits is set by a client, and the server does not support the combination, the lowest supported single bit in the combination SHALL be set and active, and all other bits SHALL indicate zero.

> For example: If RockUpDown and RockRound are both set, but this combination is not possible, then only RockUpDown becomes active.

4.4.6.10. WindSupport Attribute

This attribute is a bitmap that indicates what wind modes are supported by the server.

If this attribute is supported by the server, at least one bit SHALL be set in this attribute.

4.4.6.11. WindSetting Attribute

This attribute is a bitmap that indicates the current active fan wind feature settings. Each bit SHALL only be set to 1, if the corresponding bit in the WindSupport attribute is set to 1, otherwise a status code of CONSTRAINT_ERROR SHALL be returned.

If a combination of supported bits is set by a client, and the server does not support the combination, the lowest supported single bit in the combination SHALL be set and active, and all other bits SHALL indicate zero.

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For example: If Sleep Wind and Natural Wind are set, but this combination is not possible, then only Sleep Wind becomes active.

4.4.6.12. AirflowDirection Attribute

This attribute SHALL indicate the current airflow direction of the fan. This attribute MAY be written by a client to indicate a new airflow direction for the fan. This attribute SHALL be set to one of the values in the AirflowDirectionEnum table.

4.4.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**Step**	client ⇒ server	Y	O	STEP

4.4.7.1. Step Command

This command indirectly changes the speed-oriented attributes of the fan in steps rather than using the speed-oriented attributes, FanMode, PercentSetting, or SpeedSetting, directly. This command supports, for example, a user-operated and wall-mounted toggle switch that can be used to increase or decrease the speed of the fan by pressing the toggle switch up or down until the desired fan speed is reached. How this command is interpreted by the server and how it affects the values of the speed-oriented attributes is implementation specific.

For example, a fan supports this command, and the value of the FanModeSequence attribute is 0. The current value of the FanMode attribute is 2, or Medium. This command is received with the Direction field set to Increase. As per it's specific implementation, the server reacts to the command by setting the value of the FanMode attribute to 3, or High, which in turn sets the PercentSetting and SpeedSetting (if present) attributes to appropriate values, as defined by Section 4.4.6.3.1, "Percent Rules" and Section 4.4.6.6.1, "Speed Rules" respectively.

This command supports these fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Direction**	StepDirectionEnum				M
1	**Wrap**	bool			false	O
2	**LowestOff**	bool			true	O

4.4.7.1.1. Direction Field

This field SHALL indicate whether the speed-oriented attributes increase or decrease to the next step value.

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4.4.7.1.2. Wrap Field

This field SHALL indicate if the speed-oriented attributes wrap between highest and lowest step value.

4.4.7.1.3. LowestOff Field

This field SHALL indicate that the fan being off (FanMode = Off, PercentSetting = 0, or SpeedSetting = 0) is included as a step value.

4.4.7.1.4. When Generated

A client sends this command to change the speed-oriented attributes up or down in a step by step fashion.

4.4.7.1.5. Effect Upon Receipt

The successful receipt of this command could result in one or more of the speed-oriented attributes, FanMode, PercentSetting, or SpeedSetting, to change.

- This command SHALL be executed even if the speed-oriented attributes are not currently at an implemented step value.
- If the Direction field is Increase,
  - If the speed-oriented attributes are lower than the highest step value, the speed-oriented attributes SHALL change to the lowest step value that is higher than the current speed-oriented attributes.
  - Else if Wrap is TRUE, the speed-oriented attributes SHALL change to the lowest step value.
  - Else the speed-oriented attributes SHALL change to (or remain at) the highest step value.
- If the Direction field is Decrease,
  - If the speed-oriented attributes are higher than the lowest step value, the speed-oriented attributes SHALL change to the highest step value that is lower than the current speed-oriented attributes.
  - Else if Wrap is TRUE, the speed-oriented attributes SHALL change to the highest step value.
  - Else the speed-oriented attributes SHALL change to (or remain at) the lowest step value.
- Although the effect of the Step command is implementation specific, the effect on receipt of the Step command SHALL adhere to the conformance of the affected attributes.

4.5. Thermostat User Interface Configuration Cluster

This cluster provides an interface to allow configuration of the user interface for a thermostat, or a thermostat controller device, that supports a keypad and LCD screen.

4.5.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

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Revision	Description
1	Mandatory global ClusterRevision attribute added
2	New data model format and notation, added "Conversion of Temperature Values for Display" section

4.5.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	TSUIC

4.5.3. Cluster ID

ID	Name
0x0204	Thermostat User Interface Configuration

4.5.4. Conversion of Temperature Values for Display

See the Temperature Conversion section in the Data Model for unit conversion between Fahrenheit and Celsius.

4.5.5. Data Types

4.5.5.1. TemperatureDisplayModeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Celsius**	Temperature displayed in °C	M
1	**Fahrenheit**	Temperature displayed in °F	M

4.5.5.2. KeypadLockoutEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**NoLockout**	All functionality available to the user	M
1	**Lockout1**	Level 1 reduced functionality	M

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Value	Name	Summary	Conformance
2	**Lockout2**	Level 2 reduced functionality	M
3	**Lockout3**	Level 3 reduced functionality	M
4	**Lockout4**	Level 4 reduced functionality	M
5	**Lockout5**	Least functionality available to the user	M

The interpretation of the various levels is device-dependent.

4.5.5.3. ScheduleProgrammingVisibilityEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**ScheduleProgrammingPermitted**	Local schedule programming functionality is enabled at the thermostat	M
1	**ScheduleProgrammingDenied**	Local schedule programming functionality is disabled at the thermostat	M

4.5.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**TemperatureDisplayMode**	TemperatureDisplayModeEnum	all			RW VO	M
0x0001	**KeypadLockout**	KeypadLockoutEnum	all			RW VM	M
0x0002	**ScheduleProgrammingVisibility**	ScheduleProgrammingVisibilityEnum	all		ScheduleProgrammingPermitted	RW VM	O

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4.5.6.1. TemperatureDisplayMode Attribute

This attribute SHALL indicate the units of the temperature displayed on the thermostat screen.

4.5.6.2. KeypadLockout Attribute

This attribute SHALL indicate the level of functionality that is available to the user via the keypad.

4.5.6.3. ScheduleProgrammingVisibility Attribute

This attribute is used to hide the weekly schedule programming functionality or menu on a thermostat from a user to prevent local user programming of the weekly schedule. The schedule programming MAY still be performed via a remote interface, and the thermostat MAY operate in schedule programming mode.

This attribute is designed to prevent local tampering with or disabling of schedules that MAY have been programmed by users or service providers via a more capable remote interface. The programming schedule SHALL continue to run even though it is not visible to the user locally at the thermostat.

4.6. Valve Configuration and Control Cluster

This cluster is used to configure a valve.

4.6.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

4.6.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	VALCC

4.6.3. Cluster ID

ID	Name
0x0081	Valve Configuration and Control

4.6.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

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Bit	Code	Feature	Conformance	Summary
0	**TS**	TimeSync	desc	UTC time is used for time indications
1	**LVL**	Level	O	Device supports setting the specific position of the valve

4.6.4.1. TimeSync Feature

This feature SHALL indicate that the valve uses Time Synchronization and UTC time to indicate duration and auto close time.

This feature SHALL NOT be supported unless the device supports the Time Synchronization cluster.

4.6.4.2. Level Feature

This feature SHALL indicate that the valve is capable of being adjusted to a specific position, as a percentage, of its full range of motion.

4.6.5. Data Types

4.6.5.1. ValveFaultBitmap Type

This data type is derived from map16.

Bit	Name	Summary	Conformance
0	**GeneralFault**	Unspecified fault detected	M
1	**Blocked**	Valve is blocked	M
2	**Leaking**	Valve has detected a leak	M
3	**NotConnected**	No valve is connected to controller	M
4	**ShortCircuit**	Short circuit is detected	M
5	**CurrentExceeded**	The available current has been exceeded	M

4.6.5.2. ValveStateEnum Type

This data type is derived from enum8.

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Value	Name	Summary	Conformance
0	**Closed**	Valve is in closed position	M
1	**Open**	Valve is in open position	M
2	**Transitioning**	Valve is transitioning between closed and open positions or between levels	M

4.6.6. Status Codes

4.6.6.1. StatusCodeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0x02	**FailureDueToFault**	The requested action could not be performed due to a fault on the valve.	M

4.6.7. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**OpenDuration**	elapsed-s	min 1	X		R V	M
0x0001	**DefaultOpenDuration**	elapsed-s	min 1	N X		RW VO	M
0x0002	**AutoCloseTime**	epoch-us	all	X		R V	TS
0x0003	**RemainingDuration**	elapsed-s	all	Q X		R V	M
0x0004	**CurrentState**	ValveStateEnum	all	X		R V	M
0x0005	**TargetState**	ValveStateEnum	all	X		R V	M
0x0006	**CurrentLevel**	percent	all	X		R V	LVL

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0007	**TargetLevel**	percent	all	X		R V	LVL
0x0008	**DefaultOpenLevel**	percent	1 to 100	N	100	RW VO	[LVL]
0x0009	**ValveFault**	ValveFaultBitmap	all		0	R V	O
0x000A	**LevelStep**	uint8	1 to 50	F	1	R V	[LVL]

4.6.7.1. OpenDuration Attribute

This attribute SHALL indicate the total duration, in seconds, for which the valve will remain open for this current opening.

A value of null SHALL indicate the duration is not set, meaning that the valve will remain open until closed by the user or some other automation.

4.6.7.2. DefaultOpenDuration Attribute

This attribute SHALL indicate the default duration, in seconds, for which the valve will remain open, if the OpenDuration field is not present in the Open command.

A value of null SHALL indicate the duration is not set, meaning that the valve will remain open until closed by the user or some other automation.

4.6.7.3. AutoCloseTime Attribute

This attribute SHALL indicate the UTC time when the valve will close, depending on value of the OpenDuration attribute.

This attribute SHALL be null:

- When OpenDuration is null, or
- When the valve does not have a synchronized UTCTime in the Time Synchronization cluster, or
- When the valve is closed.

When the value of this attribute is earlier or equal to the current UTC time, the valve SHALL automatically transition to its closed position. The behavior of transitioning to the closed position, SHALL match the behavior described in the Close command.

If this attribute is not null and the Time Synchronization cluster receives a SetUTCTime command, modifying the current UTC time of the device, the value of this attribute SHALL be adjusted to match the new UTC time plus the value of the RemainingDuration attribute.

4.6.7.4. RemainingDuration Attribute

This attribute SHALL indicate the remaining duration, in seconds, until the valve closes.

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This attribute SHALL be null:

- When OpenDuration is null, or
- When the valve is closed.

The value of this attribute SHALL only be reported in the following cases:

- When it changes from null to any other value and vice versa, or
- When it changes to 0, or
- When it increases, or
- When the closing time changes.

Meaning that clients SHOULD NOT rely on the reporting of this attribute in order to keep track of the remaining duration, due to this attribute not being reported during regular countdown.

When reading this attribute it SHALL return the remaining duration, in seconds, until the valve closes.

When the value of this attribute counts down to 0, the valve SHALL automatically transition to its closed position. The behavior of transitioning to the closed position SHALL match the behavior described in the Close command.

4.6.7.5. CurrentState Attribute

This attribute SHALL indicate the current state of the valve.

A value of null SHALL indicate that the current state is not known.

4.6.7.6. TargetState Attribute

This attribute SHALL indicate the target state, while changing the state, of the valve.

A value of null SHALL indicate that no target position is set, since the change in state is either done or failed.

4.6.7.7. CurrentLevel Attribute

This attribute SHALL indicate the current level of the valve as a percentage value, between fully closed and fully open. During a transition from one level to another level, the valve SHOULD keep this attribute updated to the best of its ability, in order to represent the actual level of the valve during the movement.

A value of 100 percent SHALL indicate the fully open position.

A value of 0 percent SHALL indicate the fully closed position.

A value of null SHALL indicate that the current state is not known.

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4.6.7.8. TargetLevel Attribute

This attribute SHALL indicate the target level of the valve as a percentage value, between fully closed and fully open.

The interpretation of the percentage value is the same as for the CurrentLevel attribute.

A value of null SHALL indicate that no target position is set, since the change of level is either done or failed.

4.6.7.9. DefaultOpenLevel Attribute

This attribute SHALL indicate the default value used for the TargetLevel attribute, when a valve transitions from the closed to the open state, caused by an Open command, if a TargetLevel field is not present in the Open command.

If the LevelStep attribute is present and the value of a write interaction to this attribute field is not 100, the value SHALL be a supported value as defined by the LevelStep attribute, such that (Value received in the write interaction) % (Value of LevelStep attribute) equals 0. If the resulting value is not 0, the requested DefaultOpenLevel value is considered an unsupported value and a CONSTRAINT_ERROR status SHALL be returned.

4.6.7.10. ValveFault Attribute

This attribute SHALL indicate any faults registered by the valve.

4.6.7.11. LevelStep Attribute

This attribute SHALL indicate the step size the valve can support.

The step size defined by this attribute is counted from 0 and the final step towards 100 MAY be different than what is defined in this attribute. For example, if the value of this attribute is 15, it results in these target values being supported; 0, 15, 30, 45, 60, 75, 90 and 100.

The values of 0 and 100 SHALL always be supported, regardless of the value of this attribute.

4.6.8. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**Open**	client ⇒ server	Y	O	M
0x01	**Close**	client ⇒ server	Y	O	M

4.6.8.1. Open Command

This command is used to set the valve to its open position.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**OpenDuration**	elapsed-s	min 1	X		O
1	**TargetLevel**	percent	min 1			[LVL]

4.6.8.1.1. OpenDuration Field

This field SHALL indicate the duration that the valve will remain open for this specific Open command.

A value of null SHALL indicate the duration is not set, meaning that the valve will remain open until closed by the user or some other automation.

4.6.8.1.2. TargetLevel Field

This field SHALL indicate the target level used for this specific Open command.

4.6.8.1.3. Effect on Receipt

If the device has registered a fault, that prevents it from performing the requested action, the command SHALL be ignored and a FailureDueToFault status SHALL be returned.

The device SHALL set the TargetState attribute to the Open value and set the CurrentState attribute to the Transitioning value.

If the OpenDuration field is present, the value of the OpenDuration attribute SHALL be set to the value of the OpenDuration field.
If the OpenDuration field is not present, the value of OpenDuration attribute SHALL be set to the value of the DefaultOpenDuration attribute.

If the OpenDuration attribute is null, it SHALL indicate that there is no auto close defined for the current Open action and the device SHALL set the RemainingDuration attribute to null. If the device supports the TimeSync feature, the device SHALL set the AutoCloseTime attribute to null.
If the OpenDuration attribute is not null, it indicates that an auto close duration is defined for the current open action and the device SHALL set the value of the RemainingDuration attribute equal to the value of the OpenDuration attribute. If the device supports the TimeSync feature, the device SHALL set the AutoCloseTime attribute to the UTC value of the time when command was received plus the value of the OpenDuration attribute.

If the LevelStep attribute and the TargetLevel field are both present and the value of the TargetLevel field is not 100, the value of the TargetLevel field SHALL be a supported value as defined by the LevelStep attribute, such that (Value of TargetLevel field) % (Value of LevelStep attribute) equals 0. If the resulting value is not 0, the requested TargetLevel value is considered an unsupported value and a CONSTRAINT_ERROR status SHALL be returned.

If the device supports the Level feature, the TargetLevel attribute SHALL be set to the value of the TargetLevel field, if present. If the TargetLevel field is not present, the TargetLevel attribute SHALL be set to the value of the DefaultOpenLevel attribute, if implemented. If the DefaultOpenLevel

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attribute is not present, the TargetLevel attribute SHALL be set to 100.

When the relevant target and duration attributes have been set, the device SHALL start the movement towards the target value and start the countdown of the RemainingDuration attribute. If the device supports the Level feature, the device SHALL update the CurrentLevel attribute at the start of the movement, and SHOULD update it as appropriate during movement, especially if it is slow. When the movement is complete, the device SHALL set the CurrentState attribute to the Open value.

4.6.8.2. Close Command

This command is used to set the valve to its closed position.

4.6.8.2.1. Effect on Receipt

If the device has registered a fault that causes it to prevent the valve to perform the requested action, the command SHALL be ignored and a FailureDueToFault status SHALL be returned.

The OpenDuration and RemainingDuration attribute SHALL be set to null.

If the device supports the TimeSync feature, the AutoCloseTime attribute SHALL be set to null.

The device SHALL set the TargetState attribute to the Closed value and set the CurrentState attribute to the Transitioning value.

If the device supports the Level feature, it SHALL set the TargetLevel attribute to 0.

When the relevant target attributes have been set, the device SHALL start the movement towards the target value. If the device supports the Level feature, the device SHALL update the CurrentLevel attribute, accordingly during the movement. When the movement is complete, the device SHALL set the CurrentState attribute to the Closed value.

4.6.9. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**ValveStateChanged**	INFO		V	O
0x01	**ValveFault**	INFO		V	O

4.6.9.1. ValveStateChanged Event

This event SHALL be generated when the valve state changed. For level changes, after the end of movement, for state changes when the new state has been reached.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ValveState**	ValveStateEnum	all			M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
1	**ValveLevel**	percent	all			LVL

4.6.9.1.1. ValveState Field

This field SHALL indicate the new state of the valve.

4.6.9.1.2. ValveLevel Field

This field SHALL indicate the new level of the valve.

4.6.9.2. ValveFault Event

This event SHALL be generated when the valve registers or clears a fault, e.g. not being able to transition to the requested target level or state.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ValveFault**	ValveFault-Bitmap	all			M

4.6.9.2.1. ValveFault Field

This field SHALL indicate the value of the ValveFault attribute, at the time this event is generated.

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Chapter 5. Entry Control

The Cluster Library is made of individual chapters such as this one. References between chapters are made using a *X.Y* notation where *X* is the chapter and *Y* is the sub-section within that chapter.

5.1. General Description

5.1.1. Introduction

The clusters specified in this section are for use in applications involving "entry control" for people or elements (light, air, etc.), for example shades, windows, doors, cabinets, door locks, but MAY be used in any application domain.

5.1.2. Cluster List

This section lists the closures specific clusters as specified in this chapter.

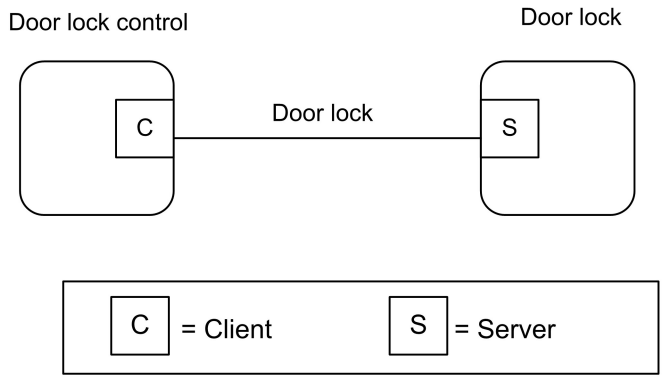
Table 9. Overview of the Closures Clusters

Cluster ID	Cluster Name	Description
0x0101	Door Lock	An interface to a generic way to secure a door
0x0102	Window Covering	Commands and attributes for controlling a window covering
0x0104	Closure Control	An interface for controlling a closure device
0x0105	Closure Dimension	An interface for controlling a ClosurePanel device

5.2. Door Lock Cluster

The door lock cluster provides an interface to a generic way to secure a door. The physical object that provides the locking functionality is abstracted from the cluster. The cluster has a small list of mandatory attributes and functions and a list of optional features.

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Note: Device names are examples for illustration purposes only

Figure 18. Typical Usage of the Door Lock Cluster

5.2.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added; CCB 1811 1812 1821
2	CCB 2430
3	CCB 2629 2630
4	All Hubs changes and added feature map
5	CCB 3472 3474 3338
6	New data model format and notation. Added User features. General cleanup of functionality
7	Added support for European door locks (unbolt feature)
8	Removed LOG and NOT feature, fixed constraints on NumberOf*Schedules attributes and added support for Aliro credential provisioning
9	Removed AlarmMask attribute
10	Removed Zigbee related elements and P quality

5.2.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	DRLK

5.2.3. Cluster ID

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ID	Name
0x0101	Door Lock

5.2.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**PIN**	PINCredential	O	Lock supports PIN credentials (via keypad, or over-the-air)
1	**RID**	RFIDCredential	P, O	Lock supports RFID credentials
2	**FGP**	FingerCredentials	P, O	Lock supports finger related credentials (fingerprint, finger vein)
4	**WDSCH**	WeekDayAccessSchedules	O	Lock supports week day user access schedules
5	**DPS**	DoorPositionSensor	O	Lock supports a door position sensor that indicates door's state
6	**FACE**	FaceCredentials	P, O	Lock supports face related credentials (face, iris, retina)
7	**COTA**	CredentialOverTheAirAccess	O	PIN codes over-the-air supported for lock/unlock operations
8	**USR**	User	PIN | RID | FGP | FACE | ALIRO	Lock supports the user commands and database
10	**YDSCH**	YearDayAccessSchedules	O	Lock supports year day user access schedules
11	**HDSCH**	HolidaySchedules	O	Lock supports holiday schedules
12	**UBOLT**	Unbolting	O	Lock supports unbolting

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Bit	Code	Feature	Conformance	Summary
13	**ALIRO**	AliroProvisioning	O	Lock supports Aliro credential provisioning as defined in Aliro
14	**ALBU**	AliroBLEUWB	[ALIRO]	Lock supports the Bluetooth LE + UWB Access Control Flow as defined in Aliro

5.2.4.1. PINCredential Feature

If the User Feature is also supported then any PIN Code stored in the lock SHALL be associated with a User.

A lock MAY support multiple credential types so if the User feature is supported the UserType, UserStatus and Schedules are all associated with a User index and not directly with a PIN index. A User index may have several credentials associated with it.

5.2.4.2. RFIDCredential Feature

If the User Feature is also supported then any RFID credential stored in the lock SHALL be associated with a User.

A lock MAY support multiple credential types so if the User feature is supported the UserType, UserStatus and Schedules are all associated with a User index and not directly with a RFID index. A User Index may have several credentials associated with it.

5.2.4.3. FingerCredentials Feature

Currently the cluster only defines the metadata format for notifications when a fingerprint/ finger vein credential is used to access the lock and doesn't describe how to create fingerprint/finger vein credentials. If the Users feature is also supported then the User that a fingerprint/finger vein is associated with can also have its UserType, UserStatus and Schedule modified.

A lock MAY support multiple credential types so if the User feature is supported the UserType, UserStatus and Schedules are all associated with a User index and not directly with a Finger index. A User Index may have several credentials associated with it.

5.2.4.4. WeekDayAccessSchedules Feature

If the User feature is supported then Week Day Schedules are applied to a User and not a credential.

Week Day Schedules are used to restrict access to a specified time window on certain days of the week. The schedule is repeated each week.

The lock MAY automatically adjust the UserType when a schedule is created or cleared.

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Support for WeekDayAccessSchedules requires that the lock has the capability of keeping track of local time.

5.2.4.5. DoorPositionSensor Feature

If this feature is supported this indicates that the lock has the ability to determine the position of the door which is separate from the state of the lock.

5.2.4.6. FaceCredentials Feature

Currently the cluster only defines the metadata format for notifications when a face recognition, iris, or retina credential is used to access the lock and doesn't describe how to create face recognition, iris, or retina credentials. If the Users feature is also supported then the User that a face recognition, iris, or retina credential is associated with can also have its UserType, UserStatus and Schedule modified.

A lock MAY support multiple credential types so if the User feature is supported the UserType, UserStatus and Schedules are all associated with a User and not directly with a credential.

5.2.4.7. CredentialOverTheAirAccess Feature

If this feature is supported then the lock supports the ability to verify a credential provided in a lock/unlock command. Currently the cluster only supports providing the PIN credential to the lock/unlock commands. If this feature is supported then the PIN Credential feature SHALL also be supported.

5.2.4.8. User Feature

If the User Feature is supported then a lock employs a User database. A User within the User database is used to associate credentials and schedules to single user record within the lock. This also means the UserType and UserStatus fields are associated with a User and not a credential.

5.2.4.9. YearDayAccessSchedules Feature

If the User feature is supported then Year Day Schedules are applied to a User and not a credential.

Year Day Schedules are used to restrict access to a specified date and time window.

The lock MAY automatically adjust the UserType when a schedule is created or cleared.

Support for YearDayAccessSchedules requires that the lock has the capability of keeping track of local time.

5.2.4.10. HolidaySchedules Feature

This feature is used to setup Holiday Schedule in the lock device. A Holiday Schedule sets a start and stop end date/time for the lock to use the specified operating mode set by the Holiday Schedule.

Support for HolidaySchedules requires that the lock has the capability of keeping track of local time.

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5.2.4.11. Unbolting Feature

Locks that support this feature differentiate between unbolting and unlocking. The Unbolt Door command retracts the bolt without pulling the latch. The Unlock Door command fully unlocks the door by retracting the bolt and briefly pulling the latch. While the latch is pulled, the lock state changes to Unlatched. Locks without unbolting support don't differentiate between unbolting and unlocking and perform the same operation for both commands.

5.2.4.12. AliroProvisioning Feature

Locks that support this feature implement the Aliro specification as defined in [Aliro] and support Matter as a method for provisioning Aliro credentials.

5.2.4.13. AliroBLEUWB Feature

Locks that support this feature implement the Bluetooth LE + UWB Access Control Flow as defined in [Aliro].

5.2.5. Recommended steps for creating a new User

It is RECOMMENDED that the Administrator query the door lock for what users already exist in its database to find an available UserIndex for creating a new User (see Section 5.2.10.17, "GetUser Command").

1. Use SetUser command with an available UserIndex to set the user record fields as applicable.
2. Use SetCredential command with same UserIndex to add one or more credentials as applicable.
3. Use SetWeekDaySchedule command or SetYearDaySchedule command with same UserIndex to add one or more schedule restrictions as applicable.

5.2.6. Data Types

5.2.6.1. DaysMaskBitmap Type

This data type is derived from map8.

This bitmap SHALL indicate the days of the week the Week Day schedule applies for.

Bit	Name	Summary	Conformance
0	**Sunday**	Schedule is applied on Sunday	M
1	**Monday**	Schedule is applied on Monday	M
2	**Tuesday**	Schedule is applied on Tuesday	M
3	**Wednesday**	Schedule is applied on Wednesday	M

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Bit	Name	Summary	Conformance
4	**Thursday**	Schedule is applied on Thursday	M
5	**Friday**	Schedule is applied on Friday	M
6	**Saturday**	Schedule is applied on Saturday	M

5.2.6.2. CredentialRulesBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**Single**	Only one credential is required for lock operation	M
1	**Dual**	Any two credentials are required for lock operation	M
2	**Tri**	Any three credentials are required for lock operation	M

5.2.6.3. OperatingModesBitmap Type

This data type is derived from map16.

Bit	Name	Summary	Conformance
0	**Normal**	Normal operation mode	M
1	**Vacation**	Vacation operation mode	O
2	**Privacy**	Privacy operation mode	O
3	**NoRemoteLockUnlock**	No remote lock and unlock operation mode	M
4	**Passage**	Passage operation mode	O

WARNING

For the OperatingModesBitmap, a bit SET indicates that the operating mode IS NOT supported. A bit CLEAR indicates that the operating mode IS supported. This is the inverse of most bitmaps in this specification, and it is RECOMMENDED that clients carefully take this into consideration. See SupportedOper-

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atingModes.

5.2.6.4. ConfigurationRegisterBitmap Type

This data type is derived from map16.

Bit	Name	Summary	Conformance
0	**LocalProgramming**	The state of local programming functionality	M
1	**KeypadInterface**	The state of the keypad interface	M
2	**RemoteInterface**	The state of the remote interface	M
5	**SoundVolume**	Sound volume is set to Silent value	M
6	**AutoRelockTime**	Auto relock time it set to 0	M
7	**LEDSettings**	LEDs is disabled	M

5.2.6.4.1. LocalProgramming Bit

This bit SHALL indicate the state related to local programming:

- 0 = Local programming is disabled
- 1 = Local programming is enabled

5.2.6.4.2. KeypadInterface Bit

This bit SHALL indicate the state related to keypad interface:

- 0 = Keypad interface is disabled
- 1 = Keypad interface is enabled

5.2.6.4.3. RemoteInterface Bit

This bit SHALL indicate the state related to remote interface:

- 0 = Remote interface is disabled
- 1 = Remote interface is enabled

5.2.6.4.4. SoundVolume Bit

This bit SHALL indicate the state related to sound volume:

- 0 = Sound volume value is 0 (Silent)
- 1 = Sound volume value is equal to something other than 0

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5.2.6.4.5. AutoRelockTime Bit

This bit SHALL indicate the state related to auto relock time:

- 0 = Auto relock time value is 0
- 1 = Auto relock time value is equal to something other than 0

5.2.6.4.6. LEDSettings Bit

This bit SHALL indicate the state related to LED settings:

- 0 = LED settings value is 0 (NoLEDSignal)
- 1 = LED settings value is equal to something other than 0

5.2.6.5. LocalProgrammingFeaturesBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**AddUsersCredentialsSchedules**	The state of the ability to add users, credentials or schedules on the device	M
1	**ModifyUsersCredentialsSchedules**	The state of the ability to modify users, credentials or schedules on the device	M
2	**ClearUsersCredentialsSchedules**	The state of the ability to clear users, credentials or schedules on the device	M
3	**AdjustSettings**	The state of the ability to adjust settings on the device	M

5.2.6.5.1. AddUsersCredentialsSchedules Bit

This bit SHALL indicate whether the door lock is able to add Users/Credentials/Schedules locally:

- 0 = This ability is disabled
- 1 = This ability is enabled

5.2.6.5.2. ModifyUsersCredentialsSchedules Bit

This bit SHALL indicate whether the door lock is able to modify Users/Credentials/Schedules locally:

- 0 = This ability is disabled
- 1 = This ability is enabled

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5.2.6.5.3. ClearUsersCredentialsSchedules Bit

This bit SHALL indicate whether the door lock is able to clear Users/Credentials/Schedules locally:

- 0 = This ability is disabled
- 1 = This ability is enabled

5.2.6.5.4. AdjustSettings Bit

This bit SHALL indicate whether the door lock is able to adjust lock settings locally:

- 0 = This ability is disabled
- 1 = This ability is enabled

5.2.6.6. AlarmMaskBitmap Type

This data type is derived from map16.

Bit	Name	Summary	Conformance
0	**LockJammed**	Locking Mechanism Jammed	M
1	**LockFactoryReset**	Lock Reset to Factory Defaults	O
3	**LockRadioPowerCycled**	RF Module Power Cycled	O
4	**WrongCodeEntryLimit**	Tamper Alarm - wrong code entry limit	PIN | RID
5	**FrontEscutcheonRemoved**	Tamper Alarm - front escutcheon removed from main	O
6	**DoorForcedOpen**	Forced Door Open under Door Locked Condition	O

5.2.6.7. AlarmCodeEnum Type

This data type is derived from enum8.

This enumeration SHALL indicate the alarm type.

Value	Name	Summary	Conformance
0	**LockJammed**	Locking Mechanism Jammed	M
1	**LockFactoryReset**	Lock Reset to Factory Defaults	O

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Value	Name	Summary	Conformance
3	**LockRadioPowerCycled**	Lock Radio Power Cycled	O
4	**WrongCodeEntryLimit**	Tamper Alarm - wrong code entry limit	[USR]
5	**FrontEsceutcheonRemoved**	Tamper Alarm - front escutcheon removed from main	O
6	**DoorForcedOpen**	Forced Door Open under Door Locked Condition	[DPS]
7	**DoorAjar**	Door ajar	[DPS]
8	**ForcedUser**	Force User SOS alarm	[USR]

5.2.6.8. CredentialRuleEnum Type

This data type is derived from enum8.

This enumeration SHALL indicate the credential rule that can be applied to a particular user.

Value	Name	Summary	Conformance
0	**Single**	Only one credential is required for lock operation	USR
1	**Dual**	Any two credentials are required for lock operation	[USR]
2	**Tri**	Any three credentials are required for lock operation	[USR]

5.2.6.9. CredentialTypeEnum Type

This data type is derived from enum8.

This enumeration SHALL indicate the credential type.

Value	Name	Summary	Conformance
0	**ProgrammingPIN**	Programming PIN code credential type	O
1	**PIN**	PIN code credential type	PIN

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Value	Name	Summary	Conformance
2	**RFID**	RFID identifier credential type	RID
3	**Fingerprint**	Fingerprint identifier credential type	FGP
4	**FingerVein**	Finger vein identifier credential type	FGP
5	**Face**	Face identifier credential type	FACE
6	**AliroCredentialIssuerKey**	A Credential Issuer public key as defined in Aliro	ALIRO
7	**AliroEvictableEndpointKey**	An Endpoint public key as defined in Aliro which can be evicted if space is needed for another endpoint key	ALIRO
8	**AliroNonEvictableEndpointKey**	An Endpoint public key as defined in Aliro which cannot be evicted if space is needed for another endpoint key	ALIRO

5.2.6.9.1. AliroCredentialIssuerKey Value

Credentials of this type SHALL be 65-byte uncompressed elliptic curve public keys as defined in section 2.3.3 of SEC 1.

Credentials of this type SHALL NOT be used to allow operating the lock. They SHALL be used, as defined in [Aliro], to create new credentials of type AliroEvictableEndpointKey via a step-up transaction.

When performing the step-up transaction, the lock SHALL request the data element with identifier "matter1", and SHALL attempt to create a new credential of type AliroEvictableEndpointKey if and only if the data element is returned and the Access Credential can be validated using the AliroCredentialIssuerKey.

When a new credential of type AliroEvictableEndpointKey is added in this manner, it SHALL be associated with the same user record as the AliroCredentialIssuerKey credential that allowed the new credential to be added.

If there are no available credential slots to add a new AliroEvictableEndpointKey credential (i.e. either the NumberOfCredentialsSupportedPerUser or the NumberOfAliroEndpointKeysSupported limit has been reached) but there exist credentials of type AliroEvictableEndpointKey associated with the user record, the server SHALL remove one of those credentials using the same procedure

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it would follow for the ClearCredential command before adding the new credential.

If there are no available credential slots to add a new AliroEvictableEndpointKey credential (i.e. either the NumberOfCredentialsSupportedPerUser or the NumberOfAliroEndpointKeysSupported limit has been reached) and there do not exist credentials of type AliroEvictableEndpointKey associated with the user record, a new AliroEvictableEndpointKey credential SHALL NOT be created.

If the step-up process results in addition of new credentials, the corresponding LockUserChange event SHALL have OperationSource set to Aliro.

If the step-up process results in the lock state changing (e.g. locking or unlocking), the credential associated with those changes in the LockOperation events SHALL be the newly provisioned AliroEvictableEndpointKey credential if one was created. If no new AliroEvictableEndpointKey credential was created, the credential associated with the changes in the LockOperation events SHALL be the AliroCredentialIssuerKey credential used for the step-up.

5.2.6.9.2. AliroEvictableEndpointKey Value

Credentials of this type SHALL be 65-byte uncompressed elliptic curve public keys as defined in section 2.3.3 of SEC 1.

5.2.6.9.3. AliroNonEvictableEndpointKey Value

Credentials of this type SHALL be 65-byte uncompressed elliptic curve public keys as defined in section 2.3.3 of SEC 1.

5.2.6.10. DataOperationTypeEnum Type

This data type is derived from enum8.

This enumeration SHALL indicate the data operation performed.

Value	Name	Summary	Conformance
0	**Add**	Data is being added or was added	M
1	**Clear**	Data is being cleared or was cleared	M
2	**Modify**	Data is being modified or was modified	M

5.2.6.11. DoorStateEnum Type

This data type is derived from enum8.

This enumeration SHALL indicate the current door state.

Value	Name	Summary	Conformance
0	**DoorOpen**	Door state is open	DPS

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Value	Name	Summary	Conformance
1	**DoorClosed**	Door state is closed	DPS
2	**DoorJammed**	Door state is jammed	[DPS]
3	**DoorForcedOpen**	Door state is currently forced open	[DPS]
4	**DoorUnspecifiedError**	Door state is invalid for unspecified reason	[DPS]
5	**DoorAjar**	Door state is ajar	[DPS]

5.2.6.12. LockDataTypeEnum Type

This data type is derived from enum8.

This enumeration SHALL indicate the data type that is being or has changed.

Value	Name	Summary	Conformance
0	**Unspecified**	Unspecified or manufacturer specific lock user data added, cleared, or modified.	O
1	**ProgrammingCode**	Lock programming PIN code was added, cleared, or modified.	O
2	**UserIndex**	Lock user index was added, cleared, or modified.	M
3	**WeekDaySchedule**	Lock user week day schedule was added, cleared, or modified.	WDSCH
4	**YearDaySchedule**	Lock user year day schedule was added, cleared, or modified.	YDSCH
5	**HolidaySchedule**	Lock holiday schedule was added, cleared, or modified.	HDSCH
6	**PIN**	Lock user PIN code was added, cleared, or modified.	PIN
7	**RFID**	Lock user RFID code was added, cleared, or modified.	RID

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Value	Name	Summary	Conformance
8	**Fingerprint**	Lock user fingerprint was added, cleared, or modified.	FGP
9	**FingerVein**	Lock user finger-vein information was added, cleared, or modified.	FGP
10	**Face**	Lock user face information was added, cleared, or modified.	FACE
11	**AliroCredentialIssuerKey**	An Aliro credential issuer key credential was added, cleared, or modified.	ALIRO
12	**AliroEvictableEndpointKey**	An Aliro endpoint key credential which can be evicted credential was added, cleared, or modified.	ALIRO
13	**AliroNonEvictableEndpointKey**	An Aliro endpoint key credential which cannot be evicted was added, cleared, or modified.	ALIRO

5.2.6.13. LockOperationTypeEnum Type

This data type is derived from enum8.

This enumeration SHALL indicate the type of Lock operation performed.

Value	Name	Summary	Conformance
0	**Lock**	Lock operation	M
1	**Unlock**	Unlock operation	M
2	**NonAccessUserEvent**	Triggered by keypad entry for user with User Type set to Non Access User	O
3	**ForcedUserEvent**	Triggered by using a user with UserType set to Forced User	O
4	**Unlatch**	Unlatch operation	M

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5.2.6.14. OperationErrorEnum Type

This data type is derived from enum8.

This enumeration SHALL indicate the error cause of the Lock/Unlock operation performed.

Value	Name	Summary	Conformance
0	**Unspecified**	Lock/unlock error caused by unknown or unspecified source	O
1	**InvalidCredential**	Lock/unlock error caused by invalid PIN, RFID, fingerprint or other credential	USR
2	**DisabledUserDenied**	Lock/unlock error caused by disabled USER or credential	M
3	**Restricted**	Lock/unlock error caused by schedule restriction	WDSCH | YDSCH
4	**InsufficientBattery**	Lock/unlock error caused by insufficient battery power left to safely actuate the lock	O

5.2.6.15. OperatingModeEnum Type

This data type is derived from enum8.

This enumeration SHALL indicate the lock operating mode.

Value	Name	Summary	Conformance
0	**Normal**		M
1	**Vacation**		O
2	**Privacy**		O
3	**NoRemoteLockUnlock**		M
4	**Passage**		O

The table below shows the operating mode and which interfaces are enabled, if supported, for each mode.

Name	Keypad*	Remote*	RFID*
Normal	Y	Y	Y
Vacation	N	Y	N

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Name	Keypad*	Remote*	RFID*
Privacy	N	N	N
NoRemoteLockUnlock	Y	N	Y
Passage	N/A	N/A	N/A

* Interface Operational: Yes, No or N/A

NOTE For modes that disable the remote interface, the door lock SHALL respond to Lock, Unlock, Toggle, and Unlock with Timeout commands with a response status Failure and not take the action requested by those commands. The door lock SHALL NOT disable the radio or otherwise unbind or leave the network. It SHALL still respond to all other commands and requests.

5.2.6.15.1. Normal Value

The lock operates normally. All interfaces are enabled.

5.2.6.15.2. Vacation Value

Only remote interaction is enabled. The keypad SHALL only be operable by the master user.

5.2.6.15.3. Privacy Value

This mode is only possible if the door is locked. Manual unlocking changes the mode to Normal operating mode. All external interaction with the door lock is disabled. This mode is intended to be used so that users, presumably inside the property, will have control over the entrance.

5.2.6.15.4. NoRemoteLockUnlock Value

This mode only disables remote interaction with the lock. This does not apply to any remote proprietary means of communication. It specifically applies to the Lock, Unlock, Toggle, and Unlock with Timeout Commands.

5.2.6.15.5. Passage Value

The lock is open or can be opened or closed at will without the use of a Keypad or other means of user validation (e.g. a lock for a business during work hours).

5.2.6.16. OperationSourceEnum Type

This data type is derived from enum8.

This enumeration SHALL indicate the source of the Lock/Unlock or user change operation performed.

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Value	Name	Summary	Conformance
0	**Unspecified**	Lock/unlock operation came from unspecified source	O
1	**Manual**	Lock/unlock operation came from manual operation (key, thumb-turn, handle, etc).	O
2	**ProprietaryRemote**	Lock/unlock operation came from proprietary remote source (e.g. vendor app/cloud)	O
3	**Keypad**	Lock/unlock operation came from keypad	O
4	**Auto**	Lock/unlock operation came from lock automatically (e.g. relock timer)	O
5	**Button**	Lock/unlock operation came from lock button (e.g. one touch or button)	O
6	**Schedule**	Lock/unlock operation came from lock due to a schedule	HDSCH
7	**Remote**	Lock/unlock operation came from remote node	M
8	**RFID**	Lock/unlock operation came from RFID card	RID
9	**Biometric**	Lock/unlock operation came from biometric source (e.g. face, fingerprint/fingervein)	[USR]
10	**Aliro**	Lock/unlock operation came from an interaction defined in Aliro, or user change operation was a step-up credential provisioning as defined in Aliro	ALIRO

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5.2.6.17. UserStatusEnum Type

This data type is derived from enum8.

This enumeration SHALL indicate what the status is for a specific user ID.

Value	Name	Summary	Conformance
0	**Available**	The user ID is available	M
1	**OccupiedEnabled**	The user ID is occupied and enabled	M
3	**OccupiedDisabled**	The user ID is occupied and disabled	O

5.2.6.18. UserTypeEnum Type

This data type is derived from enum8.

This enumeration SHALL indicate what the type is for a specific user ID.

Value	Name	Summary	Conformance
0	**UnrestrictedUser**	The user ID type is unrestricted	M
1	**YearDayScheduleUser**	The user ID type is schedule	O
2	**WeekDaySchedule-User**	The user ID type is schedule	O
3	**ProgrammingUser**	The user ID type is programming	O
4	**NonAccessUser**	The user ID type is non access	O
5	**ForcedUser**	The user ID type is forced	[USR]
6	**DisposableUser**	The user ID type is disposable	[USR]
7	**ExpiringUser**	The user ID type is expiring	[USR]
8	**ScheduleRestricted-User**	The user ID type is schedule restricted	WDSCH | YDSCH
9	**RemoteOnlyUser**	The user ID type is remote only	USR & COTA & PIN

5.2.6.18.1. UnrestrictedUser Value

This value SHALL indicate the user has access 24/7 provided proper PIN or RFID is supplied (e.g.,

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owner).

5.2.6.18.2. YearDayScheduleUser Value

This value SHALL indicate the user has the ability to open lock within a specific time period (e.g., guest).

When UserType is set to YearDayScheduleUser, user access SHALL be restricted as follows:

- If no YearDaySchedules are set for the user, then access SHALL be denied
- If one or more YearDaySchedules are set, user access SHALL be granted if and only if the current time falls within at least one of the YearDaySchedules. If current time is not known, user access SHALL NOT be granted.

5.2.6.18.3. WeekDayScheduleUser Value

This value SHALL indicate the user has the ability to open lock based on specific time period within a reoccurring weekly schedule (e.g., cleaning worker).

When UserType is set to WeekDayScheduleUser, user access SHALL be restricted as follows:

- If no WeekDaySchedules are set for the user, then access SHALL be denied
- If one or more WeekDaySchedules are set, user access SHALL be granted if and only if the current time falls within at least one of the WeekDaySchedules. If current time is not known, user access SHALL NOT be granted.

5.2.6.18.4. ProgrammingUser Value

This value SHALL indicate the user has the ability to both program and operate the door lock. This user can manage the users and user schedules. In all other respects this user matches the unrestricted (default) user. ProgrammingUser is the only user that can disable the user interface (keypad, remote, etc...).

5.2.6.18.5. NonAccessUser Value

This value SHALL indicate the user is recognized by the lock but does not have the ability to open the lock. This user will only cause the lock to generate the appropriate event notification to any bound devices.

5.2.6.18.6. ForcedUser Value

This value SHALL indicate the user has the ability to open lock but a ForcedUser LockOperationType and ForcedUser silent alarm will be emitted to allow a notified Node to alert emergency services or contacts on the user account when used.

5.2.6.18.7. DisposableUser Value

This value SHALL indicate the user has the ability to open lock once after which the lock SHALL change the corresponding user record UserStatus value to OccupiedDisabled automatically.

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5.2.6.18.8. ExpiringUser Value

This value SHALL indicate the user has the ability to open lock for ExpiringUserTimeout attribute minutes after the first use of the PIN code, RFID code, Fingerprint, or other credential. After ExpiringUserTimeout minutes the corresponding user record UserStatus value SHALL be set to OccupiedDisabled automatically by the lock. The lock SHALL persist the timeout across reboots such that the ExpiringUserTimeout is honored.

5.2.6.18.9. ScheduleRestrictedUser Value

This value SHALL indicate the user access is restricted by Week Day and/or Year Day schedule.

When UserType is set to ScheduleRestrictedUser, user access SHALL be restricted as follows:

- If no WeekDaySchedules and no YearDaySchedules are set for the user, then access SHALL be denied
- If one or more WeekDaySchedules are set, but no YearDaySchedules are set for the user, then user access SHALL be equivalent to the WeekDayScheduleUser UserType
- If one or more YearDaySchedules are set, but no WeekDaySchedules are set for the user, then user access SHALL be equivalent to the YearDayScheduleUser UserType
- If one or WeekDaySchedules are set AND one or more YearDaySchedules are set, then user access SHALL be granted if and only if the current time falls within at least one of the WeekDaySchedules AND the current time falls within at least one of the YearDaySchedules.

5.2.6.18.10. RemoteOnlyUser Value

This value SHALL indicate the user access and PIN code is restricted to remote lock/unlock commands only. This type of user might be useful for regular delivery services or voice assistant unlocking operations to prevent a PIN code credential created for them from being used at the keypad. The PIN code credential would only be provided over-the-air for the lock/unlock commands.

5.2.6.19. LockStateEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**NotFullyLocked**	Lock state is not fully locked	M
1	**Locked**	Lock state is fully locked	M
2	**Unlocked**	Lock state is fully unlocked	M
3	**Unlatched**	Lock state is fully unlocked and the latch is pulled	O

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5.2.6.20. LockTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**DeadBolt**	Physical lock type is dead bolt	M
1	**Magnetic**	Physical lock type is magnetic	M
2	**Other**	Physical lock type is other	M
3	**Mortise**	Physical lock type is mortise	M
4	**Rim**	Physical lock type is rim	M
5	**LatchBolt**	Physical lock type is latch bolt	M
6	**CylindricalLock**	Physical lock type is cylindrical lock	M
7	**TubularLock**	Physical lock type is tubular lock	M
8	**InterconnectedLock**	Physical lock type is interconnected lock	M
9	**DeadLatch**	Physical lock type is dead latch	M
10	**DoorFurniture**	Physical lock type is door furniture	M
11	**Eurocylinder**	Physical lock type is euro cylinder	M

5.2.6.21. LEDSettingEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**NoLEDSignal**	Never use LED for signalization	M
1	**NoLEDSignalAccessAllowed**	Use LED signalization except for access allowed events	M
2	**LEDSignalAll**	Use LED signalization for all events	M

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5.2.6.22. SoundVolumeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Silent**	Silent Mode	M
1	**Low**	Low Volume	M
2	**High**	High Volume	M
3	**Medium**	Medium Volume	M

5.2.6.23. EventTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Operation**	Event type is operation	M
1	**Programming**	Event type is programming	M
2	**Alarm**	Event type is alarm	M

5.2.6.24. CredentialStruct Type

This struct SHALL indicate the credential types and their corresponding indices (if any) for the event or user record.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**CredentialType**	CredentialTypeEnum	all				M
1	**CredentialIndex**	uint16	all				M

5.2.6.24.1. CredentialType Field

This field SHALL indicate the credential field used to authorize the lock operation.

5.2.6.24.2. CredentialIndex Field

This field SHALL indicate the index of the specific credential used to authorize the lock operation in the list of credentials identified by CredentialType (e.g. PIN, RFID, etc.). This field SHALL be set to 0 if CredentialType is ProgrammingPIN or does not correspond to a list that can be indexed into.

5.2.7. Status Codes

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5.2.7.1. StatusCodeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0x02	**Duplicate**	Entry would cause a duplicate credential/ID.	M
0x03	**Occupied**	Entry would replace an occupied slot.	M

5.2.7.2. Duplicate Code

The provided User ID, PIN or RFID code or other credential is a duplicate of an existing entry.

5.2.7.3. Occupied Code

The provided User ID, Credential ID, or entry location is already occupied. The entry might need to be deleted or a different ID or location chosen.

5.2.8. PIN/RFID Code Format

The PIN/RFID codes defined in this specification are all octet strings.

All value in the PIN/RFID code SHALL be ASCII encoded regardless if the PIN/RFID codes are number or characters. For example, code of “1, 2, 3, 4” SHALL be represented as 0x31, 0x32, 0x33, 0x34.

5.2.9. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**LockState**	LockStateEnum	desc	X		R V	M
0x0001	**LockType**	LockTypeEnum	desc			R V	M
0x0002	**ActuatorEnabled**	bool	all			R V	M
0x0003	**DoorState**	DoorStateEnum	desc	X		R V	DPS
0x0004	**DoorOpenEvents**	uint32	all			RW VM	[DPS]
0x0005	**DoorClosedEvents**	uint32	all			RW VM	[DPS]

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0006	**OpenPeriod**	uint16	all			RW VM	[DPS]
0x0011	**NumberOfTotalUsersSupported**	uint16	all	F		R V	USR
0x0012	**NumberOfPINUsersSupported**	uint16	all	F		R V	PIN
0x0013	**NumberOfRFIDUsersSupported**	uint16	all	F		R V	RID
0x0014	**NumberOfWeekDaySchedulesSupportedPerUser**	uint8	max 0xFD	F		R V	WDSCH
0x0015	**NumberOfYearDaySchedulesSupportedPerUser**	uint8	max 0xFD	F		R V	YDSCH
0x0016	**NumberOfHolidaySchedulesSupported**	uint8	max 0xFD	F		R V	HDSCH
0x0017	**MaxPINCodeLength**	uint8	all	F		R V	PIN
0x0018	**MinPINCodeLength**	uint8	all	F		R V	PIN
0x0019	**MaxRFIDCodeLength**	uint8	all	F		R V	RID

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x001A	**MinRFIDCodeLength**	uint8	all	F		R V	RID
0x001B	**CredentialRulesSupport**	CredentialRulesBitmap	all	F		R V	USR
0x001C	**NumberOfCredentialsSupportedPerUser**	uint8	all	F		R V	USR
0x0021	**Language**	string	max 3		MS	R[W] VM	O
0x0022	**LEDSettings**	LEDSettingEnum	all		0	R[W] VM	O
0x0023	**AutoRelockTime**	uint32	all		MS	R[W] VM	O
0x0024	**SoundVolume**	SoundVolumeEnum	all		0	R[W] VM	O
0x0025	**OperatingMode**	OperatingModeEnum	desc			R[W] VM	M
0x0026	**SupportedOperatingModes**	OperatingModesBitmap	all	F		R V	M
0x0027	**DefaultConfigurationRegister**	ConfigurationRegisterBitmap	all		0	R V	O
0x0028	**EnableLocalProgramming**	bool	all		1	R[W] VA	O
0x0029	**EnableOneTouchLocking**	bool	all		0	RW VM	O
0x002A	**EnableInsideStatusLED**	bool	all		0	RW VM	O

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x002B	**EnablePrivacyModeButton**	bool	all		0	RW VM	O
0x002C	**LocalProgrammingFeatures**	LocalProgrammingFeaturesBitmap	all		0	R[W] VA	O
0x0030	**WrongCodeEntryLimit**	uint8	1 to 255			R[W] VA	PIN | RID
0x0031	**UserCodeTemporaryDisableTime**	uint8	1 to 255			R[W] VA	PIN | RID
0x0032	**SendPINOverTheAir**	bool	all		0	R[W] VA	[!USR & PIN]
0x0033	**RequirePINforRemoteOperation**	bool	all			R[W] VA	COTA & PIN
0x0034	**SecurityLevel**				0	R V	D
0x0035	**ExpiringUserTimeout**	uint16	1 to 2880		MS	R[W] VA	[USR]
0x0080	**AliroReaderVerificationKey**	octstr	65	X		R A	ALIRO
0x0081	**AliroReaderGroupIdentifier**	octstr	16	X		R A	ALIRO
0x0082	**AliroReaderGroupSubIdentifier**	octstr	16	F		R A	ALIRO

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0083	**AliroExpeditedTransactionSupportedProtocolVersions**	list[octstr]	max 16[2]	F		R A	ALIRO
0x0084	**AliroGroupResolvingKey**	octstr	16	X		R A	ALBU
0x0085	**AliroSupportedBLEUWBProtocolVersions**	list[octstr]	max 16[2]	F		R A	ALBU
0x0086	**AliroBLEAdvertisingVersion**	uint8	all	F		R A	ALBU
0x0087	**NumberOfAliroCredentialIssuerKeysSupported**	uint16	all	F		R V	ALIRO
0x0088	**NumberOfAliroEndpointKeysSupported**	uint16	all	F		R V	ALIRO

5.2.9.1. LockState Attribute

This attribute may be NULL if the lock hardware does not currently know the status of the locking mechanism. For example, a lock may not know the LockState status after a power cycle until the first lock actuation is completed.

The Not Fully Locked value is used by a lock to indicate that the state of the lock is somewhere between Locked and Unlocked so it is only partially secured. For example, a deadbolt could be partially extended and not in a dead latched state.

5.2.9.2. LockType Attribute

This attribute SHALL indicate the type of door lock as defined in LockTypeEnum.

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5.2.9.3. ActuatorEnabled Attribute

This attribute SHALL indicate if the lock is currently able to (Enabled) or not able to (Disabled) process remote Lock, Unlock, or Unlock with Timeout commands.

This attribute has the following possible values:

Boolean Value	Summary
0	Disabled
1	Enabled

5.2.9.4. DoorState Attribute

This attribute SHALL indicate the current door state as defined in DoorStateEnum.

This attribute SHALL be null only if an internal error prevents the retrieval of the current door state.

5.2.9.5. DoorOpenEvents Attribute

This attribute SHALL hold the number of door open events that have occurred since it was last zeroed.

5.2.9.6. DoorClosedEvents Attribute

This attribute SHALL hold the number of door closed events that have occurred since it was last zeroed.

5.2.9.7. OpenPeriod Attribute

This attribute SHALL hold the number of minutes the door has been open since the last time it transitioned from closed to open.

5.2.9.8. NumberOfTotalUsersSupported Attribute

This attribute SHALL indicate the number of total users supported by the lock.

5.2.9.9. NumberOfPINUsersSupported Attribute

This attribute SHALL indicate the number of PIN users supported.

5.2.9.10. NumberOfRFIDUsersSupported Attribute

This attribute SHALL indicate the number of RFID users supported.

5.2.9.11. NumberOfWeekDaySchedulesSupportedPerUser Attribute

This attribute SHALL indicate the number of configurable week day schedule supported per user.

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5.2.9.12. NumberOfYearDaySchedulesSupportedPerUser Attribute

This attribute SHALL indicate the number of configurable year day schedule supported per user.

5.2.9.13. NumberOfHolidaySchedulesSupported Attribute

This attribute SHALL indicate the number of holiday schedules supported for the entire door lock device.

5.2.9.14. MaxPINCodeLength Attribute

This attribute SHALL indicate the maximum length in bytes of a PIN Code on this device.

5.2.9.15. MinPINCodeLength Attribute

This attribute SHALL indicate the minimum length in bytes of a PIN Code on this device.

5.2.9.16. MaxRFIDCodeLength Attribute

This attribute SHALL indicate the maximum length in bytes of a RFID Code on this device. The value depends on the RFID code range specified by the manufacturer, if media anti-collision identifiers (UID) are used as RFID code, a value of 20 (equals 10 Byte ISO 14443A UID) is recommended.

5.2.9.17. MinRFIDCodeLength Attribute

This attribute SHALL indicate the minimum length in bytes of a RFID Code on this device. The value depends on the RFID code range specified by the manufacturer, if media anti-collision identifiers (UID) are used as RFID code, a value of 8 (equals 4 Byte ISO 14443A UID) is recommended.

5.2.9.18. CredentialRulesSupport Attribute

This attribute SHALL contain a bitmap with the bits set for the values of CredentialRuleEnum supported on this device.

5.2.9.19. NumberOfCredentialsSupportedPerUser Attribute

This attribute SHALL indicate the number of credentials that could be assigned for each user.

Depending on the value of NumberOfRFIDUsersSupported and NumberOfPINUsersSupported it may not be possible to assign that number of credentials for a user.

For example, if the device supports only PIN and RFID credential types, NumberOfCredentialsSupportedPerUser is set to 10, NumberOfPINUsersSupported is set to 5 and NumberOfRFIDUsersSupported is set to 3, it will not be possible to actually assign 10 credentials for a user because maximum number of credentials in the database is 8.

5.2.9.20. Language Attribute

This attribute SHALL indicate the language for the on-screen or audible user interface using a 2-byte language code from ISO-639-1.

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5.2.9.21. LEDSettings Attribute

This attribute SHALL indicate the settings for the LED support, as defined by LEDSettingEnum.

5.2.9.22. AutoRelockTime Attribute

This attribute SHALL indicate the number of seconds to wait after unlocking a lock before it automatically locks again. 0=disabled. If set, unlock operations from any source will be timed. For one time unlock with timeout use the specific command.

5.2.9.23. SoundVolume Attribute

This attribute SHALL indicate the sound volume on a door lock as defined by SoundVolumeEnum.

5.2.9.24. OperatingMode Attribute

This attribute SHALL indicate the current operating mode of the lock as defined in OperatingModeEnum.

5.2.9.25. SupportedOperatingModes Attribute

This attribute SHALL contain a bitmap with all operating bits of the OperatingMode attribute supported by the lock.

A bit position set to zero SHALL indicate that the mode is supported. A bit position set to one SHALL indicate that the mode is not supported.

Any bit that is not yet defined in OperatingModesBitmap SHALL be set to 1.

The values considered valid to read or write in the OperatingMode attribute SHALL be the enum values from DoorLockOperatingModeEnum whose equivalent same-named bit from OperatingModesBitmap is set to zero in this attribute. WARNING: This is the opposite of most other semantically similar bitmaps in this specification.

5.2.9.26. DefaultConfigurationRegister Attribute

This attribute SHALL represent the default configurations as they are physically set on the device (example: hardware dip switch setting, etc...) and represents the default setting for some of the attributes within this cluster (for example: LED, Auto Lock, Sound Volume, and Operating Mode attributes).

This is a read-only attribute and is intended to allow clients to determine what changes MAY need to be made without having to query all the included attributes. It MAY be beneficial for the clients to know what the device's original settings were in the event that the device needs to be restored to factory default settings.

If the Client device would like to query and modify the door lock server's operating settings, it SHOULD send read and write attribute requests to the specific attributes.

For example, the Sound Volume attribute default value is Silent Mode. However, it is possible that the current Sound Volume is High Volume. Therefore, if the client wants to query/modify the cur-

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rent Sound Volume setting on the server, the client SHOULD read/write to the Sound Volume attribute.

5.2.9.26.1. LocalProgramming Bit

This bit SHALL indicate the default value of the EnableLocalProgramming attribute.

5.2.9.26.2. KeypadInterface Bit

This bit SHALL indicate the default state of the keypad interface.

5.2.9.26.3. RemoteInterface Bit

This bit SHALL indicate the default state of the remote interface.

5.2.9.26.4. SoundVolume Bit

This bit SHALL indicate the default value of SoundVolume attribute.

5.2.9.26.5. AutoRelockTime Bit

This bit SHALL indicate the default value of AutoRelockTime attribute.

5.2.9.26.6. LEDSettings Bit

This bit SHALL indicate the default value of LEDSettings attribute.

5.2.9.27. EnableLocalProgramming Attribute

This attribute SHALL enable/disable local programming on the door lock of certain features (see LocalProgrammingFeatures attribute). If this value is set to TRUE then local programming is enabled on the door lock for all features. If it is set to FALSE then local programming is disabled on the door lock for those features whose bit is set to 0 in the LocalProgrammingFeatures attribute. Local programming SHALL be enabled by default.

5.2.9.28. EnableOneTouchLocking Attribute

This attribute SHALL enable/disable the ability to lock the door lock with a single touch on the door lock.

5.2.9.29. EnableInsideStatusLED Attribute

This attribute SHALL enable/disable an inside LED that allows the user to see at a glance if the door is locked.

5.2.9.30. EnablePrivacyModeButton Attribute

This attribute SHALL enable/disable a button inside the door that is used to put the lock into privacy mode. When the lock is in privacy mode it cannot be manipulated from the outside.

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5.2.9.31. LocalProgrammingFeatures Attribute

This attribute SHALL indicate the local programming features that will be disabled when EnableLocalProgramming attribute is set to False. If a door lock doesn't support disabling one aspect of local programming it SHALL return CONSTRAINT_ERROR during a write operation of this attribute. If the EnableLocalProgramming attribute is set to True then all local programming features SHALL be enabled regardless of the bits set to 0 in this attribute.

The features that can be disabled from local programming are defined in LocalProgrammingFeaturesBitmap.

5.2.9.32. WrongCodeEntryLimit Attribute

This attribute SHALL indicate the number of incorrect Pin codes or RFID presentment attempts a user is allowed to enter before the lock will enter a lockout state. The value of this attribute is compared to all failing forms of credential presentation, including Pin codes used in an Unlock Command when RequirePINforRemoteOperation is set to true. Valid range is 1-255 incorrect attempts. The lockout state will be for the duration of UserCodeTemporaryDisableTime. If the attribute accepts writes and an attempt to write the value 0 is made, the device SHALL respond with CONSTRAINT_ERROR.

The lock MAY reset the counter used to track incorrect credential presentations as required by internal logic, environmental events, or other reasons. The lock SHALL reset the counter if a valid credential is presented.

5.2.9.33. UserCodeTemporaryDisableTime Attribute

This attribute SHALL indicate the number of seconds that the lock shuts down following wrong code entry. Valid range is 1-255 seconds. Device can shut down to lock user out for specified amount of time. (Makes it difficult to try and guess a PIN for the device.) If the attribute accepts writes and an attempt to write the attribute to 0 is made, the device SHALL respond with CONSTRAINT_ERROR.

5.2.9.34. SendPINOverTheAir Attribute

This attribute SHALL indicate the door locks ability to send PINs over the air. If the attribute is True it is ok for the door lock server to send PINs over the air. This attribute determines the behavior of the server's TX operation. If it is false, then it is not ok for the device to send PIN in any messages over the air.

The PIN field within any door lock cluster message SHALL keep the first octet unchanged and masks the actual code by replacing with 0xFF. For example (PIN "1234"): If the attribute value is True, 0x04 0x31 0x32 0x33 0x34 SHALL be used in the PIN field in any door lock cluster message payload. If the attribute value is False, 0x04 0xFF 0xFF 0xFF 0xFF SHALL be used.

5.2.9.35. RequirePINForRemoteOperation Attribute

This attribute SHALL indicate if the door lock requires an optional PIN. If this attribute is set to True, the door lock server requires that an optional PINs be included in the payload of remote lock operation events like Lock, Unlock, Unlock with Timeout and Toggle in order to function.

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5.2.9.36. ExpiringUserTimeout Attribute

This attribute SHALL indicate the number of minutes a PIN, RFID, Fingerprint, or other credential associated with a user of type ExpiringUser SHALL remain valid after its first use before expiring. When the credential expires the UserStatus for the corresponding user record SHALL be set to OccupiedDisabled.

5.2.9.37. AliroReaderVerificationKey Attribute

This attribute SHALL indicate the verification key component of the Reader's key pair as defined in [Aliro]. The value, if not null, SHALL be an uncompressed elliptic curve public key as defined in section 2.3.3 of SEC 1.

This attribute SHALL be null if no Reader key pair has been configured on the lock. See Section 5.2.10.26, "SetAliroReaderConfig Command".

5.2.9.38. AliroReaderGroupIdentifier Attribute

This attribute SHALL indicate the reader_group_identifier as defined in [Aliro].

This attribute SHALL be null if no reader_group_identifier has been configured on the lock. See Section 5.2.10.26, "SetAliroReaderConfig Command".

5.2.9.39. AliroReaderGroupSubIdentifier Attribute

This attribute SHALL indicate the reader_group_sub_identifier as defined in [Aliro].

5.2.9.40. AliroExpeditedTransactionSupportedProtocolVersions Attribute

This attribute SHALL indicate the list of protocol versions supported for expedited transactions as defined in [Aliro].

5.2.9.41. AliroGroupResolvingKey Attribute

This attribute SHALL indicate the Group Resolving Key as defined in [Aliro].

This attribute SHALL be null if no group resolving key has been configured on the lock. See Section 5.2.10.26, "SetAliroReaderConfig Command".

5.2.9.42. AliroSupportedBLEUWBProtocolVersions Attribute

This attribute SHALL indicate the list of protocol versions supported for the Bluetooth LE + UWB Access Control Flow as defined in [Aliro].

5.2.9.43. AliroBLEAdvertisingVersion Attribute

This attribute SHALL indicate the version of the Bluetooth LE advertisement as defined in [Aliro].

5.2.9.44. NumberOfAliroCredentialIssuerKeysSupported

This attribute SHALL indicate the maximum number of AliroCredentialIssuerKey credentials that

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can be stored on the lock.

5.2.9.45. NumberOfAliroEndpointKeysSupported

This attribute SHALL indicate the maximum number of endpoint key credentials that can be stored on the lock. This limit applies to the sum of the number of AliroEvictableEndpointKey credentials and the number of AliroNonEvictableEndpointKey credentials.

NOTE

The credential indices used for these two credential types are independent of each other, similar to all other credential types. As long as NumberOfAliroEndpointKeysSupported is at least 2 a client could add a credential of type AliroEvictableEndpointKey at any index from 1 to NumberOfAliroEndpointKeysSupported and also add a credential of type AliroNonEvictableEndpointKey at the same index, and both credentials would exist on the server.

5.2.10. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**LockDoor**	client ⇒ server	Y	O T	M
0x01	**UnlockDoor**	client ⇒ server	Y	O T	M
0x02	**Toggle**	client ⇒ server	Y	O T	X
0x03	**UnlockWithTimeout**	client ⇒ server	Y	O T	O
0x0B	**SetWeekDaySchedule**	client ⇒ server	Y	A	WDSCH
0x0C	**GetWeekDaySchedule**	client ⇒ server	GetWeekDaySchedule-uleResponse	A	WDSCH
0x0C	**GetWeekDaySchedule-uleResponse**	client ⇐ server	N		WDSCH
0x0D	**ClearWeekDaySchedule**	client ⇒ server	Y	A	WDSCH
0x0E	**SetYearDaySchedule**	client ⇒ server	Y	A	YDSCH
0x0F	**GetYearDaySchedule**	client ⇒ server	GetYearDaySchedule-uleResponse	A	YDSCH
0x0F	**GetYearDaySchedule-uleResponse**	client ⇐ server	N		YDSCH
0x10	**ClearYearDaySchedule**	client ⇒ server	Y	A	YDSCH

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ID	Name	Direction	Response	Access	Conformance
0x11	**SetHolidaySchedule**	client ⇒ server	Y	A	HDSCH
0x12	**GetHolidaySchedule**	client ⇒ server	GetHolidayScheduleResponse	A	HDSCH
0x12	**GetHolidayScheduleResponse**	client ⇐ server	N		HDSCH
0x13	**ClearHolidaySchedule**	client ⇒ server	Y	A	HDSCH
0x1A	**SetUser**	client ⇒ server	Y	A T	USR
0x1B	**GetUser**	client ⇒ server	GetUserResponse	A	USR
0x1C	**GetUserResponse**	client ⇐ server	N		USR
0x1D	**ClearUser**	client ⇒ server	Y	A T	USR
0x22	**SetCredential**	client ⇒ server	SetCredentialResponse	A T	USR
0x23	**SetCredentialResponse**	client ⇐ server	N		USR
0x24	**GetCredentialStatus**	client ⇒ server	GetCredentialStatusResponse	A	USR
0x25	**GetCredentialStatusResponse**	client ⇐ server	N		USR
0x26	**ClearCredential**	client ⇒ server	Y	A T	USR
0x27	**UnboltDoor**	client ⇒ server	Y	O T	UBOLT
0x28	**SetAliroReaderConfig**	client ⇒ server	Y	A T	ALIRO
0x29	**ClearAliroReaderConfig**	client ⇒ server	Y	A T	ALIRO

5.2.10.1. LockDoor Command

This command causes the lock device to lock the door. This command includes an optional code for the lock. The door lock MAY require a PIN depending on the value of the RequirePINForRemoteOperation attribute.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PINCode**	octstr				[COTA & PIN]

5.2.10.1.1. PINCode Field

If the RequirePINforRemoteOperation attribute is True then PINCode field SHALL be provided and the door lock SHALL NOT grant access if it is not provided.

If the PINCode field is provided, the door lock SHALL verify PINCode before granting access regardless of the value of RequirePINForRemoteOperation attribute.

When the PINCode field is provided an invalid PIN will count towards the WrongCodeEntryLimit and the UserCodeTemporaryDisableTime will be triggered if the WrongCodeEntryLimit is exceeded. The lock SHALL ignore any attempts to lock/unlock the door until the UserCodeTemporaryDisableTime expires.

5.2.10.2. UnlockDoor Command

This command causes the lock device to unlock the door. This command includes an optional code for the lock. The door lock MAY require a code depending on the value of the RequirePINForRemoteOperation attribute.

NOTE	If the attribute AutoRelockTime is supported the lock will transition to the locked state when the auto relock time has expired.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PINCode**	octstr				[COTA & PIN]

5.2.10.2.1. PINCode Field

See Section 5.2.10.1.1, “PINCode Field”.

5.2.10.3. UnlockWithTimeout Command

This command causes the lock device to unlock the door with a timeout parameter. After the time in seconds specified in the timeout field, the lock device will relock itself automatically. This timeout parameter is only temporary for this message transition and overrides the default relock time as specified in the AutoRelockTime attribute. If the door lock device is not capable of or does not want to support temporary Relock Timeout, it SHOULD NOT support this optional command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Timeout**	uint16				M
1	**PINCode**	octstr				[COTA & PIN]

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5.2.10.3.1. Timeout Field

This field SHALL indicate the timeout in seconds to wait before relocking the door lock. This value is independent of the AutoRelockTime attribute value.

5.2.10.3.2. PINCode Field

See Section 5.2.10.1.1, “PINCode Field”.

5.2.10.4. SetWeekDaySchedule Command

Set a weekly repeating schedule for a specified user.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**WeekDayIndex**	uint8	1 to NumberOfWeekDaySchedulesSupportedPerUser			M
1	**UserIndex**	uint16	1 to NumberOfTotalUsersSupported			M
2	**DaysMask**	DaysMaskBitmap				M
3	**StartHour**	uint8	max 23			M
4	**StartMinute**	uint8	max 59			M
5	**EndHour**	uint8	max 23			M
6	**EndMinute**	uint8	max 59			M

The associated UserType MAY be changed to ScheduleRestrictedUser by the lock when a Week Day schedule is set.

Return status SHALL be one of the following values:

Name	Summary
SUCCESS	Setting Week Day schedule was successful
FAILURE	Some unexpected internal error occurred setting Week Day schedule
INVALID_COMMAND	One or more fields violates constraints or is invalid

5.2.10.4.1. WeekDayIndex Field

This field SHALL indicate the index of the Week Day schedule.

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5.2.10.4.2. UserIndex Field

This field SHALL indicate the user ID.

5.2.10.4.3. DaysMask Field

This field SHALL indicate which week days the schedule is active.

5.2.10.4.4. StartHour Field

This field SHALL indicate the starting hour for the Week Day schedule.

5.2.10.4.5. StartMinute Field

This field SHALL indicate the starting minute for the Week Day schedule.

5.2.10.4.6. EndHour Field

This field SHALL indicate the ending hour for the Week Day schedule. EndHour SHALL be equal to or greater than StartHour.

5.2.10.4.7. EndMinute Field

This field SHALL indicate the ending minute for the Week Day schedule. If EndHour is equal to StartHour then EndMinute SHALL be greater than StartMinute.

If the EndHour is equal to 23 and the EndMinute is equal to 59 the Lock SHALL grant access to the user up until 23:59:59.

5.2.10.5. GetWeekDaySchedule Command

Retrieve the specific weekly schedule for the specific user.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**WeekDayIndex**	uint8	1 to NumberOfWeekDaySchedulesSupportedPerUser			M
1	**UserIndex**	uint16	1 to NumberOfTotalUsersSupported			M

5.2.10.6. GetWeekDayScheduleResponse Command

Returns the weekly repeating schedule data for the specified schedule index.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**WeekDayIndex**	uint8	1 to NumberOfWeekDaySchedulesSupportedPerUser			M
1	**UserIndex**	uint16	1 to NumberOfTotalUsersSupported			M
2	**Status**	enum8	desc			M
3	**DaysMask**	DaysMaskBitmap				O
4	**StartHour**	uint8	max 23			O
5	**StartMinute**	uint8	max 59			O
6	**EndHour**	uint8	max 23			O
7	**EndMinute**	uint8	max 59			O

5.2.10.6.1. WeekDayIndex Field

This field SHALL indicate the index of the Week Day schedule.

5.2.10.6.2. UserIndex Field

This field SHALL indicate the user ID.

5.2.10.6.3. Status Field

Status SHALL be one of the following values:

- SUCCESS if both WeekDayIndex and UserIndex are valid and there is a corresponding schedule entry.
- INVALID_COMMAND if either WeekDayIndex and/or UserIndex values are not within valid range
- NOT_FOUND if no corresponding schedule entry found for WeekDayIndex.
- NOT_FOUND if no corresponding user entry found for UserIndex.

If this field is SUCCESS, the optional fields for this command SHALL be present. For other (error) status values, only the fields up to the status field SHALL be present.

5.2.10.6.4. StartHour Field

This field SHALL indicate the starting hour for the Week Day schedule.

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5.2.10.6.5. StartMinute Field

This field SHALL indicate the starting minute for the Week Day schedule.

5.2.10.6.6. EndHour Field

This field SHALL indicate the ending hour for the Week Day schedule. EndHour SHALL be equal to or greater than StartHour.

5.2.10.6.7. EndMinute Field

This field SHALL indicate the ending minute for the Week Day schedule. If EndHour is equal to StartHour then EndMinute SHALL be greater than StartMinute.

5.2.10.7. ClearWeekDaySchedule Command

Clear the specific weekly schedule or all weekly schedules for the specific user.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**WeekDayIndex**	uint8	1 to NumberOfWeekDaySchedulesSupportedPerUser, 0xFE			M
1	**UserIndex**	uint16	1 to NumberOfTotalUsersSupported			M

Return status SHALL be one of the following values:

Name	Summary
SUCCESS	Clearing the requested WeekDaySchedule was successful
FAILURE	Some unexpected internal error occurred clearing Week Day schedule
INVALID_COMMAND	One or more fields violates constraints or is invalid

5.2.10.7.1. WeekDayIndex Field

This field SHALL indicate the Week Day schedule index to clear or 0xFE to clear all Week Day schedules for the specified user.

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5.2.10.7.2. UserIndex Field

This field SHALL indicate the user ID.

5.2.10.8. SetYearDaySchedule Command

Set a time-specific schedule ID for a specified user.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**YearDayIndex**	uint8	1 to NumberOfYearDaySchedulesSupportedPerUser			M
1	**UserIndex**	uint16	1 to NumberOfTotalUsersSupported			M
2	**LocalStartTime**	epoch-s	all			M
3	**LocalEndTime**	epoch-s	all			M

The associated UserType MAY be changed to ScheduleRestrictedUser by the lock when a Year Day schedule is set.

Return status SHALL be one of the following values:

Name	Summary
SUCCESS	Setting Year Day schedule was successful
FAILURE	Some unexpected internal error occurred setting Year Day schedule
INVALID_COMMAND	One or more fields violates constraints or is invalid

5.2.10.8.1. YearDayIndex Field

This field SHALL indicate the index of the Year Day schedule.

5.2.10.8.2. UserIndex Field

This field SHALL indicate the user ID.

5.2.10.8.3. LocalStartTime Field

This field SHALL indicate the starting time for the Year Day schedule in Epoch Time in Seconds with

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local time offset based on the local timezone and DST offset on the day represented by the value.

5.2.10.8.4. LocalEndTime Field

This field SHALL indicate the ending time for the Year Day schedule in Epoch Time in Seconds with local time offset based on the local timezone and DST offset on the day represented by the value. LocalEndTime SHALL be greater than LocalStartTime.

5.2.10.9. GetYearDaySchedule Command

Retrieve the specific year day schedule for the specific schedule and user indexes.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**YearDayIndex**	uint8	1 to NumberOfYearDaySchedulesSupportedPerUser			M
1	**UserIndex**	uint16	1 to NumberOfTotalUsersSupported			M

5.2.10.10. GetYearDayScheduleResponse Command

Returns the year day schedule data for the specified schedule and user indexes.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**YearDayIndex**	uint8	1 to NumberOfYearDaySchedulesSupportedPerUser			M
1	**UserIndex**	uint16	1 to NumberOfTotalUsersSupported			M
2	**Status**	enum8	desc			M
3	**LocalStartTime**	epoch-s	all			O
4	**LocalEndTime**	epoch-s	all			O

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5.2.10.10.1. YearDayIndex Field

This field SHALL indicate the index of the Year Day schedule.

5.2.10.10.2. UserIndex Field

This field SHALL indicate the user ID.

5.2.10.10.3. Status Field

Status SHALL be one of the following values:

- SUCCESS if both YearDayIndex and UserIndex are valid and there is a corresponding schedule entry.
- INVALID_COMMAND if either YearDayIndex and/or UserIndex values are not within valid range
- NOT_FOUND if no corresponding schedule entry found for YearDayIndex.
- NOT_FOUND if no corresponding user entry found for UserIndex.

If this field is SUCCESS, the optional fields for this command SHALL be present. For other (error) status values, only the fields up to the status field SHALL be present.

5.2.10.10.4. LocalStartTime Field

This field SHALL indicate the starting time for the Year Day schedule in Epoch Time in Seconds with local time offset based on the local timezone and DST offset on the day represented by the value. This SHALL be null if the schedule is not set for the YearDayIndex and UserIndex provided.

5.2.10.10.5. LocalEndTime Field

This field SHALL indicate the ending time for the Year Day schedule in Epoch Time in Seconds with local time offset based on the local timezone and DST offset on the day represented by the value. LocalEndTime SHALL be greater than LocalStartTime. This SHALL be null if the schedule is not set for the YearDayIndex and UserIndex provided.

5.2.10.11. ClearYearDaySchedule Command

Clears the specific year day schedule or all year day schedules for the specific user.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**YearDayIndex**	uint8	1 to NumberOfYearDaySchedulesSupportedPerUser, 0xFE			M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
1	**UserIndex**	uint16	1 to NumberOfTotalUsersSupported			M

Return status SHALL be one of the following values:

Name	Summary
SUCCESS	Clearing the requested YearDaySchedule was successful
FAILURE	Some unexpected internal error occurred clearing Year Day schedule
INVALID_COMMAND	One or more fields violates constraints or is invalid

5.2.10.11.1. YearDayIndex Field

This field SHALL indicate the Year Day schedule index to clear or 0xFE to clear all Year Day schedules for the specified user.

5.2.10.11.2. UserIndex Field

This field SHALL indicate the user ID.

5.2.10.12. SetHolidaySchedule Command

Set the holiday Schedule by specifying local start time and local end time with respect to any Lock Operating Mode.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**HolidayIndex**	uint8	1 to NumberOfHolidaySchedulesSupported			M
1	**LocalStartTime**	epoch-s	all			M
2	**LocalEndTime**	epoch-s	all			M
3	**OperatingMode**	OperatingModeEnum	all			M

Return status SHALL be one of the following values:

Name	Summary
SUCCESS	Setting Holiday schedule was successful
FAILURE	Some unexpected internal error occurred setting Holiday schedule
INVALID_COMMAND	One or more fields violates constraints or is invalid

5.2.10.12.1. HolidayIndex Field

This field SHALL indicate the index of the Holiday schedule.

5.2.10.12.2. LocalStartTime Field

This field SHALL indicate the starting time for the Holiday Day schedule in Epoch Time in Seconds with local time offset based on the local timezone and DST offset on the day represented by the value.

5.2.10.12.3. LocalEndTime Field

This field SHALL indicate the ending time for the Holiday Day schedule in Epoch Time in Seconds with local time offset based on the local timezone and DST offset on the day represented by the value. LocalEndTime SHALL be greater than LocalStartTime.

5.2.10.12.4. OperatingMode Field

This field SHALL indicate the operating mode to use during this Holiday schedule start/end time.

5.2.10.13. GetHolidaySchedule Command

Get the holiday schedule for the specified index.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**HolidayIndex**	uint8	1 to NumberOfHolidaySchedulesSupported			M

5.2.10.14. GetHolidayScheduleResponse Command

Returns the Holiday Schedule Entry for the specified Holiday ID.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**HolidayIndex**	uint8	1 to NumberOfHolidaySchedulesSupported			M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
1	**Status**	enum8	desc			M
2	**LocalStartTime**	epoch-s	all	X		O
3	**Local End Time**	epoch-s	all	X		O
4	**OperatingMode**	OperatingModeEnum	all	X		O

5.2.10.14.1. HolidayIndex Field

This field SHALL indicate the index of the Holiday schedule.

5.2.10.14.2. Status Field

Status SHALL be one of the following values:

- FAILURE if the attribute NumberOfHolidaySchedulesSupported is zero.
- SUCCESS if the HolidayIndex is valid and there is a corresponding schedule entry.
- INVALID_COMMAND if the HolidayIndex is not within valid range
- NOT_FOUND if the HolidayIndex is within the valid range, however, there is not corresponding schedule entry found.

If this field is SUCCESS, the optional fields for this command SHALL be present. For other (error) status values, only the fields up to the status field SHALL be present.

5.2.10.14.3. LocalStartTime Field

This field SHALL indicate the starting time for the Holiday schedule in Epoch Time in Seconds with local time offset based on the local timezone and DST offset on the day represented by the value. This SHALL be null if the schedule is not set for the HolidayIndex provided.

5.2.10.14.4. LocalEndTime Field

This field SHALL indicate the ending time for the Holiday schedule in Epoch Time in Seconds with local time offset based on the local timezone and DST offset on the day represented by the value. LocalEndTime SHALL be greater than LocalStartTime. This SHALL be null if the schedule is not set for the HolidayIndex provided.

5.2.10.14.5. OperatingMode Field

This field SHALL indicate the operating mode to use during this Holiday schedule start/end time. This SHALL be null if the schedule is not set for the HolidayIndex provided.

5.2.10.15. ClearHolidaySchedule Command

Clears the holiday schedule or all holiday schedules.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**HolidayIndex**	uint8	1 to NumberOfHolidaySchedulesSupported, 0xFE			M

5.2.10.15.1. HolidayIndex Field

This field SHALL indicate the Holiday schedule index to clear or 0xFE to clear all Holiday schedules.

5.2.10.16. SetUser Command

Set user into the lock.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**OperationType**	DataOperationTypeEnum	Add, Modify			M
1	**UserIndex**	uint16	1 to NumberOfTotalUsersSupported			M
2	**UserName**	string	max 10	X		M
3	**UserUniqueID**	uint32	all	X		M
4	**UserStatus**	UserStatusEnum	OccupiedEnabled, OccupiedDisabled	X		M
5	**UserType**	UserTypeEnum	UnrestrictedUser, NonAccessUser, ForcedUser, DisposableUser, ExpiringUser, ScheduleRestrictedUser, RemoteOnlyUser	X		M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
6	**CredentialRule**	CredentialRuleEnum		X		M

Fields used for different use cases:

Use Case	Description
Create a new user record	• OperationType SHALL be set to Add. • UserIndex value SHALL be set to a user record with UserType set to Available. • UserName MAY be null causing new user record to use empty string for UserName otherwise UserName SHALL be set to the value provided in the new user record. • UserUniqueID MAY be null causing new user record to use 0xFFFFFFFF for UserUniqueID otherwise UserUniqueID SHALL be set to the value provided in the new user record. • UserStatus MAY be null causing new user record to use OccupiedEnabled for UserStatus otherwise UserStatus SHALL be set to the value provided in the new user record. • UserType MAY be null causing new user record to use UnrestrictedUser for UserType otherwise UserType SHALL be set to the value provided in the new user record. • CredentialRule MAY be null causing new user record to use Single for CredentialRule otherwise CredentialRule SHALL be set to the value provided in the new user record. CreatorFabricIndex and LastModifiedFabricIndex in the new user record SHALL be set to the accessing fabric index. A LockUserChange event SHALL be generated after successfully creating a new user.

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Use Case	Description
Modify an existing user record	• OperationType SHALL be set to Modify. • UserIndex value SHALL be set for a user record with UserType NOT set to Available. • UserName SHALL be null if modifying a user record that was not created by the accessing fabric. • INVALID_COMMAND SHALL be returned if UserName is not null and the accessing fabric index doesn't match the CreatorFabricIndex in the user record otherwise UserName SHALL be set to the value provided in the user record. • UserUniqueID SHALL be null if modifying the user record that was not created by the accessing fabric. • INVALID_COMMAND SHALL be returned if UserUniqueID is not null and the accessing fabric index doesn't match the CreatorFabricIndex in the user record otherwise UserUniqueID SHALL be set to the value provided in the user record. • UserStatus MAY be null causing no change to UserStatus in user record otherwise UserStatus SHALL be set to the value provided in the user record. • UserType MAY be null causing no change to UserType in user record otherwise UserType SHALL be set to the value provided in the user record. • CredentialRule MAY be null causing no change to CredentialRule in user record otherwise CredentialRule SHALL be set to the value provided in the user record. CreatorFabricIndex SHALL NOT be changed in the user record. LastModifiedFabricIndex in the new user record SHALL be set to the accessing fabric index. A LockUserChange event SHALL be generated after successfully modifying a user.

Return status is a global status code or a cluster-specific status code from the Status Codes table and

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SHALL be one of the following values:

- SUCCESS, if setting User was successful.
- FAILURE, if some unexpected internal error occurred setting User.
- Occupied, if OperationType is Add and UserIndex points to an occupied slot.
- INVALID_COMMAND, if one or more fields violate constraints or are invalid or if OperationType is Modify and UserIndex points to an available slot.

5.2.10.16.1. OperationType Field

This field SHALL indicate the type of operation.

5.2.10.16.2. UserIndex Field

This field SHALL indicate the user ID.

5.2.10.16.3. UserName Field

This field SHALL contain a string to use as a human readable identifier for the user.

If UserName is null then:

- If the OperationType is Add, the UserName in the resulting user record SHALL be set to an empty string.
- If the OperationType is Modify, the UserName in the user record SHALL NOT be changed from the current value.

If UserName is not null, the UserName in the user record SHALL be set to the provided value.

5.2.10.16.4. UserUniqueID Field

This field SHALL indicate the fabric assigned number to use for connecting this user to other users on other devices from the fabric’s perspective.

If UserUniqueID is null then:

- If the OperationType is Add, the UserUniqueID in the resulting user record SHALL be set to default value specified above.
- If the OperationType is Modify, the UserUniqueID in the user record SHALL NOT be changed from the current value.

If UserUniqueID is not null, the UserUniqueID in the user record SHALL be set to the provided value.

5.2.10.16.5. UserStatus Field

This field SHALL indicate the UserStatus to assign to this user when created or modified.

If UserStatus is null then:

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- If the OperationType is Add, the UserStatus in the resulting user record SHALL be set to default value specified above.
- If the OperationType is Modify, the UserStatus in the user record SHALL NOT be changed from the current value.

If UserStatus is not null, the UserStatus in the user record SHALL be set to the provided value.

5.2.10.16.6. UserType Field

This field SHALL indicate the UserType to assign to this user when created or modified.

If UserType is null then:

- If the OperationType is Add, the UserType in the resulting user record SHALL be set to default value specified above.
- If the OperationType is Modify, the UserType in the user record SHALL NOT be changed from the current value.

If UserType is not null, the UserType in the user record SHALL be set to the provided value.

5.2.10.16.7. CredentialRule Field

This field SHALL indicate the CredentialRule to use for this user.

The valid CredentialRule enumeration values depends on the bits in the CredentialRulesBitmap map. Each bit in the map identifies a valid CredentialRule that can be used.

If CredentialRule is null then:

- If the OperationType is Add, the CredentialRule in the resulting user record SHALL be set to default value specified above.
- If the OperationType is Modify, the CredentialRule in the user record SHALL NOT be changed from the current value.

If CredentialRule is not null, the CredentialRule in the user record SHALL be set to the provided value.

5.2.10.17. GetUser Command

Retrieve user.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**UserIndex**	uint16	1 to NumberOfTotalUsersSupported			M

An InvokeResponse command SHALL be sent with an appropriate error (e.g. FAILURE, INVALID_-

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COMMAND, etc.) as needed otherwise the GetUserResponse Command SHALL be sent implying a status of SUCCESS.

5.2.10.18. GetUserResponse Command

Returns the user for the specified UserIndex.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**UserIndex**	uint16	1 to NumberOfTotalUsersSupported			M
1	**UserName**	string	max 10	X		M
2	**UserUniqueID**	uint32	all	X		M
3	**UserStatus**	UserStatusEnum	all	X		M
4	**UserType**	UserTypeEnum	all	X		M
5	**CredentialRule**	CredentialRuleEnum	desc	X		M
6	**Credentials**	list[CredentialStruct]	0 to NumberOfCredentialsSupportedPerUser	X		M
7	**CreatorFabricIndex**	fabric-idx	all	X		M
8	**LastModifiedFabricIndex**	fabric-idx	all	X		M
9	**NextUserIndex**	uint16	1 to NumberOfTotalUsersSupported	X		M

If the requested UserIndex is valid and the UserStatus is Available for the requested UserIndex then UserName, UserUniqueID, UserStatus, UserType, CredentialRule, Credentials, CreatorFabricIndex, and LastModifiedFabricIndex SHALL all be null in the response.

5.2.10.18.1. UserIndex Field

This field SHALL indicate the user ID.

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5.2.10.18.2. UserName Field

This field SHALL contain a string to use as a human readable identifier for the user.

5.2.10.18.3. UserUniqueID Field

See UserUniqueID.

5.2.10.18.4. UserStatus Field

This field SHALL indicate the UserStatus assigned to the user when created or modified.

5.2.10.18.5. UserType Field

This field SHALL indicate the UserType assigned to this user when created or modified.

5.2.10.18.6. CredentialRule Field

This field SHALL indicate the CredentialRule set for this user.

5.2.10.18.7. Credentials Field

This field SHALL contain a list of credentials for this user.

5.2.10.18.8. CreatorFabricIndex Field

This field SHALL indicate the user's creator fabric index. CreatorFabricIndex SHALL be null if UserStatus is set to Available or when the creator fabric cannot be determined (for example, when user was created outside the Interaction Model) and SHALL NOT be null otherwise. This value SHALL be set to 0 if the original creator fabric was deleted.

5.2.10.18.9. LastModifiedFabricIndex Field

This field SHALL indicate the user's last modifier fabric index. LastModifiedFabricIndex SHALL be null if UserStatus is set to Available or when the modifier fabric cannot be determined (for example, when user was modified outside the Interaction Model) and SHALL NOT be null otherwise. This value SHALL be set to 0 if the last modifier fabric was deleted.

5.2.10.18.10. NextUserIndex Field

This field SHALL indicate the next occupied UserIndex in the database which is useful for quickly identifying occupied user slots in the database. This SHALL NOT be null if there is at least one occupied entry after the requested UserIndex in the User database and SHALL be null if there are no more occupied entries.

5.2.10.19. ClearUser Command

Clears a user or all Users.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**UserIndex**	uint16	1 to NumberOfTotalUsersSupported, 0xFFFE			M

For each user to clear, all associated credentials (e.g. PIN, RFID, fingerprint, etc.) SHALL be cleared and the user entry values SHALL be reset to their default values (e.g. UserStatus SHALL be Available, UserType SHALL be UnrestrictedUser) and all associated schedules SHALL be cleared.

A LockUserChange event with the provided UserIndex SHALL be generated after successfully clearing users.

5.2.10.19.1. UserIndex Field

This field SHALL specify a valid User index or 0xFFFE to indicate all user slots SHALL be cleared.

5.2.10.20. SetCredential Command

Set a credential (e.g. PIN, RFID, Fingerprint, etc.) into the lock for a new user, existing user, or ProgrammingUser.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**OperationType**	DataOperationTypeEnum	Add, Modify			M
1	**Credential**	CredentialStruct				M
2	**CredentialData**	octstr	desc			M
3	**UserIndex**	uint16	1 to NumberOfTotalUsersSupported	X		M
4	**UserStatus**	UserStatusEnum	OccupiedEnabled, OccupiedDisabled	X		M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
5	**UserType**	UserTypeEnum	UnrestrictedUser, ProgrammingUser, NonAccessUser, ForcedUser, DisposableUser, ExpiringUser, RemoteOnlyUser	X		M

Fields used for different use cases:

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Use Case	Description
Create a new credential and a new user record	• OperationType SHALL be set to Add. • UserIndex SHALL be set to null and the lock will find a user record with a UserStatus value of Available and associate its UserIndex with the CredentialIndex in CredentialStruct provided. • CredentialIndex in CredentialStruct SHALL be for an unoccupied credential slot. • UserStatus MAY be null. If it is null, the new user record SHALL have UserStatus set to OccupiedEnabled. Otherwise the new user record SHALL have UserStatus set to the provided value. • UserType MAY be null. If it is null, the new user record SHALL have UserType set to UnrestrictedUser. Otherwise the new user record SHALL have UserType set to the provided value. • UserType SHALL NOT be set to ProgrammingUser for this use case. CreatorFabricIndex and LastModifiedFabricIndex in new user and credential records SHALL be set to the accessing fabric index. A LockUserChange event SHALL be generated after successfully creating a new credential and a new user. The UserIndex of this LockUserChange event SHALL be the UserIndex that was used to create the user. The DataIndex of this LockUserChange event SHALL be the CredentialIndex that was used to create the credential.

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Use Case	Description
Add a new credential to existing user record	• OperationType SHALL be set to Add. • UserIndex SHALL NOT be null and SHALL NOT already be associated with the CredentialIndex in CredentialStruct provided otherwise INVALID_COMMAND status response SHALL be returned. • INVALID_COMMAND SHALL be returned if the accessing fabric index doesn't match the CreatorFabricIndex in the user record pointed to by UserIndex. • CredentialIndex in CredentialStruct provided SHALL be for an available credential slot. • UserStatus SHALL be null. • UserType SHALL be null. CreatorFabricIndex SHALL NOT be changed in the user record. LastModifiedFabricIndex in the user record SHALL be set to the accessing fabric index. CreatorFabricIndex and LastModifiedFabricIndex in the new credential record SHALL be set to the accessing fabric index. A LockUserChange event SHALL be generated after successfully adding a new credential.

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Use Case	Description
Modify credential for an existing user record	• OperationType SHALL be set to Modify. • UserIndex value SHALL already be associated with the CredentialIndex in CredentialStruct provided otherwise INVALID_COMMAND status response SHALL be returned. • INVALID_COMMAND SHALL be returned if the accessing fabric index doesn't match the CreatorFabricIndex in the user record pointed to by UserIndex. • INVALID_COMMAND SHALL be returned if the accessing fabric index doesn't match the CreatorFabricIndex in the credential record pointed to by the CredentialIndex field value of the Credential parameter. • CredentialIndex in CredentialStruct provided SHALL be for an occupied credential slot • UserStatus SHALL be null. • UserType SHALL be null. CreatorFabricIndex SHALL NOT be changed in user and credential records. LastModifiedFabricIndex in user and credential records SHALL be set to the accessing fabric index. A LockUserChange event SHALL be generated after successfully modifying a credential.

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Use Case	Description
Modify credential for a Programming User	• OperationType SHALL be set to Modify. • UserIndex SHALL be null. • INVALID_COMMAND SHALL be returned if the accessing fabric index doesn't match the CreatorFabricIndex in the credential record pointed to by the CredentialIndex field value of the Credential parameter. • CredentialType in CredentialStruct SHALL be set to ProgrammingPIN. • CredentialIndex in CredentialStruct SHALL be 0. • UserStatus SHALL be null. • UserType SHALL be set to ProgrammingUser. CreatorFabricIndex SHALL NOT be changed in the credential record. LastModifiedFabricIndex in the credential record SHALL be set to the accessing fabric index. A LockUserChange event SHALL be generated after successfully modifying a ProgrammingUser PIN code.

5.2.10.20.1. OperationType Field

This field SHALL indicate the set credential operation type requested.

5.2.10.20.2. Credential Field

This field SHALL contain a credential structure that contains the CredentialTypeEnum and the credential index (if applicable or 0 if not) to set.

5.2.10.20.3. CredentialData Field

This field SHALL indicate the credential data to set for the credential being added or modified. The length of the credential data SHALL conform to the limits of the CredentialType specified in the Credential structure otherwise an INVALID_COMMAND status SHALL be returned in the SetCredentialResponse command.

5.2.10.20.4. UserIndex Field

This field SHALL indicate the user index to the user record that corresponds to the credential being added or modified. This SHALL be null if OperationType is add and a new credential and user is being added at the same time.

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5.2.10.20.5. UserStatus Field

This field SHALL indicate the user status to use in the new user record if a new user is being created. This SHALL be null if OperationType is Modify. This MAY be null when adding a new credential and user.

5.2.10.20.6. UserType Field

This field SHALL indicate the user type to use in the new user record if a new user is being created. This SHALL be null if OperationType is Modify. This MAY be null when adding a new credential and user.

5.2.10.21. SetCredentialResponse Command

Returns the status for setting the specified credential.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	status	desc			M
1	**UserIndex**	uint16	1 to NumberOfTotalUsersSupported	X		M
2	**NextCredentialIndex**	uint16	desc	X		O

5.2.10.21.1. Status Field

Status comes from the Status Codes table and SHALL be one of the following values:

- SUCCESS, if setting user credential was successful.
- FAILURE, if some unexpected internal error occurred setting user credential.
- Occupied, if OperationType is Add and CredentialIndex in Credential structure points to an occupied slot.
- Occupied, if OperationType is Modify and CredentialIndex in Credential structure does not match the CredentialIndex that is already associated with the provided UserIndex.
- Duplicate, if CredentialData provided is a duplicate of another credential with the same CredentialType (e.g. duplicate PIN code).
- RESOURCE_EXHAUSTED, if OperationType is Add and the new credential cannot be added due to resource constraints such as:
  - The user referred to by UserIndex already has NumberOfCredentialsSupportedPerUser credentials associated.
  - The credential is of type AliroEvictableEndpointKey or AliroNonEvictableEndpointKey, and adding it would cause the total number of credentials of those two types to exceed NumberOfAliroEndpointKeysSupported.

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- INVALID_COMMAND, if one or more fields violate constraints or are invalid.
- INVALID_COMMAND, if the CredentialIndex in the Credential provided exceeds the number of credentials of the provided CredentialType supported by the lock.
- INVALID_COMMAND, if OperationType is Modify and UserIndex points to an available slot.

5.2.10.21.2. UserIndex Field

This field SHALL indicate the user index that was created with the new credential. If the status being returned is not success then this SHALL be null. This SHALL be null if OperationType was Modify; if the OperationType was Add and a new User was created this SHALL NOT be null and SHALL provide the UserIndex created. If the OperationType was Add and an existing User was associated with the new credential then this SHALL be null.

5.2.10.21.3. NextCredentialIndex Field

This field SHALL indicate the next available index in the database for the credential type set, which is useful for quickly identifying available credential slots in the database. This SHALL NOT be null if there is at least one available entry after the requested credential index in the corresponding database and SHALL be null if there are no more available entries. The NextCredentialIndex reported SHALL NOT exceed the maximum number of credentials for a particular credential type.

5.2.10.22. GetCredentialStatus Command

Retrieve the status of a particular credential (e.g. PIN, RFID, Fingerprint, etc.) by index.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Credential**	CredentialStruct				M

An InvokeResponse command SHALL be sent with an appropriate error (e.g. FAILURE, INVALID_COMMAND, etc.) as needed otherwise the GetCredentialStatusResponse command SHALL be sent implying a status of SUCCESS.

5.2.10.22.1. Credential Field

This field SHALL contain a credential structure that contains the CredentialTypeEnum and the credential index (if applicable or 0 if not) to retrieve the status for.

5.2.10.23. GetCredentialStatusResponse Command

Returns the status for the specified credential.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**CredentialExists**	bool	all			M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
1	**UserIndex**	uint16	1 to NumberOfTotalUsersSupported	X		M
2	**CreatorFabricIndex**	fabric-idx	all	X		M
3	**LastModifiedFabricIndex**	fabric-idx	all	X		M
4	**NextCredentialIndex**	uint16	desc	X		O
5	**CredentialData**	octstr	desc	X		[ALIRO]

5.2.10.23.1. CredentialExists Field

This field SHALL indicate if the requested credential type and index exists and is populated for the requested user index.

5.2.10.23.2. UserIndex Field

This field SHALL indicate the credential's corresponding user index value if the credential exists. If CredentialType requested was ProgrammingPIN then UserIndex SHALL be null; otherwise, UserIndex SHALL be null if CredentialExists is set to False and SHALL NOT be null if CredentialExists is set to True.

5.2.10.23.3. CreatorFabricIndex Field

This field SHALL indicate the credential's creator fabric index. CreatorFabricIndex SHALL be null if CredentialExists is set to False or when the creator fabric cannot be determined (for example, when credential was created outside the Interaction Model) and SHALL NOT be null otherwise. This value SHALL be set to 0 if the original creator fabric was deleted.

5.2.10.23.4. LastModifiedFabricIndex Field

This field SHALL indicate the credential's last modifier fabric index. LastModifiedFabricIndex SHALL be null if CredentialExists is set to False or when the modifier fabric cannot be determined (for example, when credential was modified outside the Interaction Model) and SHALL NOT be null otherwise. This value SHALL be set to 0 if the last modifier fabric was deleted.

5.2.10.23.5. NextCredentialIndex Field

This field SHALL indicate the next occupied index in the database for the credential type requested, which is useful for quickly identifying occupied credential slots in the database. This SHALL NOT be null if there is at least one occupied entry after the requested credential index in the corresponding

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database and SHALL be null if there are no more occupied entries. The NextCredentialIndex reported SHALL NOT exceed the maximum number of credentials for a particular credential type.

5.2.10.23.6. CredentialData Field

This field SHALL indicate the credential data for the requested user index.

If the CredentialType in the GetCredentialStatus command was not AliroCredentialIssuerKey, AliroEvictableEndpointKey, or AliroNonEvictableEndpointKey, this field SHALL NOT be included.

Otherwise, if CredentialExists is false this field SHALL be null.

Otherwise, the value of this field SHALL be the value of the relevant credential, as a 65-byte uncompressed elliptic curve public key as defined in section 2.3.3 of SEC 1.

NOTE	Since the Aliro credentials are public keys, there is no security risk in allowing them to be read. Possession of the credential octet string does not allow operating the lock.

5.2.10.24. ClearCredential Command

Clear one, one type, or all credentials except ProgrammingPIN credential.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Credential**	CredentialStruct	desc	X		M

Fields used for different use cases:

Use Case	Description
Clear a single credential	• CredentialType in Credential structure SHALL be set to the credential type to be cleared. • CredentialType in Credential structure SHALL NOT be set to ProgrammingPIN. • CredentialIndex in Credential structure SHALL be set to the credential index to be cleared. A LockUserChange event SHALL be generated after successfully clearing a credential.

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Use Case	Description
Clear all credentials of one type	• CredentialType in Credential structure SHALL be set to the credential type to be cleared. • CredentialType in Credential structure SHALL NOT be set to ProgrammingPIN. • CredentialIndex in Credential structure SHALL be set to 0xFFFE to indicate all credentials of that type SHALL be cleared. A single LockUserChange event SHALL be generated after successfully clearing credentials. This event SHALL have DataIndex set to the CredentialIndex in the Credential structure.
Clear all credentials of all types	• Credential field SHALL be null. The ProgrammingPIN credential SHALL NOT be cleared. For each credential type cleared, a LockUserChange event with the corresponding LockDataType SHALL be generated. This event SHALL have DataIndex set to 0xFFFE.

For each credential cleared whose user doesn't have another valid credential, the corresponding user record SHALL be reset back to default values and its UserStatus value SHALL be set to Available and UserType value SHALL be set to UnrestrictedUser and all schedules SHALL be cleared. In this case a LockUserChange event SHALL be generated for the user being cleared.

Return status SHALL be one of the following values:

Name	Summary
SUCCESS	Clearing the requested credential was successful
FAILURE	Some unexpected internal error occurred clearing the requested credential
INVALID_COMMAND	One or more fields violate constraints or are invalid

5.2.10.24.1. Credential Field

This field SHALL contain a credential structure that contains the CredentialTypeEnum and the credential index (0xFFFE for all credentials or 0 if not applicable) to clear. This SHALL be null if clearing all credential types otherwise it SHALL NOT be null.

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5.2.10.25. UnboltDoor Command

This command causes the lock device to unlock the door without pulling the latch. This command includes an optional code for the lock. The door lock MAY require a code depending on the value of the RequirePINForRemoteOperation attribute.

NOTE	If the attribute AutoRelockTime is supported, the lock will transition to the locked state when the auto relock time has expired.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PINCode**	octstr				[COTA & PIN]

5.2.10.25.1. PINCode Field

See Section 5.2.10.1.1, “PINCode Field”.

5.2.10.26. SetAliroReaderConfig Command

This command allows communicating an Aliro Reader configuration, as defined in [Aliro], to the lock.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**SigningKey**	octstr	32			M
1	**VerificationKey**	octstr	65			M
2	**GroupIdentifier**	octstr	16			M
3	**GroupResolvingKey**	octstr	16			ALBU

5.2.10.26.1. SigningKey Field

This field SHALL indicate the signing key component of the Reader’s key pair.

5.2.10.26.2. VerificationKey Field

This field SHALL indicate the verification key component of the Reader’s key pair. This SHALL be an uncompressed elliptic curve public key as defined in section 2.3.3 of SEC 1.

5.2.10.26.3. GroupIdentifier Field

This field SHALL indicate the reader group identifier for the lock.

5.2.10.26.4. GroupResolvingKey Field

This field SHALL indicate the group resolving key for the lock.

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5.2.10.26.5. Effect on Receipt

1. If the lock already has an Aliro Reader configuration defined, (i.e. the AliroReaderVerificationKey attribute is not null), the response SHALL be INVALID_IN_STATE.

NOTE	This avoids accidentally overwriting values that were just set by a different administrator. If overwriting those is desired, they should be explicitly cleared with the ClearAliroReaderConfig command.

2. Otherwise:
   a. The door lock server SHALL save the SigningKey internally.
   b. The AliroReaderVerificationKey attribute SHALL be set to the value of VerificationKey.
   c. The AliroReaderGroupIdentifier attribute SHALL be set to the value of GroupIdentifier.
   d. If the AliroBLEUWB feature is supported, the AliroGroupResolvingKey attribute SHALL be set to the value of GroupResolvingKey.
   e. The response SHALL be SUCCESS.

5.2.10.27. ClearAliroReaderConfig Command

This command allows clearing an existing Aliro Reader configuration for the lock.

Administrators SHALL NOT clear an Aliro Reader configuration without explicit user permission.

NOTE	Using this command will revoke the ability of all existing Aliro user devices that have the old verification key to interact with the lock. This effect is not restricted to a single fabric or otherwise scoped in any way.

5.2.10.27.1. Effect on Receipt

On receipt of this command:

1. The door lock server SHALL clear out the stored SigningKey.
2. The AliroReaderVerificationKey and AliroReaderGroupIdentifier attributes SHALL be set to null.
3. If the AliroBLEUWB feature is supported, the AliroGroupResolvingKey attribute SHALL be set to null.

5.2.11. Events

This cluster SHALL support these events:

ID	Name	Priority	Quality	Access	Conformance
0x00	**DoorLock-Alarm**	CRITICAL		V	M
0x01	**DoorState-Change**	desc		V	DPS

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ID	Name	Priority	Quality	Access	Conformance
0x02	**LockOperation**	desc		V	M
0x03	**LockOperationError**	desc		V	M
0x04	**LockUserChange**	INFO		V	USR

The Events specified in this cluster are not intended to define the user experience. The events are only intended to define the metadata format used to notify any nodes that have subscribed for updates.

If the DoorState reported in the DoorStateChange event is not DoorClosed then the priority SHALL be CRITICAL; otherwise it MAY be INFO.

If the LockOperationType reported in the LockOperation event is Unlock or ForcedUserEvent then the priority SHALL be CRITICAL; otherwise it MAY be INFO.

If the OperationError reported in the LockOperationError event is DisabledUserDenied or the LockOperationType is Lock the priority SHALL be CRITICAL; otherwise it MAY be INFO.

5.2.11.1. DoorLockAlarm Event

The door lock server provides several alarms which can be sent when there is a critical state on the door lock. The alarms available for the door lock server are listed in AlarmCodeEnum.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**AlarmCode**	AlarmCodeEnum	all			M

5.2.11.1.1. AlarmCode Field

This field SHALL indicate the alarm code of the event that has happened.

5.2.11.2. DoorStateChange Event

The door lock server sends out a DoorStateChange event when the door lock door state changes.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**DoorState**	DoorStateEnum	all			M

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5.2.11.2.1. DoorState Field

This field SHALL indicate the new door state for this door event.

5.2.11.3. LockOperation Event

The door lock server sends out a LockOperation event when the event is triggered by the various lock operation sources.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**LockOperationType**	LockOperationTypeEnum	all			M
1	**OperationSource**	OperationSourceEnum	all			M
2	**UserIndex**	uint16	all	X		M
3	**FabricIndex**	fabric-idx	all	X		M
4	**SourceNode**	node-id	all	X		M
5	**Credentials**	list[CredentialStruct]	1 to NumberOfCredentialsSupportedPerUser	X		[USR]

- If the door lock server supports the Unbolt Door command, it SHALL generate a LockOperation event with LockOperationType set to Unlock after an Unbolt Door command succeeds.
- If the door lock server supports the Unbolting feature and an Unlock Door command is performed, it SHALL generate a LockOperation event with LockOperationType set to Unlatch when the unlatched state is reached and a LockOperation event with LockOperationType set to Unlock when the lock successfully completes the unlock → hold latch → release latch and return to unlock state operation.
- If the command fails during holding or releasing the latch but after passing the unlocked state, the door lock server SHALL generate a LockOperationError event with LockOperationType set to Unlatch and a LockOperation event with LockOperationType set to Unlock.
  - If it fails before reaching the unlocked state, the door lock server SHALL generate only a LockOperationError event with LockOperationType set to Unlock.
- Upon manual actuation, a door lock server that supports the Unbolting feature:
  - SHALL generate a LockOperation event of LockOperationType Unlatch when it is actuated from the outside.
  - MAY generate a LockOperation event of LockOperationType Unlatch when it is actuated

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from the inside.

5.2.11.3.1. LockOperationType Field

This field SHALL indicate the type of the lock operation that was performed.

5.2.11.3.2. OperationSource Field

This field SHALL indicate the source of the lock operation that was performed.

5.2.11.3.3. UserIndex Field

This field SHALL indicate the UserIndex who performed the lock operation. This SHALL be null if there is no user index that can be determined for the given operation source. This SHALL NOT be null if a user index can be determined. In particular, this SHALL NOT be null if the operation was associated with a valid credential.

5.2.11.3.4. FabricIndex Field

This field SHALL indicate the fabric index of the fabric that performed the lock operation. This SHALL be null if there is no fabric that can be determined for the given operation source. This SHALL NOT be null if the operation source is "Remote".

5.2.11.3.5. SourceNode Field

This field SHALL indicate the Node ID of the node that performed the lock operation. This SHALL be null if there is no Node associated with the given operation source. This SHALL NOT be null if the operation source is "Remote".

5.2.11.3.6. Credentials Field

This field SHALL indicate the list of credentials used in performing the lock operation. This SHALL be null if no credentials were involved.

5.2.11.4. LockOperationError Event

The door lock server sends out a LockOperationError event when a lock operation fails for various reasons.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**LockOperationType**	LockOperationType-Enum	all			M
1	**OperationSource**	OperationSourceEnum	all			M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
2	**OperationError**	OperationErrorEnum	all			M
3	**UserIndex**	uint16	all	X		M
4	**FabricIndex**	fabric-idx	all	X		M
5	**SourceNode**	node-id	all	X		M
6	**Credentials**	list[CredentialStruct]	1 to NumberOfCredentialsSupportedPerUser	X		[USR]

5.2.11.4.1. LockOperationType Field

This field SHALL indicate the type of the lock operation that was performed.

5.2.11.4.2. OperationSource Field

This field SHALL indicate the source of the lock operation that was performed.

5.2.11.4.3. OperationError Field

This field SHALL indicate the lock operation error triggered when the operation was performed.

5.2.11.4.4. UserIndex Field

This field SHALL indicate the lock UserIndex who performed the lock operation. This SHALL be null if there is no user id that can be determined for the given operation source.

5.2.11.4.5. FabricIndex Field

This field SHALL indicate the fabric index of the fabric that performed the lock operation. This SHALL be null if there is no fabric that can be determined for the given operation source. This SHALL NOT be null if the operation source is "Remote".

5.2.11.4.6. SourceNode Field

This field SHALL indicate the Node ID of the node that performed the lock operation. This SHALL be null if there is no Node associated with the given operation source. This SHALL NOT be null if the operation source is "Remote".

5.2.11.4.7. Credentials Field

This field SHALL indicate the list of credentials used in performing the lock operation. This SHALL be null if no credentials were involved.

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5.2.11.5. LockUserChange Event

The door lock server sends out a LockUserChange event when a lock user, schedule, or credential change has occurred.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**LockDataType**	LockDataTypeEnum	all			M
1	**DataOperationType**	DataOperationTypeEnum	all			M
2	**OperationSource**	OperationSourceEnum	Aliro, Unspecified, Keypad, Remote			M
3	**UserIndex**	uint16	all	X		M
4	**FabricIndex**	fabric-idx	all	X		M
5	**SourceNode**	node-id	all	X		M
6	**DataIndex**	uint16	all	X		M

5.2.11.5.1. LockDataType Field

This field SHALL indicate the lock data type that was changed.

5.2.11.5.2. DataOperationType Field

This field SHALL indicate the data operation performed on the lock data type changed.

5.2.11.5.3. OperationSource Field

This field SHALL indicate the source of the user data change.

5.2.11.5.4. UserIndex Field

This field SHALL indicate the lock UserIndex associated with the change (if any). This SHALL be null if there is no specific user associated with the data operation. This SHALL be 0xFFFE if all users are affected (e.g. Clear Users).

5.2.11.5.5. FabricIndex Field

This field SHALL indicate the fabric index of the fabric that performed the change (if any). This SHALL be null if there is no fabric that can be determined to have caused the change. This SHALL NOT be null if the operation source is "Remote".

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5.2.11.5.6. SourceNode Field

This field SHALL indicate the Node ID that performed the change (if any). The Node ID of the node that performed the change. This SHALL be null if there was no Node involved in the change. This SHALL NOT be null if the operation source is "Remote".

5.2.11.5.7. DataIndex Field

This field SHALL indicate the index of the specific item that was changed (e.g. schedule, PIN, RFID, etc.) in the list of items identified by LockDataType. This SHALL be null if the LockDataType does not correspond to a list that can be indexed into (e.g. ProgrammingUser). This SHALL be 0xFFFE if all indices are affected (e.g. ClearPINCode, ClearRFIDCode, ClearWeekDaySchedule, ClearYearDaySchedule, etc.).

5.3. Window Covering Cluster

The window covering cluster provides an interface for controlling and adjusting automatic window coverings such as drapery motors, automatic shades, curtains and blinds.

5.3.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added; CCB 1994 1995 1996 1997 2086 2094 2095 2096 2097
2	CCB 2328
3	CCB 2477 2555 2845 3028
4	All Hubs changes with FeatureMap & OperationalStatus attribute
5	New data model format and notation. Created plus clarified PositionAware and AbsolutePosition features. General cleanup of functionality.
6	Marked AbsolutePosition feature and associated elements provisional.
7	Removed ABS feature and all associated elements from the cluster.
8	Removed P quality.

5.3.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	WNCV

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5.3.3. Cluster ID

ID	Name
0x0102	Window Covering

5.3.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**LF**	Lift	O.a+	Lift control and behavior for lifting/sliding window coverings
1	**TL**	Tilt	O.a+	Tilt control and behavior for tilting window coverings
2	**PA_LF**	PositionAwareLift	[LF]	Position aware lift control is supported.
4	**PA_TL**	PositionAwareTilt	[TL]	Position aware tilt control is supported.

Due to backward compatibility reasons this feature map SHALL match the advertised Type Attribute Supported Features.

5.3.4.1. Lift Feature

The Lift feature applies to window coverings that lift up and down (e.g. for a roller shade, Up and Down is lift Open and Close) or slide left to right (e.g. for a sliding curtain, Left and Right is lift Open and Close).

5.3.4.2. Tilt Feature

The Tilt feature applies to window coverings with vertical or horizontal strips.

5.3.4.3. PositionAware Features

Relative positioning with *percent100ths* (min step 0.01%) attribute is mandatory, e.g. Max 10000 equals 100.00% and relative positioning with *percent* (min step 1%) attribute is for backward compatibility.

The *CurrentPosition* attributes SHALL always reflects the physical position of an actuator and the *TargetPosition* attribute SHALL reflect the requested position of an actuator once a positioning command is received.

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5.3.4.3.1. PositionAwareLift Feature

Relative positioning for lift attributes and commands.

5.3.4.3.2. PositionAwareTilt Feature

Relative positioning for tilt attributes and commands.

5.3.4.4. AbsolutePosition Feature

The percentage attributes SHALL indicate the position as a percentage between the InstalledOpenLimits and InstalledClosedLimits attributes of the window covering starting at the open (0.00%).

As a general rule, absolute positioning (in centimeters or tenth of a degrees) SHOULD NOT be supported for new implementations.

5.3.5. Data Types

5.3.5.1. ConfigStatusBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**Operational**	Device is operational.	M
1	**OnlineReserved**		D
2	**LiftMovementReversed**	The lift movement is reversed.	LF
3	**LiftPositionAware**	Supports the PositionAwareLift feature (PA_LF).	PA_LF
4	**TiltPositionAware**	Supports the PositionAwareTilt feature (PA_TL).	PA_TL
5	**LiftEncoderControlled**	Uses an encoder for lift.	PA_LF
6	**TiltEncoderControlled**	Uses an encoder for tilt.	PA_TL

5.3.5.1.1. Operational Bit

This bit SHALL indicate whether the window covering is operational for regular use:

- 0 = Not Operational
- 1 = Operational

5.3.5.1.2. LiftMovementReversed Bit

This bit SHALL indicate whether the lift movement is reversed:

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- 0 = Lift movement is normal
- 1 = Lift movement is reversed

5.3.5.1.3. LiftPositionAware Bit

This bit SHALL indicate whether the window covering supports the PositionAwareLift feature:

- 0 = Lift control is not position aware
- 1 = Lift control is position aware (PA_LF)

5.3.5.1.4. TiltPositionAware Bit

This bit SHALL indicate whether the window covering supports the PositionAwareTilt feature:

- 0 = Tilt control is not position aware
- 1 = Tilt control is position aware (PA_TL)

5.3.5.1.5. LiftEncoderControlled Bit

This bit SHALL indicate whether a position aware controlled window covering is employing an encoder for positioning the height of the window covering:

- 0 = Timer Controlled
- 1 = Encoder Controlled

5.3.5.1.6. TiltEncoderControlled Bit

This bit SHALL indicate whether a position aware controlled window covering is employing an encoder for tilting the window covering:

- 0 = Timer Controlled
- 1 = Encoder Controlled

5.3.5.2. ModeBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**MotorDirectionReversed**	Reverse the lift direction.	M
1	**CalibrationMode**	Perform a calibration.	M
2	**MaintenanceMode**	Freeze all motions for maintenance.	M
3	**LedFeedback**	Control the LEDs feedback.	M

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5.3.5.2.1. MotorDirectionReversed Bit

This bit SHALL control the motor direction:

- 0 = Lift movement is normal
- 1 = Lift movement is reversed

5.3.5.2.2. CalibrationMode Bit

This bit SHALL set the window covering into calibration mode:

- 0 = Normal mode
- 1 = Calibration mode

5.3.5.2.3. MaintenanceMode Bit

This bit SHALL set the window covering into maintenance mode:

- 0 = Normal mode
- 1 = Maintenance mode

5.3.5.2.4. LedFeedback Bit

This bit SHALL control feedback LEDs:

- 0 = LEDs are off
- 1 = LEDs will display feedback

5.3.5.3. OperationalStatusBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
1..0	**Global**	Global operational state.	M
3..2	**Lift**	Lift operational state.	LF
5..4	**Tilt**	Tilt operational state.	TL

The OperationalStatusBitmap is using several internal operational state fields (composed of 2 bits) following this definition:

- 00b = Currently not moving
- 01b = Currently opening (e.g. moving from closed to open).
- 10b = Currently closing (e.g. moving from open to closed).
- 11b = Reserved

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5.3.5.3.1. Global Bits

These bits SHALL indicate in which direction the covering is currently moving or if it has stopped. Global operational state SHALL always reflect the overall motion of the device.

5.3.5.3.2. Lift Bits

These bits SHALL indicate in which direction the covering's lift is currently moving or if it has stopped.

5.3.5.3.3. Tilt Bits

These bits SHALL indicate in which direction the covering's tilt is currently moving or if it has stopped.

5.3.5.4. SafetyStatusBitmap Type

This data type is derived from map16.

Bit	Name	Summary	Conformance
0	**RemoteLockout**	Movement commands are ignored (locked out). e.g. not granted authorization, outside some time/date range.	M
1	**TamperDetection**	Tampering detected on sensors or any other safety equipment. Ex: a device has been forcedly moved without its actuator(s).	M
2	**FailedCommunication**	Communication failure to sensors or other safety equipment.	M
3	**PositionFailure**	Device has failed to reach the desired position. e.g. with position aware device, time expired before TargetPosition is reached.	M
4	**ThermalProtection**	Motor(s) and/or electric circuit thermal protection activated.	M
5	**ObstacleDetected**	An obstacle is preventing actuator movement.	M

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Bit	Name	Summary	Conformance
6	**Power**	Device has power related issue or limitation e.g. device is running w/ the help of a backup battery or power might not be fully available at the moment.	M
7	**StopInput**	Local safety sensor (not a direct obstacle) is preventing movements (e.g. Safety EU Standard EN60335).	M
8	**MotorJammed**	Mechanical problem related to the motor(s) detected.	M
9	**HardwareFailure**	PCB, fuse and other electrics problems.	M
10	**ManualOperation**	Actuator is manually operated and is preventing actuator movement (e.g. actuator is disengaged/decoupled).	M
11	**Protection**	Protection is activated.	M

5.3.5.5. TypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**RollerShade**	RollerShade	M
1	**RollerShade2Motor**	RollerShade - 2 Motor	M
2	**RollerShadeExterior**	RollerShade - Exterior	M
3	**RollerShadeExterior2Motor**	RollerShade - Exterior - 2 Motor	M
4	**Drapery**	Drapery (curtain)	M
5	**Awning**	Awning	M
6	**Shutter**	Shutter	M
7	**TiltBlindTiltOnly**	Tilt Blind - Tilt Only	M
8	**TiltBlindLiftAndTilt**	Tilt Blind - Lift & Tilt	M

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Value	Name	Summary	Conformance
9	**ProjectorScreen**	Projector Screen	M
255	**Unknown**	Unknown	M

5.3.5.6. EndProductTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**RollerShade**	Simple Roller Shade	M
1	**RomanShade**	Roman Shade	M
2	**BalloonShade**	Balloon Shade	M
3	**WovenWood**	Woven Wood	M
4	**PleatedShade**	Pleated Shade	M
5	**CellularShade**	Cellular Shade	M
6	**LayeredShade**	Layered Shade	M
7	**LayeredShade2D**	Layered Shade 2D	M
8	**SheerShade**	Sheer Shade	M
9	**TiltOnlyInteriorBlind**	Tilt Only Interior Blind	M
10	**InteriorBlind**	Interior Blind	M
11	**VerticalBlindStripCurtain**	Vertical Blind, Strip Curtain	M
12	**InteriorVenetianBlind**	Interior Venetian Blind	M
13	**ExteriorVenetianBlind**	Exterior Venetian Blind	M
14	**LateralLeftCurtain**	Lateral Left Curtain	M
15	**LateralRightCurtain**	Lateral Right Curtain	M
16	**CentralCurtain**	Central Curtain	M
17	**RollerShutter**	Roller Shutter	M
18	**ExteriorVerticalScreen**	Exterior Vertical Screen	M
19	**AwningTerracePatio**	Awning Terrace (Patio)	M
20	**AwningVerticalScreen**	Awning Vertical Screen	M
21	**TiltOnlyPergola**	Tilt Only Pergola	M
22	**SwingingShutter**	Swinging Shutter	M
23	**SlidingShutter**	Sliding Shutter	M
255	**Unknown**	Unknown	M

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5.3.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**Type**	TypeEnum	desc	F	0	R V	M
0x0005	**NumberOfActuationsLift**	uint16	all	N	0	R V	[LF]
0x0006	**NumberOfActuationsTilt**	uint16	all	N	0	R V	[TL]
0x0007	**ConfigStatus**	ConfigStatusBitmap	desc	N	desc	R V	M
0x0008	**CurrentPositionLiftPercentage**[1]	percent		X N	null	R V	[LF & PA_LF]
0x0009	**CurrentPositionTiltPercentage**[1]	percent	all	X N	null	R V	[TL & PA_TL]
0x000A	**OperationalStatus**	OperationalStatusBitmap	max 63		0	R V	M
0x000B	**TargetPositionLiftPercent100ths**[2]	percent100ths		X	null	R V	LF & PA_LF
0x000C	**TargetPositionTiltPercent100ths**[2]	percent100ths		X	null	R V	TL & PA_TL
0x000D	**EndProductType**	EndProductTypeEnum	desc	F	0	R V	M
0x000E	**CurrentPositionLiftPercent100ths**[1]	percent100ths	max 10000	X N	null	R V	LF & PA_LF

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x000F	**CurrentPositionTiltPercent100ths** [1]	percent100ths	max 10000	X N	null	R V	TL & PA_TL
0x0017	**Mode**	ModeBitmap	max 15	N	0	RW VM	M
0x001A	**SafetyStatus**	SafetyStatusBitmap	desc		0	R V	O

NOTE

Nullable positions

1 - The null value indicates that the current position is unknown, e.g. calibration is needed.

2 - The null value indicates that the value is unavailable, e.g. no target position has been set.

5.3.6.1. General Usage Notes for Lift Percentages

Percentage values for Lift represent the amount of closure, measured as travel from the fully-open position toward the fully-closed position. A value of 0% indicates fully open; a value of 100% indicates fully closed.

5.3.6.2. Type Attribute

This attribute SHALL identify the type of window covering.

If the window covering supports the LF feature and not the TL feature, the following types SHALL be used as the constraint for this attribute:

Value	Name
0	**RollerShade**
1	**RollerShade2Motor**
2	**RollerShadeExterior**
3	**RollerShadeExterior2Motor**
4	**Drapery**
5	**Awning**
6	**Shutter**
9	**ProjectorScreen**
255	**Unknown**

If the window covering supports the TL feature and not the LF feature, the following types SHALL

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be used as the constraint for this attribute:

Value	Name
6	**Shutter**
7	**TiltBlindTiltOnly**
255	**Unknown**

If the window covering supports both the LF and TL features, the following types are allowed to be used:

Value	Name
8	**TiltBlindLiftAndTilt**
255	**Unknown**

5.3.6.3. NumberOfActuationsLift Attribute

This attribute SHALL indicate the total number of lift/slide actuations applied to the window covering since the device was installed.

5.3.6.4. NumberOfActuationsTilt Attribute

This attribute SHALL indicate the total number of tilt actuations applied to the window covering since the device was installed.

5.3.6.5. ConfigStatus Attribute

This attribute specifies the configuration and status information of the window covering.

To change settings, devices SHALL write to the Mode attribute. The behavior causing the setting or clearing of each bit is vendor specific.

5.3.6.5.1. Operational Bit

The SafetyStatus & Mode attributes might affect this bit state.

5.3.6.5.2. LiftMovementReversed Bit

This bit identifies if the directions of the lift/slide movements have been reversed in order for commands (e.g: Open, Close, GoTos) to match the physical installation conditions

This bit can be adjusted by setting the MotorDirectionReversed bit in the Mode attribute.

5.3.6.5.3. LiftEncoderControlled Bit

This bit is ignored if the device does not support the PositionAwareLift feature (PA_LF).

5.3.6.5.4. TiltEncoderControlled Bit

This bit is ignored if the device does not support the PositionAwareTilt feature (PA_TL).

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5.3.6.6. CurrentPositionLiftPercent100ths Attribute

This attribute SHALL indicate the actual position as a percentage with a minimal step of 0.01%. E.g Max 10000 equals 100.00%.

5.3.6.7. CurrentPositionTiltPercent100ths Attribute

This attribute SHALL indicate the actual position as a percentage with a minimal step of 0.01%. E.g Max 10000 equals 100.00%.

5.3.6.8. CurrentPositionLiftPercentage Attribute

This attribute SHALL indicate the actual position as a percentage from 0% to 100% with 1% default step. This attribute is equal to CurrentPositionLiftPercent100ths attribute divided by 100.

5.3.6.9. CurrentPositionTiltPercentage Attribute

This attribute SHALL indicate the actual position as a percentage from 0% to 100% with 1% default step. This attribute is equal to CurrentPositionTiltPercent100ths attribute divided by 100.

5.3.6.10. TargetPositionLiftPercent100ths Attribute

This attribute SHALL indicate the position where the window covering lift will go or is moving to as a percentage (Unit *0.01%*).

5.3.6.11. TargetPositionTiltPercent100ths Attribute

This attribute SHALL indicate the position where the window covering tilt will go or is moving to as a percentage (Unit *0.01%*).

5.3.6.12. OperationalStatus Attribute

This attribute SHALL indicate the currently ongoing operations and applies to all type of devices.

5.3.6.13. EndProductType Attribute

This attribute SHOULD provide more detail about the product type than can be determined from the main category indicated by the Type attribute.

If the window covering supports the LF feature and not the TL feature, the following types SHALL be used as the constraint for this attribute:

Value	Name
0	**RollerShade**
1	**RomanShade**
2	**BalloonShade**
3	**WovenWood**
4	**PleatedShade**

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Value	Name
5	**CellularShade**
6	**LayeredShade**
7	**LayeredShade2D**
14	**LateralLeftCurtain**
15	**LateralRightCurtain**
16	**CentralCurtain**
17	**RollerShutter**
18	**ExteriorVerticalScreen**
19	**AwningTerracePatio**
20	**AwningVerticalScreen**
21	**TiltOnlyPergola**
22	**SwingingShutter**
23	**SlidingShutter**
255	**Unknown**

If the window covering supports the TL feature and not the LF feature, the following types SHALL be used as the constraint for this attribute:

Value	Name
9	**TiltOnlyInteriorBlind**
21	**TiltOnlyPergola**
22	**SwingingShutter**
23	**SlidingShutter**
255	**Unknown**

If the window covering supports both the LF and TL features, the following types are allowed to be used:

Value	Name
8	**SheerShade**
10	**InteriorBlind**
11	**VerticalBlindStripCurtain**
12	**InteriorVenetianBlind**
13	**ExteriorVenetianBlind**
255	**Unknown**

The table below helps to match the EndProductType attribute with the Type attribute.

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Value	Name	Indoor Outdoor	Indicative Dimension	Recommended Type Attribute
0	**RollerShade**	I	1D	RollerShade
1	**RomanShade**	I	1D	RollerShade
2	**BalloonShade**	I	1D	RollerShade
3	**WovenWood**	I	1D	RollerShade
4	**PleatedShade**	I	1D	RollerShade
5	**CellularShade**	I	1D	RollerShade
6	**LayeredShade**	I	1D	RollerShade
7	**LayeredShade2D**	I	2D	RollerShade2Motor
8	**SheerShade**	I	2D	TiltBlindLiftAndTilt
9	**TiltOnlyInteriorBlind**	I	1D	TiltBlindTiltOnly
10	**InteriorBlind**	I	2D	TiltBlindLiftAndTilt
11	**VerticalBlindStripCurtain**	I	2D	TiltBlindLiftAndTilt
12	**InteriorVenetianBlind**	I	2D	TiltBlindLiftAndTilt
13	**ExteriorVenetianBlind**	O	2D	TiltBlindLiftAndTilt
14	**LateralLeftCurtain**	I	1D	Drapery
15	**LateralRightCurtain**	I	1D	Drapery
16	**CentralCurtain**	I	1D	Drapery
17	**RollerShutter**	O	1D	RollerShadeExterior
18	**ExteriorVerticalScreen**	O	1D	RollerShadeExterior
19	**AwningTerracePatio**	O	1D	Awning
20	**AwningVerticalScreen**	O	1D	Awning
21	**TiltOnlyPergola**	O	1D	Shutter
22	**SwingingShutter**	O	1D	Shutter

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Value	Name	Indoor Outdoor	Indicative Dimension	Recommended Type Attribute
23	**SlidingShutter**	O	1D	Shutter
255	**Unknown**			Unknown

5.3.6.14. Mode Attribute

The Mode attribute allows configuration of the window covering, such as: reversing the motor direction, placing the window covering into calibration mode, placing the motor into maintenance mode, disabling the network, and disabling status LEDs.

In the case a device does not support or implement a specific mode, e.g. the device has a specific installation method and reversal is not relevant or the device does not include a maintenance mode, any write interaction to the Mode attribute, with an unsupported mode bit or any out of bounds bits set, must be ignored and a response containing the status of CONSTRAINT_ERROR will be returned.

5.3.6.14.1. MotorDirectionReversed Bit

This bit SHALL control the LiftMovementReversed bit in the ConfigStatus attribute.

5.3.6.14.2. CalibrationMode Bit

If in calibration mode, all commands (e.g: UpOrOpen, DownOrClose, GoTos) that can result in movement, could be accepted and result in a self-calibration being initiated before the command is executed.

In case the window covering does not have the ability or is not able to perform a self-calibration, the command SHOULD be ignored and a FAILURE status SHOULD be returned.

In a write interaction, setting this bit to 0, while the device is in calibration mode, is not allowed and SHALL generate a FAILURE error status. In order to leave calibration mode, the device must perform its calibration routine, either as a self-calibration or assisted by external tool(s), depending on the device/manufacturer implementation.

A manufacturer might choose to set the Operational bit of the ConfigStatus attribute to its not operational value, if applicable during calibration mode.

5.3.6.14.3. MaintenanceMode Bit

While in maintenance mode, all commands (e.g: UpOrOpen, DownOrClose, GoTos) or local inputs that can result in movement, must be ignored and respond with a BUSY status. Additionally, the Operational bit of the ConfigStatus attribute should be set to its not operational value.

5.3.6.15. SafetyStatus Attribute

The SafetyStatus attribute reflects the state of the safety sensors and the common issues preventing movements. By default for nominal operation all flags are cleared (0). A device might support none, one or several bit flags from this attribute (all optional).

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5.3.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**UpOrOpen**	client ⇒ server	Y	O	M
0x01	**DownOrClose**	client ⇒ server	Y	O	M
0x02	**StopMotion**	client ⇒ server	Y	O	M
0x05	**GoToLiftPercentage**	client ⇒ server	Y	O	(LF & PA_LF), [LF]
0x08	**GoToTiltPercentage**	client ⇒ server	Y	O	(TL & PA_TL), [TL]

5.3.7.1. UpOrOpen Command

Upon receipt of this command, the window covering will adjust its position so the physical lift/slide and tilt is at the maximum open/up position. This will happen as fast as possible. The server attributes SHALL be updated as follows:

if the PositionAware feature is supported:

- TargetPositionLiftPercent100ths attribute SHALL be set to 0.00%.
- TargetPositionTiltPercent100ths attribute SHALL be set to 0.00%.

The server positioning attributes will follow the movements, once the movement has successfully finished, the server attributes SHALL be updated as follows:

if the PositionAware feature is supported:

- CurrentPositionLiftPercent100ths attribute SHALL be 0.00%.
- CurrentPositionLiftPercentage attribute SHALL be 0%.
- CurrentPositionTiltPercent100ths attribute SHALL be 0.00%.
- CurrentPositionTiltPercentage attribute SHALL be 0%.

if the AbsolutePosition feature is supported:

- CurrentPositionLift attribute SHALL be equal to the InstalledOpenLimitLift attribute.
- CurrentPositionTilt attribute SHALL be equal to the InstalledOpenLimitTilt attribute.

5.3.7.2. DownOrClose Command

Upon receipt of this command, the window covering will adjust its position so the physical lift/slide and tilt is at the maximum closed/down position. This will happen as fast as possible. The server attributes supported SHALL be updated as follows:

if the PositionAware feature is supported:

- TargetPositionLiftPercent100ths attribute SHALL be set to 100.00%.

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- TargetPositionTiltPercent100ths attribute SHALL be set to 100.00%.

The server positioning attributes will follow the movements, once the movement has successfully finished, the server attributes SHALL be updated as follows:

if the PositionAware feature is supported:

- CurrentPositionLiftPercent100ths attribute SHALL be 100.00%.
- CurrentPositionLiftPercentage attribute SHALL be 100%.
- CurrentPositionTiltPercent100ths attribute SHALL be 100.00%.
- CurrentPositionTiltPercentage attribute SHALL be 100%.

if the AbsolutePosition feature is supported:

- CurrentPositionLift attribute SHALL be equal to the InstalledClosedLimitLift attribute.
- CurrentPositionTilt attribute SHALL be equal to the InstalledClosedLimitTilt attribute.

5.3.7.3. StopMotion Command

Upon receipt of this command, the window covering will stop any adjusting to the physical tilt and lift/slide that is currently occurring. The server attributes supported SHALL be updated as follows:

- TargetPositionLiftPercent100ths attribute will be set to CurrentPositionLiftPercent100ths attribute value.
- TargetPositionTiltPercent100ths attribute will be set to CurrentPositionTiltPercent100ths attribute value.

5.3.7.4. GoToLiftPercentage Command

This command is used to set the target lift position of the window covering to the percentage value specified in the command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**LiftPercent100thsValue**	percent100ths	desc			M

Upon receipt of this command, the server will adjust the window covering to the lift/slide percentage specified in the payload of this command.

If the command includes LiftPercent100thsValue, then TargetPositionLiftPercent100ths attribute SHALL be set to LiftPercent100thsValue. Otherwise the TargetPositionLiftPercent100ths attribute SHALL be set to LiftPercentageValue * 100.

If a client includes LiftPercent100thsValue in the command, the LiftPercentageValue SHALL be set to LiftPercent100thsValue / 100, so a legacy server which only supports LiftPercentageValue (not LiftPercent100thsValue) has a value to set the target position.

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If the server does not support the PositionAware feature, then a zero percentage SHALL be treated as a UpOrOpen command and a non-zero percentage SHALL be treated as an DownOrClose command. If the device is only a tilt control device, then the command SHOULD be ignored and a UNSUPPORTED_COMMAND status SHOULD be returned.

5.3.7.5. GoToTiltPercentage Command

This command is used to set the target tilt position of the window covering to the percentage value specified in the command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**TiltPercent100thsValue**	percent100ths	desc			M

Upon receipt of this command, the server will adjust the window covering to the tilt percentage specified in the payload of this command.

If the command includes TiltPercent100thsValue, then TargetPositionTiltPercent100ths attribute SHALL be set to TiltPercent100thsValue. Otherwise the TargetPositionTiltPercent100ths attribute SHALL be set to TiltPercentageValue * 100.

If a client includes TiltPercent100thsValue in the command, the TiltPercentageValue SHALL be set to TiltPercent100thsValue / 100, so a legacy server which only supports TiltPercentageValue (not TiltPercent100thsValue) has a value to set the target position.

If the server does not support the PositionAware feature, then a zero percentage SHALL be treated as a UpOrOpen command and a non-zero percentage SHALL be treated as an DownOrClose command. If the device is only a tilt control device, then the command SHOULD be ignored and a UNSUPPORTED_COMMAND status SHOULD be returned.

5.4. Closure Control Cluster

This cluster provides an interface for controlling a Closure.

5.4.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

5.4.2. Classification

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Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	CLCTRL

5.4.3. Cluster ID

ID	Name
0x0104	Closure Control

5.4.4. Association Between Closure Control and Closure Dimension Clusters

Some device types MAY define that a Closure Dimension cluster instance is associated with a Closure Control cluster instance on the same device. In such cases, the Closure Control cluster acts as the supervisory and coordination logic for the closure system. Where this association is specified, commands originating from Closure Dimension cluster instances MAY impact the operational logic and state management of the associated Closure Control cluster. Implementers SHALL ensure that, where such an association exists, all relevant behaviors governed by the Closure Control cluster are maintained, and that interactions initiated by Closure Dimension cluster commands are properly integrated and reflected in the Closure Control cluster's logic.

5.4.5. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**PS**	Positioning	O.a+	Supports Positioning with at least Fully Opened (0%) and Fully Closed (100%) discrete positions
1	**LT**	MotionLatching	O.a+	Supports a latch (securing a position, or a state)
2	**IS**	Instantaneous	O	Supports the Instantaneous feature
3	**SP**	Speed	[PS & !IS]	Supports Speed motion throttling
4	**VT**	Ventilation	[PS]	Supports Ventilation discrete state
5	**PD**	Pedestrian	[PS]	Supports Pedestrian discrete state
6	**CL**	Calibration	[PS]	Supports the Calibration feature

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Bit	Code	Feature	Conformance	Summary
7	**PT**	Protection	O	Supports the Protection feature
8	**MO**	ManuallyOperable	O	Supports the manual operation feature

5.4.5.1. Positioning Feature

This feature SHALL indicate that the closure can be set to discrete positions, including at minimum the FullyOpen and FullyClosed positions.

5.4.5.2. MotionLatching Feature

This feature SHALL indicate that the closure can be latched and unlatched. When latched, the feature secures an axis, preventing associated actuators from moving components along that axis.

5.4.5.3. Instantaneous Feature

This feature SHALL indicate that the closure is capable of changing its position or state instantaneously. As a result, the Speed feature is not applicable, and the Stop command is not usable. In such closures, the OverallCurrentState attribute SHALL immediately follow the OverallTargetState attribute. The state transition diagram remains applicable, however, transitions involving the Stop state SHALL be omitted.

5.4.5.4. Speed Feature

This feature SHALL indicate that the closure supports configurable speed during motion toward a target position. The feature uses the values in ThreeLevelAutoEnum to set the supported speed levels.

5.4.5.4.1. Low Value

This setting prioritizes quiet operation and is typically used in silent mode. It minimizes noise from the closure product, making it ideal for environments where discretion or minimal disturbance is important, such as during nighttime operation or in quiet settings.

5.4.5.4.2. Medium Value

The most commonly used speed for everyday use. It offers a balanced performance with reliable operation and moderate noise levels, suitable for general tasks in typical working conditions.

5.4.5.4.3. High Value

Designed for time-critical situations, this setting allows the closure to open or close as quickly as possible. It prioritizes speed over noise or smoothness, ensuring rapid response when urgent access or sealing is needed.

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5.4.5.4.4. Auto Value

In auto the speed of the movement is entirely determined by the closure. The behavior when using this value is manufacturer specific.

NOTE	The server might not support three different speed values. The meaning of Low, Medium and High speeds is manufacturer-specific, as long as `Low <= Medium <= High` in terms of actual speed of motion. The Speed Feature requires at least two distinct speeds to be supported.

5.4.5.5. Ventilation Feature

This feature SHALL indicate that the closure can be set to a designated Ventilation position (e.g., partially open).

5.4.5.6. Pedestrian Feature

This feature SHALL indicate that the closure can be set to a dedicated Pedestrian position. The Pedestrian position provides a clear walkway through the closure.

5.4.5.7. Calibration Feature

This feature SHALL indicate the capability to trigger a calibration procedure. The calibration can either be fully automated, or require manual steps not described in this specification.

5.4.5.8. Protection Feature

This feature SHALL indicate that the closure is capable of activating a form of protection, such as protection against wind. A protection is manufacturer-specific, and it could be a simple software limitation, or a mechanical system deployed by the closure.

5.4.5.9. ManuallyOperable Feature

This feature SHALL indicate that the closure can be operated manually by a user, such as to open a window.

5.4.6. Data Types

5.4.6.1. CurrentPositionEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**FullyClosed**	Fully closed state	M
1	**FullyOpened**	Fully opened state	M
2	**PartiallyOpened**	Partially opened state (closure is not fully opened or fully closed)	M

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Value	Name	Summary	Conformance
3	**OpenedForPedestrian**	Closure is in the Pedestrian position	PD
4	**OpenedForVentilation**	Closure is in the Ventilation position	VT
5	**OpenedAtSignature**	Closure is in its "Signature position"	M

5.4.6.1.1. OpenedAtSignature Value

The Signature allows a pre-recorded manufacturer- or installer-defined position to be reached.

This Signature position depends on the closure type. Some examples include:

- Gate, Garage Door → Pedestrian and pets, or one leaf only.
- Venetian Blind → Lowered down with flat slats.
- Door → Position to open for a person or someone in a wheelchair.
- Window → Position to 10% for tilt position.
- Roller Shutter → Closed with maximum gap between the slats.
- By default the Signature position MAY apply the same outcome as PartiallyOpened.

If no such separately defined position exists on the closure, the Signature value SHALL have the same position meaning as FullyOpened.

5.4.6.2. TargetPositionEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**MoveToFullyClosed**	Move to a fully closed state	M
1	**MoveToFullyOpen**	Move to a fully open state	M
2	**MoveToPedestrianPosition**	Move to the Pedestrian position	PD
3	**MoveToVentilationPosition**	Move to the Ventilation position	VT
4	**MoveToSignaturePosition**	Move to the Signature position	M

5.4.6.3. MainStateEnum Type

This data type is derived from enum8.

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Value	Name	Summary	Conformance
0	**Stopped**	Closure is stopped	M
1	**Moving**	Closure is actively moving	M
2	**WaitingForMotion**	Closure is waiting before a motion (e.g. pre-heat, pre-check)	M
3	**Error**	Closure is in an error state	M
4	**Calibrating**	Closure is currently calibrating its Opened and Closed limits to determine effective physical range	CL
5	**Protected**	Some protective measures are activated to prevent damage to the closure. Commands MAY be rejected.	PT
6	**Disengaged**	Closure has a disengaged element preventing any actuator movements	MO
7	**SetupRequired**	Movement commands are ignored since the closure is not operational and requires further setup and/or calibration	M

5.4.6.4. ClosureErrorEnum Type

This data type is derived from enum8.

The following ranges are reserved for general and manufacturer specified closure errors.

Value	Name	Summary
0x00 to 0x7F	**GeneralErrors**	Generally applicable values for error, defined herein
0x80 to 0xBF	**ManufacturerError**	Vendor specific errors

The manufacturer-specific error definitions SHALL NOT duplicate the general error definitions. Such manufacturer-specific error definitions SHALL be scoped in the context of the Vendor ID present in the Basic Information cluster.

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The general closure error values are defined in the table below.

Value	Name	Summary	Conformance
0	**PhysicallyBlocked**	An obstacle is blocking the closure movement	M
1	**BlockedBySensor**	The closure is unsafe to move, as determined by a sensor (e.g. photoelectric sensor) before attempting movement	M
2	**TemperatureLimited**	A warning raised by the closure that indicates an over-temperature, e.g. due to excessive drive or stall current	M
3	**MaintenanceRequired**	Some malfunctions that are not easily recoverable are detected, or urgent servicing is needed	M
4	**InternalInterference**	An internal element is prohibiting motion, e.g. an integrated door within a bigger garage door is open and prevents motion	M

5.4.6.5. OverallCurrentStateStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Position**	CurrentPositionEnum	desc	X	null		PS
1	**Latch**	bool	desc	X	null		LT
2	**Speed**	ThreeLevelAutoEnum	desc		MS		SP
3	**SecureState**	bool	desc	X	null		M

5.4.6.5.1. Position field

This field SHALL indicate the current Position state of the closure, as defined in the CurrentPositionEnum.

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When the Positioning (PS) feature flag is set, the rules for setting the value of the Position field are:

- If the closure doesn't know accurately its current state the value null SHALL be used.
- Otherwise, the most appropriate supported value SHALL be used.

Clients which only consider the binary opened/closed states of a closure SHOULD consider the closure to be closed if the value of this field is *FullyClosed*. Otherwise those clients SHOULD consider the closure opened (non-closed).

5.4.6.5.2. Latch field

This field SHALL indicate the current latching state of the closure.

When the MotionLatching (LT) feature flag is set, the rules for setting the value of the Latch field are:

- If the closure doesn't know its current state, the value SHALL be null.
- Else, if the closure is partially latched or not latched, the value SHALL be false.
- Otherwise, if the closure is fully latched, the value SHALL be true.

NOTE	Some products exposing the MotionLatching (LT) feature might not be able to drive an actuator to achieve a latched state. Such products are built with springs or similar mechanisms to unlatch but require the user to latch manually.

5.4.6.5.3. Speed field

This field SHALL indicate the current speed of the closure, as defined in the ThreeLevelAutoEnum.

When the Speed (SP) feature flag is set, the rules for setting the value of the Speed field are:

If the closure's MainState attribute is currently either in *WaitingForMotion* or *Moving* state, the closure's most accurate current overall speed SHALL be used. Otherwise, the value used SHALL be the most appropriate default supported speed value.

5.4.6.5.4. SecureState field

This field SHALL indicate the current secure state of the closure.

A secure state requires the closure to meet all of the following conditions defined by the OverallCurrentState Struct based on feature support:

- If the Positioning feature is supported, then the Position field of OverallCurrentState is FullyClosed.
- If the MotionLatching feature is supported, then the Latch field of OverallCurrentState is True.

The rules for setting the value of the SecureState field SHALL be:

- *True* if the closure meets the required conditions for a secure state, preventing unauthorized or undetectable access.

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- *False* if the closure does not meet these conditions and unauthorized or undetectable access is possible.
- null if the closure's current secure state is unknown.

This field provides no additional details regarding mechanical properties of the closure mechanism. It is intended only as supplementary information and not as a replacement for a comprehensive security system. It is primarily useful for closures on the outer shell of objects, such as garage doors, windows, or doors.

5.4.6.6. OverallTargetStateStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Position**	TargetPositionEnum	all	X	null		PS
1	**Latch**	bool	all	X	null		LT
2	**Speed**	ThreeLevelAutoEnum	all		MS		SP

5.4.6.6.1. Position field

This field SHALL indicate the target position that the closure is moving to. It SHALL be null if there is no target position.

5.4.6.6.2. Latch field

This field SHALL indicate the desired latching state of the closure. It SHALL be null if there is no desired latching state.

5.4.6.6.3. Speed field

This field SHALL indicate the desired speed at which the closure should perform the movement toward the target position. If no speed value has yet been set, the server SHALL select and set one of the speed values defined in the ThreeLevelAutoEnum.

5.4.6.7. LatchControlModesBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**RemoteLatching**	Remote latching capability	LT
1	**RemoteUnlatching**	Remote unlatching capability	LT

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5.4.6.7.1. RemoteLatching Bit

This bit SHALL indicate whether the latch supports remote latching or not:

- 0 = the latch can only be latched through manual, physical operation.
- 1 = the latch can be latched via remote control (e.g., electronic or remote actuation).

5.4.6.7.2. RemoteUnlatching Bit

This bit SHALL indicate whether the latch supports remote unlatching or not:

- 0 = the latch can only be unlatched through manual, physical operation.
- 1 = the latch can be unlatched via remote control (e.g., electronic or remote actuation).

5.4.7. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**CountdownTime**	elapsed-s	max 259200	Q X	null	R V	[PS & !IS]
0x0001	**MainState**	MainStateEnum	all			R V	M
0x0002	**CurrentErrorList**	list[ClosureErrorEnum]	max 10[all]			R V	M
0x0003	**OverallCurrentState**	OverallCurrentStateStruct	all	X	null	R V	M
0x0004	**OverallTargetState**	OverallTargetStateStruct	all	X	null	R V	M
0x0005	**LatchControlModes**	LatchControlModesBitmap	all	F		R V	LT

5.4.7.1. CountdownTime Attribute

This attribute SHALL indicate the estimated time left before the operation is completed, in seconds.

A value of 0 (zero) means that the operation has completed.

A value of null indicates that there is no time currently defined until operation completion. This MAY happen because the completion time is unknown.

Changes to this attribute SHALL only be marked as reportable in the following cases:

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- If the tracked operation has changed due to a change in the MainState attribute, or
- When it changes from 0 to any other value and vice versa, or
- When it changes from null to any other value and vice versa, or
- When it increases, or
- When there is any increase or decrease in the estimated time remaining that was due to progressing insight of the server's control logic.

Changes to this attribute merely due to the normal passage of time with no other dynamic change of closure state SHALL NOT be reported.

As this attribute is not being reported during a regular countdown, clients SHOULD NOT rely on the reporting of this attribute in order to keep track of the remaining duration.

5.4.7.2. MainState Attribute

This attribute SHALL indicate the current operational state of the closure associated with the server.

NOTE The MainState diagram is provided exclusively for informational purposes only and is an exemplary design of the internals of a closure implementation to help illustrate the aspects of function that are considered by the cluster's normative text.

5.4.7.2.1. MainState Diagram

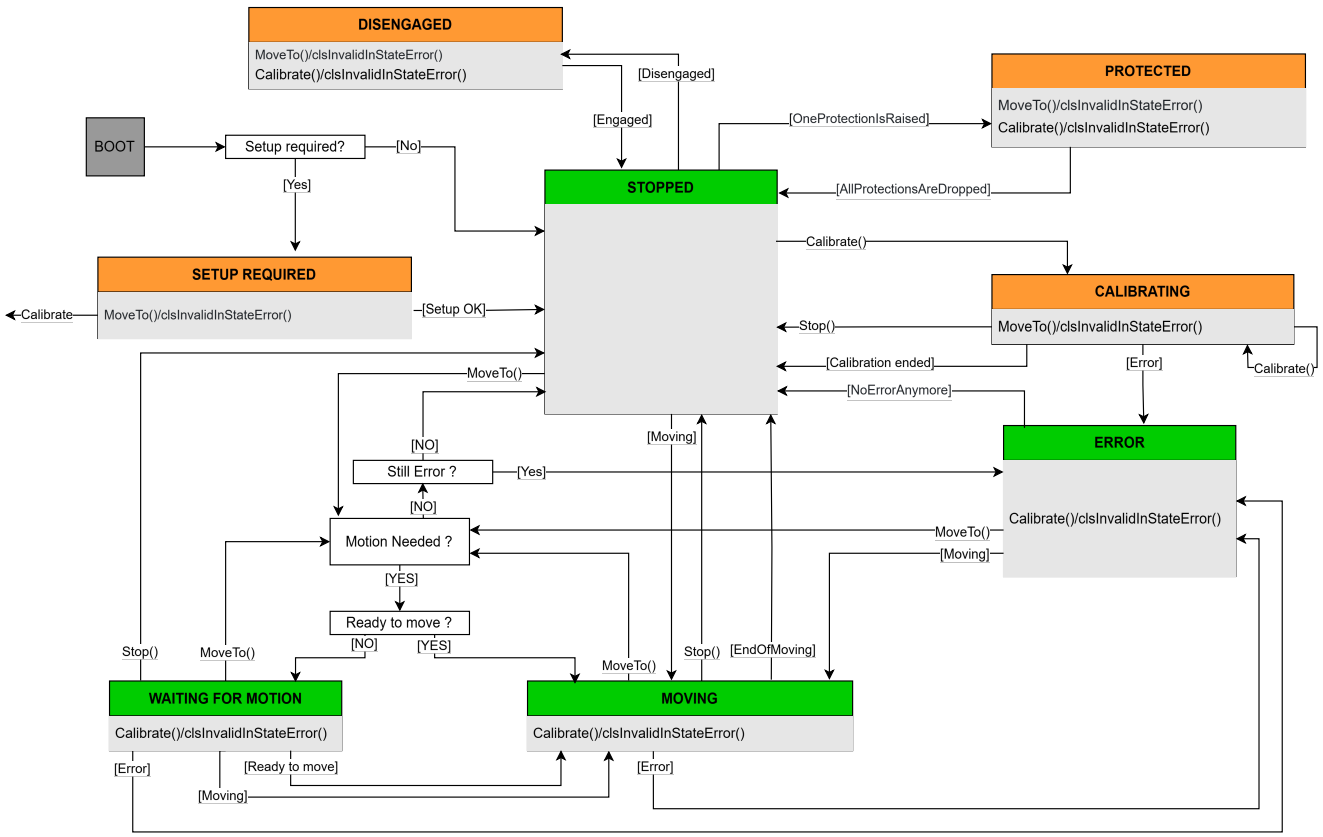


Figure 19. State diagram for Closure Control's MainState

NOTE The Calibrating state is exited upon completion of the calibration routine (whether

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through self-calibration or assisted by external tools, depending on the closure or manufacturer implementation), or if an error occurs, or if a stop command is received.

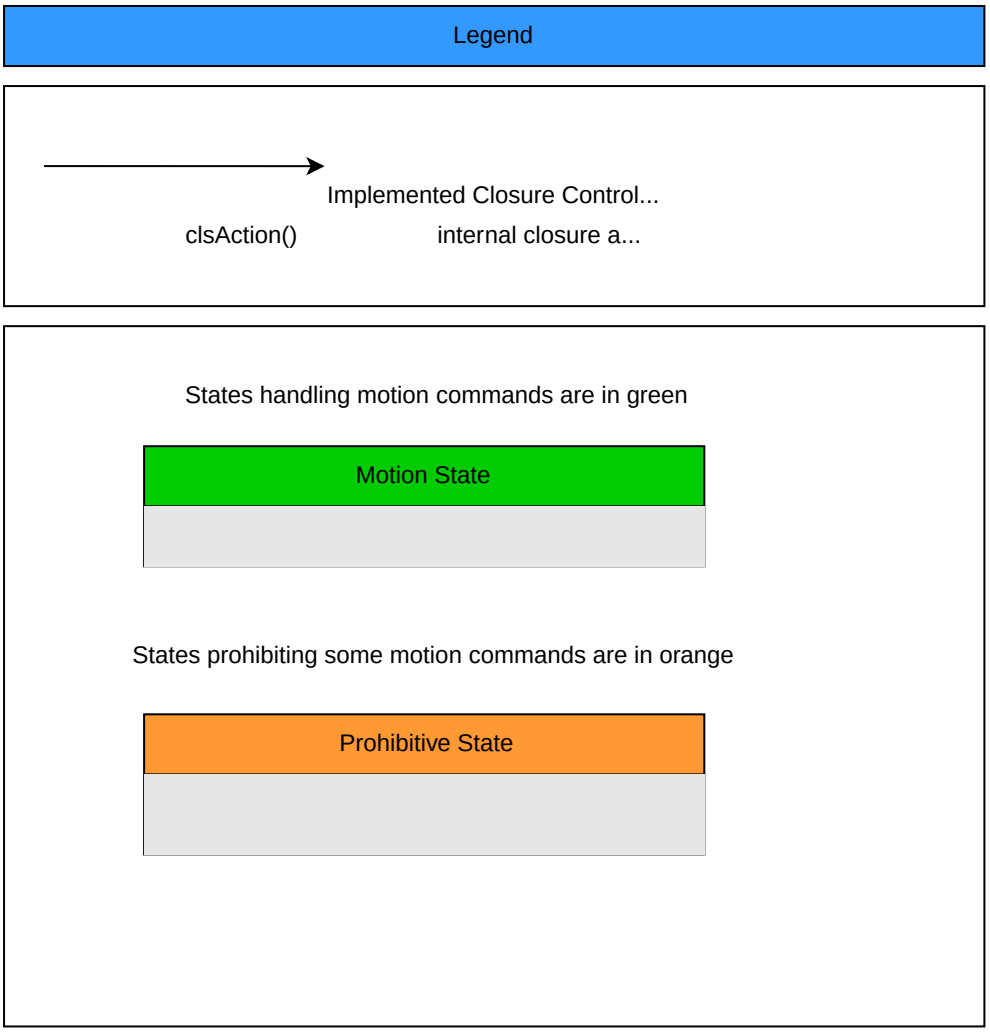


Figure 20. Legend

5.4.7.3. CurrentErrorList Attribute

This attribute SHALL indicate the currently active errors.

An empty list SHALL indicate that there are no active errors.

There SHALL NOT be duplicate values of ClosureErrorEnum within the CurrentErrorList.

5.4.7.4. OverallCurrentState Attribute

This attribute SHALL indicate the current Position, Latch and/or Speed states, whichever are applicable according to the feature flags set.

This attribute SHALL be null, if the state is unknown. Examples could be, but are not limited to:

- The state of Position/Latch is not known yet because the closure is not calibrated.
- The product has lost its Position/Latch state after manual motion during a shutdown.

The values of the different fields within the structure of this attribute depend on:

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- The effects of MoveTo commands.
- The effects of SetTarget and Step commands in a Closure Dimension Cluster associated with this cluster, as described in Section 5.4.4, “Association Between Closure Control and Closure Dimension Clusters”.
- The Stop command.

5.4.7.5. OverallTargetState Attribute

This attribute SHALL indicate the TargetPosition, TargetLatch and/or TargetSpeed values, whichever are applicable according to the feature flags set.

This attribute SHALL be null, if the state is unknown. For example after a reboot.

5.4.7.6. LatchControlModes Attribute

This attribute SHALL specify whether the latch mechanism can be latched or unlatched remotely.

5.4.8. Commands

The table below lists the changes relative to the Operational State cluster for the commands. A blank field indicates no change.

ID	Name	Direction	Response	Access	Conformance
0x00	**Stop**	client ⇒ server	Y	O	!IS
0x01	**MoveTo**	client ⇒ server	Y	O T	M
0x02	**Calibrate**	client ⇒ server	Y	M T	CL

5.4.8.1. Stop Command

On receipt of this command, the closure SHALL stop its movement as fast as the closure is able too.

If the server’s MainState attribute has one of the following values:

- Moving
- WaitingForMotion
- Calibrating

then any motions SHALL be stopped and the MainState attribute SHALL be set to Stopped.

A status code of SUCCESS SHALL always be returned, regardless of the value of the MainState attribute when this command is received.

5.4.8.2. MoveTo Command

On receipt of this command, the closure SHALL operate to update its position, latch state and/or motion speed.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Position**	TargetPositionEnum			desc	O.a+
1	**Latch**	bool			desc	O.a+
2	**Speed**	ThreeLevelAutoEnum			Auto	O.a+

The rationale behind defining the conformance as being purely optional in the table above is to ensure that commands containing one or more fields related to unsupported features are still accepted, rather than being rejected outright. For example, if a simple client, which is not able to determine the capabilities of the server, invokes a command that includes both position and speed, a server that does not support the speed feature would simply ignore the speed field while still adjusting its position as requested.

5.4.8.2.1. Position Field

This field SHALL indicate the position where the closure is moving to, as defined in the TargetPositionEnum.

If the field is present but its value is not within constraints, then:

- If the Positioning (PS) feature is not supported, the value of this field SHALL be ignored.
- If the Positioning (PS) feature is supported, a status code of CONSTRAINT_ERROR SHALL be returned.

If the Positioning (PS) feature is supported and the Position field is not present, the closure SHALL use the value of the Position field in the OverallTargetState attribute as the fallback. If the Position field in the OverallTargetState attribute is NULL, the position is unknown and the fallback is not applicable; in this case the field SHALL be ignored and the closure SHALL NOT change its position.

5.4.8.2.2. Latch Field

This field SHALL indicate the desired latching state of the closure, as defined in the OverallCurrentStateStruct Latch Field.

If the field is present but its value is not within constraints, then:

- If the MotionLatching (LT) feature is not supported, the value of this field SHALL be ignored.
- If the MotionLatching (LT) feature is supported, a status code of CONSTRAINT_ERROR SHALL be returned.

If the MotionLatching (LT) feature is supported and the Latch field is not present, the closure SHALL use the value of the Latch field in the OverallTargetState attribute as the fallback. If the Latch field in the OverallTargetState attribute is NULL, the latch state is unknown and the fallback is not applicable, in this case, the field SHALL be ignored and the closure SHALL NOT change its latch state.

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If the closure supports the MotionLatching (LT) feature, it SHALL either fulfill the latch request and update OverallTargetState.Latch or, if the "LatchControlModes" attribute specifies that manual intervention is required to latch - respond with INVALID_IN_STATE and remain in its current state.

5.4.8.2.3. Speed Field

This field SHALL indicate a desired overall speed of motion, as defined in the ThreeLevelAutoEnum.

If the field is present but its value is not within constraints, then:

- If the Speed(SP) feature is not supported, the value of this field SHALL be ignored.
- If the Speed(SP) feature is supported, a status code of CONSTRAINT_ERROR SHALL be returned.

If the closure supports the Speed(SP) feature, it SHALL set the Speed field of the OverallCurrentState attribute to the new speed.

5.4.8.2.4. Effect on Receipt

If this command is received in any state other than Moving, WaitingForMotion, or Stopped, a status code of INVALID_IN_STATE SHALL be returned.

If a field is present in the command, but its associated feature is not supported, that field SHALL be treated as absent for the purposes of command processing. Specifically:

- If the Position field is present but the Positioning (PS) feature is not supported, the Position field SHALL be ignored and have no effect.
- If the Latch field is present but the MotionLatching (LT) feature is not supported, the Latch field SHALL be ignored and have no effect.
- If the Speed field is present but the Speed (SP) feature is not supported, the Speed field SHALL be ignored and have no effect.

In all such cases, the presence of an unsupported field SHALL NOT trigger errors or state changes.

If this command requests a position change while the Latch field of the OverallCurrentState attribute is True (Latched), and the Latch field of this command is not set to False (Unlatched), a status code of INVALID_IN_STATE SHALL be returned.

Otherwise, on success:

- A status code of SUCCESS SHALL be returned.
- The OverallTargetState Attribute SHALL be updated with the requested values for Position, Latch and/or Speed.
- If all field values in the command match the corresponding field values in OverallCurrentState Attribute, the MainState Attribute SHALL be set to Stopped, and no further action SHALL be taken.
- If the closure is not able to move e.g. due to power limitation, the MainState attribute SHALL be set to WaitingForMotion, else the MainState Attribute SHALL be set to Moving.

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5.4.8.3. Calibrate Command

This command is used to trigger a calibration of the closure.

5.4.8.3.1. Effect on Receipt

Upon receipt, this SHALL initiate a calibration of the closure.

Procedures of how to achieve the calibration are manufacturer-specific.

If this command is received when already in the *Calibrating* state, the server SHALL respond with a status code of SUCCESS. Else, if this command is invoked in any state other than *Stopped* or *SetupRequired*, the server SHALL respond with INVALID_IN_STATE and there SHALL be no other effect.

Otherwise:

- The MainState attribute SHALL be set to *Calibrating*.
- The server SHALL respond with a status code equal to SUCCESS.

5.4.9. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**OperationalError**	CRITICAL		V	M
0x01	**MovementCompleted**	INFO		V	!IS
0x02	**EngageStateChanged**	INFO		V	MO
0x03	**SecureStateChanged**	INFO		V	M

5.4.9.1. OperationalError Event

This event SHALL be generated when a reportable error condition is detected. A closure that generates this event SHALL also set the MainState attribute to Error, indicating an error condition.

This event SHALL contain the following fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ErrorState**	list[ClosureErrorEnum]	1 to 10[all]			M

5.4.9.2. MovementCompleted Event

This event, if supported, SHALL be generated when the overall operation ends, either successfully or otherwise. For example, the event is sent upon the completion of a movement operation in a

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blind.

5.4.9.3. EngageStateChanged Event

This event, if supported, SHALL be generated when the MainStateEnum attribute changes state to and from disengaged, indicating if the actuator is Engaged or Disengaged.

This event SHALL contain the following fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**EngageValue**	bool	all			M

- *True*, when the actuator is in a Engaged state, actuator movements possible.
- *False*, when the actuator is in a Disengaged state, preventing any actuator movements.

5.4.9.4. SecureStateChanged Event

This event, if supported, SHALL be generated when the SecureState field in the OverallCurrentState attribute changes. It is used to indicate whether a closure is securing a space against possible unauthorized entry.

This event SHALL contain the following fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**SecureValue**	bool	all			M

- *True*, when the closure is in a secure state, e.g. unauthorized/undetectable access is not possible.
- *False*, when the closure is in an insecure state, e.g. unauthorized/undetectable access is possible.

5.5. Closure Dimension Cluster

This cluster provides an interface for controlling a single degree of freedom (also referred to as a "dimension" or an "axis" below) of a composed closure. This could be a degree of freedom in a 6-axis framework for the overall closure or a panel in the closure, or some other degree of freedom.

An instance of this cluster may represent position along a single axis, as specified by the FeatureMap, which can be one of:

- Translation : panel translates along one axis
- Rotation : panel rotates around an axis of rotation
- Modulation : panel modifies its aspect to modulate a flow

This cluster may also be used to latch or anchor a panel in a position, if the panel supports that.

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5.5.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

5.5.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	CLDIM

5.5.3. Cluster ID

ID	Name
0x0105	Closure Dimension

5.5.4. Association Between Closure Control and Closure Dimension Clusters

Some device types MAY define that a Closure Dimension cluster instance is associated with a corresponding Closure Control cluster instance on the same device. In such cases, the Closure Dimension cluster does not operate independently, rather, it relies on the Closure Control cluster for supervisory logic, coordination, and overall state management of the closure system. Where this association is specified, implementers SHALL ensure that the behaviors, commands, and states of each Closure Dimension cluster are properly integrated with and governed by the logic of the associated Closure Control cluster.

5.5.5. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**PS**	Positioning	O.a+	Supports Positioning in the range from 0.00% to 100.00%
1	**LT**	MotionLatching	O.a+	Supports a latch to secure the closure to a position or state

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Bit	Code	Feature	Conformance	Summary
2	**UT**	Unit	[PS]	Specifies the relevant unit and range for this dimension (mm, degrees etc.)
3	**LM**	Limitation	[PS]	Supports limitation of the operating range
4	**SP**	Speed	[PS]	Supports speed motion throttling.
5	**TR**	Translation	[PS].b	Drives a translation motion
6	**RO**	Rotation	[PS].b	Drives a rotation motion
7	**MD**	Modulation	[PS].b	Modulates a particular flow level (light, air, privacy …)

5.5.5.1. Positioning Feature

This feature SHALL indicate support for position percentage over the range of 0.00% to 100.00%, with a resolution of 0.01%.

NOTE In most of the products 0.00% = fully opened and 100.00% = fully closed but this is not always the case. For example, if the Modulation feature is supported and the ModulationType is SlatsOrientation or StripesAlignment, the panel can be fully closed at 0.00% and 100.00%, and be fully opened at 50.00%.

5.5.5.2. MotionLatching Feature

This feature SHALL indicate that the closure can be latched and unlatched. An axis with the MotionLatching feature has capabilities that will prevent actuators from moving along parts of that axis if the dimension is latched.

5.5.5.3. Unit Feature

This feature SHALL indicate additional information about the closure dimension’s possible range of movement.

5.5.5.4. Limitation Feature

This feature SHALL indicate that the closure dimension supports degradation of its functioning. Reachable Position range MAY be limited compared to full scope of a nominal behavior.

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5.5.5.5. Speed Feature

This feature SHALL indicate that the closure dimension can be driven at different speed levels: Low, Medium, and High. Please refer to Section 5.4.5.4, “Speed Feature” for more details.

NOTE	The server might not support three different speed values. The manufacturer SHALL select speed values linked to Low, Medium and High such that `Low <= Medium <= High`.

5.5.5.6. Translation Feature

This feature SHALL indicate that the panel can move along a single axis. The possible directions include downward, upward, leftward, rightward, forward, and backward. The Translation feature is used to control the position of the panel along the specified axis.

5.5.5.7. Rotation Feature

This feature SHALL indicate that the panel can rotate around a single axis. The possible axes include left, right, top, bottom, centered vertical, and centered horizontal. The Rotation feature is used to control the orientation of the panel around the specified axis.

5.5.5.8. Modulation Feature

This feature SHALL indicate that the panel can modify its aspect to control a particular flow, such as light, air, or privacy. The possible modulation types include slats orientation, slats openwork, stripes alignment, opacity, and ventilation. The Modulation feature is used to adjust the panel’s properties to achieve the desired effect.

5.5.6. Data Types

5.5.6.1. TranslationDirectionEnum Type

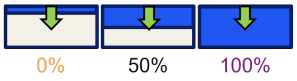
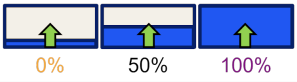
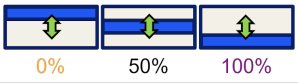
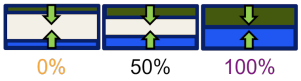
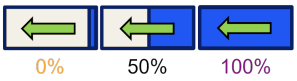
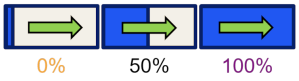
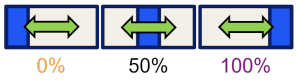
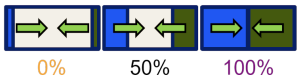
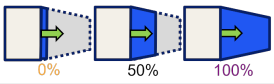
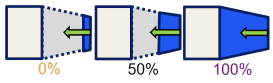
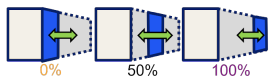
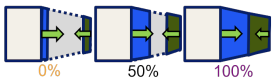
This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Downward**	Downward translation	M
1	**Upward**	Upward translation	M
2	**VerticalMask**	Vertical mask translation	M
3	**VerticalSymmetry**	Vertical symmetry translation	M
4	**Leftward**	Leftward translation	M
5	**Rightward**	Rightward translation	M
6	**HorizontalMask**	Horizontal mask translation	M

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Value	Name	Summary	Conformance
7	**HorizontalSymmetry**	Horizontal symmetry translation	M
8	**Forward**	Forward translation	M
9	**Backward**	Backward translation	M
10	**DepthMask**	Depth mask translation	M
11	**DepthSymmetry**	Depth symmetry translation	M

The following table provides a graphical representation detailing the enumeration values:

Value	Name	Axes	Drawing
0	**Downward**	Y	
1	**Upward**	Y	
2	**VerticalMask**	Y	
3	**VerticalSymmetry**	Y	
4	**RightToLeft**	X	
5	**LeftToRight**	X	
6	**HorizontalMask**	X	
7	**HorizontalSymmetry**	X	
8	**FrontToBack**	Z	
9	**BackToFront**	Z	
10	**DepthMask**	Z	
11	**DepthSymmetry**	Z	

Legend: Open Closed

5.5.6.2. RotationAxisEnum Type

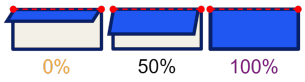
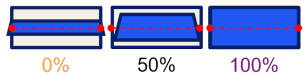
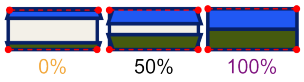
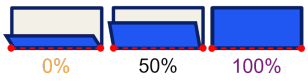
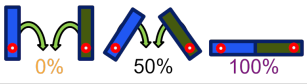
This data type is derived from enum8.

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Value	Name	Summary	Conformance
0	**Left**	The panel rotates around a vertical axis located on the left side of the panel	M
1	**CenteredVertical**	The panel rotates around a vertical axis located in the center of the panel	M
2	**LeftAndRight**	The panels rotates around vertical axes located on the left and right sides of the panel	M
3	**Right**	The panel rotates around a vertical axis located on the right side of the panel	M
4	**Top**	The panel rotates around a horizontal axis located on the top of the panel	M
5	**CenteredHorizontal**	The panel rotates around a horizontal axis located in the center of the panel	M
6	**TopAndBottom**	The panels rotates around horizontal axes located on the top and bottom of the panel	M
7	**Bottom**	The panel rotates around a horizontal axis located on the bottom of the panel	M
8	**LeftBarrier**	The barrier tilts around an axis located at the left end of the barrier	M
9	**LeftAndRightBarriers**	The dual barriers tilt around axes located at each side of the composite barrier	M
10	**RightBarrier**	The barrier tilts around an axis located at the right end of the barrier	M

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The following table provides a graphical representation detailing the enumeration values:

Value	Name	Axes	Drawing
0	**Left**	Y	
1	**CenteredVertical**	Y	
2	**LeftAndRight**	Y	
3	**Right**	Y	
4	**Top**	X	
5	**CenteredHorizontal**	X	
6	**TopAndBottom**	X	
7	**Bottom**	X	
8	**LeftBarrier**	Z (CW)	
9	**LeftAndRightBarriers**	Z (CW + CCW)	
10	**RightBarrier**	Z (CCW)	

Legend: Open Closed

5.5.6.3. OverflowEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**NoOverflow**	No overflow	M
1	**Inside**	Inside overflow	M
2	**Outside**	Outside overflow	M
3	**TopInside**	Top inside overflow	M
4	**TopOutside**	Top outside overflow	M
5	**BottomInside**	Bottom inside overflow	M
6	**BottomOutside**	Bottom outside overflow	M
7	**LeftInside**	Left inside overflow	M
8	**LeftOutside**	Left outside overflow	M

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Value	Name	Summary	Conformance
9	**RightInside**	Right inside overflow	M
10	**RightOutside**	Right outside overflow	M

The following table provides a description detailing the enumeration values:

Value	Name	Description
1	**Inside**	The panel opens inward, pivoting from the outside toward the interior side.
2	**Outside**	The panel opens outward, pivoting from the inside away from the interior side.
3	**TopInside**	The panel pivots along its top edge, opening inward towards the interior of the room.
4	**TopOutside**	The panel pivots along its top edge, opening outward away from the interior of the room.
5	**BottomInside**	The panel pivots along its bottom edge, opening inward towards the interior of the room.
6	**BottomOutside**	The panel pivots along its bottom edge, opening outward away from the interior of the room.
7	**LeftInside**	The left-hung panel opens inward, pivoting from the right towards the interior of the room.
8	**LeftOutside**	The panel opens outward, pivoting from the right towards the exterior of the building.
9	**RightInside**	The right-hung panel opens inward, pivoting from the left towards the interior of the room.
10	**RightOutside**	The panel opens outward, pivoting from the left towards the exterior of the building.

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5.5.6.4. ModulationTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**SlatsOrientation**	Orientation of the slats	M
1	**SlatsOpenwork**	Aperture of the slats	M
2	**StripesAlignment**	Alignment of blind stripes (Zebra)	M
3	**Opacity**	Opacity of a surface	M
4	**Ventilation**	Ventilation control	M

5.5.6.5. ClosureUnitEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Millimeter**	Millimeter used as unit	M
1	**Degree**	Degree used as unit	M

5.5.6.6. StepDirectionEnum Type

This data type is derived from enum8 and used for the Step command to indicate the direction of the steps.

Value	Name	Summary	Conformance
0	**Decrease**	Decrease towards 0.00%	M
1	**Increase**	Increase towards 100.00%	M

5.5.6.7. RangePercent100thsStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Min**	percent100ths	all				M
1	**Max**	percent100ths	Min to 100.00%				M

5.5.6.8. UnitRangeStruct Type

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Min**	int16	all				M
1	**Max**	int16	Min to 32767				M

5.5.6.9. DimensionStateStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Position**	percent100ths	desc	X	null		PS
1	**Latch**	bool	desc	X	null		LT
2	**Speed**	ThreeLevelAutoEnum	desc		MS		SP

5.5.6.9.1. Position Field

This field SHALL indicate the position of the closure, expressed as a percentage from 0.00% to 100.00%.

A null value indicates that the position is not known or is not set.

5.5.6.9.2. Latch Field

This field SHALL indicate the latching state of the closure, as defined in the OverallCurrentStateStruct Latch Field.

A null value indicates that the latch is not known or is not set.

5.5.6.9.3. Speed Field

This field SHALL indicate the current speed of the closure, as defined in the ThreeLevelAutoEnum.

If no speed value has yet been set, the server SHALL select and set one of the speed values defined in the ThreeLevelAutoEnum.

5.5.6.10. LatchControlModesBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**RemoteLatching**	Remote latching capability	LT
1	**RemoteUnlatching**	Remote unlatching capability	LT

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5.5.6.10.1. RemoteLatching Bit

This bit SHALL indicate whether the latch supports remote latching or not:

- 0 = the latch can only be latched through manual, physical operation.
- 1 = the latch can be latched via remote control (e.g., electronic or remote actuation).

5.5.6.10.2. RemoteUnlatching Bit

This bit SHALL indicate whether the latch supports remote unlatching or not:

- 0 = the latch can only be unlatched through manual, physical operation.
- 1 = the latch can be unlatched via remote control (e.g., electronic or remote actuation).

5.5.7. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**CurrentState**	DimensionStateStruct	desc	Q X	null	R V	M
0x0001	**TargetState**	DimensionStateStruct	desc	X	null	R V	M
0x0002	**Resolution**	percent100ths	min 0.01%	F	0.01%	R V	PS
0x0003	**StepValue**	percent100ths	min 0.01%	F	0.01%	R V	PS
0x0004	**Unit**	ClosureUnitEnum	all	F		R V	UT
0x0005	**UnitRange**	UnitRangeStruct	desc	X	null	R V	UT
0x0006	**LimitRange**	RangePercent100thsStruct	desc			R V	LM
0x0007	**TranslationDirection**	TranslationDirectionEnum	all	F		R V	TR
0x0008	**RotationAxis**	RotationAxisEnum	all	F		R V	RO
0x0009	**Overflow**	OverflowEnum	all	F		R V	RO

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x000A	**ModulationType**	ModulationTypeEnum	all	F		R V	MD
0x000B	**LatchControlModes**	LatchControlModesBitmap	all	F		R V	LT

5.5.7.1. CurrentState Attribute

This attribute SHALL indicate the current Position, Latching and/or Speed, based on the feature flags set.

A value of null SHALL indicate that:

- The position and latching state are both not known yet because the closure is not calibrated, or
- The product has lost its position and latching state after manual motion during a shutdown.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- When the attribute changes from null to any other value and vice versa, or
- When the Position changes from null to any other value and vice versa, or
- At most once every 5 seconds when the Position changes from one non-null value to another non-null value, or
- When `TargetState.Position` is reached, or
- When `CurrentState.Speed` changes, or
- When `CurrentState.Latch` changes.

The value of the different fields within the structure of this attribute depends on:

- The last SetTarget or Step commands.
- The impact of a MoveTo command received on the Closure Control Cluster instance associated with this cluster, if such a Closure Control instance exists, as described in Section 5.5.4, “Association Between Closure Control and Closure Dimension Clusters”.

5.5.7.2. TargetState Attribute

This attribute SHALL indicate the target Position, Latching and/or Speed, based on the feature flags set.

A value of null SHALL indicate that the TargetState fields are unknown (typically after a reboot, no target has been yet requested).

Each field SHALL be present only when its corresponding feature is supported. If the feature is not supported the field SHALL NOT be present.

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The value of `TargetState.Position` SHALL be set to a value that is an integer multiple of the Resolution attribute.

The value of the different fields within the structure of this attribute depends on:

- The last SetTarget or Step commands.
- The impact of a MoveTo command received on the Closure Control Cluster instance associated with this cluster, if such a Closure Control instance exists, as described in Section 5.5.4, “Association Between Closure Control and Closure Dimension Clusters”.

5.5.7.3. Resolution Attribute

This attribute SHALL indicate the minimal acceptable change to the Position field of the TargetState and CurrentState attributes.

Resolution should not be confused with an accuracy, such as Current = Target +/- Accuracy. This cluster does not provide accuracy.

Resolution gives a collection of valid Current position points over a linear ruler.

NOTE	A resolution of **100%** means that the associated dimension cannot be placed in an intermediate position - its position is binary.

5.5.7.4. StepValue Attribute

This attribute SHALL indicate the size of a single step, expressed in percent100ths. When the Step command is used, each step changes the position by this amount. The value of this attribute SHALL be an integer multiple of the Resolution attribute.

NOTE	The StepValue should be large enough to cause a visible change in the closure’s position when a Step command is invoked.

5.5.7.5. Unit Attribute

This attribute SHALL indicate the unit related to the Position.

5.5.7.6. UnitRange Attribute

This attribute SHALL indicate the minimum and the maximum values expressed by Position following the unit indicated by "Unit".

The value of this attribute MAY be null if the product has not been set up.

5.5.7.6.1. Constraint for values in Millimeters

Range values SHALL contain unsigned values from 0 to 32767 only.

5.5.7.6.2. Constraint for values in Degrees

The maximum span range is 360 degrees.

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Some examples to express the 360 degrees range:

- 0 to 359 degrees
- -180 to 179 degrees
- -90 to 269 degrees

Values in degrees can also be used to express the orientation range of slats even if slats filter a vertical closure such as vertical blinds or a horizontal closure such as a bioclimatic pergola or venetian blinds. The 0° value orientation is the value corresponding to the perpendicular axis to the Closure panel and means that all slats are parallel and offer the maximum opening.

Horizontal Slats:

- Negative values indicate that the user-facing edge is higher than the opposing edge.
- Positive values indicate that the user-facing edge is lower than the opposing edge.

Vertical Slats:

- Negative values indicate that the user-facing edge is to the left of the opposing edge.
- Positive values indicate that the user-facing edge is to the right of the opposing edge.

For example :

Slats travel range	Min Value	Max Value
0°/90° slats travel	0	90
-90°/90° slats travel	-90	90
-30°/90° slats travel	-30	90

5.5.7.7. LimitRange Attribute

This attribute SHALL indicate the range of possible values for the Position field in the CurrentState attribute.

This range MAY evolve dynamically.

`LimitRange.Min` and `LimitRange.Max` SHALL be equal to an integer multiple of the Resolution attribute.

5.5.7.8. TranslationDirection Attribute

This attribute SHALL indicate the direction of the translation.

A properly configured closure dimension SHOULD reflect as best as possible the translation as seen by the user. This attribute is not supposed to change once the installation is finalized.

5.5.7.9. RotationAxis Attribute

This attribute SHALL indicate the axis of the rotation.

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A properly configured closure dimension SHOULD reflect as best as possible the rotation axis as perceived by the user. This attribute is not supposed to change once the installation is finalized.

5.5.7.10. Overflow Attribute

This attribute SHALL indicate the overflow related to Rotation(RO).

A closure that rotates following an axis (with Rotation(RO) feature declared in FeatureMap), could overflow Inside and/or Outside. If the axis is centered, one part goes Outside and the other part goes Inside. In this case, this attribute SHALL use Top/Bottom/Left/Right Inside or Top/Bottom/Left/Right Outside enumerated value.

5.5.7.11. ModulationType Attribute

This attribute SHALL indicate the modulation type related to Modulation(MD).

The server SHOULD reflect, as best as possible, the modulation type as perceived by the user. This attribute is not supposed to change once the installation is finalized.

5.5.7.12. LatchControlModes Attribute

This attribute SHALL specify whether the latch mechanism can be latched or unlatched remotely.

5.5.8. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**SetTarget**	client ⇒ server	Y	O T	M
0x01	**Step**	client ⇒ server	Y	O T	PS

5.5.8.1. SetTarget Command

This command is used to move a dimension of the closure to a target position.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Position**	percent100ths			desc	O.a+
1	**Latch**	bool			desc	O.a+
2	**Speed**	ThreeLevelAutoEnum			Auto	O.a+

The rationale behind defining the conformance as being purely optional in the table above is to ensure that commands containing one or more fields related to unsupported features are still accepted, rather than being rejected outright. For example, if a simple client, which is not able to determine the capabilities of the server, invokes a command that includes both position and speed, a server that does not support the speed feature would simply ignore the speed field while still adjusting its position as requested.

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5.5.8.1.1. Position Field

This field SHALL indicate the position where the closure is moving to.

If the field is present but its value is not within constraints, then:

- If the Positioning (PS) feature is not supported, the value of this field SHALL be ignored.
- If the Positioning (PS) feature is supported, a status code of CONSTRAINT_ERROR SHALL be returned.

If the Positioning (PS) feature is supported and the Position field is not present, the closure SHALL use the value of the Position field in the TargetState attribute as the fallback. If the Position field in the TargetState attribute is NULL, the position is unknown and the fallback is not applicable, in this case the field SHALL be ignored and the closure SHALL NOT change its position.

When the closure supports the Positioning(PS) feature:

If a new position value is requested, the closure SHALL set the Position field of the TargetState attribute to the nearest valid position (an integer multiple of the Resolution attribute), whether by rounding the requested value or using it directly if already valid, and then initiate the motion procedure.

5.5.8.1.2. Latch Field

This field SHALL indicate the desired latching state of the closure, as defined in the OverallCurrentStateStruct Latch Field.

If the field is present but its value is not within constraints, then:

- If the MotionLatching (LT) feature is not supported, the value of this field SHALL be ignored.
- If the MotionLatching (LT) feature is supported, a status code of CONSTRAINT_ERROR SHALL be returned.

If the MotionLatching (LT) feature is supported and the Latch field is not present, the closure SHALL use the value of the Latch field in the TargetState attribute as the fallback. If the Latch field in the TargetState attribute is NULL, the latch state is unknown and the fallback is not applicable, in this case, the field SHALL be ignored and the closure SHALL NOT change its latch state.

If the server supports the MotionLatching (LT) feature, it SHALL either fulfill the latch request and update `TargetState.Latch`, or - if the "LatchControlModes" attribute specifies that manual intervention is required to latch - respond with INVALID_IN_STATE and remain in its current state.

5.5.8.1.3. Speed Field

This field SHALL indicate a desired overall speed of motion, as defined in the ThreeLevelAutoEnum.

If the field is present but its value is not within constraints, then:

- If the Speed(SP) feature is not supported, the value of this field SHALL be ignored.
- If the Speed(SP) feature is supported, a status code of CONSTRAINT_ERROR SHALL be returned.

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If the Speed (SP) feature is supported, the TargetState.Speed attribute SHALL be updated with the value of the Speed field.

A closure dimension SHOULD attempt to follow the indicated Speed value as closely as possible (meant to override its default speed).

If a SetTarget command is sent with only the Speed field:

- It SHOULD adjust the speed of the ongoing motion of the closure (i.e., allow speed change on the fly).

NOTE	The internal default speed SHALL not be affected by this command field. The priorities and cross capabilities to achieve, skip or ignore Position + Latch + Speed combinations are product/manufacturer specific. If the Limitation feature is supported, the closure will automatically offset the TargetState.Position value to fit within LimitRange.Min and LimitRange.Max.

5.5.8.1.4. Effect on Receipt

If this command is received while the MainState attribute of the Closure Control Cluster that is associated with this cluster has any of the following values:

- Disengaged
- Protected
- Calibrating
- SetupRequired
- Error

then a status code of INVALID_IN_STATE SHALL be returned.

If a field is present in the command, but its associated feature is not supported, that field SHALL be treated as absent for the purposes of command processing. Specifically:

- If the Position field is present but the Positioning (PS) feature is not supported, the Position field SHALL be ignored and have no effect.
- If the Latch field is present but the MotionLatching (LT) feature is not supported, the Latch field SHALL be ignored and have no effect.
- If the Speed field is present but the Speed (SP) feature is not supported, the Speed field SHALL be ignored and have no effect.

In all such cases, the presence of an unsupported field SHALL NOT trigger errors or state changes.

If the Positioning (PS) feature and the MotionLatching (LT) feature are supported, and the command requests a position change, then:

- If the Latch field of the CurrentState attribute is True (Latched), and
- The Latch field in this command is either not present or not explicitly set to False (Unlatched),

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the command SHALL return a status code of INVALID_IN_STATE.

If all field values in the command match the corresponding field values in CurrentState, the command SHALL have no effect.

Otherwise, the TargetState attribute SHALL be updated with the new values for Position, Latch and/or Speed, and a status code of SUCCESS SHALL be returned.

5.5.8.1.5. Impact of Unit(UT) feature

The Unit (UT) feature defines the relationship between the physical unit range (e.g., millimeters, degrees) and the normalized percent100ths value. This allows clients to convert native unit values into percentage values for commands and attribute updates.

The mapping is as follows:

- The Min field of the "Range" attribute corresponds to 0.00%.
- The Max field of the "Range" attribute corresponds to 100.00%.

The formulas below show how to perform this conversion.

5.5.8.1.5.1. Specific case of Degrees

In case of Slat orientation with range -90°/90°, both 0.00% and 100.00% target value create a Closed Position. This is the only accepted exception.

Example of Closure Dimension used for Modulation feature with Slats orientation type for different types of range.

Min in degrees	Max in degrees	Needed value in degrees	Request Target value in %
0°	90°	0°	0%
		45°	50%
		90°	100%
-90°	90°	-90°	0%
		-45°	25%
		0°	50%
		45°	75%
		90°	100%
-30°	90°	-30°	0%
		0°	25%
		30°	50%
		60°	75%
		90°	100%

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5.5.8.2. Step Command

This command is used to move a dimension of the closure to a target position by a number of steps.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Direction**	StepDirectionEnum				M
1	**NumberOfSteps**	uint16	min 1			M
2	**Speed**	ThreeLevelAutoEnum	desc		Auto	[SP]

5.5.8.2.1. Direction Field

This field SHALL indicate whether the Position field of the TargetState attribute must be:

- Increased toward 100.00%.
- Decreased toward 0.00%.

5.5.8.2.2. NumberOfSteps Field

This field SHALL indicate the number of steps by which the position should be changed. The size of one step, expressed in percent100ths, is determined by the "StepValue" attribute.

5.5.8.2.3. Speed Field

This field SHALL indicate the desired speed of motion.

5.5.8.2.4. Effect on Receipt

Upon receipt of this command:

- If this command is received while the Latch field of the CurrentState attribute is True (Latched), a status code of INVALID_IN_STATE SHALL be returned.
- If this command is received while the MainState attribute of the Closure Control Cluster that is associated with this cluster has any of the following values:
  - Disengaged
  - Protected
  - Calibrating
  - SetupRequired
  - Error

  then a status code of INVALID_IN_STATE SHALL be returned.
- Otherwise, the server SHALL respond with a status code of SUCCESS and the TargetState attribute value SHALL be changed as follows:

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- If the Direction field is set to Decrease, the Position field of the TargetState attribute SHALL be set to: `TargetState.Position = CurrentState.Position - NumberOfSteps * StepValue` and the value SHALL be clamped to 0.00% if the Limitation feature is not supported or `LimitRange.Min` if the Limitation feature is supported.
- If the Direction field is set to Increase, the Position field of the TargetState attribute SHALL be set to: `TargetState.Position = CurrentState.Position + NumberOfSteps * StepValue` and the value SHALL be clamped to 100.00% if the Limitation feature is not supported or `LimitRange.Max` if the Limitation feature is supported.
- If the Speed field of the command is present, the Speed field of the TargetState attribute SHALL be set to the Speed field of the command, otherwise the Speed field of the TargetState attribute SHALL remain unchanged.

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Chapter 6. Media

The Cluster Library is made of individual chapters such as this one. See Document Control in the Cluster Library for a list of all chapters and documents. References between chapters are made using a *X.Y* notation where *X* is the chapter and *Y* is the sub-section within that chapter. References to external documents are contained in Chapter 1 and are made using [*Rn*] notation.

6.1. General Description

6.1.1. Introduction

The clusters specified in this document are for use typically in applications involving media (e.g., Video Players, Content Apps, Speakers), but MAY be used in any application domain.

6.1.2. Cluster List

This section lists the media specific clusters as specified in this chapter.

Table 10. Overview of the Media Clusters

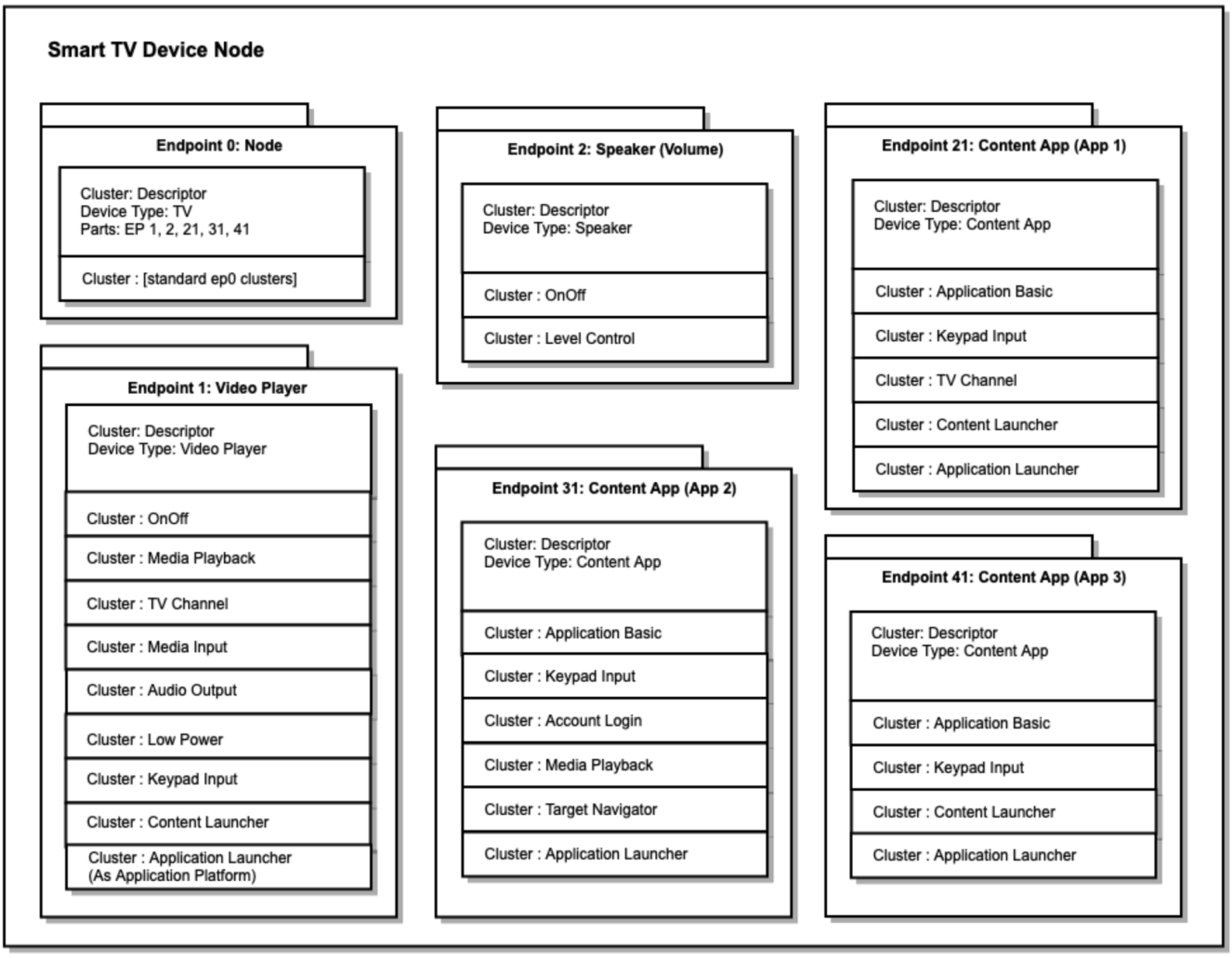
Cluster ID	Cluster Name	Description
0x050E	Account Login	This cluster provides an interface for facilitating user account login on an application or a node.
0x050D	Application Basic	Provides information about a Content App running on a Video Player device which is represented as an endpoint.
0x050C	Application Launcher	This cluster provides an interface for launching content on a Video Player device.
0x050B	Audio Output	This cluster provides an interface for controlling the Output on a Video Player device.
0x0504	Channel	This cluster provides an interface for controlling the current Channel on an endpoint.
0x0510	Content App Observer	This cluster provides an interface for sending targeted messages from a Content App to a Client of that Content App.

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Cluster ID	Cluster Name	Description
0x050A	Content Launcher	This cluster provides an interface for launching content on a Video Player device or a Content App.
0x0509	Keypad Input	This cluster provides an interface for controlling a Video Player or a Content App using action commands such as UP, DOWN, and SELECT.
0x0507	Media Input	This cluster provides an interface for controlling the Input Selector on a Video Player device.
0x0506	Media Playback	This cluster provides an interface for controlling Media Playback (PLAY, PAUSE, etc) on a Video Player device.
0x0505	Target Navigator	This cluster provides an interface for UX navigation within a set of targets on a Video Player device or Content App endpoint.
0x050F	Content Control	This cluster provides an interface for controlling Content controls such as turning on/off Content Control feature on a Video Player device.

Example Usage of the Media Clusters

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6.2. Account Login Cluster

This cluster provides commands that facilitate user account login on a Content App or a node. For example, a Content App running on a Video Player device, which is represented as an endpoint (see Device Type Library document), can use this cluster to help make the user account on the Content App match the user account on the Client.

Often a fabric administrator will facilitate commissioning of a Client (such as a Casting Video Client), and invoke commands on the Account Login cluster on the Content App associated with that client. Specifically:

1. GetSetupPIN in order to attempt to obtain the Passcode for commissioning.
2. Login in order to let the Content App know that commissioning has completed. The Content App can use information provided in this command in order to determine the user account associated with the client, and potentially assume that user account.
3. Logout in order to let the Content App know that client access has been removed, and potentially clear the current user account.

The cluster server for this cluster may be supported on each endpoint that represents a Content App on a Video Player device.

See Device Type Library document for details of how a Content App, represented as an endpoint on

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the Video Player device, may implement the cluster server for this cluster to simplify account login for its users.

6.2.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Add support for 8 character Passcode in GetSetupPIN and Login, Add Node to Login, Logout commands. Add LoggedOut event

6.2.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	ALOGIN

6.2.3. Cluster ID

Cluster ID	Cluster Name
0x050E	Account Login

6.2.4. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**GetSetupPIN**	client ⇒ server	GetSetupPIN-Response	A F T	M
0x01	**GetSetupPIN-Response**	client ⇐ server	N	F	M
0x02	**Login**	client ⇒ server	Y	A F T	M
0x03	**Logout**	client ⇒ server	Y	O F T	M

6.2.4.1. GetSetupPIN Command

The purpose of this command is to determine if the active user account of the given Content App matches the active user account of a given Commissionee, and when it does, return a Setup PIN which can be used for password-authenticated session establishment (PASE) with the Commissionee.

For example, a Video Player with a Content App Platform may invoke this command on one of its Content App endpoints to facilitate commissioning of a Phone App made by the same vendor as the Content App. If the accounts match, then the Content App may return a Setup PIN that can be used

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by the Video Player to commission the Phone App without requiring the user to physically input a Setup PIN.

The account match is determined by the Content App using a method which is outside the scope of this specification and will typically involve a central service which is in communication with both the Content App and the Commissionee. The GetSetupPIN command is needed in order to provide the Commissioner/Admin with a Setup PIN when this Commissioner/Admin is operated by a different vendor from the Content App.

This method is used to facilitate Setup PIN exchange (for PASE) between Commissioner and Commissionee when the same user account is active on both nodes. With this method, the Content App satisfies proof of possession related to commissioning by requiring the same user account to be active on both Commissionee and Content App, while the Commissioner/Admin ensures user consent by prompting the user prior to invocation of the command.

Upon receipt of this command, the Content App checks if the account associated with the Temporary Account Identifier sent by the client is the same account that is active on itself. If the accounts are the same, then the Content App returns the GetSetupPIN Response which includes a Setup PIN that may be used for PASE with the Commissionee.

The Temporary Account Identifier for a Commissionee may be populated with the Rotating ID field of the client's commissionable node advertisement (see Rotating Device Identifier section in [MatterCore]) encoded as an octet string where the octets of the Rotating Device Identifier are encoded as 2-character sequences by representing each octet's value as a 2-digit hexadecimal number, using uppercase letters.

The Setup PIN is a character string so that it can accommodate different future formats, including alpha-numeric encodings. For a Commissionee it SHALL be populated with the Manual Pairing Code (see Manual Pairing Code section in [MatterCore]) encoded as a string (11 characters) or the Passcode portion of the Manual Pairing Code (when less than 11 characters).

The server SHALL implement rate limiting to prevent brute force attacks. No more than 10 unique requests in a 10 minute period SHALL be allowed; a command response status of FAILURE should sent for additional commands received within the 10 minute period. Because access to this command is limited to nodes with Admin-level access, and the user is prompted for consent prior to Commissioning, there are in place multiple obstacles to successfully mounting a brute force attack. A Content App that supports this command SHALL ensure that the Temporary Account Identifier used by its clients is not valid for more than 10 minutes.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**TempAccountIdentifier**	string	16 to 100			M

6.2.4.1.1. TempAccountIdentifier Field

This field SHALL specify the client's Temporary Account Identifier. The length of this field SHALL be at least 16 characters to protect the account holder against password guessing attacks.

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6.2.4.2. GetSetupPINResponse Command

This message is sent in response to the GetSetupPIN command, and contains the Setup PIN, or null when the account identified in the request does not match the active account of the running Content App.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**SetupPIN**	string	desc			M

6.2.4.2.1. SetupPIN Field

This field SHALL provide the Setup PIN as a text string at least 8 characters in length or empty string to indicate that the accounts do not match.

NOTE	Newer cluster clients should be aware that Account Login cluster version 1 specified an 11 digit minimum length.

6.2.4.3. Login Command

The purpose of this command is to allow the Content App to assume the user account of a given Commissionee by leveraging the Setup PIN input by the user during the commissioning process.

For example, a Video Player with a Content App Platform may invoke this command on one of its Content App endpoints after the commissioning has completed of a Phone App made by the same vendor as the Content App. The Content App may determine whether the Temporary Account Identifier maps to an account with a corresponding Setup PIN and, if so, it may automatically login to the account for the corresponding user. The end result is that a user performs commissioning of a Phone App to a Video Player by inputting the Setup PIN for the Phone App into the Video Player UX. Once commissioning has completed, the Video Player invokes this command to allow the corresponding Content App to assume the same user account as the Phone App.

The verification of Setup PIN for the given Temporary Account Identifier is determined by the Content App using a method which is outside the scope of this specification and will typically involve a central service which is in communication with both the Content App and the Commissionee. Implementations of such a service should impose aggressive time outs for any mapping of Temporary Account Identifier to Setup PIN in order to prevent accidental login due to delayed invocation.

Upon receipt, the Content App checks if the account associated with the client's Temp Account Identifier has a current active Setup PIN with the given value. If the Setup PIN is valid for the user account associated with the Temp Account Identifier, then the Content App MAY make that user account active.

The Temporary Account Identifier for a Commissionee may be populated with the Rotating ID field of the client's commissionable node advertisement encoded as an octet string where the octets of the Rotating Device Identifier are encoded as 2-character sequences by representing each octet's value as a 2-digit hexadecimal number, using uppercase letters.

The Setup PIN for a Commissionee may be populated with the Manual Pairing Code encoded as a

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string of decimal numbers (11 characters) or the Passcode portion of the Manual Pairing Code encoded as a string of decimal numbers (8 characters).

The server SHALL implement rate limiting to prevent brute force attacks. No more than 10 unique requests in a 10 minute period SHALL be allowed; a command response status of FAILURE should sent for additional commands received within the 10 minute period. Because access to this command is limited to nodes with Admin-level access, and the user is involved when obtaining the Setup PIN, there are in place multiple obstacles to successfully mounting a brute force attack. A Content App that supports this command SHALL ensure that the Temporary Account Identifier used by its clients is not valid for more than 10 minutes.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**TempAccountIdentifier**	string	16 to 100			M
1	**SetupPIN**	string	min 8			M
2	**Node**	node-id				O

6.2.4.3.1. TempAccountIdentifier Field

This field SHALL specify the client's temporary account identifier.

6.2.4.3.2. SetupPIN Field

This field SHALL provide the Setup PIN as a text string at least 8 characters in length.

NOTE	Newer cluster clients should be aware that Account Login cluster revision 1 specified an 11 digit minimum length.

6.2.4.3.3. Node Field

This optional field SHALL provide the Node ID of the Client. This field can be used by the Content App to keep track of Nodes which currently have access to it.

6.2.4.4. Logout Command

The purpose of this command is to instruct the Content App to clear the current user account. This command SHOULD be used by clients of a Content App to indicate the end of a user session.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Node**	node-id				O

6.2.4.4.1. Node Field

This optional field SHALL provide the Node ID of the Client. This field can be used by the Content App to keep track of Nodes which currently have access to it.

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6.2.5. Events

ID	Name	Priority	Quality	Access	Conformance
0	**LoggedOut**	CRITICAL		A S	O

6.2.5.1. LoggedOut Event

This event can be used by the Content App to indicate that the current user has logged out. In response to this event, the Fabric Admin SHALL remove access to this Content App by the specified Node. If no Node is provided, then the Fabric Admin SHALL remove access to all non-Admin Nodes.

The data of this event SHALL contain the following information:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Node**	node-id	all			O

6.2.5.1.1. Node Field

This field SHALL provide the Node ID corresponding to the user account that has logged out, if that Node ID is available. If it is NOT available, this field SHALL NOT be present in the event.

6.3. Application Basic Cluster

This cluster provides information about a Content App running on a Video Player device which is represented as an endpoint (see Device Type Library document).

The cluster server for this cluster should be supported on each endpoint that represents a Content App on a Video Player device. This cluster provides identification information about the Content App such as vendor and product.

6.3.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

6.3.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	APBSC

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6.3.3. Cluster ID

ID	Name
0x050D	Application Basic

6.3.4. Data Types

6.3.4.1. ApplicationStatusEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Stopped**	Application is not running.	M
1	**ActiveVisibleFocus**	Application is running, is visible to the user, and is the active target for input.	M
2	**ActiveHidden**	Application is running but not visible to the user.	M
3	**ActiveVisibleNotFocus**	Application is running and visible, but is not the active target for input.	M

6.3.4.2. ApplicationStruct Type

This indicates a global identifier for an Application given a catalog.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**CatalogVendorID**	uint16	all				M
1	**ApplicationID**	string	all				M

6.3.4.2.1. CatalogVendorID Field

This field SHALL indicate the Connectivity Standards Alliance issued vendor ID for the catalog. The DIAL registry SHALL use value 0x0000.

It is assumed that Content App Platform providers (see Video Player Architecture section in [Matter-DevLib]) will have their own catalog vendor ID (set to their own Vendor ID) and will assign an ApplicationID to each Content App.

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6.3.4.2.2. ApplicationID Field

This field SHALL indicate the application identifier, expressed as a string, such as "123456-5433", "PruneVideo" or "Company X". This field SHALL be unique within a catalog.

For the DIAL registry catalog, this value SHALL be the DIAL prefix (see [DIAL Registry]).

6.3.5. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**VendorName**	string	max 32	F	empty	R V	O
0x0001	**VendorID**	vendor-id	all	F		R V	O
0x0002	**ApplicationName**	string	max 256	F		R V	M
0x0003	**ProductID**	uint16	all	F		R V	O
0x0004	**Application**	ApplicationStruct	desc	F		R V	M
0x0005	**Status**	ApplicationStatusEnum	desc		MS	R V	M
0x0006	**ApplicationVersion**	string	max 32	F		R V	M
0x0007	**AllowedVendorList**	list[vendor-id]		F		R A	M

6.3.5.1. VendorName Attribute

This attribute SHALL specify a human readable (displayable) name of the vendor for the Content App.

6.3.5.2. VendorID Attribute

This attribute, if present, SHALL specify the Connectivity Standards Alliance assigned Vendor ID for the Content App.

6.3.5.3. ApplicationName Attribute

This attribute SHALL specify a human readable (displayable) name of the Content App assigned by the vendor. For example, "NPR On Demand".

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6.3.5.4. ProductID Attribute

This attribute, if present, SHALL specify a numeric ID assigned by the vendor to identify a specific Content App made by them. If the Content App is certified by the Connectivity Standards Alliance, then this would be the Product ID as specified by the vendor for the certification.

6.3.5.5. Application Attribute

This attribute SHALL specify a Content App which consists of an Application ID using a specified catalog.

6.3.5.6. Status Attribute

This attribute SHALL specify the current running status of the application.

6.3.5.7. ApplicationVersion Attribute

This attribute SHALL specify a human readable (displayable) version of the Content App assigned by the vendor.

6.3.5.8. AllowedVendorList Attribute

This attribute is a list of vendor IDs. Each entry is a vendor-id.

6.4. Application Launcher Cluster

This cluster provides an interface for launching applications on a Video Player device such as a TV.

This cluster is supported on endpoints that can launch Applications, such as a Casting Video Player device with a Content App Platform. It supports identifying an Application by global identifier from a given catalog, and launching it. It also supports tracking the currently in-focus Application.

Depending on the support for the Application Platform feature, the cluster can either support launching the application corresponding to the endpoint on which the cluster is supported (AP feature not supported) or it can support launching any application (AP feature supported).

6.4.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Addition of new states to StatusEnum

6.4.2. Classification

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Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	APPLAUNCHER

6.4.3. Cluster ID

ID	Name
0x050C	Application Launcher

6.4.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**AP**	ApplicationPlatform	O	Support for attributes and commands required for endpoint to support launching any application within the supported application catalogs

6.4.5. Data Types

6.4.5.1. StatusEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Success**	Command succeeded	M
1	**AppNotAvailable**	Requested app is not available	M
2	**SystemBusy**	Video platform unable to honor command	M
3	**PendingUserApproval**	User approval for app download is pending	M
4	**Downloading**	Downloading the requested app	M
5	**Installing**	Installing the requested app	M

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6.4.5.2. ApplicationStruct Type

This indicates a global identifier for an Application given a catalog.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**CatalogVendorID**	uint16	all				M
1	**ApplicationID**	string	all				M

6.4.5.2.1. CatalogVendorID Field

This field SHALL indicate the Connectivity Standards Alliance-issued vendor ID for the catalog. The DIAL registry SHALL use value 0x0000.

Content App Platform providers will have their own catalog vendor ID (set to their own Vendor ID) and will assign an ApplicationID to each Content App.

6.4.5.2.2. ApplicationID Field

This field SHALL indicate the application identifier, expressed as a string, such as "PruneVideo" or "Company X". This field SHALL be unique within a catalog.

For the DIAL registry catalog, this value SHALL be the DIAL prefix (see [DIAL Registry]).

6.4.5.3. ApplicationEPStruct Type

This specifies an app along with its corresponding endpoint.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Application**	ApplicationStruct	all				M
1	**Endpoint**	endpoint-no	all		MS		O

6.4.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**CatalogList**	list[uint16]		N		R V	AP
0x0001	**CurrentApp**	ApplicationEPStruct	desc	X	null	R V	O

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6.4.6.1. CatalogList Attribute

This attribute SHALL specify the list of supported application catalogs, where each entry in the list is the Connectivity Standards Alliance-issued vendor ID for the catalog. The DIAL registry (see [DIAL Registry]) SHALL use value 0x0000.

It is expected that Content App Platform providers will have their own catalog vendor ID (set to their own Vendor ID) and will assign an ApplicationID to each Content App.

6.4.6.2. CurrentApp Attribute

This attribute SHALL specify the current in-focus application, identified using an Application ID, catalog vendor ID and the corresponding endpoint number when the application is represented by a Content App endpoint. A null SHALL be used to indicate there is no current in-focus application.

6.4.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**LaunchApp**	client ⇒ server	LauncherResponse	O	M
0x01	**StopApp**	client ⇒ server	LauncherResponse	O	M
0x02	**HideApp**	client ⇒ server	LauncherResponse	O	M
0x03	**LauncherResponse**	client ⇐ server	N		M

6.4.7.1. LaunchApp Command

Upon receipt of this command, the server SHALL launch the application with optional data. The application SHALL be either

- the specified application, if the Application Platform feature is supported;
- otherwise the application corresponding to the endpoint.

The endpoint SHALL launch and bring to foreground the requisite application if the application is not already launched and in foreground. The Status attribute SHALL be updated to ActiveVisibleFocus on the Application Basic cluster of the Endpoint corresponding to the launched application. The Status attribute SHALL be updated on any other application whose Status MAY have changed as a result of this command. The CurrentApp attribute, if supported, SHALL be updated to reflect the new application in the foreground.

This command returns a Launcher Response.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Application**	Application-Struct	desc			AP
1	**Data**	octstr			MS	O

6.4.7.1.1. Application Field

This field SHALL specify the Application to launch.

6.4.7.1.2. Data Field

This field SHALL specify optional app-specific data to be sent to the app.

NOTE	This format and meaning of this value is proprietary and outside the specification. It provides a transition path for device makers that use other protocols (like DIAL) which allow for proprietary data. Apps that are not yet Matter aware can be launched via Matter, while retaining the existing ability to launch with proprietary data.

6.4.7.2. StopApp Command

Upon receipt of this command, the server SHALL stop the application if it is running. The application SHALL be either

- the specified application, if the Application Platform feature is supported;
- otherwise the application corresponding to the endpoint.

The Status attribute SHALL be updated to Stopped on the Application Basic cluster of the Endpoint corresponding to the stopped application. The Status attribute SHALL be updated on any other application whose Status MAY have changed as a result of this command.

This command returns a Launcher Response.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Application**	Application-Struct	desc		MS	AP

6.4.7.2.1. Application Field

This field SHALL specify the Application to stop.

6.4.7.3. HideApp Command

Upon receipt of this command, the server SHALL hide the application. The application SHALL be either

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- the specified application, if the Application Platform feature is supported;
- otherwise the application corresponding to the endpoint.

The endpoint MAY decide to stop the application based on manufacturer specific behavior or resource constraints if any. The Status attribute SHALL be updated to ActiveHidden or Stopped, depending on the action taken, on the Application Basic cluster of the Endpoint corresponding to the application on which the action was taken. The Status attribute SHALL be updated on any other application whose Status MAY have changed as a result of this command.

This command returns a Launcher Response.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Application**	Application-Struct	desc		MS	AP

6.4.7.3.1. Application Field

This field SHALL specify the Application to hide.

6.4.7.4. LauncherResponse Command

This command SHALL be generated in response to LaunchApp/StopApp/HideApp commands.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	StatusEnum	all			M
1	**Data**	octstr			MS	O

6.4.7.4.1. Status Field

This field SHALL indicate the status of the command which resulted in this response.

6.4.7.4.2. Data Field

This field SHALL specify Optional app-specific data.

6.5. Audio Output Cluster

This cluster provides an interface for controlling the Output on a Video Player device such as a TV.

This cluster would be supported on a device with audio outputs like a Video Player device (Smart TV, TV Setup Top Box, Smart Speaker, etc).

This cluster provides the list of available outputs and provides commands for selecting and renaming them.

The cluster server for Audio Output is implemented by a device that has configurable audio output.

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6.5.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

6.5.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	AUDIOOUTPUT

6.5.3. Cluster ID

Cluster ID	Name
0x050B	Audio Output

6.5.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**NU**	NameUpdates	O	Supports updates to output names

6.5.5. Data Types

6.5.5.1. OutputTypeEnum Type

This data type is derived from enum8.

The type of output, expressed as an enum, with the following values:

Value	Name	Summary	Conformance
0	**HDMI**	HDMI	M
1	**BT**		M
2	**Optical**		M
3	**Headphone**		M
4	**Internal**		M
5	**Other**		M

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6.5.5.2. OutputInfoStruct Type

This contains information about an output.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Index**	uint8	all				M
1	**OutputType**	OutputTypeEnum	desc				M
2	**Name**	string	max 32				M

6.5.5.2.1. Index Field

This field SHALL indicate the unique index into the list of outputs.

6.5.5.2.2. OutputType Field

This field SHALL indicate the type of output.

6.5.5.2.3. Name Field

The device defined and user editable output name, such as “Soundbar”, “Speakers”. This field may be blank, but SHOULD be provided when known.

6.5.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**OutputList**	list[OutputInfoStruct]				R V	M
0x0001	**CurrentOutput**	uint8	all			R V	M

6.5.6.1. OutputList Attribute

This attribute provides the list of outputs supported by the device.

6.5.6.2. CurrentOutput Attribute

This attribute contains the value of the index field of the currently selected OutputInfoStruct.

6.5.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**SelectOutput**	client ⇒ server	Y	O	M

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ID	Name	Direction	Response	Access	Conformance
0x01	**RenameOutput**	client ⇒ server	Y	M	NU

6.5.7.1. SelectOutput Command

Upon receipt, this SHALL change the output on the device to the output at a specific index in the Output List.

Note that when the current output is set to an output of type HDMI, adjustments to volume via a Speaker endpoint on the same node MAY cause HDMI volume up/down commands to be sent to the given HDMI output.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Index**	uint8	all			M

6.5.7.1.1. Index Field

This SHALL indicate the index field of the OutputInfoStruct from the OutputList attribute in which to change to.

6.5.7.2. RenameOutput Command

Upon receipt, this SHALL rename the output at a specific index in the Output List.

Updates to the output name SHALL appear in the device's settings menus. Name updates MAY automatically be sent to the actual device to which the output connects.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Index**	uint8	all			M
1	**Name**	string	max 32			M

6.6. Channel Cluster

This cluster provides an interface for controlling the current Channel on a device or endpoint.

This cluster server would be supported on Video Player devices or endpoints that allow Channel control such as a Content App. This cluster provides a list of available channels and provides commands for absolute and relative channel changes. Some of these commands and/or their responses MAY be large (see Large Message Quality under Data Model section in [MatterCore]), but they do not have the Large quality indicator (L) because they can also be transferred over MRP (see Message Reliability Protocol in [MatterCore]) in pages that fit within the MRP MTU limit. However, an implementation MAY leverage a transport like TCP that allows large payloads, if available, to minimize the number of messages required to transfer the corresponding payload.

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The cluster server for Channel is implemented by an endpoint that controls the current Channel.

6.6.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Add EG and RP features, Identifier and Type to Channel Info for Over-the-Top (OTT) channel support

6.6.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	CHANNEL

6.6.3. Cluster ID

ID	Name
0x0504	Channel

6.6.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**CL**	ChannelList	O	Provides list of available channels
1	**LI**	LineupInfo	O	Provides lineup info, which is a reference to an external source of lineup information
2	**EG**	ElectronicGuide	O	Provides electronic program guide information
3	**RP**	RecordProgram	O	Provides ability to record program

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6.6.5. Data Types

6.6.5.1. RecordingFlagBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**Scheduled**	The program is scheduled for recording.	M
1	**RecordSeries**	The program series is scheduled for recording.	M
2	**Recorded**	The program is recorded and available to be played.	M

6.6.5.2. LineupInfoTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**MSO**	Multi System Operator	M

6.6.5.3. StatusEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Success**	Command succeeded	M
1	**MultipleMatches**	More than one equal match for the ChannelInfoStruct passed in.	M
2	**NoMatches**	No matches for the ChannelInfoStruct passed in.	M

6.6.5.4. ChannelTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Satellite**	The channel is sourced from a satellite provider.	M

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Value	Name	Summary	Conformance
1	**Cable**	The channel is sourced from a cable provider.	M
2	**Terrestrial**	The channel is sourced from a terrestrial provider.	M
3	**OTT**	The channel is sourced from an OTT provider.	M

6.6.5.5. ChannelInfoStruct Type

This indicates a channel in a channel lineup.

While the major and minor numbers in the ChannelInfoStruct support use of ATSC channel format, a lineup MAY use other formats which can map into these numeric values.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**MajorNumber**	uint16	all				M
1	**MinorNumber**	uint16	all				M
2	**Name**	string			empty		O
3	**CallSign**	string			empty		O
4	**AffiliateCallSign**	string			empty		O
5	**Identifier**	string			empty		O
6	**Type**	ChannelTypeEnum			empty		O

6.6.5.5.1. MajorNumber Field

This field SHALL indicate the channel major number value (for example, using ATSC format). When the channel number is expressed as a string, such as "13.1" or "256", the major number would be 13 or 256, respectively. This field is required but SHALL be set to 0 for channels such as over-the-top channels that are not represented by a major or minor number.

6.6.5.5.2. MinorNumber Field

This field SHALL indicate the channel minor number value (for example, using ATSC format). When the channel number is expressed as a string, such as "13.1" or "256", the minor number would be 1 or 0, respectively. This field is required but SHALL be set to 0 for channels such as over-the-top channels that are not represented by a major or minor number.

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6.6.5.5.3. Name Field

This field SHALL indicate the marketing name for the channel, such as “The CW" or "Comedy Central". This field is optional, but SHOULD be provided when known.

6.6.5.5.4. CallSign Field

This field SHALL indicate the call sign of the channel, such as "PBS". This field is optional, but SHOULD be provided when known.

6.6.5.5.5. AffiliateCallSign Field

This field SHALL indicate the local affiliate call sign, such as "KCTS". This field is optional, but SHOULD be provided when known.

6.6.5.5.6. Identifier Field

This SHALL indicate the unique identifier for a specific channel. This field is optional, but SHOULD be provided when MajorNumber and MinorNumber are not available.

6.6.5.5.7. Type Field

This SHALL indicate the type or grouping of a specific channel. This field is optional, but SHOULD be provided when known.

6.6.5.6. LineupInfoStruct Type

The Lineup Info allows references to external lineup sources like Gracenote. The combination of OperatorName, LineupName, and PostalCode SHALL uniquely identify a lineup.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**OperatorName**	string					M
1	**LineupName**	string			empty		O
2	**PostalCode**	string			empty		O
3	**LineupInfoType**	LineupInfoType-Enum	desc				M

6.6.5.6.1. OperatorName Field

This field SHALL indicate the name of the operator, for example “Comcast”.

6.6.5.6.2. LineupName Field

This field SHALL indicate the name of the provider lineup, for example "Comcast King County". This field is optional, but SHOULD be provided when known.

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6.6.5.6.3. PostalCode Field

This field SHALL indicate the postal code (zip code) for the location of the device, such as "98052". This field is optional, but SHOULD be provided when known.

6.6.5.6.4. LineupInfoType Field

This field SHALL indicate the type of lineup. This field is optional, but SHOULD be provided when known.

6.6.5.7. ProgramStruct Type

This indicates a program within an electronic program guide (EPG).

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Identifier**	string	max 255				M
1	**Channel**	ChannelInfoStruct					M
2	**StartTime**	epoch-s					M
3	**EndTime**	epoch-s					M
4	**Title**	string	max 255				M
5	**Subtitle**	string	max 255		empty		O
6	**Description**	string	max 8192		empty		O
7	**AudioLanguages**	list[string]	max 10[max 50]		empty		O
8	**Ratings**	list[string]	max 255		empty		O
9	**ThumbnailUrl**	string	max 8192		empty		O
10	**PosterArtUrl**	string	max 8192		empty		O
11	**DvbiUrl**	string	max 8192		empty		O
12	**ReleaseDate**	string	max 30		empty		O
13	**ParentalGuidanceText**	string	max 255		empty		O
14	**RecordingFlag**	RecordingFlagBitmap					RP
15	**SeriesInfo**	SeriesInfoStruct		X	null		O

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
16	**CategoryList**	list[ProgramCategoryStruct]	max 255		empty		O
17	**CastList**	list[ProgramCastStruct]	max 255		empty		O
18	**ExternalIDList**	list[AdditionalInfoStruct]	max 255		empty		O

6.6.5.7.1. Identifier Field

This field SHALL indicate a unique identifier for a program within an electronic program guide list. The identifier SHALL be unique across multiple channels.

6.6.5.7.2. Channel Field

This field SHALL indicate the channel associated to the program.

6.6.5.7.3. StartTime Field

This field SHALL indicate an epoch time in seconds indicating the start time of a program, as a UTC time. This field can represent a past or future value.

6.6.5.7.4. EndTime Field

This field SHALL indicate an epoch time in seconds indicating the end time of a program, as a UTC time. This field can represent a past or future value but SHALL be greater than the StartTime.

6.6.5.7.5. Title Field

This field SHALL indicate the title or name for the specific program. For example, “MCIS: Los Angeles”.

6.6.5.7.6. Subtitle Field

This field SHALL indicate the subtitle for the specific program. For example, “Maybe Today" which is an episode name for “MCIS: Los Angeles”. This field is optional but SHALL be provided if applicable and known.

6.6.5.7.7. Description Field

This field SHALL indicate the brief description for the specific program. For example, a description of an episode. This field is optional but SHALL be provided if known.

6.6.5.7.8. AudioLanguages Field

This field SHALL indicate the audio language for the specific program. The value is a string contain-

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ing one of the standard Tags for Identifying Languages [RFC5646]. This field is optional but SHALL be provided if known.

6.6.5.7.9. Ratings Field

This field SHALL be used for indicating the level of parental guidance recommended for of a particular program. This can be any rating system used in the country or region where the program is broadcast. For example, in the United States “TV-PG” may contain material that parents can find not suitable for younger children but can be accepted in general for older children. This field is optional but SHALL be provided if known.

6.6.5.7.10. ThumbnailUrl Field

This field SHALL represent a URL of a thumbnail that clients can use to render an image for the program. The syntax of this field SHALL follow the syntax as specified in [RFC1738] and SHALL use the `https` scheme.

6.6.5.7.11. PosterArtUrl Field

This field SHALL represent a URL of a poster that clients can use to render an image for the program on the detail view. The syntax of this field SHALL follow the syntax as specified in [RFC1738] and SHALL use the `https` scheme.

6.6.5.7.12. DvbiUrl Field

This field SHALL represent the DVB-I URL associated to the program. The syntax of this field SHALL follow the syntax as specified in [RFC1738] and SHALL use the `https` scheme.

6.6.5.7.13. ReleaseDate Field

This field SHALL be a string, in ISO 8601 format, representing the date on which the program was released. This field is optional but when provided, the year SHALL be provided as part of the string.

6.6.5.7.14. ParentalGuidanceText Field

This field SHALL represent a string providing additional information on the parental guidance. This field is optional.

6.6.5.7.15. RecordingFlag Field

This field SHALL represent the recording status of the program. This field is required if the RecordProgram feature is set.

6.6.5.7.16. SeriesInfo Field

This field SHALL represent the information of a series such as season and episode number. This field is optional but SHOULD be provided if the program represents a series and this information is available.

6.6.5.7.17. CategoryList Field

This field SHALL represent the category of a particular program. This field is optional but SHALL be

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provided if known.

6.6.5.7.18. CastList Field

This field SHALL represent a list of the cast or the crew on the program. A single cast member may have more than one role. This field is optional but SHALL be provided if known.

6.6.5.7.19. ExternalIDList Field

This field SHALL indicate the list of additional external content identifiers.

6.6.5.8. ProgramCategoryStruct Type

This object defines the category associated to a program.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Category**	string	max 256				M
1	**SubCategory**	string	max 256		empty		O

6.6.5.8.1. Category Field

This field SHALL represent the category or genre of the program. Ex. News.

6.6.5.8.2. SubCategory Field

This field SHALL represent the sub-category or sub-genre of the program. Ex. Local.

6.6.5.9. SeriesInfoStruct Type

This object provides the episode information related to a program.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Season**	string	max 256				M
1	**Episode**	string	max 256				M

6.6.5.9.1. Season Field

This field SHALL represent the season of the series associated to the program.

6.6.5.9.2. Episode Field

This field SHALL represent the episode of the program.

6.6.5.10. ProgramCastStruct Type

This object provides the cast information related to a program.

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Name**	string	max 256				M
1	**Role**	string	max 256				M

6.6.5.10.1. Name Field

This field SHALL represent the name of the cast member.

6.6.5.10.2. Role Field

This field SHALL represent the role of the cast member. Ex. Actor, Director.

6.6.5.11. PageTokenStruct Type

This object defines the pagination structure.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Limit**	uint16	all		0		O
1	**After**	string	max 8192		empty		O
2	**Before**	string	max 8192		empty		O

6.6.5.11.1. Limit Field

This field SHALL indicate the maximum number of entries that should be retrieved from the program guide in a single response. It allows clients to specify the size of the paginated result set based on their needs.

6.6.5.11.2. After Field

This field SHALL indicate the cursor that pinpoints the start of the upcoming data page. In a Cursor-based pagination system, the field acts as a reference point, ensuring the set of results corresponds directly to the data following the specified cursor. In a Offset-based pagination system, the field, along with limit, indicate the offset from which entries in the program guide will be retrieved.

6.6.5.11.3. Before Field

This field SHALL indicate the cursor that pinpoints the end of the upcoming data page. In a Cursor-based pagination system, the field acts as a reference point, ensuring the set of results corresponds directly to the data preceding the specified cursor. In a Offset-based pagination system, the field, along with limit, indicate the offset from which entries in the program guide will be retrieved.

6.6.5.12. ChannelPagingStruct Type

This object defines the paging structure that includes the previous and next pagination tokens.

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**PreviousToken**	PageTokenStruct		X	null		O
1	**NextToken**	PageTokenStruct		X	null		O

6.6.5.12.1. PreviousToken Field

This field SHALL indicate the token to retrieve the preceding page. Absence of this field denotes the response as the initial page.

6.6.5.12.2. NextToken Field

This field SHALL indicate the token to retrieve the next page. Absence of this field denotes the response as the last page.

6.6.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**ChannelList**	list[ChannelInfoStruct]			empty	R V	CL
0x0001	**Lineup**	LineupInfoStruct	desc	X	null	R V	LI
0x0002	**CurrentChannel**	ChannelInfoStruct	desc	X	null	R V	O

6.6.6.1. ChannelList Attribute

This attribute SHALL provide the list of supported channels.

6.6.6.2. Lineup Attribute

This attribute SHALL identify the channel lineup using external data sources.

6.6.6.3. CurrentChannel Attribute

This attribute SHALL contain the current channel. When supported but a channel is not currently tuned to (if a content application is in foreground), the value of the field SHALL be null.

6.6.7. Commands

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ID	Name	Direction	Response	Access	Quality	Conformance
0x00	**ChangeChannel**	client ⇒ server	ChangeChannelResponse	O		CL | LI
0x01	**ChangeChannelResponse**	client ⇐ server	N			CL | LI
0x02	**ChangeChannelByNumber**	client ⇒ server	Y	O		M
0x03	**SkipChannel**	client ⇒ server	Y	O		M
0x04	**GetProgramGuide**	client ⇒ server	ProgramGuideResponse	O		EG
0x05	**ProgramGuideResponse**	client ⇐ server	N			EG
0x06	**RecordProgram**	client ⇒ server	Y	O		RP & EG
0x07	**CancelRecordProgram**	client ⇒ server	Y	O		RP & EG

6.6.7.1. ChangeChannel Command

Change the channel to the channel case-insensitive exact matching the value passed as an argument.

The match priority order SHALL be: Identifier, AffiliateCallSign, CallSign, Name, Number. In the match string, the Channel number should be presented in the `Major.Minor` format, such as `13.1`.

Upon receipt, this SHALL generate a ChangeChannelResponse command.

Upon success, the CurrentChannel attribute, if supported, SHALL be updated to reflect the change.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Match**	string				M

6.6.7.1.1. Match Field

This field SHALL contain a user-input string to match in order to identify the target channel.

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6.6.7.2. ChangeChannelResponse Command

This command SHALL be generated in response to a ChangeChannel command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	StatusEnum	desc			M
1	**Data**	string	any			O

6.6.7.2.1. Status Field

This field SHALL indicate the status of the command which resulted in this response.

6.6.7.2.2. Data Field

This field SHALL indicate Optional app-specific data.

6.6.7.3. ChangeChannelByNumber Command

Change the channel to the channel with the given Number in the *ChannelList* attribute.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**MajorNumber**	uint16	all			M
1	**MinorNumber**	uint16	all			M

6.6.7.3.1. MajorNumber Field

This field SHALL indicate the channel major number value (ATSC format) to which the channel should change.

6.6.7.3.2. MinorNumber Field

This field SHALL indicate the channel minor number value (ATSC format) to which the channel should change.

6.6.7.4. SkipChannel Command

This command provides channel up and channel down functionality, but allows channel index jumps of size *Count*.

Jumps are relative to the available list of channels. For example, when the current channel is 100.0 and the list of available channels is [100.0, 200.0, 201.0, 305.1], a SkipChannel command with jump value of 2 SHALL change the channel to 201.0.

When the value of the increase or decrease is larger than the number of channels remaining in the given direction, then the behavior SHALL be to return to the beginning (or end) of the channel list

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and continue. For example, if the current channel is at index 0 and count value of -1 is given, then the current channel should change to the last channel.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Count**	int16	all			M

6.6.7.4.1. Count Field

This field SHALL indicate the number of steps to increase (Count is positive) or decrease (Count is negative) the current channel.

6.6.7.5. GetProgramGuide Command

This command retrieves the program guide. It accepts several filter parameters to return specific schedule and program information from a content app. The command shall receive in response a ProgramGuideResponse. Standard error codes SHALL be used when arguments provided are not valid. For example, if StartTime is greater than EndTime, the status code INVALID_ACTION SHALL be returned.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**StartTime**	epoch-s				M
1	**EndTime**	epoch-s				M
2	**ChannelList**	list[ChannelInfoStruct]	max 255		empty	O
3	**PageToken**	PageTokenStruct		X	null	O
5	**RecordingFlag**	RecordingFlagBitmap		X	null	O
6	**ExternalIDList**	list[AdditionalInfoStruct]	max 255		empty	O
7	**Data**	octstr	max 8092		MS	O

6.6.7.5.1. StartTime Field

This field SHALL indicate the beginning of the time window for which program guide entries are to be retrieved, as a UTC time. Entries with a start time on or after this value will be included in the results.

6.6.7.5.2. EndTime Field

This field SHALL indicate the end of the time window for which program guide entries are to be retrieved, as a UTC time. Entries with an end time on or before this value will be included in the

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results. This field can represent a past or future value but SHALL be greater than the StartTime.

6.6.7.5.3. ChannelList Field

This field SHALL indicate the set of channels for which program guide entries should be fetched. By providing a list of channels in this field, the response will only include entries corresponding to the specified channels.

6.6.7.5.4. PageToken Field

This field SHALL indicate the pagination token used for managing pagination progression.

6.6.7.5.5. RecordingFlag Field

This field SHALL indicate the flags of the programs for which entries should be fetched.

6.6.7.5.6. ExternalIDList Field

This field SHALL indicate the list of additional external content identifiers.

6.6.7.5.7. Data Field

This field SHALL indicate Optional app-specific data.

6.6.7.6. ProgramGuideResponse Command

This command is a response to the GetProgramGuide command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Paging**	ChannelPagingStruct				M
1	**ProgramList**	list[ProgramStruct]			empty	M

6.6.7.6.1. Paging Field

This field SHALL indicate the necessary pagination attributes that define information for both the succeeding and preceding data pages.

6.6.7.6.2. ProgramList Field

This field SHALL indicate the list of programs.

6.6.7.7. RecordProgram Command

Record a specific program or series when it goes live. This functionality enables DVR recording features.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ProgramIdentifier**	string	max 255			M
1	**ShouldRecordSeries**	bool				M
2	**ExternalIDList**	list[AdditionalInfoStruct]	max 255		empty	O
3	**Data**	octstr	max 8092		MS	O

6.6.7.7.1. ProgramIdentifier Field

This field SHALL indicate the program identifier for the program that should be recorded. This value is provided by the identifier field in ProgramStruct.

6.6.7.7.2. ShouldRecordSeries Field

This field SHALL indicate whether the whole series associated to the program should be recorded. For example, invoking record program on an episode with that flag set to true, the target should schedule record the whole series.

6.6.7.7.3. ExternalIDList Field

This field, if present, SHALL indicate the list of additional external content identifiers.

6.6.7.7.4. Data Field

This field, if present, SHALL indicate app-specific data.

6.6.7.8. CancelRecordProgram Command

Cancel recording for a specific program or series.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ProgramIdentifier**	string	max 255			M
1	**ShouldRecordSeries**	bool				M
2	**ExternalIDList**	list[AdditionalInfoStruct]	max 255		empty	O
3	**Data**	octstr	max 8092		MS	O

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6.6.7.8.1. ProgramIdentifier Field

This field SHALL indicate the program identifier for the program that should be cancelled from recording. This value is provided by the identifier field in ProgramStruct.

6.6.7.8.2. ShouldRecordSeries Field

This field SHALL indicate whether the whole series associated to the program should be cancelled from recording. For example, invoking record program on an episode with that flag set to true, the target should schedule record the whole series.

6.6.7.8.3. ExternalIDList Field

This field, if present, SHALL indicate the list of additional external content identifiers.

6.6.7.8.4. Data Field

This field, if present, SHALL indicate app-specific data.

6.7. Content Launcher Cluster

This cluster provides an interface for launching content on a Video Player device such as a Streaming Media Player, Smart TV or Smart Screen.

This cluster would be supported on a Video Player device or devices that can playback content, such as a Streaming Media Player, Smart TV or Smart Screen. This cluster supports playing back content referenced by URL. It supports finding content by type and global identifier, and either playing the content or displaying the search results.

The cluster server for Content Launcher is implemented by an endpoint that can launch content, such as a Video Player, or an endpoint representing a Content App on such a device.

When this cluster is implemented for an Content App Endpoint (Endpoint with type “Content App” and having an Application Basic cluster), the Video Player device SHALL launch the application when a client invokes the LaunchContent or LaunchURL commands.

6.7.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Add Seasons, Episode, Any and CurrentContext to search, Added Text/Audio tracks support with PlaybackPreferences field

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6.7.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	CONTENTLAUNCHER

6.7.3. Cluster ID

ID	Name
0x050A	Content Launcher

6.7.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**CS**	ContentSearch	O	Device supports content search (non-app specific)
1	**UP**	URLPlayback	O	Device supports basic URL-based file playback
2	**AS**	AdvancedSeek	O	Enables clients to implement more advanced media seeking behavior in their user interface, such as for example a "seek bar"
3	**TT**	TextTracks	O	Device or app supports Text Tracks
4	**AT**	AudioTracks	O	Device or app supports Audio Tracks

6.7.5. Data Types

6.7.5.1. SupportedProtocolsBitmap Type

This data type is derived from map32.

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Bit	Name	Summary	Conformance
0	**DASH**	Device supports Dynamic Adaptive Streaming over HTTP (DASH)	M
1	**HLS**	Device supports HTTP Live Streaming (HLS)	M

6.7.5.2. StatusEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Success**	Command succeeded	M
1	**URLNotAvailable**	Requested URL could not be reached by device	M
2	**AuthFailed**	Requested URL returned 401 error code.	M
3	**TextTrackNotAvailable**	Requested Text Track (in PlaybackPreferences) not available	TT
4	**AudioTrackNotAvailable**	Requested Audio Track (in PlaybackPreferences) not available	AT

6.7.5.3. ParameterEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Actor**	Actor represents an actor credited in video media content; for example, “Gaby Hoffman”	M
1	**Channel**	Channel represents the identifying data for a television channel; for example, "PBS"	M

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Value	Name	Summary	Conformance
2	**Character**	A character represented in video media content; for example, “Snow White”	M
3	**Director**	A director of the video media content; for example, “Spike Lee”	M
4	**Event**	An event is a reference to a type of event; examples would include sports, music, or other types of events; for example, searching for "Football games" would search for a 'game' event entity and a 'football' sport entity	M
5	**Franchise**	A franchise is a video entity which can represent a number of video entities, like movies or TV shows; for example, take the fictional franchise "Intergalactic Wars" which represents a collection of movie trilogies, as well as animated and live action TV shows	M
6	**Genre**	Genre represents the genre of video media content such as action, drama or comedy	M
7	**League**	League represents the categorical information for a sporting league; for example, "NCAA"	M
8	**Popularity**	Popularity indicates whether the user asks for popular content	M

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Value	Name	Summary	Conformance
9	**Provider**	The provider (MSP) the user wants this media to be played on; for example, "Netflix"	M
10	**Sport**	Sport represents the categorical information of a sport; for example, football	M
11	**SportsTeam**	SportsTeam represents the categorical information of a professional sports team; for example, "University of Washington Huskies"	M
12	**Type**	The type of content requested. Supported types are "Movie", "MovieSeries", "TVSeries", "TVSeason", "TVEpisode", "Trailer", "SportsEvent", "LiveEvent", and "Video"	M
13	**Video**	Video represents the identifying data for a specific piece of video content; for example, "Manchester by the Sea"	M
14	**Season**	Season represents the specific season number within a TV series	O
15	**Episode**	Episode represents a specific episode number within a Season in a TV series	O
16	**Any**	Represents a search text input across many parameter types or even outside of the defined param types	O

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6.7.5.3.1. Franchise Value

This value SHALL indicate a franchise, this entity type was introduced to account for requests by customers such as "Find Intergalactic Wars movies", which would search for all 'Intergalactic Wars' programs of the MOVIE MediaType, rather than attempting to match to a single title.

6.7.5.4. MetricTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Pixels**	Dimensions defined in a number of Pixels	M
1	**Percentage**	Dimensions defined as a percentage	M

6.7.5.4.1. Pixels Value

This value is used for dimensions defined in a number of Pixels.

6.7.5.4.2. Percentage Value

This value is for dimensions defined as a percentage of the overall display dimensions. For example, if using a Percentage Metric type for a Width measurement of 50.0, against a display width of 1920 pixels, then the resulting value used would be 960 pixels (50.0% of 1920) for that dimension. Whenever a measurement uses this Metric type, the resulting values SHALL be rounded ("floored") towards 0 if the measurement requires an integer final value.

6.7.5.5. AdditionalInfoStruct Type

This object defines additional name=value pairs that can be used for identifying content.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Name**	string	max 256				M
1	**Value**	string	max 8192				M

6.7.5.5.1. Name Field

This field SHALL indicate the name of external id, ex. "musicbrainz".

6.7.5.5.2. Value Field

This field SHALL indicate the value for external id, ex. "ST0000000666661".

6.7.5.6. ParameterStruct Type

This object defines inputs to a search for content for display or playback.

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Type**	ParameterEnum	all				M
1	**Value**	string	max 1024				M
2	**ExternalIDList**	list[AdditionalInfoStruct]	all		empty		O

6.7.5.6.1. Type Field

This field SHALL indicate the entity type.

6.7.5.6.2. Value Field

This field SHALL indicate the entity value, which is a search string, ex. “Manchester by the Sea”.

6.7.5.6.3. ExternalIDList Field

This field SHALL indicate the list of additional external content identifiers.

6.7.5.7. ContentSearchStruct Type

This object defines inputs to a search for content for display or playback.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**ParameterList**	list[ParameterStruct]	all		0		M

6.7.5.7.1. ParameterList Field

This field SHALL indicate the list of parameters comprising the search. If multiple parameters are provided, the search parameters SHALL be joined with 'AND' logic. e.g. action movies with Tom Cruise will be represented as [{Actor: 'Tom Cruise'}, {Type: 'Movie'}, {Genre: 'Action'}]

6.7.5.8. DimensionStruct Type

This object defines dimension which can be used for defining Size of background images.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Width**	double			MS		M
1	**Height**	double			MS		M
2	**Metric**	MetricTypeEnum					M

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6.7.5.8.1. Width Field

This field SHALL indicate the width using the metric defined in Metric

6.7.5.8.2. Height Field

This field SHALL indicate the height using the metric defined in Metric

6.7.5.8.3. Metric Field

This field SHALL indicate metric used for defining Height/Width.

6.7.5.9. StyleInformationStruct Type

This object defines style information which can be used by content providers to change the Media Player's style related properties.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**ImageURL**	string	max 8192		MS		O
1	**Color**	string	7, 9		MS		O
2	**Size**	DimensionStruct			MS		O

6.7.5.9.1. ImageURL Field

This field SHALL indicate the URL of image used for Styling different Video Player sections like Logo, Watermark etc. The syntax of this field SHALL follow the syntax as specified in [RFC1738] and SHALL use the `https` scheme.

6.7.5.9.2. Color Field

This field SHALL indicate the color, in RGB or RGBA, used for styling different Video Player sections like Logo, Watermark, etc. The value SHALL conform to the 6-digit or 8-digit format defined for CSS sRGB hexadecimal color notation [https://www.w3.org/TR/css-color-4/#hex-notation]. Examples:

- #76DE19 for R=0x76, G=0xDE, B=0x19, A absent
- #76DE1980 for R=0x76, G=0xDE, B=0x19, A=0x80

6.7.5.9.3. Size Field

This field SHALL indicate the size of the image used for Styling different Video Player sections like Logo, Watermark etc.

6.7.5.10. BrandingInformationStruct Type

This object defines Branding Information which can be provided by the client in order to customize the skin of the Video Player during playback.

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**ProviderName**	string	max 256				M
1	**Background**	StyleInformationStruct			MS		O
2	**Logo**	StyleInformationStruct			MS		O
3	**ProgressBar**	StyleInformationStruct			MS		O
4	**Splash**	StyleInformationStruct			MS		O
5	**WaterMark**	StyleInformationStruct			MS		O

6.7.5.10.1. ProviderName Field

This field SHALL indicate name of the provider for the given content.

6.7.5.10.2. Background Field

This field SHALL indicate background of the Video Player while content launch request is being processed by it. This background information MAY also be used by the Video Player when it is in idle state.

6.7.5.10.3. Logo Field

This field SHALL indicate the logo shown when the Video Player is launching. This is also used when the Video Player is in the idle state and Splash field is not available.

6.7.5.10.4. ProgressBar Field

This field SHALL indicate the style of progress bar for media playback.

6.7.5.10.5. Splash Field

This field SHALL indicate the screen shown when the Video Player is in an idle state. If this property is not populated, the Video Player SHALL default to logo or the provider name.

6.7.5.10.6. Watermark Field

This field SHALL indicate watermark shown when the media is playing.

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6.7.5.11. PlaybackPreferencesStruct Type

PlaybackPreferencesStruct defines the preferences sent by the client to the receiver in the Content-Launcher LaunchURL or LaunchContent commands.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**PlaybackPosition**	uint64		X			AS
1	**TextTrack**	TrackPreferenceStruct		X			TT
2	**AudioTracks**	list[TrackPreferenceStruct]		X			AT

6.7.5.11.1. PlaybackPosition Field

This field SHALL indicate the preferred position (in milliseconds) in the media to launch playback from. In case the position falls in the middle of a frame, the server SHALL set the position to the beginning of that frame and set the `SampledPosition` attribute on the MediaPlayback cluster accordingly. A value of null SHALL indicate that playback position is not applicable for the current state of the media playback. (For example : Live media with no known duration and where seek is not supported).

6.7.5.11.2. TextTrack Field

This field SHALL indicate the user's preferred Text Track. A value of null SHALL indicate that the user did not specify a preferred Text Track on the client. In such a case, the decision to display and select a Text Track is up to the server.

6.7.5.11.3. AudioTracks Field

This field SHALL indicate the list of the user's preferred Audio Tracks. If the list contains multiple values, each AudioTrack must also specify a unique audioOutputIndex to play the track on. A value of null SHALL indicate that the user did not specify a preferred Audio Track on the client. In such a case, the decision to play and select an Audio Track is up to the server.

6.7.5.12. TrackPreferenceStruct Type

This structure defines Text/Audio Track preferences.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**LanguageCode**	string	max 32				M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
1	**Characteristics**	list[CharacteristicEnum]		X	null		O
2	**AudioOutputIndex**	uint8		X			AT

6.7.5.12.1. LanguageCode Field

This field SHALL contain one of the standard Tags for Identifying Languages [RFC5646], which identifies the primary language used in the Track.

6.7.5.12.2. Characteristics Field

This field SHALL contain a list of enumerated CharacteristicEnum values that indicate a purpose, trait or feature associated with the Track. A value of null SHALL indicate that there are no Characteristics corresponding to the Track.

6.7.5.12.3. AudioOutputIndex Field

This field if present SHALL indicate the index of the OutputInfoStruct from the OutputList attribute (from the AudioOutput cluster) and indicates which audio output the Audio Track should be played on.

This field SHALL NOT be present if the track is not an audio track.

If the track is an audio track, this field SHALL be present. A value of null SHALL indicate that the server can choose the audio output(s) to play the Audio Track on.

6.7.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**AcceptHeader**	list[string]	max 100[max 1024]	N	empty	R V	UP
0x0001	**SupportedStreamingProtocols**	SupportedProtocolsBitmap		N	0	R V	UP

6.7.6.1. AcceptHeader Attribute

This attribute SHALL provide a list of content types supported by the Video Player or Content App in the form of entries in the HTTP "Accept" request header.

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6.7.6.2. SupportedStreamingProtocols Attribute

This attribute SHALL provide information about supported streaming protocols.

6.7.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**LaunchContent**	client ⇒ server	LauncherResponse	O	CS
0x01	**LaunchURL**	client ⇒ server	LauncherResponse	O	UP
0x02	**LauncherResponse**	client ⇐ server	N		CS | UP

6.7.7.1. LaunchContent Command

Upon receipt, this SHALL launch the specified content with optional search criteria.

This command returns a Launch Response.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Search**	ContentSearchStruct	desc			M
1	**AutoPlay**	bool	desc			M
2	**Data**	string			MS	O
3	**PlaybackPreferences**	PlaybackPreferencesStruct	all		MS	O
4	**UseCurrentContext**	bool	desc		TRUE	O

6.7.7.1.1. Search Field

This field SHALL indicate the content to launch.

6.7.7.1.2. AutoPlay Field

This field SHALL indicate whether to automatically start playing content, where:

- TRUE means best match should start playing automatically.
- FALSE means matches should be displayed on screen for user selection.

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6.7.7.1.3. Data Field

This field, if present, SHALL indicate app-specific data.

6.7.7.1.4. PlaybackPreferences Field

This field, if present, SHALL indicate the user's preferred Text/AudioTracks and playbackPosition for the media, sent from the client to the server. If the server does not find an available track for the title being played exactly matching a Track requested here, in the list of available tracks, it may default to picking another track that closely matches the requested track. Alternately, it may go with user preferences set on the server side (it will use this option if these PlaybackPreferences are not specified). In the case of text tracks, that may mean that the subtitle text is not displayed at all. In the cases where the preferred Text/AudioTracks are not available, the server SHALL return the TextTrackNotAvailable and/or AudioTrackNotAvailable Status(es) in the LauncherResponse.

6.7.7.1.5. UseCurrentContext Field

This field, if present, SHALL indicate whether to consider the context of current ongoing activity on the receiver to fulfill the request. For example if the request only includes data in ContentSearch that specifies an Episode number, and UseCurrentContent is set to TRUE, if there is a TV series on going, the request refers to the specific episode of the ongoing season of the TV series. TRUE means current activity context MAY be considered FALSE means current activity context SHALL NOT be considered

6.7.7.2. LaunchURL Command

Upon receipt, this SHALL launch content from the specified URL.

The content types supported include those identified in the AcceptHeader and SupportedStreamingProtocols attributes.

A check SHALL be made to ensure the URL is secure (uses HTTPS).

When playing a video stream in response to this command, an indication (ex. visual) of the identity of the origin node of the video stream SHALL be provided. This could be in the form of a friendly name label which uniquely identifies the node to the user. This friendly name label is typically assigned by the Matter Admin (ex. TV) at the time of commissioning and, when it's a device, is often editable by the user. It might be a combination of a company name and friendly name, for example, "Acme" or "Acme Streaming Service on Alice's Phone".

This command returns a Launch Response.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ContentURL**	string	max 8192			M
1	**DisplayString**	string	max 256		MS	O

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
2	**BrandingInformation**	BrandingInformationStruct	any		MS	O
3	**PlaybackPreferences**	PlaybackPreferencesStruct	any		MS	O

6.7.7.2.1. ContentURL Field

This field SHALL indicate the URL of content to launch. The syntax of this field SHALL follow the syntax as specified in [RFC1738] and SHALL use the `https` scheme.

6.7.7.2.2. DisplayString Field

This field, if present, SHALL provide a string that MAY be used to describe the content being accessed at the given URL.

6.7.7.2.3. BrandingInformation Field

This field, if present, SHALL indicate the branding information that MAY be displayed when playing back the given content.

6.7.7.2.4. PlaybackPreferences Field

This field, if present, SHALL indicate the user's preferred Text/AudioTracks and playbackPosition for the media, sent from the client to the server. If the server does not find an available track for the title being played exactly matching a Track requested here, in the list of available tracks, it may default to picking another track that closely matches the requested track. Alternately, it may go with user preferences set on the server side (it will use this option if these PlaybackPreferences are not specified). In the case of text tracks, that may mean that the subtitle text is not displayed at all. In the cases where the preferred Text/AudioTracks are not available, the server SHALL return the TextTrackNotAvailable and/or AudioTrackNotAvailable Status(es) in the LauncherResponse.

6.7.7.3. LauncherResponse Command

This command SHALL be generated in response to LaunchContent and LaunchURL commands.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	StatusEnum	all			M
1	**Data**	string			MS	O

6.7.7.3.1. Status Field

This field SHALL indicate the status of the command which resulted in this response.

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6.7.7.3.2. Data Field

This field SHALL indicate Optional app-specific data.

6.7.8. Events

6.8. Keypad Input Cluster

This cluster provides an interface for key code based input and control on a device like a Video Player or an endpoint like a Content App. This may include text or action commands such as UP, DOWN, and SELECT.

This cluster would be supported on Video Player devices as well as devices that support remote control input from a keypad or remote. This cluster provides the list of supported keypad inputs and provides a command for sending them.

The cluster server for Keypad Input is implemented by a device that can receive keypad input, such as a Video Player, or an endpoint that can receive keypad input, such as a Content App.

The key codes used are those defined in the HDMI CEC specification (see HDMI).

Devices MAY understand a subset of these key codes. Feature flags are used to indicate a specific subset that is supported. Device MAY support additional codes beyond what is indicated in feature flags.

6.8.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

6.8.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	KEYPADINPUT

6.8.3. Cluster ID

ID	Name
0x0509	Keypad Input

6.8.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

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Bit	Code	Feature	Conformance	Summary
0	**NV**	NavigationKeyCodes	O	Supports UP, DOWN, LEFT, RIGHT, SELECT, BACK, EXIT, MENU
1	**LK**	LocationKeys	O	Supports CEC keys 0x0A (Settings) and 0x09 (Home)
2	**NK**	NumberKeys	O	Supports numeric input 0..9

6.8.5. Data Types

6.8.5.1. StatusEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Success**	Succeeded	M
1	**UnsupportedKey**	Key code is not supported.	M
2	**InvalidKeyInCurrentState**	Requested key code is invalid in the context of the responder's current state.	M

6.8.5.2. CecKeyCodeEnum Type

This data type is derived from enum8.

Value	Name	Conformance
0x00	**Select**	M
0x01	**Up**	M
0x02	**Down**	M
0x03	**Left**	M
0x04	**Right**	M
0x05	**RightUp**	M
0x06	**RightDown**	M
0x07	**LeftUp**	M
0x08	**LeftDown**	M
0x09	**RootMenu**	M

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Value	Name	Conformance
0x0A	**SetupMenu**	M
0x0B	**ContentsMenu**	M
0x0C	**FavoriteMenu**	M
0x0D	**Exit**	M
0x10	**MediaTopMenu**	M
0x11	**MediaContextSensitiveMenu**	M
0x1D	**NumberEntryMode**	M
0x1E	**Number11**	M
0x1F	**Number12**	M
0x20	**Number0OrNumber10**	M
0x21	**Numbers1**	M
0x22	**Numbers2**	M
0x23	**Numbers3**	M
0x24	**Numbers4**	M
0x25	**Numbers5**	M
0x26	**Numbers6**	M
0x27	**Numbers7**	M
0x28	**Numbers8**	M
0x29	**Numbers9**	M
0x2A	**Dot**	M
0x2B	**Enter**	M
0x2C	**Clear**	M
0x2F	**NextFavorite**	M
0x30	**ChannelUp**	M
0x31	**ChannelDown**	M
0x32	**PreviousChannel**	M
0x33	**SoundSelect**	M
0x34	**InputSelect**	M
0x35	**DisplayInformation**	M
0x36	**Help**	M
0x37	**PageUp**	M
0x38	**PageDown**	M
0x40	**Power**	M

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Value	Name	Conformance
0x41	**VolumeUp**	M
0x42	**VolumeDown**	M
0x43	**Mute**	M
0x44	**Play**	M
0x45	**Stop**	M
0x46	**Pause**	M
0x47	**Record**	M
0x48	**Rewind**	M
0x49	**FastForward**	M
0x4A	**Eject**	M
0x4B	**Forward**	M
0x4C	**Backward**	M
0x4D	**StopRecord**	M
0x4E	**PauseRecord**	M
0x4F	**Reserved**	M
0x50	**Angle**	M
0x51	**SubPicture**	M
0x52	**VideoOnDemand**	M
0x53	**ElectronicProgramGuide**	M
0x54	**TimerProgramming**	M
0x55	**InitialConfiguration**	M
0x56	**SelectBroadcastType**	M
0x57	**SelectSoundPresentation**	M
0x60	**PlayFunction**	M
0x61	**PausePlayFunction**	M
0x62	**RecordFunction**	M
0x63	**PauseRecordFunction**	M
0x64	**StopFunction**	M
0x65	**MuteFunction**	M
0x66	**RestoreVolumeFunction**	M
0x67	**TuneFunction**	M
0x68	**SelectMediaFunction**	M
0x69	**SelectAvInputFunction**	M

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Value	Name	Conformance
0x6A	**SelectAudioInputFunction**	M
0x6B	**PowerToggleFunction**	M
0x6C	**PowerOffFunction**	M
0x6D	**PowerOnFunction**	M
0x71	**F1Blue**	M
0x72	**F2Red**	M
0x73	**F3Green**	M
0x74	**F4Yellow**	M
0x75	**F5**	M
0x76	**Data**	M

6.8.6. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**SendKey**	client ⇒ server	SendKeyResponse	O	M
0x01	**SendKeyResponse**	client ⇐ server	N		M

6.8.6.1. SendKey Command

Upon receipt, this SHALL process a keycode as input to the media endpoint.

If a device has multiple media endpoints implementing this cluster, such as a casting video player endpoint with one or more content app endpoints, then only the endpoint receiving the command SHALL process the keycode as input. In other words, a specific content app endpoint SHALL NOT process a keycode received by a different content app endpoint.

If a second SendKey request with the same KeyCode value is received within 200 ms, then the endpoint will consider the first key press to be a press and hold. When such a repeat KeyCode value is not received within 200 ms, then the endpoint will consider the last key press to be a release.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**KeyCode**	CecKeyCodeEnum	all			M

6.8.6.1.1. KeyCode Field

This field SHALL indicate the key code to process.

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6.8.6.2. SendKeyResponse Command

This command SHALL be generated in response to a SendKey command. The data for this command SHALL be as follows:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	StatusEnum	all			M

6.8.6.2.1. Status Field

This field SHALL indicate the status of the request.

6.9. Media Input Cluster

This cluster provides an interface for controlling the Input Selector on a media device such as a Video Player.

This cluster would be implemented on TV and other media streaming devices, as well as devices that provide input to or output from such devices.

This cluster provides the list of available inputs and provides commands for selecting and renaming them.

The cluster server for Media Input is implemented by a device that has selectable input, such as a Video Player device.

6.9.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

6.9.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	MEDIAINPUT

6.9.3. Cluster ID

ID	Name
0x0507	Media Input

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6.9.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**NU**	NameUpdates	O	Supports updates to the input names

6.9.5. Data Types

6.9.5.1. InputTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Internal**	Indicates content not coming from a physical input.	M
1	**Aux**		M
2	**Coax**		M
3	**Composite**		M
4	**HDMI**		M
5	**Input**		M
6	**Line**		M
7	**Optical**		M
8	**Video**		M
9	**SCART**		M
10	**USB**		M
11	**Other**		M

6.9.5.2. InputInfoStruct Type

This contains information about an input.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Index**	uint8	all				M
1	**InputType**	InputTypeEnum	desc				M
2	**Name**	string					M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
3	**Description**	string					M

6.9.5.2.1. Index Field

This field SHALL indicate the unique index into the list of Inputs.

6.9.5.2.2. InputType Field

This field SHALL indicate the type of input

6.9.5.2.3. Name Field

This field SHALL indicate the input name, such as “HDMI 1”. This field may be blank, but SHOULD be provided when known.

6.9.5.2.4. Description Field

This field SHALL indicate the user editable input description, such as “Living room Playstation”. This field may be blank, but SHOULD be provided when known.

6.9.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**InputList**	list[InputInfoStruct]				R V	M
0x0001	**CurrentInput**	uint8	all			R V	M

6.9.6.1. InputList Attribute

This attribute SHALL provide a list of the media inputs supported by the device.

6.9.6.2. CurrentInput Attribute

This attribute SHALL contain the value of the index field of the currently selected InputInfoStruct.

6.9.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**SelectInput**	client ⇒ server	Y	O	M
0x01	**ShowInputStatus**	client ⇒ server	Y	O	M

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ID	Name	Direction	Response	Access	Conformance
0x02	**HideInputStatus**	client ⇒ server	Y	O	M
0x03	**RenameInput**	client ⇒ server	Y	M	NU

6.9.7.1. SelectInput Command

Upon receipt, this command SHALL change the media input on the device to the input at a specific index in the Input List.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Index**	uint8	all			M

6.9.7.1.1. Index Field

This field SHALL indicate the index field of the InputInfoStruct from the InputList attribute in which to change to.

6.9.7.2. ShowInputStatus Command

Upon receipt, this command SHALL display the active status of the input list on screen.

6.9.7.3. HideInputStatus Command

Upon receipt, this command SHALL hide the input list from the screen.

6.9.7.4. RenameInput Command

Upon receipt, this command SHALL rename the input at a specific index in the Input List.

Updates to the input name SHALL appear in the device's settings menus.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Index**	uint8	all			M
1	**Name**	string	all			M

6.10. Media Playback Cluster

This cluster provides an interface for controlling Media Playback (PLAY, PAUSE, etc) on a media device such as a TV, Set-top Box, or Smart Speaker.

This cluster server would be supported on Video Player devices or endpoints that provide media playback, such as a Content App. This cluster provides an interface for controlling Media Playback.

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6.10.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Added Text/Audio tracks support, Audio-while-advancing (AA) feature, StateChanged event

6.10.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	MEDIAPLAYBACK

6.10.3. Cluster ID

ID	Name
0x0506	Media Playback

6.10.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**AS**	AdvancedSeek	O	Advanced media seeking
1	**VS**	VariableSpeed	O	Variable speed playback
2	**TT**	TextTracks	O	Text Tracks
3	**AT**	AudioTracks	O	Audio Tracks
4	**AA**	AudioAdvance	O	Can play audio during fast and slow playback speeds

6.10.4.1. AdvancedSeek Feature

This feature provides access to the time offset location within current playback media and allows for jumping to a specific location using time offsets. This enables clients to implement more advanced media seeking behavior in their user interface, for instance a "seek bar".

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6.10.4.2. VariableSpeed Feature

This feature is for a device which supports variable speed playback on media that supports it.

6.10.4.3. TextTracks Feature

This feature is for a device or app that supports Text Tracks.

6.10.4.4. AudioTracks Feature

This feature is for a device or app that supports Audio Tracks.

6.10.4.5. AudioAdvance Feature

This feature is for a device or app that supports playing audio during fast and slow advance and rewind (e.g., while playback speed is not 1). A device that supports this feature MAY only support playing audio during certain speeds.

A cluster implementing AA SHALL implement AS.

6.10.5. Data Types

6.10.5.1. PlaybackStateEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Playing**	Media is currently playing (includes FF and REW)	M
1	**Paused**	Media is currently paused	M
2	**NotPlaying**	Media is not currently playing	M
3	**Buffering**	Media is not currently buffering and playback will start when buffer has been filled	M

6.10.5.2. StatusEnum Type

Status Data Type is derived from enum8.

Value	Name	Summary	Conformance
0	**Success**	Succeeded	M

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Value	Name	Summary	Conformance
1	**InvalidStateForCommand**	Requested playback command is invalid in the current playback state.	M
2	**NotAllowed**	Requested playback command is not allowed in the current playback state For example, attempting to fast-forward during a commercial might return NotAllowed.	M
3	**NotActive**	This endpoint is not active for playback.	M
4	**SpeedOutOfRange**	The FastForward or Rewind Command was issued but the media is already playing back at the fastest speed supported by the server in the respective direction.	VS
5	**SeekOutOfRange**	The Seek Command was issued with a value of position outside of the allowed seek range of the media.	AS

6.10.5.3. CharacteristicEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**ForcedSubtitles**	Textual information meant for display when no other text representation is selected. It is used to clarify dialogue, alternate languages, texted graphics or location/person IDs that are not otherwise covered in the dubbed/localized audio	M

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Value	Name	Summary	Conformance
1	**DescribesVideo**	Textual or audio media component containing a textual description (intended for audio synthesis) or an audio description describing a visual component	M
2	**EasyToRead**	Simplified or reduced captions as specified in [United States Code Title 47 CFR 79.103(c)(9)]	M
3	**FrameBased**	A media characteristic that indicates that a track selection option includes frame-based content	M
4	**MainProgram**	Main media component(s) which is/are intended for presentation if no other information is provided	M
5	**OriginalContent**	A media characteristic that indicates that a track or media selection option contains original content	M
6	**VoiceOverTranslation**	A media characteristic that indicates that a track or media selection option contains a language translation and verbal interpretation of spoken dialog	M
7	**Caption**	Textual media component containing transcriptions of spoken dialog and auditory cues such as sound effects and music for the hearing impaired	M
8	**Subtitle**	Textual transcriptions of spoken dialog.	M

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Value	Name	Summary	Conformance
9	**Alternate**	Textual media component containing transcriptions of spoken dialog and auditory cues such as sound effects and music for the hearing impaired	M
10	**Supplementary**	Media content component that is supplementary to a media content component of a different media component type	M
11	**Commentary**	Experience that contains a commentary (e.g. director's commentary) (typically audio)	M
12	**DubbedTranslation**	Experience that contains an element that is presented in a different language from the original (e.g. dubbed audio, translated captions)	M
13	**Description**	Textual or audio media component containing a textual description (intended for audio synthesis) or an audio description describing a visual component	M
14	**Metadata**	Media component containing information intended to be processed by application specific elements	M
15	**EnhancedAudioIntelligibility**	Experience containing an element for improved intelligibility of the dialogue.	M

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Value	Name	Summary	Conformance
16	**Emergency**	Experience that provides information, about a current emergency, that is intended to enable the protection of life, health, safety, and property, and may also include critical details regarding the emergency and how to respond to the emergency	M
17	**Karaoke**	Textual representation of a songs' lyrics, usually in the same language as the associated song as specified in [SMPTE ST 2067-2]	M

6.10.5.4. PlaybackPositionStruct Type

This structure defines a playback position within a media stream being played.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**UpdatedAt**	epoch-us	all				M
1	**Position**	uint64	all	X			M

6.10.5.4.1. UpdatedAt Field

This field SHALL indicate the time when the position was last updated.

6.10.5.4.2. Position Field

This field SHALL indicate the associated discrete position within the media stream, in milliseconds from the beginning of the stream, being associated with the time indicated by the UpdatedAt field. The Position SHALL NOT be greater than the duration of the media if duration is specified. The Position SHALL NOT be greater than the time difference between current time and start time of the media when start time is specified.

A value of null SHALL indicate that playback position is not applicable for the current state of the media playback (For example : Live media with no known duration and where seek is not supported).

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6.10.5.5. TrackStruct Type

This structure defines a uniquely identifiable Text Track or Audio Track.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**ID**	string	max 32				M
1	**TrackAttributes**	TrackAttributesStruct	all				M

6.10.5.5.1. ID Field

This field SHALL indicate the Identifier for the Track which is unique within the Track catalog. The Track catalog contains all the Text/Audio tracks corresponding to the main media content.

6.10.5.5.2. TrackAttributes Field

This field SHALL indicate the Attributes associated to the Track, like languageCode.

6.10.5.6. TrackAttributesStruct Type

This structure includes the attributes associated with a Text/Audio Track

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**LanguageCode**	string	max 32				M
1	**Characteristics**	list[CharacteristicEnum]	all	X	null		O
2	**DisplayName**	string	max 256	X	null		O

6.10.5.6.1. LanguageCode Field

The value is a String containing one of the standard Tags for Identifying Languages [RFC5646], which identifies the primary language used in the Track.

6.10.5.6.2. Characteristics Field

This is a list of enumerated CharacteristicEnum values that indicate a purpose, trait or feature associated with the Track. A value of null SHALL indicate that there are no Characteristics corresponding to the Track.

6.10.5.6.3. DisplayName Field

The value is a String containing a user displayable name for the Track. A value of null SHALL indicate that there is no DisplayName corresponding to the Track.

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6.10.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**CurrentState**	PlaybackStateEnum	desc			R V	M
0x0001	**StartTime**	epoch-us	desc	X	null	R V	AS
0x0002	**Duration**	uint64	desc	X	null	R V	AS
0x0003	**SampledPosition**	PlaybackPositionStruct	desc	X	null	R V	AS
0x0004	**PlaybackSpeed**	single	desc		0	R V	AS
0x0005	**SeekRangeEnd**	uint64	desc	X	null	R V	AS
0x0006	**SeekRangeStart**	uint64	desc	X	null	R V	AS
0x0007	**ActiveAudioTrack**	TrackStruct	desc	X	null	R V	AT
0x0008	**AvailableAudioTracks**	list[TrackStruct]	desc	X	null	R V	AT
0x0009	**ActiveTextTrack**	TrackStruct	desc	X	null	R V	TT
0x000A	**AvailableTextTracks**	list[TrackStruct]	desc	X	null	R V	TT

6.10.6.1. CurrentState Attribute

This attribute SHALL indicate the current playback state of media.

During fast-forward, rewind, and other seek operations; this attribute SHALL be set to PLAYING.

6.10.6.2. StartTime Attribute

This attribute SHALL indicate the start time of the media, in case the media has a fixed start time (for example, live stream or television broadcast), or null when start time does not apply to the current media (for example, video-on-demand). This time is a UTC time. The client needs to handle conversion to local time, as required, taking in account time zone and possible local DST offset.

6.10.6.3. Duration Attribute

This attribute SHALL indicate the duration, in milliseconds, of the current media being played back

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or null when duration is not applicable (for example, in live streaming content with no known duration). This attribute SHALL never be 0.

6.10.6.4. SampledPosition Attribute

This attribute SHALL indicate the position of playback (Position field) at the time (UpdateAt field) specified in the attribute. The client MAY use the SampledPosition attribute to compute the current position within the media stream based on the PlaybackSpeed, PlaybackPositionStruct.UpdatedAt and PlaybackPositionStruct.Position fields. To enable this, the SampledPosition attribute SHALL be updated whenever a change in either the playback speed or the playback position is triggered outside the normal playback of the media. The events which MAY cause this to happen include:

- Starting or resumption of playback
- Seeking
- Skipping forward or backward
- Fast-forwarding or rewinding
- Updating of playback speed as a result of explicit request, or as a result of buffering events

6.10.6.5. PlaybackSpeed Attribute

This attribute SHALL indicate the speed at which the current media is being played. The new PlaybackSpeed SHALL be reflected in this attribute whenever any of the following occurs:

- Starting of playback
- Resuming of playback
- Fast-forwarding
- Rewinding

The PlaybackSpeed SHALL reflect the ratio of time elapsed in the media to the actual time taken for the playback assuming no changes to media playback (for example buffering events or requests to pause/rewind/forward).

- A value for PlaybackSpeed of 1 SHALL indicate normal playback where, for example, playback for 1 second causes the media to advance by 1 second within the duration of the media.
- A value for PlaybackSpeed which is greater than 0 SHALL indicate that as playback is happening the media is currently advancing in time within the duration of the media.
- A value for PlaybackSpeed which is less than 0 SHALL indicate that as playback is happening the media is currently going back in time within the duration of the media.
- A value for PlaybackSpeed of 0 SHALL indicate that the media is currently not playing back. When the CurrentState attribute has the value of PAUSED, NOT_PLAYING or BUFFERING, the PlaybackSpeed SHALL be set to 0 to reflect that the media is not playing.

Following examples illustrate the PlaybackSpeed attribute values in various conditions.

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Seconds of Media Played	Actual Time Taken in Seconds	Direction of playback	PlaybackSpeed
2	2	Forward	1.0
2	1	Forward	2.0
1	2	Forward	0.5
2	2	Reverse	-1.0
2	1	Reverse	-2.0
1	2	Reverse	-0.5

6.10.6.6. SeekRangeStart Attribute

This attribute SHALL indicate the earliest valid position to which a client MAY seek back, in milliseconds from start of the media. A value of NULL SHALL indicate that seeking backwards is not allowed.

6.10.6.7. SeekRangeEnd Attribute

This attribute SHALL indicate the furthest forward valid position to which a client MAY seek forward, in milliseconds from the start of the media. When the media has an associated StartTime, a value of null SHALL indicate that a seek forward is valid only until the current time within the media, using a position computed from the difference between the current time offset and StartTime, in milliseconds from start of the media, truncating fractional milliseconds towards 0. A value of NULL when StartTime is not specified SHALL indicate that seeking forward is not allowed.

6.10.6.8. ActiveAudioTrack Attribute

This attribute SHALL indicate the Audio track currently set and being used for the streaming media. A value of null SHALL indicate that no Audio Track corresponding to the current media is currently being played.

6.10.6.9. AvailableAudioTracks Attribute

This attribute SHALL indicate the list of Audio tracks available for the current title being played. A value of null SHALL indicate that no Audio Tracks corresponding to the current media are selectable by the client.

6.10.6.10. ActiveTextTrack Attribute

This attribute SHALL indicate the Text track currently set and being used for the streaming media. This can be nil. A value of null SHALL indicate that no Text Track corresponding to the current media is currently being displayed.

6.10.6.11. AvailableTextTracks Attribute

This attribute SHALL indicate the list of Text tracks available for the current title being played. This can be an empty list. A value of null SHALL indicate that no Text Tracks corresponding to the current media are selectable by the client.

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6.10.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**Play**	client ⇒ server	PlaybackResponse	O	M
0x01	**Pause**	client ⇒ server	PlaybackResponse	O	M
0x02	**Stop**	client ⇒ server	PlaybackResponse	O	M
0x03	**StartOver**	client ⇒ server	PlaybackResponse	O	O
0x04	**Previous**	client ⇒ server	PlaybackResponse	O	O
0x05	**Next**	client ⇒ server	PlaybackResponse	O	O
0x06	**Rewind**	client ⇒ server	PlaybackResponse	O	VS
0x07	**FastForward**	client ⇒ server	PlaybackResponse	O	VS
0x08	**SkipForward**	client ⇒ server	PlaybackResponse	O	O
0x09	**SkipBackward**	client ⇒ server	PlaybackResponse	O	O
0x0A	**PlaybackResponse**	client ⇐ server	N		M
0x0B	**Seek**	client ⇒ server	PlaybackResponse	O	AS
0x0C	**ActivateAudioTrack**	client ⇒ server	Y	O	AT
0x0D	**ActivateTextTrack**	client ⇒ server	Y	O	TT
0x0E	**DeactivateTextTrack**	client ⇒ server	Y	O	TT

6.10.7.1. Play Command

This command is used to start playback of the media.

Upon receipt, this SHALL play media. If content is currently in a FastForward or Rewind state. Play SHALL return media to normal playback speed.

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6.10.7.2. Pause Command

This command is used to pause playback of the media.

Upon receipt, this SHALL pause playback of the media.

6.10.7.3. Stop Command

This command is used to stop playback of the media.

Upon receipt, this SHALL stop playback of the media. User-visible outcome is context-specific. This MAY navigate the user back to the location from where the media was originally launched.

6.10.7.4. StartOver Command

This command is used to start playback of the media from the beginning.

Upon receipt, this SHALL Start Over with the current media playback item.

6.10.7.5. Previous Command

This command is used to go back to the previous media playback item.

Upon receipt, this SHALL cause the handler to be invoked for "Previous". User experience is context-specific. This will often Go back to the previous media playback item.

6.10.7.6. Next Command

This command is used to go to the next media playback item.

Upon receipt, this SHALL cause the handler to be invoked for "Next". User experience is context-specific. This will often Go forward to the next media playback item.

6.10.7.7. Rewind Command

This command is used to rewind the media.

Upon receipt, this SHALL start playback of the media backward in case the media is currently playing in the forward direction or is not playing. If the playback is already happening in the backwards direction receipt of this command SHALL increase the speed of the media playback backwards.

Different "rewind" speeds MAY be reflected on the media playback device based upon the number of sequential calls to this function and the capability of the device. This is to avoid needing to define every speed (multiple fast, slow motion, etc). If the PlaybackSpeed attribute is supported it SHALL be updated to reflect the new speed of playback. If the playback speed cannot be changed for the media being played(for example, in live streaming content not supporting seek), the status of NOT_ALLOWED SHALL be returned. If the playback speed has reached the maximum supported speed for media playing backwards, the status of SPEED_OUT_OF_RANGE SHALL be returned.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**AudioAdvanceUnmuted**	bool	all		false	AA

6.10.7.7.1. AudioAdvanceUnmuted Field

This field SHALL indicate whether audio should be unmuted by the player during rewind.

A value of true does not guarantee that audio can be heard by the user since the speaker may be muted, turned down to a low level and/or unplugged.

6.10.7.8. FastForward Command

This command is used to fast forward the media.

Upon receipt, this SHALL start playback of the media in the forward direction in case the media is currently playing in the backward direction or is not playing. If the playback is already happening in the forward direction receipt of this command SHALL increase the speed of the media playback.

Different "fast-forward" speeds MAY be reflected on the media playback device based upon the number of sequential calls to this function and the capability of the device. This is to avoid needing to define every speed (multiple fast, slow motion, etc). If the PlaybackSpeed attribute is supported it SHALL be updated to reflect the new speed of playback. If the playback speed cannot be changed for the media being played(for example, in live streaming content not supporting seek), the status of NOT_ALLOWED SHALL be returned. If the playback speed has reached the maximum supported speed for media playing forward, the status of SPEED_OUT_OF_RANGE SHALL be returned.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**AudioAdvanceUnmuted**	bool	all		false	AA

6.10.7.8.1. AudioAdvanceUnmuted Field

This field SHALL indicate whether audio should be unmuted by the player during fast forward.

A value of true does not guarantee that audio can be heard by the user since the speaker may be muted, turned down to a low level and/or unplugged.

6.10.7.9. SkipForward Command

This command is used to skip forward in the media.

Upon receipt, this SHALL Skip forward in the media by the given number of milliseconds, using the data as follows:

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**DeltaPositionMilliseconds**	uint64	all			M

6.10.7.9.1. DeltaPositionMilliseconds Field

This field SHALL indicate the duration of the time span to skip forward in the media, in milliseconds. In case the resulting position falls in the middle of a frame, the server SHALL set the position to the beginning of that frame and set the SampledPosition attribute on the cluster accordingly. If the resultant position falls beyond the furthest valid position in the media the client MAY seek forward to, the position should be set to that furthest valid position. If the SampledPosition attribute is supported it SHALL be updated on the cluster accordingly.

6.10.7.10. SkipBackward Command

This command is used to skip backward in the media.

Upon receipt, this SHALL Skip backward in the media by the given number of milliseconds, using the data as follows:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**DeltaPositionMilliseconds**	uint64	all			M

6.10.7.10.1. DeltaPositionMilliseconds Field

This field SHALL indicate the duration of the time span to skip backward in the media, in milliseconds. In case the resulting position falls in the middle of a frame, the server SHALL set the position to the beginning of that frame and set the SampledPosition attribute on the cluster accordingly. If the resultant position falls before the earliest valid position to which a client MAY seek back to, the position should be set to that earliest valid position. If the SampledPosition attribute is supported it SHALL be updated on the cluster accordingly.

6.10.7.11. Seek Command

This command is used to seek to a specific position in the media.

Upon receipt, this SHALL change the playback position in the media to the given position using data as follows:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Position**	uint64	all			M

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6.10.7.11.1. Position Field

This field SHALL indicate the position (in milliseconds) in the media to seek to. In case the position falls in the middle of a frame, the server SHALL set the position to the beginning of that frame and set the SampledPosition attribute on the cluster accordingly. If the position falls before the earliest valid position or beyond the furthest valid position to which a client MAY seek back or forward to respectively, the status of SEEK_OUT_OF_RANGE SHALL be returned and no change SHALL be made to the position of the playback.

6.10.7.12. PlaybackResponse Command

This command is used to indicate the status of the command that was issued by the client.

This command SHALL be generated in response to various Playback Commands. The data for this command SHALL be as follows:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	StatusEnum	desc			M
1	**Data**	string	any			O

6.10.7.12.1. Status Field

This field SHALL indicate the status of the command which resulted in this response.

6.10.7.12.2. Data Field

This field SHALL indicate Optional app-specific data.

6.10.7.13. ActivateAudioTrack Command

This command is used to activate a specific Audio Track for the media being played.

Upon receipt, the server SHALL set the active Audio Track to the one identified by the TrackID in the Track catalog for the streaming media. If the TrackID does not exist in the Track catalog, OR does not correspond to the streaming media OR no media is being streamed at the time of receipt of this command, the server will return an error status of INVALID_ARGUMENT.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**TrackID**	string	max 32			M
1	**AudioOutputIndex**	uint8	all	X		AT

6.10.7.13.1. TrackID Field

This field SHALL indicate the Audio Track to activate.

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6.10.7.13.2. AudioOutputIndex Field

This value is the index field of the OutputInfoStruct from the OutputList attribute (from the AudioOutput cluster) and indicates which audio output the Audio Track should be played on. This field is absent for Text Tracks and only present for Audio Tracks. A value of null SHALL indicate that the server can choose the audio output(s) to play the Audio Track on.

6.10.7.14. ActivateTextTrack Command

This command is used to activate a specific Text Track for the media being played.

Upon receipt, the server SHALL set the active Text Track to the one identified by the TrackID in the Track catalog for the streaming media. If the TrackID does not exist in the Track catalog, OR does not correspond to the streaming media OR no media is being streamed at the time of receipt of this command, the server SHALL return an error status of INVALID_ARGUMENT.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**TrackID**	string	max 32			M

6.10.7.14.1. TrackID Field

This field SHALL indicate the Text Track to activate.

6.10.7.15. DeactivateTextTrack Command

This command is used to deactivate a specific Text Track for the media being played.

If a Text Track is active (i.e. being displayed), upon receipt of this command, the server SHALL stop displaying it.

6.10.8. Events

ID	Name	Priority	Access	Conformance
0x00	**StateChanged**	INFO	V	O

6.10.8.1. StateChanged Event

If supported, this event SHALL be generated when there is a change in any of the supported attributes of the Media Playback cluster. This event SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0x0000	**CurrentState**	PlaybackStateEnum	desc			M
0x0001	**StartTime**	epoch-us	desc			AS
0x0002	**Duration**	uint64	desc			AS

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0x0003	**SampledPosition**	PlaybackPositionStruct	desc			AS
0x0004	**PlaybackSpeed**	single	desc			AS
0x0005	**SeekRangeEnd**	uint64	desc			AS
0x0006	**SeekRangeStart**	uint64	desc			AS
0x0007	**Data**	octstr	max 900			O
0x0008	**AudioAdvanceUnmuted**	bool	desc		false	AA

6.10.8.1.1. CurrentState Field

This field SHALL indicate the updated playback state as defined by the CurrentState attribute, and has the same constraint as that attribute.

6.10.8.1.2. StartTime Field

This field SHALL indicate the updated start time as defined by the StartTime attribute, and has the same constraint as that attribute.

This field value SHALL be 0 when the value of the StartTime attribute is NULL.

6.10.8.1.3. Duration Field

This field SHALL indicate the updated duration as defined by the Duration attribute, and has the same constraint as that attribute.

This field value SHALL be 0 when the value of the Duration attribute is NULL.

6.10.8.1.4. SampledPosition Field

This field SHALL indicate the updated position of playback as defined by the SampledPosition attribute, and has the same constraint as that attribute.

The UpdatedAt field value of the PlaybackPositionStruct SHALL be 0, and the Position field value of the PlaybackPositionStruct SHALL be NULL, when the value of the SampledPosition attribute is NULL.

6.10.8.1.5. PlaybackSpeed Field

This field SHALL indicate the updated speed at which the current media is being played as defined by the PlaybackSpeed attribute, and has the same constraint as that attribute.

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6.10.8.1.6. SeekRangeStart Field

This field SHALL indicate the updated start of the seek range start as defined by the SeekRangeStart attribute, and has the same constraint as that attribute.

This field value SHALL be 0 when the value of the SeekRangeStart attribute is NULL.

6.10.8.1.7. SeekRangeEnd Field

This field SHALL indicate the updated start of the seek range end as defined by the SeekRangeEnd attribute, and has the same constraint as that attribute.

This field value SHALL be 0 when the value of the SeekRangeEnd attribute is NULL.

6.10.8.1.8. Data Field

This field SHALL indicate Optional app-specific data.

6.10.8.1.9. AudioAdvanceUnmuted Field

This field SHALL indicate whether audio is unmuted by the player due to a FF or REW command. This field is only meaningful when the PlaybackSpeed is present and not equal to 0 (paused) or 1 (normal playback). Typically the value will be false (muted), however, some players will play audio during certain fast forward and rewind speeds, and in these cases, the value will be true (not muted).

A value of true does not guarantee that audio can be heard by the user since the speaker may be muted, turned down to a low level and/or unplugged.

6.11. Target Navigator Cluster

This cluster provides an interface for UX navigation within a set of targets on a device or endpoint.

This cluster would be supported on Video Player devices or devices with navigable user interfaces. This cluster would also be supported on endpoints with navigable user interfaces such as a Content App. It supports listing a set of navigation targets, tracking and changing the current target.

The cluster server for Target Navigator is implemented by endpoints on a device that support UX navigation.

When this cluster is implemented for a Content App endpoint, the Video Player device containing the endpoint SHALL launch the Content App when a client invokes the NavigateTarget command.

6.11.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

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Revision	Description
2	Add TargetUpdated event

6.11.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	TGTNAV

6.11.3. Cluster ID

ID	Name
0x0505	Target Navigator

6.11.4. Data Types

6.11.4.1. StatusEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Success**	Command succeeded	M
1	**TargetNotFound**	Requested target was not found in the TargetList	M
2	**NotAllowed**	Target request is not allowed in current state.	M

6.11.4.2. TargetInfoStruct Type

This indicates an object describing the navigable target.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Identifier**	uint8	max 254				M
1	**Name**	string					M

6.11.4.2.1. Identifier Field

This field SHALL contain an unique id within the TargetList.

6.11.4.2.2. Name Field

This field SHALL contain a name string for the TargetInfoStruct.

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6.11.5. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**TargetList**	list[TargetInfoStruct]				R V	M
0x0001	**CurrentTarget**	uint8	desc		0xFF	R V	O

6.11.5.1. TargetList Attribute

This attribute SHALL represent a list of targets that can be navigated to within the experience presented to the user by the Endpoint (Video Player or Content App). The list SHALL NOT contain any entries with the same Identifier in the TargetInfoStruct object.

6.11.5.2. CurrentTarget Attribute

This attribute SHALL represent the Identifier for the target which is currently in foreground on the corresponding Endpoint (Video Player or Content App), or 0xFF to indicate that no target is in the foreground.

When not 0xFF, the CurrentTarget SHALL be an Identifier value contained within one of the TargetInfoStruct objects in the TargetList attribute.

6.11.6. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**NavigateTarget**	client ⇒ server	NavigateTargetResponse	O	M
0x01	**NavigateTargetResponse**	client ⇐ server	N		M

6.11.6.1. NavigateTarget Command

Upon receipt, this SHALL navigation the UX to the target identified.

ID	Field	Type	Constraint	Quality	Fallback	Conformance
0	**Target**	uint8	all			M
1	**Data**	string			MS	O

6.11.6.1.1. Target Field

This field SHALL indicate the Identifier for the target for UX navigation. The Target SHALL be an Identifier value contained within one of the TargetInfoStruct objects in the TargetList attribute.

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6.11.6.1.2. Data Field

This field SHALL indicate Optional app-specific data.

6.11.6.2. NavigateTargetResponse Command

This command SHALL be generated in response to NavigateTarget command.

ID	Field	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	StatusEnum	all			M
1	**Data**	string	any		MS	O

6.11.6.2.1. Status Field

This field SHALL indicate the of the command.

6.11.6.2.2. Data Field

This field SHALL indicate Optional app-specific data.

6.11.7. Events

ID	Name	Priority	Access	Conformance
0x00	**TargetUpdated**	INFO	V	O

6.11.7.1. TargetUpdated Event

This event SHALL be generated when there is a change in either the active target or the list of available targets or both. This event SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**TargetList**	list[TargetInfoStruct]				O
1	**CurrentTarget**	uint8	desc		0xFF	O
2	**Data**	octstr	max 900			O

6.11.7.2. TargetList Field

This field SHALL indicate the updated target list as defined by the TargetList attribute if there is a change in the list of targets. Otherwise this field can be omitted from the event.

6.11.7.3. CurrentTarget Field

This field SHALL indicate the updated target that is in foreground as defined by the CurrentTarget attribute if supported (see CurrentTarget attribute for constraints).

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6.11.7.4. Data Field

This field SHALL indicate Optional app-specific data.

6.12. Content App Observer Cluster

This cluster provides an interface for sending targeted commands to an Observer of a Content App on a Video Player device such as a Streaming Media Player, Smart TV or Smart Screen.

The cluster server for Content App Observer is implemented by an endpoint that communicates with a Content App, such as a Casting Video Client.

The cluster client for Content App Observer is implemented by a Content App endpoint.

A Content App is informed of the NodeId of an Observer when a binding is set on the Content App. For a Content App Platform, the binding is set by the platform when a CastingVideoClient is granted access to the Content App, and the CastingVideoClient supports the Content App Observer cluster. The Content App can then send the ContentAppMessage to the Observer (server cluster), and the Observer responds with a ContentAppMessageResponse.

The Data and EncodingHint fields of the ContentAppMessage and ContentAppMessageResponse contain content app-specific values, the format and interpretation of which is defined by the Content App vendor, analogous to the custom message features offered by other popular casting protocols. Standardized cluster and commands are used here rather than manufacturer-specific cluster and commands because of the role that the Content App Platform plays in creating the ACLs and Bindings on both sides of the communication between the Content App Observer endpoint and the Content App endpoint.

By using standard cluster and commands:

1. The Content App Platform is able to easily determine that a binding is needed on the Content App endpoint because it can recognize the Content App Observer cluster implemented by a client node.
2. The Content App Platform is able to easily identify commands that are allowed to be sent by the Content App to a client node because those commands use the Content App Observer cluster.
3. The Content App is able to easily determine that a node supports the Content App Observer cluster because it has received a binding which specifies the Content App Observer cluster.
4. The Casting Video Client is able to support a single cluster for receiving commands from any Content App and does not need to explicitly list every Content App it understands.

A Content App Observer SHOULD ignore the Data and EncodingHint field values in commands from a Content App it does not recognize. A Content App SHOULD ignore the Data field values in responses when the EncodingHint value is blank or not recognized.

6.12.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

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Revision	Description
1	Initial revision

6.12.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	APPOBSERVER

6.12.3. Cluster ID

ID	Name
0x0510	Content App Observer

6.12.4. Data Types

6.12.4.1. StatusEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Success**	Command succeeded	M
1	**UnexpectedData**	Data field in command was not understood by the Observer	M

6.12.5. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**ContentAppMessage**	client ⇒ server	ContentAppMessageResponse	O	M
0x01	**ContentAppMessageResponse**	client ⇐ server	N		M

6.12.5.1. ContentAppMessage Command

Upon receipt, the data field MAY be parsed and interpreted. Message encoding is specific to the Content App. A Content App MAY when possible read attributes from the Basic Information Cluster on the Observer and use this to determine the Message encoding.

This command returns a ContentAppMessage Response.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Data**	string	max 500			M
1	**EncodingHint**	string	max 100			O

6.12.5.1.1. Data Field

This field SHALL indicate content app-specific data.

6.12.5.1.2. EncodingHint Field

This optional field SHALL indicate a content app-specific hint to the encoding of the data.

6.12.5.2. ContentAppMessageResponse Command

This command SHALL be generated in response to ContentAppMessage command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Status**	StatusEnum	all			M
1	**Data**	string	max 500			O
2	**EncodingHint**	string	max 100			O

6.12.5.2.1. Status Field

This field SHALL indicate the status of the command which resulted in this response.

6.12.5.2.2. Data Field

This optional field SHALL indicate content app-specific data.

6.12.5.2.3. EncodingHint Field

This optional field SHALL indicate a content app-specific hint to the encoding of the data.

6.13. Content Control Cluster

This cluster is used for managing the content control (including "parental control") settings on a media device such as a TV, or Set-top Box.

This cluster allows to configure content control settings by clients with the Management privilege. It is responsibility of the end product to enforce appropriate right access (for example, to prevent a child from disabling this feature).

NOTE	Support for Content Control cluster is provisional.

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6.13.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

6.13.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	CONCON

6.13.3. Cluster ID

ID	Name	Conformance
0x050F	Content Control	P

6.13.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	ST	ScreenTime	O	Supports managing screen time limits
1	PM	PINManagement	O	Supports managing a PIN code which is used for restricting access to configuration of this feature
2	BU	BlockUnrated	O	Supports managing content controls for unrated content
3	OCR	OnDemandContentRating	O	Supports managing content controls based upon rating threshold for on demand content

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Bit	Code	Feature	Conformance	Summary
4	SCR	ScheduledContentRating	O	Supports managing content controls based upon rating threshold for scheduled content
5	BC	BlockChannels	O	Supports managing a set of channels that are prohibited
6	BA	BlockApplications	O	Supports managing a set of applications that are prohibited
7	BTW	BlockContentTimeWindow	O	Supports managing content controls based upon setting time window in which all contents and applications SHALL be blocked

6.13.5. Data Types

6.13.5.1. DayOfWeekBitmap type

This data type is derived from map8.

Bit	Name	Summary
0	**Sunday**	Sunday
1	**Monday**	Monday
2	**Tuesday**	Tuesday
3	**Wednesday**	Wednesday
4	**Thursday**	Thursday
5	**Friday**	Friday
6	**Saturday**	Saturday

6.13.5.2. RatingNameStruct Type

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**RatingName**	string	max 8				M
1	**RatingNameDesc**	string	max 64				O

6.13.5.2.1. RatingName Field

This field SHALL indicate the name of the rating level of the applied rating system. The applied rating system is dependent upon the region or country where the Node has been provisioned, and may vary from one country to another.

6.13.5.2.2. RatingNameDesc Field

This field SHALL specify a human readable (displayable) description for RatingName.

6.13.5.3. BlockChannelStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**BlockChannelIndex**	uint16	all	X			M
1	**MajorNumber**	uint16	all				M
2	**MinorNumber**	uint16	all				M
3	**Identifier**	string	all				O

6.13.5.3.1. BlockChannelIndex Field

This field SHALL indicate a unique index value for a blocked channel. This value may be used to indicate one selected channel which will be removed from BlockChannelList attribute.

6.13.5.3.2. MajorNumber Field

This field SHALL indicate the channel major number value (for example, using ATSC format). When the channel number is expressed as a string, such as "13.1" or "256", the major number would be 13 or 256, respectively. This field is required but SHALL be set to 0 for channels such as over-the-top channels that are not represented by a major or minor number.

6.13.5.3.3. MinorNumber Field

This field SHALL indicate the channel minor number value (for example, using ATSC format). When the channel number is expressed as a string, such as "13.1" or "256", the minor number would be 1 or 0, respectively. This field is required but SHALL be set to 0 for channels such as over-the-top channels that are not represented by a major or minor number.

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6.13.5.3.4. Identifier

This field SHALL indicate the unique identifier for a specific channel. This field is optional, but SHOULD be provided when MajorNumber and MinorNumber are not available.

6.13.5.4. AppInfoStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**CatalogVendorID**	uint16	all				M
1	**ApplicationID**	string	all				M

6.13.5.4.1. CatalogVendorID Field

This field SHALL indicate the Connectivity Standards Alliance-issued vendor ID for the catalog. The DIAL registry SHALL use value 0x0000.

Content App Platform providers will have their own catalog vendor ID (set to their own Vendor ID) and will assign an ApplicationID to each Content App.

6.13.5.4.2. ApplicationID field

This field SHALL indicate the application identifier, expressed as a string, such as "PruneVideo" or "Company X". This field SHALL be unique within a catalog.

6.13.5.5. TimeWindowStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**TimeWindowIndex**	uint16	all	X			M
1	**DayOfWeek**	DayOfWeekBitmap	desc				M
2	**TimePeriod**	list[TimePeriodStruct]	desc				M

6.13.5.5.1. TimeWindowIndex Field

This field SHALL indicate a unique index of a specific time window. This value may be used to indicate a selected time window which will be removed from the BlockContentTimeWindow attribute.

6.13.5.5.2. DayOfWeek Field

This field SHALL indicate a day of week.

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6.13.5.5.3. TimePeriod Field

This field SHALL indicate one or more discrete time periods.

6.13.5.6. TimePeriodStruct type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**StartHour**	uint8	0 to 23				M
1	**StartMinute**	uint8	0 to 59				M
2	**EndHour**	uint8	0 to 23				M
3	**EndMinute**	uint8	0 to 59				M

6.13.5.6.1. StartHour Field

This field SHALL indicate the starting hour.

6.13.5.6.2. StartMinute Field

This field SHALL indicate the starting minute.

6.13.5.6.3. EndHour Field

This field SHALL indicate the ending hour. EndHour SHALL be equal to or greater than StartHour

6.13.5.6.4. EndMinute Field

This field SHALL indicate the ending minute. If EndHour is equal to StartHour then EndMinute SHALL be greater than StartMinute. If the EndHour is equal to 23 and the EndMinute is equal to 59, all contents SHALL be blocked until 23:59:59.

6.13.6. Status Codes

6.13.6.1. StatusCodeEnum Type

This data type is derived from enum8.

Value	Name	Summary
0x02	**InvalidPINCode**	Provided PIN Code does not match the current PIN code
0x03	**InvalidRating**	Provided Rating is out of scope of the corresponding Rating list
0x04	**InvalidChannel**	Provided Channel(s) is invalid
0x05	**ChannelAlreadyExist**	Provided Channel(s) already exists

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Value	Name	Summary
0x06	**ChannelNotExist**	Provided Channel(s) doesn't exist in BlockChannelList attribute
0x07	**UnidentifiableApplication**	Provided Application(s) is not identified
0x08	**ApplicationAlreadyExist**	Provided Application(s) already exists
0x09	**ApplicationNotExist**	Provided Application(s) doesn't exist in BlockApplicationList attribute
0x0A	**TimeWindowAlreadyExist**	Provided time Window already exists in BlockContentTimeWindow attribute
0x0B	**TimeWindowNotExist**	Provided time window doesn't exist in BlockContentTimeWindow attribute

6.13.7. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**Enabled**	bool	all			R V	M
0x0001	**OnDemandRatings**	list[RatingNameStruct]	all			R V	OCR
0x0002	**OnDemandRatingThreshold**	string	max 8			R V	OCR
0x0003	**ScheduledContentRatings**	list[RatingNameStruct]	all			R V	SCR
0x0004	**ScheduledContentRatingThreshold**	string	max 8			R V	SCR
0x0005	**ScreenDailyTime**	elapsed-s	max 86400			R V	ST

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0006	**RemainingScreenTime**	elapsed-s	max 86400			R V	ST
0x0007	**BlockUnrated**	bool	all			R V	BU
0x0008	**BlockChannelList**	list[BlockChannelStruct]	all			R V	BC
0x0009	**BlockApplicationList**	list[AppInfoStruct]	all			R V	BA
0x000A	**BlockContentTimeWindow**	list[TimeWindowStruct]	max 7			R V	BTW

6.13.7.1. Enabled Attribute

This attribute SHALL indicate whether the Content Control feature implemented on a media device is turned off (FALSE) or turned on (TRUE).

6.13.7.2. OnDemandRatings Attribute

This attribute SHALL provide the collection of ratings that are currently valid for this media device. The items should honor the metadata of the on-demand content (e.g. Movie) rating system for one country or region where the media device has been provisioned. For example, for the MPAA system, RatingName may be one value out of "G", "PG", "PG-13", "R", "NC-17".

The media device SHALL have a way to determine which rating system applies for the on-demand content and then populate this attribute. For example, it can do it through examining the Location attribute in the Basic Information cluster, and then determining which rating system applies.

The ratings in this collection SHALL be in order from a rating for the youngest viewers to the one for the oldest viewers. Each rating in the list SHALL be unique.

6.13.7.3. OnDemandRatingThreshold Attribute

This attribute SHALL indicate a threshold rating as a content filter which is compared with the rating for on-demand content. For example, if the on-demand content rating is greater than or equal to OnDemandRatingThreshold, for a rating system that is ordered from lower viewer age to higher viewer age, then on-demand content is not appropriate for the User and the Node SHALL prevent the playback of content.

This attribute SHALL be set to one of the values present in the OnDemandRatings attribute.

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When this attribute changes, the device SHOULD make the user aware of any limits of this feature. For example, if the feature does not control content within apps, then the device should make this clear to the user when the attribute changes.

6.13.7.4. ScheduledContentRatings Attribute

This attribute SHALL indicate a collection of ratings which ScheduledContentRatingThreshold can be set to. The items should honor metadata of the scheduled content rating system for the country or region where the media device has been provisioned.

The media device SHALL have a way to determine which scheduled content rating system applies and then populate this attribute. For example, this can be done by examining the Location attribute in Basic Information cluster, and then determining which rating system applies.

The ratings in this collection SHALL be in order from a rating for the youngest viewers to the one for the oldest viewers. Each rating in the list SHALL be unique.

6.13.7.5. ScheduledContentRatingThreshold Attribute

This attribute SHALL indicate a threshold rating as a content filter which is used to compare with the rating for scheduled content. For example, if the scheduled content rating is greater than or equal to ScheduledContentRatingThreshold for a rating system that is ordered from lower viewer age to higher viewer age, then the scheduled content is not appropriate for the User and SHALL be blocked.

This attribute SHALL be set to one of the values present in the ScheduledContentRatings attribute.

When this attribute changes, the device SHOULD make the user aware of any limits of this feature. For example, if the feature does not control content within apps, then the device should make this clear to the user when the attribute changes.

6.13.7.6. ScreenDailyTime Attribute

This attribute SHALL indicate the amount of time (in seconds) which the User is allowed to spend watching TV within one day when the Content Control feature is activated.

6.13.7.7. RemainingScreenTime Attribute

This attribute SHALL indicate the remaining screen time (in seconds) which the User is allowed to spend watching TV for the current day when the Content Control feature is activated. When this value equals 0, the media device SHALL terminate the playback of content.

This attribute SHALL be updated when the AddBonusTime command is received and processed successfully (with the correct PIN).

6.13.7.8. BlockUnrated Attribute

This attribute SHALL indicate whether the playback of unrated content is allowed when the Content Control feature is activated. If this attribute equals FALSE, then playback of unrated content SHALL be permitted. Otherwise, the media device SHALL prevent the playback of unrated content.

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When this attribute changes, the device SHOULD make the user aware of any limits of this feature. For example, if the feature does not control content within apps, then the device should make this clear to the user when the attribute changes.

6.13.7.9. BlockChannelList Attribute

This attribute SHALL indicate a set of channels that SHALL be blocked when the Content Control feature is activated.

6.13.7.10. BlockApplicationList Attribute

This attribute SHALL indicate a set of applications that SHALL be blocked when the Content Control feature is activated.

6.13.7.11. BlockContentTimeWindow Attribute

This attribute SHALL indicate a set of periods during which the playback of content on media device SHALL be blocked when the Content Control feature is activated. The media device SHALL reject any request to play content during one period of this attribute. If it is entering any one period of this attribute, the media device SHALL block content which is playing and generate an event EnteringBlockContentTimeWindow. There SHALL NOT be multiple entries in this attribute list for the same day of week.

6.13.8. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**UpdatePIN**	client ⇒ server	Y	M T	PM
0x01	**ResetPIN**	client ⇒ server	ResetPINResponse	A T	PM
0x02	**ResetPINResponse**	client ⇐ server	N		PM
0x03	**Enable**	client ⇒ server	Y	M T	M
0x04	**Disable**	client ⇒ server	Y	M T	M
0x05	**AddBonusTime**	client ⇒ server	Y	O	ST
0x06	**SetScreenDailyTime**	client ⇒ server	Y	M	ST
0x07	**BlockUnratedContent**	client ⇒ server	Y	M	BU
0x08	**UnblockUnratedContent**	client ⇒ server	Y	M	BU
0x09	**SetOnDemandRatingThreshold**	client ⇒ server	Y	M	OCR

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ID	Name	Direction	Response	Access	Conformance
0x0A	**SetScheduledContentRatingThreshold**	client ⇒ server	Y	M	SCR
0x0B	**AddBlockChannels**	client ⇒ server	Y	M	BC
0x0C	**RemoveBlockChannels**	client ⇒ server	Y	M	BC
0x0D	**AddBlockApplications**	client ⇒ server	Y	M	BA
0x0E	**RemoveBlockApplications**	client ⇒ server	Y	M	BA
0x0F	**SetBlockContentTimeWindow**	client ⇒ server	Y	M	BTW
0x10	**RemoveBlockContentTimeWindow**	client ⇒ server	Y	M	BTW

6.13.8.1. UpdatePIN Command

The purpose of this command is to update the PIN used for protecting configuration of the content control settings. Upon success, the old PIN SHALL no longer work.

The PIN is used to ensure that only the Node (or User) with the PIN code can make changes to the Content Control settings, for example, turn off Content Controls or modify the ScreenDailyTime. The PIN is composed of a numeric string of up to 6 human readable characters (displayable) .

Upon receipt of this command, the media device SHALL check if the OldPIN field of this command is the same as the current PIN. If the PINs are the same, then the PIN code SHALL be set to NewPIN. Otherwise a response with InvalidPINCode error status SHALL be returned.

The media device MAY provide a default PIN to the User via an out of band mechanism. For security reasons, it is recommended that a client encourage the user to update the PIN from its default value when performing configuration of the Content Control settings exposed by this cluster. The ResetPIN command can also be used to obtain the default PIN.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**OldPIN**	string	max 6			M
1	**NewPIN**	string	max 6			M

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6.13.8.1.1. OldPIN Field

This field SHALL specify the original PIN. Once the UpdatePIN command is performed successfully, it SHALL be invalid.

6.13.8.1.2. NewPIN Field

This field SHALL indicate a new PIN for the Content Control feature.

6.13.8.2. ResetPIN Command

The purpose of this command is to reset the PIN.

If this command is executed successfully, a ResetPINResponse command with a new PIN SHALL be returned.

6.13.8.3. ResetPINResponse Command

This command SHALL be generated in response to a ResetPIN command. The data for this command SHALL be as follows:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PINCode**	string	max 6			M

6.13.8.3.1. PINCode Field

This field SHALL indicate a new PIN of the Content Control feature.

6.13.8.4. Enable Command

The purpose of this command is to turn on the Content Control feature on a media device.

Upon receipt of the Enable command, the media device SHALL set the Enabled attribute to TRUE.

6.13.8.5. Disable Command

The purpose of this command is to turn off the Content Control feature on a media device.

On receipt of the Disable command, the media device SHALL set the Enabled attribute to FALSE.

6.13.8.6. AddBonusTime Command

The purpose of this command is to add the extra screen time for the user.

If a client with Operate privilege invokes this command, the media device SHALL check whether the PINCode passed in the command matches the current PINCode value. If these match, then the RemainingScreenTime attribute SHALL be increased by the specified BonusTime value.

If the PINs do not match, then a response with InvalidPINCode error status SHALL be returned, and no changes SHALL be made to RemainingScreenTime.

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If a client with Manage privilege or greater invokes this command, the media device SHALL ignore the PINCode field and directly increase the RemainingScreenTime attribute by the specified BonusTime value.

A server that does not support the PM feature SHALL respond with InvalidPINCode to clients that only have Operate privilege unless:

- It has been provided with the PIN value to expect via an out of band mechanism, and
- The client has provided a PINCode that matches the expected PIN value.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PINCode**	string	max 6			O
1	**BonusTime**	elapsed-s	desc		300s	M

6.13.8.6.1. PINCode Field

This field SHALL indicate the PIN.

This field SHALL be optional for clients with Manage or greater privilege but SHALL be mandatory for clients with Operate privilege. The PIN provided in this field SHALL be used to guarantee that a client with Operate permission is allowed to invoke this command only if the PIN passed in this command is equal to the current PIN value.

6.13.8.6.2. BonusTime Field

This field SHALL indicate the amount of extra time (in seconds) to increase RemainingScreenTime. This field SHALL NOT exceed the remaining time of this day.

6.13.8.7. SetScreenDailyTime Command

The purpose of this command is to set the ScreenDailyTime attribute.

Upon receipt of the SetScreenDailyTime command, the media device SHALL set the ScreenDailyTime attribute to the ScreenTime value.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ScreenTime**	elapsed-s	max 86400			M

6.13.8.7.1. ScreenTime Field

This field SHALL indicate the time (in seconds) which the User is allowed to spend watching TV on this media device within one day.

6.13.8.8. BlockUnratedContent Command

The purpose of this command is to specify whether programs with no Content rating must be blocked by this media device.

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Upon receipt of the BlockUnratedContent command, the media device SHALL set the BlockUnrated attribute to TRUE.

6.13.8.9. UnblockUnratedContent Command

The purpose of this command is to specify whether programs with no Content rating must be blocked by this media device.

Upon receipt of the UnblockUnratedContent command, the media device SHALL set the BlockUnrated attribute to FALSE.

6.13.8.10. SetOnDemandRatingThreshold Command

The purpose of this command is to set the OnDemandRatingThreshold attribute.

Upon receipt of the SetOnDemandRatingThreshold command, the media device SHALL check if the Rating field is one of values present in the OnDemandRatings attribute. If not, then a response with InvalidRating error status SHALL be returned.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Rating**	string	max 8			M

6.13.8.10.1. Rating

This field indicates a threshold rating for filtering on-demand content. This field SHALL be set to one of the values present in the OnDemandRatings attribute

6.13.8.11. SetScheduledContentRatingThreshold Command

The purpose of this command is to set ScheduledContentRatingThreshold attribute.

Upon receipt of the SetScheduledContentRatingThreshold command, the media device SHALL check if the Rating field is one of values present in the ScheduledContentRatings attribute. If not, then a response with InvalidRating error status SHALL be returned.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Rating**	string	max 8			M

6.13.8.11.1. Rating

This field indicates a threshold rating for filtering scheduled content. This field SHALL be set to one of the values present in the ScheduledContentRatings attribute.

6.13.8.12. AddBlockChannels command

The purpose of this command is to set BlockChannelList attribute.

Upon receipt of the AddBlockChannels command, the media device SHALL check if the channels

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passed in this command are valid. If the channel is invalid, then a response with InvalidChannel error Status SHALL be returned.

If there is at least one channel in Channels field which is not in the BlockChannelList attribute, the media device SHALL process the request by adding these new channels into the BlockChannelList attribute and return a successful Status Response. During this process, the media device SHALL assign one unique index to BlockChannelIndex field for every channel passed in this command.

If all channels in Channel field already exist in the BlockChannelList attribute, then a response with ChannelAlreadyExist error Status SHALL be returned.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Channels**	list[BlockChannelStruct]	all			M

6.13.8.12.1. Channels Field

This field indicates a set of channels that SHALL be blocked when the Content Control feature is activated. This field SHALL be set to values present in ChannelList attribute in the Channel cluster. The BlockChannelIndex field passed in this command SHALL be NULL.

6.13.8.13. RemoveBlockChannels command

The purpose of this command is to remove channels from the BlockChannelList attribute.

Upon receipt of the RemoveBlockChannels command, the media device SHALL check if the channels indicated by ChannelIndexes passed in this command are present in BlockChannelList attribute. If one or more channels indicated by ChannelIndexes passed in this command field are not present in the BlockChannelList attribute, then a response with ChannelNotExist error Status SHALL be returned.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ChannelIndexes**	list[uint16]	all			M

6.13.8.13.1. ChannelIndexes Field

This field SHALL specify a set of indexes indicating Which channels SHALL be removed from the BlockChannelList attribute.

6.13.8.14. AddBlockApplications Command

The purpose of this command is to set applications to the BlockApplicationList attribute.

Upon receipt of the AddBlockApplications command, the media device SHALL check if the Applications passed in this command are installed. If there is an application in Applications field which is not identified by media device, then a response with UnidentifiableApplication error Status MAY be

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returned.

If there is one or more applications which are not present in BlockApplicationList attribute, the media device SHALL process the request by adding the new application to the BlockApplicationList attribute and return a successful Status Response.

If all applications in Applications field are already present in BlockApplicationList attribute, then a response with ApplicationAlreadyExist error Status SHALL be returned.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Applications**	list[AppInfoStruct]	all			M

6.13.8.14.1. Applications Field

This field indicates a set of applications that SHALL be blocked when the Content Control feature is activated.

6.13.8.15. RemoveBlockApplications Command

The purpose of this command is to remove applications from the BlockApplicationList attribute.

Upon receipt of the RemoveBlockApplications command, the media device SHALL check if the applications passed in this command present in the BlockApplicationList attribute. If one or more applications in Applications field which are not present in the BlockApplicationList attribute, then a response with ApplicationNotExist error Status SHALL be returned.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Applications**	list[AppInfoStruct]	all			M

6.13.8.15.1. Applications Field

This field indicates a set of applications which SHALL be removed from BlockApplicationList attribute.

6.13.8.16. SetBlockContentTimeWindow Command

The purpose of this command is to set the BlockContentTimeWindow attribute.

Upon receipt of the SetBlockContentTimeWindow command, the media device SHALL check if the TimeWindowIndex field passed in this command is NULL. If the TimeWindowIndex field is NULL, the media device SHALL check if there is an entry in the BlockContentTimeWindow attribute which matches with the TimePeriod and DayOfWeek fields passed in this command. * If Yes, then a response with TimeWindowAlreadyExist error status SHALL be returned. * If No, then the media device SHALL assign one unique index for this time window and add it into the BlockContentTimeWindow list attribute.

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If the TimeWindowIndex field is not NULL and presents in the BlockContentTimeWindow attribute, the media device SHALL replace the original time window with the new time window passed in this command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**TimeWindow**	TimeWindowStruct				M

6.13.8.16.1. TimeWindow Field

This field SHALL indicate a time window requested to set to the BlockContentTimeWindow attribute.

6.13.8.17. RemoveBlockContentTimeWindow Command

The purpose of this command is to remove the selected time windows from the BlockContentTimeWindow attribute.

Upon receipt of the RemoveBlockContentTimeWindow command, the media device SHALL check if the time window index passed in this command presents in the BlockContentTimeWindow attribute.

If one or more time window indexes passed in this command are not present in BlockContentTimeWindow attribute, then a response with TimeWindowNotExist error status SHALL be returned.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**TimeWindowIndexes**	list[uint16]				M

6.13.8.17.1. TimeWindowIndexes Field

This field SHALL specify a set of time window indexes indicating which time windows will be removed from the BlockContentTimeWindow attribute.

6.13.9. Events

ID	Name	Priority	Access	Conformance
0x00	**RemainingScreenTimeExpired**	INFO	V	ST
0x01	**EnteringBlockContentTimeWindow**	INFO	V	BTW

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6.13.9.1. RemainingScreenTimeExpired Event

This event SHALL be generated when the RemainingScreenTime equals 0.

6.13.9.2. EnteringBlockContentTimeWindow Event

This event SHALL be generated when entering a period of blocked content as configured in the BlockContentTimeWindow attribute.

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Chapter 7. Robots

The Cluster Library is made of individual chapters such as this one. See Document Control in the Cluster Library for a list of all chapters and documents. References between chapters are made using a *X.Y* notation where *X* is the chapter and *Y* is the sub-section within that chapter.

7.1. General Description

7.1.1. Introduction

The clusters specified in this section define the operation of robotic devices, such as Robotic Vacuum Cleaners (RVCs).

7.1.2. Cluster List

This section lists the RVC specific clusters as specified in this chapter.

Table 11. Overview of the RVC Clusters

Cluster ID	Cluster Name	Description
0x0054	RVC Run Mode	Commands and attributes for controlling the running mode of an RVC device.
0x0055	RVC Clean Mode	Commands and attributes for controlling the cleaning mode of an RVC device.
0x0061	RVC Operational State	Commands and attributes for monitoring and controlling the operational state of an RVC device.

7.2. RVC Run Mode Cluster

This cluster is derived from the Mode Base cluster and defines additional mode tags and namespaced enumerated values for the running modes of robotic vacuum cleaner devices.

7.2.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

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Revision	Description
2	Add constraint about switching from non-Idle to non-Idle modes. ChangeToModeResponse command: StatusText must be provided for InvalidInMode status. Deprecate the OnMode attribute and the related feature map bit. Add the Mapping mode tag.
3	Remove constraint on changing cleaning modes while the RVC Run Mode cluster is in a non-Idle mode. Continue to allow InvalidInMode response for devices that do not support such mode changes.
4	DirectModeChange made non-provisional

7.2.2. Classification

Hierarchy	Role	Scope	PICS Code
Mode Base	Application	Endpoint	RVCRUNM

7.2.3. Cluster ID

ID	Name
0x0054	RVC Run Mode

7.2.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**DEPONOFF**	OnOff	X	Dependency with the OnOff cluster
20	**DIRECTMODECH**	DirectModeChange	O	Cluster supports changing run modes from non-Idle states

7.2.4.1. DirectModeChange Feature

This feature indicates whether the cluster implementation supports changing the run modes while the RVC Run Mode cluster's CurrentMode attribute is set to a mode without the Idle mode tag. If the implementation does not support such a change, the ChangeToModeResponse command SHALL have the StatusCode field set to the InvalidInMode value.

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7.2.5. Data Types

7.2.5.1. ModeOptionStruct Type

The table below lists the changes relative to the Mode Base cluster for the fields of the ModeOptionStruct type. A blank field indicates no change.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Label**						M
1	**Mode**						M
2	**ModeTags**		1 to 8				M

7.2.6. Attributes

The table below lists the changes relative to the Mode Base cluster for the attributes. A blank field indicates no change.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SupportedModes**						
0x0001	**CurrentMode**						
0x0002	**StartUpMode**						X
0x0003	**OnMode**						X

7.2.6.1. SupportedModes Attribute

At least one entry in the SupportedModes attribute SHALL include the Idle mode tag in the ModeTags field.

At least one entry in the SupportedModes attribute (different from the one above) SHALL include the Cleaning mode tag in the ModeTags field.

The Mapping, Cleaning, and Idle mode tags are mutually exclusive and SHALL NOT be used together in a mode's ModeTags.

7.2.7. Derived Cluster Namespace

This namespace includes definitions for data associated exclusively with the derived cluster.

7.2.7.1. ChangeToModeResponse Command Namespace Definitions

The following table defines the derived cluster specific StatusCode values.

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Status Code Value	Name
0x41	**Stuck**
0x42	**DustBinMissing**
0x43	**DustBinFull**
0x44	**WaterTankEmpty**
0x45	**WaterTankMissing**
0x46	**WaterTankLidOpen**
0x47	**MopCleaningPadMissing**
0x48	**BatteryLow**

7.2.7.2. Mode Tags

The following table defines the base and derived-cluster-specific ModeTag values.

Mode Tag Value	Name
0x0000	**Auto**
0x0001	**Quick**
0x0002	**Quiet**
0x0003	**LowNoise**
0x0004	**LowEnergy**
0x0005	**Vacation**
0x0006	**Min**
0x0007	**Max**
0x0008	**Night**
0x0009	**Day**
0x4000	**Idle**
0x4001	**Cleaning**
0x4002	**Mapping**

7.2.7.2.1. Idle Tag

The device is not performing any of the main operations of the other modes. However, auxiliary actions, such as seeking the charger or charging, may occur.

For example, the device has completed cleaning, successfully or not, on its own or due to a command, or has not been asked to clean after a restart.

7.2.7.2.2. Cleaning Tag

The device was asked to clean so it may be actively running, or paused due to an error, due to a pause command, or for recharging etc. If currently paused and the device can resume it will con-

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tinue to clean.

7.2.7.2.3. Mapping Tag

The device was asked to create a map of the space it is located in, so it may be actively running, or paused due to an error, due to a pause command, or for recharging etc. If currently paused and the device can resume, it will continue to map.

NOTE	this mode is intended to be used so the current space can be mapped by the device if the robot has not previously done that, or if the layout has substantially changed, for an optimal subsequent cleaning experience.

7.2.8. Mode Use

If the DirectModeChange feature is not present:

- Clients SHALL start a cleaning cycle by switching from a mode with the Idle mode tag to a mode with the Cleaning mode tag.
- Clients SHALL stop a cleaning cycle by switching from a mode with the Cleaning mode tag to a mode with the Idle mode tag.
- Clients SHALL start a mapping cycle by switching from a mode with the Idle mode tag to a mode with the Mapping mode tag.
- Clients SHALL stop a mapping cycle by switching from a mode with the Mapping mode tag to a mode with the Idle mode tag.

If the DirectModeChange feature is present, clients MAY change between run modes without needing to set an Idle mode first.

The RVC Run Mode cluster's SupportedModes attribute MAY include multiple modes with the Cleaning mode tag, and/or modes with the Mapping mode tag.

7.2.9. Mode Change Restrictions

The server MAY refuse attempts to switch the RVC Run Mode from a mode without the Idle mode tag to another non-Idle mode, as indicated by the DirectModeChange feature.

7.3. RVC Clean Mode Cluster

This cluster is derived from the Mode Base cluster and defines additional mode tags and namespaced enumerated values for the cleaning type of robotic vacuum cleaner devices.

7.3.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

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Revision	Description
1	Initial revision
2	Add constraint about changing cleaning modes while the RVC Run Mode cluster is in a non-Idle mode. ChangeToModeResponse command: StatusText must be provided for InvalidInMode status. Deprecate the OnMode attribute and the related feature map bit.
3	Remove constraint on changing cleaning modes while the RVC Run Mode cluster is in a non-Idle mode. Continue to allow InvalidInMode response for devices that do not support such mode changes.
4	Add Vacuum then Mop cleaning mode
5	DirectModeChange made non-provisional

7.3.2. Classification

Hierarchy	Role	Scope	PICS Code
Mode Base	Application	Endpoint	RVCCLEANM

7.3.3. Cluster ID

ID	Name
0x0055	RVC Clean Mode

7.3.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**DEPONOFF**	OnOff	X	Dependency with the OnOff cluster
20	**DIRECTMODECH**	DirectModeChange	O	Cluster supports changing clean modes from non-Idle states

7.3.4.1. DirectModeChange Feature

This feature indicates whether the cluster implementation supports changing the clean modes while the RVC Run Mode cluster's CurrentMode attribute is set to a mode without the Idle mode tag. If the implementation does not support such a change, the ChangeToModeResponse command SHALL have the StatusCode field set to the InvalidInMode value.

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7.3.5. Data Types

7.3.5.1. ModeOptionStruct Type

The table below lists the changes relative to the Mode Base cluster for the fields of the ModeOptionStruct type. A blank field indicates no change.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Label**						M
1	**Mode**						M
2	**ModeTags**		1 to 8				M

7.3.6. Attributes

The table below lists the changes relative to the Mode Base cluster for the attributes. A blank field indicates no change.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SupportedModes**						
0x0001	**CurrentMode**						
0x0002	**StartUpMode**						X
0x0003	**OnMode**						X

7.3.6.1. SupportedModes Attribute

At least one entry in the SupportedModes attribute SHALL include the Vacuum and/or the Mop mode tag in the ModeTags field list.

7.3.7. Derived Cluster Namespace

This namespace includes definitions for data associated exclusively with the derived cluster.

7.3.7.1. ChangeToModeResponse Command Namespace Definitions

The following table defines the derived cluster specific StatusCode values.

Status Code Value	Name
0x40	**CleaningInProgress**

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7.3.7.2. Mode Tags

The following table defines the base and derived-cluster-specific ModeTag values.

Mode Tag Value	Name
0x0000	**Auto**
0x0001	**Quick**
0x0002	**Quiet**
0x0003	**LowNoise**
0x0004	**LowEnergy**
0x0005	**Vacation**
0x0006	**Min**
0x0007	**Max**
0x0008	**Night**
0x0009	**Day**
0x4000	**DeepClean**
0x4001	**Vacuum**
0x4002	**Mop**
0x4003	**Vacuum then Mop**

7.3.7.2.1. Deep Clean Tag

While in this mode, the device is optimizing for improved cleaning.

7.3.7.2.2. Vacuum Tag

The device’s vacuuming feature is enabled in this mode.

7.3.7.2.3. Mop Tag

The device’s mopping feature is enabled in this mode.

7.3.7.2.4. Vacuum then Mop Tag

The device will vacuum then mop the relevant areas in this mode.

7.3.8. Mode Change Restrictions

The server MAY refuse attempts to switch the RVC Run Mode from a mode without the Idle mode tag to another non-Idle mode, as indicated by the DirectModeChange feature.

7.3.9. Mode Examples

A few examples of modes and their mode tags are provided below.

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For the "Turbo, Vacuum Only" mode, tags: 0x4000 (Deep Clean), 0x4001 (Vacuum).

For the "Mop Only" mode, tags: 0x4002 (Mop), 0x0003 (Low Noise).

For the "Rapid Vacuum and Mop" mode, tags: 0x0001 (Quick), 0x4001 (Vacuum), 0x4002 (Mop).

Note that the "Low Noise" and "Quick" mode tags are defined in the generic Mode Base cluster specification.

7.4. RVC Operational State Cluster

This cluster is derived from the Operational State cluster and provides an interface for monitoring the operational state of a robotic vacuum cleaner.

7.4.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	The Pause and Resume commands are usable in all compatible states. Deprecate the Start and Stop commands. Add the GoHome command.
3	Add several operational states and errors

7.4.2. Classification

Hierarchy	Role	Scope	PICS Code
Operational State	Application	Endpoint	RVCOPSTATE

7.4.3. Cluster ID

ID	Name
0x0061	RVC Operational State

7.4.4. Data Types

7.4.4.1. OperationalStateEnum Type

This data type is derived from enum8.

The values defined herein are applicable to this derived cluster of Operational State only and are additional to the set of values defined in Operational State itself.

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Value	Name	Summary	Conformance
0x00	**Stopped**	The device is stopped	M
0x01	**Running**	The device is operating	M
0x02	**Paused**	The device is paused during an operation	M
0x03	**Error**	The device is in an error state	M
0x40	**SeekingCharger**	The device is en route to the charging dock	M
0x41	**Charging**	The device is charging	M
0x42	**Docked**	The device is on the dock, not charging	M
0x43	**EmptyingDustBin**	The device is automatically emptying its own dust bin, such as to a dock	O
0x44	**CleaningMop**	The device is automatically cleaning its own mopping device, such as on a dock	O
0x45	**FillingWaterTank**	The device is automatically filling its own clean water tank for use when mopping, such as from a dock	O
0x46	**UpdatingMaps**	The device is processing acquired data to update its maps	O

RVC Pause Compatibility defines the compatibility of the states this cluster defines with the Pause command.

Table 12. RVC Pause Compatibility

State Value	State Name	Pause-Compatible	Notes
0x40	**SeekingCharger**	Y	
0x41	**Charging**	N	
0x42	**Docked**	N	
0x43	**EmptyingDustBin**	N	
0x44	**CleaningMop**	N	
0x45	**FillingWaterTank**	N	

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State Value	State Name	Pause-Compatible	Notes
0x46	**UpdatingMaps**	N	

RVC Resume Compatibility defines the compatibility of the states this cluster defines with the Resume command.

Table 13. RVC Resume Compatibility

State Value	State Name	Resume-Compatible	Notes
0x40	**SeekingCharger**	N	
0x41	**Charging**	Y	
0x42	**Docked**	Y	
0x43	**EmptyingDustBin**	N	
0x44	**CleaningMop**	N	
0x45	**FillingWaterTank**	N	
0x46	**UpdatingMaps**	N	

While in the Charging or Docked states, the device SHALL NOT attempt to resume unless it transitioned to those states while operating and can resume, such as, for example, if it is recharging while in a cleaning cycle. Else, if the operational state is Charging or Docked but there's no operation to resume or the operation can't be resumed, the device SHALL respond with an OperationalCommandResponse command with an ErrorStateID of CommandInvalidInState but take no further action.

7.4.4.2. ErrorStateEnum Type

This data type is derived from enum8.

The values defined herein are applicable to this derived cluster of Operational State only and are additional to the set of values defined in Operational State itself.

Value	Name	Summary	Conformance
0x00	**NoError**	The device is not in an error state	M
0x01	**UnableToStartOrResume**	The device is unable to start or resume operation	M
0x02	**UnableToCompleteOperation**	The device was unable to complete the current operation	M
0x03	**CommandInvalidInState**	The device cannot process the command in its current state	M

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Value	Name	Summary	Conformance
0x40	**FailedToFindChargingDock**	The device has failed to find or reach the charging dock	M
0x41	**Stuck**	The device is stuck and requires manual intervention	M
0x42	**DustBinMissing**	The device has detected that its dust bin is missing	M
0x43	**DustBinFull**	The device has detected that its dust bin is full	M
0x44	**WaterTankEmpty**	The device has detected that its clean water tank is empty	M
0x45	**WaterTankMissing**	The device has detected that its clean water tank is missing	M
0x46	**WaterTankLidOpen**	The device has detected that its water tank lid is open	M
0x47	**MopCleaningPadMissing**	The device has detected that its cleaning pad is missing	M
0x48	**LowBattery**	The device is unable to start or to continue operating due to a low battery	M
0x49	**CannotReachTargetArea**	The device is unable to move to an area where it was asked to operate, such as by setting the ServiceArea cluster's SelectedAreas attribute, due to an obstruction. For example, the obstruction might be a closed door or objects blocking the mapped path.	M
0x4A	**DirtyWaterTankFull**	The device has detected that its dirty water tank is full	M

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Value	Name	Summary	Conformance
0x4B	**DirtyWaterTankMissing**	The device has detected that its dirty water tank is missing	M
0x4C	**WheelsJammed**	The device has detected that one or more wheels are jammed by an object	M
0x4D	**BrushJammed**	The device has detected that its brush is jammed by an object	M
0x4E	**NavigationSensorObscured**	The device has detected that one of its sensors, such as LiDAR, infrared, or camera is obscured and needs to be cleaned	M

7.4.5. Commands

The table below lists the changes relative to the Operational State cluster for the commands. A blank field indicates no change.

ID	Name	Direction	Response	Access	Conformance
0x00	**Pause**				
0x01	**Stop**				X
0x02	**Start**				X
0x03	**Resume**				
0x04	**OperationalCommandResponse**				
0x80	**GoHome**	client ⇒ server	OperationalCommandResponse	O	O

7.4.5.1. GoHome Command

On receipt of this command, the device SHALL start seeking the charging dock, if possible in the current state of the device.

If this command is received when already in the SeekingCharger state the device SHALL respond with an OperationalCommandResponse command with an ErrorStateID of NoError but the command SHALL have no other effect.

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A device that receives this command in any state which does not allow seeking the charger, such as Charging or Docked, SHALL respond with an OperationalCommandResponse command with an ErrorStateID of CommandInvalidInState and SHALL have no other effect.

Otherwise, on success:

- The OperationalState attribute SHALL be set to SeekingCharger.
- The device SHALL respond with an OperationalCommandResponse command with an ErrorStateID of NoError.
- After the device reaches the dock and completes its docking activities, the RVC Run Mode cluster's CurrentMode attribute SHALL be set to an Idle mode.

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Chapter 8. Home Appliances

The Cluster Library is made of individual chapters such as this one. See Document Control in the Cluster Library for a list of all chapters and documents. References between chapters are made using a *X.Y* notation where *X* is the chapter and *Y* is the sub-section within that chapter.

8.1. General Description

8.1.1. Introduction

The clusters specified in this section are typically used in control of home appliances, such as laundry washers, refrigerators etc.

8.1.2. Cluster List

This section lists the home appliance specific clusters as specified in this chapter.

Table 14. Overview of the Home Appliance Clusters

Cluster ID	Cluster Name	Description
Common Clusters		
0x0056	Temperature Control	Commands and attributes for control of a temperature set point
Dishwasher Clusters		
0x0059	Dishwasher Mode	Commands and attributes for controlling a dishwasher
0x005D	Dishwasher Alarm	Alarm definitions for Dishwasher devices
Laundry Clusters (General)		
0x0051	Laundry Washer Mode	Commands and attributes for controlling a laundry washer or dryer
Laundry Dryer Clusters		
0x004A	Laundry Dryer Controls	Commands and attributes for the control of options on a device that does laundry drying
Laundry Washer Clusters		
0x0053	Laundry Washer Controls	Commands and attributes for the control of options on a device that does laundry washing
Microwave Oven Clusters		

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Cluster ID	Cluster Name	Description
0x005E	Microwave Oven Mode	Modes associated with a Microwave Oven
0x005F	Microwave Oven Control	Commands and attributes for controlling a Microwave Oven
Oven Clusters		
0x0048	Oven Operational State	Operational states and phases associated with an Oven
0x0049	Oven Mode	Modes associated with an Oven
Refrigerator Clusters		
0x0052	Refrigerator And Temperature Controlled Cabinet Mode	Commands and attributes for controlling a refrigerator or a temperature controlled cabinet
0x0057	Refrigerator Alarm	Alarm definitions for Refrigerator devices

8.2. Temperature Control Cluster

This cluster provides an interface to the setpoint temperature on devices such as washers, refrigerators, and water heaters. The setpoint temperature is the temperature to which a device using this cluster would attempt to control to. This cluster does not provide access to the actual or physical temperature associated with any device using this cluster. Access to the physical temperature associated with a device using this cluster would be provided by other clusters as part of that devices device type definition.

The values and constraints of the attributes communicated to clients SHOULD match the controls on any physical interface on a device implementing this server. For example, the value of the Step attribute SHOULD match the incremental value by which the temperature setpoint can be changed on the physical device.

8.2.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

8.2.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	TCTL

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8.2.3. Cluster ID

ID	Name
0x0056	Temperature Control

8.2.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**TN**	TemperatureNumber	O.a	Use actual temperature numbers
1	**TL**	TemperatureLevel	O.a	Use temperature levels
2	**STEP**	TemperatureStep	[TN]	Use step control with temperature numbers

8.2.4.1. TemperatureNumber Feature

For devices that use an actual temperature value for the temperature setpoint, such as some water heaters, the feature TN SHALL be used. Note that this cluster provides and supports temperatures in degrees Celsius via the `temperature` data type.

8.2.4.2. TemperatureLevel Feature

For devices that use vendor-specific temperature levels for the temperature setpoint, such as some washers, the feature TL SHALL be used.

8.2.4.3. TemperatureStep Feature

For devices that support discrete temperature setpoints that are larger than the temperature resolution imposed via the `temperature` data type, the Step feature MAY be used.

8.2.5. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**TemperatureSetpoint**	temperature	MinTemperature to MaxTemperature			R V	TN
0x0001	**MinTemperature**	temperature	max (MaxTemperature - 1)	F		R V	TN

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0002	**MaxTemperature**	temperature	desc	F		R V	TN
0x0003	**Step**	temperature	1 to (MaxTemperature - MinTemperature)	F		R V	STEP
0x0004	**SelectedTemperatureLevel**	uint8	max 31			R V	TL
0x0005	**SupportedTemperatureLevels**	list[string]	max 32[max 16]			R V	TL

NOTE See the `temperature` data type, in the data model, for encoding units. The scalar values are uint16 counts of 0.01C. For example, MinTemperature of -1800 is -18.00C.

8.2.5.1. TemperatureSetpoint Attribute

This attribute SHALL represent the desired Temperature Setpoint on the device.

8.2.5.2. MinTemperature Attribute

This attribute SHALL represent the minimum temperature to which the TemperatureSetpoint attribute MAY be set.

8.2.5.3. MaxTemperature Attribute

This attribute SHALL represent the maximum temperature to which the TemperatureSetpoint attribute MAY be set.

If the Step attribute is supported, this attribute SHALL be such that MaxTemperature = MinTemperature + (Step * n), where n is an integer and n > 0. If the Step attribute is not supported, this attribute SHALL be such that MaxTemperature > MinTemperature.

8.2.5.4. Step Attribute

This attribute SHALL represent the discrete value by which the TemperatureSetpoint attribute can be changed via the SetTemperature command.

> For example, if the value of MinTemperature is 25.00C (2500) and the Step value is 0.50C (50), valid values of the TargetTemperature field of the SetTemperature command would be 25.50C

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(2550), 26.00C (2600), 26.50C (2650), etc.

8.2.5.5. SelectedTemperatureLevel Attribute

This attribute SHALL represent the currently selected temperature level setting of the server. This attribute SHALL be the positional index of the list item in the SupportedTemperatureLevels list that represents the currently selected temperature level setting of the server.

8.2.5.6. SupportedTemperatureLevels Attribute

This attribute SHALL represent the list of supported temperature level settings that may be selected via the TargetTemperatureLevel field in the SetTemperature command. Each string is readable text that describes each temperature level setting in a way that can be easily understood by humans. For example, a washing machine can have temperature levels like "Cold", "Warm", and "Hot". Each string is specified by the manufacturer.

Each item in this list SHALL represent a unique temperature level. Each entry in this list SHALL have a unique value. The entries in this list SHALL appear in order of increasing temperature level with list item 0 being the setting with the lowest temperature level.

8.2.6. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**SetTemperature**	client ⇒ server	Y	O	M

8.2.6.1. SetTemperature Command

This command is used to set the temperature setpoint.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**TargetTemperature**	temperature	desc			TN
1	**TargetTemperatureLevel**	uint8	desc			TL

8.2.6.1.1. TargetTemperature Field

This field SHALL specify the desired temperature setpoint that the server is to be set to.

The TargetTemperature SHALL be from MinTemperature to MaxTemperature inclusive. If the Step attribute is supported, TargetTemperature SHALL be such that (TargetTemperature - MinTemperature) % Step == 0.

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8.2.6.1.2. TargetTemperatureLevel Field

This field SHALL specify the index of the list item in the SupportedTemperatureLevels list that represents the desired temperature level setting of the server. The value of this field SHALL be between 0 and the length of the SupportedTemperatureLevels list -1.

8.2.6.1.3. Effect on Receipt

If the TargetTemperature or TargetTemperatureLevel fields of the command meet all constraints but the server is unable to execute the command at the time the command is received by the server (e.g. due to the design of a device it cannot accept a change in its temperature setting after it has begun operation), then the server SHALL respond with a status code of INVALID_IN_STATE, and discard the command.

If the TN feature is supported, on receipt of this command:

- If the STEP feature is supported and the TargetTemperature field does not meet the equality (TargetTemperature - MinTemperature) % Step == 0, then the server SHALL respond with a status code of CONSTRAINT_ERROR, and the value of the TemperatureSetpoint attribute SHALL remain unchanged.
- Otherwise, the server SHALL set the TemperatureSetpoint attribute to the value of the TargetTemperature field and the response SHALL have a status code of SUCCESS.

If the TL feature is supported, on receipt of this command,

- If value of the TargetTemperatureLevel field meets all constraints, the server SHALL set its SelectedTemperatureLevel attribute to the value of TargetTemperatureLevel field and respond with status SUCCESS.
- Otherwise (e.g. if the value of the TargetTemperatureLevel field is out of bounds of the SupportedTemperatureLevels list), the status of the response SHALL be CONSTRAINT_ERROR, and the value of SelectedTemperatureLevel SHALL remain unchanged.

8.3. Dishwasher Mode Cluster

This cluster is derived from the Mode Base cluster and defines additional mode tags and namespaced enumerated values for dishwasher devices.

8.3.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	ChangeToModeResponse command: StatusText must be provided for InvalidInMode status

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Revision	Description
3	Set OnOff feature as disallowed (previously a Device Type override), Set StartUpMode and OnMode as disallowed (previously provisional)

8.3.2. Classification

Hierarchy	Role	Scope	PICS Code
Mode Base	Application	Endpoint	DISHM

8.3.3. Cluster ID

ID	Name
0x0059	Dishwasher Mode

8.3.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**DEPONOFF**	OnOff	X	Dependency with the OnOff cluster

8.3.5. Data Types

8.3.5.1. ModeOptionStruct Type

The table below lists the changes relative to the Mode Base cluster for the fields of the ModeOption-Struct type. A blank field indicates no change.

ID	Name	Type	Constraint	Quality	Fallback	Access	Confor-mance
0	**Label**						M
1	**Mode**						M
2	**ModeTags**		1 to 8				M

8.3.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Confor-mance
0x0000	**Support-edModes**						M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0001	**CurrentMode**						M
0x0002	**StartUpMode**						X
0x0003	**OnMode**						X

8.3.6.1. SupportedModes Attribute

At least one entry in the SupportedModes attribute SHALL include the Normal mode tag in the ModeTags field list.

8.3.7. Derived Cluster Namespace

This namespace includes definitions for data associated exclusively with the derived cluster.

8.3.7.1. Mode Tags

The following table defines the base and derived-cluster-specific ModeTag values.

Mode Tag Value	Name
0x0000	**Auto**
0x0001	**Quick**
0x0002	**Quiet**
0x0003	**LowNoise**
0x0004	**LowEnergy**
0x0005	**Vacation**
0x0006	**Min**
0x0007	**Max**
0x0008	**Night**
0x0009	**Day**
0x4000	**Normal**
0x4001	**Heavy**
0x4002	**Light**

8.3.7.1.1. Normal Tag

The normal regime of operation.

8.3.7.1.2. Heavy Tag

Mode optimized for washing heavily-soiled dishes.

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8.3.7.1.3. Light Tag

Mode optimized for light washing.

8.4. Dishwasher Alarm Cluster

This cluster is a derived cluster of the Alarm Base cluster and provides the alarm definition related to dishwasher devices.

8.4.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

8.4.2. Classification

Hierarchy	Role	Scope	PICS Code
Alarm Base	Application	Endpoint	DISHALM

8.4.3. Cluster ID

ID	Name
0x005D	Dishwasher Alarm

8.4.4. Data Types

8.4.4.1. AlarmBitmap Type

This data type is derived from map32.

Bit	Name	Summary	Conformance
0	**InflowError**	Water inflow is abnormal	P, O.a+
1	**DrainError**	Water draining is abnormal	P, O.a+
2	**DoorError**	Door or door lock is abnormal	O.a+
3	**TempTooLow**	Unable to reach normal temperature	P, O.a+
4	**TempTooHigh**	Temperature is too high	P, O.a+

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Bit	Name	Summary	Conformance
5	**WaterLevelError**	Water level is abnormal	P, O.a+

8.5. Laundry Washer Mode Cluster

This cluster is derived from the Mode Base cluster and defines additional mode tags and namespaced enumerated values for laundry washer as well as laundry dryer devices.

8.5.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	ChangeToModeResponse command: StatusText must be provided for InvalidInMode status
3	Set OnOff feature, StartUpMode, and OnMode as disallowed (previously a Device Type override)

8.5.2. Classification

Hierarchy	Role	Scope	PICS Code
Mode Base	Application	Endpoint	LWM

8.5.3. Cluster ID

ID	Name
0x0051	Laundry Washer Mode

8.5.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**DEPONOFF**	OnOff	X	Dependency with the OnOff cluster

8.5.5. Data Types

8.5.5.1. ModeOptionStruct Type

The table below lists the changes relative to the Mode Base cluster for the fields of the ModeOption-

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Struct type. A blank field indicates no change.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Label**						M
1	**Mode**						M
2	**ModeTags**		1 to 8				M

8.5.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SupportedModes**						M
0x0001	**CurrentMode**						M
0x0002	**StartUpMode**						X
0x0003	**OnMode**						X

8.5.6.1. SupportedModes Attribute

At least one entry in the SupportedModes attribute SHALL include the Normal mode tag in the ModeTags field list.

8.5.7. Derived Cluster Namespace

This namespace includes definitions for data associated exclusively with the derived cluster.

8.5.7.1. Mode Tags

The following table defines the base and derived-cluster-specific ModeTag values.

Mode Tag Value	Name
0x0000	**Auto**
0x0001	**Quick**
0x0002	**Quiet**
0x0003	**LowNoise**
0x0004	**LowEnergy**
0x0005	**Vacation**
0x0006	**Min**
0x0007	**Max**

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Mode Tag Value	Name
0x0008	**Night**
0x0009	**Day**
0x4000	**Normal**
0x4001	**Delicate**
0x4002	**Heavy**
0x4003	**Whites**

8.5.7.1.1. Normal Tag

The normal regime of operation.

8.5.7.1.2. Delicate Tag

Mode optimized for washing delicate garments.

8.5.7.1.3. Heavy Tag

Mode optimized for heavy washing.

8.5.7.1.4. Whites Tag

Mode optimized for stain removal on white fabrics.

8.5.8. Mode Examples

A few examples of Laundry modes and their mode tags are provided below.

- For the "Heavy Wash, Whites" mode, tags: 0x4002 (Heavy), 0x4003 (Whites).
- For the "Fast" mode, tags: 0x0001 (Quick), 0x4000 (Normal).

8.6. Laundry Washer Controls Cluster

This cluster provides a way to access options associated with the operation of a laundry washer device type.

8.6.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Updated Feature Map Conformance

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8.6.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	WASHERCTRL

8.6.3. Cluster ID

ID	Name
0x0053	Laundry Washer Controls

8.6.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**SPIN**	Spin	O.a+	Multiple spin speeds supported
1	**RINSE**	Rinse	O.a+	Multiple rinse cycles supported

8.6.4.1. Spin Feature

This feature indicates multiple spin speeds are supported in at least one supported mode. Note that some modes may not support multiple spin speeds even if this feature is supported.

8.6.4.2. Rinse Feature

This feature indicates multiple rinse cycles are supported in at least one supported mode. Note that some modes may not support selection of the number of rinse cycles even if this feature is supported.

8.6.5. Data Types

8.6.5.1. NumberOfRinsesEnum Type

This data type is derived from enum8.

The NumberOfRinsesEnum provides a representation of the number of rinses that will be performed for a selected mode. NumberOfRinsesEnum is derived from enum8. It is up to the device manufacturer to determine the mapping between the enum values and the corresponding numbers of rinses.

Value	Name	Summary	Conformance
0	**None**	This laundry washer mode does not perform rinse cycles	RINSE

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Value	Name	Summary	Conformance
1	**Normal**	This laundry washer mode performs normal rinse cycles determined by the manufacturer	RINSE
2	**Extra**	This laundry washer mode performs an extra rinse cycle	RINSE
3	**Max**	This laundry washer mode performs the maximum number of rinse cycles determined by the manufacturer	RINSE

8.6.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SpinSpeeds**	list[string]	max 16[max 64]			R V	SPIN
0x0001	**SpinSpeedCurrent**	uint8	max 15	X		RW VO	SPIN
0x0002	**NumberOfRinses**	NumberOfRinsesEnum	desc			RW VO	RINSE
0x0003	**SupportedRinses**	list[NumberOfRinsesEnum]	max 4			R V	RINSE

8.6.6.1. SpinSpeeds Attribute

This attribute SHALL indicate the list of spin speeds available to the appliance in the currently selected mode. The spin speed values are determined by the manufacturer. At least one spin speed value SHALL be provided in the SpinSpeeds list. The list of spin speeds MAY change depending on the currently selected Laundry Washer mode. For example, Quick mode might have a completely different list of SpinSpeeds than Delicates mode.

8.6.6.2. SpinSpeedCurrent Attribute

This attribute SHALL indicate the currently selected spin speed. It is the index into the SpinSpeeds list of the selected spin speed, as such, this attribute can be an integer between 0 and the number of entries in SpinSpeeds - 1. If a value is received that is outside of the defined constraints, a CONSTRAINT_ERROR SHALL be sent as the response. If a value is attempted to be written that doesn't match a valid index (e.g. an index of 5 when the list has 4 values), a CONSTRAINT_ERROR SHALL be

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sent as the response. If null is written to this attribute, there will be no spin speed for the selected cycle. If the value is null, there will be no spin speed on the current mode.

8.6.6.3. NumberOfRinses Attribute

This attribute SHALL indicate how many times a rinse cycle SHALL be performed on a device for the current mode of operation. A value of None SHALL indicate that no rinse cycle will be performed. This value may be set by the client to adjust the number of rinses that are performed for the current mode of operation. If the device is not in a compatible state to accept the provided value, an INVALID_IN_STATE error SHALL be sent as the response.

8.6.6.4. SupportedRinses Attribute

This attribute SHALL indicate the amount of rinses allowed for a specific mode. Each entry SHALL indicate a NumberOfRinsesEnum value that is possible in the selected mode on the device. The value of this attribute MAY change at runtime based on the currently selected mode. Each entry SHALL be distinct.

8.7. Refrigerator And Temperature Controlled Cabinet Mode Cluster

This cluster is derived from the Mode Base cluster and defines additional mode tags and namespaced enumerated values for refrigerator and temperature controlled cabinet devices.

8.7.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	ChangeToModeResponse command: StatusText must be provided for InvalidInMode status
3	Set OnOff feature, StartUpMode, and OnMode as disallowed (previously a Device Type override)

8.7.2. Classification

Hierarchy	Role	Scope	PICS Code
Mode Base	Application	Endpoint	TCCM

8.7.3. Cluster ID

ID	Name
0x0052	Refrigerator And Temperature Controlled Cabinet Mode

8.7.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**DEPONOFF**	OnOff	X	Dependency with the OnOff cluster

8.7.5. Data Types

8.7.5.1. ModeOptionStruct Type

The table below lists the changes relative to the Mode Base cluster for the fields of the ModeOptionStruct type. A blank field indicates no change.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Label**						M
1	**Mode**						M
2	**ModeTags**		1 to 8				M

8.7.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SupportedModes**						M
0x0001	**CurrentMode**						M
0x0002	**StartUpMode**						X
0x0003	**OnMode**						X

8.7.6.1. SupportedModes Attribute

At least one entry in the SupportedModes attribute SHALL include the Auto mode tag in the ModeTags field list.

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8.7.7. Derived Cluster Namespace

This namespace includes definitions for data associated exclusively with the derived cluster.

8.7.7.1. Mode Tags

The following table defines the base and derived-cluster-specific ModeTag values.

Mode Tag Value	Name
0x0000	**Auto**
0x0001	**Quick**
0x0002	**Quiet**
0x0003	**LowNoise**
0x0004	**LowEnergy**
0x0005	**Vacation**
0x0006	**Min**
0x0007	**Max**
0x0008	**Night**
0x0009	**Day**
0x4000	**RapidCool**
0x4001	**RapidFreeze**

8.7.7.1.1. RapidCool Tag

This mode reduces the temperature rapidly, typically above freezing grade.

8.7.7.1.2. RapidFreeze Tag

This mode reduces the temperature rapidly, below freezing grade.

8.7.8. Mode Examples

A few examples of Refrigerator and Temperature Controlled Cabinet modes and their mode tags are provided below.

- For the "Normal" mode, tags: 0x0000 (Auto)
- For the "Energy Save" mode, tags: 0x0004 (LowEnergy)
- For the "Rapid Cool" mode, tags: 0x4000 (RapidCool)

8.8. Refrigerator Alarm Cluster

This cluster is a derived cluster of Alarm Base cluster and provides the alarm definition related to refrigerator and temperature controlled cabinet devices.

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8.8.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

8.8.2. Classification

Hierarchy	Role	Scope	PICS Code
Alarm Base	Application	Endpoint	REFALM

8.8.3. Cluster ID

ID	Name
0x0057	Refrigerator Alarm

8.8.4. Features

Bit	Code	Feature	Conformance	Summary
0	**RESET**	Reset	X	Supports the ability to reset alarms

8.8.5. Data Types

8.8.5.1. AlarmBitmap Type

This data type is derived from map32.

Bit	Name	Summary	Conformance
0	**DoorOpen**	The cabinet's door has been open for a vendor defined amount of time.	M

8.8.6. Attributes

8.8.6.1. Mask Attribute

If the generation of the alarm has not been suppressed at the device itself, then this attribute SHALL have these fixed values.

Bit	Name	Value
0	**DoorOpen**	1

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This alarm SHALL be cleared only when the door is closed (manual action).

If the generation of the alarm is suppressed at the device itself, then bit 0 SHALL have a value of 0. It SHALL be re-set to 1 if the alarm is re-enabled at the device itself.

8.8.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x01	**ModifyEnabledAlarms**				X

8.9. Laundry Dryer Controls Cluster

This cluster provides a way to access options associated with the operation of a laundry dryer device type.

8.9.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

8.9.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	DRYERCTRL

8.9.3. Cluster ID

ID	Name
0x004A	Laundry Dryer Controls

8.9.4. Data Types

8.9.4.1. DrynessLevelEnum Type

This data type is derived from enum8.

This enum provides a representation of the level of dryness that will be used while drying in a selected mode.

It is up to the device manufacturer to determine the mapping between the enum values and the corresponding temperature level.

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Value	Name	Summary	Conformance
0	**Low**	Provides a low dryness level for the selected mode	M
1	**Normal**	Provides the normal level of dryness for the selected mode	M
2	**Extra**	Provides an extra dryness level for the selected mode	M
3	**Max**	Provides the max dryness level for the selected mode	M

8.9.5. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SupportedDrynessLevels**	list[DrynessLevelEnum]	1 to 4			R V	M
0x0001	**SelectedDrynessLevel**	DrynessLevelEnum	desc	X		RW VO	M

8.9.5.1. SupportedDrynessLevels Attribute

This attribute SHALL indicate the list of supported dryness levels available to the appliance in the currently selected mode. The dryness level values are determined by the manufacturer. At least one dryness level value SHALL be provided in the SupportedDrynessLevels list. The list of dryness levels MAY change depending on the currently-selected Laundry Dryer mode.

8.9.5.2. SelectedDrynessLevel Attribute

This attribute SHALL indicate the currently-selected dryness level and it SHALL be the index into the SupportedDrynessLevels list of the selected dryness level.

If an attempt is made to write this attribute with a value other than null or a value contained in SupportedDrynessLevels, a CONSTRAINT_ERROR response SHALL be sent as the response. If an attempt is made to write this attribute while the device is not in a state that supports modifying the dryness level, an INVALID_IN_STATE error SHALL be sent as the response. A value of null SHALL indicate that there will be no dryness level setting for the current mode.

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8.10. Oven Cavity Operational State Cluster

This cluster is derived from the Operational State cluster and provides an interface for monitoring the operational state of an oven.

8.10.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Set Pause and Resume commands as disallowed (previously a Device Type override)

8.10.2. Classification

Hierarchy	Role	Scope	PICS Code
Operational State	Application	Endpoint	OVENOPSTATE

8.10.3. Cluster ID

ID	Name
0x0048	Oven Cavity Operational State

8.10.4. Attributes

8.10.4.1. PhaseList Attribute

As defined in the base cluster, this attribute indicates a list of names of different phases that the device can go through for the selected function or mode.

For this derived cluster, only these pre-defined strings may be used in the PhaseList attribute:

"pre-heating", "pre-heated", and "cooling down".

Other values SHALL NOT be used.

As defined in the base cluster, a null value indicates that the device does not present phases during its operation. When this attribute's value is null, the CurrentPhase attribute SHALL also be set to null.

8.10.5. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**Pause**				X

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ID	Name	Direction	Response	Access	Conformance
0x01	**Stop**				
0x02	**Start**				
0x03	**Resume**				X
0x04	**Operational-Comman-dResponse**				

8.11. Oven Mode Cluster

This cluster is derived from the Mode Base cluster and defines additional mode tags and name-spaced enumerated values for oven devices.

8.11.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Set OnOff feature, StartUpMode, and OnMode as disallowed (previously a Device Type override)

8.11.2. Classification

Hierarchy	Role	Scope	PICS Code
Mode Base	Application	Endpoint	OTCCM

8.11.3. Cluster ID

ID	Name
0x0049	Oven Mode

8.11.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**DEPONOFF**	OnOff	X	Dependency with the OnOff cluster

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8.11.5. Data Types

8.11.5.1. ModeOptionStruct Type

The table below lists the changes relative to the Mode Base cluster for the fields of the ModeOptionStruct type. A blank field indicates no change.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Label**						M
1	**Mode**						M
2	**ModeTags**		1 to 8				M

8.11.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SupportedModes**						
0x0001	**CurrentMode**						
0x0002	**StartUpMode**						X
0x0003	**OnMode**						X

8.11.6.1. SupportedModes Attribute

At least one entry in the SupportedModes attribute SHALL include the Bake mode tag in the ModeTags field list.

8.11.7. Derived Cluster Namespace

This namespace includes definitions for data associated exclusively with the derived cluster.

8.11.7.1. Mode Tags

The following table defines the base and derived-cluster-specific ModeTag values.

Mode Tag Value	Name
0x0000	**Auto**
0x0001	**Quick**
0x0002	**Quiet**
0x0003	**LowNoise**
0x0004	**LowEnergy**

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Mode Tag Value	Name
0x0005	**Vacation**
0x0006	**Min**
0x0007	**Max**
0x0008	**Night**
0x0009	**Day**
0x4000	**Bake**
0x4001	**Convection**
0x4002	**Grill**
0x4003	**Roast**
0x4004	**Clean**
0x4005	**Convection Bake**
0x4006	**Convection Roast**
0x4007	**Warming**
0x4008	**Proofing**
0x4009	**Steam**
0x400A	**Air Fry**
0x400B	**Air Sous Vide**
0x400C	**Frozen Food**

8.11.7.1.1. Bake Tag

This mode sets the device into baking mode for baking food items.

8.11.7.1.2. Convection Tag

This mode sets the device into convection mode which creates an airflow within the device during the cooking duration.

8.11.7.1.3. Grill Tag

This mode sets the device into grill mode for grilling food items. This is the same as Broil for many regions.

8.11.7.1.4. Roast Tag

This mode sets the device into roast mode for roasting food items.

8.11.7.1.5. Clean Tag

This mode sets the device into cleaning mode to clean the internal components of the appliance.

NOTE Depending on regulatory or other requirements, the following restrictions could

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apply to the use of this mode:

- It MAY NOT be possible to start this mode from a client. It MAY need to be started at the device due to the temperatures used during this mode.
- The setting of the oven temperature or other features MAY be limited or not available while this mode is running.
- Oven temperature measurement MAY NOT be available while this mode is running.

8.11.7.1.6. Convection Bake Tag

This mode sets the device into convection bake mode which creates an airflow within the device during the baking duration.

8.11.7.1.7. Convection Roast Tag

This mode sets the device into convection roast mode which creates an airflow within the device during the roasting duration.

8.11.7.1.8. Warming Tag

This mode sets the device into a warming mode which begins warming the cavity.

8.11.7.1.9. Proofing Tag

This mode sets the device into proofing mode which creates an environment ready for proofing.

8.11.7.1.10. Air Fry Tag

This mode sets the device into a cooking mode at a high temperature with high airflow to mimic deep frying.

8.11.7.1.11. Air Sous Vide Tag

This mode sets the device into a precisely regulated cooking mode at a low temperature.

8.11.7.1.12. Frozen Food Tag

This mode sets the device into a mode suitable for heating or cooking frozen food.

8.11.8. Mode Examples

A few examples of Oven modes and their mode tags are provided below.

- For the "Convection" mode, tags: 0x4001 (Convection)
- For the "Bake" mode, tags: 0x4000 (Bake)
- For the "Bake and Warm" mode, tags: 0x4000 (Bake), 0x4007 (Warming)
- For the "Convection Cook and Clean" mode, tags: 0x4001 (Convection), 0x4004 (Clean)

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8.12. Microwave Oven Mode Cluster

This cluster is derived from the Mode Base cluster and defines additional mode tags and name-spaced enumerated values for microwave oven devices.

8.12.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Set OnOff feature as disallowed

8.12.2. Classification

Hierarchy	Role	Scope	PICS Code
Mode Base	Application	Endpoint	MWOM

8.12.3. Cluster ID

ID	Name
0x005E	Microwave Oven Mode

8.12.4. Features

Bit	Code	Feature	Conformance	Summary
0	**DEPONOFF**	OnOff	X	Dependency with the OnOff cluster

8.12.5. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SupportedModes**						
0x0001	**CurrentMode**						
0x0002	**StartUpMode**						X
0x0003	**OnMode**						X

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8.12.5.1. SupportedModes Attribute

Exactly one entry in the SupportedModes attribute SHALL include the Normal mode tag in the ModeTags field.

The Normal and Defrost mode tags are mutually exclusive and SHALL NOT both be used together in a mode's ModeTags.

8.12.6. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**ChangeToMode**				X
0x01	**ChangeToModeResponse**				X

8.12.7. Derived Cluster Namespace

This namespace includes definitions for data associated exclusively with the derived cluster.

8.12.7.1. Mode Tags

The following table defines the base and derived-cluster-specific ModeTag values.

Mode Tag Value	Name
0x0000	**Auto**
0x0001	**Quick**
0x0002	**Quiet**
0x0003	**LowNoise**
0x0004	**LowEnergy**
0x0005	**Vacation**
0x0006	**Min**
0x0007	**Max**
0x0008	**Night**
0x0009	**Day**
0x4000	**Normal**
0x4001	**Defrost**

8.12.7.1.1. Normal tag

This is the normal mode of operation for general cooking of food.

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8.12.7.1.2. Defrost tag

This is a mode optimized for defrosting food.

8.13. Microwave Oven Control Cluster

This cluster defines the requirements for the Microwave Oven Control cluster.

This cluster has dependencies with the Operational State and Microwave Oven Mode clusters. The Operational State cluster and the Microwave Oven Mode clusters, or derivatives of those clusters SHALL appear on the same endpoint as this cluster.

8.13.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

8.13.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	MWOCTRL

8.13.3. Cluster ID

ID	Name
0x005F	Microwave Oven Control

8.13.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**PWRNUM**	PowerAsNumber	O.a	Power is specified as a unitless number or a percentage
1	**WATTS**	PowerInWatts	P, O.a	Power is specified in Watts
2	**PWRLMTS**	PowerNumberLimits	[PWRNUM]	Supports the limit attributes used with the PWRNUM feature

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8.13.5. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**CookTime**	elapsed-s	1 to MaxCookTime			R V	M
0x0001	**MaxCookTime**	elapsed-s	1 to 86400	F		R V	M
0x0002	**PowerSetting**	uint8	desc			R V	PWRNUM
0x0003	**MinPower**	uint8	1 to 99	F		R V	PWRLMTS
0x0004	**MaxPower**	uint8	(MinPower + 1) to 100	F		R V	PWRLMTS
0x0005	**PowerStep**	uint8	desc	F		R V	PWRLMTS
0x0006	**SupportedWatts**	list[uint16]	1 to 10	F		R V	P, WATTS
0x0007	**SelectedWattIndex**	uint8	desc			R V	P, WATTS
0x0008	**WattRating**	uint16	all	F	MS	R V	O

8.13.5.1. CookTime Attribute

This attribute SHALL indicate the total cook time associated with the operation of the device.

This attribute SHALL remain unchanged during the operation of the oven unless the value is changed via a command or out-of-band action.

8.13.5.2. MaxCookTime Attribute

This attribute SHALL indicate the maximum value to which the CookTime attribute can be set.

8.13.5.3. PowerSetting Attribute

This attribute SHALL indicate the power level associated with the operation of the device.

If the MinPower, MaxPower, and PowerStep attributes are not supported:

- The minimum value of this attribute SHALL be 10,
- The maximum value of this attribute SHALL be 100,
- The value SHALL be in even multiples of 10,
- The default value SHALL be 100.

If the MinPower, MaxPower, and PowerStep attributes are supported:

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- The value of this attribute SHALL be between MinPower and MaxPower inclusive.
- The value of this attribute SHALL be such that `(PowerSetting - MinPower) % PowerStep == 0`

8.13.5.4. MinPower Attribute

This attribute SHALL indicate the minimum value to which the PowerSetting attribute that can be set on the server.

8.13.5.5. MaxPower Attribute

This attribute SHALL indicate the maximum value to which the PowerSetting attribute that can be set on the server.

8.13.5.6. PowerStep Attribute

This attribute SHALL indicate the increment of power that can be set on the server.

The value of this attribute SHALL be between 1 and MaxPower inclusive.

The value of this attribute SHALL be such that `(MaxPower - MinPower) % PowerStep == 0`

For example, if MinPower is 1, MaxPower is 10, and PowerSetting can be set to any integer between MinPower and MaxPower, PowerStep would be set to 1.

8.13.5.7. SupportedWatts Attribute

This attribute SHALL indicate the list of power levels (in W) supported by the server.

8.13.5.8. SelectedWattIndex Attribute

This attribute SHALL indicate the index into the list of SupportedWatts of the currently selected power setting.

The index SHALL be a valid index into the SupportedWatts list.

8.13.5.9. WattRating Attribute

This attribute SHALL indicate the rating, in Watts, of the microwave power of the oven.

Supporting this attribute can assist clients in suggesting cooking settings for various foods and beverages.

8.13.6. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**SetCookingParameters**	client ⇒ server	Y	O	M
0x01	**AddMoreTime**	client ⇒ server	Y	O	O

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8.13.6.1. Command Responses Impacted By the Operational State Cluster

When the Operational State cluster or a cluster derived from it is included on the same endpoint as this cluster, the server MAY respond to commands defined in this cluster with an INVALID_IN_STATE response if the server is unable to accept those command due to restrictions imposed by the current operational state of the device or other factors.

8.13.6.2. SetCookingParameters Command

This command is used to set the cooking parameters associated with the operation of the device. This command supports the following fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**CookMode**	uint8	desc		desc	O.b+
1	**CookTime**	elapsed-s	1 to MaxCookTime		30	O.b+
2	**PowerSetting**	uint8	MinPower to MaxPower		MaxPower	[PWRNUM].b+
3	**WattSettingIndex**	uint8	desc		MS	[WATTS].b+
4	**StartAfterSetting**	bool	all		false	O

8.13.6.2.1. CookMode Field

This field SHALL indicate the value to which the CurrentMode attribute of the Microwave Oven Mode cluster should be set. The value of this field SHALL be one from the list of SupportedModes from the Microwave Oven Mode cluster.

If this field is missing, the CurrentMode attribute SHALL be set to a mode having the Normal mode tag.

8.13.6.2.2. CookTime Field

This field SHALL indicate the CookTime associated with the operation of the device. The value of this field SHALL be subject to the constraints of the CookTime attribute of this cluster.

If this field is missing, the CookTime attribute SHALL be set to 30 seconds by the server.

8.13.6.2.3. PowerSetting Field

This field SHALL indicate the PowerSetting associated with the operation of the device. The value of this field SHALL be subject to the constraints of the PowerSetting attribute of this cluster. If the PowerSetting field does not conform to the constraints of the PowerSetting attribute, the server SHALL return a CONSTRAINT_ERROR status.

If this field is missing, the PowerSetting attribute SHALL be set to 100 if MaxPower is not supported

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by the server, otherwise it SHALL be set to MaxPower if the MaxPower attribute is supported by the server.

8.13.6.2.4. WattSettingIndex Field

This field SHALL indicate the value to which the SelectedWattIndex attribute is set. If the value of this field is greater than or equal to the length of the SupportedWatts attribute list, the server SHALL return a CONSTRAINT_ERROR status and the value of the SelectedWattIndex attribute SHALL be unchanged.

If this field is missing, the SelectedWattIndex attribute SHALL be set by the server to the index associated with the highest Watt setting for the selected CookMode.

8.13.6.2.5. StartAfterSetting Field

This field SHALL indicate whether or not oven operation SHALL be started when the command is received.

8.13.6.2.6. Effect on Receipt

If this command is received while the operational state of the server cannot support the command in that state, the server SHALL respond with an INVALID_IN_STATE response and the attributes and state SHALL remain unchanged. See Operational State Cluster for details on the operational states.

If this command is received and any fields sent with the command do not meet the constraints of any of the associated attributes (i.e. bearing the same name as the field or as described in the field description), the server SHALL respond with a response of CONSTRAINT_ERROR and the attributes and state SHALL remain unchanged.

If the StartAfterSetting field is present in the command but the Start command of the Operational State cluster or one of its derivatives on the same endpoint as this cluster is not supported, the server SHALL respond with a response of INVALID_COMMAND and the attributes and state SHALL remain unchanged.

Otherwise:

- The attributes associated with any included fields SHALL be set to the values of those fields.
- The attributes associated with any missing fields SHALL be set to the values as specified in descriptions of the missing fields.
- If the StartAfterSetting field is included:
  - If the value of StartAfterSetting is TRUE, oven operation SHALL start.
  - If the value of StartAfterSetting is FALSE, oven operation SHALL NOT start.

8.13.6.3. AddMoreTime Command

This command is used to add more time to the CookTime attribute of the server.

This command supports these fields:

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**TimeToAdd**	elapsed-s	1 to MaxCookTime			M

8.13.6.3.1. TimeToAdd Field

This field SHALL indicate the number of seconds to be added to the CookTime attribute.

8.13.6.3.2. Effect on Receipt

Upon receipt of this command, if the sum of the value of the TimeToAdd field and the current value of the CookTime attribute is greater than the MaxCookTime attribute, the server SHALL respond with a response of CONSTRAINT_ERROR and the command SHALL be ignored.

If this command is received while the operational state of the server cannot support the command in that state, the server SHALL respond with an INVALID_IN_STATE response. See Operational State Cluster for details on the operational states.

Otherwise, the server SHALL add the value of the TimeToAdd field to the value of the CookTime attribute of this cluster and the server SHALL add the value of the TimeToAdd field to the value of the CountdownTime attribute of the Operational State cluster if that cluster or a derivative is on the same endpoint as this cluster.

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Chapter 9. Energy Management

The Cluster Library is made of individual chapters such as this one. See Document Control in the Cluster Library for a list of all chapters and documents. References between chapters are made using a *X.Y* notation where *X* is the chapter and *Y* is the sub-section within that chapter.

9.1. General Description

9.1.1. Introduction

The clusters specified in this chapter are for use typically in Energy Management applications with associated security controls at the application layer. These clusters may be used in any application domain.

9.1.2. Data Types

9.1.3. TariffUnitEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**kWh**	Kilowatt Hours	M
1	**kVAh**	Kilovolt-amp Hours	M

9.1.4. PowerThresholdSourceEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0x00	**Contract**	The power threshold comes from a signed contract	M
0x01	**Regulator**	The power threshold comes from a legal regulator	M
0x02	**Equipment**	The power threshold comes from a certified limits of the meter	M

9.1.5. TariffPriceTypeEnum Type

This data type is derived from enum8.

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Value	Name	Summary	Conformance
0x00	**Standard**	Standard tariff price	M
0x01	**Critical**	Price during CPP events	M
0x02	**Virtual**	Price during VPP events	M
0x03	**Incentive**	Price incentives	M
0x04	**IncentiveSignal**	Price incentive signals	M

9.1.5.1. Standard Value

This value SHALL indicate that a price comes from a standard tariff rate.

9.1.5.2. Critical Value

This value SHALL indicate that a price comes from a critical peak pricing event.

9.1.5.3. Virtual Value

This value SHALL indicate that a price comes from a virtual power plant event.

9.1.5.4. Incentive Value

This value SHALL indicate that a price comes from a incentive event.

9.1.5.5. IncentiveSignal Value

This value SHALL indicate that a price is synthesized from a non-tariff source; e.g. gCO2e/kWh.

9.1.6. PowerThresholdStruct Type

This struct represents information about a power threshold.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**PowerThreshold**	power-mW	all				O.b
1	**ApparentPowerThreshold**	power-mVA	all				O.b
2	**PowerThresholdSource**	PowerThresholdSourceEnum		X			M

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9.1.6.1. PowerThreshold Field

This field SHALL indicate the instantaneous power demand that can be distributed to the customer without any risk of overload. The value is in mW and could be provided by the contract or Distribution Network Operator (DNO).

9.1.6.2. ApparentPowerThreshold Field

This field SHALL indicate the instantaneous apparent power demand that can be distributed to the customer without any risk of overload. The value is in mVA and could be provided by the contract or Distribution Network Operator (DNO).

9.1.6.3. PowerThresholdSource Field

This field SHALL indicate the reason why the PowerThreshold field was set. If the reason is unavailable, this field SHALL be null.

9.1.7. Cluster List

This section lists the Energy Management specific clusters as specified in this chapter.

Table 15. Overview of the Energy Management Clusters

Cluster ID	Cluster Name	Description
0x0098	Device Energy Management	Generic cluster for enabling power adjustment, sharing power forecasts and modifying power forecasts for devices
0x0099	Energy EVSE	Commands and attributes for controlling an EVSE
0x009D	Energy EVSE Mode	Commands and attributes for setting the mode of an EVSE
0x0094	Water Heater Management	Commands and attributes for controlling a Water Heater
0x009E	Water Heater Mode	Commands and attributes for setting the mode of a Water Heater
0x009B	Energy Preference	Attributes and commands for expressing user preferences around energy consumption
0x009F	Device Energy Management Mode	Commands and attributes for setting the mode of devices with energy management capability
0x0095	Commodity Price	Commands and attributes for reporting price

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Cluster ID	Cluster Name	Description
0x0B01	Meter Identification	Attributes for advanced utility metering device information
0x0700	Commodity Tariff	Attributes for commodity tariff reporting
0x0B07	Commodity Metering	Attributes for commodity metering
0x00A0	Electrical Grid Conditions	Attributes for reporting conditions on the electricity grid

9.2. Device Energy Management Cluster

This cluster allows a client to manage the power draw of a device. An example of such a client could be an Energy Management System (EMS) which controls an Energy Smart Appliance (ESA).

In most deployments the EMS will be the client, and the ESA will host the Device Energy Management Cluster server.

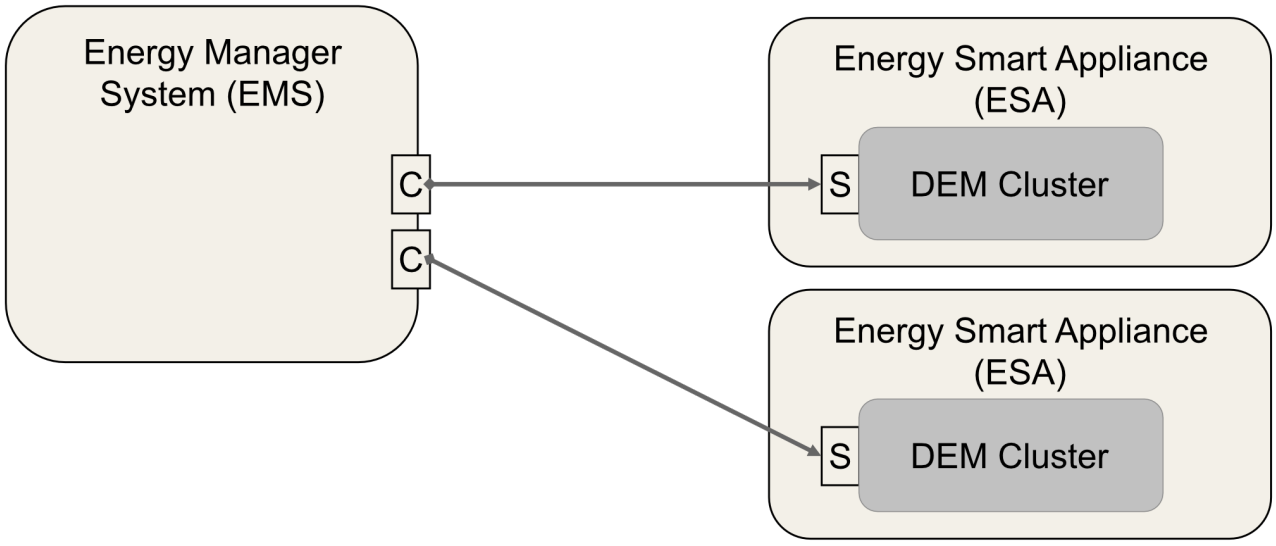


Figure 21. Example of the how an EMS is a client of multiple ESAs Device Energy Management clusters.

This cluster is intended to be generic in nature and could apply to any electrical load or generator (e.g. a Battery Electric Storage System - BESS, solar PV inverter, EVSE, HVAC, heat pump, hot water heater, white goods appliances etc).

It consists of the following areas which SHALL be supported by all devices implementing this cluster:

- Description of ESA and its capabilities & power limits (sometimes referred to as a nameplate)
- Current state of operation (including user opt-out, safety limitations / alarms)

There are some optional capabilities that some ESAs may be able to offer:

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- Ability to control the load or generation
- Forecast data, including when it can be flexible (i.e. modify the power or time period)
- The ability to have their power profile adjusted by an EMS, and to provide an updated Forecast back to the EMS.

This allows the EMS to manage multiple home loads and where ESAs can be flexible, continuously optimizing the home energy to minimize cost, reduce CO2 impact, maximize self-consumption of solar PV and provide Demand Side Response (DSR) Grid services.

It is likely that the ESA may also use the Pricing Cluster to obtain incentive signals such as 'grid carbon intensity', 'time of use' or 'type of use' tariffs to schedule its operation to run at the cheapest and greenest times.

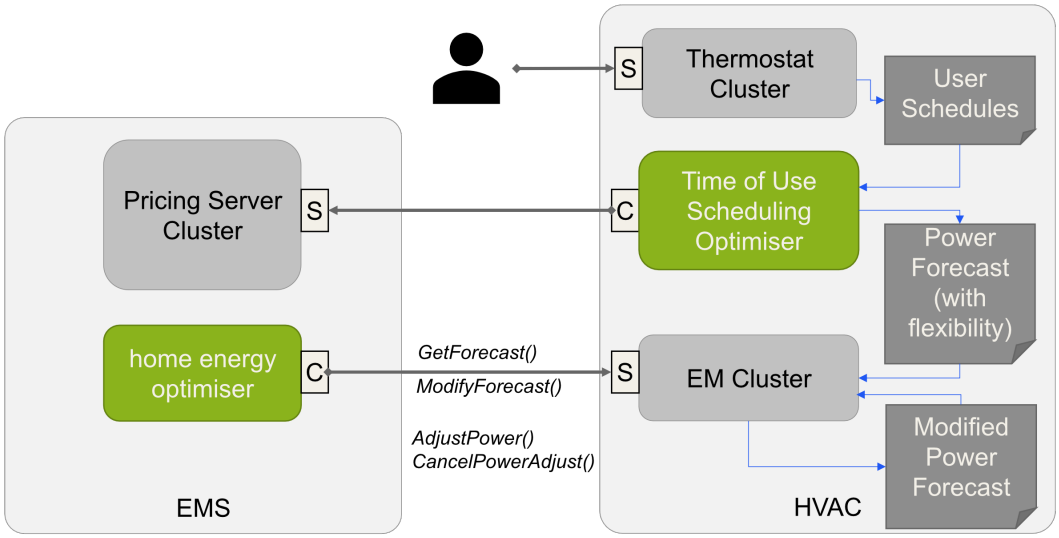


Figure 22. Example of the how an HVAC may use multiple clusters

NOTE Grid Services are market dependent and will use other protocols ([OpenADR] / [IEEE2030.5]) to communicate grid events to the EMS. These are outside the scope of Matter.

NOTE Different markets may follow different approaches, but the UK [PAS1878] and [EUCodeOfConduct] give examples of how ESAs may be mandated to support these features in the future.

9.2.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Updates after 0.7 Ballot review
3	Updates to match cluster spec updates

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Revision	Description
4	Updates to feature conformance. Corrected ForecastID type

9.2.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	DEM

9.2.3. Cluster ID

Cluster ID	Cluster Name
0x0098	Device Energy Management

9.2.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**PA**	PowerAdjustment	O	Allows an EMS to make a temporary power adjustment (within the limits offered by the ESA).
1	**PFR**	PowerForecastReporting	[!PA].a, (STA | PAU | FA | CON), O	Allows an ESA to advertise its indicative future power consumption vs time.
2	**SFR**	StateForecastReporting	[!PA].a	Allows an ESA to advertise its indicative future state vs time.
3	**STA**	StartTimeAdjustment	O	Allows an EMS to delay an ESA's planned operation.
4	**PAU**	Pausable	O	Allows an EMS to pause an ESA's planned operation.

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Bit	Code	Feature	Conformance	Summary
5	**FA**	ForecastAdjustment	O	Allows an EMS to adjust an ESA's planned operation.
6	**CON**	ConstraintBasedAdjustment	O	Allows an EMS to request constraints to an ESA's planned operation.

The conformance rules specified above can be described as:

- At least one of the features SHALL be supported.
- At most one of the 'SFR' and 'PFR' features SHALL be supported.
- If one or more of the 'STA', 'PAU', 'FA', or 'CON' features are supported, then either the 'PFR' or the 'SFR' feature SHALL also be supported.
- If PA is supported, SFR SHALL NOT be supported.

9.2.4.1. PowerAdjustment Feature

For Energy Smart Appliances (ESA) the definition of being 'smart' mandates that they can report their current power adjustment capability and have an EMS request a temporary adjustment. This may typically be to curtail power requirements during peak periods, but can also be used to turn on an ESA if there is excess renewable or local generation (Solar PV).

For example, a home may have solar PV which often produces more power than the home requires, resulting in the excess power flowing into the grid. This excess power naturally fluctuates when clouds pass overhead and other loads in the home are switched on and off.

EVSE Example: An EMS may therefore be able to turn on the EVSE (if the vehicle is plugged in) and can start charging the vehicle, and periodically modify the charging power depending on PV generation and other home loads, so as to minimize import and export to the grid. An EMS may also use this feature to control the discharging (and re-charging) of the vehicle if the EVSE and vehicle support the V2X feature of the EVSE cluster of the associated EVSE device.

9.2.4.2. PowerForecastReporting Feature

For Energy Smart Appliances (ESA) the definition of being 'smart' implies that they can report their indicative forecast power demands or generation, to a greater or lesser extent. For some ESAs this is highly predictable (in terms of both power and time), in other appliances this is more challenging and only a basic level of forecast is possible.

Forecasts are defined from a current time, using a slot format, where the slot is akin to a relatively constant operating mode.

Washing machine example: a washing machine may have stages of a washing cycle: heating, tum-

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bling, rinse and spin stages. At each stage, the approximate minimum and maximum power consumption may be known, as well as the duration of that stage.

In some circumstances the ESA may allow the stage to be delayed or paused (subject to safety and manufacturer's discretion and user preferences).

Typically, appliances with a heating element cannot have their power consumption adjusted and can only be paused or delayed.

Some ESAs may not be flexible other than a delayed cycle start (for example, once the washing cycle has been started then they run continuously until the cycle completes).

Appliances that only support the PowerForecastReporting and not any of the adjustment features may indicate that they are not flexible in the forecast slot format.

The PowerForecastReporting and the adjustment features aim to align to the [SAREF4ENER] ontology.

Inverter driven ESAs: some inverter driven ESAs can consume or generate a variable amount of power.

For example, a single phase EVSE can be adjusted in the range of 6-32Amps in 0.6 Amp steps in EU or on a hardwired 120V supply in the range of 6-15 Amps in US.

For example, a home battery may be adjusted to charge or discharge in steps of 1W.

For example, a heat pump may be able to modulate its compressor inverter between 20-100% of its rated power.

The ESA indicates its power adjustment range and its nominal power consumption as part of its Forecast.

9.2.4.3. StateForecastReporting Feature

Some ESAs do not know their actual power consumption, but do know the state of operation. Like the PowerForecastingReporting feature, this uses the same slot structure mechanism to indicate a change in state vs time.

An external observing EMS may have access to real-time meter readings, and could learn the typical power consumption based on the advertised internal state of the ESA.

To enable this capability, the ESA SHALL report its internal operational state using an manufacturer specific value.

Once the EMS has built a model of the state vs observed power consumption, it may request a forecast adjustment for particular times of the day, encouraging the ESA to use power at alternative times.

9.2.4.4. StartTimeAdjustment Feature

ESAs which support the Start Time Adjustment feature, allow an EMS to recommend a change to

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the start time of the energy transfer that the ESA has previously suggested it would use.

Washing machine example: A Washing Machine may have been set to start a wash cycle at 9pm when the variable tariff normally reduces.

However, the EMS is aware that a grid event has occurred, making it cheaper to run the cycle at a later time, but the washing machine is not aware of this.

The EMS first requests the Forecast data from each of its registered ESAs. It determines that the washing machine has a power profile suggesting it will start the wash cycle at 9pm, but the EMS now knows that the grid event means it will be cheaper to delay the start until 11pm.

The EMS can then optimize the cost by asking the washing machine to delay starting the wash cycle until 11pm.

It does this by sending a StartTimeAdjustRequest to the washing machine to request delaying the start of the washing cycle.

9.2.4.5. Pausable Feature

ESAs which support the Pausable feature, allow an EMS to recommend a pause in the middle of a forecast power profile that the ESA is currently using.

Washing machine example: A Washing Machine is in operation, and starting its water heating step.

However, the EMS becomes aware from the smart meter that the total home load on the grid is close to exceeding its allowed total grid load.

The EMS first requests the Forecast data from each of its registered ESAs. It determines that the washing machine has a power profile suggesting its current step in the wash cycle is using power to heat the water, but that this step can be paused.

The EMS can then reduce the grid load by asking the washing machine to pause the wash cycle for a short duration.

It does this by sending a PauseRequest to the washing machine to request pausing the current step of the forecast power usage for a period to allow other home loads to finish before resuming the washing cycle.

9.2.4.6. ForecastAdjustment Feature

ESAs which support the Forecast adjustment feature, allow an EMS to recommend a change to the start, duration and/or power level limits of the steps of the power profile that the ESA has previously suggested it would use.

Heat pump and Solar PV example: A heat pump may have the ability to heat hot water as well as heating the home. The heat pump scheduling system may have determined that the home will be unoccupied during the day, or that the indoor temperature is above the set-point and so it knows that it will not need to heat the home.

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However, the hot water tank is likely to need to be reheated before the homeowner comes home in the evening. The heat pump is not aware that the property also has a solar PV inverter which is also an ESA that is communicating with the EMS.

The EMS first requests the Forecast data from each of its registered ESAs. It determines that the heat pump has a power profile suggesting it needs to heat hot water around 6pm. The solar PV inverter has forecast that it will generate 3.6kW of power during the middle of the day and into the afternoon before the sun goes down.

The EMS can then optimize the home considering other non-ESA loads and can ask the heat pump to heat the hot water around 3pm when it has forecast that excess solar power will be available.

It does this by sending a ModifyForecastRequest to the heat pump and asks the heat pump to expect to run at a lower power consumption (within the solar excess power) which requires the heat pump to run for a longer duration to achieve its required energy demand.

9.2.4.7. ConstraintBasedAdjustment Feature

ESAs which support the Constraint-Based Adjustment feature allow an EMS to inform the ESA of periods during which power usage should be modified (for example when the EMS has been made aware that the grid supplier has requested reduced energy usage due to overall peak grid demand) and may cause the ESA to modify the intended power profile has previously suggested it would use.

EVSE example: An EVSE scheduling system may have determined that the vehicle would be charged starting at a moderate rate at 1am, so that it has enough charge by the time it is needed later that morning.

However, the DSR service provider has informed the EMS that due to high forecast winds it is now forecast that there will be very cheap energy available from wind generation between 2am and 3am.

The EMS first requests the Forecast data from each of its registered ESAs. It determines that the EVSE has a power profile suggesting it plans to start charging the vehicle at 1am.

The EMS can then try to reduce the cost of charging the EV by informing the EVSE of the desire to increase the charging between scheduled times.

It does this by sending a RequestConstraintBasedForecast to the EVSE and asks it to run at a higher NominalPower consumption during the constraint period, which may require it to decrease its charge rate outside the constraint period to achieve its required energy demand.

9.2.5. Dependencies

This cluster does not report electrical power and electrical energy. Devices that use this cluster SHALL also support the Electrical Power Measurement and optionally support the Electrical Energy Measurement cluster to allow an energy management system to perform its role. See Device Type library for device specific details.

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9.2.6. Definitions

9.2.6.1. Power

Power is defined in the main specification (see Data Model) in units of milliwatts. It is a signed value, where positive values indicate the direction of power flow into the node. A negative value indicates that the device is supplying power, and thus reducing the overall load of the premises on the grid supply.

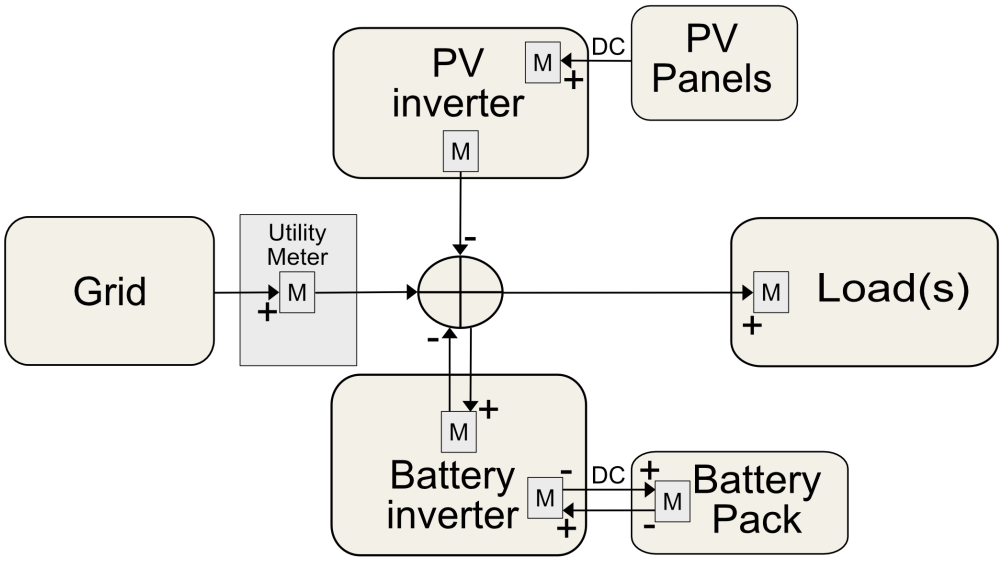


Figure 23. Power flows into devices

Solar PV inverter example: A solar PV inverter is normally connected to the premises' wiring, so its generation power will be indicated as a negative integer, since power is flowing out of the node towards the mains connection. Note that at night (when there is no solar production) the PV inverter will consume some standby power, and this will result in a small positive power reading.

Grid Power: Power from the grid is measured by a utility meter which is imported into the node, so positive power values indicate that power is flowing into the premises. Negative values indicate power is flowing back towards the grid.

EVSE example: An EVSE provides power to the EV (when charging) so a positive value indicates that the power is flowing into the EVSE from the mains supply when the EV is charging, and is acting as a load on the grid supply, and negative value indicates that the EV is discharging.

BESS example: A battery storage inverter normally provides power to loads when discharging, so negative power indicates discharging (power is flowing back to the mains supply) therefore reducing the load on the grid, and positive power indicates power flowing from the mains to the battery during charging (adding load to the grid).

Washing Machine example: A washing machine only consumes power (i.e. is a load), so it will always have a positive power value.

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9.2.6.2. Energy

Energy is defined in the main specification (see Data Model). This is defined in units of mWh (milli-watt-hours). It is signed value, where positive values indicate the direction of current flow towards a load.

9.2.7. Data Types

9.2.7.1. CostTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Financial**	Financial cost	M
1	**GHGEmissions**	Grid CO2e grams cost	M
2	**Comfort**	Consumer comfort impact cost	M
3	**Temperature**	Temperature impact cost	M

9.2.7.1.1. Financial Value

This value SHALL indicate that the cost is related to the financial cost to provide the energy.

9.2.7.1.2. GHGEmissions Value

This value SHALL indicate that the cost is related to greenhouse gas emissions (in grams of CO2e).

9.2.7.1.3. Comfort Value

This value SHALL indicate that the cost is related to some abstract sense of comfort expressed by the consumer; a higher value indicates more discomfort. For example, a consumer may be more comfortable knowing that their EV is charged earlier in the day in case there is a sudden need to depart and drive to the hospital. Or the consumer may feel inconvenienced by the fact that they need to wait for the washing machine to finish its load so that they can use it again.

9.2.7.1.4. Temperature Value

This value SHALL indicate that the cost is related to the temperature of the home or water being at its setpoint. Some consumers may be more sensitive to being too hot or too cold.

This is expressed in degrees Celsius.

9.2.7.2. ESATypeEnum Type

This data type is derived from enum8.

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Value	Name	Summary	Conformance
0	**EVSE**	EV Supply Equipment	O
1	**SpaceHeating**	Space heating appliance	O
2	**WaterHeating**	Water heating appliance	O
3	**SpaceCooling**	Space cooling appliance	O
4	**SpaceHeatingCooling**	Space heating and cooling appliance	O
5	**BatteryStorage**	Battery Electric Storage System	O
6	**SolarPV**	Solar PV inverter	O
7	**FridgeFreezer**	Fridge / Freezer	O
8	**WashingMachine**	Washing Machine	O
9	**Dishwasher**	Dishwasher	O
10	**Cooking**	Cooking appliance	O
11	**HomeWaterPump**	Home water pump (e.g. drinking well)	O
12	**IrrigationWaterPump**	Irrigation water pump	O
13	**PoolPump**	Pool pump	O
255	**Other**	Other appliance type	O

9.2.7.3. ESAStateEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Offline**	The ESA is not available to the EMS (e.g. start-up, maintenance mode)	M
1	**Online**	The ESA is working normally and can be controlled by the EMS	M
2	**Fault**	The ESA has developed a fault and cannot provide service	M
3	**PowerAdjustActive**	The ESA is in the middle of a power adjustment event	PA

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Value	Name	Summary	Conformance
4	**Paused**	The ESA is currently paused by a client using the PauseRequest command	PAU

9.2.7.4. OptOutStateEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**NoOptOut**	The user has not opted out of either local or grid optimizations	M
1	**LocalOptOut**	The user has opted out of local EMS optimizations only	M
2	**GridOptOut**	The user has opted out of grid EMS optimizations only	M
3	**OptOut**	The user has opted out of all external optimizations	M

9.2.7.5. CauseEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**NormalCompletion**	The ESA completed the adjustment as requested	M
1	**Offline**	The ESA was set to offline	M
2	**Fault**	The ESA has developed a fault could not complete the adjustment	M
3	**UserOptOut**	The user has disabled the ESA's flexibility capability	M
4	**Cancelled**	The adjustment was cancelled by a client	M

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9.2.7.6. AdjustmentCauseEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**LocalOptimization**	The adjustment is to optimize the local energy usage	M
1	**GridOptimization**	The adjustment is to optimize the grid energy usage	M

9.2.7.7. ForecastUpdateReasonEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**InternalOptimization**	The update was due to internal ESA device optimization	M
1	**LocalOptimization**	The update was due to local EMS optimization	M
2	**GridOptimization**	The update was due to grid optimization	M

9.2.7.8. PowerAdjustReasonEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**NoAdjustment**	There is no Power Adjustment active	M
1	**LocalOptimizationAdjustment**	There is PowerAdjustment active due to local EMS optimization	M
2	**GridOptimizationAdjustment**	There is PowerAdjustment active due to grid optimization	M

9.2.7.9. CostStruct Type

This indicates a generic mechanism for expressing cost to run an appliance, in terms of financial, GHG emissions, comfort value etc.

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**CostType**	CostTypeEnum	all				M
1	**Value**	int32	all				M
2	**DecimalPoints**	uint8	all				M
3	**Currency**	uint16	max 999		0		O

9.2.7.9.1. CostType Field

This field SHALL indicate the type of cost being represented (see CostTypeEnum).

9.2.7.9.2. Value Field

This field SHALL indicate the value of the cost. This may be negative (indicating that it is not a cost, but a free benefit).

For example, if the Value was -302 and DecimalPoints was 2, then this would represent a benefit of 3.02.

9.2.7.9.3. DecimalPoints Field

This field SHALL indicate the number of digits to the right of the decimal point in the Value field. For example, if the Value was 102 and DecimalPoints was 2, then this would represent a cost of 1.02.

9.2.7.9.4. Currency Field

Indicates the currency for the value in the Value field. The value of the currency field SHALL match the values defined by [ISO 4217].

This is an optional field. It SHALL be included if CostType is Financial.

9.2.7.10. PowerAdjustStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**MinPower**	power-mW	all				M
1	**MaxPower**	power-mW	min MinPower				M
2	**MinDuration**	elapsed-s	all				M
3	**MaxDuration**	elapsed-s	min MinDuration				M

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9.2.7.10.1. MinPower Field

This field SHALL indicate the minimum power that the ESA can have its power adjusted to.

Note that this is a signed value. Negative values indicate power flows out of the node (e.g. discharging a battery).

9.2.7.10.2. MaxPower Field

This field SHALL indicate the maximum power that the ESA can have its power adjusted to.

Note that this is a signed value. Negative values indicate power flows out of the node (e.g. discharging a battery).

For example, if the charging current of an EVSE can be adjusted within the range of 6A to 32A on a 230V supply, then the power adjustment range is between 1380W and 7360W. Here the MinPower would be 1380W, and MaxPower would be 7360W.

For example, if a battery storage inverter can discharge between 0 to 3000W towards a load, then power is flowing out of the node and is therefore negative. Its MinPower would be -3000W and its MaxPower would be 0W.

In another example, if a battery storage inverter can charge its internal battery, between 0W and 2000W. Here power is flowing into the node when charging. As such the MinPower becomes 0W and MaxPower becomes 2000W.

9.2.7.10.3. MinDuration field

This field SHALL indicate the minimum duration, in seconds, that a controller may invoke an ESA power adjustment. Manufacturers may use this to as an anti-cycling capability to avoid controllers from rapidly making power adjustments.

9.2.7.10.4. MaxDuration field

This field SHALL indicate the maximum duration, in seconds, that a controller may invoke an ESA power adjustment. Manufacturers may use this to protect the user experience, to avoid over heating of the ESA, ensuring that there is sufficient headroom to use or store energy in the ESA or for any other reason.

9.2.7.11. PowerAdjustCapabilityStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**PowerAdjustCapability**	list[PowerAdjustStruct]	max 8	X			M
1	**Cause**	PowerAdjustReasonEnum	all				M

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9.2.7.12. PowerAdjustCapability

This field SHALL indicate how the ESA can be adjusted at the current time.

For example, a battery storage inverter may need to regulate its internal temperature, or the charging rate of the battery may be limited due to cold temperatures, or a change in the state of charge of the battery may mean that the maximum charging or discharging rate is limited.

An empty list SHALL indicate that no power adjustment is currently possible.

Multiple entries in the list allow indicating that permutations of scenarios may be possible.

For example, a 10kWh battery could be at 80% state of charge. If charging at 2kW, then it would be full in 1 hour. However, it could be discharged at 2kW for 4 hours.

In this example the list of PowerAdjustStructs allows multiple scenarios to be offered as follows:

Entry [0] - Charging option:		
MinPower	0	Watts
MaxPower	2000	Watts
MinDuration	60	seconds
MaxDuration	3600	seconds (1 hour)
Entry [1] - Discharging option:		
MinPower	-2000	Watts
MaxPower	0	Watts
MinDuration	60	seconds
MaxDuration	14400	seconds (4 hours)

9.2.7.12.1. Cause field

This field SHALL indicate the cause of the currently active power adjustment.

9.2.7.13. ForecastStruct Type

This indicates a list of 'slots' describing the overall timing of the ESA's planned energy and power use, with different power and energy demands per slot. For example, slots might be used to describe the distinct stages of a washing machine cycle.

Where an ESA does not know the actual power and energy use of the system, it may support the SFR feature and instead report its internal state.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**ForecastID**	uint32	all				M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
1	**ActiveSlotNumber**	uint16	all	X			M
2	**StartTime**	epoch-s	all				M
3	**EndTime**	epoch-s	all				M
4	**EarliestStartTime**	epoch-s	all	X			STA
5	**LatestEndTime**	epoch-s	all				STA
6	**IsPausable**	bool	all				M
7	**Slots**	list[SlotStruct]	max 10				M
8	**ForecastUpdateReason**	ForecastUpdateReasonEnum	all				M

9.2.7.13.1. ForecastID Field

This field SHALL indicate the sequence number for the current forecast. If the ESA updates a forecast, it shall monotonically increase this value.

The ESA does not need to persist this value across reboots, since the EMS SHOULD be able to detect that any previous subscriptions are lost if a device reboots. The loss of a subscription and subsequent re-subscription allows the EMS to learn about any new forecasts.

The value of ForecastID is allowed to wrap.

9.2.7.13.2. ActiveSlotNumber Field

This field SHALL indicate which element of the Slots list is currently active in the Forecast sequence. A null value indicates that the sequence has not yet started.

9.2.7.13.3. StartTime Field

This field SHALL indicate the planned start time, in UTC, for the entire Forecast.

9.2.7.13.4. EndTime Field

This field SHALL indicate the planned end time, in UTC, for the entire Forecast.

9.2.7.13.5. EarliestStartTime Field

This field SHALL indicate the earliest start time, in UTC, that the entire Forecast can be shifted to.

A null value indicates that it can be started immediately.

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9.2.7.13.6. LatestEndTime Field

This field SHALL indicate the latest end time, in UTC, for the entire Forecast.

e.g. for an EVSE charging session, this may indicate the departure time for the vehicle, by which time the charging session must end.

9.2.7.13.7. IsPausable Field

This field SHALL indicate that some part of the Forecast can be paused. It aims to allow a client to read this flag and if it is false, then none of the slots contain SlotIsPausable set to true. This can save a client from having to check each slot in the list.

9.2.7.13.8. Slots Field

This field SHALL contain a list of SlotStructs.

It SHALL contain at least 1 entry, and a maximum of 10.

9.2.7.13.9. ForecastUpdateReason Field

This field SHALL contain the reason the current Forecast was generated.

9.2.7.14. SlotStruct Type

This indicates a specific stage of an ESA's operation.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**MinDuration**	elapsed-s	all				M
1	**MaxDuration**	elapsed-s	all				M
2	**DefaultDuration**	elapsed-s	all				M
3	**ElapsedSlotTime**	elapsed-s	all				M
4	**RemainingSlotTime**	elapsed-s	all				M
5	**SlotIsPausable**	bool	all				PAU
6	**MinPauseDuration**	elapsed-s	all				PAU

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
7	**MaxPauseDuration**	elapsed-s	all				PAU
8	**ManufacturerESAState**	uint16	all				SFR
9	**NominalPower**	power-mW	all				PFR
10	**MinPower**	power-mW	all				PFR
11	**MaxPower**	power-mW	all				PFR
12	**NominalEnergy**	energy-mWh	all				PFR
13	**Costs**	list[CostStruct]	max 5				O
14	**MinPowerAdjustment**	power-mW	all				FA & PFR
15	**MaxPowerAdjustment**	power-mW	all				FA & PFR
16	**MinDurationAdjustment**	elapsed-s	all				FA
17	**MaxDurationAdjustment**	elapsed-s	all				FA

9.2.7.14.1. MinDuration Field

This field SHALL indicate the minimum time (in seconds) that the appliance expects to be in this slot for.

9.2.7.14.2. MaxDuration Field

This field SHALL indicate the maximum time (in seconds) that the appliance expects to be in this slot for.

9.2.7.14.3. DefaultDuration Field

This field SHALL indicate the expected time (in seconds) that the appliance expects to be in this slot for.

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9.2.7.14.4. ElapsedSlotTime Field

This field SHALL indicate the time (in seconds) that has already elapsed whilst in this slot. If the slot has not yet been started, then it SHALL be 0. Once the slot has been completed, then this reflects how much time was spent in this slot.

When subscribed to, a change in this field value SHALL NOT cause the Forecast attribute to be updated since this value may change every 1 second.

When the Forecast attribute is read, then this value SHALL be the most recent value.

9.2.7.14.5. RemainingSlotTime Field

This field SHALL indicate the time (in seconds) that is estimated to be remaining.

Note that it may not align to the DefaultDuration - ElapsedSlotTime since an appliance may have revised its planned operation based on conditions.

When subscribed to, a change in this field value SHALL NOT cause the Forecast attribute to be updated, since this value may change every 1 second.

Note that if the ESA is currently paused, then this value SHALL NOT change.

When the Forecast attribute is read, then this value SHALL be the most recent value.

9.2.7.14.6. SlotIsPausable Field

This field SHALL indicate whether this slot can be paused.

9.2.7.14.7. MinPauseDuration Field

This field SHALL indicate the shortest period that the slot can be paused for. This can be set to avoid controllers trying to pause ESAs for short periods and then resuming operation in a cyclic fashion which may damage or cause excess energy to be consumed with restarting of an operation.

9.2.7.14.8. MaxPauseDuration Field

This field SHALL indicate the longest period that the slot can be paused for.

9.2.7.14.9. ManufacturerESAState Field

This field SHALL indicate a manufacturer defined value indicating the state of the ESA.

This may be used by an observing EMS which also has access to the metering data to ascertain the typical power drawn when the ESA is in a manufacturer defined state.

Some appliances, such as smart thermostats, may not know how much power is being drawn by the HVAC system, but do know what they have asked the HVAC system to do.

Manufacturers can use this value to indicate a variety of states in an unspecified way. For example, they may choose to use values between 0-100 as a percentage of compressor modulation, or could use these values as Enum states meaning heating with fan, heating without fan etc.

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NOTE An ESA SHALL always use the same value to represent the same operating state.

By providing this information a smart EMS may be able to learn the observed power draw when the ESA is put into a specific state. It can potentially then use the ManufacturerESAState field in the Forecast attribute along with observed power drawn to predict the power draw from the appliance and potentially ask it to modify its timing via one of the adjustment request commands, or adjust other ESAs power to compensate.

9.2.7.14.10. NominalPower Field

This field SHALL indicate the expected power that the appliance will use during this slot. It may be considered the average value over the slot, and some variation from this would be expected (for example, as it is ramping up).

9.2.7.14.11. MinPower Field

This field SHALL indicate the lowest power that the appliance expects to use during this slot. (e.g. during a ramp up it may be 0W).

Some appliances (e.g. battery inverters which can charge and discharge) may have a negative power.

9.2.7.14.12. MaxPower Field

This field SHALL indicate the maximum power that the appliance expects to use during this slot. (e.g. during a ramp up it may be 0W). This field ignores the effects of short-lived inrush currents.

Some appliances (e.g. battery inverters which can charge and discharge) may have a negative power.

9.2.7.14.13. NominalEnergy Field

This field SHALL indicate the expected energy that the appliance expects to use or produce during this slot.

Some appliances (e.g. battery inverters which can charge and discharge) may have a negative energy.

9.2.7.14.14. Costs Field

This field SHALL indicate the current estimated cost for operating.

For example, if the device has access to an Energy pricing server it may be able to use the tariff to estimate the cost of energy for this slot in the power forecast.

When an Energy Management System requests a change in the schedule, then the device MAY suggest a change in the cost as a result of shifting its energy. This can allow a demand side response service to be informed of the relative cost to use energy at a different time.

The Costs field is a list of CostStruct structures which allows multiple CostTypeEnum and Values to be shared by the energy appliance. These could be based on GHG emissions, comfort value for the

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consumer etc.

For example, comfort could be expressed in abstract units or in currency. A water heater that is heated earlier in the day is likely to lose some of its heat before it is needed, which could require a top-up heating event to occur later in the day (which may incur additional cost).

If the ESA cannot calculate its cost for any reason (such as losing its connection to a Price server) it may omit this field. This is treated as extra meta data that an EMS may use to optimize a system.

9.2.7.14.15. MinPowerAdjustment Field

This field SHALL indicate the minimum power that the appliance can be requested to use.

For example, some EVSEs cannot be switched on to charge below 6A which may equate to ~1.3kW in EU markets. If the slot indicates a NominalPower of 0W (indicating it is expecting to be off), this allows an ESA to indicate it could be switched on to charge, but this would be the minimum power limit it can be set to.

9.2.7.14.16. MaxPowerAdjustment Field

This field SHALL indicate the maximum power that the appliance can be requested to use.

For example, an EVSE may be limited by its electrical supply to 32A which would be ~7.6kW in EU markets. If the slot indicates a NominalPower of 0W (indicating it is expecting to be off), this allows an ESA to indicate it could be switched on to charge, but this would be the maximum power limit it can be set to.

9.2.7.14.17. MinDurationAdjustment Field

This field SHALL indicate the minimum time, in seconds, that the slot can be requested to shortened to.

For example, if the slot indicates a NominalPower of 0W (indicating it is expecting to be off), this would allow an ESA to specify the minimum time it could be switched on for. This is to help protect the appliance from being damaged by short cycling times.

For example, a heat pump compressor may have a minimum cycle time of order a few minutes.

9.2.7.14.18. MaxDurationAdjustment Field

This field SHALL indicate the maximum time, in seconds, that the slot can be requested to extended to.

For example, if the slot indicates a NominalPower of 0W (indicating it is expecting to be off), this allows an ESA to specify the maximum time it could be switched on for. This may allow a battery or water heater to indicate the maximum duration that it can charge for before becoming full. In the case of a battery inverter which can be discharged, it may equally indicate the maximum time the battery could be discharged for (at the MaxPowerAdjustment power level).

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9.2.7.15. SlotAdjustmentStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**SlotIndex**	uint8	desc				M
1	**NominalPower**	power-mW	desc				PFR
2	**Duration**	elapsed-s	desc				M

9.2.7.15.1. SlotIndex Field

This field SHALL indicate the index into the Slots list within the Forecast that is to be modified. It SHALL be less than the actual length of the Slots list (implicitly it must be in the range 0 to 9 based on the maximum length of the Slots list constraint).

9.2.7.15.2. NominalPower Field

This field SHALL indicate the new requested power that the ESA SHALL operate at. It SHALL be between the AbsMinPower and AbsMaxPower attributes as advertised by the ESA if it supports PFR.

This is a signed value and can be used to indicate charging or discharging.

If the ESA does NOT support PFR this value SHALL be ignored by the ESA.

9.2.7.15.3. Duration Field

This field SHALL indicate the new requested duration, in seconds, that the ESA SHALL extend or shorten the slot duration to. It SHALL be between the MinDurationAdjustment and MaxDurationAdjustment for the slot as advertised by the ESA.

9.2.7.16. ConstraintsStruct Type

The ConstraintsStruct allows a client to inform an ESA about a constraint period (such as a grid event, or perhaps excess solar PV). The format allows the client to suggest that the ESA can either turn up its energy consumption, or turn down its energy consumption during this period.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**StartTime**	epoch-s	desc				M
1	**Duration**	elapsed-s	max 86400				M
2	**NominalPower**	power-mW	desc				PFR
3	**MaximumEnergy**	energy-mWh	all				PFR
4	**LoadControl**	int8	all				SFR

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9.2.7.16.1. StartTime Field

This field SHALL indicate the start time of the constraint period that the client wishes the ESA to compute a new Forecast.

This value is in UTC and SHALL be in the future.

9.2.7.16.2. Duration Field

This field SHALL indicate the duration of the constraint in seconds.

9.2.7.16.3. NominalPower Field

This field SHALL indicate the nominal power that client wishes the ESA to operate at during the constrained period. It SHALL be between the AbsMinPower and AbsMaxPower attributes as advertised by the ESA if it supports PFR.

This is a signed value and can be used to indicate charging or discharging.

9.2.7.16.4. MaximumEnergy Field

This field SHALL indicate the maximum energy that can be transferred to or from the ESA during the constraint period.

This is a signed value and can be used to indicate charging or discharging.

9.2.7.16.5. LoadControl Field

This field SHALL indicate the turn up or turn down nature that the grid wants as the outcome by the ESA during the constraint period.

This is expressed as a signed value between -100 to +100. A value of 0 would indicate no bias to using more or less energy. A negative value indicates a request to use less energy. A positive value indicates a request to use more energy.

Note that the mapping between values and operation is manufacturer specific.

9.2.8. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**ESAType**	ESATypeEnum	all	F		R V	M
0x0001	**ESACanGenerate**	bool	all	F		R V	M
0x0002	**ESAState**	ESAStateEnum	desc			R V	M
0x0003	**AbsMinPower**	power-mW	all			R V	M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0004	**AbsMaxPower**	power-mW	min AbsMinPower			R V	M
0x0005	**PowerAdjustmentCapability**	PowerAdjustCapabilityStruct	all	Q X		R V	PA
0x0006	**Forecast**	ForecastStruct	all	Q X		R V	PFR | SFR
0x0007	**OptOutState**	OptOutStateEnum	desc			R V	PA | STA | PAU | FA | CON

9.2.8.1. ESAType Attribute

This attribute SHALL indicate the type of ESA.

This attribute enables an EMS to understand some of the basic properties about how the energy may be consumed, generated, and stored by the ESA.

For example, the heat energy converted by a heat pump will naturally be lost through the building to the outdoor environment relatively quickly, compared to storing heat in a well-insulated hot water tank. Similarly, battery storage and EVs can store electrical energy for much longer durations.

This attribute can also help the EMS display information to a user and to make basic assumptions about typical best use of energy. For example, an EVSE may not always have an EV plugged in, so knowing the type of ESA that is being controlled can allow advanced energy management strategies.

9.2.8.2. ESACanGenerate Attribute

This attribute SHALL indicate whether the ESA is classed as a generator or load. This allows an EMS to understand whether the power values reported by the ESA need to have their sign inverted when dealing with forecasts and adjustments.

For example, a solar PV inverter (being a generator) may produce negative values to indicate generation (since power is flowing out of the node into the home), however a display showing the power to the consumers may need to present a positive solar production value to the consumer.

For example, a home battery storage system (BESS) which needs to charge the battery and then discharge to the home loads, would be classed as a generator. These types of devices SHALL have this field set to true. When generating its forecast or advertising its PowerAdjustmentCapability, the power values shall be negative to indicate discharging to the loads in the home, and positive to indicate when it is charging its battery.

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```
GRID meter = Σ LoadPowers + Σ GeneratorPowers
```

Example:

Home has the following loads:	
Water Heater:	3000W
TV:	200W
FridgeFreezer:	200W
Total:	**3400W**

Home has the following generators (ESACanGenerate = true)	
Solar:	-2500W
BESS:	-900W
Total:	**-3400W**

GRID Meter:	
Σ Loads:	3400W
Σ Generators:	-3400W
Total:	**0W**

9.2.8.3. ESAState Attribute

This attribute SHALL indicate the current state of the ESA.

If the ESA is in the Offline or Fault state it cannot be controlled by an EMS, and may not be able to report its Forecast information. An EMS may subscribe to the ESAState to get notified about changes in operational state.

The ESA may have a local user interface to allow a service technician to put the ESA into Offline mode, for example to avoid the EMS accidentally starting or stopping the appliance when it is being serviced or tested.

9.2.8.4. AbsMinPower Attribute

This attribute SHALL indicate the minimum electrical power that the ESA can consume when switched on. This does not include when in power save or standby modes.

NOTE	For Generator ESAs that can discharge an internal battery (such as a battery storage inverter) to loads in the home, the AbsMinPower will be a negative number representing the maximum power that the ESA can discharge its internal battery.

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9.2.8.5. AbsMaxPower Attribute

This attribute SHALL indicate the maximum electrical power that the ESA can consume when switched on.

Note that for Generator ESAs that can charge a battery by importing power into the node (such as a battery storage inverter), the AbsMaxPower will be a positive number representing the maximum power at which the ESA can charge its internal battery.

For example, a battery storage inverter that can charge its battery at a maximum power of 2000W and can discharge the battery at a maximum power of 3000W, would have a AbsMinPower: -3000, AbsMaxPower: 2000W.

9.2.8.6. PowerAdjustmentCapability Attribute

This attribute SHALL indicate how the ESA can be adjusted at the current time, and the state of any active adjustment.

A null value indicates that no power adjustment is currently possible, and nor is any adjustment currently active.

This attribute SHOULD be updated periodically by ESAs to reflect any changes in internal state, for example temperature or stored energy, which would affect the power or duration limits.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- At most once every 10 seconds on changes, or
- When it changes from null to any other value and vice versa.

9.2.8.7. Forecast Attribute

This attribute allows an ESA to share its intended forecast with a client (such as an Energy Management System).

A null value indicates that there is no forecast currently available (for example, a program has not yet been selected by the user).

A server MAY reset this value attribute to null on a reboot, and it does not need to persist any previous forecasts.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- At most once every 10 seconds on changes, or
- When it changes from null to any other value and vice versa, or
- As a result of a command which causes the forecast to be updated, or
- As a result of a change in the opt-out status which in turn may cause the ESA to recalculate its forecast.

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9.2.8.8. OptOutState Attribute

This attribute SHALL indicate the current Opt-Out state of the ESA. The ESA may have a local user interface to allow the user to control this OptOutState. An EMS may subscribe to the OptOutState to get notified about changes in operational state.

If the ESA is in the LocalOptOut or OptOut states, so it cannot be controlled by an EMS for local optimization reasons, it SHALL reject any commands which have the AdjustmentCauseEnum value LocalOptimization. If the ESA is in the GridOptOut or OptOut states, so it cannot be controlled by an EMS for grid optimization reasons, it SHALL reject any commands which have the AdjustmentCauseEnum value GridOptimization.

If the user changes the Opt-Out state of the ESA which is currently operating with a Forecast that is due to a previous StartTimeAdjustRequest, ModifyForecastRequest or RequestConstraintBasedForecast command that would now not be permitted due to the new Opt-out state (i.e. the Forecast attribute ForecastUpdateReason field currently contains a reason which is now opted out), the ESA SHALL behave as if it had received a CancelRequest command.

If the user changes the Opt-Out state of the ESA which currently has the ESAStateEnum with value `Paused` due to a previous PauseRequest command that would now not be permitted due to the new Opt-out state, and the ESA supports the `PFR` or `SFR` features (i.e. the Forecast attribute ForecastUpdateReason field currently contains a reason which is now opted out), the ESA SHALL behave as if it had received a ResumeRequest command.

If the user changes the Opt-Out state of the ESA which currently has the ESAStateEnum with value `PowerAdjustActive` due to a previous PowerAdjustRequest command that would now not be permitted due to the new Opt-out state (i.e. the Forecast attribute ForecastUpdateReason field currently contains a reason which is now opted out), the ESA SHALL behave as if it had received a CancelPowerAdjustRequest command.

If the ESA is in the LocalOptOut, GridOptOut, or NoOptOut states, the device is still permitted to optimize its own energy usage, for example, using tariff information it may obtain.

9.2.9. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**PowerAdjustRequest**	client ⇒ server	Y	O	PA
0x01	**CancelPowerAdjustRequest**	client ⇒ server	Y	O	PA
0x02	**StartTimeAdjustRequest**	client ⇒ server	Y	O	STA
0x03	**PauseRequest**	client ⇒ server	Y	O	PAU
0x04	**ResumeRequest**	client ⇒ server	Y	O	PAU
0x05	**ModifyForecastRequest**	client ⇒ server	Y	O	FA

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ID	Name	Direction	Response	Access	Conformance
0x06	**RequestConstraintBasedForecast**	client ⇒ server	Y	O	CON
0x07	**CancelRequest**	client ⇒ server	Y	O	STA | FA | CON

9.2.9.1. PowerAdjustRequest Command

Allows a client to request an adjustment in the power consumption of an ESA for a specified duration.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Power**	power-mW	desc			M
1	**Duration**	elapsed-s	desc			M
2	**Cause**	AdjustmentCauseEnum	desc			M

9.2.9.1.1. Power Field

This field SHALL indicate the power that the ESA SHALL use during the adjustment period.

This value SHALL be between the MinPower and MaxPower fields of the PowerAdjustStruct in the PowerAdjustmentCapability attribute.

9.2.9.1.2. Duration Field

This field SHALL indicate the duration that the ESA SHALL maintain the requested power for.

This value SHALL be between the MinDuration and MaxDuration fields of the PowerAdjustStruct in the PowerAdjustmentCapability attribute.

9.2.9.1.3. Cause Field

This field SHALL indicate the cause of the request from the EMS.

9.2.9.1.4. Effect on receipt

On receipt of this command, the ESA SHALL validate that the Power and Duration specified in the command are within the limits of its current operation and advertised PowerAdjustmentCapability attribute, the OptOutState permits the specified AdjustmentCauseEnum (see OptOutState for details), and the ESAState is Online.

If the PowerAdjustRequest command is accepted, then the ESA SHALL change its ESAState to PowerAdjustActive, and the PowerAdjustmentCapability attribute SHALL be updated to set the Cause value from the Cause field of this command.

The command status returned SHALL be SUCCESS if the adjustment is accepted; otherwise the com-

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mand SHALL be rejected with CONSTRAINT_ERROR if the Power, Duration, or Cause values are not permitted as above, or with FAILURE otherwise.

Start of adjustment

The ESA SHALL generate a PowerAdjustStart Event when it begins to adjust its power.

The ESA then begins to adjust its power consumption or generation to the power level commanded in the Power field.

End of adjustment (normal completion)

After the elapsed duration, the ESA SHALL revert to normal (or idle) power levels.

The ESA SHALL also generate a PowerAdjustEnd Event with a cause code to indicate a 'Normal completion', the ESAState SHALL be restored to Online, and the PowerAdjustmentCapability attribute SHALL be updated to set the Cause value to 'NoAdjustment'.

Expected behavior when the PowerAdjustmentCapability changes while PowerAdjustment is active

If a power adjustment is active, the value of the PowerAdjustmentCapability attribute changes such that the requested Power or Duration is no longer valid, then the ESA SHALL generate a PowerAdjustEnd Event to indicate the end of the session, with the cause set to Invalid.

The ESAState SHALL be updated to reflect the new state, and the PowerAdjustmentCapability attribute SHALL be updated to set the Cause value to 'NoAdjustment'.

Failure or Opt-out during adjustment

If during the power adjustment session a failure or other condition occurs (such as the user deciding to opt-out by updating the OptOutState) then the ESA SHALL generate a PowerAdjustEnd Event to indicate the end of the session, with the appropriate cause code.

The ESAState SHALL be updated to reflect the new state, and the PowerAdjustmentCapability attribute SHALL be updated to set the Cause value to 'NoAdjustment'.

Receiving a new PowerAdjustRequest command when adjustment is on-going

If an existing power adjustment is already happening, and a new PowerAdjustRequest command is received, then if the ESA allows it, the ESA SHALL return SUCCESS, and the PowerAdjustmentCapability attribute SHALL be updated to set the Cause value from the Cause field of the new command. If the ESA does not permit this new PowerAdjustmentRequest command to interrupt the adjustment that is in progress, it SHALL return BUSY.

Note that if the new command is accepted, then the ESA SHALL NOT generate a new PowerAdjustEnd Event until the new duration has elapsed. This is to avoid generating too many events in cases where a device is routinely commanded with overlapping power adjustments.

For example, a battery inverter ESA may be sent a new request every 5 seconds to adjust its discharge power based on real-time meter readings. Each command may have a 60 second duration,

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but this command is superseded after 5 seconds by a new request.

After the client has sent its last command and after this command duration expires then the last active power adjustment has been completed. This final command causes the PowerAdjustEnd Event to be generated when the ESAState is restored to Online, and the PowerAdjustmentCapability attribute SHALL be updated to set the Cause value to 'NoAdjustment'.

9.2.9.2. CancelPowerAdjustRequest Command

Allows a client to cancel an ongoing PowerAdjustmentRequest operation.

9.2.9.2.1. Effect on receipt

On receipt of this command, the ESA SHALL end the active power adjustment session and return to normal (or idle) power levels.

If the ESAState is not PowerAdjustActive, then the command SHALL be rejected with INVALID_IN_STATE.

If the command is accepted, the ESA SHALL generate an PowerAdjustEnd Event and the ESAState SHALL be restored to Online, the PowerAdjustmentCapability attribute SHALL be updated to set the Cause value to 'NoAdjustment', and the command status returned SHALL be SUCCESS.

9.2.9.3. StartTimeAdjustRequest Command

Allows a client to adjust the start time of a Forecast sequence that has not yet started operation (i.e. where the current Forecast StartTime is in the future).

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**RequestedStartTime**	epoch-s	desc			M
1	**Cause**	AdjustmentCauseEnum	all			M

9.2.9.3.1. RequestedStartTime Field

This field SHALL indicate the requested start time, in UTC, that the client would like the appliance to shift its Forecast to. This value SHALL be in the future.

A client can estimate the entire Forecast sequence duration by computing the EndTime - StartTime fields from the Forecast attribute, and therefore avoid scheduling the start time too late.

This value SHALL be after the EarliestStartTime in the Forecast attribute. The new EndTime, that can be computed from the RequestedStartTime and the Forecast sequence duration, SHALL be before the LatestEndTime.

9.2.9.3.2. Cause Field

This field SHALL indicate the cause of the request from the EMS.

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9.2.9.3.3. Effect on receipt

On receipt of this command, if the ESA supports STA, and the OptOutState permits the specified AdjustmentCauseEnum (see OptOutState for details), and the RequestedStartTime adjustment would be within constraints described in RequestedStartTime then ESA SHALL accept the request to modify the Start Time.

If the RequestedStartTime value resulted in a time shift which is outside the time constraints of EarliestStartTime and LatestEndTime, or the Cause is not permitted, then the command SHALL be rejected with CONSTRAINT_ERROR; in other failure scenarios the command SHALL be rejected with FAILURE.

The command status returned SHALL be SUCCESS if the StartTime in the Forecast is updated. The ESA SHALL update its Forecast attribute using the new RequestedStartTime including a new ForecastID, StartTime, EndTime, and ForecastUpdateReason.

9.2.9.4. PauseRequest Command

Allows a client to temporarily pause an operation and reduce the ESAs energy demand.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Duration**	elapsed-s	desc			M
1	**Cause**	AdjustmentCauseEnum	all			M

9.2.9.4.1. Duration Field

This field SHALL indicate the duration that the ESA SHALL be paused for. This value SHALL be between the MinPauseDuration and MaxPauseDuration indicated in the ActiveSlotNumber index in the Slots list in the Forecast.

9.2.9.4.2. Cause Field

This field SHALL indicate the cause of the request from the EMS.

9.2.9.4.3. Effect on receipt

On receipt of this command, if the ESA supports PAU and the SlotIsPausable field is true in the ActiveSlotNumber index in the Slots list, the OptOutState permits the specified AdjustmentCauseEnum (see OptOutState for details), and the OptOutState is Online or PowerAdjustActive, the ESA SHALL allow its current operation to be Paused. If the Cause is not permitted, then the command SHALL be rejected with CONSTRAINT_ERROR.

If the ESA SlotIsPausable field is false for the ActiveSlotNumber, then the command SHALL be rejected with FAILURE.

The ESA SHALL validate that the Duration field is within the range of MinPauseDuration and MaxPauseDuration. If it is outside of this range then the command SHALL be rejected with CONSTRAINT_ERROR.

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The command status returned SHALL be SUCCESS if the ESA is paused.

Once the command has been accepted, the ESA SHALL also generate a Paused Event and the ESAState SHALL be set to Paused.

The ESA SHALL update its update its Forecast to account for the pause, including updating the ForecastUpdateReason from the Cause field.

Pause timer start The ESA SHALL start a Pause timer based on the value of the Duration field.

Paused State Whilst in the Paused state, the ESA SHALL NOT consume or produce significant power (other than required to keep its basic control system operational).

Pause timer expiry When the Pause timer expires the ESA SHALL automatically resume operation. When it does this, then it SHALL also generate a Resumed Event and the ESAState SHALL be updated accordingly to reflect its current state (for example, it may still have PowerAdjustActive now it has resumed from the Pause). The ESA SHALL update its Forecast including setting ForecastUpdateReason to `InternalOptimization` to account for the resumption after the pause.

Receiving a further pause request command whilst paused If a further Pause Request is received in the same forecast slot whilst already in the paused state, the ESA SHALL again validate that the Duration value is within the range of the (possibly revised) MinPauseDuration and MaxPauseDuration. If it is outside of this range then the command SHALL be rejected with CONSTRAINT_ERROR.

If the command is accepted the pause timer SHALL be extended by the new Duration.

ResumeRequest A client MAY send a ResumeRequest command, before the Pause timer has expired, to request the ESA to return to its normal operating state. See ResumeRequest for details.

Failure or Opt-out during PauseRequest A consumer MAY decide to opt-out of the remotely commanded PauseRequest by updating the OptOutState, and the ESA SHALL also change the ESAState to the new state.

If the ESA develops a fault whilst Paused, the ESAState SHALL be set to Fault.

On change of ESAState (from Paused to another state), the ESA SHALL generate a Resumed Event.

9.2.9.5. ResumeRequest Command

Allows a client to cancel the PauseRequest command and enable earlier resumption of operation.

9.2.9.5.1. Effect on receipt

On receipt of this command, if the ESA supports `PAU` and it is currently Paused, the ESA MAY decide not to resume immediately if the MinPauseDuration has not yet elapsed. This behavior is manufacturer specific.

If accepted, the ESA SHALL resume its operation. The ESA SHALL also generate a Resumed Event and the ESAState SHALL be updated accordingly to reflect its current state.

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The command status returned SHALL be SUCCESS if the operation is resumed, or the command SHALL be rejected with the response INVALID_IN_STATE if the ESA is not currently Paused, otherwise the command SHALL be rejected with FAILURE.

The ESA SHALL update its Forecast including setting ForecastUpdateReason to `InternalOptimization` to account for the resumption after the pause.

9.2.9.6. ModifyForecastRequest Command

Allows a client to modify a Forecast within the limits allowed by the ESA.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ForecastID**	uint32	all			M
1	**SlotAdjustments**	list[SlotAdjustmentStruct]	max 10			M
2	**Cause**	AdjustmentCauseEnum	all			M

9.2.9.6.1. ForecastID Field

This field SHALL indicate the ForecastID that is to be modified.

9.2.9.6.2. SlotAdjustments Field

This field SHALL contain a list of SlotAdjustment parameters that should be modified in the corresponding Forecast with matching ForecastID.

9.2.9.6.3. Cause Field

This field SHALL indicate the cause of the request from the EMS.

9.2.9.6.4. Effect on receipt

On receipt of this command, if the ESA supports `FA`, and the OptOutState permits the specified AdjustmentCauseEnum (see OptOutState for details), it SHALL attempt to adjust its forecast.

The client may be an energy management system which has retrieved the forecasts from multiple ESA and then performed some optimization (for example, taking advantage of local solar PV generation).

The ESA SHALL inspect the requested forecast and ensure that it does not exceed the limits set in each of its slots. The ESA manufacturer may also reject the request if it could cause the user's preferences to be breached (for example, if it may cause the home to be too hot or too cold, or a battery to be insufficiently charged).

Note that the client may make an adjustment when the ESA's program has already been started. In this case the ActiveSlotNumber may have moved through the Forecast sequence. Any attempts to

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modify slots which have already been run SHALL result in the entire command being rejected.

The command allows a single modification to a particular slot, or to multiple slots in a single command by sending a list of modifications.

If the ESA accepts the requested Forecast then it SHALL update its Forecast attribute (incrementing its ForecastID, and updating its ForecastUpdateReason from the Cause value) and run the revised Forecast as its new intended operation.

Note that an ESA may adapt its Forecast based on a slot modification. For example, if an ESA was switched on earlier in the day to take account of available solar PV, then it may not need to use energy later in the day. It may add or remove slots as it chooses. This may result in several iterations for the Energy Management System to reprocess updated forecasts until it has stabilized on an agreed set of changes.

If for any reason the ESA cannot accept the entire requested forecast adjustments then it SHALL reject the entire command.

If the Cause is not permitted, or the requested forecast exceeds any of the limits in the respective slots, then the command SHALL be rejected with CONSTRAINT_ERROR, otherwise if the ForecastID is valid, and the entire list of SlotAdjustmentStruct are accepted, the command status returned SHALL be SUCCESS, otherwise the command SHALL be rejected with FAILURE.

9.2.9.7. RequestConstraintBasedForecast Command

Allows a client to ask the ESA to recompute its Forecast based on power and time constraints.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Constraints**	list[ConstraintsStruct]	max 10			M
1	**Cause**	AdjustmentCauseEnum	all			M

9.2.9.7.1. Constraints Field

This field SHALL indicate the series of turn up or turn down power requests that the ESA is being asked to constrain its operation within. These requests SHALL be in the future, SHALL be in chronological order, starting with the earliest start time, and SHALL NOT overlap in time.

For example, a grid event which requires devices to reduce power (turn down) between 4pm and 6pm and due to excess power on the grid overnight, may request ESAs to increase their power demand (turn up) between midnight and 6am.

If this ESA supports PFR this would have 2 entries in the list as follows:

Entry [0] - Turn Down		
StartTime	2023-10-23T16:00:00	

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Duration	7200	(2 hours)
NominalPower	20	W
MaximumEnergy	40	Wh

Entry [1] - Turn Up		
StartTime	2023-10-24T00:00:00	
Duration	21,600	(6 hours)
NominalPower	7000	W
MaximumEnergy	42000	Wh

If this ESA supports SFR where it does not know the actual power, but has an understanding of the functions that use more energy, it could be requested to use more or less energy using the LoadControl field as follows:

Entry [0] - Turn Down		
StartTime	2023-10-23T16:00:00	
Duration	7200	(2 hours)
LoadControl	-100	(Use less power)

Entry [1] - Turn Up		
StartTime	2023-10-24T00:00:00	
Duration	21,600	(6 hours)
LoadControl	+100	(Use more power)

9.2.9.7.2. Cause Field

This field SHALL indicate the cause of the request from the EMS.

9.2.9.7.3. Effect on receipt

On receipt of this command, if the ESA supports CON, and the OptOutState permits the specified AdjustmentCauseEnum (see OptOutState for details), it may be requested to generate a new forecast by a client.

For example the client may be an energy management system which has determined that the peak load on the home should be reduced (for example a grid event) or that there is more local generation available and the solar export power needs to be consumed.

The EMS may not be best placed to make the forecast adjustment because the ESA knows more about its internal operation and system efficiencies.

For example the total energy used when charging a battery faster may result in increased heat losses in the inverter, so the total energy required to charge a battery with 10kWh of stored energy may vary based on charging power.

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The ESA SHALL inspect the constraints field to ensure that the constraints defined for that field and its elements are met. The ESA manufacturer may also reject the request if it could cause the user's preferences to be breached (e.g. may cause the home to be too hot or too cold, or a battery to be insufficiently charged).

If the ESA can meet the requested power limits, it SHALL regenerate a new Power Forecast with a new ForecastID and ForecastUpdateReason from the Cause value.

If an appliance has already begun running a program (and agreed to modify its schedule), it may continue to run the same program and meet the same user settings (e.g. ECO mode), but may take more or less time to complete the cycle. The new reported Forecast will start from the current time, i.e. the slots that have already been completed SHALL NOT be included in the new forecast.

The command status returned SHALL be SUCCESS, otherwise if the Cause is not permitted, or the constraints field value does not meet the defined constraints, the command SHALL be rejected with CONSTRAINT_ERROR, otherwise the command SHALL be rejected with FAILURE.

9.2.9.8. CancelRequest Command

Allows a client to request cancellation of a previous adjustment request in a StartTimeAdjustRequest, ModifyForecastRequest or RequestConstraintBasedForecast command.

9.2.9.8.1. Effect on receipt

On receipt of this command, the ESA SHALL attempt to cancel the effects of any previous adjustment request commands, and re-evaluate its forecast for intended operation ignoring those previous requests.

If the ESA ForecastUpdateReason was already `Internal Optimization`, then the command SHALL be rejected with INVALID_IN_STATE.

If the command is accepted, the ESA SHALL update its ESAState if required, and the command status returned SHALL be SUCCESS. The ESA SHALL update its Forecast attribute to match its new intended operation, and update the ForecastUpdateReason to `Internal Optimization`.

9.2.10. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**PowerAdjustStart**	INFO		V	PA
0x01	**PowerAdjustEnd**	INFO		V	PA
0x02	**Paused**	INFO		V	PAU
0x03	**Resumed**	INFO		V	PAU

9.2.10.1. PowerAdjustStart Event

This event SHALL be generated when the Power Adjustment session is started.

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9.2.10.2. PowerAdjustEnd Event

This event SHALL be generated when the Power Adjustment session ends.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Cause**	CauseEnum	all			M
1	**Duration**	elapsed-s	all			M
2	**EnergyUse**	energy-mWh	all			M

9.2.10.2.1. Cause Field

This field SHALL indicate the reason why the power adjustment session ended.

9.2.10.2.2. Duration Field

This field SHALL indicate the number of seconds that the power adjustment session lasted before ending.

9.2.10.2.3. EnergyUse Field

This field SHALL indicate the approximate energy used by the ESA during the session.

For example, if the ESA was on and was adjusted to be switched off, then this SHALL be 0 mWh. If this was a battery inverter that was requested to discharge it would have a negative EnergyUse value. If this was a normal load that was turned on, then it will have positive value.

9.2.10.3. Paused Event

This event SHALL be generated when the ESA enters the Paused state.

There is no data for this event.

9.2.10.4. Resumed Event

This event SHALL be generated when the ESA leaves the Paused state and resumes operation.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Cause**	CauseEnum	all			M

9.2.10.4.1. Cause Field

This field SHALL indicate the reason why the pause ended.

9.3. Energy EVSE Cluster

Electric Vehicle Supply Equipment (EVSE) is equipment used to charge an Electric Vehicle (EV) or Plug-In Hybrid Electric Vehicle. This cluster provides an interface to the functionality of Electric

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Vehicle Supply Equipment (EVSE) management.

Devices targeted by this cluster include Electric Vehicle Supply Equipment (EVSE). The cluster generically assumes a signaling protocol (J1772 in NA and IEC61851 in Europe and Asia) between the EVSE and Electric Vehicle (EV) that utilizes a pilot signal to manage the states of the charging process. [SAE J2847/3_202311] version and IEC61841 define Pilot signal as a modulated DC voltage on a single wire.

Power Line Communication (PLC) is supported by some EVSEs (e.g. for support of ISO 15118 in Europe and SAE J2931/4 in NA) and may enable features such as Vehicle to Grid (V2G) or Vehicle to Home (V2H) that allows for bi-directional charging/discharging of electric vehicles.

More modern EVSE devices may optionally support ISO 15118-20 in Europe and SAE J2836/3 for NA to support bi-directional charging (Vehicle to Grid - V2G) and Plug and Charge capabilities.

This cluster definition assumes AC charging only. DC charging options may be added in future revisions of this cluster.

This cluster supports a safety mechanism that may lockout remote operation until the initial latching conditions have been met. Some of the fault conditions defined in SAE J1772, such as Ground-Fault Circuit Interrupter (GFCI) or Charging Circuit Interrupting Device (CCID), may require clearing by an operator by, for example, pressing a button on the equipment or breaker panel.

This EVSE cluster is written around support of a single EVSE. Having multiple EVSEs at home or a business is managed by backend system and outside scope of this cluster.

Note that in many deployments the EVSE may be outside the home and may suffer from intermittent network connections (e.g. a weak Wi-Fi signal). It also allows for a charging profile to be preconfigured, in case there is a temporary communications loss during a charging session.

9.3.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Updates after 0.7 Ballot review
3	Added Q quality for SessionDuration, SessionEnergyCharged and SessionEnergyDischarged, updated to V2X and associated areas and made PREF mandatory
4	Sync with SDK revision number

9.3.2. Classification

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Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	EEVSE

9.3.3. Cluster ID

ID	Name
0x0099	Energy EVSE

9.3.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**PREF**	ChargingPreferences	M	EVSE supports storing user charging preferences
1	**SOC**	SoCReporting	O	EVSE supports reporting of vehicle State of Charge (SoC)
2	**PNC**	PlugAndCharge	O	EVSE supports PLC to support Plug and Charge
3	**RFID**	RFID	O	EVSE is fitted with an RFID reader
4	**V2X**	V2X	O	EVSE supports bidirectional charging / discharging

9.3.4.1. ChargingPreferences Feature

Since some EVSEs cannot obtain the SoC from the vehicle, some EV charging solutions allow the consumer to specify a daily charging target (for adding energy to the EV's battery). This feature allows the consumer to specify how many miles or km of additional range they need for their typical daily commute. This range requirement can be converted into a daily energy demand with a target charging completion time.

The EVSE itself can use this information (or may allow a controller such as an EMS) to compute an optimized charging schedule. For example, an optimizer may be able to use the Commodity Price cluster to determine the cheapest and lowest GHG based charging schedule for the vehicle.

An EVSE device which includes a Device Energy Management device with the Device Energy Management cluster `PFR` (PowerForecastReporting) feature can use the charging preferences information to produce its power forecast.

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EVSE devices that support the Device Energy Management cluster's FA feature can have their charging profiles set by a controller device such as an EMS. For example, if the EVSE advertises a simple power forecast which allows the EMS to adjust over a wide range of power and time durations, then the EVSE may allow the EMS to propose a revised optimized forecast (which is the charging profile). For example, a solar PV ESA may also share its Forecast, so enabling an EMS to adjust the EVSE forecast to the best time to charge so that any excess solar generation is used to charge the EV.

See the Device Energy Management Cluster for more details.

9.3.4.2. SoCReporting Feature

Vehicles and EVSEs which support ISO 15118 may allow the vehicle to report its battery size and state of charge. If the EVSE supports PLC it may have a vehicle connected which optionally supports reporting of its battery size and current State of Charge (SoC).

If the EVSE supports reporting of State of Charge this feature will only work if a compatible EV is connected.

Note some EVSEs may use other undefined mechanisms to obtain vehicle State of Charge outside the scope of this cluster.

9.3.4.3. PlugAndCharge Feature

If the EVSE supports PLC, it may be able to support the Plug and Charge feature. e.g. this may allow the vehicle ID to be obtained which may allow an energy management system to track energy usage per vehicle (e.g. to give the owner an indicative cost of charging, or for work place charging).

If the EVSE supports the Plug and Charge feature, it will only work if a compatible EV is connected.

9.3.4.4. RFID Feature

If the EVSE is fitted with an RFID reader, it may be possible to obtain the User or Vehicle ID from an RFID card. This may be used to record a charging session against a specific charging account, and may optionally be used to authorize a charging session.

An RFID event can be generated when a user taps an RFID card onto the RFID reader. The event must be subscribed to by the EVSE Management cluster client. This client may use this to enable the EV to charge or discharge. The lookup and authorization of RIFD UID is outside the scope of this cluster.

9.3.4.5. V2X Feature

If the EVSE can support bi-directional charging, it may be possible to request that the vehicle can discharge to the home or grid.

The charging and discharging may be controlled by a home Energy Management System (EMS) using the Device Energy Management cluster of the associated Device Energy Management device. For example, an EMS may use the PA (PowerAdjustment) feature to continually adjust the charge/discharge current to/from the EV so as to minimise the energy flow from/to the grid as the demand in the home and the solar supply to the home both fluctuate.

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9.3.5. Dependencies

The server side of this cluster SHALL depend on setting time from another device or using a real-time clock.

This can either use a built-in real-time clock or non-Matter time source, or can be derived using the Time Synchronization cluster.

9.3.5.1. Diagnostic Event logs

It is quite common that users may experience issues charging their vehicle, which, without logs to understand what happened, makes it very difficult to resolve the root cause.

Matter supports events (see Matter specification section 7.14) which includes a buffered log of previous events including a Timestamp (see Matter specification 7.14.2.2).

This Timestamp can be the System Time (since boot) or Epoch Time. It is recommended to use Epoch time (which can be set using the Time Synchronization cluster), which would allow remote support operatives to retrieve the event logs and analyze them against a user reported actual time.

9.3.6. Definitions

EVSE Session - An EVSE session starts when an EV is plugged in. It ends when it is unplugged.

9.3.7. Data Types

9.3.7.1. TargetDayOfWeekBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**Sunday**	Sunday	M
1	**Monday**	Monday	M
2	**Tuesday**	Tuesday	M
3	**Wednesday**	Wednesday	M
4	**Thursday**	Thursday	M
5	**Friday**	Friday	M
6	**Saturday**	Saturday	M

9.3.7.2. StateEnum Type

This data type is derived from enum8.

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Value	Name	Summary	Conformance
0	**NotPluggedIn**	The EV is not plugged in.	M
1	**PluggedInNoDemand**	The EV is plugged in, but not demanding current.	M
2	**PluggedInDemand**	The EV is plugged in and is demanding current, but EVSE is not allowing current to flow.	M
3	**PluggedInCharging**	The EV is plugged in, charging is in progress, and current is flowing	M
4	**PluggedInDischarging**	The EV is plugged in, discharging is in progress, and current is flowing	V2X
5	**SessionEnding**	The EVSE is transitioning from any plugged-in state to NotPluggedIn	M
6	**Fault**	There is a fault, further details in the FaultState attribute	M

9.3.7.3. SupplyStateEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Disabled**	The EV is not currently allowed to charge or discharge	M
1	**ChargingEnabled**	The EV is currently allowed to charge	M
2	**DischargingEnabled**	The EV is currently allowed to discharge	[V2X]
3	**DisabledError**	The EV is not currently allowed to charge or discharge due to an error, the error must be cleared before operation can continue	M

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Value	Name	Summary	Conformance
4	**DisabledDiagnostics**	The EV is not currently allowed to charge or discharge due to self-diagnostics mode.	M
5	**Enabled**	The EV is currently allowed to charge and discharge	[V2X]

9.3.7.4. FaultStateEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**NoError**	The EVSE is not in an error state.	M
1	**MeterFailure**	The EVSE is unable to obtain electrical measurements.	M
2	**OverVoltage**	The EVSE input voltage level is too high.	M
3	**UnderVoltage**	The EVSE input voltage level is too low.	M
4	**OverCurrent**	The EVSE detected charging current higher than allowed by charger.	M
5	**ContactWetFailure**	The EVSE detected voltage on charging pins when the contactor is open.	M
6	**ContactDryFailure**	The EVSE detected absence of voltage after enabling contactor.	M
7	**GroundFault**	The EVSE has an unbalanced current supply.	M
8	**PowerLoss**	The EVSE has detected a loss in power.	M
9	**PowerQuality**	The EVSE has detected another power quality issue (e.g. phase imbalance).	M

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Value	Name	Summary	Conformance
10	**PilotShortCircuit**	The EVSE pilot signal amplitude short circuited to ground.	M
11	**EmergencyStop**	The emergency stop button was pressed.	M
12	**EVDisconnected**	The EVSE detected that the cable has been disconnected.	M
13	**WrongPowerSupply**	The EVSE could not determine proper power supply level.	M
14	**LiveNeutralSwap**	The EVSE detected Live and Neutral are swapped.	M
15	**OverTemperature**	The EVSE internal temperature is too high.	M
255	**Other**	Any other reason.	M

9.3.7.5. EnergyTransferStoppedReasonEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**EVStopped**	The EV decided to stop	M
1	**EVSEStopped**	The EVSE decided to stop	M
2	**Other**	An other unknown reason	M

9.3.7.6. ChargingTargetStruct Type

This represents a single user specified charging target for an EV.

An EVSE or EMS system optimizer may use this information to take the Time of Use Tariff, grid carbon intensity, local generation (solar PV) into account to provide the cheapest and cleanest energy to the EV.

The optimization strategy is not defined here, however in simple terms, the AddedEnergy requirement can be fulfilled by knowing the charging Power (W) and the time needed to charge.

> To compute the Charging Time: Required Energy (Wh) = Power (W) x ChargingTime (s) / 3600
>
> Therefore: ChargingTime (s) = (3600 x RequiredEnergy (wH)) / Power (W)

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To compute the charging time: Charging StartTime = TargetTimeMinutesPastMidnight - ChargingTime

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**TargetTimeMinutesPastMidnight**	uint16	max 1439				M
1	**TargetSoC**	percent			0		SOC, O.a+
2	**AddedEnergy**	energy-mWh	min 0		0		[SOC], O.a+

9.3.7.6.1. TargetTimeMinutesPastMidnight Field

This field SHALL indicate the desired charging completion time of the associated day. The time will be represented by a 16 bits unsigned integer to designate the minutes since midnight. For example, 6am will be represented by 360 minutes since midnight and 11:30pm will be represented by 1410 minutes since midnight.

This field is based on local wall clock time. In case of Daylight Savings Time transition which may result in an extra hour or one hour less in the day, the charging algorithm should take into account the shift appropriately.

Note that if the TargetTimeMinutesPastMidnight values are too close together (e.g. 2 per day) these may overlap. The EVSE may have to coalesce the charging targets into a single target. e.g. if the 1st charging target cannot be met in the time available, the EVSE may be forced to begin working towards the 2nd charging target and immediately continue until both targets have been satisfied (or the vehicle becomes full).

The EVSE itself cannot predict the behavior of the vehicle (i.e. if it cannot obtain the SoC from the vehicle), so should attempt to perform a sensible operation based on these targets. It is recommended that the charging schedule is pessimistic (i.e. starts earlier) since the vehicle may charge more slowly than the electrical supply may provide power (especially if it is cold).

If the user configures large charging targets (e.g. high values of AddedEnergy or SoC) then it is expected that the EVSE may need to begin charging immediately, and may not be able to guarantee that the vehicle will be able to reach the target.

9.3.7.6.2. TargetSoC Field

This field represents the target SoC that the vehicle should be charged to before the TargetTimeMinutesPastMidnight.

If the EVSE supports the SOC feature and can obtain the SoC of the vehicle:

- the TargetSoC field SHALL take precedence over the AddedEnergy field.
- the EVSE SHOULD charge to the TargetSoC and then stop the charging automatically when it

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reaches that point.

- if the TargetSoC value is set to 100% then the EVSE SHOULD continue to charge the vehicle until the vehicle decides to stop charging.

If the EVSE does not support the SOC feature or cannot obtain the SoC of the vehicle:

- the AddedEnergy field SHALL take precedence over the TargetSoC field, and if the EVSE does not support the SOC feature then the TargetSoC field may only take the values null or 100%.
- if the AddedEnergy field has not been provided, the EVSE SHOULD assume the vehicle is empty and charge until the vehicle stops demanding a charge.

9.3.7.6.3. AddedEnergy Field

This field represents the amount of energy that the user would like to have added to the vehicle before the TargetTimeMinutesPastMidnight.

This represents a positive value in mWh that SHOULD be added during the session (i.e. if the vehicle charging is stopped and started several times, this equates to the total energy since the vehicle has been plugged in).

The maximum value (500kWh) is much larger than most EV batteries on the market today. If the client tries to set this value too high then the EVSE will need to start charging immediately and continue charging until the vehicle stops demanding charge (i.e. it is full). Therefore the maximum value should be set based on typical battery size of the vehicles on the market (e.g. 70000Wh), however this is up to the client to carefully choose a value.

NOTE	If the EVSE can obtain the Battery Capacity of the vehicle, it SHOULD NOT limit this AddedEnergy value to the Battery Capacity of the vehicle, since the EV may also require energy for heating and cooling of the battery during charging, or for heating or cooling the cabin.

9.3.7.7. ChargingTargetScheduleStruct Type

This represents a set of user specified charging targets for an EV for a set of specified days.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**DayOfWeekForSequence**	TargetDayOfWeekBitmap	all				M
1	**ChargingTargets**	list[ChargingTargetStruct]	max 10				M

9.3.7.8. DayOfWeekForSequence Field

This field SHALL indicate the days of the week that the charging targets SHOULD be associated to.

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This field is a bitmap and therefore the associated targets could be applied to multiple days.

9.3.7.9. ChargingTargets Field

This field SHALL indicate a list of up to 10 charging targets for each of the associated days of the week.

9.3.8. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**State**	StateEnum	all	X		R V	M
0x0001	**SupplyState**	SupplyStateEnum	all			R V	M
0x0002	**FaultState**	FaultStateEnum	all			R V	M
0x0003	**ChargingEnabledUntil**	epoch-s	all	N X		R V	M
0x0004	**DischargingEnabledUntil**	epoch-s	all	N X		R V	V2X
0x0005	**CircuitCapacity**	amperage-mA	min 0	N		R V	M
0x0006	**MinimumChargeCurrent**	amperage-mA	min 0	N		R V	M
0x0007	**MaximumChargeCurrent**	amperage-mA	min 0	N		R V	M
0x0008	**MaximumDischargeCurrent**	amperage-mA	min 0	N		R V	V2X
0x0009	**UserMaximumChargeCurrent**	amperage-mA	desc	N	0	RW VM	O

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x000A	**RandomizationDelayWindow**	elapsed-s	max 86400	N	600	RW VM	O
0x0023	**NextChargeStartTime**	epoch-s	all	X		R V	PREF
0x0024	**NextChargeTargetTime**	epoch-s	all	X		R V	PREF
0x0025	**NextCharge-eRequiredEnergy**	energy-mWh	min 0	X		R V	PREF
0x0026	**NextChargeTargetSoC**	percent		X		R V	PREF
0x0027	**ApproximateEVEfficiency**	uint16	desc	N X	null	RW VM	[PREF]
0x0030	**StateOfCharge**	percent		X		R V	SOC
0x0031	**BatteryCapacity**	energy-mWh	min 0	X		R V	SOC
0x0032	**VehicleID**	string	max 32	X		R V	PNC
0x0040	**SessionID**	uint32	all	N X		R V	M
0x0041	**SessionDuration**	elapsed-s	all	N Q X		R V	M
0x0042	**SessionEnergyCharged**	energy-mWh	min 0	N Q X		R V	M
0x0043	**SessionEnergyDischarged**	energy-mWh	min 0	N Q X		R V	V2X

9.3.8.1. State Attribute

This attribute SHALL indicate the current status of the EVSE. This higher-level status is partly derived from the signaling protocol as communicated between the EVSE and the vehicle through the pilot signal.

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The State attribute SHALL change when the EVSE detects change of condition of the EV (plugged in or unplugged, whether the vehicle is asking for demand or not, and if it is charging or discharging).

NOTE	SessionEnding is not really a state but a transition. However, the transition period may take a few seconds and is useful for some clean up purposes.

The Fault state is used to indicate that the FaultState attribute is not NoError.

A null value SHALL indicate that the state cannot be determined.

9.3.8.2. SupplyState Attribute

This attribute SHALL indicate whether the EV is currently allowed to charge from or discharge to the EVSE.

9.3.8.3. FaultState Attribute

This attribute SHALL indicate the type of fault detected by the EVSE (internally or as detected in the pilot signal).

When the SupplyState attribute is DisabledError, the FaultState attribute will be one of the values listed in FaultStateEnum, except NoError. For all values of SupplyState other than DisabledError, the FaultState attribute SHALL be NoError.

9.3.8.4. ChargingEnabledUntil Attribute

This attribute SHALL indicate the time, in UTC, that the EVSE will automatically stop current flow to the EV.

A null value indicates the EVSE is always enabled for charging.

A value in the past or 0x0 indicates that EVSE charging SHALL be disabled. The attribute is only set via the payload of the EnableCharging command.

This attribute SHALL be persisted, for example a temporary power failure should not stop the vehicle from being charged.

9.3.8.5. DischargingEnabledUntil Attribute

This attribute SHALL indicate the time, in UTC, that the EVSE will automatically stop current flow from the EV.

A null value indicates the EVSE is always enabled for discharging.

A value in the past or 0x0 indicates that EVSE discharging SHALL be disabled. The attribute is only set via the payload of the EnableDischarging command.

This attribute SHALL be persisted, for example a temporary power failure should not stop the vehicle from being discharged.

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9.3.8.6. CircuitCapacity Attribute

This attribute SHALL indicate the capacity that the circuit that the EVSE is connected to can provide. It is intended to allow implementation of a self-managed network of EVSEs. It is assumed that the device will allow the setting of such values by an installer.

9.3.8.7. MinimumChargeCurrent Attribute

This attribute SHALL indicate the minimum current that can be delivered by the EVSE to the EV.

The attribute can be set using the EnableCharging command.

9.3.8.8. MaximumChargeCurrent Attribute

This attribute SHALL indicate the maximum current that can be delivered by the EVSE to the EV.

This SHALL represent the actual maximum current offered to the EV at any time. Note that the EV can draw less current than this value. For example, the EV may be limiting its power draw based on the operating conditions of the battery, such as temperature and state of charge.

The attribute can be initially set using the EnableCharging command or by adjusting the UserMaximumChargeCurrent attribute.

This attribute value SHALL be the minimum of:

- CircuitCapacity - Electrician's installation setting
- CableAssemblyCurrentLimit (detected by the EVSE when the cable is plugged in)
- MaximumChargeCurrent field in the EnableCharging command
- UserMaximumChargeCurrent attribute

9.3.8.9. MaximumDischargeCurrent Attribute

This attribute SHALL indicate the maximum current that can be received by the EVSE from the EV.

This attribute can be set using the EnableDischarging command.

This attribute value SHALL be the minimum of:

- CircuitCapacity - Electrician's installation setting
- CableAssemblyCurrentLimit (detected by the EVSE when the cable is plugged in)
- MaximumDischargeCurrent field in the EnableDischarging command

9.3.8.10. UserMaximumChargeCurrent Attribute

This attribute SHALL indicate a maximum current that can set by the consumer (e.g. via an app) as a preference to further reduce the charging rate. This may be desirable if the home owner has a

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solar PV or battery storage system which may only be able to deliver a limited amount of power. The consumer can manually control how much they allow the EV to take.

This attribute value SHALL be limited by the EVSE to be in the range of:

```
MinimumChargeCurrent <= UserMaximumChargeCurrent <= MaximumChargeCurrent
```

where MinimumChargeCurrent and MaximumChargeCurrent are the values received in the EnableCharging command.

Its default value SHOULD be initialized to the same as the CircuitCapacity attribute. This value SHALL be persisted across reboots to ensure it does not cause charging issues during temporary power failures.

9.3.8.11. RandomizationDelayWindow Attribute

This attribute SHALL indicate the size of a random window over which the EVSE will randomize the start of a charging session. This value is in seconds.

This is a feature that is mandated in some markets (such as UK) where the EVSE should by default randomize its start time within the randomization window. By default in the UK this should be 600s.

For example, if the RandomizationDelayWindow is 600s (i.e. 10 minutes) and if there was a cheap rate energy starting at 00:30, then the EVSE must compute a random delay between 0-599s and add this to its initial planned start time.

9.3.8.12. NextChargeStartTime Attribute

This attribute SHALL indicate the time, in UTC, when the EVSE plans to start the next scheduled charge based on the charging preferences.

A null value indicates that there is no scheduled charging (for example, the EVSE Mode is set to use Manual mode tag), or that the vehicle is not plugged in with the SupplyState indicating that charging is enabled.

9.3.8.13. NextChargeTargetTime Attribute

This attribute SHALL indicate the time, in UTC, when the EVSE SHOULD complete the next scheduled charge based on the charging preferences.

A null value indicates that there is no scheduled charging (for example, the EVSE Mode is set to use Manual mode tag), or that the vehicle is not plugged in with the SupplyState indicating that charging is enabled.

9.3.8.14. NextChargeRequiredEnergy Attribute

This attribute SHALL indicate the amount of energy that the EVSE is going to attempt to add to the vehicle in the next charging target.

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A null value indicates that there is no scheduled charging (for example, the EVSE Mode is set to use Manual mode tag), or that the vehicle is not plugged in with the SupplyState indicating that charging is enabled, or that the next ChargingTargetStruct is using the TargetSoC value to charge the vehicle.

9.3.8.15. NextChargeTargetSoC Attribute

This attribute SHALL indicate the target SoC the EVSE is going to attempt to reach when the vehicle is next charged.

A null value indicates that there is no scheduled charging (for example, the EVSE Mode is set to use Manual mode tag), or that the vehicle is not plugged in with the SupplyState indicating that charging is enabled, or that the next ChargingTargetStruct is using the AddedEnergy value to charge the vehicle.

If the SOC feature is not supported, only the values null and 100% are supported.

9.3.8.16. ApproximateEVEfficiency Attribute

This attribute SHALL indicate the vehicle efficiency rating for a connected vehicle.

This can be used to help indicate to the user *approximately* how many miles or km of range will be added. It allows user interfaces to display to the user simpler terms that they can relate to compared to kWh.

This is value is stored in km per kWh multiplied by a scaling factor of 1000.

A null value indicates that the EV efficiency is unknown and the NextChargeRequiredEnergy attribute cannot be converted from Wh to miles or km.

To convert from Wh into Range:

```
AddedRange (km)    = AddedEnergy (Wh) x ApproxEVEfficiency (km/kWh x 1000)
AddedRange (Miles) = AddedEnergy (Wh) x ApproxEVEfficiency (km/kWh x 1000) x
0.6213
```

Example:

```
ApproxEVEfficiency (km/kWh x 1000): 4800 (i.e. 4.8km/kWh x 1000)
AddedEnergy (Wh): 10,000
```

```
AddedRange (km)    = 10,000 x 4800 / 1,000,000 = 48 km
AddedRange (Miles) = AddedEnergy (Wh) x ApproxEVEfficiency (km/kWh x 1000) x
0.6213
                     = 29.82 Miles
```

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9.3.8.17. StateOfCharge Attribute

This attribute SHALL indicate the state of charge of the EV battery in steps of 1%. The values are in the 0-100%. This attribute is only available on EVSEs which can read the state of charge from the vehicle and that support the SOC feature. If the StateOfCharge cannot be read from the vehicle it SHALL be returned with a NULL value.

9.3.8.18. BatteryCapacity Attribute

This attribute SHALL indicate the capacity of the EV battery in mWh. This value is always positive.

9.3.8.19. VehicleID Attribute

This attribute SHALL indicate the vehicle ID read by the EVSE via ISO-15118 using the PNC feature, if the EVSE supports this capability.

The field may be based on the e-Mobility Account Identifier (EMAID).

A null value SHALL indicate that this is unknown.

9.3.8.20. Session Attributes

The following set of attributes provides information about a charging session, defined as the period from the EVSE detecting an EV plug-in to when it is unplugged.

NOTE	the session attributes hold their values from the previous session and are reset when the next plug-in event happens.

Whenever the State leaves the NotPluggedIn state and moves to any ‘Plugged’ state (see the State attribute), the SessionID attribute SHALL be incremented. The SessionDuration, SessionEnergyCharged and SessionEnergyDischarged attributes SHALL be reset to zero.

9.3.8.20.1. SessionID Attribute

This attribute SHALL indicate a unique identifier for the current or last session. A value of null indicates no sessions have occurred. The SessionID SHALL be incremented each time a plugin is detected. A session begins when the vehicle is plugged in and ends when the vehicle is unplugged.

SessionIDs are allowed to roll over, although the range of SessionID is a large number and it is unlikely that the EVSE will have this many sessions in its lifetime unless there is a electrical fault causes sessions to be detected at a rapid rate.

If there is no session in progress, the Session ID attribute will remain at the value for the last session.

9.3.8.20.2. SessionDuration Attribute

This attribute SHALL indicate the duration in seconds for the current or last charging session. A default value of null indicates no sessions have occurred. A charge session begins when the vehicle is plugged in and ends when the vehicle is unplugged.

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The SessionDuration can be calculated from the start time of the session. Manufacturers SHALL ensure this can be correctly calculated if there has been an power failure or reboot since the start of the session.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- At most once every 10 seconds on changes, or
- When it changes from null to any other value and vice versa, such as at the beginning and end of a charging session.

9.3.8.20.3. SessionEnergyCharged Attribute

This attribute SHALL indicate the energy, in mWh, delivered by the EVSE to the EV for the current or last charging session. A default value of null indicates no sessions have occurred.

The SessionEnergyCharged value can be calculated by knowing the initial value of the energy meter at the start time of the session. Manufacturers SHALL ensure this can be correctly calculated if there has been an power failure or reboot since the start of the session.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- At most once every 10 seconds on changes, or
- When it changes from null to any other value and vice versa, such as at the beginning and end of a charging session.

9.3.8.20.4. SessionEnergyDischarged Attribute

This attribute SHALL indicate the energy, in mWh, received by the EVSE from the EV for the current or last charging session. A default value of null indicates no sessions have occurred.

The SessionEnergyDischarged value can be calculated by knowing the initial value of the energy meter at the start time of the session. Manufacturers SHALL ensure this can be correctly calculated if there has been an power failure or reboot since the start of the session.

Changes to this attribute SHALL only be marked as reportable in the following cases:

- At most once every 10 seconds on changes, or
- When it changes from null to any other value and vice versa, such as at the beginning and end of a charging session.

9.3.9. Commands

ID	Name	Direction	Response	Access	Conformance
0x01	**Disable**	client ⇒ server	Y	O T	M
0x02	**EnableCharging**	client ⇒ server	Y	O T	M
0x03	**EnableDischarging**	client ⇒ server	Y	O T	V2X

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ID	Name	Direction	Response	Access	Conformance
0x04	**StartDiagnostics**	client ⇒ server	Y	O T	O
0x05	**SetTargets**	client ⇒ server	Y	O T	PREF
0x06	**GetTargets**	client ⇒ server	GetTargetsResponse	O T	PREF
0x07	**ClearTargets**	client ⇒ server	Y	O T	PREF
0x00	**GetTargetsResponse**	client ⇐ server	N		PREF

9.3.9.1. Disable Command

Allows a client to disable the EVSE from charging and discharging.

9.3.9.1.1. Effect on Receipt

On receipt of this command, the EVSE will stop any power flow between the EV and EVSE.

If the command cannot be handled, a status of FAILURE SHALL be returned.

If the SupplyState attribute is already Disabled, a response with status of SUCCESS SHALL be returned.

Otherwise, if successful, the ChargingEnabledUntil and DischargingEnabledUntil attributes SHALL be set to 0x0, the SupplyState attribute changed to Disabled, and a response with status of SUCCESS SHALL be returned.

If any energy was being transferred, then a corresponding EnergyTransferStopped event SHALL be generated.

9.3.9.2. EnableCharging Command

This command allows a client to enable the EVSE to charge an EV, and to provide or update the maximum and minimum charge current.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ChargingEnabledUntil**	epoch-s	all	X		M
1	**MinimumChargeCurrent**	amperage-mA	min 0			M
2	**MaximumChargeCurrent**	amperage-mA	min 0			M

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9.3.9.2.1. ChargingEnabledUntil Field

This field SHALL indicate the expiry time, in UTC, when charging will be automatically disabled.

A value in the past in this field SHALL disable the EVSE charging whereas a null value SHALL enable it permanently.

9.3.9.2.2. MinimumChargeCurrent Field

This field SHALL indicate the minimum current that can be delivered by the EVSE to the EV in trickle mode. The EVSE current limit can be advertised to an EV in 0.6A steps.

The value of the MinimumChargeCurrent attribute SHALL be set to the value of this field (see Section 9.3.8.7, “MinimumChargeCurrent Attribute” for further details).

9.3.9.2.3. MaximumChargeCurrent Field

This field SHALL indicate the maximum current that can be delivered by the EVSE to the EV. The EVSE current limit can be advertised to an EV in 0.6A steps.

The value of the this field SHALL be stored by the EVSE to determine the value of MaximumChargeCurrent attribute. For example, if the UserMaximumChargeCurrent attribute is adjusted below then this value, and then later adjusted above this value, the resulting MaximumChargeCurrent attribute will be limited to this value.

9.3.9.2.4. Effect on Receipt

On receipt of this command, this SHALL allow the EVSE to charge the EV until the specified timestamp within the current limits specified in the fields of the command.

If there is currently an error present on the EVSE, or Diagnostics are currently active, then the command SHALL be ignored and a response with a status of FAILURE SHALL be returned.

If successful, the SupplyState attribute SHALL be set to ChargingEnabled (if previously Disabled) or Enabled (if previously DischargingEnabled), and a response with status of SUCCESS SHALL be returned.

The timestamp indicated in the ChargingEnabledUntil attribute SHALL be updated to the timestamp of the ChargingEnabledUntil field if the command is successful.

If the ChargingEnabledUntil is null (i.e. should be permanently enabled), then the EVSE should enable charging indefinitely.

If the ChargingEnabledUntil time is not null, then when this time expires then the EVSE SHALL stop charging and SHALL update the State attribute to indicate it is no longer charging, and SHALL update the SupplyState attribute to Disabled (if DischargingEnabledUntil is also in the past) or DischargingEnabled (if DischargingEnabledUntil is in the future or null).

9.3.9.3. EnableDischarging Command

Upon receipt, this SHALL allow a client to enable the discharge of an EV, and to provide or update the maximum discharge current.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**DischargingEnabledUntil**	epoch-s	all	X		M
1	**MaximumDischargeCurrent**	amperage-mA	min 0			M

9.3.9.3.1. DischargingEnabledUntil Field

This field SHALL indicate the expiry time, in UTC, when discharging will be automatically disabled.

A value in the past in this field SHALL disable the EVSE discharging whereas a null value SHALL enable EVSE discharging permanently.

9.3.9.3.2. MaximumDischargeCurrent Field

This field SHALL indicate the maximum current that can be received by the EVSE from the EV. The EVSE current limit can be advertised to an EV in 0.6A steps. The value of the MaximumDischargeCurrent attribute SHALL be stored and persisted across reboots by the EVSE to the value of this field.

9.3.9.3.3. Effect on Receipt

On receipt of this command, this SHALL allow the EVSE to discharge the EV until the specified timestamp, and allows a maximum current limit to be specified in the fields of the command.

If there is currently an error present on the EVSE, or Diagnostics are currently active, then the command SHALL be ignored and a response with a status of FAILURE SHALL be returned.

If successful, the SupplyState attribute SHALL be set to DischargingEnabled (if previously Disabled) or Enabled (if previously ChargingEnabled), and a response with status of SUCCESS SHALL be returned.

The timestamp indicated in the DischargingEnabledUntil attribute SHALL be updated to the timestamp of the DischargingEnabledUntil field if the command is successful.

If the DischargingEnabledUntil is null (i.e. should be permanently enabled), then the EVSE should enable discharging indefinitely.

If the DischargingEnabledUntil time is not null, then when this time expires then the EVSE SHALL stop discharging and SHALL update the State attribute to indicate it is no longer discharging, and SHALL update the SupplyState attribute to Disabled (if ChargingEnabledUntil is also in the past) or ChargingEnabled (if ChargingEnabledUntil is in the future or null).

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9.3.9.4. StartDiagnostics Command

Allows a client to put the EVSE into a self-diagnostics mode.

9.3.9.4.1. Effect on Receipt

On receipt of this command, the EVSE SHALL enter a Diagnostics state only if the SupplyState attribute is in the Disabled state.

If the EVSE cannot start diagnostics, a response SHALL be generated with a status of FAILURE.

If successful, the SupplyState attribute SHALL be set to DisabledDiagnostics, and a response with status of SUCCESS SHALL be returned.

The diagnostics are at the discretion of the manufacturer and usually include internal checks. Upon completion of the diagnostics, the EVSE SHALL restore SupplyState to the Disabled state (see Section 9.3.8.2, “SupplyState Attribute” for further details).

9.3.9.5. SetTargets Command

Allows a client to set the user specified charging targets.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ChargingTargetSchedules**	list[ChargingTargetScheduleStruct]	max 7			M

9.3.9.5.1. ChargingTargetSchedules Field

This field SHALL indicate a list of up to 7 sets of daily charging targets together with their associated days of the week. Each of the days of the week may only be included in a single ChargingTargetSchedule within this list field.

9.3.9.5.2. Effect on Receipt

On receipt of this command, the EVSE SHALL validate that all of the charging targets are within a valid range, and that each day of the week is included in at most one of the ChargingTargetSchedule, if they are not then the response SHALL be CONSTRAINT_ERROR.

When a command is received that requires a total number of charging targets greater than the device supports, the status of the response SHALL be RESOURCE_EXHAUSTED.

If the charging targets are accepted, then the charging targets SHALL be stored for all of days that are set in the DayOfWeekForSequence bitmap fields in all the ChargingTargetSchedules. If a ChargingTargetSchedule is defined with no ChargingTargets then the ChargingTargets are cleared for those days defined in the DayOfWeekForSequence

The SetTargets command is used to update the EVSE’s weekly charging schedule. If the EVSE already has some stored weekly charging targets, then it SHALL replace each daily charging target

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as it receives the updates from the client. For example, if the EVSE has 2 charging targets for every day of the week and is sent a SetTargets command with one target for Saturday then the EVSE SHALL remove both charging targets for Saturday and replace those with the updated charging target but leave all other days unchanged.

NOTE	the EVSE may not be able to compute the schedules by itself, or may rely upon an EMS or other optimizer to do this.

In a standalone mode (e.g. without using a ToU tariff from the Pricing Cluster) the EVSE should be able to compute the latest NextChargeStartTime based on the preset MaximumChargeCurrent attribute and local grid voltage to determine the charging duration required. It should automatically begin charging and following the charging schedule if the EVSE is plugged in (and the EVSE is enabled for charging).

If the EVSE supports the PFR (PowerForecastReporting) feature in the Device Energy Management cluster, it SHALL auto update the Forecast with the indicative power forecast.

If the EVSE supports the FA (ForecastAdjustment) feature in the Device Energy Management cluster, it SHALL indicate its adjustment capability in each of the Forecast slots (see Device Energy Management Cluster for more details).

The EVSE SHALL be responsible for updating the NextChargeEndTime, NextChargeRequiredEnergy and/or NextChargeTargetSoC attributes as it runs through its internal schedule. For example, as a scheduled charging session is completed or as it transitions to the next day, it should compute the next time it expects to start charging, and then update the corresponding attributes accordingly.

9.3.9.6. GetTargets Command

Allows a client to retrieve the current set of charging targets.

9.3.9.6.1. Effect on Receipt

On receipt of this command, the EVSE SHALL send the GetTargetsResponse command to the client.

9.3.9.7. GetTargetsResponse Command

The GetTargetsResponse is sent in response to the GetTargets Command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ChargingTargetSchedules**	list[ChargingTargetScheduleStruct]	max 7			M

9.3.9.7.1. ChargingTargetSchedules Field

This field SHALL indicate a list of up to 7 sets of daily charging targets together with their associated days of the week.

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9.3.9.8. ClearTargets Command

Allows a client to clear all stored charging targets.

9.3.9.8.1. Effect on Receipt

On receipt of this command, all weekly targets that are currently stored SHALL be cleared and a default response of SUCCESS SHALL be sent in response. There are no error responses to this command.

If the EVSE is currently charging based on being in automatic mode, then it SHALL stop the EVSE charging.

9.3.10. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**EVConnected**	INFO		V	M
0x01	**EVNotDetected**	INFO		V	M
0x02	**EnergyTransferStarted**	INFO		V	M
0x03	**EnergyTransferStopped**	INFO		V	M
0x04	**Fault**	CRITICAL		V	M
0x05	**RFID**	INFO		V	[RFID]

9.3.10.1. EVConnected Event

This event SHALL be generated when the EV is plugged in.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**SessionID**	uint32	all			M

9.3.10.1.1. SessionID Field

This is the new session ID created after the vehicle is plugged in.

9.3.10.2. EVNotDetected Event

This event SHALL be generated when the EV is unplugged or not detected (having been previously plugged in). When the vehicle is unplugged then the session is ended.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**SessionID**	uint32	all			M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
1	**State**	StateEnum	all			M
2	**SessionDuration**	elapsed-s	all			M
3	**SessionEnergyCharged**	energy-mWh	min 0			M
4	**SessionEnergyDischarged**	energy-mWh	min 0			V2X

9.3.10.2.1. SessionID Field

This field SHALL indicate the current value of the SessionID attribute.

9.3.10.2.2. State Field

This field SHALL indicate the value of the State attribute prior to the EV not being detected.

9.3.10.2.3. SessionDuration Field

This field SHALL indicate the total duration of the session, from the start of the session when the EV was plugged in, until it was unplugged.

9.3.10.2.4. SessionEnergyCharged Field

This field SHALL indicate the total amount of energy transferred from the EVSE to the EV during the session.

Note that if bi-directional charging occurs during the session, then this value SHALL only include the sum of energy transferred from the EVSE to the EV, and SHALL NOT be a net value of charging and discharging energy.

9.3.10.2.5. SessionEnergyDischarged Field

This field SHALL indicate the total amount of energy transferred from the EV to the EVSE during the session.

Note that if bi-directional discharging occurs during the session, then this value SHALL only include the sum of energy transferred from the EV to the EVSE, and SHALL NOT be a net value of charging and discharging energy.

9.3.10.3. EnergyTransferStarted Event

This event SHALL be generated whenever the EV starts charging or discharging, except when an EV has switched between charging and discharging under the control of the PowerAdjustment feature of the Device Energy Management cluster of the associated Device Energy Management device.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**SessionID**	uint32	all			M
1	**State**	StateEnum	all			M
2	**MaximumCurrent**	amperage-mA	min 0			M
3	**MaximumDischargeCurrent**	amperage-mA	min 0			V2X

9.3.10.3.1. SessionID Field

This field SHALL indicate the value of the SessionID attribute at the time the event was generated.

9.3.10.3.2. State Field

This field SHALL indicate the value of the State attribute at the time the event was generated.

9.3.10.3.3. MaximumCurrent Field

This field SHALL indicate the value of the maximum charging current at the time the event was generated.

A non-zero value indicates that the EV has been enabled for charging and the value is taken directly from the MaximumChargeCurrent attribute. A zero value indicates that the EV has not been enabled for charging.

9.3.10.3.4. MaximumDischargeCurrent Field

This field SHALL indicate the value of the maximum discharging current at the time the event was generated.

A non-zero value indicates that the EV has been enabled for discharging and the value is taken directly from the MaximumDischargeCurrent attribute. A zero value indicates that the EV has not been enabled for discharging.

9.3.10.4. EnergyTransferStopped Event

This event SHALL be generated whenever the EV stops charging or discharging, except when an EV has switched between charging and discharging under the control of the PowerAdjustment feature of the Device Energy Management cluster of the associated Device Energy Management device.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**SessionID**	uint32	all			M
1	**State**	StateEnum	all			M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
2	**Reason**	Energy-Transfer-StoppedReasonEnum	all			M
4	**Energy-Transferred**	energy-mWh	min 0			M
5	**EnergyDischarged**	energy-mWh	min 0			V2X

9.3.10.4.1. SessionID Field

This field SHALL indicate the value of the SessionID attribute prior to the energy transfer stopping.

9.3.10.4.2. State Field

This field SHALL indicate the value of the State attribute prior to the energy transfer stopping.

9.3.10.4.3. Reason Field

This field SHALL indicate the reason why the energy transferred stopped.

9.3.10.4.4. EnergyTransferred Field

This field SHALL indicate the amount of energy transferred from the EVSE to the EV since the previous EnergyTransferStarted event, in mWh.

9.3.10.4.5. EnergyDischarged Field

This field SHALL indicate the amount of energy transferred from the EV to the EVSE since the previous EnergyTransferStarted event, in mWh.

9.3.10.5. Fault Event

If the EVSE detects a fault it SHALL generate a Fault Event. The SupplyState attribute SHALL be set to DisabledError and the type of fault detected by the EVSE SHALL be stored in the FaultState attribute.

This event SHALL be generated when the FaultState changes from any error state. i.e. if it changes from NoError to any other state and if the error then clears, this would generate 2 events.

It is assumed that the fault will be cleared locally on the EVSE device. When all faults have been cleared, the EVSE device SHALL set the FaultState attribute to NoError and the SupplyState attribute SHALL be set back to its previous state.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**SessionID**	uint32	all	X		M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
1	**State**	StateEnum	all			M
2	**FaultStatePreviousState**	FaultStateEnum	all			M
4	**FaultStateCurrentState**	FaultStateEnum	all			M

9.3.10.5.1. SessionID Field

This field SHALL indicate the value of the SessionID attribute prior to the Fault State being changed. A value of null indicates no sessions have occurred before the fault occurred.

9.3.10.5.2. State Field

This field SHALL indicate the value of the State attribute prior to the Fault State being changed.

9.3.10.5.3. FaultStatePreviousState Field

This field SHALL indicate the value of the FaultState attribute prior to the Fault State being changed.

9.3.10.5.4. FaultStateCurrentState Field

This field SHALL indicate the current value of the FaultState attribute.

9.3.10.6. RFID Event

This event SHALL be generated when a RFID card has been read. This allows a controller to register the card ID and use this to authenticate and start the charging session.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**UID**	octstr	max 10			M

9.3.10.6.1. UID Field

The UID field (ISO 14443A UID) is either 4, 7 or 10 bytes.

9.4. Energy EVSE Mode Cluster

This cluster is derived from the Mode Base cluster and defines additional mode tags and namespaced enumerated values for EVSE devices.

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9.4.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Disallowed DEPONOFF feature and OnMode and StartUpMode attributes

9.4.2. Classification

Hierarchy	Role	Scope	PICS Code
Mode Base	Application	Endpoint	EEVSEM

9.4.3. Cluster ID

ID	Name
0x009D	Energy EVSE Mode

9.4.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**DEPONOFF**	OnOff	X	Dependency with the OnOff cluster

9.4.5. Data Types

9.4.5.1. ModeOptionStruct Type

The table below lists the changes relative to the Mode Base cluster for the fields of the ModeOptionStruct type. A blank field indicates no change.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Label**						M
1	**Mode**						M
2	**ModeTags**		1 to 8				M

9.4.6. Attributes

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SupportedModes**						
0x0001	**CurrentMode**						
0x0002	**StartUpMode**						X
0x0003	**OnMode**						X

9.4.6.1. SupportedModes Attribute

At least one entry in the SupportedModes attribute SHALL include the Manual mode tag in the ModeTags field list.

Modes with entries in the SupportedModes attribute which contain multiple mode tags permitting charging or discharging under different conditions SHALL permit the charging or discharging to occur if any of the conditions are satisfied.

Modes SHALL NOT have both the Manual tag and the TimeOfUse or SolarCharging tags defined in the SupportedModes attribute.

9.4.7. Derived Cluster Namespace

This namespace includes definitions for data associated exclusively with the derived cluster.

9.4.7.1. Mode Tags

The following table defines the base and derived-cluster-specific ModeTag values.

Mode Tag Value	Name
0x0000	**Auto**
0x0001	**Quick**
0x0002	**Quiet**
0x0003	**LowNoise**
0x0004	**LowEnergy**
0x0005	**Vacation**
0x0006	**Min**
0x0007	**Max**
0x0008	**Night**
0x0009	**Day**
0x4000	**Manual**

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Mode Tag Value	Name
0x4001	**TimeOfUse**
0x4002	**SolarCharging**
0x4003	**V2X**

9.4.7.1.1. Manual Tag

While in modes with this tag, and once enabled with the EnableCharging command, the EVSE will permit charging based on demand from the EV.

9.4.7.1.2. TimeOfUse Tag

While in modes with this tag, and once enabled with the EnableCharging command, the EVSE will attempt to automatically start charging based on the user's charging targets (for example, set based on a Time of Use tariff to charge at the cheapest times of the day).

9.4.7.1.3. SolarCharging Tag

While in modes with this tag, and once enabled with the EnableCharging, the EVSE will attempt to automatically start charging based on available excess solar PV generation, limiting the charging power to avoid importing energy from the grid.

9.4.7.1.4. V2X Tag

While in modes with this tag, and once enabled with the EnableDischarging command, the EVSE will permit discharging based on the current charge state of the EV, and its control from an associated Device Energy Management cluster.

NOTE	being in a mode with this tag set or not does not affect the handling of the EnableDischarging command by the Energy EVSE cluster, but once enabled, only modes with this tag enable the discharging to actually occur.

9.4.8. Mode Examples

A few examples of EVSE modes and their mode tags are provided below.

- For the "Manual" mode, tags: 0x4000 (Manual)
- For the "Auto-scheduled" mode, tags: 0x4001 (TimeOfUse)
- For the "Solar" mode, tags: 0x4002 (SolarCharging)
- For the "Auto-scheduled with Solar charging" mode, tags: 0x4001 (TimeOfUse), 0x4002 (SolarCharging)
- For the "Vehicle-to-home with Solar charging" mode, tags: 0x4002 (SolarCharging), 0x4003 (V2X)

9.5. Water Heater Management Cluster

This cluster is used to allow clients to control the operation of a hot water heating appliance so that

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it can be used with energy management.

Heating of hot water is one of the main energy uses in homes, and when coupled with the Energy Management cluster, it can help consumers save cost (e.g. using power at cheaper times or from local solar PV generation).

9.5.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Added events and simplified data types

9.5.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	EWATERHTR

9.5.3. Cluster ID

ID	Name
0x0094	Water Heater Management

9.5.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**EM**	EnergyManagement	O	Allows energy management control of the tank
1	**TP**	TankPercent	O	Supports monitoring the percentage of hot water in the tank

9.5.5. Dependencies

This cluster SHALL be on an endpoint that also includes the Thermostat cluster as this cluster is dependent on the Thermostat cluster.

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9.5.6. Data Types

9.5.6.1. WaterHeaterHeatSourceBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**ImmersionElement1**	Immersion Heating Element 1	M
1	**ImmersionElement2**	Immersion Heating Element 2	M
2	**HeatPump**	Heat pump Heating	M
3	**Boiler**	Boiler Heating (e.g. Gas or Oil)	M
4	**Other**	Other Heating	M

9.5.6.2. BoostStateEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Inactive**	Boost is not currently active	M
1	**Active**	Boost is currently active	M

9.5.6.3. WaterHeaterBoostInfoStruct Type

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Duration**	elapsed-s	min 1				M
1	**OneShot**	bool	all		False		[!TP], [TP].a-
2	**EmergencyBoost**	bool	all		False		O
3	**TemporarySetpoint**	temperature	desc				O
4	**TargetPercentage**	percent	all				TargetReheat, [TP]
5	**TargetReheat**	percent	max TargetPercentage				[TP].a-

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9.5.6.3.1. Duration Field

This field SHALL indicate the time period, in seconds, for which the boost state is activated.

9.5.6.3.2. OneShot Field

This field SHALL indicate whether the boost state SHALL be automatically canceled once the hot water has reached either:

- the set point temperature (from the thermostat cluster)
- the TemporarySetpoint temperature (if specified)
- the TargetPercentage (if specified).

9.5.6.3.3. EmergencyBoost Field

This field SHALL indicate that the consumer wants the water to be heated quickly. This MAY cause multiple heat sources to be activated (e.g. a heat pump and direct electric immersion heating element).

The choice of which heat sources are activated is manufacturer specific.

9.5.6.3.4. TemporarySetpoint Field

This field SHALL indicate the target temperature to which the water will be heated.

If included, it SHALL be used instead of the thermostat cluster set point temperature whilst the boost state is activated.

The value of this field SHALL be within the constraints of the MinHeatSetpointLimit and MaxHeatSetpointLimit attributes (inclusive), of the thermostat cluster.

9.5.6.3.5. TargetPercentage Field

This field SHALL indicate the target percentage of hot water in the tank that the TankPercentage attribute must reach before the heating is switched off.

9.5.6.3.6. TargetReheat Field

This field SHALL indicate the percentage to which the hot water in the tank SHALL be allowed to fall before again beginning to reheat it.

For example if the TargetPercentage was 80%, and the TargetReheat was 40%, then after initial heating to 80% hot water, the tank may have hot water drawn off until only 40% hot water remains. At this point the heater will begin to heat back up to 80% of hot water. If this field and the OneShot field were both omitted, heating would begin again after any water draw which reduced the TankPercentage below 80%.

This field SHALL be less than or equal to the TargetPercentage field.

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9.5.7. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**HeaterTypes**	WaterHeaterHeatSourceBitmap	all	F		R V	M
0x0001	**HeatDemand**	WaterHeaterHeatSourceBitmap	all			R V	M
0x0002	**TankVolume**	uint16	all			R V	EM
0x0003	**EstimatedHeatRequired**	energy-mWh	min 0			R V	EM
0x0004	**TankPercentage**	percent	all			R V	TP
0x0005	**BoostState**	BoostStateEnum	all			R V	M

9.5.7.1. HeaterTypes Attribute

This attribute SHALL indicate the heat sources that the water heater can call on for heating. If a bit is set then the water heater supports the corresponding heat source.

9.5.7.2. HeatDemand Attribute

This attribute SHALL indicate if the water heater is heating water. If a bit is set then the corresponding heat source is active.

9.5.7.3. TankVolume Attribute

This attribute SHALL indicate the volume of water that the hot water tank can hold (in units of Litres). This allows an energy management system to estimate the required heating energy needed to reach the target temperature.

9.5.7.4. EstimatedHeatRequired Attribute

This attribute SHALL indicate the estimated heat energy needed to raise the water temperature to the target setpoint. This can be computed by taking the specific heat capacity of water (4182 J/kg °C) and by knowing the current temperature of the water, the tank volume and target temperature.

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For example, if the target temperature was 60°C, the current temperature was 20°C and the tank volume was 100L:

```
Mass of water          = 1kg per Litre
Total Mass             = 100 x 1kg = 100kg
Δ Temperature          = (target temperature - current temperature)
                       = (60°C - 20°C) = 40°C
```

```
Energy required to
heat the water to 60°C = 4182 x 40 x 100 = 16,728,000 J
```

```
Converting Joules in to Wh of heat (divide by 3600):
                       = 16,728,000 J / 3600
                       = 4647 Wh (4.65kWh)
```

If the TankPercent feature is supported, then this estimate SHALL also take into account the percentage of the water in the tank which is already hot.

NOTE

The electrical energy required to heat the water depends on the heating system used to heat the water. For example, a direct electric immersion heating element can be close to 100% efficient, so the electrical energy needed to heat the hot water is nearly the same as the EstimatedHeatEnergyRequired. However some forms of heating, such as an air-source heat pump which extracts heat from ambient air, requires much less electrical energy to heat hot water. Heat pumps can be produce 3kWh of heat output for 1kWh of electrical energy input. The conversion between heat energy and electrical energy is outside the scope of this cluster.

9.5.7.5. TankPercentage Attribute

This attribute SHALL indicate an approximate level of hot water stored in the tank, which might help consumers understand the amount of hot water remaining in the tank. The accuracy of this attribute is manufacturer specific.

In most hot water tanks, there is a stratification effect where the hot water layer rests above a cooler layer of water below. For this reason cold water is fed in at the bottom of the tank and the hot water is drawn from the top.

Some water tanks might use multiple temperature probes to estimate the level of the hot water layer. A water heater with multiple temperature probes is likely to implement an algorithm to estimate the hot water tank percentage by taking into account the temperature values of each probe to determine the height of the hot water.

However it might be possible with a single temperature probe to estimate how much hot water is left using a simpler algorithm:

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For example, if the target temperature was 60°C, the CurrentTemperature was 40°C from a single temperature probe measuring the average water temperature and the temperature of incoming cold water (COLD_WATER_TEMP) was assumed to be 20°C:

```
TankPercentage = int(((current temperature - COLD_WATER_TEMP) / (target
temperature - COLD_WATER_TEMP)) * 100)
TankPercentage = min( max(TankPercentage,0), 100)
```

```
TankPercentage = 50%
```

9.5.7.6. BoostState Attribute

This attribute SHALL indicate whether the Boost, as triggered by a Boost command, is currently `Active` or `Inactive`.

See Section 9.5.8.1, “Boost Command” and Section 9.5.8.2, “CancelBoost Command” for more details.

9.5.8. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**Boost**	client ⇒ server	Y	M	M
0x01	**CancelBoost**	client ⇒ server	Y	M	M

9.5.8.1. Boost Command

Allows a client to request that the water heater is put into a Boost state.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**BoostInfo**	WaterHeaterBoostInfoStruct	all			M

9.5.8.1.1. Effect on receipt

On receipt of this command, this SHALL allow the water heater to attempt to heat the water, which may override other settings, for example, if the Water Heater Mode is set to `Off`, or `Timed` and it is during one of the Off periods.

If the duration field is too short for the water heater to accept, for example a heat pump may take several minutes to ramp up in operation, then the boost command SHALL be rejected with a status of INVALID_IN_STATE.

If successful, the value of the BoostState attribute SHALL transition to `Active`, and a response with

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status of SUCCESS SHALL be returned, and a BoostStarted event SHALL be generated to include the field values received in the command.

The water in the tank SHALL be heated until one of the following conditions is met:

- the temperature reaches the set point in the corresponding thermostat cluster or the TemporarySetpoint (if included)
- the TankPercentage attribute value reaches the TargetPercentage (if it was included)

If OneShot is specified then once the hot water has reached the set point temperature (or the TemporarySetpoint temperature, if specified) or the TargetPercentage (if specified), or the boost command's duration times out after the specified Duration, then BoostState transitions to `Inactive`. This SHALL cause the BoostEnded event to be generated.

If the Water Heater was already in the BoostState 'Active' when this command is received, it SHALL continue in this BoostState, but SHALL discard the effect of the values of the fields from the previous Boost commands, and SHALL continue with the values of the fields from the new Boost command. A new BoostStarted event SHALL be generated to include the field values received in the new command.

9.5.8.2. CancelBoost Command

Allows a client to cancel an ongoing Boost operation.

This command has no payload.

9.5.8.2.1. Effect on receipt

If the BoostState attribute value was already `Inactive` when this command is received, the BoostState attribute value shall remain `Inactive` and the server SHALL return SUCCESS.

Otherwise the Water Heater SHALL transition the BoostState to `Inactive`, generate a BoostEnded event, and the water heating reverts to being controlled through the current Water Heater Mode (e.g. `Off`, `Manual` or `Timed`).

9.5.9. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**BoostStarted**	INFO		V	M
0x01	**BoostEnded**	INFO		V	M

9.5.9.1. BoostStarted Event

This event SHALL be generated whenever a Boost command is accepted.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**BoostInfo**	WaterHeaterBoostInfoStruct	all			M

The corresponding structure fields within the WaterHeaterBoostInfoStruct are copied from the Boost command.

9.5.9.2. BoostEnded Event

This event SHALL be generated whenever the BoostState transitions from `Active` to `Inactive`.

9.6. Water Heater Mode Cluster

This cluster is derived from the Mode Base cluster and defines additional mode tags and namespaced enumerated values for water heater devices.

9.6.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

9.6.2. Classification

Hierarchy	Role	Scope	PICS Code
Mode Base	Application	Endpoint	WHM

9.6.3. Cluster ID

ID	Name
0x009E	Water Heater Mode

9.6.4. Features

Bit	Code	Feature	Conformance	Summary
0	**DEPONOFF**	OnOff	X	Dependency with the OnOff cluster

9.6.5. Data Types

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9.6.5.1. ModeOptionStruct Type

The table below lists the changes relative to the Mode Base cluster for the fields of the ModeOptionStruct type. A blank field indicates no change.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Label**						M
1	**Mode**						M
2	**ModeTags**		1 to 8				M

9.6.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SupportedModes**						
0x0001	**CurrentMode**						
0x0002	**StartUpMode**						X
0x0003	**OnMode**						X

9.6.6.1. SupportedModes Attribute

At least one entry in the SupportedModes attribute SHALL include the Manual mode tag in the ModeTags field list.

At least one entry in the SupportedModes attribute SHALL include the Off mode tag in the ModeTags field list.

An entry in the SupportedModes attribute that includes one of an Off, Manual, or Timed tag SHALL NOT also include an additional instance of any one of these tag types.

9.6.7. Derived Cluster Namespace

This namespace includes definitions for data associated exclusively with the derived cluster.

9.6.7.1. Mode Tags

The following table defines the base and derived-cluster-specific ModeTag values.

Mode Tag Value	Name
0x0000	**Auto**
0x0001	**Quick**

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Mode Tag Value	Name
0x0002	**Quiet**
0x0003	**LowNoise**
0x0004	**LowEnergy**
0x0005	**Vacation**
0x0006	**Min**
0x0007	**Max**
0x0008	**Night**
0x0009	**Day**
0x4000	**Off**
0x4001	**Manual**
0x4002	**Timed**

9.6.7.1.1. Off Tag

While in modes with this tag, the device will not attempt to keep the water warm.

9.6.7.1.2. Manual Tag

While in modes with this tag, the device will attempt to keep the water warm based on the OccupiedHeatingSetpoint attribute of the associated Thermostat cluster.

9.6.7.1.3. Timed Tag

While in modes with this tag, the device will attempt to keep the water warm based on the Schedules attribute of the associated Thermostat cluster.

9.6.8. Mode Examples

A few examples of Water Heater modes and their mode tags are provided below.

- For the "Off" mode, tags: 0x4000 (Off)
- For the "Manual" mode, tags: 0x4001 (Manual)
- For the "Timed" mode, tags: 0x4002 (Timed)

9.7. Energy Preference Cluster

This cluster provides an interface to specify preferences for how devices should consume energy.

9.7.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

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Revision	Description
1	Initial revision

9.7.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	EPREF

9.7.3. Cluster ID

ID	Name
0x009B	Energy Preference

9.7.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**BALA**	EnergyBalance	O.a+	Device can balance energy consumption vs. another priority
1	**LPMS**	LowPowerModeSensitivity	O.a+	Device can adjust the conditions for entering a low power mode

9.7.4.1. EnergyBalance Feature

This feature allows a user to select from a list of energy balances with associated descriptions of which strategies a device will use to target the specified balance.

9.7.4.2. LowPowerModeSensitivity Feature

This feature allows the user to select a condition or set of conditions which will cause the device to switch to a mode using less power. For example, a device might provide a scale of durations that must elapse without user interaction before it goes to sleep.

9.7.5. Data Types

9.7.5.1. EnergyPriorityEnum Type

This data type is derived from enum8.

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Value	Name	Summary	Conformance
0	**Comfort**	User comfort	M
1	**Speed**	Speed of operation	M
2	**Efficiency**	Amount of Energy consumed by the device	M
3	**WaterConsumption**	Amount of water consumed by the device	M

9.7.5.1.1. Comfort Value

This value SHALL emphasize user comfort; e.g. local temperature for a thermostat.

9.7.5.1.2. Speed Value

This value SHALL emphasize how quickly a device accomplishes its targeted use; e.g. how quickly a robot vacuum completes a cleaning cycle.

9.7.5.1.3. Efficiency Value

This value SHALL emphasize how much energy a device uses; e.g. electricity usage for a Pump.

9.7.5.1.4. Water consumption Value

This value SHALL emphasize how much water is consumed during device use; e.g. how much water a dishwasher uses during a cleaning cycle.

9.7.5.2. BalanceStruct Type

This represents a step along a scale of preferences.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Step**	percent	all	F			M
1	**Label**	string	max 64	F			O

9.7.5.2.1. Step Field

This field SHALL indicate the relative value of this step.

9.7.5.2.2. Label Field

This field SHALL indicate an optional string explaining which actions a device might take at the given step value.

9.7.6. Attributes

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**EnergyBalances**	list[BalanceStruct]	2 to 10	F		R V	BALA
0x0001	**CurrentEnergyBalance**	uint8	all	N		RW VO	BALA
0x0002	**EnergyPriorities**	list[EnergyPriorityEnum]	2	F		R V	BALA
0x0003	**LowPowerModeSensitivities**	list[BalanceStruct]	2 to 10	F		R V	LPMS
0x0004	**CurrentLowPowerModeSensitivity**	uint8	all	N		RW VO	LPMS

9.7.6.1. EnergyBalances Attribute

This attribute SHALL indicate a list of BalanceStructs, each representing a step along a linear scale of relative priorities. A Step field with a value of zero SHALL indicate that the device SHOULD entirely favor the priority specified by the first element in EnergyPriorities; whereas a Step field with a value of 100 SHALL indicate that the device SHOULD entirely favor the priority specified by the second element in EnergyPriorities. The midpoint value of 50 SHALL indicate an even split between the two priorities.

This SHALL contain at least two BalanceStructs.

Each BalanceStruct SHALL have a Step field larger than the Step field on the previous BalanceStruct in the list.

The first BalanceStruct SHALL have a Step value of zero, and the last BalanceStruct SHALL have a Step value of 100.

9.7.6.2. CurrentEnergyBalance Attribute

This attribute SHALL indicate the current preference of the user for balancing different priorities during device use. The value of this attribute is the index, 0-based, into the EnergyBalances attribute for the currently selected balance.

If an attempt is made to set this attribute to an index outside the maximum index for EnergyBalances, a response with the status code CONSTRAINT_ERROR SHALL be returned.

If the value of EnergyBalances changes after an update, the device SHALL migrate the value of the CurrentEnergyBalance attribute to the index which the manufacturer specifies most closely

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matches the previous value, while preserving extreme preferences as follows:

1. If the previous value of CurrentEnergyBalance was zero, indicating a total preference for the priority specified by the first element in EnergyPriorities, the new value of CurrentEnergyBalance SHALL also be zero.
2. If the previous value of CurrentEnergyBalance was the index of the last BalanceStruct in the previous value of EnergyBalances, indicating a total preference for the priority specified by the last element in EnergyPriorities, the new value of CurrentEnergyBalance SHALL be the index of the last element in the updated value of EnergyBalances.

9.7.6.3. EnergyPriorities Attribute

This attribute SHALL indicate two extremes for interpreting the values in the EnergyBalances attribute. These two priorities SHALL be in opposition to each other; e.g. Comfort vs. Efficiency or Speed vs. WaterConsumption.

If the value of EnergyPriorities changes after an update to represent a new balance between priorities, the value of the CurrentEnergyBalance attribute SHALL be set to its default.

9.7.6.4. LowPowerModeSensitivities Attribute

This attribute SHALL indicate a list of BalanceStructs, each representing a condition or set of conditions for the device to enter a low power mode.

This SHALL contain at least two BalanceStructs.

Each BalanceStruct SHALL have a Step field larger than the Step field on the previous BalanceStruct in the list.

9.7.6.5. CurrentLowPowerModeSensitivity Attribute

This attribute SHALL indicate the current preference of the user for determining when the device should enter a low power mode. The value of this attribute is the index, 0-based, into the LowPowerModeSensitivities attribute for the currently selected preference.

If an attempt is made to set this attribute to an index outside the maximum index for LowPowerModeSensitivities, a response with the status code CONSTRAINT_ERROR SHALL be returned.

If the value of LowPowerModeSensitivities changes after an update, the device SHALL migrate the value of the LowPowerModeSensitivity attribute to the index which the manufacturer specifies most closely matches the previous value.

9.8. Device Energy Management Mode Cluster

This cluster is derived from the Mode Base cluster and defines additional mode tags and namespaced enumerated values for Device Energy Management devices.

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9.8.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision
2	Disallowed DEPONOFF feature and OnMode and StartUpMode attributes

9.8.2. Classification

Hierarchy	Role	Scope	PICS Code
Mode Base	Application	Endpoint	DEMM

9.8.3. Cluster ID

ID	Name
0x009F	Device Energy Management Mode

9.8.4. Features

Bit	Code	Feature	Conformance	Summary
0	**DEPONOFF**	OnOff	X	Dependency with the OnOff cluster

9.8.5. Data Types

9.8.5.1. ModeOptionStruct Type

The table below lists the changes relative to the Mode Base cluster for the fields of the ModeOptionStruct type. A blank field indicates no change.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Label**						M
1	**Mode**						M
2	**ModeTags**		1 to 8				M

9.8.6. Attributes

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SupportedModes**						
0x0001	**CurrentMode**						
0x0002	**StartUpMode**						X
0x0003	**OnMode**						X

9.8.6.1. SupportedModes Attribute

At least one entry in the SupportedModes attribute SHALL include the NoOptimization mode tag in the ModeTags field.

At least one entry in the SupportedModes attribute SHALL include the LocalOptimization mode tag in the ModeTags field list.

At least one entry in the SupportedModes attribute SHALL include the GridOptimization mode tag in the ModeTags field list.

An entry in the SupportedModes attribute that includes one of an DeviceOptimization, LocalOptimization, or GridOptimization tags SHALL NOT also include NoOptimization tag.

9.8.7. Derived Cluster Namespace

This namespace includes definitions for data associated exclusively with the derived cluster.

9.8.7.1. Mode Tags

The following table defines the base and derived-cluster-specific ModeTag values.

Mode Tag Value	Name
0x0000	**Auto**
0x0001	**Quick**
0x0002	**Quiet**
0x0003	**LowNoise**
0x0004	**LowEnergy**
0x0005	**Vacation**
0x0006	**Min**
0x0007	**Max**
0x0008	**Night**
0x0009	**Day**

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Mode Tag Value	Name
0x4000	**NoOptimization**
0x4001	**DeviceOptimization**
0x4002	**LocalOptimization**
0x4003	**GridOptimization**

9.8.7.1.1. NoOptimization Tag

The device prohibits optimization of energy usage management: its energy usage is determined only by the user configuration and internal device needs.

9.8.7.1.2. DeviceOptimization Tag

The device is permitted to manage its own energy usage. For example, using tariff information it may obtain.

9.8.7.1.3. LocalOptimization Tag

The device permits management of energy usage by an energy manager to optimize the local energy usage.

9.8.7.1.4. GridOptimization Tag

The device permits management of energy usage by an energy manager to optimize the grid energy usage.

9.8.8. Mode Examples

A few examples of Device Energy Management modes and their mode tags are provided below.

- For the "No Energy Management (Forecast reporting only)" mode, tags: 0x4000 (NoOptimization).
- For the "Device Energy Management" mode, tags: 0x4001 (DeviceOptimization).
- For the "Home Energy Management" mode, tags: 0x4001 (DeviceOptimization), 0x4002 (LocalOptimization).
- For the "Grid Energy Management" mode, tags: 0x4003 (GridOptimization).
- For the "Full Energy Management" mode, tags: 0x4001 (DeviceOptimization), 0x4002 (LocalOptimization), 0x4003 (GridOptimization).

9.9. Commodity Price Cluster

The Commodity Price Cluster provides the mechanism for communicating Gas, Energy, or Water pricing information within the premises.

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9.9.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Mandatory global ClusterRevision attribute added
2	Updated from SE1.4 version; CCB 1447 2964 2965
3	CCBs 2043 2706 2810 2912 3566
4	Initial Matter revision

9.9.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	SEPR

9.9.3. Cluster ID

ID	Name
0x0095	Commodity Price

9.9.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**FORE**	Forecasting	O	Forecasts upcoming pricing

9.9.5. Data Types

9.9.5.1. CommodityPriceDetailBitmap Type

This data type is derived from map16.

Bit	Name	Summary	Conformance
0	**Description**	A textual description of a price; e.g. the name of a rate plan.	M
1	**Components**	A breakdown of the component parts of a price; e.g. generation, delivery, etc.	M

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9.9.5.2. CommodityPriceComponentStruct Type

This represents a component of a given price; it is only used in the Components field.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Price**	money	all				M
1	**Source**	TariffPriceTypeEnum	all				M
2	**Description**	string	max 32				O
3	**TariffComponentID**	uint32	all				O

9.9.5.2.1. Price Field

This field SHALL indicate the component price of the commodity per TariffUnit, with the currency indicated by the currency of the Price field of the parent CommodityPriceStruct.

9.9.5.2.2. PriceLevel Field

This field SHALL indicate the tariff component price level.

9.9.5.2.3. Source Field

This field SHALL indicate the source of the price component.

9.9.5.2.4. Description Field

This field SHALL indicate a description of the pricing plan yielding the value of the Price field. For example, this field may contain the name of the current block of the selected billing plan, or the name of the time of usage tier.

9.9.5.2.5. TariffComponentID Field

This field SHALL indicate the ID of the associated TariffComponent for this price component. If there is no associated TariffComponent, this field SHALL be omitted.

9.9.5.3. CommodityPriceStruct Type

This represents a price over a given period.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**PeriodStart**	epoch-s	all				M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
1	**PeriodEnd**	epoch-s	min (PeriodStart + 1)	X			M
2	**Price**	money	all				O.b+
3	**PriceLevel**	int16	all				O.b+
4	**Description**	string	max 32				O
5	**Components**	list[CommodityPriceComponentStruct]	max 10		empty		[Price]

9.9.5.3.1. PeriodStart Field

This field SHALL indicate the beginning timestamp in UTC of the period covered by the price indicated in the Price field, or the price level indicated in the Price Level field, or both.

9.9.5.3.2. PeriodEnd Field

This field SHALL indicate the ending timestamp in UTC of the period covered by the price indicated in the Price field, or the price level indicated in the Price Level field, or both.

If this field is null, then the period has no definite end.

9.9.5.3.3. Price Field

This field SHALL indicate the price of the commodity per TariffUnit.

9.9.5.3.4. PriceLevel Field

This field SHALL indicate the tariff price level.

9.9.5.3.5. Description Field

This field SHALL indicate a description of the pricing plan yielding the value of the Price field, or the Price Level field. For example, this field may contain the name of the selected billing plan.

9.9.5.3.6. Components Field

This field SHALL indicate a list of the components that comprise the value in the Price field. For example, if a pricing plan has a base price and a surcharge for a given time of day, it may have two entries in the Components field.

If this field is not empty, the Price fields in the list SHALL sum to the value in the Price field.

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9.9.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**TariffUnit**	TariffUnitEnum	all			R V	M
0x0001	**Currency**	currency	all	X		R V	M
0x0002	**CurrentPrice**	CommodityPriceStruct	all	X		R V	M
0x0003	**PriceForecast**	list[CommodityPriceStruct]	max 56			R V	FORE

9.9.6.1. TariffUnit Attribute

This attribute SHALL indicate the unit of measure for all pricing reported by this cluster.

9.9.6.2. Currency Attribute

This attribute SHALL indicate the currency for all pricing reported by this cluster. If the current currency is unknown, or cannot be determined, the value SHALL be null.

9.9.6.3. CurrentPrice Attribute

This attribute SHALL indicate the current price. If the current price is unknown, or cannot be determined, the value SHALL be null.

The Description and Components fields SHALL be omitted in this attribute's value.

9.9.6.4. PriceForecast Attribute

This attribute SHALL indicate the forecast of upcoming price changes. If the forecast is unable to be determined, this list SHALL be empty.

The list entries SHALL be in time order:

- All entries except the last one SHALL have a non-null PeriodEnd.
- For all entries except the first one, PeriodStart SHALL be greater than the previous entry's PeriodEnd.

The Description and Components fields SHALL be omitted from CommodityPriceStructs in this attribute's value.

If the PeriodEnd field is null on the value of the CurrentPrice attribute, then this list SHALL be empty.

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9.9.7. Commands

ID	Name	Direction	Response	Access	Quality	Conformance
0x00	**GetDetailedPriceRequest**	client ⇒ server	GetDetailedPriceResponse	O		O
0x01	**GetDetailedPriceResponse**	client ⇐ server	N			GetDetailedPriceRequest
0x02	**GetDetailedForecastRequest**	client ⇒ server	GetDetailedForecastResponse	O	L	[FORE]
0x03	**GetDetailedForecastResponse**	client ⇐ server	N		L	GetDetailedForecastRequest

9.9.7.1. GetDetailedPriceRequest Command

Upon receipt, this SHALL generate a GetDetailedPrice Response command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Details**	CommodityPriceDetailBitmap	all			M

9.9.7.1.1. Details Field

This field SHALL indicate which fields on the CommodityPriceStruct in the GetDetailedPriceResponseCurrentPrice field will be included.

9.9.7.2. GetDetailedPriceResponse Command

This command SHALL be generated in response to a GetDetailedPrice Request command. The data for this command SHALL be as follows:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**CurrentPrice**	CommodityPriceStruct	all	X		M

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9.9.7.2.1. CurrentPrice Field

This field SHALL indicate the current price. Unlike the value returned from the CurrentPrice attribute, the Description and Components fields MAY be populated based on the value of the GetDetailedPriceRequestDetails field.

If the current price is unknown, or cannot be determined, the value SHALL be null.

9.9.7.3. GetDetailedForecastRequest Command

Upon receipt, this SHALL generate a GetDetailedForecast Response command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Details**	CommodityPriceDetailBitmap	all			M

9.9.7.3.1. Details Field

This field SHALL indicate which fields on the CommodityPriceStructs in the GetDetailedForecastResponsePriceForecast field will be included.

9.9.7.4. GetDetailedForecastResponse Command

This command SHALL be generated in response to a GetDetailedForecast Request command. The data for this command SHALL be as follows:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PriceForecast**	list[CommodityPriceStruct]	max 56			M

9.9.7.4.1. PriceForecast Field

This field SHALL indicate the current forecast of upcoming price changes. Unlike the value returned from the PriceForecast attribute, the Description and Components fields MAY be populated on each CommodityPriceStruct based on the value of the GetDetailedPriceRequestDetails field.

If the forecast is unable to be determined, this list SHALL be empty.

9.9.8. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**PriceChange**	INFO		V	O

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9.9.8.1. PriceChange Event

This event SHALL be generated when the value of the CurrentPrice attribute changes.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**CurrentPrice**	CommodityPriceStruct		X		M

9.9.8.1.1. CurrentPrice Field

This field SHALL be the new value of the CurrentPrice attribute.

9.10. Meter Identification Cluster

This Meter Identification Cluster provides attributes for determining advanced information about utility metering device.

9.10.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial version

9.10.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	MTRID

9.10.3. Cluster ID

ID	Name
0x0B06	Meter Identification

9.10.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**PWRTHLD**	PowerThreshold	O	Supports information about power threshold

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9.10.5. Data Types

9.10.5.1. MeterTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0x00	**Utility**	Utility Meter	M
0x01	**Private**	Private Meter	M
0x02	**Generic**	Generic Meter	M

9.10.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**MeterType**	MeterTypeEnum	all	X		R V	M
0x0001	**PointOfDelivery**	string	max 64	X		R V	M
0x0002	**MeterSerialNumber**	string	max 64	X		R V	M
0x0003	**ProtocolVersion**	string	max 64	X		R V	O
0x0004	**PowerThreshold**	PowerThresholdStruct	all	X		R V	PWRTHLD

9.10.6.1. MeterType Attribute

This attribute SHALL indicate the Meter type features, decided by manufacturer. If the type is unavailable, this attribute SHALL be null.

9.10.6.2. PointOfDelivery Attribute

This attribute SHALL indicate the unique identification of the connection point for the premises for billing purposes. If the point of delivery is unavailable, this attribute SHALL be null.

9.10.6.3. MeterSerialNumber Attribute

This attribute SHALL indicate the serial number of the meter. If the serial number is unavailable, this attribute SHALL be null.

9.10.6.4. ProtocolVersion Attribute

This attribute SHALL indicate the underlying protocol version to express local market features. If the protocol version is unavailable, this attribute SHALL be null.

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9.10.6.4.1. PowerThreshold Attribute

This attribute SHALL indicate the maximum instantaneous power demand that can be distributed to the customer without any risk of overload. If the maximum instantaneous power demand is unavailable, this attribute SHALL be null.

9.11. Commodity Metering Cluster

The Commodity Metering Cluster provides the mechanism for communicating commodity consumption information within a premises.

9.11.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial Release

9.11.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	COMMTR

9.11.3. Cluster ID

ID	Name
0x0B07	Commodity Metering

9.11.4. Data Types

9.11.4.1. MeteredQuantityStruct Type

Provides access to the Electric Metering device's readings.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**TariffComponentIDs**	list[uint32]	max 128				M
1	**Quantity**	int64					M

9.11.4.1.1. TariffComponentIDs Field

Indicates the specific TariffComponentStructs associated with the metered commodity.

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9.11.4.1.2. Quantity Field

This field indicates the amount of a commodity metered during the associated TariffComponentStructs.

9.11.5. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**MeteredQuantity**	list[MeteredQuantityStruct]	max MaximumMeteredQuantities	X		R V	M
0x0001	**MeteredQuantityTimestamp**	epoch-s	all	X		R V	M
0x0002	**TariffUnit**	TariffUnitEnum	all	X		R V	M
0x0003	**MaximumMeteredQuantities**	uint16	min 1	X		R V	M

9.11.5.1. MeteredQuantity Attribute

The most recent summed value of a commodity delivered to and consumed in the premises. A null value indicates that metering data is currently unavailable.

9.11.5.2. MeteredQuantityTimestamp Attribute

The timestamp in UTC for when the value of the MeteredQuantity attribute was last updated. A null value indicates that metering data is currently unavailable.

9.11.5.3. TariffUnit Attribute

This attribute SHALL indicate the unit for the Quantity field on all MeteredQuantityStructs in the MeteredQuantity attribute. A null value indicates that metering data is currently unavailable.

9.11.5.4. MaximumMeteredQuantities Attribute

This attribute SHALL indicate the maximum number of MeteredQuantityStructs in the MeteredQuantity attribute. A null value indicates that metering data is currently unavailable.

9.12. Commodity Tariff Cluster

The CommodityTariffCluster provides the mechanism for communicating Commodity Tariff infor-

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mation within the premises.

9.12.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial Matter revision

9.12.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	SETRF

9.12.3. Cluster ID

ID	Name
0x0700	Commodity Tariff

9.12.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**PRICE**	Pricing	O.a+	Supports information about commodity pricing
1	**FCRED**	FriendlyCredit	O.a+	Supports information about when friendly credit periods begin and end
2	**AUXLD**	AuxiliaryLoad	O.a+	Supports information about when auxiliary loads should be enabled or disabled
3	**PEAKP**	PeakPeriod	O	Supports information about peak periods
4	**PWRTHLD**	PowerThreshold	O	Supports information about power threshold

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Bit	Code	Feature	Conformance	Summary
5	**RNDM**	Randomization	O	Supports information about randomization of calendar day entries

9.12.5. Data Types

9.12.5.1. DayPatternDayOfWeekBitmap Type

This data type is derived from map8.

Bit	Name	Summary	Conformance
0	**Sunday**	Sunday	M
1	**Monday**	Monday	M
2	**Tuesday**	Tuesday	M
3	**Wednesday**	Wednesday	M
4	**Thursday**	Thursday	M
5	**Friday**	Friday	M
6	**Saturday**	Saturday	M

9.12.5.2. AuxiliaryLoadSettingEnum Type

This data type is derived from enum8.

This enumeration SHALL indicated the required state of an auxiliary switch.

Value	Name	Summary	Conformance
0	**Off**	The switch should be in the OFF state	M
1	**On**	The switch should be in the ON state	M
2	**None**	No state is required	M

9.12.5.3. DayTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Standard**	Standard	M
1	**Holiday**	Holiday	M
2	**Dynamic**	Dynamic Pricing	M

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Value	Name	Summary	Conformance
3	**Event**	Individual Events	M

9.12.5.4. PeakPeriodSeverityEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Unused**	Unused	M
1	**Low**	Low	M
2	**Medium**	Medium	M
3	**High**	High	M

9.12.5.5. DayEntryRandomizationTypeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**None**	No randomization applied	M
1	**Fixed**	An unchanging offset	M
2	**Random**	A random value	M
3	**RandomPositive**	A random positive value	M
4	**RandomNegative**	A random negative value	M

9.12.5.6. BlockModeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**NoBlock**	Tariff has no usage blocks	M
1	**Combined**	Usage is metered in combined blocks	M
2	**Individual**	Usage is metered separately by tariff component	M

9.12.5.6.1. NoBlock Value

This value SHALL indicate that the tariff has no usage blocks. All Threshold fields on TariffCompo-

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nentStruct in a tariff whose BlockMode field is set to NoBlock SHALL be null.

9.12.5.6.2. Combined Value

This value SHALL indicate that all Threshold fields apply to combined usage across all TariffComponentStruct during the billing period.

9.12.5.6.3. Individual Value

This value SHALL indicate that all Threshold fields apply only to usage during the active period of the associated TariffComponentStruct in the billing period.

9.12.5.7. AuxiliaryLoadSwitchSettingsStruct Type

This represents the settings for a given auxiliary load switch in a tariff component.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Number**	uint8	all				M
1	**RequiredState**	AuxiliaryLoadSettingEnum					M

9.12.5.8. AuxiliaryLoadSwitchesSettingsStruct Type

This represents the set of auxiliary load settings in a tariff component.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**SwitchStates**	list[AuxiliaryLoadSwitchSettingsStruct]	max 8				M

9.12.5.9. CalendarPeriodStruct Type

This represents a sub period of a calendar, commencing on its StartDate.

NOTE	A 'Calendar Period', while normally considered to be a 3 or 6 month period, could be used for other arbitrary periods e.g. monthly or quarterly. The minimum resolution is 1 day, although a week would normally be the smallest interval.

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**StartDate**	epoch-s	min StartDate	X			M
1	**DayPatternIDs**	list[uint32]	1 to 7				M

9.12.5.9.1. StartDate Field

This field SHALL indicate the timestamp in UTC when the calendar period becomes active.

A null value SHALL indicate the calendar period becomes active immediately. See CalendarPeriods attribute.

9.12.5.9.2. DayPatternIDs Field

This field SHALL indicate a list of DayPatternIDs for the DayPatternStructs in use during this calendar period.

If any of the DayPatternStructs referenced by this list has no bits set in DaysOfWeek, then none of the DayPatternStructs referenced by this list SHALL have any bits set, and the list SHALL be treated as a rotating set of days.

If at least one of the DayPatternStructs referenced by this list has a bit set in DaysOfWeek, then:

1. All of the DayPatternStructs referenced by this list SHALL have at least one bit set in DaysOfWeek
2. No two DayPatternStructs referenced by this list SHALL have the same bit set in DaysOfWeek.
3. Every bit defined in DayPatternDayOfWeekBitmap referenced by this list SHALL be set in DaysOfWeek on one of the DayPatternStructs.

Meeting these constraints ensures that every day of the week during this week has specified day entries, and no day of the week has more than one set of day entries.

9.12.5.10. DayEntryStruct Type

This struct represents a day entry at a particular time of day, along with an optional duration and randomization parameters.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**DayEntryID**	uint32	all				M
1	**StartTime**	uint16	max 1499				M
2	**Duration**	uint16	max (1500 - StartTime)				O

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
3	**RandomizationOffset**	int16	all				[RNDM]
4	**RandomizationType**	DayEntryRandomizationTypeEnum	all		None		[RNDM]

9.12.5.10.1. DayEntryID Field

This field SHALL indicate an identifier for the day entry which is unique on the server to a set of values:

1. If this identifier is included in the DayEntryIDs associated with a DayPatternStruct, then the identifier SHALL be unique to the combination of:
   a. the StartTime
   b. the Duration, if indicated
   c. the DaysOfWeek field in the containing DayPatternStruct
2. Otherwise, if this identifier is included in the DayEntryIDs associated with a DayStruct, then the identifier SHALL be unique to the combination of:
   a. the StartTime
   b. the Duration, if indicated
   c. the Date field in the containing DayStruct

Once an identifier has been used for a given combination above, it SHALL never be used for any other combination of these values.

9.12.5.10.2. StartTime Field

This field SHALL indicate the start time of the DayEntryStruct, expressed as the number of minutes that have elapsed since midnight on the associated days.

For example, 6am will be represented by 360 minutes since midnight and 11:30pm will be represented by 1410 minutes since midnight.

This will differ on days with a day entry in or out of daylight saving time. For example, if the implementation of daylight saving time in a region means that the hour between 2am and 3am will be skipped on the day entry into daylight saving time, then 2am cannot be represented, and 3am will be represented by 120 minutes.

Similarly, if the implementation of daylight saving time in a region means that the hour between 2am and 3am will be repeated on the date of day entry out of daylight saving time, then the first 2am will be represented by 120 minutes, while the second 2am will be represented by 180 minutes,

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and 4am will be represented by 300 minutes. As such, the maximum start time on this day is 1499, 60 minutes longer than all other days.

DayEntryStruct on daylight saving time days SHOULD be handled by a DayStruct in the IndividualDays attribute whose Date field indicates the date of the day entry.

9.12.5.10.3. Duration Field

This field SHALL indicate the duration of the day entry, expressed as the number of minutes since the time indicated by the StartTime field.

If this field is omitted, the day entry SHALL end at the StartTime of the DayEntryStruct identified by the next DayEntryID in the containing list. If the DayEntryStruct is the last item in the containing list, then the day entry SHALL last until the end of the day.

9.12.5.10.4. RandomizationOffset Field

This field SHALL indicate a randomization offset for this particular day entry in seconds. If this field is not indicated, randomization SHALL use the value in the DefaultRandomizationOffset attribute.

1. If the value of the RandomizationType field is Fixed, then the value of this field MAY be negative
2. Otherwise if the value of the RandomizationType field is RandomNegative, then the value of this field SHALL be less than or equal to zero
3. Otherwise the value of this field SHALL be greater than or equal to zero

9.12.5.10.5. RandomizationType Field

This field SHALL indicate a randomization type for this particular day entry. If this field is not indicated, randomization SHALL use the value in the DefaultRandomizationType attribute.

If the calculated value for this field is of type None, no randomization SHALL be applied to the start of this day entry.

If the calculated value for this field is of type Fixed, then the calculated value of the RandomizationOffset field SHALL be added to the start time of the day entry.

If the calculated value for this field is of type Random, then a random number of whole seconds whose absolute value is less than or equal to the calculated value of the RandomizationOffset field SHALL be added to the start time of the day entry.

If the calculated value for this field is of type RandomPositive, then a random non-negative number of whole seconds whose value is less than or equal to the calculated value of the RandomizationOffset field SHALL be added to the start time of the day entry.

If the calculated value for this field is of type RandomNegative, then a random negative number of whole seconds whose value is greater than or equal to the calculated value of the RandomizationOffset field SHALL be added to the start time of the day entry.

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9.12.5.11. DayStruct Type

This represents a series of day entries over the course of a day for a specific date.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Date**	epoch-s					M
1	**DayType**	DayTypeEnum	any				M
2	**DayEntryIDs**	list[uint32]	1 to 96				M

9.12.5.11.1. Date Field

This field SHALL indicate the date the associated set of DayEntryStructs applies to.

9.12.5.11.2. DayType Field

This field SHALL indicate the type of day represented by the struct.

9.12.5.11.3. DayEntryIDs Field

This field SHALL indicate a list of DayEntryIDs for the DayEntryStructs to apply during the date specified by Date, ordered by the value of the StartTime field.

This list SHALL NOT contain two DayEntryIDs for DayEntryStructs with the same value of the StartTime field.

If the Randomization feature is supported, every DayEntryStruct whose DayEntryID is included in this field SHALL have its StartTime field set to a value less than the following DayEntryStruct's StartTime field minus the calculated value of its RandomizationOffset field. In other words, it should not be possible for a random offset to cause a day entry to begin before the preceding day entry.

If the DayType field is not Event:

1. The DayEntryStruct referenced by the first DayEntryID in this list SHALL have its StartTime field set to zero.
2. Every DayEntryStruct referenced by this list SHALL have its Duration field omitted.

Otherwise, if the DayType field is Event:

1. Every DayEntryStruct referenced by this list SHALL NOT have a Duration field whose value, added to the value of StartTime field, is greater than the value of StartTime field on any subsequent DayEntryStruct referenced by this list. In other words, day entries with a set duration can not overlap any other day entries.

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9.12.5.12. DayPatternStruct Type

This represents a series of day entries over the course of a day for a given set of days of the week.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**DayPatternID**	uint32	all				M
1	**DaysOfWeek**	DayPatternDayOfWeekBitmap	all				M
2	**DayEntryIDs**	list[uint32]	1 to 96				M

9.12.5.12.1. DayPatternID Field

This field SHALL indicate an identifier for the day pattern. It SHALL be a unique identifier for the combination of values of the DaysOfWeek and DayEntryIDs fields.

Once an identifier has been used for this combination, it SHALL NOT be used to represent any other combination of these values.

9.12.5.12.2. DaysOfWeek Field

This field SHALL indicate which days of the week the associated set of DayEntryStructs applies to. If no bits are set, then this SHALL be a rotating day. See DayPatternIDs.

9.12.5.12.3. DayEntryIDs Field

This field SHALL indicate a list of DayEntryIDs for the DayEntryStructs to apply during the days specified by DaysOfWeek, ordered by StartTime.

This list SHALL NOT contain two DayEntryIDs for the DayEntryStructs with the same value of the StartTime field.

9.12.5.12.3.1. Day Coverage

The DayEntryStruct referenced by the first DayEntryID in this list SHALL have its StartTime field set to zero.

Every DayEntryStruct referenced by this list SHALL have its Duration field omitted.

Meeting these constraints ensures that there are no periods during the day which are not covered by a DayEntryStruct.

9.12.5.13. PeakPeriodStruct Type

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Severity**	PeakPeriodSeverityEnum	all				M
1	**PeakPeriod**	uint16	min 1				M

9.12.5.13.1. Severity Field

This field SHALL indicate the severity of the peak period.

9.12.5.13.2. PeakPeriod Field

This field SHALL indicate the PeakPeriod number.

9.12.5.14. TariffInformationStruct Type

This represents particular Tariff Price information.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**TariffLabel**	string	max 128	X			M
1	**ProviderName**	string	max 128	X			M
2	**Currency**	currency	all	X			PRICE
3	**BlockMode**	BlockModeEnum	all	X			M

9.12.5.14.1. TariffLabel Field

This field SHALL indicate a label for the tariff.

9.12.5.14.2. ProviderName Field

This field SHALL indicate the name of the commodity provider for this tariff.

9.12.5.14.3. Currency Field

This field SHALL indicate the currency for the value of the Price field on all TariffPriceStruct.

9.12.5.14.4. BlockMode Field

This field SHALL indicate the mode for metering blocks of usage.

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9.12.5.15. TariffPriceStruct Type

This indicates a price or price level for a given tariff component, as well as what type of pricing it represents

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**PriceType**	TariffPriceTypeEnum					M
1	**Price**	money	all				O.c+
2	**PriceLevel**	int16	all				O.c+

9.12.5.15.1. PriceType Field

This field SHALL indicate the type of price for the Price or PriceLevel fields.

9.12.5.15.2. Price Field

This field SHALL indicate the tariff price.

9.12.5.15.3. PriceLevel Field

This field SHALL indicate the tariff price level.

9.12.5.16. TariffComponentStruct Type

This represents components of a tariff.

Tariff components typically represent a price or price level, though they may also specify other aspects of a tariff. For example, a tariff component may indicate changes in power thresholds, friendly credit status, or the expected state of auxiliary switches.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**TariffComponentID**	uint32	all				M
1	**Price**	TariffPriceStruct	all	X			[PRICE].b+
2	**FriendlyCredit**	bool	all				[FCRED].b+
3	**AuxiliaryLoad**	AuxiliaryLoadSwitchSettingsStruct	all				[AUXLD].b+
4	**PeakPeriod**	PeakPeriodStruct	all		0		[PEAKP].b+

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
5	**PowerThresholdStruct**	PowerThresholdStruct	all		0		[PWRTHLD].b+
6	**Threshold**	int64	all	X			M
7	**Label**	string	max 128	X			O
8	**Predicted**	bool	all				O

9.12.5.16.1. TariffComponentID Field

This field SHALL indicate an identifier for the tariff component. If the tariff component is not a prediction, this field SHALL be a unique identifier for the combination of values of the Price, Threshold, FriendlyCredit, AuxiliaryLoad, and PeakPeriod fields with the value of the DayEntryIDs field on the encompassing TariffPeriodStruct.

Once an identifier has been used for this combination, it SHALL NOT be used to represent any other combination of these values.

9.12.5.16.2. Price Field

This field SHALL indicate the price when the tariff component is active.

When the Predicted field is set to TRUE, a null value SHALL indicate that the price and/or price level is not yet available.

When the Predicted field is set to FALSE, a null value SHALL indicate that the server is unable to provide a specific price or price level.

9.12.5.16.3. FriendlyCredit Field

This field SHALL indicate whether the calendar is entering a friendly credit period or not.

NOTE	When a meter enters into a Friendly Credit Period with a usable positive credit balance, the consumer will be allowed to consume energy for the duration of the Friendly Credit Period, regardless of their credit status while in that period. If, however, the consumer had already run out of credit and supply was interrupted before entering into the Friendly Credit Period, they will not be allowed to reconnect without first adding suitable additional credit.
NOTE	At the end of the Friendly Credit Period, the normal delivery rules connected with the accounting functions of the meter will be resumed, and if the meter's credit balance has dropped below the disablement threshold during the Friendly Credit Period, then the meter will disconnect upon resuming normal delivery rules

9.12.5.16.4. AuxiliaryLoad Field

This field SHALL indicate the required state of auxiliary load switches.

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9.12.5.16.5. PeakPeriod Field

This field SHALL indicate whether the tariff component represents a peak period.

9.12.5.16.6. PowerThreshold Field

This field SHALL indicate whether the tariff component represents a power threshold.

9.12.5.16.7. Threshold Field

This field SHALL indicate the maximum level of consumption this tariff component applies to, in units specified by the TariffUnit attribute. If the tariff component applies to any level of consumption, this field SHALL be null.

9.12.5.16.8. Label Field

A free-form label for the tariff component.

9.12.5.16.9. Predicted Field

This field SHALL indicate whether the tariff component represents a price prediction.

9.12.5.17. TariffPeriodStruct Type

This represents the tariff components in effect for a set of calendar day entry IDs.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Label**	string	max 128	X			M
1	**DayEntryIDs**	list[uint32]	1 to 20				M
2	**TariffComponentIDs**	list[uint32]	1 to 20				M

9.12.5.17.1. Label Field

A free-form label for the tariff period.

9.12.5.17.2. DayEntryIDs Field

This field SHALL indicate a list of DayEntryIDs for the DayEntryStructs during which the tariff components are active.

Each DayEntryID SHALL be included in at most one DayEntryIDs field. In other words, there SHALL be only one TariffPeriodStruct for each DayEntryID.

9.12.5.17.3. TariffComponentIDs Field

This field SHALL indicate a list of TariffComponentIDs for the TariffComponentStructs active during the specified day entries.

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9.12.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**TariffInfo**	TariffInformationStruct		X		R V	M
0x0001	**TariffUnit**	TariffUnitEnum	all	X		R V	M
0x0002	**StartDate**	epoch-s	desc	X		R V	M
0x0003	**DayEntries**	list[DayEntryStruct]	max 672	X		R V	M
0x0004	**DayPatterns**	list[DayPatternStruct]	max 28	X		R V	M
0x0005	**CalendarPeriods**	list[CalendarPeriodStruct]	1 to 4	X		R V	M
0x0006	**IndividualDays**	list[DayStruct]	max 50	X		R V	M
0x0007	**CurrentDay**	DayStruct		X		R V	M
0x0008	**NextDay**	DayStruct		X		R V	M
0x0009	**CurrentDayEntry**	DayEntryStruct		X		R V	M
0x000A	**CurrentDayEntryDate**	epoch-s	all	X		R V	M
0x000B	**NextDayEntry**	DayEntryStruct		X		R V	M
0x000C	**NextDayEntryDate**	epoch-s	all	X		R V	M
0x000D	**TariffComponents**	list[TariffComponentStruct]	1 to 672	X		R V	M
0x000E	**TariffPeriods**	list[TariffPeriodStruct]	1 to 672	X		R V	M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x000F	**CurrentTariffComponents**	list[TariffComponentStruct]	max 20	X		R V	M
0x0010	**NextTariffComponents**	list[TariffComponentStruct]	max 20	X		R V	M
0x0011	**DefaultRandomizationOffset**	int16	desc	X		R V	RNDM
0x0012	**DefaultRandomizationType**	DayEntryRandomizationTypeEnum	all	X		R V	RNDM

9.12.6.1. TariffInfo Attribute

This attribute SHALL indicate basic information about the tariff.

If the tariff is unavailable, this attribute SHALL be null.

9.12.6.2. TariffUnit Attribute

This attribute SHALL indicate the unit of the commodity for pricing.

If the tariff is unavailable, this attribute SHALL be null.

9.12.6.3. StartDate Attribute

This attribute SHALL indicate a timestamp in UTC to denote the time at which the published calendar becomes active. A start date/time of 0x00000000 shall indicate that the calendar should become active immediately.

If the tariff is unavailable, this attribute SHALL be null.

9.12.6.4. DayEntries Attribute

This attribute SHALL indicate the list of DayEntryStructs included in this calendar.

The maximum constraint of this attribute is intended to allow representation of seven days of tariff information at 15 minute intervals.

If the tariff is unavailable, this attribute SHALL be null.

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9.12.6.5. DayPatterns Attribute

This attribute SHALL indicate the list of DayPatternStructs used in the CalendarPeriodStruct in this calendar.

Each day pattern in this list SHALL define the specific schedule that applies to the days it covers.

9.12.6.6. CalendarPeriods Attribute

This attribute SHALL indicate the list of CalendarPeriodStructs comprising this calendar. The CalendarPeriodStructs in this list SHALL be arranged in increasing order by the value of StartDate.

If and only if the value of the StartDate attribute is null, the value of the StartDate field on the first CalendarPeriodStruct in the CalendarPeriods attribute SHALL also be null, indicating that the period begins immediately. The value of the StartDate field on any subsequent CalendarPeriodStruct in CalendarPeriods SHALL NOT be null.

The active calendar period SHALL be in effect until the StartDate of the next calendar period.

If the calendar is unavailable, this attribute SHALL be null.

9.12.6.7. IndividualDays Attribute

This attribute SHALL indicate the list of days to overlay on this calendar.

The DayStruct in this list SHALL be arranged in increasing order by the value of Date. The DayStruct in this list SHALL not overlap.

If the calendar is unavailable, this attribute SHALL be null.

9.12.6.8. CurrentDay Attribute

This attribute SHALL indicate the current day's day entries.

If the tariff is not active or CurrentDay information is not available, this attribute SHALL be null.

9.12.6.9. NextDay Attribute

This attribute SHALL indicate the next day's day entries.

If the tariff is not active or NextDay information is not available, this attribute SHALL be null.

9.12.6.10. CurrentDayEntry Attribute

This attribute SHALL indicate the currently active DayEntryStruct.

If the tariff is not active or day entry information is not available, this attribute SHALL be null.

9.12.6.11. CurrentDayEntryDate Attribute

This attribute SHALL indicate the UTC date when the CurrentDay, CurrentDayEntry, and CurrentTariffComponents attributes were last updated.

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If the tariff is not active or day entry information is not available, this attribute SHALL be null.

9.12.6.12. NextDayEntry Attribute

This attribute SHALL indicate the predicted next active DayEntryStruct.

If the tariff is not active or is not available, this attribute SHALL be null.

9.12.6.13. NextDayEntryDate Attribute

This attribute SHALL indicate the predicted UTC date when the CurrentDay, CurrentDayEntry, and CurrentTariffComponents attributes will update to the values in the NextDay, NextDayEntry, and NextTariffComponents attributes, respectively.

If the tariff is not active or is not available, this attribute SHALL be null.

9.12.6.14. TariffComponents Attribute

This attribute SHALL indicate a list of TariffComponentStructs for the tariff.

If the tariff is unavailable, this attribute SHALL be empty.

9.12.6.15. TariffPeriods Attribute

This attribute SHALL indicate a list of TariffPeriodStructs for the tariff.

If the tariff is unavailable, this attribute SHALL be empty.

9.12.6.16. CurrentTariffComponents Attribute

This attribute SHALL indicate a list of the currently active TariffComponentStructs.

If the tariff is unavailable, this attribute SHALL be empty.

9.12.6.17. NextTariffComponents Attribute

This attribute SHALL indicate the predicted next active TariffComponentStructs.

If the tariff is unavailable, this attribute SHALL be null.

9.12.6.18. DefaultRandomizationOffset Attribute

This attribute SHALL indicate a default randomization offset for DayEntryStructs in this tariff. See RandomizationOffset for details.

9.12.6.19. DefaultRandomizationType Attribute

This attribute SHALL indicate a default randomization type for DayEntryStruct in this tariff. See RandomizationType for details.

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9.12.7. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**GetTariffComponent**	client ⇒ server	GetTariffComponentResponse	O	M
0x00	**GetTariffComponentResponse**	client ⇐ server	N		M
0x01	**GetDayEntry**	client ⇒ server	GetDayEntryResponse	O	M
0x01	**GetDayEntryResponse**	client ⇐ server	N		M

9.12.7.1. GetTariffComponent Command

The GetTariffComponent command allows a client to request information for a tariff component identifier that may no longer be available in the TariffPeriods attributes.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**TariffComponentID**	uint32	all			M

9.12.7.1.1. TariffComponentID Field

This field SHALL be used to indicate the TariffComponentID of the DayEntryStruct to be returned.

9.12.7.1.2. Effect on Receipt

If the TariffComponentStruct indicated by the GetTariffComponentTariffComponentID field is no longer available, the server SHALL return a status code of NOT_FOUND.

Otherwise, the server SHALL return information about the TariffComponentStruct.

See GetTariffComponentResponse for a description of the response command.

9.12.7.2. GetTariffComponentResponse Command

The GetTariffComponentResponse command is sent in response to a GetTariffComponent command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Label**	string	max 128	X		M
1	**DayEntryIDs**	list[uint32]	1 to 96			M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
2	**TariffComponent**	TariffComponentStruct				M

9.12.7.2.1. Label Field

A free-form label for the tariff period.

9.12.7.2.2. DayEntryIDs Field

This field SHALL indicate a list of DayEntryIDs for the DayEntryStructs during which the tariff component is active.

9.12.7.2.3. TariffComponent Field

This field SHALL indicate the TariffComponentStruct whose TariffComponentID field matches the requested GetTariffComponentTariffComponentID.

9.12.7.3. GetDayEntry Command

The GetDayEntry command allows a client to request information for a calendar day entry identifier that may no longer be available in the CalendarPeriods or IndividualDays attributes.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**DayEntryID**	uint32	all			M

9.12.7.3.1. DayEntryID Field

This field SHALL be used to indicate the DayEntryID of the DayEntryStruct to be returned.

9.12.7.3.2. Effect on Receipt

If the DayEntryStruct indicated by the DayEntryID field is no longer available, the server SHALL return a status code of NOT_FOUND.

Otherwise, the server SHALL return information about the DayEntryStruct.

See GetDayEntryResponse for a description of the response command.

9.12.7.4. GetDayEntryResponse Command

The GetDayEntryResponse command is sent in response to a GetDayEntry command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**DayEntry**	DayEntryStruct				M

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9.12.7.4.1. DayEntryField

This field SHALL indicate the DayEntryStruct whose DayEntryID field matches the requested DayEntryID.

9.12.8. Tariff Examples

9.12.8.1. Flat Rate

A flat rate tariff has the simplest structure: the consumer pays a single price at all times of year.

For example if a tariff charges:

1. $0.1485 per unit, all day, every day, starting on 2025/01/01

This can be represented by:

1. One DayEntryStruct starting at midnight; in this example, the DayEntryID is 0x19
2. One DayPatternStruct with all DaysOfWeek set and containing the DayEntryID 0x19; in this example, the DayPatternID is 0x15
3. One CalendarPeriodStruct starting on 2025/01/01 and containing the DayPatternID 0x15
4. One TariffComponentStruct with a standard price of $0.1485; in this example, the TariffComponentID is 0x29
5. One TariffPeriodStruct linking the TariffComponentID of 0x29 to the DayEntryID of 0x19

TariffPeriods

Label	TariffComponentIDs	DayEntryIDs
All Day	[0x29]	[0x019]

TariffComponents

TariffComponentID	Price		Threshold
0x29	*PriceType*	Standard	0
	Price	$0.1485	

DayEntries

DayEntryID	StartTime
0x19	0

DayPatterns

DayPatternID	DaysOfWeek	DayEntryIDs
0x15	0x7F	[0x19]

CalendarPeriods

StartDate	DayPatternIDs
2025/01/01	[0x15]

Figure 24. Example of a flat rate commodity tariff

9.12.8.2. Flat Rate with Blocks

Tariffs may have thresholds of consumption; when an account has consumed more than the threshold, a new rate applies.

For example, if a tariff charges:

1. $0.40 per unit, all day, every day, starting on 2025/01/01
2. $0.50 per unit, once 100 units have been consumed

This can be represented by:

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1. One DayEntryStruct starting at midnight; in this example, the DayEntryID is 0x806
2. One DayPatternStruct with all DaysOfWeek set and containing the DayEntryID 0x806; in this example, the DayPatternID is 0x62
3. One CalendarPeriodStruct starting on 2025/01/01 and containing the DayPatternID 0x62
4. One TariffComponentStruct with a standard price of $0.40 and a Threshold of zero, indicating that it takes effect after zero consumption (immediately). In this example, this component's TariffComponentID is 0x59.
5. One second TariffComponentStruct with a standard price of $0.50 and a Threshold of 100, indicating that it takes effect after consumption reaches 100. In this example, this component's TariffComponentID is 0x5A.
6. One TariffPeriodStruct linking the TariffComponentIDs of 0x59 and 0x5A to the DayEntryID of 0x806

TariffPeriods

Label	TariffComponentIDs	DayEntryIDs
All Day	[0x59, 0x5A]	[0x806]

TariffComponents

TariffComponentID	Price		Threshold
0x59	*PriceType*	Standard	0
	Price	$0.40	
0x5A	*PriceType*	Standard	100
	Price	$0.50	

DayEntries

DayEntryID	StartTime
0x806	0

DayPatterns

DayPatternID	DaysOfWeek	DayEntryIDs
0x62	0x7F	[0x806]

CalendarPeriods

StartDate	DayPatternIDs
2025/01/01	[0x62]

Figure 25. Example of a flat rate commodity tariff with usage blocks

9.12.8.3. Seasonal Flat Rate

Tariffs may have periods of the year during which different rates apply.

For example, if a tariff charges:

1. $0.1295 per unit, all day, every day, from 2025/01/01 until 2025/05/01
2. $0.1331 per unit, all day, every day, from 2025/05/01 until 2025/06/01
3. $0.2126 per unit, all day, every day, from 2025/06/01 until 2025/10/01
4. $0.1331 per unit, all day, every day, after 2025/10/01

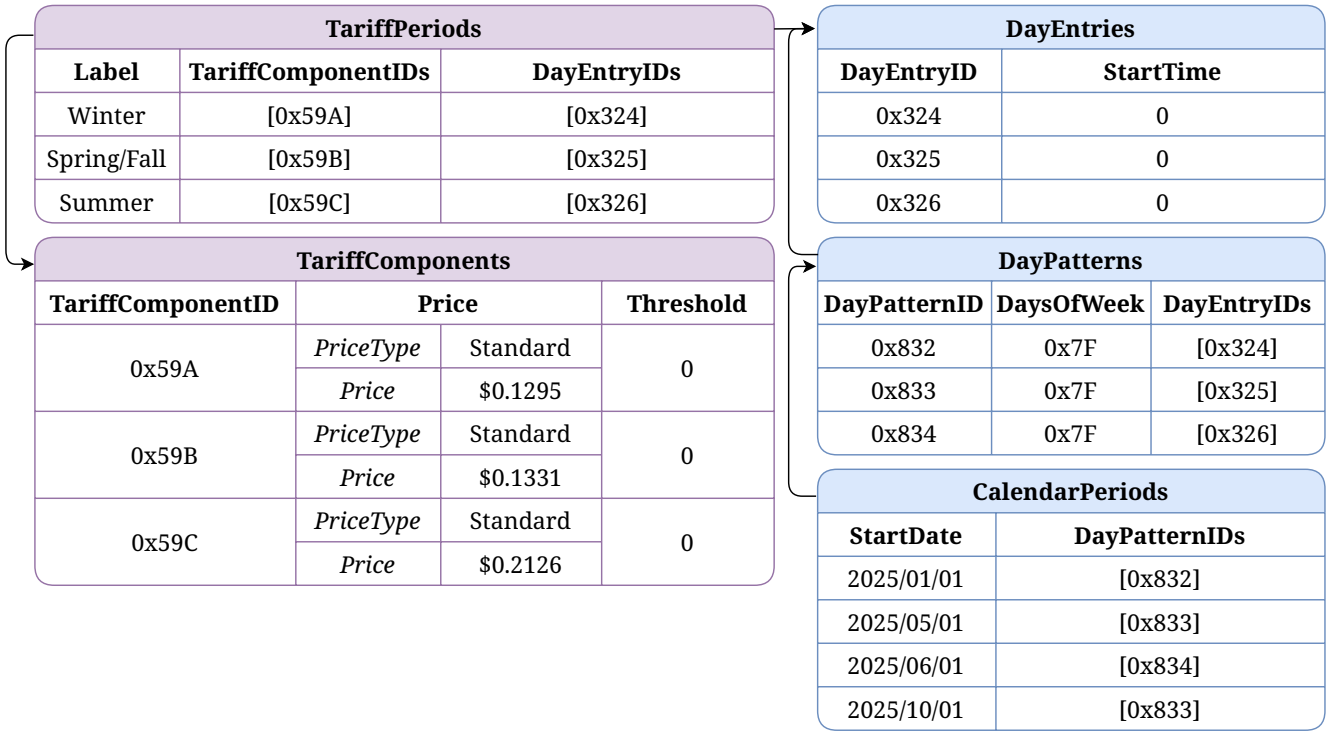
This can be represented by:

1. Three DayEntryStructs starting at midnight:
   a. One with a DayEntryID of 0x324; this will represent days between 2025/01/01 and 2025/05/01
   b. One with a DayEntryID of 0x325; this will represent days between 2025/05/01 and 2025/06/01 as well as days after 2025/10/01
   c. One with a DayEntryID of 0x326; this will represent days between 2025/06/01 and 2025/10/01
2. Three DayPatternStructs with all DaysOfWeek set:

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a. One with a DayPatternID of 0x832, containing the DayEntryID 0x324
b. One with a DayPatternID of 0x833, containing the DayEntryID 0x325
c. One with a DayPatternID of 0x834, containing the DayEntryID 0x326

3. Four CalendarPeriodStructs:
   a. One starting on 2025/01/01, containing the DayPatternID 0x832
   b. One starting on 2025/05/01, containing the DayPatternID 0x833
   c. One starting on 2025/06/01, containing the DayPatternID 0x834
   d. One starting on 2025/10/01, containing the DayPatternID 0x833
4. Three TariffComponentStructs:
   a. One with a TariffComponentID of 0x59A, and a standard price of $0.1295
   b. One with a TariffComponentID of 0x59B, and a standard price of $0.1331
   c. One with a TariffComponentID of 0x59C, and a standard price of $0.2126
5. Three TariffPeriodStructs
   a. One linking the TariffComponentID of 0x59A to the DayEntryID of 0x324
   b. One linking the TariffComponentID of 0x59B to the DayEntryID of 0x325
   c. One linking the TariffComponentID of 0x59C to the DayEntryID of 0x326



TariffPeriods

Label	TariffComponentIDs	DayEntryIDs
Winter	[0x59A]	[0x324]
Spring/Fall	[0x59B]	[0x325]
Summer	[0x59C]	[0x326]

TariffComponents

TariffComponentID	Price		Threshold
0x59A	*PriceType*	Standard	0
	Price	$0.1295	
0x59B	*PriceType*	Standard	0
	Price	$0.1331	
0x59C	*PriceType*	Standard	0
	Price	$0.2126	

DayEntries

DayEntryID	StartTime
0x324	0
0x325	0
0x326	0

DayPatterns

DayPatternID	DaysOfWeek	DayEntryIDs
0x832	0x7F	[0x324]
0x833	0x7F	[0x325]
0x834	0x7F	[0x326]

CalendarPeriods

StartDate	DayPatternIDs
2025/01/01	[0x832]
2025/05/01	[0x833]
2025/06/01	[0x834]
2025/10/01	[0x833]

Figure 26. Example of a seasonal flat rate commodity tariff

9.12.8.4. Time of Use Rate

Tariffs may have different rates at different times of day, for different days of the week.

For example, if a tariff starting on 2025/01/01 charges:

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1. $0.40 per unit from 00:00 to 15:00, Monday to Friday
2. $0.50 per unit from 15:00 to 18:00, Monday to Friday
3. $0.60 per unit from 18:00 to 21:00, Monday to Friday
4. $0.40 per unit from 21:00 to 00:00, Monday to Friday
5. $0.25 per unit from 00:00 to 17:00, Saturday and Sunday
6. $0.35 per unit from 17:00 to 20:00, Saturday and Sunday
7. $0.25 per unit from 20:00 to 00:00, Saturday and Sunday

This can be represented by:

1. Seven DayEntryStructs:
    a. Four representing the schedule on weekdays:
        i. One with a start time of zero (00:00) with a DayEntryID of 0x6
        ii. One with a start time of 900 (15:00) with a DayEntryID of 0x7
        iii. One with a start time of 1080 (18:00) with a DayEntryID of 0x8
        iv. One with a start time of 1260 (21:00) with a DayEntryID of 0x9
    b. Three representing the schedule on weekends
        i. One with a start time of zero (00:00) with a DayEntryID of 0xA
        ii. One with a start time of 1020 (17:00) with a DayEntryID of 0xB
        iii. One with a start time of 1200 (20:00) with a DayEntryID of 0xC
2. Two DayPatternStructs:
    a. One with a DayPatternID of 0x73
        i. Containing the DayEntryIDs 0x6, 0x7, 0x8, and 0x9
        ii. DaysOfWeek set to Monday, Tuesday, Wednesday, Thursday, and Friday
    b. One with a DayPatternID of 0x74
        i. Containing the DayEntryIDs 0xA, 0xB, and 0xC
        ii. DaysOfWeek set to Saturday and Sunday
3. A CalendarPeriodStruct starting on 2025/01/01 and containing the DayPatternIDs 0x73 and 0x74.
4. Five TariffComponentStructs:
    a. One with a TariffComponentID of 0x491, and a standard price of $0.40
    b. One with a TariffComponentID of 0x492, and a standard price of $0.50
    c. One with a TariffComponentID of 0x493, and a standard price of $0.60
    d. One with a TariffComponentID of 0x494, and a standard price of $0.25
    e. One with a TariffComponentID of 0x495, and a standard price of $0.35
5. Five TariffPeriodStructs
    a. One linking the TariffComponentID of 0x491 to the DayEntryIDs of 0x6 and 0x9

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b. One linking the TariffComponentID of 0x492 to the DayEntryID of 0x7

c. One linking the TariffComponentID of 0x493 to the DayEntryID of 0x8

d. One linking the TariffComponentID of 0x494 to the DayEntryIDs of 0xA and 0xC

e. One linking the TariffComponentID of 0x495 to the DayEntryID of 0xB

TariffPeriods

Label	TariffComponentIDs	DayEntryIDs
M-F *Off-Peak*	[0x491]	[0x6, 0x9]
M-F *Mid-Peak*	[0x492]	[0x7]
M-F *Peak*	[0x493]	[0x8]
Sa-Su *Off-Peak*	[0x494]	[0xA, 0xC]
Sa-Su *Peak*	[0x495]	[0xB]

TariffComponents

TariffComponentID	Price		Threshold
0x491	*PriceType*	Standard	0
	Price	$0.40	
0x492	*PriceType*	Standard	0
	Price	$0.50	
0x493	*PriceType*	Standard	0
	Price	$0.60	
0x494	*PriceType*	Standard	0
	Price	$0.25	
0x495	*PriceType*	Standard	0
	Price	$0.35	

DayEntries

DayEntryID	StartTime
0x6	0
0x7	900
0x8	1080
0x9	1260
0xA	0
0xB	1020
0xC	1200

DayPatterns

DayPatternID	DoW	DayEntryIDs
0x73	0x3E	[0x6, 0x7, 0x8, 0x9]
0x74	0x41	[0xA, 0xB, 0xC]

CalendarPeriods

StartDate	DayPatternIDs
2025/01/01	[0x73, 0x74]

Figure 27. Example of a time-of-use rate commodity tariff

9.12.8.5. Seasonal Time of Use Rate

Tariffs with a time-of-use structure may vary the times of day during different times of the year.

For example, if a tariff starting on 2025/01/01 charges:

1. From 2025/01/01 until 2025/05/01, and after 2025/10/01:
   a. $0.40 per unit from 00:00 to 16:00, every day
   b. $0.50 per unit from 16:00 to 21:00, every day
   c. $0.40 per unit from 21:00 to 00:00, every day
2. From 2025/05/01 until 2025/10/01:
   a. $0.40 per unit from 00:00 to 15:00, every day
   b. $0.60 per unit from 15:00 to 21:00, every day
   c. $0.40 per unit from 21:00 to 00:00, every day

This can be represented by:

1. Six DayEntryStructs:

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a. Three representing the schedule from 2025/01/01 until 2025/05/01, and after 2025/10/01:
   i. One with a start time of zero (00:00) with a DayEntryID of 0x71
   ii. One with a start time of 960 (16:00) with a DayEntryID of 0x72
   iii. One with a start time of 1260 (21:00) with a DayEntryID of 0x73
b. Three representing the schedule from 2025/05/01 until 2025/10/01:
   i. One with a start time of zero (00:00) with a DayEntryID of 0x74
   ii. One with a start time of 900 (15:00) with a DayEntryID of 0x75
   iii. One with a start time of 1260 (21:00) with a DayEntryID of 0x76

2. Two DayPatternStructs with all DaysOfWeek set:
   a. One with a DayPatternID of 0x56A containing the DayEntryIDs 0x71, 0x72, and 0x73
   b. One with a DayPatternID of 0x56B containing the DayEntryIDs 0x74, 0x75, and 0x76
3. Three CalendarPeriodStructs:
   a. One starting on 2025/01/01, containing the DayPatternID 0x56A
   b. One starting on 2025/05/01, containing the DayPatternID 0x56B
   c. One starting on 2025/10/01, containing the DayPatternID 0x56A
4. Four TariffComponentStructs:
   a. One with a TariffComponentID of 0x891, and a standard price of $0.40
   b. One with a TariffComponentID of 0x892, and a standard price of $0.50
   c. One with a TariffComponentID of 0x893, and a standard price of $0.40
   d. One with a TariffComponentID of 0x894, and a standard price of $0.60
5. Four TariffPeriodStructs
   a. One linking the TariffComponentID of 0x891 to the DayEntryIDs of 0x71 and 0x73
   b. One linking the TariffComponentID of 0x892 to the DayEntryID of 0x72
   c. One linking the TariffComponentID of 0x893 to the DayEntryID of 0x74 and 0x76
   d. One linking the TariffComponentID of 0x894 to the DayEntryIDs of 0x75

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TariffPeriods

Label	TariffComponentIDs	DayEntryIDs
Winter $_{Off\text{-}Peak}$	[0x891]	[0x71, 0x73]
Winter $_{Peak}$	[0x892]	[0x72]
Summer $_{Off\text{-}Peak}$	[0x893]	[0x74, 0x76]
Summer $_{Peak}$	[0x894]	[0x75]

TariffComponents

TariffComponentID	Price		Threshold
0x891	*PriceType*	Standard	0
	Price	$0.40	
0x892	*PriceType*	Standard	0
	Price	$0.50	
0x893	*PriceType*	Standard	0
	Price	$0.40	
0x894	*PriceType*	Standard	0
	Price	$0.60	

DayEntries

DayEntryID	StartTime
0x71	0
0x72	960
0x73	1260
0x74	0
0x75	900
0x76	1260

DayPatterns

DayPatternID	DoW	DayEntryIDs
0x56A	0x7F	[0x71, 0x72, 0x73]
0x56B	0x7F	[0x74, 0x75, 0x76]

CalendarPeriods

StartDate	DayPatternIDs
2025/01/01	[0x56A]
2025/05/01	[0x56B]
2025/10/01	[0x56A]

Figure 28. Example of a seasonal time-of-use rate commodity tariff

9.12.8.6. Individual Days

Tariffs can override scheduled rates on specific days using the IndividualDays attribute.

For example, if a tariff starting on 2025/01/01 charges:

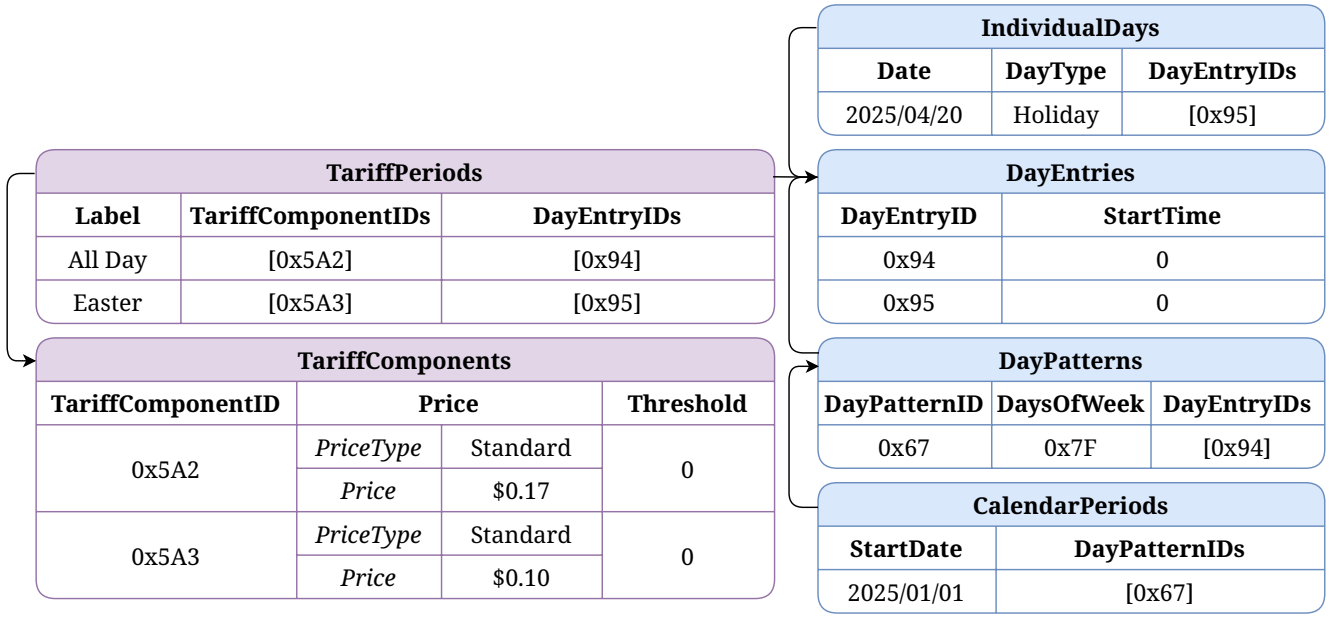
1. $0.17 per unit, all day, every day
2. $0.10 per unit, all day on Easter (2025/04/20)

This can be represented by:

1. Two DayEntryStructs starting at midnight:
   a. One with a DayEntryID of 0x94; this will represent the regular schedule
   b. One with a DayEntryID of 0x95; this will represent the schedule on Easter
2. One DayPatternStruct with a DayPatternID of 0x67, with all DaysOfWeek set and containing the DayEntryID 0x94
3. One CalendarPeriodStruct starting on 2025/01/01 and containing the DayPatternID 0x67
4. One DayStruct with a DayStructDate of 2025/04/20, a DayType of Holiday, and containing the DayEntryID 0x95
5. Two TariffComponentStructs:
   a. One with a TariffComponentID of 0x5A2, and a standard price of $0.17
   b. One with a TariffComponentID of 0x5A3, and a standard price of $0.10
6. Two TariffPeriodStructs
   a. One linking the TariffComponentID of 0x5A2 to the DayEntryID of 0x94

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b. One linking the TariffComponentID of 0x5A3 to the DayEntryID of 0x95



IndividualDays

Date	DayType	DayEntryIDs
2025/04/20	Holiday	[0x95]

TariffPeriods

Label	TariffComponentIDs	DayEntryIDs
All Day	[0x5A2]	[0x94]
Easter	[0x5A3]	[0x95]

DayEntries

DayEntryID	StartTime
0x94	0
0x95	0

TariffComponents

TariffComponentID	Price		Threshold
0x5A2	*PriceType*	Standard	0
	Price	$0.17	
0x5A3	*PriceType*	Standard	0
	Price	$0.10	

DayPatterns

DayPatternID	DaysOfWeek	DayEntryIDs
0x67	0x7F	[0x94]

CalendarPeriods

StartDate	DayPatternIDs
2025/01/01	[0x67]

Figure 29. Example of a flat rate commodity tariff with holiday pricing

9.12.8.7. Individual Days with Durations

Tariffs can override scheduled rates on specific days during specific times using the IndividualDays attribute.

For example, if a tariff starting on 2025/01/01 charges:

1. $0.23 per unit, all day, every day
2. $0.40 per unit from 16:00 to 19:00 on 2025/02/14 as part of a Critical Peak Pricing event

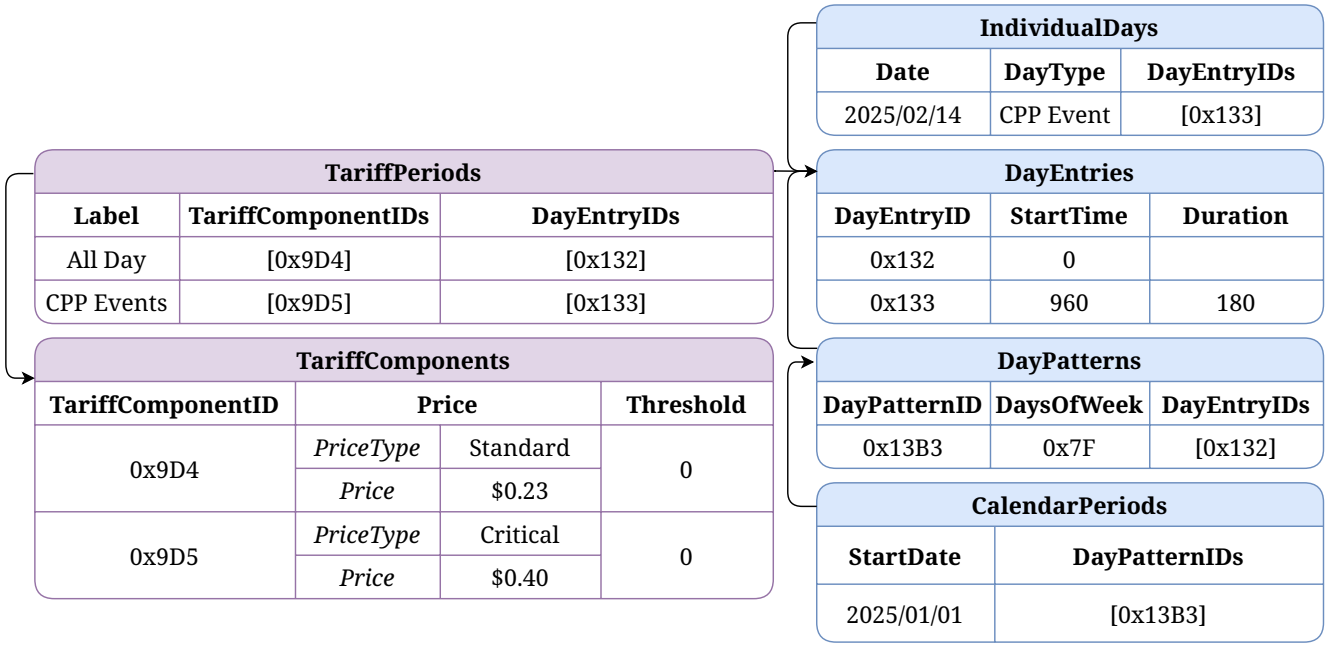
This can be represented by:

1. Two DayEntryStructs:
   a. One with a DayEntryID of 0x132, starting at midnight; this will represent the regular schedule
   b. One with a DayEntryID of 0x133, starting and 960 (16:00) with a duration of 180 (three hours); this will represent the CPP event
2. One DayPatternStruct with a DayPatternID of 0x13B3, with all DaysOfWeek set and containing the DayEntryID 0x132
3. One CalendarPeriodStruct starting on 2025/01/01 and containing the DayPatternID 0x67
4. One DayStruct with a DayStructDate of 2025/02/14, a DayType of Dynamic, and containing the DayEntryID 0x133
5. Two TariffComponentStructs:
   a. One with a TariffComponentID of 0x9D4, and a standard price of $0.23
   b. One with a TariffComponentID of 0x9D5, and a critical price of $0.40
6. Two TariffPeriodStructs:

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a. One linking the TariffComponentID of 0x9D4 to the DayEntryID of 0x132

b. One linking the TariffComponentID of 0x9D5 to the DayEntryID of 0x133



IndividualDays

Date	DayType	DayEntryIDs
2025/02/14	CPP Event	[0x133]

TariffPeriods

Label	TariffComponentIDs	DayEntryIDs
All Day	[0x9D4]	[0x132]
CPP Events	[0x9D5]	[0x133]

DayEntries

DayEntryID	StartTime	Duration
0x132	0	
0x133	960	180

TariffComponents

TariffComponentID	Price		Threshold
0x9D4	*PriceType*	Standard	0
	Price	$0.23	
0x9D5	*PriceType*	Critical	0
	Price	$0.40	

DayPatterns

DayPatternID	DaysOfWeek	DayEntryIDs
0x13B3	0x7F	[0x132]

CalendarPeriods

StartDate	DayPatternIDs
2025/01/01	[0x13B3]

Figure 30. Example of a flat rate commodity tariff with a critical peak pricing event

9.12.8.8. Dynamic Pricing

Tariffs can also use the IndividualDays attribute to implement fully dynamic pricing.

For example, if a tariff charges a different rate every hour of the day, publishing the current day's rates as well as predictions of the next six days rates, the state of the tariff on 2025/03/15 can be represented by:

1. 168 DayEntryStructs:
   a. 24 for today, with DayEntryIDs ranging from 0x200 to 0x217
   b. 24 for tomorrow, with DayEntryIDs ranging from 0x218 to 0x22F
   c. 120 for the remaining five days, with DayEntryIDs ranging from 0x230 to 0x2A7
2. Seven DayStructs:
   a. One for 2025/03/15, with a DayType of Dynamic, containing DayEntryIDs ranging from 0x200 to 0x217
   b. One for 2025/03/16, with a DayType of Dynamic, containing DayEntryIDs ranging from 0x218 to 0x22F
   c. Five more for the remaining days
3. 168 TariffComponentStructs:
   a. 24 for today, with TariffComponentIDs ranging from 0x400 to 0x417, prices for each hour, and Predicted set to false
   b. 24 for tomorrow, with TariffComponentIDs ranging from 0x418 to 0x42F, prices for each hour, and Predicted set to true
   c. 120 for the remaining five days, with TariffComponentIDs ranging from 0x430 to 0x4A7,

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prices for each hour, and Predicted set to true

4. 168 TariffPeriodStructs:
    a. 24 linking the 0x400 to 0x417 range of TariffComponentIDs to the 0x200 to 0x217 range of DayEntryIDs
        i. These components link the price for each hour of today to the day entry representing that hour
    b. 24 linking the 0x418 to 0x42F range of TariffComponentIDs to the 0x218 to 0x22F range of DayEntryIDs
        i. These components link the price for each hour of tomorrow to the day entry representing that hour
    c. 120 linking the 0x430 to 0x4A7 range of TariffComponentID to the 0x230 to 0x2A7 range of DayEntryIDs
        i. These components link the price for each hour of the remaining five days to the day entry representing that hour

As new rates are provided, the IndividualDays, TariffComponents, and TariffPeriods attributes can drop expired values while adding new predictions. TariffComponentStructs which previously had Predicted set to true can be updated with actual prices and Predicted set to false.

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TariffPeriods

Label	TariffComponentIDs	DayEntryIDs	
3/15 00:00	[0x400]	[0x200]	
3/15 01:00	[0x401]	[0x201]	
3/15 02:00	[0x402]	[0x202]	*3/15*
...			
3/15 23:00	[0x417]	[0x217]	
3/16 00:00	[0x418]	[0x218]	
...			*3/16*
3/16 23:00	[0x42F]	[0x22F]	
3/17 00:00	[0x430]	[0x230]	
...			*3/17*
3/17 23:00	[0x447]	[0x247]	
3/18 00:00	[0x448]	[0x248]	
...			*3/18*
3/18 23:00	[0x45F]	[0x25F]	
3/19 00:00	[0x460]	[0x260]	
...			*3/19*
3/19 23:00	[0x477]	[0x277]	
3/20 00:00	[0x478]	[0x278]	
...			*3/20*
3/20 23:00	[0x48F]	[0x28F]	
3/21 00:00	[0x490]	[0x290]	
...			*3/21*
3/21 23:00	[0x4A7]	[0x2A7]	

DayEntries

DayEntryID	StartTime	
0x200	0	
0x201	60	
0x202	120	*3/15*
...	...	
0x217	1380	
0x218	0	
...	...	*3/16*
0x22F	1380	
0x230	0	
...	...	*3/17*
0x247	1380	
0x248	0	
...	...	*3/18*
0x25F	1380	
0x260	0	
...	...	*3/19*
0x277	1380	
0x278	0	
...	...	*3/20*
0x28F	1380	
0x290	0	
...	...	*3/21*
0x2A7	1380	

TariffComponents

TariffComponentID	Price		Predicted	
0x400	*PriceType*	Standard	False	
	Price	$0.10		
0x401	*PriceType*	Standard	False	
	Price	$0.12		
0x402	*PriceType*	Standard	False	*3/15*
	Price	$0.15		
...				
0x417	*PriceType*	Standard	False	
	Price	$0.06		
0x418	*PriceType*	Standard	False	
	Price	$0.08		
...				*3/16*
0x42F	*PriceType*	Standard	False	
	Price	$0.07		
...				

IndividualDays

Date	DayType	DayEntryIDs
2025/03/15	Dynamic	[0x200, ... 0x217]
2025/03/16	Dynamic	[0x218, ... 0x22F]
2025/03/17	Dynamic	[0x230, ... 0x247]
2025/03/18	Dynamic	[0x248,... 0x25F]
2025/03/19	Dynamic	[0x260, ... 0x277]
2025/03/20	Dynamic	[0x278, ... 0x28F]
2025/03/21	Variable	[0x290, ... 0x2A7]

Figure 31. Example of a fully dynamic rate commodity tariff

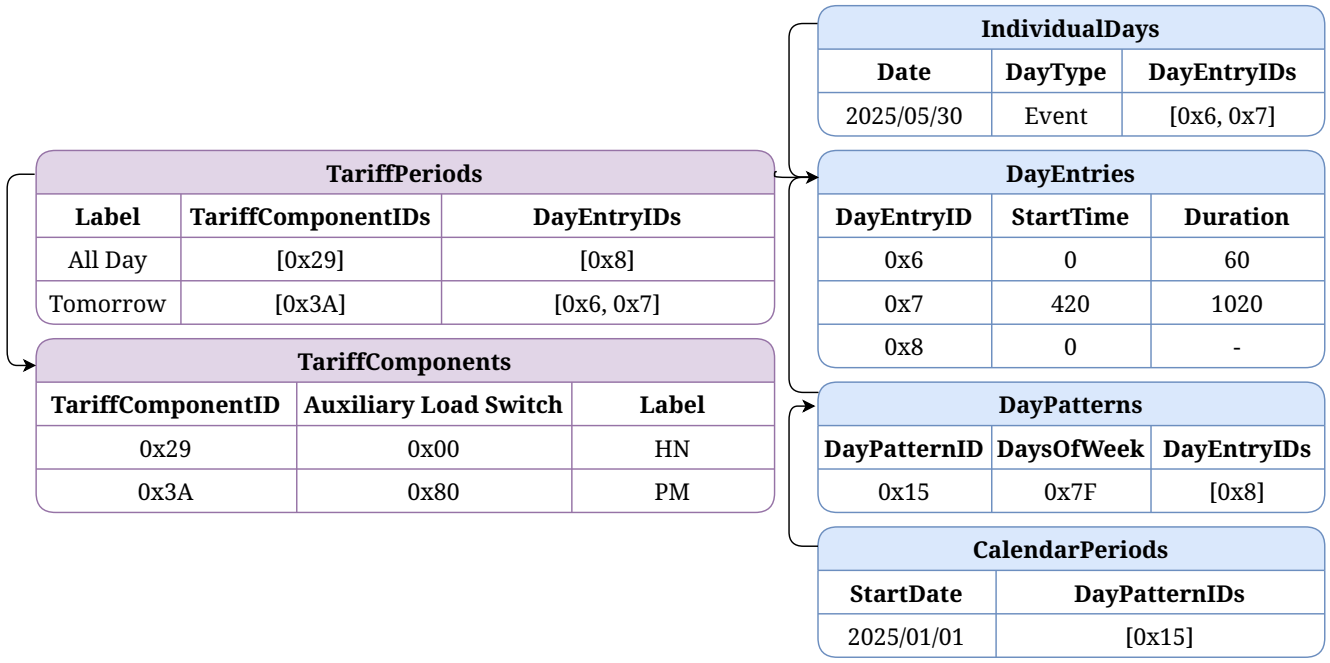
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9.12.8.9. Other Tariff Components

Tariffs components can represent changes to other metering states than price.

For example, a tariff that controls a set of auxiliary load switches might only switch auxiliary switch 8 to ON from 00:00 to 01:00 and 07:00 to 24:00 on 2025/05/30. This tariff could be represented by:

1. Three DayEntryStructs:
   a. One with a DayEntryID of 0x06; this will represent the 00:00 to 01:00 period on 2025/05/30
   b. One with a DayEntryID of 0x07; this will represent the 07:00 to 24:00 period on 2025/05/30
   c. One with a DayEntryID of 0x08; this will represent the all-day regular schedule
2. One DayPatternStruct with a DayPatternID of 0x15, with all DaysOfWeek set and containing the DayEntryID 0x08
3. One CalendarPeriodStruct starting on 2025/01/01 and containing the DayPatternID 0x15
4. One DayStruct with a DayStructDate of 2025/05/30, a DayType of Event, and containing the DayEntryIDs 0x06 and 0x07
5. Two TariffComponentStructs:
   a. One with a TariffComponentID of 0x29, with all auxiliary switches disabled
   b. One with a TariffComponentID of 0x3A, with the eighth auxiliary switch enabled
6. Two TariffPeriodStructs
   a. One linking the TariffComponentID of 0x29 to the DayEntryID of 0x08
   b. One linking the TariffComponentID of 0x3A to the DayEntryIDs of 0x06 and 0x07



IndividualDays

Date	DayType	DayEntryIDs
2025/05/30	Event	[0x6, 0x7]

TariffPeriods

Label	TariffComponentIDs	DayEntryIDs
All Day	[0x29]	[0x8]
Tomorrow	[0x3A]	[0x6, 0x7]

DayEntries

DayEntryID	StartTime	Duration
0x6	0	60
0x7	420	1020
0x8	0	-

TariffComponents

TariffComponentID	Auxiliary Load Switch	Label
0x29	0x00	HN
0x3A	0x80	PM

DayPatterns

DayPatternID	DaysOfWeek	DayEntryIDs
0x15	0x7F	[0x8]

CalendarPeriods

StartDate	DayPatternIDs
2025/01/01	[0x15]

Figure 32. Example of a commodity tariff for auxiliary load switches

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9.12.9. Active Tariff Components for a Timestamp

The currently active tariff components are indicated by the value of the CurrentTariffComponents attribute. Additionally, the upcoming tariff components are indicated by the value of the NextTariffComponents attribute; the value of the CurrentTariffComponents attribute will next update at the timestamp indicated by the NextDayEntryDate attribute.

Finding the active tariff components for an arbitrary timestamp requires examining the contents of the DayEntries, DayPatterns, CalendarPeriods, IndividualDays, TariffComponents, and TariffPeriods attributes, and applying the following steps:

1. Determine the minutes since elapsed since midnight in the meter's timezone for the timestamp
2. For each DayStruct in the IndividualDays attribute:
   a. If the Date field matches the timestamp:
      i. Find the DayEntryStruct referenced in DayStructDayEntryIDs with the largest StartTime that is less than or equal to the minutes since midnight and either:
         A. Has no Duration, or
         B. StartTime + Duration is greater than or equal to the minutes since midnight
3. If no matching DayEntryStruct was found:
   a. Determine the day of the week for the timestamp
   b. Find the CalendarPeriodStruct in the CalendarPeriods attribute with the largest StartDate that is less than or equal to the timestamp
   c. If a matching CalendarPeriodStruct was found:
      i. For each DayPatternStruct in the DayPatterns attribute referenced in DayPatternIDs:
         A. If the bit matching the day of the week is set in DaysOfWeek:
            I. Find the DayEntryStruct referenced in DayStructDayEntryIDs with the largest StartTime that is less than or equal to the minutes since midnight
4. If a matching DayEntryStruct was found:
   a. For each TariffPeriodStruct in the TariffPeriods attribute:
      i. If the DayEntryIDs field contains the DayEntryID of the matched DayEntryStruct, then the TariffComponentIDs field contains the TariffComponentIDs of the active TariffComponentStructs.
5. Otherwise, there is no available information in the tariff for determining the tariff components active at the timestamp

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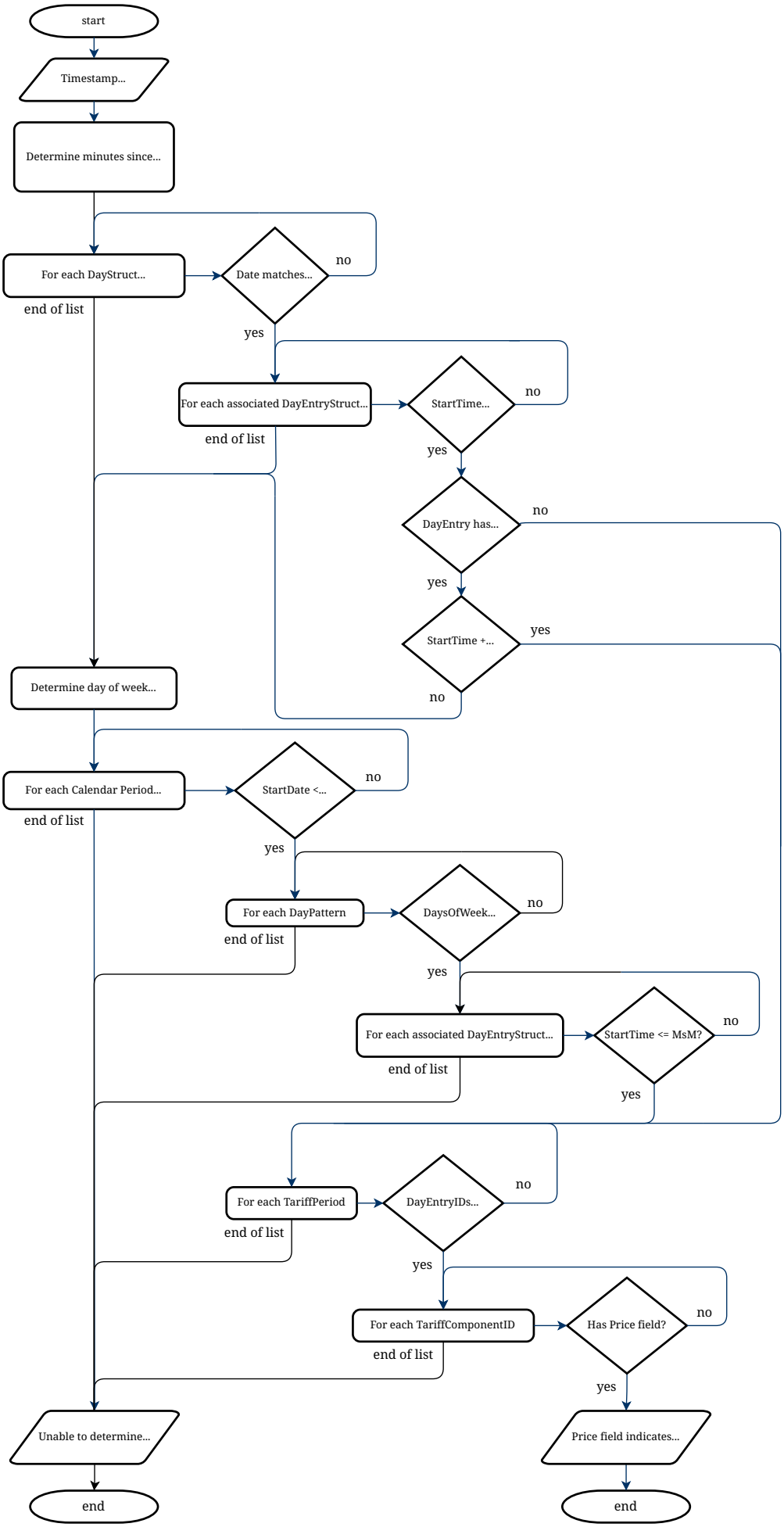


Figure 33. Flowchart for determining tariff components for a given timestamp

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9.13. Electrical Grid Conditions Cluster

The Electrical Grid Conditions Cluster provides the mechanism for communicating electricity grid carbon intensity to devices within the premises in units of Grams of CO2e per kWh.

This is an important mechanism to allow energy appliances decide when to operate to help consumers reduce their carbon footprint, when the pricing may be the same throughout the day.

When homes have local generation (for example Solar PV) this may mean that the premises at the local grid connection point has effectively no green house gas emissions, and so this cluster allows devices to understand both the grid and local conditions.

9.13.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial Matter revision

9.13.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	EGC

9.13.3. Cluster ID

ID	Name
0x00A0	Electrical Grid Conditions

9.13.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**FORE**	Forecasting	O	Forecasts upcoming

9.13.4.1. Forecasting Feature

The feature indicates the server is capable of providing a forecast of grid and local conditions for several hours in the future.

9.13.5. Data Types

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9.13.5.1. ThreeLevelEnum Type

This data type is derived from enum8 and is used for indicating three levels: Low, Medium, High.

Value	Name	Summary	Conformance
0	**Low**	Low	M
1	**Medium**	Medium	M
2	**High**	High	M

9.13.5.2. ElectricalGridConditionsStruct Type

This represents the greenhouse gas carbon intensity over a given period.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**PeriodStart**	epoch-s	all				M
1	**PeriodEnd**	epoch-s	min (PeriodStart + 1)	X			M
2	**GridCarbonIntensity**	int16	all				M
3	**GridCarbonLevel**	ThreeLevelEnum	all				M
4	**LocalCarbonIntensity**	int16	all				M
5	**LocalCarbonLevel**	ThreeLevelEnum	all				M

9.13.5.2.1. PeriodStart Field

This field SHALL indicate the beginning timestamp in UTC of the period.

9.13.5.2.2. PeriodEnd Field

This field SHALL indicate the ending timestamp in UTC of the period. This SHALL be greater than PeriodStart. If this field is null, then the period has no definite end.

9.13.5.2.3. GridCarbonIntensity Field

This field SHALL indicate the estimated carbon intensity in grams of CO2 equivalent per kWh of the grid. This is not impacted by any local generation.

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9.13.5.2.4. GridCarbonLevel Field

This field SHALL indicate the relative level of carbon intensity of the grid. This is not impacted by any local generation.

It is up to the cluster server to determine the thresholds of High, Medium or Low based upon typical grid carbon levels for this region or market, since this can vary significantly between countries across the world.

9.13.5.2.5. LocalCarbonIntensity Field

This field SHALL indicate the estimated carbon intensity in grams of CO2 equivalent per kWh of the premises mains supply. This value SHALL take into account the impact of any local generation.

For example, if an EMS can forecast that excess generation will occur in a period or the premises are currently generating excess power to the grid, then this could assume a value of 0 grams CO2 equivalent per kWh for this period.

When solar PV is not being exported to the grid then this value is typically the same as the GridCarbonIntensity.

Clients are expected to use this value when computing or displaying the local premises carbon intensity to users.

If there is no local generation, this value SHALL be the same as the GridCarbonIntensity at all times.

9.13.5.2.6. LocalCarbonLevel Field

This field SHALL indicate the relative level of carbon intensity of the premises mains supply. This level SHALL take into account impact of any local generation.

It is up to the cluster server to determine the thresholds of High, Medium or Low based upon typical grid carbon levels for this region or market, since this can vary significantly between countries across the world.

When local power generation (for example from solar PV) is not being exported to the grid then this level is the same as the GridCarbonLevel.

Clients are expected to use this value when displaying the local premises carbon intensity to users.

If there is no local generation, this value SHALL be the same as the GridCarbonLevel at all times.

9.13.6. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**LocalGenerationAvailable**	bool	all	X		RW VO	M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0001	**CurrentConditions**	ElectricalGridConditionsStruct	all	X		R V	M
0x0002	**ForecastConditions**	list[ElectricalGridConditionsStruct]	max 56			R V	FORE

9.13.6.1. LocalGenerationAvailable Attribute

This SHALL indicate if there is known to be local generation (for example Solar PV or Battery Storage) at the premises.

If the presence of any local generation is unknown, or cannot be determined, the value SHALL be null.

9.13.6.2. CurrentConditions Attribute

This SHALL indicate the current electricity supply conditions. If the current conditions are unknown, or cannot be determined, the value SHALL be null.

9.13.6.3. ForecastConditions Attribute

This SHALL indicate the forecast of upcoming electricity supply conditions. If the forecast is unable to be determined, this list SHALL be empty.

The list entries SHALL be in time order:

- All entries except the last one SHALL have a non-null PeriodEnd.
- For all entries except the first one, PeriodStart SHALL be greater than the previous entry's PeriodEnd.

9.13.7. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**CurrentConditionsChanged**	INFO		V	O

9.13.7.1. CurrentConditionsChanged Event

This event SHALL be generated when the value of the CurrentConditions attribute changes.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**CurrentConditions**	ElectricalGridConditionsStruct		X		M

9.13.7.1.1. CurrentConditions Field

This field SHALL be the new value of the CurrentConditions attribute.

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Chapter 10. Network Infrastructure

The Cluster Library is made of individual chapters such as this one. See Document Control in the Cluster Library for a list of all chapters and documents. References between chapters are made using a *X.Y* notation where *X* is the chapter and *Y* is the sub-section within that chapter.

10.1. General Description

10.1.1. Introduction

The clusters specified in this section are used to control network infrastructure devices, such as home internet gateways, Wi-Fi access points, Thread Border Routers, etc.

10.1.2. Cluster List

This section lists the clusters specified in this document, and gives examples of typical usage for the purpose of clarification.

Table 16. Clusters Specified in the Network Infrastructure Functional Domain

Cluster ID	Cluster Name	Description
0x0451	Wi-Fi Network Management	Request Wi-Fi network details
0x0452	Thread Border Router Management	Management of a Thread Border Router / Network
0x0453	Thread Network Directory	Directory of Thread networks and Border Routers

10.2. Wi-Fi Network Management Cluster

This cluster provides an interface for getting information about the Wi-Fi network that a Network Infrastructure Manager device type provides. Privileged nodes within the same fabric as a Network Infrastructure Manager can use these interfaces to request information related to the Wi-Fi Network such as SSID and Passphrase.

10.2.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

10.2.2. Classification

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Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	WIFINM

10.2.3. Cluster ID

ID	Name
0x0451	Wi-Fi Network Management

10.2.4. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SSID**	octstr	1 to 32	N X		R V	M
0x0001	**PassphraseSurrogate**	uint64	all	N X		R M	M

10.2.4.1. SSID Attribute

This attribute SHALL indicate the SSID of the primary Wi-Fi network provided by this device.

A value of null SHALL indicate that no primary Wi-Fi network is available (e.g. because the Wi-Fi network has not yet been configured by the user).

NOTE

The SSID in Wi-Fi is a collection of 1-32 bytes, the text encoding of which is not specified. Implementations must be careful to support transferring these byte strings without requiring a particular encoding. The most common encoding is UTF-8, however this is just a convention. Some configurations may use Latin-1 or other character sets.

10.2.4.2. PassphraseSurrogate Attribute

This attribute SHALL contain an arbitrary numeric value; this value SHALL increase whenever the passphrase or PSK associated with the primary Wi-Fi network provided by this device changes.

A value of null SHALL indicate that no primary Wi-Fi network is available.

Clients can subscribe to this attribute or compare its value to a locally cached copy to detect if a cached passphrase value has become stale.

It is RECOMMENDED that servers implement this attribute as either a timestamp or a counter. When implemented as a counter it SHOULD be initialized with a random value.

NOTE

The passphrase itself is not exposed as an attribute to avoid its unintentional retrieval or caching by clients that use wildcard reads or otherwise routinely read all available attributes. It can be retrieved using the NetworkPassphraseRequest

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command.

10.2.5. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**NetworkPassphraseRequest**	client ⇒ server	NetworkPassphraseResponse	M	M
0x01	**NetworkPassphraseResponse**	client ⇐ server	N		M

10.2.5.1. NetworkPassphraseRequest Command

This command is used to request the current WPA-Personal passphrase or PSK associated with the Wi-Fi network provided by this device.

If the command is not executed via a CASE session, the command SHALL be rejected with a status of UNSUPPORTED_ACCESS.

If no primary Wi-Fi network is available (the SSID attribute is null), the command SHALL be rejected with a status of INVALID_IN_STATE.

Otherwise a NetworkPassphraseResponse SHALL be generated.

10.2.5.2. NetworkPassphraseResponse Command

This command SHALL be generated in response to a NetworkPassphraseRequest command. The data for this command SHALL be as follows:

ID	Name	Type	Constraint	Conformance
0x00	**Passphrase**	octstr	max 64	M

10.2.5.2.1. Passphrase Field

This field SHALL indicate the current WPA-Personal passphrase or PSK associated with the primary Wi-Fi network provided by this device, in one of the following formats:

- 8..63 bytes: WPA/WPA2/WPA3 passphrase.
- 64 bytes: WPA/WPA2/WPA3 raw hex PSK. Each byte SHALL be a ASCII hexadecimal digit.

This matches the formats defined for WPA networks by the Credentials field in the Network Commissioning cluster (see [MatterCore]).

NOTE WPA3-Personal permits passphrases shorter than 8 or longer than 63 characters, however the Network Commissioning cluster does not currently support configuring Matter devices to connect to operational networks utilizing such a passphrase.

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10.3. Thread Border Router Management Cluster

This cluster provides an interface for managing a Thread Border Router and the Thread network that it belongs to. Privileged nodes within the same fabric as a Thread Border Router can use these interfaces to request and set credentials information to the Thread network.

10.3.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

10.3.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	TBRM

10.3.3. Cluster ID

ID	Name
0x0452	Thread Border Router Management

10.3.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**PC**	PANChange	O	The ability to change PAN configuration with pending dataset setting request.

10.3.4.1. PANChange Feature

This feature SHALL indicate the ability of the Border Router to change its already configured PAN to another, by setting a pending dataset.

NOTE	This feature flag can be used to protect an already-configured network from accidental configuration change, e.g. when the Thread Border Router serves non-Matter devices that do not support PAN change for an implementation-specific reason.

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10.3.5. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**BorderRouterName**	string	1 to 63			R V	M
0x0001	**BorderAgentID**	octstr	16			R V	M
0x0002	**ThreadVersion**	uint16	all	F		R V	M
0x0003	**InterfaceEnabled**	bool	all	N		R V	M
0x0004	**ActiveDatasetTimestamp**	uint64	all	N X		R V	M
0x0005	**PendingDatasetTimestamp**	uint64	all	N X		R V	M

10.3.5.1. BorderRouterName Attribute

This attribute SHALL indicate a user-friendly name identifying the device model or product of the Border Router in MeshCOP (DNS-SD service name) as defined in the Thread specification, and has the following recommended format: `<VendorName> <ProductName>._meshcop._udp`. An example name would be `ACME Border Router (74be)._meshcop._udp`.

10.3.5.2. BorderAgentID Attribute

This attribute SHALL indicate a 16-byte globally unique ID for a Thread Border Router device. This ID is manufacturer-specific, and it is created and managed by the border router's implementation.

10.3.5.3. ThreadVersion Attribute

This attribute SHALL indicate the Thread version supported by the Thread interface configured by the cluster instance.

The format SHALL match the value mapping defined in the "Version TLV" section of the Thread specification. For example, Thread 1.3.0 would have ThreadVersion set to 4.

10.3.5.4. InterfaceEnabled Attribute

This attribute SHALL indicate whether the associated IEEE 802.15.4 Thread interface is enabled or disabled.

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10.3.5.5. ActiveDatasetTimestamp Attribute

This attribute SHALL be null if the Thread Border Router has no dataset configured, otherwise it SHALL be the timestamp value extracted from the Active Dataset value configured by the Thread Node to which the border router is connected. This attribute SHALL be updated when a new Active dataset is configured on the Thread network to which the border router is connected.

10.3.5.6. PendingDatasetTimestamp Attribute

This attribute SHALL be null if the Thread Border Router has no Pending dataset configured, otherwise it SHALL be the timestamp value extracted from the Pending Dataset value configured by the Thread Node to which the border router is connected. This attribute SHALL be updated when a new Pending dataset is configured on the Thread network to which the border router is connected.

10.3.6. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**GetActiveDatasetRequest**	client ⇒ server	DatasetResponse	M	M
0x01	**GetPendingDatasetRequest**	client ⇒ server	DatasetResponse	M	M
0x02	**DatasetResponse**	client ⇐ server	N		M
0x03	**SetActiveDatasetRequest**	client ⇒ server	Y	M T	M
0x04	**SetPendingDatasetRequest**	client ⇒ server	Y	M T	PC

10.3.6.1. GetActiveDatasetRequest Command

This command SHALL be used to request the active operational dataset of the Thread network to which the border router is connected.

If the command is not executed via a CASE session, the command SHALL fail with a status code of UNSUPPORTED_ACCESS.

If an internal error occurs, then this command SHALL fail with a FAILURE status code sent back to the initiator.

Otherwise, this SHALL generate a DatasetResponse command.

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10.3.6.2. GetPendingDatasetRequest Command

This command SHALL be used to request the pending dataset of the Thread network to which the border router is connected.

If the command is not executed via a CASE session, the command SHALL fail with a status code of UNSUPPORTED_ACCESS.

If an internal error occurs, then this command SHALL fail with a FAILURE status code sent back to the initiator.

Otherwise, this SHALL generate a DatasetResponse command.

10.3.6.3. DatasetResponse Command

This command is sent in response to GetActiveDatasetRequest or GetPendingDatasetRequest command. The data for this command SHALL be as follows:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Dataset**	octstr	max 254			M

10.3.6.3.1. Dataset field

If no dataset (active or pending as requested) is configured, this field SHALL be set to empty.

Otherwise, this field SHALL contain the active or pending dataset of the Thread network to which the Border Router is connected as an octet string containing the raw Thread TLV value of the dataset, as defined in the Thread specification.

10.3.6.4. SetActiveDatasetRequest Command

This command SHALL be used to set the active Dataset of the Thread network to which the Border Router is connected, when there is no active dataset already.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ActiveDataset**	octstr	max 254			M
1	**Breadcrumb**	uint64	all			O

10.3.6.4.1. ActiveDataset Field

This field SHALL contain the active dataset to set of the Thread network to configure in the Border Router as an octet string containing the raw Thread TLV value of the dataset, as defined in the Thread specification.

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10.3.6.4.2. Breadcrumb Field

See Breadcrumb Attribute section of General Commissioning Cluster in [MatterCore] for usage.

10.3.6.4.3. Effect on receipt

If the command is not executed via a CASE session, the command SHALL fail with a status code of UNSUPPORTED_ACCESS.

If this command is received without a Fail Safe Context (see ArmFailSafe Command of General Commissioning Cluster in [MatterCore]), then this command SHALL fail with a FAILSAFE_REQUIRED status code sent back to the initiator.

If any of the parameters in the ActiveDataset is invalid, the command SHALL fail with a status code of INVALID_COMMAND.

If this command is invoked when the ActiveDatasetTimestamp attribute is not null, the command SHALL fail with a status code of INVALID_IN_STATE. The administrator MAY use the SetPendingDatasetRequest command to modify the future active dataset.

If this command is invoked and the ActiveDatasetTimestamp attribute is null, the Thread Border Router SHALL configure and activate its active dataset using the ActiveDataset parameter.

The activation process SHALL be a time-bound process that completes before expiration of a fail-safe timer. The fail-safe timer SHALL be set at the beginning of the activation process.

If the fail-safe timer expires prior to activation process completion, the command SHALL respond with TIMEOUT, and the Border Router state SHALL revert to the configuration set prior to the fail-safe timer being armed, see ArmFailSafe Command of General Commissioning Cluster in [MatterCore].

After successfully creating or joining a Thread network with the new active dataset, the InterfaceEnabled attribute SHALL be updated to TRUE, indicating that the associated Thread Interface is active and functioning. The ActiveDatasetTimestamp attribute SHALL be updated with the ActiveDataset timestamp. If the Adjacent Infrastructure Link of the Border Router is connected, then the device SHOULD also enable and operate Thread Border Routing functionality using the Thread and Adjacent Infrastructure Link interfaces.

If this command is received during the activation process triggered by a previous SetActiveDatasetRequest command, then the command SHALL fail with a status code of BUSY.

10.3.6.5. SetPendingDatasetRequest Command

This command SHALL be used to set or update the pending Dataset of the Thread network to which the Border Router is connected, if the Border Router supports PAN Change.

If the command is not executed via a CASE session, the command SHALL fail with a status code of UNSUPPORTED_ACCESS.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PendingDataset**	octstr	max 254			M

This PendingDataset field SHALL contain the pending dataset to which the Thread network should be updated. The format of the data SHALL be an octet string containing the raw Thread TLV value of the pending dataset, as defined in the Thread specification.

If any of the parameters in the PendingDataset is invalid, the command SHALL fail with a status of INVALID_COMMAND.

Otherwise, this command SHALL configure the pending dataset of the Thread network to which the Border Router is connected, with the value given in the PendingDataset parameter. The Border Router will manage activation of the pending dataset as defined in the Thread specification.

10.4. Thread Network Directory Cluster

This cluster stores a list of Thread networks (including the credentials required to access each network), as well as a designation of the user's preferred network, to facilitate the sharing of Thread networks across fabrics.

10.4.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial revision

10.4.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	THNETDIR

10.4.3. Cluster ID

ID	Name
0x0453	Thread Network Directory

10.4.4. Data Types

10.4.4.1. ThreadNetworkStruct Type

Represents the data associated with a Thread Network.

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**ExtendedPanID**	octstr	8				M
1	**NetworkName**	string	1 to 16				M
2	**Channel**	uint16	all				M
3	**ActiveTimestamp**	uint64	all				M

10.4.4.1.1. ExtendedPanID Field

This field SHALL indicate the Extended PAN ID from the OperationalDataset for the given Thread network.

10.4.4.1.2. NetworkName Field

This field SHALL indicate the Network Name from the OperationalDataset for the given Thread network.

10.4.4.1.3. Channel Field

This field SHALL indicate the Channel from the OperationalDataset for the given Thread network.

10.4.4.1.4. ActiveTimestamp Field

This field SHALL indicate the Active Timestamp from the OperationalDataset for the given Thread network.

10.4.5. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**PreferredExtendedPanID**	octstr	8	N X		RW VM	M
0x0001	**ThreadNetworks**	list[ThreadNetworkStruct]	max ThreadNetworkTableSize	N		R V	M
0x0002	**ThreadNetworkTableSize**	uint8	desc	F		R V	M

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10.4.5.1. PreferredExtendedPanID Attribute

This attribute SHALL represent the Thread Extended PAN ID value for the Thread network designated by the user to be their preferred network for commissioning of Thread devices. If not null, the value of this attribute SHALL match the ExtendedPanID of a network in the ThreadNetworks attribute. A write operation with a non-null value that does not match any network in the ThreadNetworks list SHALL be rejected with a status of CONSTRAINT_ERROR.

The purpose of designating one Thread network as preferred is to help a commissioner to select a Thread network when a Thread device is within suitable range of more than one Thread network which appears in the ThreadNetworks list. A value of null indicates that there is no current preferred network: All networks MAY be treated as equally preferred by a commissioner with access to this cluster.

This attribute MAY be automatically set to the ExtendedPanID of the first Thread network added to the ThreadNetworks list.

A client SHALL obtain user consent before changing the value of this attribute from a non-null value.

On a factory reset this attribute SHALL be reset to null.

10.4.5.2. ThreadNetworks Attribute

This attribute SHALL indicate the list of Thread Networks known about by this cluster. If the node hosting this cluster includes a Thread Border Router, then an entry for its Thread Network SHALL be included in this list.

The list can be modified via the AddNetwork and RemoveNetwork commands.

For each entry in the list, the cluster server also stores a Thread Operational Dataset. Clients use the GetOperationalDataset command to obtain the Operational Dataset for an entry in this list.

On a factory reset this list SHALL be cleared, and any Thread Operational datasets previously stored SHALL be removed from the Node.

10.4.5.3. ThreadNetworkTableSize Attribute

This attribute SHALL indicate the maximum number of entries that can be held in the ThreadNetworks list; it SHALL be at least 2 times the number of SupportedFabrics advertised in the Operational Credentials Cluster on the root endpoint of this node.

10.4.6. Commands

ID	Name	Direction	Response	Access	Conformance
0x00	**AddNetwork**	client ⇒ server	Y	M T	M
0x01	**RemoveNetwork**	client ⇒ server	Y	M T	M

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ID	Name	Direction	Response	Access	Conformance
0x02	**GetOperationalDataset**	client ⇒ server	OperationalDatasetResponse	M	M
0x03	**OperationalDatasetResponse**	server ⇒ client	N		M

10.4.6.1. AddNetwork Command

Adds an entry to the ThreadNetworks attribute with the specified Thread Operational Dataset.

If there is an existing entry with the Extended PAN ID then the Thread Operational Dataset for that entry is replaced. As a result, changes to the network parameters (e.g. Channel, Network Name, PSKc, ...) of an existing entry with a given Extended PAN ID can be made using this command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**OperationalDataset**	octstr	max 254			M

10.4.6.1.1. OperationalDataset Field

This field SHALL represent the Operational Dataset for the network, using the encoding defined in the Thread specification. It SHALL contain at least the following sub-TLVs: Active Timestamp, Channel, Channel Mask, Extended PAN ID, Network Key, Network Mesh-Local Prefix, Network Name, PAN ID, PSKc, and Security Policy.

10.4.6.1.2. Effect on Receipt

1. If the TLV structure of the Operational Dataset is invalid, or if any of the required sub-TLVs are missing, the command SHALL be rejected with a status of CONSTRAINT_ERROR. However, inclusion of unknown sub-TLVs SHALL NOT be treated as an error, as future versions of the Thread specification MAY define additional sub-TLVs.
2. If there is an existing ThreadNetworks entry with an Extended PAN ID matching the received dataset, then that entry SHALL be replaced with the received dataset. However, if the received dataset has an Active Timestamp that is less than or equal to that of the existing entry, then the update SHALL be rejected with a status of INVALID_IN_STATE.
3. Otherwise, a new entry SHALL be added based on the received dataset. If this would cause the size of the ThreadNetworks list to exceed the ThreadNetworkTableSize, then a RESOURCE_EXHAUSTED status SHALL be returned.

10.4.6.2. RemoveNetwork Command

Removes the network with the given Extended PAN ID from the ThreadNetworks attribute.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ExtendedPanID**	octstr	8			M

10.4.6.2.1. Effect on Receipt

1. If no network with the given Extended PAN ID exists in the ThreadNetworks attribute, the command SHALL be rejected with a status of NOT_FOUND.
2. If the given Extended PAN ID matches the PreferredExtendedPanID attribute, the command SHALL be rejected with a status of CONSTRAINT_ERROR.
3. Otherwise, the network with the matching Extended PAN ID SHALL be removed from the ThreadNetworks attribute.

10.4.6.3. GetOperationalDataset Command

Retrieves the Thread Operational Dataset with the given Extended PAN ID.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ExtendedPanID**	octstr	8			M

10.4.6.3.1. Effect on Receipt

1. If the command is not executed via a CASE session, the command SHALL be rejected with a status of UNSUPPORTED_ACCESS.
2. If no network with the given Extended PAN ID exists in the ThreadNetworks attribute, the command SHALL be rejected with a status of NOT_FOUND.
3. Otherwise, an OperationalDatasetResponse command containing the operational dataset for the requested network SHALL be generated.

10.4.6.4. OperationalDatasetResponse Command

Contains the Thread Operational Dataset for the Extended PAN specified in GetOperationalDataset.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**OperationalDataset**	octstr	max 254			M

10.4.7. Guidance for Fabrics / Commissioners

This cluster allows fabrics (via their Administrators or Nodes with Manage privilege) to collaboratively share and manage a directory of ThreadNetworks, as well as a designation of one of those

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networks as the user's preferred network via the PreferredExtendedPanID attribute.

It is RECOMMENDED that fabrics only interact with instances of this cluster hosted on endpoints that declare a device type that explicitly allows this cluster; currently this includes the Network Infrastructure Manager and Thread Border Router device types. Networks from the ThreadNetworks list of any such directory instance, as well as the Active Dataset made available by any Thread Border Router via the Thread Border Router Management Cluster, SHOULD be considered as potentially available networks.

In determining the user's preferred network, a directory instance on a Network Infrastructure Manager device SHOULD take precedence over other instances. Where a fabric has access to multiple Network Infrastructure Manager devices, it MAY give precedence to the directory instance on the Network Infrastructure Manager device considered to be acting as the primary Internet gateway device for the network.

Due to the sensitive nature of Thread network credentials, it is RECOMMENDED that fabrics obtain user consent before starting to contribute credentials to a particular directory instance. Contributions to a directory instance SHOULD be restricted to those Thread networks that are likely to be reachable from the infrastructure network to which the device hosting that directory is connected.

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Chapter 11. Cameras

The Cluster Library is made of individual chapters such as this one. See Document Control in the Cluster Library for a list of all chapters and documents. References between chapters are made using a *X.Y* notation where *X* is the chapter and *Y* is the sub-section within that chapter.

11.1. General Description

11.1.1. Introduction

The clusters specified in this section are used to set up and control camera devices.

11.1.2. Cluster List

This section lists the clusters specified in this document, and gives examples of typical usage for the purpose of clarification.

Table 17. Clusters Specified in the Camera Functional Domain

Cluster ID	Cluster Name	Description
0x0551	Camera AV Stream Management	Management of Camera AV Streams
0x0552	Camera AV Settings User-Level Management	Management of Camera User-Level AV Settings
0x0553	WebRTC Transport Provider	Camera side of WebRTC
0x0554	WebRTC Transport Requestor	Viewer side of WebRTC
0x0555	Push AV Stream Transport	Setup and use of Push Transport
0x0556	Chime	Selection and playing of preinstalled chime sounds

11.1.3. Data Types

11.1.3.1. StreamUsageEnum Type

This data type is derived from enum8.

This data type provides an enumeration of the different usages of streams supported by the camera. This categorization indicates the use-case of a specific stream and thus factors into its priority in terms of resource allocation by the camera. For example, a Recording stream MAY be given higher priority than a LiveView stream to maintain a higher quality in terms of resolution, bitrate, etc.

Value	Name	Summary	Conformance
0	**Internal**	Internal video stream.	M

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Value	Name	Summary	Conformance
1	**Recording**	Stream for recording clips.	O
2	**Analysis**	Stream for analysis and entity detection.	O
3	**LiveView**	Stream for liveview.	M

11.1.3.1.1. Internal Value

Stream is used for internal purposes, e.g., robotic vacuum using camera for navigation. While the primary use-case for a stream of this usage-type is for an internal purpose, it MAY be re-used for other camera functions, e.g., LiveView. However, this stream SHALL not be modified.

11.1.3.1.2. Recording Value

Stream is used for clip upload and archival.

11.1.3.1.3. Analysis Value

Stream is used for automated audio/video analysis without archival.

11.1.3.1.4. LiveView Value

Stream is used for live watching through client devices.

11.1.3.2. ViewportStruct Type

This struct is used to encode a bounding rectangle of the viewport on the image sensor

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**X1**	uint16	all				M
1	**Y1**	uint16	all				M
2	**X2**	uint16	all				M
3	**Y2**	uint16	all				M

11.1.3.2.1. X1 Field

This field SHALL represent the position of the starting vertex along the horizontal (x) axis.

11.1.3.2.2. Y1 Field

This field SHALL represent the position of the starting vertex along the vertical (y) axis.

11.1.3.2.3. X2 Field

This field SHALL represent the position of the ending vertex along the horizontal (x) axis.

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11.1.3.2.4. Y2 Field

This field SHALL represent the position of the ending vertex along the vertical (y) axis.

11.2. Camera AV Stream Management Cluster

This cluster is used to allow clients to manage, control, and configure various audio, video, and snapshot streams on a camera. Broadly, this cluster would be used to:

1. Create and manage all streams on the camera.
2. Manage the static and dynamic characteristics of all streams on the camera.
3. Manage settings attributes that are globally applicable across fabrics and controlled by administrators.

11.2.1. Lifecycle and Resource Management

11.2.1.1. Stream Lifecycle

It is RECOMMENDED that Commissioners allocate streams when the device is commissioned, based on the expected Stream Usages. Streams would therefore have very long lifetimes and only change if the expected use of the camera would change. Stream allocation is a complex and resource intensive task for devices to perform, and thus should be done as infrequently as possible.

When temporary disabling of streams is needed for things such as geo-fencing, scheduled recordings, alarms, etc; Controllers SHOULD use the SoftRecordingPrivacyModeEnabled or SoftLivestreamPrivacyModeEnabled attributes instead of deallocating and reallocating streams.

11.2.1.2. Resource Management and Stream Priorities

Allocated streams require camera resources including but not limited to CPU, memory and network bandwidth. With multiple streams allocated on the camera that can be shared among clients (potentially in different fabrics), management of these resources allocated to the streams and the stream priorities is of critical importance. The camera maintains a ranked list of stream-usage priorities that SHALL be configurable by an administrator via the SetStreamPriorities command.

11.2.1.2.1. Stream Sharing and Reuse

Whenever possible, the camera SHOULD attempt to reuse an existing stream when serving to multiple clients. If for a newly requested stream, the parameter requirements around bitrate, framerate, resolution, etc. are the same as an already existing stream, the camera SHALL reuse the existing one. For multiple available closely matching streams, one with an identical StreamUsage as the request SHOULD be selected.

If the video encoder produces a single stream, then the camera would share this stream with all receiving endpoints. However, if the video encoder is capable of producing multiple streams, the camera MAY allocate, for example, a high resolution, high bitrate stream for recording and, lower resolution, lower bitrate streams for real-time viewing on various devices.

Cameras with multiple video encoders can support simultaneous high-quality streams, with each

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encoder dedicated to a specific stream, allowing for simultaneous recording and live viewing at the highest possible quality.

11.2.1.2.2. Multiple Stream Resource Conflict Resolution

The camera SHOULD utilize the highest resolution and bitrate the network can support for the stream that is at the top of the video stream priority list. For multiple simultaneous liveview clients, the camera MAY automatically adjust the resolution and bitrate while still maintaining a good quality of experience for all viewers.

If the accommodation of a new stream request requires an adjustment of video parameters (resolution, bitrate, etc) for an existing stream, it SHALL abide by the minimum configuration requirements of the existing stream. Otherwise, the new stream request SHALL be rejected with a FAILED notification to the requester.

11.2.2. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial release
2	Add attribute ImageRotationDiscreteAngles

11.2.3. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	AVSM

11.2.4. Cluster ID

ID	Name
0x0551	Camera AV Stream Management

11.2.5. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Table 18. Camera AV Stream Management Features

Bit	Code	Feature	Conformance	Description
0	**ADO**	Audio	O.a+	Audio Streams supported
1	**VDO**	Video	O.a+	Video Streams supported

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Bit	Code	Feature	Conformance	Description
2	**SNP**	Snapshot	O.a+	Snapshot Streams supported
3	**PRIV**	Privacy	O	Privacy supported
4	**SPKR**	Speaker	[ADO]	Speaker supported
5	**ICTL**	ImageControl	[VDO | SNP]	Image control supported
6	**WMARK**	Watermark	[VDO | SNP]	Watermark supported
7	**OSD**	OnScreenDisplay	[VDO | SNP]	OSD supported
8	**STOR**	LocalStorage	O	Local Storage available
9	**HDR**	HighDynamicRange	[VDO | SNP]	High Dynamic Range supported
10	**NV**	NightVision	[VDO | SNP]	Night Vision mode supported

11.2.5.1. Audio Feature

The Audio feature indicates the ability of the node to support audio streams.

11.2.5.2. Video Feature

The Video feature indicates the ability of the node to support video streams. The video streams could be for either live streaming or recording stream transfer, or both.

11.2.5.3. Snapshot Feature

The Snapshot feature indicates the ability of the node to support snapshot streams.

11.2.5.4. Privacy Feature

The Privacy feature indicates the ability of the node to support privacy settings.

11.2.5.5. Speaker Feature

The Speaker feature indicates the ability of the node to support audio playback via a speaker. The Audio feature SHALL be supported if the Speaker feature is supported. Nodes which support this feature SHALL have the ability to perform playback audio mixing in software or hardware.

11.2.5.6. Image Control Feature

The Image Control feature indicates the ability of the node to support Image Control. The Video or Snapshot feature SHALL be supported if the Image Control feature is supported.

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11.2.5.7. Watermark Feature

The Watermark feature indicates the ability of the node to apply a manufacturer watermark logo on a video stream.

11.2.5.8. OnScreenDisplay Feature

The On Screen Display (OSD) feature indicates the ability of the node to display text such as date, time, timezone, and/or device name, etc. for a video stream.

11.2.5.9. LocalStorage Feature

The Local Storage feature indicates that this device has the ability to store recordings and/or snapshots on this device itself. While this specification defines the ability to have this feature and enable or disable it only, it does not currently define any way to access or manage this storage.

11.2.5.10. HighDynamicRange Feature

The High Dynamic Range feature indicates that the sensor on this device supports operating in High Dynamic Range mode, in addition to a normal operating mode.

11.2.5.11. NightVision Feature

The Night Vision feature indicates the ability to operate in a low light environment mode, in addition to a normal operating mode.

11.2.6. Data Types

11.2.6.1. VideoCodecEnum Type

This data type is derived from enum8.

This data type provides an enumeration of the video codecs supported by the camera.

Value	Name	Summary	Conformance
0	**H264**	Advanced Video Coding (H.264) codec.	M
1	**HEVC**	High efficiency Video Coding (H.265) codec.	O
2	**VVC**	Versatile Video Coding (H.266) codec.	O
3	**AV1**	AOMedia Video 1 codec.	O

11.2.6.1.1. H264 Value

Advanced Video Coding (AVC) standard, also referred to as MPEG-4 Part 10, is a widely used codec offering good balance of compression and quality.

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11.2.6.1.2. HEVC Value

High Efficiency Video Coding, also known as H.265, is a successor to H.264. It offers significantly better compression at the same quality level of its predecessors.

11.2.6.1.3. VVC Value

Versatile Video Coding, also known as H.266, is a successor to H.265. It builds upon the capabilities of its predecessors to achieve even more efficient compression.

11.2.6.1.4. AV1 Value

An open-source and royalty-free codec that provides high compression efficiency.

11.2.6.2. AudioCodecEnum Type

This data type is derived from enum8.

This data type provides an enumeration of the audio codecs supported by the camera.

Value	Name	Summary	Conformance
0	**OPUS**	Open source IETF standard codec.	M
1	**AAC-LC**	Advanced Audio Coding codec-Low Complexity	O

11.2.6.2.1. OPUS Value

An open-source and royalty-free audio codec that is versatile and designed for both speech and music.

11.2.6.2.2. AAC-LC Value

Proprietary audio codec providing high-quality audio at moderate bit rates that is used widely across different products. AAC-LC is the low-complexity profile of the audio format.

11.2.6.3. ImageCodecEnum Type

This data type is derived from enum8.

This data type provides an enumeration of the image codecs supported by the camera.

Value	Name	Summary	Conformance
0	**JPEG**	JPEG image codec.	M
1	**HEIC**	HEIC image codec.	O

11.2.6.3.1. JPEG Value

The JPEG image codec as defined by ITU-T T.81 and ISO/IEC 10918.

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11.2.6.3.2. HEIC Value

The HEIC image codec as defined in Annex B of ISO/IEC 23008-12 also known as the HEVC Image File Format. Support for HEIC is optional, but RECOMMENDED if the HEVC video codec is supported.

11.2.6.4. TwoWayTalkSupportTypeEnum Type

This data type is derived from enum8.

This data type provides an enumeration of the different modes of bi-directional audio communication that are supported by the camera.

Value	Name	Summary	Conformance
0	**NotSupported**	Two-way Talk support is absent.	M
1	**HalfDuplex**	Audio in one direction at a time.	M
2	**FullDuplex**	Audio in both directions simultaneously.	M

11.2.6.5. TriStateAutoEnum Type

This data type is derived from enum8 and is used for tri-state settings on a device, where a setting can be in one of three states, i.e., On, Off, or Automatic.

Value	Name	Summary	Conformance
0	**Off**	Off	M
1	**On**	On	M
2	**Auto**	Automatic Operation	M

11.2.6.6. VideoSensorParamsStruct Type

This struct is used to define a video sensor and its characteristics.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**SensorWidth**	uint16	min 64				M
1	**SensorHeight**	uint16	min 64				M
2	**MaxFPS**	uint16	min 1				M
3	**MaxHDRFPS**	uint16	1 to MaxFPS				HDR

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11.2.6.6.1. SensorWidth Field

This field SHALL indicate the practical width of the video sensor in pixels. This value is used for various purposes such as resolution control, boundaries for the Zone Management Cluster and digital Pan/Tilt/Zoom commands in the Camera AV Settings User Level Management.

11.2.6.6.2. SensorHeight Field

This field SHALL indicate the practical height of the video sensor in pixels. This value is used for various purposes such as resolution control, boundaries for the Zone Management Cluster and digital Pan/Tilt/Zoom commands in the Camera AV Settings User Level Management.

11.2.6.6.3. MaxFPS Field

This field SHALL indicate the maximum frame rate, in frames per second, that the video sensor is capable of supporting.

11.2.6.6.4. MaxHDRFPS Field

This field SHALL indicate the maximum frame rate, in frames per second, that the video sensor is capable of supporting when HDR is enabled. The value MAY be less than or equal to the MaxFPS.

11.2.6.7. VideoResolutionStruct Type

This object defines the resolution parameters in pixels which can be used for defining the resolutions of different video streams.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Width**	uint16	min 1				M
1	**Height**	uint16	min 1				M

11.2.6.7.1. Width Field

This field SHALL indicate the width, in number of pixels, for a frame.

11.2.6.7.2. Height Field

This field SHALL indicate the height, in number of pixels, for a frame.

11.2.6.8. RateDistortionTradeOffPointsStruct Type

This struct is used to define a set of parameters of the hardware video encoder that alter the rate distortion trade-off points. The points are expressed as the minimum bitrate and resolution for each supported codec type.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Codec**	VideoCodecEnum	all				M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
1	**Resolution**	VideoResolutionStruct					M
2	**MinBitRate**	uint32	min 1				M

11.2.6.8.1. Codec Field

This field SHALL indicate the type of video codec based on the supported VideoCodecEnum types.

11.2.6.8.2. Resolution Field

This field SHALL indicate the resolution in pixels for a specific rate distortion trade-off point.

11.2.6.8.3. MinBitRate Field

This field SHALL indicate the minimum bitrate for a specific rate distortion trade-off point expressed as bits per second.

11.2.6.9. SnapshotCapabilitiesStruct Type

This struct is used to define the set of parameters that characterize a snapshot image that is used to build a snapshot stream.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Resolution**	VideoResolutionStruct					M
1	**MaxFrameRate**	uint16	min 1				M
2	**ImageCodec**	ImageCodecEnum	all				M
3	**RequiresEncodedPixels**	bool	all				M
4	**RequiresHardwareEncoder**	bool	all				RequiresEncodedPixels == True

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11.2.6.9.1. Resolution Field

This field SHALL indicate the resolution in pixels of the snapshot image.

11.2.6.9.2. MaxFrameRate Field

This field SHALL indicate the maximum frame rate in frames per second of the snapshot stream.

11.2.6.9.3. ImageCodec Field

This field SHALL indicate the format of the snapshot image, e.g., JPEG, as specified in ImageCodecEnum.

11.2.6.9.4. RequiresEncodedPixels Field

This field SHALL indicate if this entry requires using any resources from the available MaxEncodedPixelRate. If true, clients need to include this entry's Resolution and MaxFrameRate in the calculation for determining overall stream allocation resources.

11.2.6.9.5. RequiresHardwareEncoder Field

This field SHALL indicate if this entry requires using a hardware encoder and thus needs to be considered when determining overall stream allocation resources.

If true, the device requires an encoder from MaxConcurrentEncoders for this combination of ImageCodec, Resolution, and MaxFrameRate.

If false, the device can produce this combination without needing a dedicated encoder.

This field is only considered if RequiresEncodedPixels is true.

11.2.6.10. AudioCapabilitiesStruct Type

This struct is used to express the audio capabilities of the camera.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x00	**MaxNumberOfChannels**	uint8	min 1				M
0x01	**SupportedCodecs**	list[AudioCodecEnum]	min 1				M
0x02	**SupportedSampleRates**	list[uint32]	min 1				M
0x03	**SupportedBitDepths**	list[uint8]	min 1				M

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11.2.6.10.1. MaxNumberOfChannels Field

This field SHALL indicate the maximum number of channels supported by an audio stream from the camera.

11.2.6.10.2. SupportedCodecs Field

This field SHALL indicate the list of audio codecs from AudioCodecEnum that are supported by the camera.

11.2.6.10.3. SupportedSampleRates Field

This field SHALL indicate the list of sample rates that are supported by the audio stream from the camera expressed in Hz, e.g., (48000, 32000, 16000).

11.2.6.10.4. SupportedBitDepths Field

This field SHALL indicate the list of bit depths that are supported by the audio stream, e.g., (16-bit, 24-bit).

11.2.6.11. VideoStreamStruct Type

This struct is used to capture all constituent parameters of a video stream in order to fully characterize it.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**VideoStreamID**	VideoStreamID	all				M
0x0001	**StreamUsage**	StreamUsageEnum	all				M
0x0002	**VideoCodec**	VideoCodecEnum	all				M
0x0003	**MinFrameRate**	uint16	1 to MaxFrameRate				M
0x0004	**MaxFrameRate**	uint16	min 1				M
0x0005	**MinResolution**	VideoResolutionStruct					M
0x0006	**MaxResolution**	VideoResolutionStruct					M
0x0007	**MinBitRate**	uint32	1 to MaxBitRate				M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0008	**MaxBitRate**	uint32	min 1				M
0x0009	**KeyFrameInterval**	uint16	0 to 65500				M
0x000A	**WatermarkEnabled**	bool	all		0		WMARK
0x000B	**OSDEnabled**	bool	all		0		OSD
0x000C	**ReferenceCount**	uint8	all		0		M

11.2.6.11.1. VideoStreamID Field

This field SHALL indicate the uniquely allocated identifier for the video stream.

11.2.6.11.2. StreamUsage Field

This field SHALL indicate the usage of the stream as described in StreamUsageEnum.

11.2.6.11.3. VideoCodec Field

This field SHALL indicate the type of video codec being used by the corresponding video stream as described in VideoCodecEnum.

11.2.6.11.4. MinFrameRate Field

This field SHALL indicate the minimum frame rate in frames per second for the corresponding video stream.

11.2.6.11.5. MaxFrameRate Field

This field SHALL indicate the maximum frame rate in frames per second for the corresponding video stream.

11.2.6.11.6. MinResolution Field

This field SHALL indicate the minimum resolution for the corresponding video stream.

11.2.6.11.7. MaxResolution Field

This field SHALL indicate the maximum resolution for the corresponding video stream.

11.2.6.11.8. MinBitRate Field

This field SHALL indicate the minimum bitrate for the corresponding video stream in bits per second.

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11.2.6.11.9. MaxBitRate Field

This field SHALL indicate the maximum bitrate for the corresponding video stream in bits per second.

11.2.6.11.10. KeyFrameInterval Field

This field SHALL indicate the duration in milliseconds before a regular key-frame SHALL be generated. A value of 0 SHALL mean that no regular key-frames are generated. When using push transports with a stream, it is RECOMMENDED to use a value of 4000 (4 seconds). If the value requested does not exactly align with the framerate, then the next frame after the requested value SHALL be a regular key-frame.

11.2.6.11.11. WatermarkEnabled Field

This field indicates the status of an applied watermark for the specific video stream. An Enabled value of TRUE means that watermarking has been enabled for that stream.

11.2.6.11.12. OSDEnabled Field

This field indicates the status of the OSD (On-Screen Display) for the specific video stream. An Enabled value of TRUE means that OSD has been enabled for that stream.

11.2.6.11.13. ReferenceCount Field

This field SHALL indicate the number of entities currently using this video stream. The node SHALL recompute this field to reflect the correct value at runtime (e.g., when restored from a persisted value after a reboot).

11.2.6.12. AudioStreamStruct Type

This struct is used to capture all constituent parameters of an audio stream in order to fully characterize it.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x00	**AudioStreamID**	AudioStreamID	all				M
0x01	**StreamUsage**	StreamUsageEnum	all				M
0x02	**AudioCodec**	AudioCodecEnum	all				M
0x03	**ChannelCount**	uint8	1 to 8				M
0x04	**SampleRate**	uint32	min 1				M
0x05	**BitRate**	uint32	min 1				M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x06	**BitDepth**	uint8	8, 16, 24, 32				M
0x07	**ReferenceCount**	uint8	all				M

11.2.6.12.1. AudioStreamID Field

This field SHALL indicate the uniquely allocated identifier for the audio stream.

11.2.6.12.2. StreamUsage Field

This field SHALL indicate the usage of stream as described in StreamUsageEnum.

11.2.6.12.3. AudioCodec Field

This field SHALL indicate the type of audio codec being used by the corresponding audio stream as described in AudioCodecEnum.

11.2.6.12.4. ChannelCount Field

This field SHALL indicate the number of independent channels or tracks being used by the corresponding audio stream. e.g., 1 for mono, 2 for stereo

11.2.6.12.5. SampleRate Field

This field SHALL indicate the audio sample rate, in hertz (Hz).

11.2.6.12.6. BitRate Field

This field SHALL indicate the target bit rate in bits per second of the audio stream.

11.2.6.12.7. BitDepth Field

This field SHALL indicate the bit depth (8, 16, 24 or 32 bits) of the audio stream. It represents the number of bits of information used to represent each sample of the audio signal, and affects the resolution and dynamic range of the audio.

11.2.6.12.8. ReferenceCount Field

This field SHALL indicate the number of entities currently using this audio stream. The node SHALL recompute this field to reflect the correct value at runtime (e.g., when restored from a persisted value after a reboot).

11.2.6.13. SnapshotStreamStruct Type

This struct is used to capture all constituent parameters of a snapshot stream in order to fully characterize it.

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x00	**SnapshotStreamID**	SnapshotStreamID	all				M
0x01	**ImageCodec**	ImageCodecEnum	all				M
0x02	**FrameRate**	uint16	min 1				M
0x03	**MinResolution**	VideoResolutionStruct					M
0x04	**MaxResolution**	VideoResolutionStruct					M
0x05	**Quality**	uint8	1 to 100				M
0x06	**ReferenceCount**	uint8	all				M
0x07	**EncodedPixels**	bool	all				M
0x08	**HardwareEncoder**	bool	all				M
0x09	**WatermarkEnabled**	bool	all				WMARK
0x0A	**OSDEnabled**	bool	all				OSD

11.2.6.13.1. SnapshotStreamID Field

This field SHALL indicate the uniquely allocated identifier for the snapshot stream.

11.2.6.13.2. ImageCodec Field

This field SHALL indicate the type of image codec being used by the corresponding snapshot stream as described in ImageCodecEnum.

11.2.6.13.3. FrameRate Field

This field SHALL indicate the frame rate as frames per second of the snapshot stream.

11.2.6.13.4. MinResolution Field

This field SHALL indicate the minimum resolution for the corresponding snapshot stream.

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11.2.6.13.5. MaxResolution Field

This field SHALL indicate the maximum resolution for the corresponding snapshot stream.

11.2.6.13.6. Quality Field

This field SHALL indicate a generic quality metric (integer between 1 and 100) as an input parameter to the image codec. A lower number indicates lower image quality. A higher value indicates higher image quality but larger file size and higher bit rate.

11.2.6.13.7. ReferenceCount Field

This field SHALL indicate the number of entities currently using this snapshot stream. The node SHALL recompute this field to reflect the correct value at runtime (e.g., when restored from a persisted value after a reboot).

11.2.6.13.8. EncodedPixels Field

This field SHALL indicate if this entry counts in the system encoded pixel rate calculation.

This SHALL be true if the SnapshotCapabilitiesStruct for the selected ImageCodec and MaxResolution has RequiresEncodedPixels set to true.

11.2.6.13.9. HardwareEncoder Field

This field SHALL indicate if one of the system hardware encoders is used by this snapshot stream.

This SHALL be true if the SnapshotCapabilitiesStruct for the selected ImageCodec and MaxResolution has RequiresHardwareEncoder set to true.

11.2.6.13.10. WatermarkEnabled Field

This field indicates the status of an applied watermark for the specific snapshot stream. A value of TRUE means that watermarking has been enabled for that stream.

11.2.6.13.11. OSDEnabled Field

This field indicates the status of the OSD (On-Screen Display) for the specific snapshot stream. A value of TRUE means that OSD has been enabled for that stream.

11.2.6.14. AVMetadataStruct Type

This struct is used to encode the metadata generated by the device and is included by the various transports that handle audio and video streams.

When encoded in TLV binary format and placed inside other standards, this SHALL be represented using the [RFC8141] compliant string `urn:csa:matter:av-metadata`.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x01	**UTCTime**	epoch-us	all	X			M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x02	**MotionZonesActive**	list[ZoneID]	min 1				O
0x03	**BlackAndWhiteActive**	bool	all				O
0x04	**UserDefined**	octstr	max 256				O

11.2.6.14.1. UTCTime Field

This field SHALL represent the UTC time that this metadata belongs to. The field is sourced from the Time Synchronization cluster's UTCTime attribute.

If null, the device has no current source of wall clock time.

11.2.6.14.2. MotionZonesActive Field

This field SHALL represent the list of Motion Zones that are currently triggered.

11.2.6.14.3. BlackAndWhiteActive Field

This field SHALL indicate if the sensor is currently active in black and white only mode. A value of true means the sensor and video encode process is operating in black and white only mode.

11.2.6.14.4. UserDefined Field

This field SHALL be an octet string representing arbitrary format user defined metadata attached. The UserDefined field of the ManuallyTriggerTransport command can be used to populate this field. The format and meaning of this field is not defined in this specification and is up to the users, vendors, or ecosystems deploying it.

11.2.6.15. VideoStreamID Type

This data type is derived from uint16 and represents an allocated video stream. This value starts at 0 and monotonically increases by 1 with each new allocation provisioned by the Node. A value incremented past 65534 SHALL wrap to 0. The Node SHALL verify that the incremented ID does not match any other ID. If such a match is found, the ID SHALL be incremented until a unique ID is found.

11.2.6.16. AudioStreamID Type

This data type is derived from uint16 and represents an allocated audio stream. This value starts at 0 and monotonically increases by 1 with each new allocation provisioned by the Node. A value incremented past 65534 SHALL wrap to 0. The Node SHALL verify that the incremented ID does not match any other ID. If such a match is found, the ID SHALL be incremented until a unique ID is found.

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11.2.6.17. SnapshotStreamID Type

This data type is derived from uint16 and represents an allocated snapshot stream. This value starts at 0 and monotonically increases by 1 with each new allocation provisioned by the Node. A value incremented past 65534 SHALL wrap to 0. The Node SHALL verify that the incremented ID does not match any other ID. If such a match is found, the ID SHALL be incremented until a unique ID is found.

11.2.7. Attributes

Table 19. Camera AV Stream Management Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**MaxConcurrentEncoders**	uint8	all	F		R V	VDO | SNP
0x0001	**MaxEncodedPixelRate**	uint32	all	F		R V	VDO | SNP
0x0002	**VideoSensorParams**	VideoSensorParamsStruct	all	F		R V	VDO
0x0003	**NightVisionUsesInfrared**	bool	all	F		R V	NV
0x0004	**MinViewportResolution**	VideoResolutionStruct	all	F		R V	VDO
0x0005	**RateDistortionTradeOffPoints**	list[RateDistortionTradeOffPointsStruct]	all	F		R V	VDO
0x0006	**MaxContentBufferSize**	uint32	all	F		R V	M
0x0007	**MicrophoneCapabilities**	AudioCapabilitiesStruct	all	F		R V	ADO
0x0008	**SpeakerCapabilities**	AudioCapabilitiesStruct	all	F		R V	SPKR

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0009	**TwoWayTalkSupport**	TwoWayTalkSupportTypeEnum	all	F		R V	SPKR
0x000A	**SnapshotCapabilities**	list[SnapshotCapabilitiesStruct]	all	F		R V	SNP
0x000B	**MaxNetworkBandwidth**	uint32	all	F		R V	M
0x000C	**CurrentFrameRate**	uint16	all			R V	VDO
0x000D	**HDRModeEnabled**	bool	all	N		RW M	HDR
0x000E	**SupportedStreamUsages**	list[StreamUsageEnum]	all	F		R V	M
0x000F	**AllocatedVideoStreams**	list[VideoStreamStruct]	all	N		R V	VDO
0x0010	**AllocatedAudioStreams**	list[AudioStreamStruct]	all	N		R V	ADO
0x0011	**AllocatedSnapshotStreams**	list[SnapshotStreamStruct]	all	N		R V	SNP
0x0012	**StreamUsagePriorities**	list[StreamUsageEnum]	all	N		R V	M
0x0013	**SoftRecordingPrivacyModeEnabled**	bool	all	N		RW VO	PRIV

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0014	**SoftLivestreamPrivacyModeEnabled**	bool	all	N		RW VO	PRIV
0x0015	**HardPrivacyModeOn**	bool	all		0	R V	O
0x0016	**NightVision**	TriStateAutoEnum	all	N		RW M	NV
0x0017	**NightVisionIllum**	TriStateAutoEnum	all	N		RW M	[NV]
0x0018	**Viewport**	ViewportStruct	all	N		RW M	VDO
0x0019	**SpeakerMuted**	bool	all	N		RW M	SPKR
0x001A	**SpeakerVolumeLevel**	uint8	SpeakerMinLevel to SpeakerMaxLevel	N		RW M	SPKR
0x001B	**SpeakerMaxLevel**	uint8	SpeakerMinLevel to 254			R M	SPKR
0x001C	**SpeakerMinLevel**	uint8	max SpeakerMaxLevel			R M	SPKR
0x001D	**MicrophoneMuted**	bool	all	N		RW M	ADO
0x001E	**MicrophoneVolumeLevel**	uint8	MicrophoneMinLevel to MicrophoneMaxLevel	N		RW M	ADO
0x001F	**MicrophoneMaxLevel**	uint8	MicrophoneMinLevel to 254			R M	ADO

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0020	**MicrophoneMinLevel**	uint8	max MicrophoneMaxLevel			R M	ADO
0x0021	**MicrophoneAGCEnabled**	bool	all	N	1	RW M	[ADO]
0x0022	**ImageRotation**	uint16	max 359	N		RW M	[ICTL].b+
0x0023	**ImageFlipHorizontal**	bool	all	N	0	RW M	[ICTL].b+
0x0024	**ImageFlipVertical**	bool	all	N	0	RW M	[ICTL].b+
0x0025	**LocalVideoRecordingEnabled**	bool	all	N		RW M	VDO & STOR
0x0026	**LocalSnapshotRecordingEnabled**	bool	all	N		RW M	SNP & STOR
0x0027	**StatusLightEnabled**	bool	all	N	1	RW M	O
0x0028	**StatusLightBrightness**	ThreeLevelAutoEnum	all	N		RW M	O
0x0029	**ImageRotationDiscreteAngles**	uint16	0, 90, 180, 270	N	0	RW M	P, [(Rev >= v2) & ICTL & (!ImageRotation)].b+

11.2.7.1. MaxConcurrentEncoders Attribute

This attribute SHALL indicate the maximum number of concurrent encoders supported by the camera.

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11.2.7.2. MaxEncodedPixelRate Attribute

This attribute SHALL indicate the maximum data rate in encoded pixels per second that the camera can produce given the hardware encoder resources it has. This value is manufacturer specified.

If the camera supports Snapshots and requires hardware encoder resources to produce those Snapshots, then this attribute SHALL be present, and a manufacturer specific value SHALL be present in each SnapshotCapabilities MaxFrameRate entry that requires hardware resources to produce.

11.2.7.3. VideoSensorParams Attribute

This attribute SHALL indicate the set of video sensor parameters for the camera. These include the video sensor dimensions, its frame rate and HDR capabilities.

11.2.7.4. NightVisionUsesInfrared Attribute

This attribute SHALL indicate if the night vision mode is infrared based or not. A value of TRUE indicates infrared mode with a cut filter being used. A value of FALSE indicates color is used. When infrared is active and the resulting content is in black and white, the BlackAndWhiteActive field inside any produced AVMetadataStruct SHALL be TRUE.

11.2.7.5. MinViewportResolution Attribute

This attribute SHALL indicate the minimum resolution (width and height) in pixels that the camera allows for its viewport.

The choice of the minimum viewport width and height is, typically, directed towards maintaining the best image quality (reduced distortion) for a given size of the video sensor, for different camera functions, e.g., digital zoom. Furthermore, the minimum viewport size and the video sensor size also dictate the upscaling capabilities and requirements of the image processor.

11.2.7.6. RateDistortionTradeOffPoints Attribute

This attribute SHALL list the set of rate distortion trade-off points between resolution, frame rate and bitrate for each supported hardware encoder.

11.2.7.7. MaxContentBufferSize Attribute

This attribute SHALL indicate the maximum size of the content buffer in bytes. This buffer holds the compressed and/or raw content for audio/video pre-roll, queued transmissions, the current frame for each snapshot stream, and the metadata context for recording events. For devices which support more than one encoder, the device SHALL evenly allocate this buffer space amongst all streams that utilize pre-roll content such as the Push AV Stream Transport Cluster.

11.2.7.8. MicrophoneCapabilities Attribute

This attribute SHALL indicate the audio capabilities of the microphone in terms of the codec used, supported sample rates and the number of channels.

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11.2.7.9. SpeakerCapabilities Attribute

This attribute SHALL indicate the audio capabilities of the speaker in terms of the supported codecs, sample rates, and the number of channels when a speaker is present.

11.2.7.10. TwoWayTalkSupport Attribute

This attribute SHALL indicate the type of two-way talk support the device has, e.g., NotSupported, HalfDuplex, or FullDuplex.

11.2.7.11. SnapshotCapabilities Attribute

This attribute SHALL indicate the list of supported snapshot capabilities the device has. This list is a set of entries for image codec, resolution, maximum frame rate, hardware encoder, and encoded pixels.

11.2.7.12. MaxNetworkBandwidth Attribute

This attribute SHALL indicate the maximum network bandwidth in bits per second that the device would consume for the transmission of its media streams.

11.2.7.13. CurrentFrameRate Attribute

This attribute SHALL indicate the current logical frame rate of the sensor in frames per second.

11.2.7.14. HDRModeEnabled Attribute

This attribute indicates the currently selected High Dynamic Range (HDR) mode. A value of TRUE indicates that HDR video capturing is enabled. Otherwise, HDR video capturing is disabled.

11.2.7.15. SupportedStreamUsages Attribute

This attribute SHALL indicate the list of Stream Usages that are supported by the camera. Manufacturers SHALL provide a usages list that is appropriate to their product. If a usage is found in this list, then it can be used in the StreamUsagePriorities attribute. The ordering and values of this list SHALL match the values found in StreamUsagePriorities after a factory reset.

11.2.7.16. AllocatedVideoStreams Attribute

This attribute SHALL indicate the list of allocated video streams on the device.

11.2.7.17. AllocatedAudioStreams Attribute

This attribute SHALL indicate the list of allocated audio streams on the device.

11.2.7.18. AllocatedSnapshotStreams Attribute

This attribute SHALL indicate the list of allocated snapshot streams on the device.

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11.2.7.19. StreamUsagePriorities Attribute

This attribute SHALL indicate a list of the video stream usages represented in a ranked order of their priorities, starting with Index 0 having the stream usage type with the highest priority. See Resource Management and Stream Priorities for further details. Only usages found in the SupportedStreamUsages attribute can be included. To change the contents, use the SetStreamPriorities command. Manufacturers SHALL provide a default ranked priorities list that is appropriate to their product and this default ranking SHALL exactly match the contents of SupportedStreamUsages. Clients can use the contents of the SupportedStreamUsages to restore this default state if the contents have been changed by the SetStreamPriorities command.

11.2.7.20. SoftRecordingPrivacyModeEnabled Attribute

This attribute indicates the current value of the soft privacy mode for transports using the Stream Usage types Recording and Analysis. A value of TRUE indicates that delivery of video frames and audio samples from any streams to these transports is skipped. A value of TRUE also indicates that no new transports using these stream usage values can be created or started. When FALSE, these transports can be resumed or started, and have video frames and audio samples delivered.

When this attribute is set to TRUE, any active WebRTC transports using these stream usage types SHALL terminate the session by calling End using WebRTCEndReasonEnum PrivacyMode.

11.2.7.21. SoftLivestreamPrivacyModeEnabled Attribute

This attribute indicates the current value of the soft privacy mode for transports using the Stream Usage type LiveView. A value of TRUE indicates that delivery of video frames and audio samples from any streams to these transports is skipped. A value of TRUE also indicates that no new transports using this stream usage type can be created or started. When FALSE, these transports can be resumed or started, and have video frames and audio samples delivered.

When this attribute is set to TRUE, any active WebRTC transports using this stream usage type SHALL terminate the session by calling End using WebRTCEndReasonEnum PrivacyMode.

11.2.7.22. HardPrivacyModeOn Attribute

This attribute indicates the current value of the hard privacy mode for all streams. This is controlled via a physical button or switch, potentially. A value of TRUE indicates that all streams are currently paused. When FALSE, the streams MAY resume if they are not already paused by their corresponding soft privacy mode.

11.2.7.23. NightVision Attribute

This attribute indicates the currently selected Night Vision mode. A value of Off means the device will never activate its Night Vision mode of operation. A value of On means the Night Vision mode of operation is always active. A value of Auto means the device will automatically move between active and inactive based on the light level it detects.

11.2.7.24. NightVisionIllum Attribute

This attribute indicates the currently selected the Night Vision Illumination mode. A value of Off

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means the device will never activate its built-in Night Vision Illumination. A value of On means the built-in Night Vision Illumination is always active. A value of Auto means the device will automatically enable its built-in Night Vision Illumination based on the light level it detects.

11.2.7.25. Viewport Attribute

This attribute SHALL be a ViewportStruct representing the viewport to apply to all streams.

The coordinate values represent the upper left corner and lower right corner coordinates of the source rectangle on the sensor. The coordinate values are within the two-dimensional Cartesian plane of size `SensorWidth` and `SensorHeight` (See VideoSensorParamsStruct in the VideoSensorParams) with the origin (0,0) being the upper left corner, positive X and Y values moving right and down across the Cartesian plane respectively, and (SensorWidth, SensorHeight) being the lower right corner.

When changing the Viewport, the aspect ratio of the sensor as indicated in the VideoSensorParams attribute SHOULD be preserved.

After a factory reset, this SHALL default to {0, 0, SensorWidth,SensorHeight}, using the SensorWidth and SensorHeight fields from the VideoSensorParams attribute.

When this attribute is changed, all Viewport values found in DPTZStreams SHALL be updated to the new values set here.

11.2.7.25.1. Example

The following figure shows an example of a 1920x1080 viewport on a 2592x1944 (i.e. 5MP) sensor.

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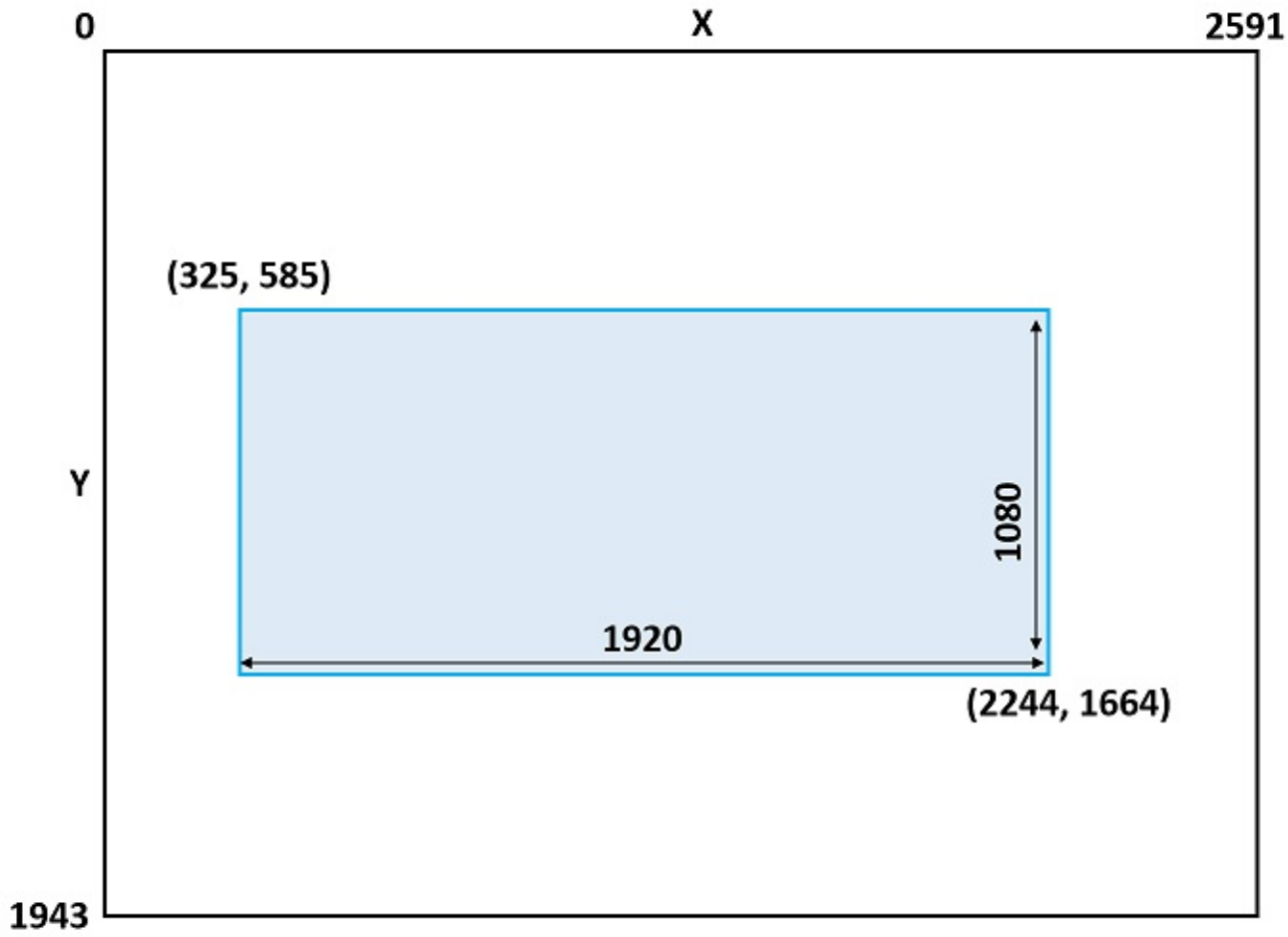


Figure 34. Example of a 1920x1080 viewport (colored in blue) on a 5MP sensor.

11.2.7.26. SpeakerMuted Attribute

This attribute indicates whether the speaker is currently muted or not. A value of TRUE indicates that the speaker has been muted and SHALL not play anything. A value of FALSE indicates that the Speaker is enabled.

11.2.7.27. SpeakerVolumeLevel Attribute

This attribute indicates the current volume level of the speaker.

11.2.7.28. SpeakerMaxLevel Attribute

This attribute indicates the maximum value of the SpeakerVolumeLevel that can be assigned.

11.2.7.29. SpeakerMinLevel Attribute

This attribute indicates the minimum value of the SpeakerVolumeLevel that can be assigned.

11.2.7.30. MicrophoneMuted Attribute

This attribute indicates whether the microphone is currently muted or not. A value of TRUE indicates that the microphone has been muted. In this state, the microphone data SHALL be replaced with all 0 bits, representing silence. A value of FALSE indicates that the microphone is On and is capable of transmitting audio.

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11.2.7.31. MicrophoneVolumeLevel Attribute

This attribute indicates the current gain or volume level of the microphone.

11.2.7.32. MicrophoneMaxLevel Attribute

This attribute indicates the maximum value of the MicrophoneVolumeLevel that can be assigned.

11.2.7.33. MicrophoneMinLevel Attribute

This attribute indicates the minimum value of the MicrophoneVolumeLevel that can be assigned.

11.2.7.34. MicrophoneAGCEnabled Attribute

This attribute indicates the currently selected AGC (Automatic Gain Control) mode for the microphone. A value of TRUE indicates that microphone AGC is enabled. Otherwise, it is disabled.

11.2.7.35. ImageRotation Attribute

This attribute indicates the amount of clockwise rotation in degrees that the image has been subjected to.

This attribute MAY be present if the underlying hardware allows for arbitrary angle rotation within the full 360 degree range. If this attribute is not present, then discrete angle rotation MAY be supported via the ImageRotationDiscreteAngles. A value of 0 means no rotation has been applied.

11.2.7.36. ImageFlipHorizontal Attribute

This attribute indicates whether the image has been flipped horizontally or not. A value of TRUE indicates that the image has been flipped horizontally.

11.2.7.37. ImageFlipVertical Attribute

This attribute indicates whether the image has been flipped vertically or not. A value of TRUE indicates that the image has been flipped vertically.

11.2.7.38. LocalVideoRecordingEnabled Attribute

This attribute indicates whether local storage based video recording is enabled. A value of TRUE indicates that local storage based video recording has been enabled.

11.2.7.39. LocalSnapshotRecordingEnabled Attribute

This attribute indicates whether local storage based snapshot recording is enabled. A value of TRUE indicates that local storage based snapshot recording has been enabled.

11.2.7.40. StatusLightEnabled Attribute

This attribute indicates whether the status light has been enabled or not. A value of TRUE indicates the status light has been enabled. When enabled, the camera MAY use it for visual signaling purposes to indicate various states of the camera.

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11.2.7.41. StatusLightBrightness Attribute

This attribute indicates the brightness level of the status light.

11.2.7.42. ImageRotationDiscreteAngles Attribute

This attribute indicates the amount of clockwise rotation in specific angles of 0, 90, 180, and 270 degrees. This attribute is used by cameras that do not have the capability of full arbitrary rotation angles that can be set via the ImageRotation attribute. Note that the ImageFlipHorizontal and the ImageFlipVertical attributes only allow for a lateral inversion or mirror reflection of the image along the horizontal and/or vertical axes. A value of 0 means no rotation has been applied.

11.2.8. Commands

ID	Name	Direction	Response	Access	Quality	Conformance
0x00	**AudioStreamAllocate**	client ⇒ server	AudioStreamAllocateResponse	M		ADO
0x01	**AudioStreamAllocateResponse**	server ⇒ client	N			ADO
0x02	**AudioStreamDeallocate**	client ⇒ server	Y	M		ADO
0x03	**VideoStreamAllocate**	client ⇒ server	VideoStreamAllocateResponse	M		VDO
0x04	**VideoStreamAllocateResponse**	server ⇒ client	N			VDO
0x05	**VideoStreamModify**	client ⇒ server	Y	M		VDO & (WMARK | OSD)
0x06	**VideoStreamDeallocate**	client ⇒ server	Y	M		VDO
0x07	**SnapshotStreamAllocate**	client ⇒ server	SnapshotStreamAllocateResponse	M		SNP

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ID	Name	Direction	Response	Access	Quality	Conformance
0x08	**SnapshotStreamAllocateResponse**	server ⇒ client	N			SNP
0x09	**SnapshotStreamModify**	client ⇒ server	Y	M		SNP & (WMARK | OSD)
0x0A	**SnapshotStreamDeallocate**	client ⇒ server	Y	M		SNP
0x0B	**SetStreamPriorities**	client ⇒ server	Y	A		M
0x0C	**CaptureSnapshot**	client ⇒ server	CaptureSnapshotResponse	O	L	SNP
0x0D	**CaptureSnapshotResponse**	server ⇒ client	N		L	SNP

11.2.8.1. AudioStreamAllocate Command

This command SHALL allocate an audio stream on the camera and return an allocated audio stream identifier.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0x00	**StreamUsage**	StreamUsageEnum	Recording, Analysis, LiveView			M
0x01	**AudioCodec**	AudioCodecEnum	all			M
0x02	**ChannelCount**	uint8	1 to 8			M
0x03	**SampleRate**	uint32	min 1			M
0x04	**BitRate**	uint32	min 1			M
0x05	**BitDepth**	uint8	8, 16, 24, 32			M

11.2.8.1.1. StreamUsage Field

This field SHALL indicate the usage of the stream (Recording, LiveView, etc) that this allocation is for.

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11.2.8.1.2. AudioCodec Field

This field SHALL indicate the type of Codec used by the allocated stream.

11.2.8.1.3. ChannelCount Field

This field SHALL indicate the number of channels used by the allocated stream, e.g., Mono (1), Stereo (2), etc.

11.2.8.1.4. SampleRate Field

This field SHALL indicate the sampling rate of the audio stream in Hz. Typical values would be 48000 (48 kHz), 32000 (32 kHz) or 16000 (16 kHz).

11.2.8.1.5. BitRate Field

This field SHALL indicate the bitrate of the specified audio codec in bits per second. The default bitrate MAY vary based on the type of device and codec.

11.2.8.1.6. BitDepth Field

This field SHALL indicate the number of bits of information (8, 16, 24 or 32 bits) used to represent each sample of the audio signal.

11.2.8.1.7. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If the requested StreamUsage is not found in the StreamUsagePriorities attribute:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
- If an unsupported AudioCodec, ChannelCount, SampleRate, BitRate, or BitDepth is requested:
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- If there is an existing stream in AllocatedAudioStreams that matches this request:
  - Return the existing AudioStreamID in AudioStreamAllocateResponse, and end processing with no other side effects.
- If there are not enough resources to allocate a new audio stream:
  - Fail the command with the status code RESOURCE_EXHAUSTED, and end processing with no other side effects.
- Allocate a new AudioStreamID
- Create and store a new AudioStreamStruct for the allocated stream mapped against the new AudioStreamID.
- Append the new AudioStreamStruct to the AllocatedAudioStreams.
- Return the newly allocated AudioStreamID in AudioStreamAllocateResponse.

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11.2.8.2. AudioStreamAllocateResponse Command

This command SHALL be sent by the camera in response to the AudioStreamAllocate command, carrying the newly allocated or re-used audio stream identifier.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**AudioStreamID**	AudioStreamID	all			M

11.2.8.2.1. AudioStreamID Field

The AudioStreamID field SHALL be a AudioStreamID representing the unique audio stream identifier.

11.2.8.3. AudioStreamDeallocate Command

This command SHALL deallocate an audio stream on the camera, corresponding to the given audio stream identifier.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**AudioStreamID**	AudioStreamID	all			M

11.2.8.3.1. AudioStreamID Field

The AudioStreamID field SHALL be a AudioStreamID representing the unique audio stream identifier for the stream that needs to be deallocated.

11.2.8.3.2. Effect on Receipt

The process below SHALL be followed by the camera when receiving this command:

- If the AudioStreamID does not match an entry in AllocatedAudioStreams:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- If the reference count for the stream with AudioStreamID is greater than 0:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
- If the StreamUsage for the stream with AudioStreamID is Internal:
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- Deallocate the audio stream with AudioStreamID.
- Remove the entry for AudioStreamID from AllocatedAudioStreams.
- Respond with the status code SUCCESS.

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11.2.8.4. VideoStreamAllocate Command

This command SHALL allocate a video stream on the camera and return an allocated video stream identifier.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0x00	**StreamUsage**	StreamUsageEnum	Recording, Analysis, LiveView			M
0x01	**VideoCodec**	VideoCodecEnum	all			M
0x02	**MinFrameRate**	uint16	1 to MaxFrameRate			M
0x03	**MaxFrameRate**	uint16	min 1			M
0x04	**MinResolution**	VideoResolutionStruct				M
0x05	**MaxResolution**	VideoResolutionStruct				M
0x06	**MinBitRate**	uint32	1 to MaxBitRate			M
0x07	**MaxBitRate**	uint32	min 1			M
0x08	**KeyFrameInterval**	uint16	0 to 65500			M
0x09	**WatermarkEnabled**	bool	all			WMARK
0x0A	**OSDEnabled**	bool	all			OSD

11.2.8.4.1. StreamUsage Field

This field SHALL indicate the usage of the stream (Recording, LiveView, etc) that this allocation is for.

11.2.8.4.2. VideoCodec Field

This field SHALL indicate the type of codec used by the allocated video stream.

11.2.8.4.3. MinFrameRate Field

This field SHALL indicate the minimum frame rate in frames per second of the allocated video stream.

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11.2.8.4.4. MaxFrameRate Field

This field SHALL indicate the maximum frame rate in frames per second of the allocated video stream.

11.2.8.4.5. MinResolution Field

This field SHALL indicate the minimum resolution of the allocated video stream.

11.2.8.4.6. MaxResolution Field

This field SHALL indicate the maximum resolution of the allocated video stream.

11.2.8.4.7. MinBitRate Field

This field SHALL indicate the minimum bitrate in bits per second of the allocated video stream.

11.2.8.4.8. MaxBitRate Field

This field SHALL indicate the maximum bitrate in bits per second of the allocated video stream.

11.2.8.4.9. KeyFrameInterval Field

This field SHALL indicate the duration in milliseconds before a regular key-frame is generated. See KeyFrameInterval for further details.

11.2.8.4.10. WatermarkEnabled Field

This field SHALL indicate whether a watermark can be applied on the video stream. A value of TRUE means that watermarking has been enabled for that stream.

11.2.8.4.11. OSDEnabled Field

This field SHALL indicate whether the OSD (On-Screen Display) can be applied on the video stream. A value of TRUE means that OSD has been enabled for that stream.

11.2.8.4.12. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If the requested StreamUsage is not found in the StreamUsagePriorities:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
- If an unsupported VideoCodec, MinFrameRate, MaxFrameRate, MinResolution, MaxResolution, MinBitRate, or MaxBitRate is requested:
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- If there is an existing stream in AllocatedVideoStreams that matches this request:
  - Return the existing VideoStreamID in VideoStreamAllocateResponse, and end processing with no other side effects.

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- If there are not enough resources to allocate a new video stream:
  - Fail the command with the status code RESOURCE_EXHAUSTED, and end processing with no other side effects.
- Allocate a new VideoStreamID
- Create and store a new VideoStreamStruct for the allocated stream mapped against the new VideoStreamID.
- Append the new VideoStreamStruct to the AllocatedVideoStreams.
- If DigitalPTZ is supported in Camera AV Settings User Level Management:
  - Add a new DPTZStruct entry in DPTZStreams using the new VideoStreamID and current value of Viewport.
- Return the newly allocated VideoStreamID in VideoStreamAllocateResponse.

11.2.8.5. VideoStreamAllocateResponse Command

This command SHALL be sent by the camera in response to the VideoStreamAllocate command, carrying the newly allocated or re-used video stream identifier.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**VideoStreamID**	VideoStreamID	all			M

11.2.8.5.1. VideoStreamID Field

This field SHALL be a VideoStreamID representing the newly created unique video stream identifier.

11.2.8.6. VideoStreamModify Command

This command SHALL be used to modify a stream specified by the VideoStreamID.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**VideoStreamID**	VideoStreamID	all			M
1	**WatermarkEnabled**	bool	all			[WMARK].d+
2	**OSDEnabled**	bool	all			[OSD].d+

11.2.8.6.1. VideoStreamID Field

This field SHALL be a VideoStreamID representing the video stream to modify.

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11.2.8.6.2. WatermarkEnabled Field

This field SHALL indicate whether a watermark can be applied on the video stream. A value of TRUE means that watermarking has been requested for that stream. If this field is not present, then no change to the existing value SHALL be made.

11.2.8.6.3. OSDEnabled Field

This field SHALL indicate whether the OSD (On-Screen Display) can be applied on the video stream. A value of TRUE means that OSD has been requested for that stream. If this field is not present, then no change to the existing value SHALL be made.

11.2.8.6.4. Effect on Receipt

The process below SHALL be followed by the camera when receiving this command:

- If the VideoStreamID does not match an entry in AllocatedVideoStreams:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- If the requested change is unable to be made at this time:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
- If the StreamUsage for the stream with VideoStreamID is Internal:
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- Modify the fields of the video stream corresponding to the included VideoStreamID.
- Respond with the status code SUCCESS.

11.2.8.7. VideoStreamDeallocate Command

This command SHALL deallocate a video stream on the camera, corresponding to the given video stream identifier.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**VideoStreamID**	VideoStreamID	all			M

11.2.8.7.1. VideoStreamID Field

This field SHALL be a VideoStreamID for the stream to be deallocated.

11.2.8.7.2. Effect on Receipt

The process below SHALL be followed by the camera when receiving this command:

- If the VideoStreamID does not match an entry in AllocatedVideoStreams:

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- Fail the command with the status code NOT_FOUND, and end processing with no other side effects.

- If the reference count for the stream with VideoStreamID is greater than 0:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
- If the StreamUsage for the stream with VideoStreamID is Internal:
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- If DigitalPTZ is supported in Camera AV Settings User Level Management:
  - Remove the entry for VideoStreamID from DPTZStreams.
- Deallocate the video stream with VideoStreamID.
- Remove the entry for VideoStreamID from AllocatedVideoStreams.
- Respond with the status code SUCCESS.

11.2.8.8. SnapshotStreamAllocate Command

This command SHALL allocate a snapshot stream on the device and return an allocated snapshot stream identifier.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0x00	**ImageCodec**	ImageCodecEnum	all			M
0x01	**MaxFrameRate**	uint16	min 1			M
0x02	**MinResolution**	VideoResolutionStruct				M
0x03	**MaxResolution**	VideoResolutionStruct				M
0x04	**Quality**	uint8	1 to 100			M
0x05	**WatermarkEnabled**	bool	all			WMARK
0x06	**OSDEnabled**	bool	all			OSD

11.2.8.8.1. ImageCodec Field

This field SHALL indicate the type of image codec to be used by the allocated snapshot stream as described in ImageCodecEnum.

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11.2.8.8.2. MaxFrameRate Field

This field SHALL indicate the maximum frame rate in frames per second of the allocated snapshot stream.

11.2.8.8.3. MinResolution Field

This field SHALL indicate the minimum resolution of the allocated snapshot stream.

11.2.8.8.4. MaxResolution Field

This field SHALL indicate the maximum resolution of the allocated snapshot stream.

11.2.8.8.5. Quality Field

This field SHALL indicate a codec quality metric(integer between 1 and 100) for the allocated snapshot stream.

11.2.8.8.6. WatermarkEnabled Field

This field SHALL indicate whether a watermark can be applied on the snapshot stream. A value of TRUE means that watermarking has been requested for that stream.

This field MAY be ignored if RequiresHardwareEncoder is FALSE on the matching entry from SnapshotCapabilities. When ignored, the associated WatermarkEnabled entry for the Video Stream that is the source of this snapshot MAY be used.

11.2.8.8.7. OSDEnabled Field

This field SHALL indicate whether the OSD (On-Screen Display) can be applied on the snapshot stream. A value of TRUE means that OSD has been requested for that stream.

This field MAY be ignored if RequiresHardwareEncoder is FALSE on the matching entry from SnapshotCapabilities. When ignored, the associated OSDEnabled entry for the Video Stream that is the source of this snapshot MAY be used.

11.2.8.8.8. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If an unsupported ImageCodec, MaxFrameRate, MinResolution, MaxResolution, or Quality is requested:
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- If there is an existing stream in AllocatedSnapshotStreams that matches this request:
  - Return the existing SnapshotStreamID in SnapshotStreamAllocateResponse, and end processing with no other side effects.
- If there are not enough resources to allocate a new snapshot stream:
  - Fail the command with the status code RESOURCE_EXHAUSTED, and end processing with no other side effects.

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- Allocate a new SnapshotStreamID
- Create and store a new SnapshotStreamStruct for the allocated stream mapped against the new SnapshotStreamID.
- Append the new SnapshotStreamStruct to the AllocatedSnapshotStreams.
- Return the newly allocated SnapshotStreamID in SnapshotStreamAllocateResponse.

11.2.8.9. SnapshotStreamAllocateResponse Command

This command SHALL be sent by the device in response to the SnapshotStreamAllocate command, carrying the newly allocated or re-used snapshot stream identifier.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**SnapshotStreamID**	SnapshotStreamID	all			M

11.2.8.9.1. SnapshotStreamID Field

The SnapshotStreamID field SHALL be an unsigned 16 bit integer representing the unique snapshot stream identifier.

11.2.8.10. SnapshotStreamDeallocate Command

This command SHALL deallocate an snapshot stream on the camera, corresponding to the given snapshot stream identifier.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**SnapshotStreamID**	SnapshotStreamID	all			M

11.2.8.10.1. SnapshotStreamID Field

The SnapshotStreamID field SHALL be an unsigned 16 bit integer representing the unique snapshot stream identifier for the stream that needs to be deallocated.

11.2.8.10.2. Effect on Receipt

The process below SHALL be followed by the camera when receiving this command:

- If the SnapshotStreamID does not match an entry in AllocatedSnapshotStreams:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- If the reference count for the stream with SnapshotStreamID is greater than 0:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.

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- Deallocate the snapshot stream with SnapshotStreamID.
- Remove the entry for SnapshotStreamID from AllocatedSnapshotStreams.
- Respond with the status code SUCCESS.

11.2.8.11. SnapshotStreamModify Command

This command SHALL be used to modify a stream specified by the VideoStreamID.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**SnapshotStreamID**	SnapshotStreamID	all			M
1	**WatermarkEnabled**	bool	all			[WMARK].c+
2	**OSDEnabled**	bool	all			[OSD].c+

11.2.8.11.1. SnapshotStreamID Field

This field SHALL be a SnapshotStreamID representing the snapshot stream to modify.

11.2.8.11.2. WatermarkEnabled Field

This field SHALL indicate whether a watermark can be applied on the video stream. A value of TRUE means that watermarking has been requested for that stream.

If this field is not present, then no change to the existing value SHALL be made.

11.2.8.11.3. OSDEnabled Field

This field SHALL indicate whether the OSD (On-Screen Display) can be applied on the video stream. A value of TRUE means that OSD has been requested for that stream.

If this field is not present, then no change to the existing value SHALL be made.

11.2.8.11.4. Effect on Receipt

The process below SHALL be followed by the camera when receiving this command:

- If the SnapshotStreamID does not match an entry in AllocatedSnapshotStreams:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- If the RequiresHardwareEncoder is FALSE on the matching entry from SnapshotCapabilities
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- If the requested change is unable to be made at this time:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no

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other side effects.

- Modify the fields of the snapshot stream corresponding to the included SnapshotStreamID.
- Respond with the status code SUCCESS.

11.2.8.12. SetStreamPriorities Command

This command SHALL set the relative priorities of the various stream usages on the camera. The camera then bases its allocation of resources for each stream allocation based on the order of these stream priorities. In order to avoid the complexity of dynamically changing the configurations of currently active streams, this command SHALL NOT be invoked when there are allocated streams. If changes are required while streams are allocated, all existing streams would need to be deallocated before invoking this command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**StreamPriorities**	list[StreamUsageEnum]	desc			M

11.2.8.12.1. StreamPriorities Field

The StreamPriorities field SHALL contain a list of stream usages in decreasing order of stream priorities, starting at index 0 of the list, with no duplicate values allowed, and only containing entries found in SupportedStreamUsages.

11.2.8.12.2. Effect on Receipt

The process below SHALL be followed by the camera when receiving this command:

- If there are any entries in AllocatedSnapshotStreams, AllocatedVideoStreams, or AllocatedAudioStreams:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
- If any requested value in StreamPriorities is not found in SupportedStreamUsages:
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- If there are duplicate stream usages in StreamPriorities field:
  - Fail the command with the status code ALREADY_EXISTS, and end processing with no other side effects.
- Update the StreamUsagePriorities attribute with the contents of StreamPriorities.

11.2.8.13. CaptureSnapshot Command

This command SHALL return a Snapshot from the camera.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**SnapshotStreamID**	SnapshotStreamID	all	X		M
1	**RequestedResolution**	VideoResolutionStruct				M

11.2.8.13.1. SnapshotStreamID Field

The SnapshotStreamID field SHALL be a SnapshotStreamID representing the allocated Snapshot Stream to use, or null to allow automatic stream selection.

11.2.8.13.2. RequestedResolution Field

The RequestedResolution field SHALL be a VideoResolutionStruct representing the preferred resolution of the snapshot.

11.2.8.13.3. Effect on Receipt

The process below SHALL be followed by the camera when receiving this command:

- If there are no entries in AllocatedSnapshotStreams:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- If the SnapshotStreamID is not null, and does not match an entry in AllocatedSnapshotStreams:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- If either SoftLivestreamPrivacyModeEnabled or HardPrivacyModeOn is true:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
- If the SnapshotStreamID is null:
  - Use the closest existing snapshot stream that is greater than or equal to the RequestedResolution size.
  - If there are no allocated streams greater than or equal to the RequestedResolution:
    - Use the stream with the closest possible resolution.
- If the requested size is smaller than the stream's native size, and there are available resources to downscale:
  - Downscale the stream's snapshot to the RequestedResolution size.
- Return the snapshot in the corresponding CaptureSnapshotResponse command.

11.2.8.14. CaptureSnapshotResponse Command

This command SHALL be sent by the device in response to the CaptureSnapshot command, carrying the requested snapshot.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Data**	octstr				M
1	**ImageCodec**	ImageCodecEnum	all			M
2	**Resolution**	VideoResolutionStruct				M

11.2.8.14.1. Data Field

The Data field SHALL be an octet string representing the image data.

11.2.8.14.2. ImageCodec Field

The ImageCodec field SHALL be an ImageCodecEnum representing the codec used to encode the image. This SHALL match the value found in the corresponding allocated snapshot stream and is provided for convenience.

11.2.8.14.3. Resolution Field

The Resolution field SHALL be a VideoResolutionStruct representing the resolution of the supplied image.

11.3. Camera AV Settings User Level Management

This cluster provides an interface into controls associated with the operation of a camera that provides pan, tilt, and zoom functions, either mechanically, or against a digital image.

11.3.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial release

11.3.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	AVSUM

11.3.3. Cluster ID

ID	Name
0x0552	Camera AV Settings User Level Management

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11.3.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Table 20. Camera AV Settings User Level Management Features

Bit	Code	Feature	Conformance	Summary
0	**DPTZ**	DigitalPTZ	O.a+	Digital PTZ support
1	**MPAN**	MechanicalPan	O.a+	Mechanical Pan support
2	**MTILT**	MechanicalTilt	O.a+	Mechanical Tilt support
3	**MZOOM**	MechanicalZoom	O.a+	Mechanical Zoom support
4	**MPRESETS**	MechanicalPresets	[MPAN | MTILT | MZOOM]	Mechanical saved presets support

11.3.4.1. DigitalPTZ Feature

This feature indicates that per video stream digital pan, tilt, and zoom is supported.

11.3.4.2. MechanicalPan Feature

This feature indicates that mechanical pan is supported on the camera.

11.3.4.3. MechanicalTilt Feature

This feature indicates that mechanical tilt is supported on the camera.

11.3.4.4. MechanicalZoom Feature

This feature indicates that mechanical zoom is supported on the camera.

11.3.4.5. MechanicalPresets Feature

This feature indicates that the storage of presets is supported on the camera.

11.3.5. Data Types

11.3.5.1. PhysicalMovementEnum Type

This data type is derived from enum8.

The PhysicalMovementEnum provides an enumeration of the possible physical movement states in which the camera could be.

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Value	Name	Summary	Conformance
0	**Idle**	The camera is idle.	M
1	**Moving**	The camera is moving to a new value of Pan, Tilt, and/or Zoom.	M

11.3.5.1.1. Idle Value

The camera is idle, there is no movement active.

11.3.5.1.2. Moving Value

The camera is moving to a new value of Pan, Tilt, and/or Zoom as a result of the reception of a command that changes one or more of these values.

11.3.5.2. DPTZStruct Type

This type is used to indicate support for the per stream digital pan, tilt, and zoom values.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**VideoStreamID**	VideoStreamID	all				M
1	**Viewport**	ViewportStruct	all				M

11.3.5.2.1. VideoStreamID Field

This field SHALL indicate the video stream this applies too.

11.3.5.2.2. Viewport Field

This field SHALL indicate the per stream viewport applied to this video stream. See Viewport for details on the coordinate system.

11.3.5.3. MPTZStruct Type

This type is used to indicate the mechanical pan, tilt, and zoom values.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Pan**	int16	PanMin to PanMax				MPAN
1	**Tilt**	int16	TiltMin to TiltMax				MTILT
2	**Zoom**	uint8	1 to ZoomMax				MZOOM

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11.3.5.3.1. Pan Field

This field SHALL indicate the mechanical pan value in angular degrees of angle. A zero value SHALL indicate the home position horizontal reference for the direction of view of the camera. A negative value SHALL indicate a leftward rotation of the camera about the vertical axis of the camera coordinate system. A positive value SHALL indicate a rightward rotation of the camera about the vertical axis of the camera coordinate system.

11.3.5.3.2. Tilt Field

This field SHALL indicate the mechanical tilt value in angular degrees of angle. A zero value SHALL indicate a vertical reference for the direction of view of the camera. A negative value SHALL indicate a downward rotation of the camera about the horizontal axis of the camera coordinate system. A positive value SHALL indicate an upward rotation of the camera about the horizontal axis of the camera coordinate system.

11.3.5.3.3. Zoom Field

This field SHALL indicate the zoom value to use. A value of 1 SHALL indicate the widest possible optical field of view. A value of ZoomMax SHALL indicate the narrowest possible field of optical view.

11.3.5.4. MPTZPresetStruct Type

This type is used to save a preset location for mechanical pan, tilt and zoom.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**PresetID**	uint8	1 to MaxPresets				M
1	**Name**	string	max 32				M
2	**Settings**	MPTZStruct					M

11.3.5.4.1. PresetID Field

This SHALL be derived from uint8 and represents the ID for a saved set of preset values for mechanical pan, tilt and zoom.

11.3.5.4.2. Name Field

The SHALL be a string representing the name of the Preset.

11.3.5.4.3. Settings Field

This SHALL hold the mechanical pan, tilt and zoom values.

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11.3.6. Attributes

Table 21. Camera AV Settings User Level Management Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**MPTZPosition**	MPTZStruct	all	N		R V	MPAN | MTILT | MZOOM
1	**MaxPresets**	uint8	all	F		R V	MPRESETS
2	**MPTZPresets**	list[MPTZPresetStruct]	max MaxPresets	N		R V	MPRESETS
3	**DPTZStreams**	list[DPTZStruct]	all	N		R V	DPTZ
4	**ZoomMax**	uint8	2 to 100			R V	MZOOM
5	**TiltMin**	int16	-180 to 0			R V	MTILT
6	**TiltMax**	int16	1 to 180			R V	MTILT
7	**PanMin**	int16	-180 to 0			R V	MPAN
8	**PanMax**	int16	1 to 180			R V	MPAN
9	**MovementState**	PhysicalMovementEnum	all			R V	MPAN | MTILT | MZOOM

11.3.6.1. MPTZPosition Attribute

This attribute indicates the currently selected mechanical pan, tilt, and zoom position.

11.3.6.2. MaxPresets Attribute

This attribute indicates the maximum number of presets for the mechanical pan, tilt, zoom.

11.3.6.3. MPTZPresets Attribute

This attribute SHALL be a list of MPTZPresetStruct. Each entry in the list contains a preset for mechanical pan, tilt, and/or zoom, the values for which are represented by an instance of an MPTZStruct.

11.3.6.4. DPTZStreams Attribute

This attribute is a list of DPTZStruct. If a video stream is listed, it means digital movement is supported via DPTZSetViewport or DPTZRelativeMove. The initial values for each Viewport entry SHALL be the values found in the global Viewport.

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11.3.6.5. ZoomMax Attribute

This attribute SHALL indicate the maximum value for the mechanical zoom specified by the camera manufacturer that allows for increments of 1 to be noticeable. The handling of this value is implementation specific.

11.3.6.6. TiltMin Attribute

This attribute SHALL indicate the minimum value for the mechanical tilt specified by the camera manufacturer in angular degrees.

11.3.6.7. TiltMax Attribute

This attribute SHALL indicate the maximum value for the mechanical tilt specified by the camera manufacturer in angular degrees.

11.3.6.8. PanMin Attribute

This attribute SHALL indicate the minimum value for the mechanical pan specified by the camera manufacturer in angular degrees.

11.3.6.9. PanMax Attribute

This attribute SHALL indicate the maximum value for the mechanical pan specified by the camera manufacturer in angular degrees.

11.3.6.10. MovementState Attribute

This attribute SHALL indicate the current movement state of the camera.

11.3.7. Commands

ID	Name	Direction	Response	Access	Conformance
0	**MPTZSetPosition**	client ⇒ server	Y	O	MPAN | MTILT | MZOOM
1	**MPTZRelativeMove**	client ⇒ server	Y	O	MPAN | MTILT | MZOOM
2	**MPTZMoveToPreset**	client ⇒ server	Y	O	MPRESETS
3	**MPTZSavePreset**	client ⇒ server	Y	O	MPRESETS
4	**MPTZRemovePreset**	client ⇒ server	Y	O	MPRESETS
5	**DPTZSetViewport**	client ⇒ server	Y	O	DPTZ
6	**DPTZRelativeMove**	client ⇒ server	Y	O	[DPTZ]

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11.3.7.1. MPTZSetPosition Command

This command SHALL move the camera to the provided values for pan, tilt, and zoom in the mechanical PTZ. The data for this command SHALL be as follows:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**Pan**	int16	PanMin to PanMax			[MPAN].b+
1	**Tilt**	int16	TiltMin to TiltMax			[MTILT].b+
2	**Zoom**	uint8	1 to ZoomMax			[MZOOM].b+

11.3.7.1.1. Pan Field

This field SHALL indicate the absolute pan value in angular degrees.

11.3.7.1.2. Tilt Field

This field SHALL indicate the absolute tilt value in angular degrees.

11.3.7.1.3. Zoom Field

This field SHALL indicate the absolute zoom value.

11.3.7.1.4. Effect on Receipt

The process below SHALL be followed by the camera when receiving this command:

- If the camera is already engaged in a command that results in a physical move, or if the camera is otherwise unable to apply the changes at the time of receipt:
  - Fail the command with the status code BUSY, and end processing with no other side effects.
- Initiate movement of the camera to the new values of Pan, Tilt, or Zoom.
- Set MovementState to Moving.
- Respond with the status code SUCCESS.
- Once the camera has completed the physical movement to the target values:
  - Set MovementState to Idle.
  - Update the values in MptzPosition to the values as received. Any values that are not present in this command's fields SHALL NOT be modified.

11.3.7.2. MPTZRelativeMove Command

This command SHALL move the camera by the delta values relative to the currently defined position. The data for this command SHALL be as follows:

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PanDelta**	int16	(-1 * (PanMax - PanMin)) to (PanMax - PanMin)		0	[MPAN].c+
1	**TiltDelta**	int16	(-1 * (TiltMax - TiltMin)) to (TiltMax - TiltMin)		0	[MTILT].c+
2	**ZoomDelta**	int8	(-1 * (ZoomMax - 1)) to (ZoomMax - 1)		0	[MZOOM].c+

11.3.7.2.1. PanDelta Field

This field SHALL indicate the change in the pan value in degrees relative to the current location. A value of 0 means no movement. A negative value means move left. A positive value means move right.

11.3.7.2.2. TiltDelta Field

This field SHALL indicate the change in the tilt value in degrees relative to the current location. A value of 0 means no movement. A negative value means move down. A positive value means move up.

11.3.7.2.3. ZoomDelta Field

This field SHALL indicate the percentage change in the zoom value relative to the current zoom value on the camera. A value of 0 means no change. A negative value means zoom out. A positive value means zoom in.

11.3.7.2.4. Effect on Receipt

The process below SHALL be followed by the camera when receiving this command:

- If the camera is already engaged in a command that results in a physical move, or if the camera is otherwise unable to apply the changes at this time:
  - Fail the command with the status code BUSY, and end processing with no other side effects.
- If PanDelta is present:
  - Add PanDelta to the Pan value in MptzPosition and clip the resulting value to be between PanMin and PanMax, inclusive.
- If TiltDelta is present:
  - Add TiltDelta to the Tilt value in MptzPosition and clip the resulting value to be between Tilt-

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Min and TiltMax, inclusive.

- If ZoomDelta is present:
  - Add ZoomDelta to the Zoom value in MptzPosition and clip the resulting value to be between 1 and ZoomMax, inclusive.
- Initiate movement of the camera to the new values of Pan, Tilt, or Zoom.
- Set MovementState to Moving.
- Respond with the status code SUCCESS.
- Once the camera has completed the physical movement to the target values:
  - Set MovementState to Idle.
  - Update the values in MptzPosition to the new values of Pan, Tilt, and Zoom.

11.3.7.3. MPTZMoveToPreset Command

This command SHALL move the camera to the positions specified by the Preset passed. The data for this command SHALL be as follows:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PresetID**	uint8	1 to MaxPresets			M

11.3.7.3.1. PresetID Field

This field SHALL match the PresetID of an entry in MptzPresets.

11.3.7.3.2. Effect on Receipt

The process below SHALL be followed by the camera when receiving this command:

- If the provided PresetID does not match an entry in MptzPresets:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- If the camera is already engaged in a command that results in a physical move, or if the camera is otherwise unable to apply new Pan, Tilt, or Zoom values:
  - Fail the command with the status code BUSY, and end processing with no other side effects.
- Initiate movement of the camera to the applicable values for Pan, Tilt, and Zoom found in the entry identified by the provided PresetID.
- Set MovementState to Moving.
- Respond with the status code SUCCESS.
- Once the camera has completed the physical movement to the target values:
  - Set MovementState to Idle.
  - Update the values in MptzPosition to the new values for Pan, Tilt, and Zoom as obtained

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from the provided PresetID prior to the initiation of movement of the camera.

11.3.7.4. MPTZSavePreset Command

This command allows creating a new preset or updating the values of an existing one. The data for this command SHALL be as follows:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PresetID**	uint8	1 to MaxPresets			O
1	**Name**	string	max 32			M

11.3.7.4.1. PresetID Field

This field SHALL indicate the ID of an entry in MptzPresets.

11.3.7.4.2. Effect on Receipt

The process below SHALL be followed by the camera when receiving this command:

- If the PresetID field is not present:
  - If the size of MptzPresets is greater than or equal to MaxPresets:
    - Fail the command with the status code RESOURCE_EXHAUSTED, and end processing with no other side effects.
  - Generate a new PresetID value. This value starts at 1 and monotonically increases by 1 with each new saved preset.
  - A value incremented past MaxPresets wraps to 1.
  - The camera verifies that the incremented ID does not match any other ID. If such a match is found:
    - The ID is incremented until a unique ID is found, wrapping to 1 if the value equals MaxPresets.
  - Create a new entry in MptzPresets using the new PresetID, the provided Name field, and the current values of MptzPosition.
- Else:
  - If the PresetID matches an existing entry in MptzPresets:
    - Update that entry using the provided Name field and the current values of MptzPosition.
  - Else:
    - If the size of MptzPresets is greater than or equal to MaxPresets:
      - Fail the command with the status code RESOURCE_EXHAUSTED, and end processing with no other side effects.
    - Create a new entry in MptzPresets using the provided PresetID, the provided Name field, and the current values of MptzPosition.

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- Respond with the status code SUCCESS.

11.3.7.5. MPTZRemovePreset Command

This command SHALL remove a preset entry from the PresetMptzTable. The data for this command SHALL be as follows:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**PresetID**	uint8	1 to MaxPresets			M

11.3.7.5.1. PresetID Field

This field SHALL indicate the ID of an entry in MptzPresets.

11.3.7.5.2. Effect on Receipt

The process below SHALL be followed by the camera when receiving this command:

- If the PresetID does not exist in MptzPresets:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- Remove the entry for the provided PresetID from MptzPresets.
- Respond with the status code SUCCESS

11.3.7.6. DPTZSetViewport Command

This command allows for setting the digital viewport for a specific Video Stream. This command is a per-stream version of the Viewport Attribute. The data for this command SHALL be as follows:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**VideoStreamID**	VideoStreamID	all			M
1	**Viewport**	ViewportStruct				M

11.3.7.6.1. VideoStreamID Field

This field SHALL be a VideoStreamID representing the video stream to modify.

11.3.7.6.2. Viewport Field

This field SHALL be a ViewportStruct representing the new viewport to apply to the requested stream. The aspect ratio of the viewport SHALL match the aspect ratio of the stream requested.

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11.3.7.6.3. Effect on Receipt

The process below SHALL be followed by the camera when receiving this command:

- If a VideoStreamID entry does not exist in DPTZStreams:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side-effects.
- If the requested viewport is outside of the defined allowed range:
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side-effects.
- Update the entry in DPTZStreams with the requested Viewport.
- Apply the new Viewport to the requested video stream.
- Respond with the status code SUCCESS

11.3.7.7. DPTZRelativeMove Command

This command SHALL change the per stream viewport by the amount specified in a relative fashion. This allows for multiple users to interact with a directional arrow based user interface. It is RECOMMENDED to increment or decrement the values by 10% of the SensorWidth and SensorHeight found in VideoSensorParams.

The data for this command SHALL be as follows:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**VideoStreamID**	VideoStreamID	all			M
1	**DeltaX**	int16			0	O.d+
2	**DeltaY**	int16			0	O.d+
3	**ZoomDelta**	int8	-100 to 100		0	O.d+

11.3.7.7.1. VideoStreamID Field

This field SHALL be a VideoStreamID representing the video stream to modify.

11.3.7.7.2. DeltaX Field

This field SHALL represent the number of pixels to move the Viewport on the X axis within the SensorWidth and SensorHeight found in VideoSensorParams. A value of 0 means no movement. A negative value means move the viewport left. A positive value means move the viewport right.

11.3.7.7.3. DeltaY Field

This field SHALL represent the number of pixels to move the Viewport on the Y axis within the SensorWidth and SensorHeight found in VideoSensorParams. A value of 0 means no movement. A negative value means move the viewport down. A positive value means move the viewport up.

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11.3.7.7.4. ZoomDelta Field

This field SHALL represent a percentage change to the size of the Viewport is within the SensorWidth and SensorHeight found in VideoSensorParams. A value of 0 means no change. A negative value means make the viewport larger. A positive value means make the viewport smaller.

11.3.7.7.5. Effect on Receipt

The process below SHALL be followed by the camera when receiving this command:

- If a VideoStreamID entry does not exist in DPTZStreams:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side-effects.
- If DeltaX is present and not 0:
  - Add DeltaX to the X1 and X2 fields of the VideoStreamID viewport bounding rectangle.
- If DeltaY is present and not 0:
  - Add DeltaY to the Y1 and Y2 fields of the VideoStreamID viewport bounding rectangle.
- If ZoomDelta is present and not 0:
  - Scale the VideoStreamID viewport bounding rectangle by the percentage value; anchoring the re-scaled rectangle at the center point of the existing viewport.
- If the aspect ratio following application of the delta values does not match the aspect ratio of the video stream:
  - Clip the viewport to the aspect ratio of the video stream while maintaining the upper left location.
- If the resultant requested viewport is smaller than the MinViewportResolution
  - Use size in MinViewportResolution while preserving the upper left location if possible.
- If the requested viewport is larger than `SensorWidth` or `SensorHeight` in the VideoSensorParams:
  - Use the sensor size values.
- If any portion of requested viewport falls outside the sensor Cartesian plane:
  - Change the upper left of the requested viewport to the nearest point allowing the viewport to be within the sensor Cartesian plane.
- Apply the new viewport values to the video stream with ID VideoStreamID.
- Update the entry in DPTZStreams with the new viewport values.
- Respond with the status code SUCCESS

11.4. WebRTC Transport

11.4.1. Background Info

WebRTC (Web Real-Time Communication) is a real-time communication transport technology built on a set of standards that allows audio and video communication between two peers and also supports the transport of arbitrary data between them using an Offer/Answer model.

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It supports numerous use cases such as P2P communication and live stream broadcasting, etc. Readers unfamiliar with WebRTC should read all of [RFC8825] to understand the concepts and understand the interplay between the IETF RFCs and the W3C browser specifications.

Matter devices implementing WebRTC SHALL conform to the following:

- Implement all IETF required elements in [RFC8825] and all sub-requirements found within its dependent RFCs.
- Support at least the `WebRTC Non-Browser` type as defined in [RFC8825].
- Support [RFC7728] for per SSRC media stream control.
- Support [RFC8851] for adding resolution and bitrate values to video and audio media lines.

The list below is not exhaustive (see full details in [RFC8825]), but are the important RFCs and concepts to understand for Matter's WebRTC implementation:

- [RFC8866]: SDP - The Session Description Protocol
- [RFC3264]: An Offer/Answer Model with Session Description Protocol (SDP)
- [RFC8489]: STUN Protocol for discovering public and NAT mapped IP addresses
- [RFC8656]: TURN Protocol for relaying media through firewalls
- [RFC8834]: Media Transport and Use of RTP in WebRTC
- [RFC8445]: Interactive Connectivity Establishment (ICE): A Protocol for Network Address Translator (NAT) Traversal
- [RFC8839]: Session Description Protocol (SDP) Offer/Answer Procedures for Interactive Connectivity Establishment (ICE)
- W3C WebRTC [https://www.w3.org/TR/webrtc/]: Real-Time Communication in Browsers

Unlike [RFC8825] and W3C WebRTC [https://www.w3.org/TR/webrtc/], Matter provides a full signaling implementation for transport of all commands between Nodes and while this implementation does not follow the naming conventions of JSEP [RFC9429], it aims to provide a transparent approach to endpoints that are both JSEP and Matter so that all WebRTC signaling can be passed through.

It is RECOMMENDED that Requestors implement support for the SDP imageattr [RFC6236] attribute using the procedures defined in [RFC9429], Section 3.6 to properly indicate their screen sizes for receiving video streams.

11.4.2. Operating Roles

This section defines two roles, Provider and Requestor, which are reflected in the application clusters defined in this section. The roles are not mutually exclusive, with devices implementing one or both depending on their functionality.

11.4.2.1. Provider

The Provider role is expressed by implementing the WebRTC Transport Provider Cluster and is typically implemented by devices with video and audio capabilities that produce streams that are consumed by a Requestor. Providers don't initiate outbound sessions and only receive requests to start

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them. In the example flows shown below the Camera is the Provider.

11.4.2.2. Requestor

The Requestor role is expressed by implementing the WebRTC Transport Requestor Cluster and is implemented by devices which want to watch or listen to the streams produced by a Provider. Requestors only initiate outbound sessions. In the example flows shown below the Controller is the Requestor.

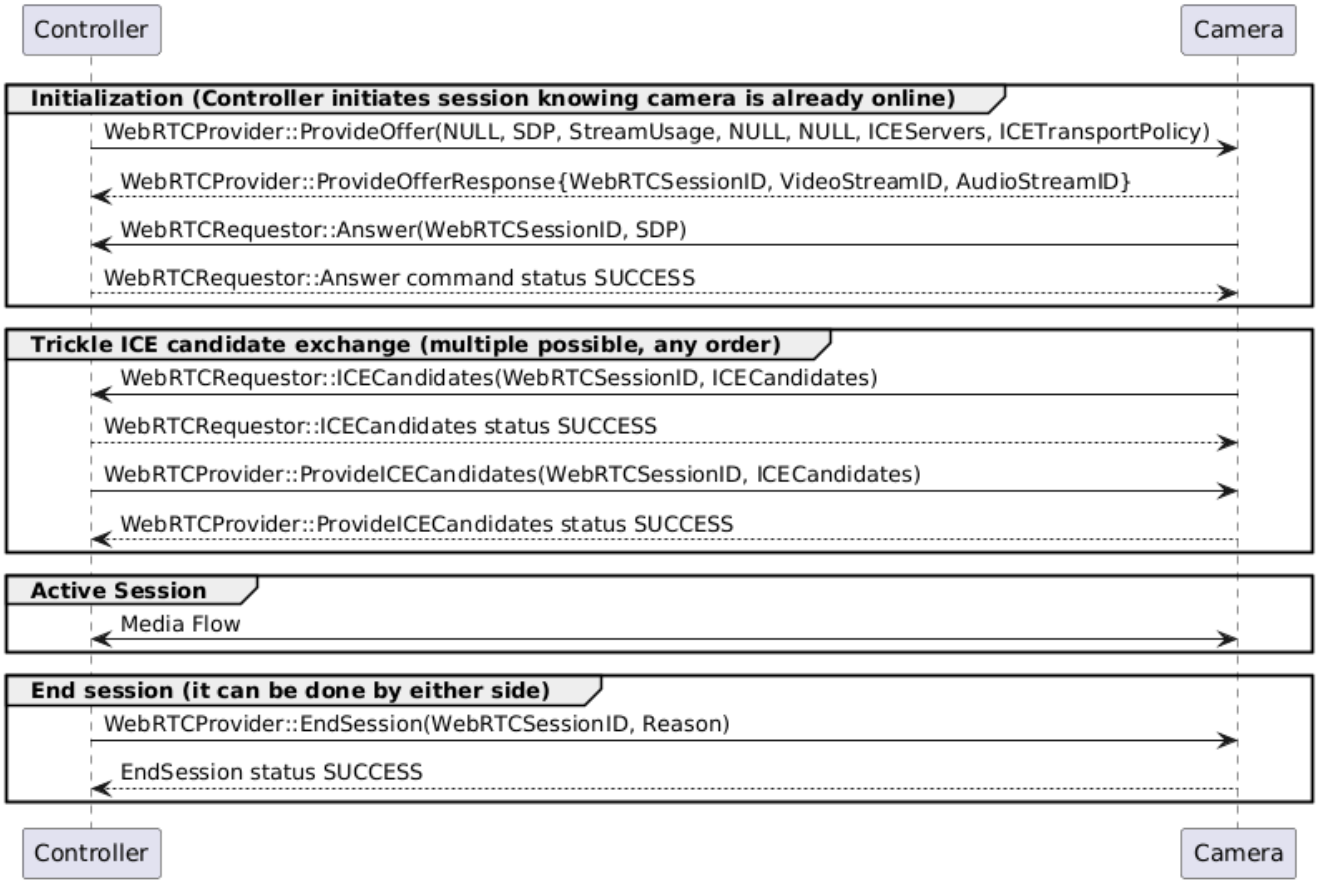
11.4.3. Signaling Flows

The following sequence diagrams show the back and forth commands, statuses, and command responses that are involved in the WebRTC session lifecycle.

11.4.3.1. Normal Flow

In this flow, the controller starts the signaling sequence by providing an SDP Offer to the camera via ProvideOffer. If the camera is successful in starting a session, it SHALL send a ProvideOfferResponse with the allocated WebRTC session ID, the allocated video stream ID, and the allocated audio stream ID. Otherwise, it SHALL respond with a StatusReport with a FAILURE status code. Subsequently, the camera SHALL follow up with an SDP Answer. The second phase is the trickle ICE candidate exchange, where both peers determine their ICE candidates and exchange them via ProvideICECandidates and ICECandidates. In the third phase, the actual connection is established and the WebRTC session becomes active.

Lastly, when the controller needs to end the session, it SHALL send the EndSession command to the camera. If the camera needs to end the session, it SHALL send the End command to the controller.



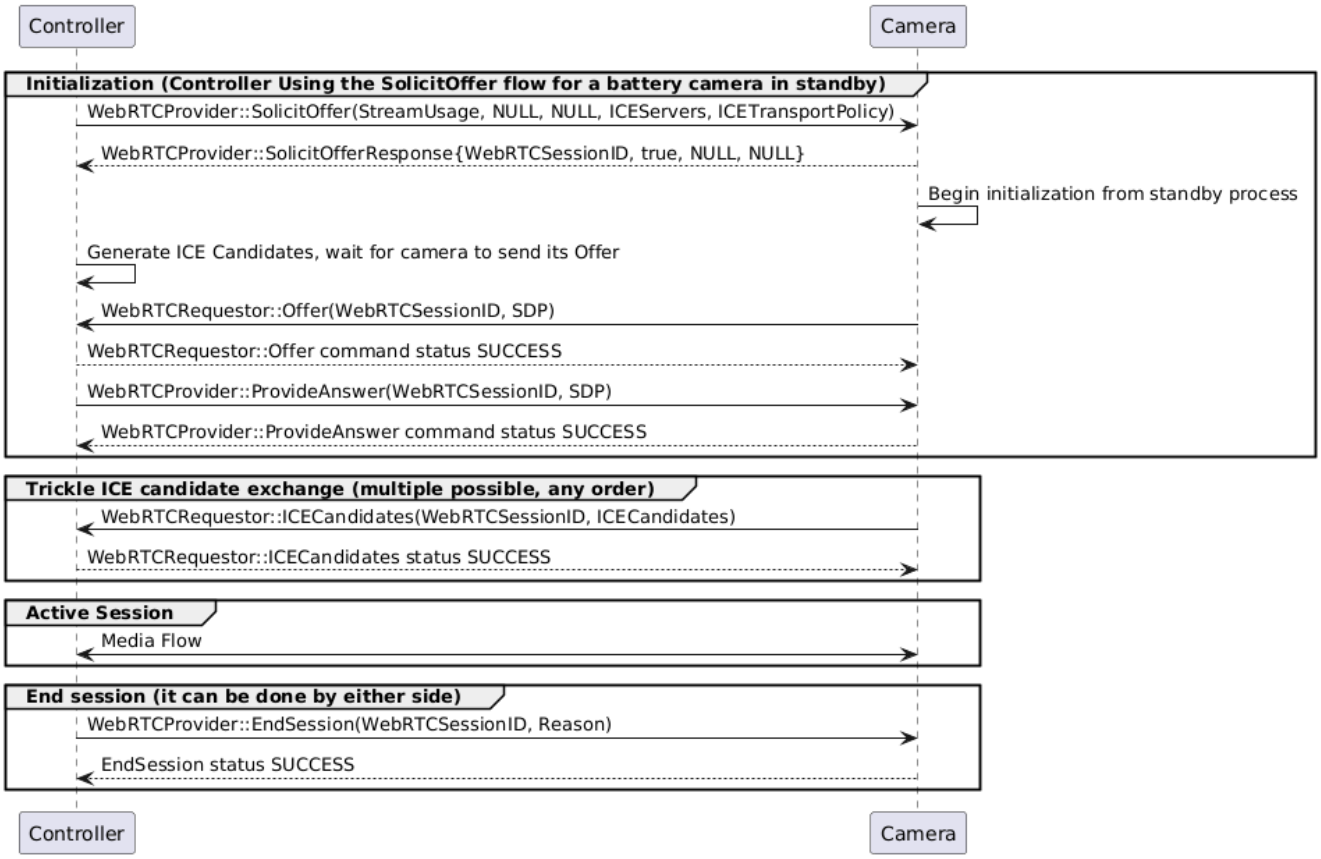
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11.4.3.2. Battery Camera in Standby Flow

In this flow, the controller sends the SolicitOffer command to the camera, which wakes up and replies with SolicitOfferResponse with an allocated WebRTCSessionID and the *DeferredOffer* parameter set to TRUE. The Video or Audio Stream IDs MAY be NULL if the camera can't determine the stream assignments this early. The Camera would then begin its process of initializing all the relevant components, which MAY take a while. For the Controller, the *DeferredOffer:TRUE* is the signal that up to 30 seconds MAY elapse before the Camera can send the Offer to the Controller.

During the time it takes for the camera to fully initialize all its components, the controller SHOULD determine its ICE candidates so that all the candidates are ready by the time the Camera calls Offer. The Controller would then respond with its Answer including ICE candidates via the ProvideAnswer command.

During the ICE candidate exchange phase of this flow, it is expected that only the Camera would send Trickle ICE candidates via ICECandidates since the Controller SHOULD have determined and sent its ICE candidates within its initial SDP Answer command.



NOTE See Determining If Battery Powered for the appropriate logic for when to use this SolicitOffer based flow.

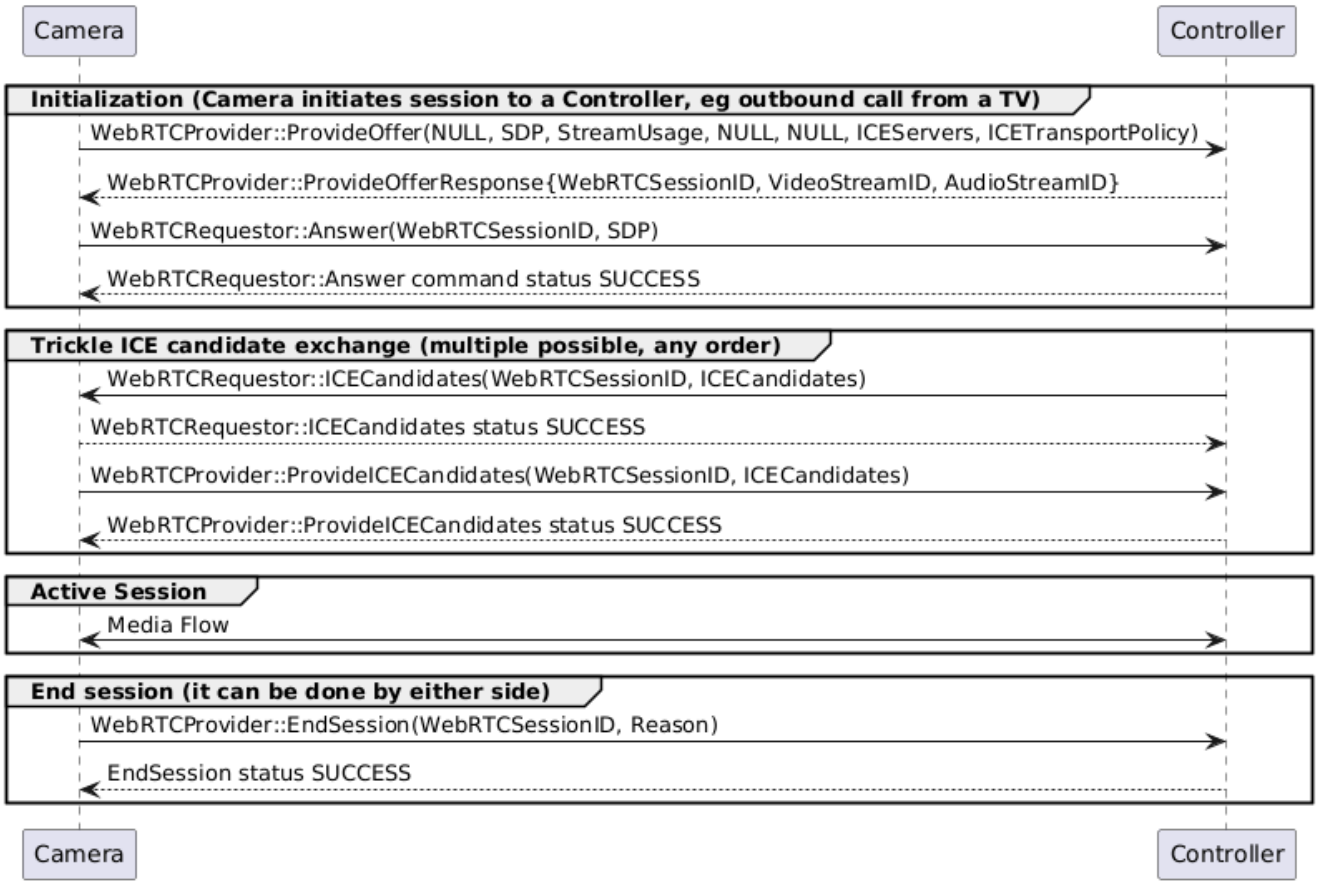
11.4.3.3. Camera Initiated Flow

In this flow, the camera is the initiator of the WebRTC session and assumes the Requestor role, with the Controller assuming the Provider role. This flow is only supported if the Camera implements WebRTC Transport Requestor Cluster and the other side (Controller) implements WebRTC Transport Provider Cluster.

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The Camera begins by sending the ProvideOffer command to the Controller and getting back a ProvideOfferResponse. The Controller then sends a Answer command to the Camera. The rest of the flow is the same as in WebRTC Normal Flow.

This flow is used when the Camera needs to start the streaming session, such as in an outbound call from a TV with a built-in camera or an Intercom style system, when the Camera has both the Provider and Requestor roles.



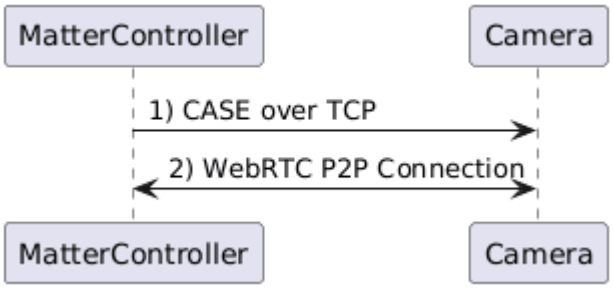
11.4.3.4. Signaling Modes

There are two signaling modes depending on the network topology of the camera, the Matter controller and the livestream viewer entities.

11.4.3.4.1. In-Home Streaming

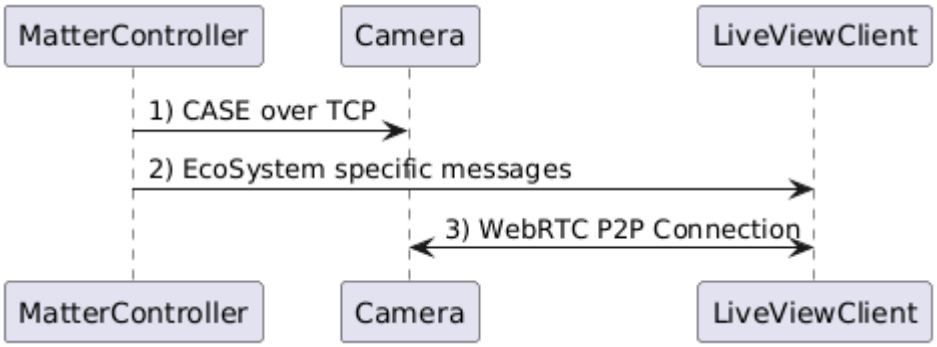
In this situation, all the entities share the same local area network and there are no firewall restrictions between any of them. There would be no use of STUN/TURN servers and the signaling SHALL be done completely over a Matter CASE session. In the example shown below both the controller and the liveview client are the same device/node. The WebRTC session SHALL be established between the Matter controller and the camera after the signaling is done between them.

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11.4.3.4.2. Out-Of-Home Streaming

In this situation, the viewer and the camera are likely not in the same network or there could be firewall restrictions that prevent direct communication between them. In this case, a Matter controller that has a CASE session with the camera is needed to coordinate the signaling phase between the camera and the viewer device. The Matter controller exchanges the necessary signaling data with the viewer through an ecosystem specific mechanism.



11.4.3.5. Determining If Battery Powered

For camera devices, Controllers SHOULD use the Power Source cluster and the features and attributes found within (See `Power Source Cluster`, in [MatterCore]) to determine if a device is currently operating on battery power. The pseudo code for this is as follows:

```
// Returns true if device is currently known to be battery powered, false otherwise
function on_battery_power
  // Create a empty list of power source instances, and then add them sorted by the
Order attribute lowest to highest
  list[power_source_instances]
  for each endpoint in device:
    If endpoint contains a 'Power Source Cluster':
      let power_source_instance be that endpoints 'Power Source Cluster'
      // Check if the power source applies to the whole device (empty list) or the
cameras endpoint
      If power_source_instance.EndpointList attribute is an empty list or
power_source_instance.EndpointList contains the endpoint-id of the camera devices
endpoint:
        power_source_instances.insert_by_rank(power_source_instance,
power_source_instance.Order)
```

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```
// Go through the resulting list
for each power_source in power_source_instances:
  If WIRED feature is present and power_source.Status == Active:
    return false
  If BAT feature is present and power_source.Status == Active:
    return true

// No definitive information was found so assume it's not battery powered
return false
```

11.4.4. Multi Stream

Cameras often have multiple concurrent viewer participants who are out of the home (off the local network) and, thus, ecosystems would want to provide infrastructure that minimizes the number of times high bandwidth streams need to be transmitted out of the home. This model, known as remote fan-out, allows for a single WebRTC session, with multiple media streams coming from the camera, to be used as the source for multiple concurrent viewers.

The standard mechanism used for remote fan-out with WebRTC involves the ecosystem deployment of Selective Forwarding Units (SFUs or media servers). SFUs retransmit each RTP packet from a single WebRTC session to multiple participants at the same time. When the camera has a MaxConcurrentEncoders of 2 or more, and is setup to have its AllocatedVideoStreams running at different resolutions and bitrates, the SFU or remote participant can use rate control and packet loss knowledge to select which stream from the camera would be forwarded to which viewing client in real-time.

In Matter's WebRTC implementation, each audio and video stream requested SHALL be added to the SDP Offer/Answer as a unique SDP `m=` media line with a unique SSRC. The SFU or remote participant, can then use the [RFC7728] defined RTCP control messages PAUSE, UNPAUSE, RESUME, and/or REFUSED sent to the camera, to control the real-time flow of media from each stream / `m=` media line. Since each RTCP message requires a SSRC, this allows for precise targeting of the stream to be controlled.

The requested streams for a session are set via the VideoStreams and AudioStreams fields in the SolicitOffer or ProvideOffer commands. Once a session has been established, the streams used in the session are found in the AudioStreams and VideoStreams fields of the WebRTCSessionStruct available in the CurrentSessions attribute in the WebRTC Transport Provider Cluster.

The initial transmission state of any given stream MAY be set to *paused* by placing an `a=inactive` attribute in a given `m=` media section in the SDP Answer. The far end MAY, subsequently, send a [RFC7728] RTCP UNPAUSE message at anytime, which upon receipt at the camera SHALL toggle this state and re-start transmission. This allows for fine-grained control of media transmission without having to do a full WebRTC re-offer flow.

In Matter's WebRTC implementation, Providers SHALL add a [RFC8851] format `a=rid` line to each video media section. This line is added to communicate the resolution, frame rate, and bitrate for each video stream in a codec-independent fashion to the far-end. The following [RFC8851] defined parameters SHALL be added to every media section for each video stream in the Offer or Answer:

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- `max-width` - This is the max width of the video stream in pixels as sourced from the Width field of the MaxResolution field.
- `max-height` - This is the max height of the video stream in pixels as sourced from the Height field of the MaxResolution field.
- `max-fps` - This is the max framerate in frames per second of the video stream as sourced from the MaxFrameRate field.
- `max-br` - This is the max bitrate in bits per second of the video stream as sourced from the MaxBitRate field.

11.4.5. Data Types

11.4.5.1. WebRTCSessionID Type

This data type is derived from uint16.

It represents an active WebRTC session. This value starts at 0 and monotonically increases by 1 with each new allocation provisioned by the Node. A value incremented past 65534 SHALL wrap to 0. The Node SHALL verify that the incremented ID does not match any other active session ID. If such a match is found, the ID SHALL be incremented until a unique ID is found.

11.4.5.2. WebRTCEndReasonEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**ICEFailed**	No media connection could be established to the other party	M
1	**ICETimeout**	The call timed out whilst waiting for ICE candidate gathering to complete	M
2	**UserHangup**	The user chose to end the call	M
3	**UserBusy**	The remote party is busy	M
4	**Replaced**	The call was replaced by another call	M
5	**NoUserMedia**	An error code when there is no local mic/camera to use. This may be because the hardware isn't plugged in, or the user has explicitly denied access	M

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Value	Name	Summary	Conformance
6	**InviteTimeout**	The call timed out whilst waiting for the offer/answer step to complete	M
7	**AnsweredElsewhere**	The call was answered from a different device	M
8	**OutOfResources**	The was unable to continue due to not enough resources or available streams	M
9	**MediaTimeout**	The call ended due to a media timeout	M
10	**LowPower**	The call ended due to hitting a low power condition	M
11	**PrivacyMode**	The call ended due to the camera being set into a privacy mode.	M
12	**UnknownReason**	Unknown or unspecified reason	M

11.4.5.3. ICEServerStruct Type

This type SHALL specify the [RFC8825] compliant ICE servers [RFC8839] used to facilitate the negotiation of peer-to-peer connections through NATs (Network Address Translators) and firewalls. It mimics the model used in the W3C WebRTC API RTCIceServer [https://www.w3.org/TR/webrtc/#dom-rtcice-server] dictionary with the addition of a Matter specific field for specifying the Root Certificate of any ICE servers that require TLS.

There are two types of ICE Servers which help to discover the public IP address of a device and relay media traffic when direct peer-to-peer communication is not possible:

- STUN [RFC8489] Servers, which help to discover the public IP address and NAT/Firewall type if any, of a device. When a WebRTC session starts, it contacts the STUN server, which returns the device's public IP and port number. This information is used to generate ICE candidates for the peer-to-peer connection setup.
- TURN [RFC8656] Servers, which are used when STUN is not sufficient to establish a peer-to-peer connection—typically such as when devices are behind symmetric NATs, which STUN cannot traverse. TURN servers act as a relay between the peers, routing the media traffic between them.

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**URLs**	list[string]	max 10 [max 2000]				M
1	**Username**	string	max 508				O
2	**Credential**	string	max 512				O
3	**CAID**	TLSCAID	0 to 65534				O

11.4.5.3.1. URLs Field

This field SHALL specify a list of URLs pointing to the STUN and/or TURN servers. The URL scheme distinguishes whether it is a STUN or TURN server (`stun:`, `stuns:`, `turn:`, or `turns:` respectively). This field maps to the RTCIceServer urls [https://www.w3.org/TR/webrtc/#dom-rtciceserver-urls] field.

11.4.5.3.2. Username Field

(Optional for STUN, usually required for TURN) The [RFC8489] compliant UTF-8 encoded username required for authentication with the STUN or TURN servers found in the URLs field. This field maps to the RTCIceServer username [https://www.w3.org/TR/webrtc/#dom-rtciceserver-username] field.

11.4.5.3.3. Credential Field

(Optional for STUN, usually required for TURN) The [RFC8489] compliant UTF-8 encoded short-term credential (password) used for authentication with the STUN or TURN servers found in the URLs field. This field maps to the RTCIceServer credential [https://www.w3.org/TR/webrtc/#dom-rtciceserver-credential] field.

11.4.5.3.4. CAID Field

This field represents the TLSRCAC via its assigned TLSCAID (see Chapter 14, `Certificate Authority ID (CAID) Mapping` and `TLS Certificate Management Commands` sections in [MatterCore]) that will validate the certificate chain presented by the entries in the urls field. It SHALL be set to a valid value if a `turns:` or `stuns:` url is present in the urls field and SHALL be used to validate those servers' presented TLS root certificates.

11.4.5.4. ICECandidateStruct Type

This type SHALL specify the [RFC8825] compliant ICE Candidate [RFC8839] used to facilitate the negotiation of peer-to-peer connections through NATs (Network Address Translators) and firewalls. It mimics the model used in the W3C WebRTC API RTCIceCandidate [https://www.w3.org/TR/webrtc/#rtcicecandidate-interface] dictionary.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Candidate**	string					M
1	**SDPMid**	string	min 1	X			M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
2	**SDPMLineIndex**	uint16		X			M

11.4.5.4.1. Candidate Field

This field SHALL specify the [RFC8825] compliant [RFC8839] `candidate-attribute` field in string form. This is the same value as the W3C WebRTC API RTCIceCandidate candidate [https://www.w3.org/TR/webrtc/#dom-rtcicecandidate-candidate] value. The RFCs define no min or max length on this value.

Note: This string is not the same string as doing a `candidate.toString()` on a RTCIceCandidate ECMAScript object itself. Some browsers and ECMAScript libraries use non-standard ways to serialize sdpMid and sdpMLineIndex into the resulting string, but this is not defined in the W3 specification. This specification requires those fields to be passed directly using the named struct fields SDPMid and SDPMLineIndex.

11.4.5.4.2. SDPMid Field

This field SHALL specify the Candidate's media stream identification tag which uniquely identifies the media stream within the component with which the candidate is associated, or null if no such association exists. This is the same value as the W3C WebRTC API RTCIceCandidate sdpMid [https://www.w3.org/TR/webrtc/#dom-rtcicecandidate-sdpmid] value. The RFCs define no max length on this value.

11.4.5.4.3. SDPMLineIndex Field

This field SHALL specify the zero-based index number of the media description (as defined in [RFC8866]) in the SDP with which the Candidate is associated or null if no such association exists. This is the same value as the W3C WebRTC API RTCIceCandidate sdpMLineIndex [https://www.w3.org/TR/webrtc/#dom-rtcicecandidate-sdpmlineindex] value.

11.4.5.5. WebRTCSessionStruct Type

This type stores all the relevant values associated with an active WebRTC session.

Access Modifier: Fabric Scoped							
ID	**Name**	**Type**	**Constraint**	**Quality**	**Fallback**	**Access**	**Conformance**
0	**ID**	WebRTCSessionID	all				M
1	**PeerNodeID**	node-id	all				M
2	**PeerEndpointID**	endpoint-no	all				M

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Access Modifier: Fabric Scoped							
3	**StreamUsage**	StreamUsageEnum	all				M
4	**VideoStreamID**	VideoStreamID	all	X			O, D
5	**AudioStreamID**	AudioStreamID	all	X			O, D
6	**MetadataEnabled**	bool	all				M
7	**VideoStreams**	list[VideoStreamID]	1 to 16				O.a+
8	**AudioStreams**	list[AudioStreamID]	1 to 16				O.a+

This values of PeerNodeID and FabricIndex are used to validate the source of, or select the correct remote target, for WebRTC session related commands. The implicit field FabricIndex exists since this structure is defined as Fabric Scoped.

11.4.5.5.1. ID Field

This field contains the WebRTC Session ID for this session.

11.4.5.5.2. PeerNodeID Field

This field contains the NodeId for the peer entity involved in this session.

11.4.5.5.3. PeerEndpointID Field

This field contains the EndpointId for the peer entity involved in this session.

11.4.5.5.4. StreamUsage Field

This field contains the StreamUsageEnum of this session.

11.4.5.5.5. VideoStreamID Field

This field is deprecated and the VideoStreams field used instead.

For compatibility with clients implementing cluster revision 1, the first video stream found in the VideoStreams field SHALL be populated here, or null if no video stream is currently associated with this session.

11.4.5.5.6. AudioStreamID Field

This field is deprecated and the AudioStreams field used instead.

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For compatibility with clients implementing cluster revision 1, the first audio stream found in the AudioStreams field SHALL be populated here, or null if no audio stream is currently associated with this session.

11.4.5.5.7. MetadataEnabled Field

This field indicates if metadata is active in this session.

11.4.5.5.8. VideoStreams Field

This field SHALL be a list of all video streams used by this session. Each VideoStreamID entry corresponds to an entry in the AllocatedVideoStreams attribute.

- If present, the specified video streams from the AllocatedVideoStreams attribute SHALL be used.
- If not present, this session has no video.

11.4.5.5.9. AudioStreams Field

This field SHALL be a list of all audio streams used by this session. Each VideoStreamID entry corresponds to an entry in the AllocatedVideoStreams attribute.

- If present, the specified audio streams from the AllocatedAudioStreams attribute SHALL be used.
- If not present, this session has no audio.

11.5. WebRTC Transport Provider Cluster

The WebRTC transport provider cluster provides a way for stream providers (e.g. Cameras) to stream or receive their data through WebRTC. Devices implementing this cluster SHALL also implement Camera AV Stream Management Cluster on the same endpoint.

11.5.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial version of the WebRTC Transport Provider Cluster
2	Add multi stream support

11.5.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	WEBRTCP

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11.5.3. Cluster ID

ID	Name
0x0553	WebRTC Transport Provider

11.5.4. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

Bit	Code	Feature	Conformance	Summary
0	**METADATA**	Metadata	P, O	Supports metadata transmission in WebRTC sessions

11.5.4.1. Metadata Feature

When this feature is supported and a session activates it, a WebRTC DataChannel using the protocol name `urn:csa:matter:av-metadata` SHALL be used for transmitting the metadata. The ability to include metadata is supported on a per session basis.

This feature is designed to be JSEP [RFC9429] compliant with the RTCDataChannel [https://www.w3.org/TR/webrtc/#rtcdatachannel] object interface and consists of AVMetadataStruct content.

11.5.5. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**CurrentSessions**	list[WebRTCSessionStruct]	all			R M S	M

11.5.5.1. CurrentSessions Attribute

This attribute SHALL be a list of WebRTCSessionStruct, which represents all the active WebRTC Sessions.

11.5.6. Commands

ID	Name	Direction	Response	Access	Quality	Conformance
0x00	**SolicitOffer**	client ⇒ server	SolicitOfferResponse	O F	L	M
0x01	**SolicitOfferResponse**	client ⇐ server	N		L	M

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ID	Name	Direction	Response	Access	Quality	Conformance
0x02	**ProvideOffer**	client ⇒ server	ProvideOfferResponse	O F	L	M
0x03	**ProvideOfferResponse**	client ⇐ server	N		L	M
0x04	**ProvideAnswer**	client ⇒ server	Y	O F	L	M
0x05	**ProvideICECandidates**	client ⇒ server	Y	O F	L	M
0x06	**EndSession**	client ⇒ server	Y	O F	L	M

11.5.6.1. SolicitOffer Command

Requests that the Provider initiates a new session with the Offer / Answer flow in a way that allows for options to be passed and work with devices needing the standby flow. This command SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**StreamUsage**	StreamUsageEnum	all			M
1	**OriginatingEndpointID**	endpoint-no	all			M
2	**VideoStreamID**	VideoStreamID	all	X		O, D
3	**AudioStreamID**	AudioStreamID	all	X		O, D
4	**ICEServers**	list[ICEServerStruct]	max 10			O
5	**ICETransportPolicy**	string	max 16			O
6	**MetadataEnabled**	bool	all			METADATA
8	**VideoStreams**	list[VideoStreamID]	1 to 16			[Rev >= v2].b+, O

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
9	**AudioStreams**	list[AudioStreamID]	1 to 16			[Rev >= v2].b+, O

11.5.6.1.1. StreamUsage Field

This field SHALL contain the StreamUsageEnum that indicates the stream usage for this session and is used per Resource Management and Stream Priorities.

11.5.6.1.2. OriginatingEndpointID Field

This field SHALL indicate the endpoint that originates this command. This endpoint SHALL be used when acting as a client to invoke the commands on the Requestor cluster.

11.5.6.1.3. AudioStreamID Field

This field is deprecated and the AudioStreams field used instead.

If this field is encountered from clients implementing cluster revision 1, then the following SHALL be done:

- If not present, no audio should be included in the resulting Offer.
- If present and null, then automatic stream assignment or creation is requested.
- If present and non-null, the specific audio stream identified by the AudioStreamID is added as an entry to the AudioStreams list.

11.5.6.1.4. VideoStreamID Field

This field is deprecated and the VideoStreams field used instead.

If this field is encountered from clients implementing cluster revision 1, then the following SHALL be done:

- If not present, no video should be included in the resulting Offer.
- If present and null, then automatic stream assignment or creation is requested.
- If present and non-null, the specific video stream identified by the VideoStreamID is added as an entry to the VideoStreams list.

11.5.6.1.5. ICEServers Field

This field SHALL be a list of ICEServerStruct which contains the ICE servers and their credentials to use for this session. See ICEServerStruct for further details.

11.5.6.1.6. ICETransportPolicy Field

This field SHALL contain a string which dictates the gathering and usage of ICE candidates. Specifically whether all candidates are used or if restrictions are applied to only use TURN server candidates. It is important for both security and connectivity behavior, as it allows applications to define

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a preferred approach to network routing and privacy. This field SHALL mirror the acceptable values in the W3C WebRTC API RTCIceTransportPolicy [https://www.w3.org/TR/webrtc/#rtcicetransportpolicy-enum] enum, which are listed below for convenience:

Value	Description
'all'	This is the default setting in most cases. The ICE agent can gather candidates from all available network interfaces (both public and private IP addresses). This setting provides the highest probability of establishing a direct peer-to-peer connection without the need for relay servers like TURN (Traversal Using Relays around NAT).
'relay'	The ICE agent will only use relay candidates provided by TURN servers. This policy enhances privacy and security because it prevents the disclosure of the user's private IP addresses and ensures that all traffic goes through a controlled relay server. However, it may increase latency and dependency on the relay infrastructure.

11.5.6.1.7. MetadataEnabled Field

This field indicates if metadata transmission SHALL be active in this session.

11.5.6.1.8. VideoStreams Field

This field SHALL be the list of requested VideoStreamID for this session. Valid values are found in the AllocatedVideoStreams attribute.

11.5.6.1.9. AudioStreams Field

This field SHALL be a list of requested AudioStreamID for this session. Valid values are found in the AllocatedAudioStreams attribute.

11.5.6.1.10. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If HardPrivacyModeOn is true:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
- If SoftLivestreamPrivacyModeEnabled is true and *StreamUsage* is LiveView:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
- If SoftRecordingPrivacyModeEnabled is true and *StreamUsage* is Recording or Analysis:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.

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- If *StreamUsage* is not found in the StreamUsagePriorities:
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- If the *VideoStreams* or *AudioStreams* fields are present, and the *VideoStreamID* or *AudioStreamID* fields are present:
  - Fail the command with INVALID_COMMAND, and end processing with no other side effects.
- If *VideoStreamID* is present:
  - If that field is null:
    - Automatically select an existing video stream per the Resource Management and Stream Priorities.
  - Create an empty *VideoStreams* and add the selected VideoStreamID as a new entry.
- If *AudioStreamID* is present:
  - If that field is null:
    - Automatically select an existing audio stream per the Resource Management and Stream Priorities.
  - Create an empty *AudioStreams* and add the selected AudioStreamID as a new entry.
- If *VideoStreams* is present:
  - If the AllocatedVideoStreams is empty:
    - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
  - If there are duplicate entries in *VideoStreams*:
    - Fail the command with the status code ALREADY_EXISTS, and end processing with no other side effects.
- For each entry in *VideoStreams*:
  - If an entry for the VideoStreamID is not found in the AllocatedVideoStreams list:
    - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- If *AudioStreams* is present:
  - If the AllocatedAudioStreams is empty:
    - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
  - If there are duplicate entries in *AudioStreams*:
    - Fail the command with the status code ALREADY_EXISTS, and end processing with no other side effects.
- For each entry in *AudioStreams*:
  - If an entry for the AudioStreamID is not found in the AllocatedAudioStreams list:
    - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end pro-

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cessing with no other side effects.

- For each URL in the URLs field of each ICEServerStruct in the passed in *ICEServers*:
  - If the URL scheme is `turns` or `stuns`, and the UTCTime attribute of the Time Synchronization cluster is null:
    - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
- Create a new WebRTCSessionID
- Create a new WebRTCSessionStruct associated with the accessing fabric, populated with the passed in values from this command, the *OriginatingEndpointID* field as the PeerEndpointID, and the accessing Peer Node ID entry stored in the Secure Session Context (See Chapter 4 Secure Channel, Secure Session Context section, in [MatterCore]) as the PeerNodeID. The PeerNodeID, PeerEndpointID and the associated fabric, SHALL be used as the destination for all commands for the corresponding WebRTCSessionID.
- If in standby mode:
  - Respond with a SolicitOfferResponse with DeferredOffer set to TRUE.
  - Perform device specific steps to exit standby mode. This SHALL take less than 30 seconds.
- Else:
  - Respond with a SolicitOfferResponse with DeferredOffer set to FALSE.
- If not able to meet the Resource Management and Stream Priorities conditions or unable to provide another WebRTC session:
  - Invoke End with Reason set to OutOfResources, and end processing with no other side effects.
- Internally increment the ReferenceCount on any used video streams.
- Internally increment the ReferenceCount on any used audio streams.
- Store the WebRTCSessionStruct in the CurrentSessions
- Generate the SDP Offer and invoke Offer, and if the passed in *ICEServers* and *ICETransportPolicy* were present, include those fields in the command.

11.5.6.2. SolicitOfferResponse Command

This command SHALL be generated in response to a SolicitOffer command.

This response contains information about the session and streams created and/or if a deferred offer will take place.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**WebRTCSessionID**	WebRTCSessionID	all			M
1	**DeferredOffer**	bool	all			M

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
2	**VideoStreamID**	VideoStreamID	all	X		SolicitOffer.VideoStreamID, D
3	**AudioStreamID**	AudioStreamID	all	X		SolicitOffer.AudioStreamID, D

Upon receipt, the client SHALL create a new WebRTCSessionStruct populated with the values from this command, along with the accessing Peer Node ID and Local Fabric Index entries stored in the Secure Session Context as the PeerNodeID and FabricIndex values, and store it in the WebRTC Transport Requestor cluster's CurrentSessions attribute.

The session establishment SHALL be considered failed unless a Offer command is received by the Requestor from the PeerEndpointID / FabricIndex within 30 seconds.

11.5.6.2.1. WebRTCSessionID Field

This field SHALL contain the ID of the established WebRTC session.

11.5.6.2.2. DeferredOffer Field

This field SHALL indicates the delayed processing hint of the provider.

If DeferredOffer is FALSE, the VideoStreamID and AudioStreamID fields SHALL NOT be null in this response and the Provider will without delay invoke the Offer command. If DeferredOffer is TRUE, the VideoStreamID and AudioStreamID fields MAY be non-null in this response, if the stream mapping logic can be done while in its low-power state, or they MAY be null, if it cannot. When TRUE, this is a hint to the Requestor that there will be a larger than normal amount of time before the Offer command will be invoked and the Requestor SHOULD use this extra time gap to begin its own ICE Candidate generation until the Offer arrives. (See further details in the section WebRTC Battery Camera in Standby Flow)

11.5.6.2.3. VideoStreamID Field

This field is deprecated and is only present when clients implementing cluster revision 1 included the VideoStreamID field in the request. When included, it SHALL contain a VideoStreamID used for the session if known or null if unknown at this time.

11.5.6.2.4. AudioStreamID Field

This field is deprecated and is only present when clients implementing cluster revision 1 included the AudioStreamID field in the request. When included, it SHALL contain a AudioStreamID used for the session if known or null if unknown at this time.

11.5.6.3. ProvideOffer Command

This command allows an SDP Offer to be set and start a new session. This command can also be

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used in the re-offer flow of an existing session to change the details of the SDP (e.g. to enable/disable two-way talk).

This command SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0x0	**WebRTCSessionID**	WebRTCSessionID	all	X		M
0x1	**SDP**	string	all			M
0x2	**StreamUsage**	StreamUsageEnum	all			WebRTCSessionID == NULL, O
0x3	**OriginatingEndpointID**	endpoint-no	all			WebRTCSessionID == NULL, O
0x4	**VideoStreamID**	VideoStreamID	all	X		O, D
0x5	**AudioStreamID**	AudioStreamID	all	X		O, D
0x6	**ICEServers**	list[ICEServerStruct]	max 10			O
0x7	**ICETransportPolicy**	string	max 16			O
0x8	**MetadataEnabled**	bool	all			METADATA & (WebRTCSessionID == NULL)
0xA	**VideoStreams**	list[VideoStreamID]	1 to 16			[(Rev >= v2) & (WebRTCSessionID == NULL)].d+, O
0xB	**AudioStreams**	list[AudioStreamID]	1 to 16			[(Rev >= v2) & (WebRTCSessionID == NULL)].d+, O

11.5.6.3.1. WebRTCSessionID Field

This field SHALL be a WebRTCSessionID and contain the ID of an established WebRTC session or null if requesting a new session.

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11.5.6.3.2. SDP Field

This field SHALL contain the string based SDP Offer [RFC8866]. See WebRTC Transport for further details on SDP and Offer/Answer semantics.

11.5.6.3.3. StreamUsage Field

This field SHALL contain the StreamUsageEnum that indicates the stream usage for this session.

11.5.6.3.4. OriginatingEndpointID Field

This field SHALL indicate the endpoint that originates this command. This endpoint SHALL be used when acting as a client to invoke the commands on the Requestor cluster.

11.5.6.3.5. VideoStreamID Field

This field is deprecated and the VideoStreams field used instead.

If this field is encountered from clients implementing cluster revision 1, then the following SHALL be done:

- If not present, no video should be included in the resulting Answer.
- If present and null, then automatic stream assignment or creation is requested.
- If present and non-null, the specific video stream identified by the VideoStreamID SHALL be added as an entry to the VideoStreams list.

11.5.6.3.6. AudioStreamID Field

This field is deprecated and the AudioStreams field used instead.

If this field is encountered from clients implementing cluster revision 1, then the following SHALL be done:

- If not present, no audio should be included in the resulting Answer.
- If present and null, then automatic stream assignment or creation is requested.
- If present and non-null, the specific video stream identified by the AudioStreamID SHALL be added as an entry to the AudioStreams list.

11.5.6.3.7. ICEServers Field

This field SHALL be a list of ICEServerStruct which contains the ICE servers and their credentials to use for this session. See ICEServerStruct for further details.

11.5.6.3.8. ICETransportPolicy Field

This field controls the gathering and usage of ICE candidates and SHALL have one of the values found in ICETransportPolicy.

11.5.6.3.9. MetadataEnabled Field

This field indicates if metadata transmission SHALL be active in this session.

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11.5.6.3.10. VideoStreams Field

This field SHALL be the list of requested VideoStreamID for this session. Valid values are found in the AllocatedVideoStreams attribute.

11.5.6.3.11. AudioStreams Field

This field SHALL be a list of requested AudioStreamID for this session. Valid values are found in the AllocatedAudioStreams attribute.

11.5.6.3.12. Effect on Receipt

The process below SHALL be followed by the server when receiving this command:

- If *WebRTCSessionID* is not null and does not match a value in CurrentSessions:
  - Respond with a response status of NOT_FOUND
- If *WebRTCSessionID* is null:
  - If HardPrivacyModeOn is true:
    - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
  - If SoftLivestreamPrivacyModeEnabled is true and *StreamUsage* is LiveView:
    - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
  - If SoftRecordingPrivacyModeEnabled is true and *StreamUsage* is Recording or Analysis:
    - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
  - If the passed in *StreamUsage* is not found in the StreamUsagePriorities:
    - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
  - If the *VideoStreams* or *AudioStreams* fields are present, and the *VideoStreamID* or *AudioStreamID* fields are present:
    - Fail the command with INVALID_COMMAND, and end processing with no other side effects.
  - If *VideoStreamID* is present:
    - If that field is null:
      - Automatically select an existing video stream per the Resource Management and Stream Priorities.
    - Create an empty *VideoStreams* and add the selected VideoStreamID as a new entry.
  - If *AudioStreamID* is present:
    - If that field is null:
      - Automatically select an existing audio stream per the Resource Management and Stream Priorities.

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        - Create an empty *AudioStreams* and add the selected AudioStreamID as a new entry.
    - If *VideoStreams* is present:
        - If the AllocatedVideoStreams is empty:
            - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
        - If there are duplicate entries in *VideoStreams*:
            - Fail the command with the status code ALREADY_EXISTS, and end processing with no other side effects.
    - For each entry in *VideoStreams*:
        - If an entry for the VideoStreamID is not found in the AllocatedVideoStreams list:
            - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
    - If *AudioStreams* is present:
        - If the AllocatedAudioStreams is empty:
            - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
        - If there are duplicate entries in *AudioStreams*:
            - Fail the command with the status code ALREADY_EXISTS, and end processing with no other side effects.
    - For each entry in *AudioStreams*:
        - If an entry for the AudioStreamID is not found in the AllocatedAudioStreams list:
            - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
    - If not able to meet the Resource Management and Stream Priorities conditions or unable to provide another WebRTC session:
        - Respond with a response status of RESOURCE_EXHAUSTED+
    - For each URL in the URLs field of each ICEServerStruct in the passed in *ICEServers*:
        - If the URL scheme is `turns` or `stuns`, and the UTCTime attribute of the Time Synchronization cluster is null:
            - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
    - Create a new WebRTCSessionID
    - Create a new WebRTCSessionStruct associated with the accessing fabric, populated with the passed in values from this command, the *OriginatingEndpointID* field as the PeerEndpointID, and the accessing Peer Node ID entry stored in the Secure Session Context (See Chapter 4 Secure Channel, Secure Session Context section, in [MatterCore]) as the PeerNodeID. The PeerNodeID, PeerEndpointID and the associated fabric, SHALL be used as the destination for all asynchronous commands for the corresponding WebRTCSessionID.

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- Internally increment the ReferenceCount on any used video streams.
- Internally increment the ReferenceCount on any used audio streams.
- Store the WebRTCSessionStruct in the CurrentSessions

- Else:
  - If the accessing Peer Node ID stored in the Secure Session Context and the accessing fabric do not match the PeerNodeID and associated fabric for the *WebRTCSessionID* entry in CurrentSessions:
    - Respond with a response status of NOT_FOUND
- Respond with a ProvideOfferResponse
- Process the SDP Offer, begin the ICE Candidate gathering phase, create the SDP Answer, and invoke Answer on the sessions PeerEndpointID / FabricIndex

11.5.6.4. ProvideOfferResponse Command

This command contains information about the session and streams created as a response to the requestor's offer.

This command SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**WebRTCSessionID**	WebRTCSessionID	all			M
1	**VideoStreamID**	VideoStreamID	all	X		WebRTCProvideOffer.VideoStreamID, D
2	**AudioStreamID**	AudioStreamID	all	X		WebRTCProvideOffer.AudioStreamID, D

Upon receipt, the client SHALL create a new WebRTCSessionStruct populated with the values from this command, along with the accessing Peer Node ID and Local Fabric Index entries stored in the Secure Session Context as the PeerNodeID and FabricIndex values, and store it in the WebRTC Transport Requestor cluster's CurrentSessions attribute.

11.5.6.4.1. WebRTCSessionID Field

This field SHALL contain the WebRTCSessionID of the established WebRTC session.

11.5.6.4.2. VideoStreamID Field

This field is deprecated and is only present when clients implementing cluster revision 1 included the VideoStreamID field in the request. When included, it SHALL contain a VideoStreamID used for

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the session if known or null if unknown at this time.

11.5.6.4.3. AudioStreamID Field

This field is deprecated and is only present when clients implementing cluster revision 1 included the AudioStreamID field in the request. When included, SHALL contain a AudioStreamID used for the session if known or null if unknown at this time.

11.5.6.5. ProvideAnswer Command

This command SHALL be initiated from a Node in response to an Offer that was previously received from a remote peer. It SHALL have the following data fields:

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**WebRTCSessionID**	WebRTCSessionID	all			M
1	**SDP**	string	all			M

This command SHALL respond with a response status of NOT_FOUND if the WebRTCSessionID does not match an entry in CurrentSessions, or if the matching entry's associated fabric and PeerNodeID do not match the accessing fabric and the Peer Node ID entry stored in the Secure Session Context of the session this command was received on.

11.5.6.5.1. WebRTCSessionID Field

This field SHALL contain the WebRTCSessionID of the established WebRTC session.

11.5.6.5.2. SDP Field

This field SHALL contain the string based SDP Answer [RFC8866]. See WebRTC Transport for further details on SDP and Offer/Answer semantics.

11.5.6.6. ProvideICECandidates Command

This command allows for string based ICE candidates [RFC8839], Section 5.1, generated after the initial Offer / Answer exchange, via a JSEP onicecandidate event [RFC9429], Section 4.1.20, a DOM rtcpeerconnectioniceevent [https://www.w3.org/TR/webrtc/#dom-rtcpeerconnectioniceevent] event, or other WebRTC compliant implementations, to be added to a session during the gathering phase. This is typically used for STUN or TURN discovered candidates, or to indicate the end of gathering state.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**WebRTCSessionID**	WebRTCSessionID	all			M
1	**ICECandidates**	list[ICECandidateStruct]	min 1			M

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This command SHALL respond with a response status of NOT_FOUND if the *WebRTCSessionID* does not match an entry in CurrentSessions, or if the matching entry's associated fabric and PeerNodeID do not match the accessing fabric and the Peer Node ID entry stored in the Secure Session Context of the session this command was received on.

11.5.6.6.1. WebRTCSessionID Field

This field SHALL contain the WebRTCSessionID of the established WebRTC session.

11.5.6.6.2. ICECandidates Field

This field SHALL contain a list of JSEP [RFC9429] compliant ICE Candidate Format objects per [RFC9429], Section 3.5.2.1.

11.5.6.7. EndSession Command

This command instructs the stream provider to end the WebRTC session.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**WebRTCSessionID**	WebRTCSessionID	all			M
1	**Reason**	WebRTCEndReasonEnum	all			M

11.5.6.7.1. WebRTCSessionID Field

This field SHALL contain the WebRTCSessionID of the established WebRTC session.

11.5.6.7.2. Reason Field

This field SHALL be one of the values in WebRTCEndReasonEnum.

11.5.6.7.3. Effect on Receipt

The process below SHALL be followed by the server when receiving this command or when the Provider detects a MEDIA_TIMEOUT:

- If *WebRTCSessionID* does not match a value in CurrentSessions:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- If the accessing fabric and Peer Node ID entry stored in the Secure Session Context do not match the associated fabric and PeerNodeID field of the *WebRTCSessionID* entry in CurrentSessions:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- Internally decrement the ReferenceCount on any used video streams.
- Internally decrement the ReferenceCount on any used audio streams.

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- Remove the entry in the CurrentSessions for *WebRTCSessionID*.

11.6. WebRTC Transport Requestor Cluster

The WebRTC transport requestor cluster provides a way for stream consumers (e.g. Matter Stream Viewer) to establish a WebRTC connection with a stream provider that implements the WebRTC Transport Provider Cluster.

11.6.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial version of the WebRTC Transport Requestor Cluster
2	Add multi stream support

11.6.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	WEBRTCR

11.6.3. Cluster ID

ID	Name
0x0554	WebRTC Transport Requestor

11.6.4. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**CurrentSessions**	list[WebRTCSessionStruct]	all			R A S	M

11.6.4.1. CurrentSessions Attribute

This attribute SHALL be a list of WebRTCSessionStruct, which represents all the active WebRTC Sessions on this Node.

11.6.5. Commands

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ID	Name	Direction	Response	Access	Quality	Conformance
0x00	**Offer**	client ⇒ server	Y	O	L	M
0x01	**Answer**	client ⇒ server	Y	O	L	M
0x02	**ICECandidates**	client ⇒ server	Y	O	L	M
0x03	**End**	client ⇒ server	Y	O	L	M

11.6.5.1. Offer Command

This command provides the stream requestor with WebRTC session details. It is sent following the receipt of a SolicitOffer command or a re-offer initiated by the Provider.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**WebRTCSessionID**	WebRTCSessionID	all			M
1	**SDP**	string	all			M
2	**ICEServers**	list[ICEServerStruct]	max 10			O
3	**ICETransportPolicy**	string	max 16			O

This command SHALL respond with a response status of NOT_FOUND if the WebRTCSessionID does not match an entry in CurrentSessions, or if the matching entry's associated fabric and PeerNodeID do not match the accessing fabric and the Peer Node ID entry stored in the Secure Session Context (see Chapter 4 Secure Channel, Secure Session Context section, in [MatterCore]) of the session this command was received on.

11.6.5.1.1. WebRTCSessionID Field

This field SHALL contain the ID of the established WebRTC session.

11.6.5.1.2. SDP Field

This field SHALL contain the string based SDP Offer [RFC8866]. See WebRTC Transport for further details on SDP and Offer/Answer semantics.

11.6.5.1.3. ICEServers Field

This field SHALL be a list of ICEServerStruct which contains the ICE servers and their credentials to use for this session. See ICEServerStruct for further details.

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11.6.5.1.4. ICETransportPolicy Field

This field controls the gathering and usage of ICE candidates and SHALL have one of the values found in ICETransportPolicy.

11.6.5.2. Answer Command

This command provides the stream requestor with the WebRTC session details (i.e. Session ID and SDP answer), It is the next command in the Offer/Answer flow to the ProvideOffer command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**WebRTCSessionID**	WebRTCSessionID	all			M
1	**SDP**	string	all			M

This command SHALL respond with a response status of NOT_FOUND if the WebRTCSessionID does not match an entry in CurrentSessions, or if the matching entry's associated fabric and PeerNodeID do not match the accessing fabric and the Peer Node ID entry stored in the Secure Session Context of the session this command was received on.

11.6.5.2.1. WebRTCSessionID Field

This field SHALL contain the WebRTCSessionID of the established WebRTC session.

11.6.5.2.2. SDP Field

This field SHALL contain the string based SDP Answer [RFC8866]. See WebRTC Transport for further details on SDP and Offer/Answer semantics.

11.6.5.3. ICECandidates Command

This command allows for the object based ICE candidates [RFC8839], Section 5.1, generated after the initial Offer / Answer exchange, via a JSEP onicecandidate event [RFC9429], Section 4.1.20, a DOM rtcpeerconnectioniceevent [https://www.w3.org/TR/webrtc/#dom-rtcpeerconnectioniceevent] event, or other WebRTC compliant implementations, to be added to a session during the gathering phase. This is typically used for STUN or TURN discovered candidates, or to indicate the end of gathering state.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**WebRTCSessionID**	WebRTCSessionID	all			M
1	**ICECandidates**	list[ICECandidateStruct]	min 1			M

This command SHALL respond with a response status of NOT_FOUND if the WebRTCSessionID does not match an entry in CurrentSessions, or if the matching entry's associated fabric and PeerNodeID

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do not match the accessing fabric and the Peer Node ID entry stored in the Secure Session Context of the session this command was received on.

11.6.5.3.1. WebRTCSessionID Field

This field SHALL contain the WebRTCSessionID of the established WebRTC session.

11.6.5.3.2. ICECandidates Field

This field SHALL contain a list of JSEP [RFC9429] compliant ICE Candidate Format objects per [RFC9429], Section 3.5.2.1.

11.6.5.4. End Command

This command notifies the stream requestor that the provider has ended the WebRTC session.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**WebRTCSessionID**	WebRTCSessionID	all			M
1	**Reason**	WebRTCEndReasonEnum	all			M

11.6.5.4.1. WebRTCSessionID Field

This field SHALL contain the WebRTCSessionID of the established WebRTC session.

11.6.5.4.2. Reason Field

This field SHALL be one of the values in WebRTCEndReasonEnum.

11.6.5.4.3. Effect on Receipt

The process below SHALL be followed by the server when receiving this command:

- If WebRTCSessionID does not match a value in CurrentSessions:
  - Respond with a response status of NOT_FOUND
- If the accessing Peer Node ID stored in the Secure Session Context and the accessing fabric do not match the PeerNodeID and associated fabric for the WebRTCSessionID entry in CurrentSessions:
  - Respond with a response status of NOT_FOUND
- Remove the entry in the CurrentSessions for WebRTCSessionID

11.7. Push AV Stream Transport Cluster

This cluster implements the upload of Audio and Video streams from the Camera AV Stream Management Cluster using suitable push-based transports. Push AV Stream Transport is the transmis-

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sion of a stream of media to a server, initiated by the activation of an associated trigger. A push transport can be triggered from a camera operating in an always-on mode for supporting 24/7 style recording, triggered by events based on motion zones defined in the Zone Management Cluster, or manually triggered using commands or bindings.

A push transport SHALL be allocated before it is intended to be used, and then it is subsequently started when its configured trigger is activated.

An allocated push transport consists of five main logical things:

- A container format, which specifies how the individual media frames are formatted or multiplexed together
- An ingestion format which specifies how it is signaled and transmitted to the far end.
- A trigger which specifies when transport will occur.
- Which media stream/s to package together in the container format.
- Various options to the above.

All push transport ingest methods SHALL use TLS as specified by the core specification. TLS Client Certificates SHALL be used for Authorization and Identification of a Node on the underlying TLS connection (see Chapter 14, TLS Certificate Management and TLS Client Management sections in [MatterCore]). Nodes supporting this cluster SHALL also support the TLS Client Management Cluster and its dependencies.

11.7.1. CMAF Ingestion

CMAF Ingestion as defined in DASH-IF Live Media Ingest Protocol version 1.2 and newer SHALL be supported.

Nodes SHALL support both Interface-1: CMAF Ingest [https://dashif.org/Ingest/#interface-1] and Interface-2: DASH and HLS Ingest [https://dashif.org/Ingest/#interface-2].

Nodes SHALL use HTTP/2 [RFC9113] or newer, for all CMAF Ingestion HTTP based operations.

Nodes using CMAFIngest as the ingestion method SHALL use CMAF as the container format.

11.7.1.1. Background

The Matter Node is considered the CMAF ingest source [https://dashif.org/Ingest/#ingest-source]. The CMAF receiving-entity [https://dashif.org/Ingest/#receiving-entity] is the server that the Matter Node will send the media stream to. The receiving-entity details along with the media streams that make up the ingest-stream [https://dashif.org/Ingest/#ingest-stream] are provisioned in a TransportOptionsStruct.

In CMAF, the basic units of the protocol are Segments, Tracks, and Sessions.

11.7.1.1.1. Segment

A Segment is a fixed duration ISOBMFF file which contains the Audio, Video, and/or Metadata samples themselves. A Segment MAY multiplex Audio and Video if configured to do so, otherwise the Segment would contain a single type of media.

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11.7.1.1.2. Track

A Track is a symbolic name applied to a sequence of one or more Segments, of the same media type, from the same source. In Matter, an Audio Stream maps to an audio Track, a Video Stream maps to a video Track, and one or more metadata content maps to a metadata Track. All Segments within the same Track utilize the same fixed duration, except the last Segment which may be less than or equal to that duration. Additionally, all Segments within the same Track are sequentially numbered per the Segment Numbering rules.

A single CMAF Track SHALL NOT contain both audio and video in the same Segment. In MP4 terms, a CMAF Track is non-interleaved audio and video. A CMAF Track MAY contain interleaved metadata content inside the video or audio MP4 Segment file.

11.7.1.1.3. Session

A Session is defined as the set of one or more Tracks, that lasts for a duration of time. The duration of a Session can be variable or fixed. The Session's duration is based on the duration of the Trigger Type that caused its transmission to occur. In Matter's CMAF usage, a Session can be considered a low level logical event, though one delineated only by Motion detection, Command invocation timings, or fixed durations.

11.7.1.2. Operation

CMAF at its core is a sequence of HTTP PUT/POST operations for each media Segment, within each Track, within each Session. Every operation is sent to a unique URL, which is known as the POST_URL [https://dashif.org/Ingest/#post_url]. Each unique URL contains information about the Session, Track, and Segment. Each unique URL is comprised of two components that are concatenated together.

The first component of the POST_URL is a static value called the publishing_point_URL [https://dashif.org/Ingest/#publishing_point_url]. The publishing_point_URL contains the scheme, hostname of the server, and whatever initial URL path components the receiving-entity would like to receive. Typically, deployments encode user specific information or other ecosystem specific items within the initial URL path values in the publishing_point_URL.

For Matter deployments, the publishing_point_URL SHALL use the `https` scheme, SHALL NOT contain the URL fragment character `#`, SHALL NOT contain the URL query character `?`, and SHALL end with a URL forward slash character `/`. The publishing_point_URL MAY contain as many URL path components as fit within the allocated space of the URL field. Any publishing_point_URL issued to a Matter device SHALL allow for one or more Sessions.

An example of a publishing_point_url is: `https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/`

The second component of the POST_URL is a dynamically computed string that includes the Session name, Track name, and Segment name. Each of these three values are placed in that order as extended paths at the end of the publishing_point_URL. The Session and Segment names also contain numerically increasing portions of their string, which allows for the receiving-entity to determine the event time bounds information along with the individual segment information.

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An example of the second component is: `session_1/media1/segment_1001.m4s`. In this example, `session_1` is the session name, `media1` is the track name, and `segment_1001.m4s` is the segment name.

For Matter deployments, the extended path SHALL be comprised of the three elements in the following order: `session_name/track_name/segment_name` and each sub-portion SHALL follow these rules:

session_name: Consists of the fixed value `session_` followed by a monotonically increasing *SessionNumber* numeric string managed by this cluster. *SessionNumber* SHALL be of type uint64 and start with value 1 when a transport is first allocated.

track_name: Consists of a provisioned string provided in the VideoStreamName or AudioStreamName fields.

segment_name: Consists of the fixed value `segment_` followed by a monotonically increasing *SegmentNumber* numeric string managed by this cluster, followed by a period `.`, followed by the media type's Segment Extension.

When the fixed and variable pieces are put together the extended paths are: `session_<SessionNumber>/<TrackName>/segment_<SegmentNumber>.<SegmentExtension>`

11.7.1.3. Segment Numbering

In Matter's usage of CMAF, *SegmentNumber* SHALL be of type uint16 and start at 1001 for each Track within each Session. In practice, a *SegmentNumber* will likely never exceed 9999 due to the Session length duration limit.

Implementations SHALL keep an internal value on the Transport which tracks the current Segment Number for each Track.

11.7.1.4. Segment Extension

In Matter's usage of CMAF, the following table SHALL determine the value for *SegmentExtension*:

Extension	Mime-Type	Usage
m4s	video/iso.segment	When the track contains video or audio

11.7.1.5. Track Naming

The CMAF Track Name is populated from the VideoStreamName or AudioStreamName fields of the associated stream entry in the VideoStreams or AudioStreams lists for that transport. CMAF Track Names are not user visible so the Controller MAY use any naming scheme they wish.

The Track Name is placed in the computed extended path fields portion of the CMAF POST_URL.

11.7.1.6. Session Numbering

In Matter's usage of CMAF, there are two types of Session durations that affect when the *SessionNumber* changes for an allocated transport.

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- Fixed duration Sessions are used with Continuous triggers. These sessions SHALL have a fixed duration of five minutes.
- Variable duration Sessions with max durations are used with Motion and Command triggers. These sessions SHALL have a minimum duration of two segments, and a maximum duration of five minutes.

Implementations SHALL persistently store the following internal values on the Transport:

- A *SessionNumber* of type uint64.
- A *SessionStartedTimestamp* of type systime-ms.

SessionNumber SHALL initially start at 1. It SHALL increment by 1, with every Trigger activation, or if the current SystemTimeStamp value at the time of a transmission is greater than 5 minutes past the *SessionStartedTimestamp*. Upon incrementing the *SessionNumber*, the *SessionStartedTimestamp* SHALL be set to the current SystemTimeStamp value.

11.7.1.7. Multiple Tracks

In some ecosystem topologies it can be advantageous to send multiple streams/resolutions/encodings of the same content. For example, content analysis could be done more efficiently on a smaller resolution, while storage uses a higher resolution.

Another possible topology is one where the ecosystem wants to store multiple sizes of the same content to allow for adaptive streaming playback. In this topology lower bandwidth viewers can receive a smaller resolution, while high bandwidth viewers receive a higher resolution.

Either style of deployment can be achieved by having provisioned Transports request more than one stream of a given type concurrently in the VideoStreams or AudioStreams lists when configuring the transport.

For a given transport, track names, video stream IDs, and audio stream IDs, SHALL all be unique.

11.7.1.8. CMAF Interface-2 Upload Deviations

This specification is deviating from and extending the base CMAF specification.

The first change dictates a specific order of HTTP operations for initial manifest/playlists uploads, init segment uploads, regular segment uploads, and final manifest/playlist uploads.

The second change specifies a particular syntax be used in DASH manifests and HLS playlists for initial uploads, and a different syntax for the final manifest/playlist upload. This is to allow the CMAF server receiving the content to definitively know when a CMAF Session begins and ends, by just parsing the contents of the DASH manifest file or the HLS media playlist.

The third change dictates a Matter specific HTTP header to be included on certain Interface-2 HLS Segment uploads to remove the need for intermediate HLS playlist uploads after each Segment has been transmitted.

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11.7.1.8.1. DASH Upload Deviations

When Interface-2 DASH mode is selected, a Session is uploaded using the following order and rules:

1. The manifest filename SHALL always be `session_name/index.mpd`.
2. The Session upload SHALL start with the initial manifest file.
3. When the initial manifest is uploaded at the start of a Session, it SHALL set the `type` attribute to `dynamic` in the root `MPD` tag of the XML and use the `SegmentTemplate` syntax for loading segments.
4. Track uploads SHALL begin only after the initial manifest file completes uploading, and Tracks SHOULD be uploaded in parallel.
5. Each Track in the Session SHALL then follow this process:
   a. The Track's init segment is uploaded as `session_name/track_name/track_name.init`
   b. Each Segment within the Track is uploaded in order, one at a time.
6. When the final Segment of the last Track completes uploading, the manifest file SHALL be uploaded again, but setting the `type` attribute to `static` in the root `MPD` tag. In this final manifest file, the `SegmentTemplate` SHALL include a `SegmentTimeline` element that accurately describes the duration of all uploaded segments.

With the ordering and rules specified above, the following state tracking can be used at the CMAF Server for DASH mode:

- The contents of a `session_name/index.mpd` manifest file for which the root tag `MPD` contains the attribute `type` set to the value `"dynamic"` indicates the start of a new CMAF Session.
- The `Representation` tags in the manifest file will indicate how many Tracks will be uploaded.
- The contents of a `session_name/index.mpd` manifest file for which the root tag `MPD` contains the attribute `type` set to the value `"static"` indicates the end of a new CMAF Session.

11.7.1.8.2. HLS Upload Deviations

When Interface-2 HLS mode is selected, a Session is uploaded using the following order and rules:

1. The multi-variant playlist filename SHALL always be `session_name/index.m3u8`.
2. The Session upload SHALL start with the multi-variant playlist file.
3. Track uploads SHALL begin only after the multi-variant playlist file completes the upload, and Tracks SHOULD be uploaded in parallel.
4. Each Track in the Session SHALL then follow this process:
   a. The Track's initial media playlist is uploaded as `session_name/track_name/index.m3u8`. This initial media playlist SHALL contain a `#EXT-X-PLAYLIST-TYPE:EVENT` tag and SHALL NOT contain `#EXT-X-ENDLIST`, `#EXTINF, or segment lines.
   b. The Track's init segment is uploaded as `session_name/track_name/track_name.init`
   c. Each Segment within the Track is uploaded in order, one at a time.
   d. Each Segment uploaded SHALL add an additional HTTP header called `X-EXTINF-duration`. This headers value is the `duration` decimal-floating-point value in seconds of the `#EXTINF:`

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segment tag in an HLS playlist (see [RFC8216bis], Section 4.4.4.1). For example: `X-EXTINF-duration: 4.00000` for a segment that is exactly 4.000000 seconds long. The CMAF Server can thus synthesize and automatically append the two lines per Segment needed (#EXTINF tag and Segment URL) to keep the live media playlist updated.

e. When the final Segment of the Track completes uploading, the media playlist SHALL be uploaded again, but this time adding all #EXTINF and Segment lines, and adding a `#EXT-X-ENDLIST` tag as the last line in the playlist.

With the ordering and rules specified above, the following state tracking can be used at the CMAF Server for HLS mode:

- The upload of the `session_name/index.m3u8` multi-variant playlist file indicates the start of a new CMAF Session.
- Counting the number of lines in the multi-variant playlist which contain the strings `EXT-X-MEDIA` or `EXT-X-STREAM-INF` will indicate how many Tracks will be uploaded.
- The contents of a `session_name/track_name/index.m3u8` media playlist file does not contain the string `EXT-X-ENDLIST` indicates the start of a CMAF Track.
- The contents of a `session_name/track_name/index.m3u8` media playlist file containing the string `EXT-X-ENDLIST` indicates that this CMAF Track has completed uploading.
- When the number of completed Tracks equals the number of Tracks listed in the multi-variant playlist file, the CMAF Session has completed uploading.

11.7.2. CMAF Ingest Examples

This section provides URL examples for Interface-1, Interface-2 HLS, and Interface-2 DASH showing the computed filenames for the manifests, playlists, init segments and media segments in their upload order.

11.7.2.1. Interface-1 Examples

11.7.2.1.1. Single Track

Using the examples and rules above the sequence of POST_URL values for a three segment session using just a video track would be:

- `https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/video.init`
- `https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1001.m4s`
- `https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1002.m4s`
- `https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1003.m4s`

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11.7.2.1.2. Separate Tracks

Using the examples and rules above the sequence of POST_URL values for a three segment session using separate tracks for audio and video would be:

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/video1.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/audio1.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1003.m4s

11.7.2.1.3. Separate Tracks with Multiple Video

Using the examples and rules above the sequence of POST_URL values for a three segment session using separate tracks for audio and having multiple video tracks would be:

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/video1.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video2/video2.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/audio1.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video2/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video2/segment_1002.m4s

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- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video2/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1003.m4s

11.7.2.1.4. Single Track over Multiple Sessions

Using the examples and rules above the sequence of POST_URL values for three sessions of three segments each using just a video track would be:

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/video.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1003.m4s

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/video/video.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/video/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/video/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/video/segment_1003.m4s

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/video/video.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/video/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/video/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/video/segment_1003.m4s

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11.7.2.2. HLS Examples

11.7.2.2.1. HLS Single Track

Using the examples and rules above the sequence of POST_URL values for a three segment session using just a video track would be:

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/index.m3u8
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/index.m3u8 - Initial upload. Does not contain #EXT-X-ENDLIST tag.
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/video.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/index.m3u8 - Final upload. Contains #EXT-X-ENDLIST tag.

11.7.2.2.2. HLS Separate Tracks

Using the examples and rules above the sequence of POST_URL values for a three segment session using separate tracks for audio and video would be:

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/index.m3u8
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/index.m3u8 - Initial upload. Does not contain #EXT-X-ENDLIST tag.
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/index.m3u8 - Initial upload. Does not contain #EXT-X-ENDLIST tag.
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/video1.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/audio1.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/

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audio1/segment_1002.m4s

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/index.m3u8 - Final upload. Contains #EXT-X-ENDLIST tag.
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/index.m3u8 - Final upload. Contains #EXT-X-ENDLIST tag.

11.7.2.2.3. HLS Separate Tracks with Multiple Video

Using the examples and rules above the sequence of POST_URL values for a three segment session using separate tracks for audio and having multiple video tracks would be:

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/index.m3u8
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/index.m3u8 - Initial upload. Does not contain #EXT-X-ENDLIST tag.
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video2/index.m3u8 - Initial upload. Does not contain #EXT-X-ENDLIST tag.
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/index.m3u8 - Initial upload. Does not contain #EXT-X-ENDLIST tag.
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/video1.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video2/video2.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/audio1.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video2/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video2/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/

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video1/segment_1003.m4s

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video2/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/index.m3u8 - Final upload. Contains #EXT-X-ENDLIST tag.
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video2/index.m3u8 - Final upload. Contains #EXT-X-ENDLIST tag.
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/index.m3u8 - Final upload. Contains #EXT-X-ENDLIST tag.

11.7.2.2.4. HLS Single Track over Multiple Sessions

Using the examples and rules above the sequence of POST_URL values for three sessions of three segments each using just a video track would be:

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/index.m3u8
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/index.m3u8 - Initial upload. Does not contain #EXT-X-ENDLIST tag.
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/video.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/index.m3u8 - Final upload. Contains #EXT-X-ENDLIST tag.

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/index.m3u8
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/video/index.m3u8 - Initial upload. Does not contain #EXT-X-ENDLIST tag.
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/video/video.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/video/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/video/segment_1002.m4s

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- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/video/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/video/index.m3u8 - Final upload. Contains #EXT-X-ENDLIST tag.

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/index.m3u8
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/video/index.m3u8 - Initial upload. Does not contain #EXT-X-ENDLIST tag.
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/video/video.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/video/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/video/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/video/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/video/index.m3u8 - Final upload. Contains #EXT-X-ENDLIST tag.

11.7.2.3. DASH Examples

11.7.2.3.1. DASH Single Track

Using the examples and rules above the sequence of POST_URL values for a three segment DASH session using just a video track would be:

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/index.mpd - Initial upload. Has `MPD@type="dynamic"`.
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/video.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/index.mpd - Final upload. Has `MPD@type="static"`.

11.7.2.3.2. DASH Separate Tracks

Using the examples and rules above the sequence of POST_URL values for a three segment DASH session using separate tracks for audio and video would be:

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- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/index.mpd - Initial upload. Has MPD@type="dynamic".
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/video1.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/audio1.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/index.mpd - Final upload. Has MPD@type="static".

11.7.2.3.3. DASH Separate Tracks with Multiple Video

Using the examples and rules above the sequence of POST_URL values for a three segment DASH session using separate tracks for audio and having multiple video tracks would be:

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/index.mpd - Initial upload. Has MPD@type="dynamic".
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/video1.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video2/video2.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/audio1.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video2/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1002.m4s

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- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video2/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video1/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video2/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/audio1/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/index.mpd - Final upload. Has MPD@type="static".

11.7.2.3.4. DASH Single Track over Multiple Sessions

Using the examples and rules above the sequence of POST_URL values for three DASH sessions of three segments each using just a video track would be:

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/index.mpd - Initial upload. Has MPD@type="dynamic".
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/video.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/video/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_1/index.mpd - Final upload. Has MPD@type="static".

- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/index.mpd - Initial upload. Has MPD@type="dynamic".
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/video/video.init
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/video/segment_1001.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/video/segment_1002.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/video/segment_1003.m4s
- https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_2/

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`index.mpd` - Final upload. Has `MPD@type="static"`.

- `https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/index.mpd` - Initial upload. Has `MPD@type="dynamic"`.
- `https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/video/video.init`
- `https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/video/segment_1001.m4s`
- `https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/video/segment_1002.m4s`
- `https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/video/segment_1003.m4s`
- `https://ingest.example.com/provider_supplied_arbitrary_data/in/base/part/of/url/session_3/index.mpd` - Final upload. Has `MPD@type="static"`.

11.7.3. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial Release
2	Add DoorbellPressed to TriggerActivationReason, augment PushTransport events with Session numbers and Event context and add multi-stream support

11.7.4. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	PAVST

11.7.5. Cluster ID

ID	Name
0x0555	Push AV Stream Transport

11.7.6. Features

This cluster SHALL support the FeatureMap bitmap attribute as defined below.

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Bit	Code	Feature	Conformance	Summary
0	**PERZONESENS**	PerZoneSensitivity	O	Supports a sensitivity value per Zone
1	**METADATA**	Metadata	P, O	Supports metadata transmission in Push transports

11.7.6.1. PerZoneSensitivity Feature

When this feature is supported, the Sensitivity for a Motion Trigger can be set per zone. When not supported, only a single sensitivity can be used for all Motion Triggers.

11.7.6.2. Metadata Feature

When this feature is supported and a transport activates it, metadata SHALL be included within the uploaded data.

11.7.7. Data Types

11.7.7.1. TransportTriggerTypeEnum Type

This data type is derived from enum8.

The Trigger Type determines the basic operation of the Push Transport and when it will actually transmit content.

Value	Name	Summary	Conformance
0	**Command**	Triggered only via a command invocation	M
1	**Motion**	Triggered via motion detection or command	M
2	**Continuous**	Triggered always when transport status is Active	M

11.7.7.1.1. Command Value

When set to this value, transport will only occur if a command is invoked to trigger it. This could be a ManuallyTriggerTransport command or an internally triggered command such as a physical button press.

11.7.7.1.2. Motion Value

When set to this value, transport will occur if either the Motion Detector becomes triggered, or ManuallyTriggerTransport is invoked. This is generally known as event driven recording.

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11.7.7.1.3. Continuous Value

When set to this value, transport will always occur so long as the TransportStatus is Active. This is generally known as 24/7 always-on recording.

11.7.7.2. TransportStatusEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**Active**	Push Transport can transport AV Streams	M
1	**Inactive**	Push Transport cannot transport AV Streams	M

11.7.7.3. ContainerFormatEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**CMAF**	CMAF container format	M

11.7.7.3.1. CMAF Value

CMAF (Common Media Application Format) is a container format based on the ISO Base Media File Format and is described in MPEGCMAF. Nodes using CMAF as the container format SHALL also use CMAFIngest as the ingestion format.

11.7.7.4. IngestMethodsEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**CMAFIngest**	CMAF ingestion format	M

11.7.7.4.1. CMAFIngest Value

This value SHALL mean that CMAF Ingestion is utilized. See CMAF Ingestion for full details.

11.7.7.5. TriggerActivationReasonEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0	**UserInitiated**	Trigger has been activated by user action	M
1	**Automation**	Trigger has been activated by automation	M

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Value	Name	Summary	Conformance
2	**Emergency**	Trigger has been activated for emergency reasons	M
3	**DoorbellPressed**	Trigger has been activated by a doorbell press	Rev >= v2

11.7.7.6. SupportedFormatStruct Type

This struct holds the combination of container format and ingest method which represents a valid combination for a transport.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**ContainerFormat**	ContainerFormatEnum	all				M
1	**IngestMethod**	IngestMethodsEnum	all				M

11.7.7.6.1. ContainerFormat Field

This field SHALL indicate a supported container format that when combined with IngestMethod, can be used in a transport.

11.7.7.6.2. IngestMethod Field

This field SHALL indicate a supported ingest method that when combined with ContainerFormat, can be used in a transport.

11.7.7.7. CMAFInterfaceEnum Type

This data type is derived from enum8. This type indicates the exact mode of CMAF that is in use.

Value	Name	Summary	Conformance
0	**Interface1**	CMAF Interface-1 Mode	M
1	**Interface2DASH**	CMAF Interface-2 Mode with DASH Support	P, M
2	**Interface2HLS**	CMAF Interface-2 Mode with HLS Support	P, M

11.7.7.7.1. Interface1 Value

This value indicates that only the operations specified in CMAF Interface-1 [https://dashif.org/Ingest/#

interface-1] will be done.

11.7.7.7.2. Interface2DASH Value

This value indicates the operations specified in CMAF Interface-2 [https://dashif.org/Ingest/#interface-2] with the DASH specific portions [https://dashif.org/Ingest/#dash-ingest-requirements] will be done.

11.7.7.7.3. Interface2HLS Value

This value indicates the operations specified in CMAF Interface-2 [https://dashif.org/Ingest/#interface-2] with the HLS specific portions [https://dashif.org/Ingest/#hls-ingest-requirements] will be done.

11.7.7.8. VideoStreamStruct Type

This struct holds a video stream id and the symbolic stream name associated with it. For CMAF based transports, this becomes a CMAF Track.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**VideoStreamName**	string	1 to 16				M
1	**VideoStreamID**	VideoStreamID	all				M

11.7.7.8.1. VideoStreamName Field

This field SHALL identify the unique name assigned to this stream. In CMAF, this value becomes the CMAF track name.

11.7.7.8.2. VideoStreamID Field

This field SHALL indicate the video stream identified by the VideoStreamID entry in the AllocatedVideoStreams list to use.

11.7.7.9. AudioStreamStruct Type

This struct holds a video stream id and the symbolic stream name associated with it. For CMAF based transports, this becomes a CMAF Track.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**AudioStreamName**	string	1 to 16				M
1	**AudioStreamID**	AudioStreamID	all				M

11.7.7.9.1. AudioStreamName Field

This field SHALL identify the unique name assigned to this stream. In CMAF, this value becomes the

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CMAF track name.

11.7.7.9.2. AudioStreamID Field

This field SHALL indicate the audio stream identified by the AudioStreamID entry in the AllocatedAudioStreams list to use.

11.7.7.10. CMAFContainerOptionsStruct Type

This struct encodes options for configuration of the CMAF container format.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**CMAFInterface**	CMAFInterfaceEnum	all				M
1	**SegmentDuration**	uint16	500 to 65500				M
2	**ChunkDuration**	uint16	max (SegmentDuration/2)				M
3	**SessionGroup**	uint8	all				O, D
4	**TrackName**	string	1 to 16				O, D
7	**MetadataEnabled**	bool	all				[METADATA]

11.7.7.10.1. CMAFInterface Field

This field SHALL indicate the selected Interface of the CMAF container. The Interface chosen determines the number and type of operations that occur within each CMAF session. See CMAFInterfaceEnum for details on CMAF Interfaces.

11.7.7.10.2. SegmentDuration Field

This field SHALL indicate the segment duration (in milliseconds) of the CMAF container. This value SHALL be a multiple of the KeyFrameInterval for the associated video stream. It is RECOMMENDED to use a value of 4000 (4 seconds).

11.7.7.10.3. ChunkDuration Field

This field SHALL indicate the chunk duration (in milliseconds) of the CMAF container. A value of 0 SHALL indicate that chunks are not used.

When chunking is used, an even divisor of SegmentDuration SHOULD be used that aligns with the video frame rate and audio frame duration. RECOMMENDED values are 1/2, 1/4, or 1/8 of the Seg-

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mentDuration depending on the end to end latency requirements needed. Each chunk results in an additional 144 bytes of overhead in the resulting file.

11.7.7.10.4. SessionGroup Field

This field is deprecated and has been replaced with the VideoStreams and AudioStreams fields.

11.7.7.10.5. TrackName Field

This field is deprecated and has been replaced with the VideoStreamName or AudioStreamName fields of the associated stream entry in the VideoStreams or AudioStreams lists

11.7.7.10.6. MetadataEnabled Field

This field, if present and true, indicates that AVMetadataStruct based Metadata tracks and boxes MAY be included in the CMAF segments. If this field is not present or is false, metadata tracks and boxes SHALL NOT be included.

Metadata within CMAF segments when used, SHALL be encoded as follows:

- Use `urim` as the codec-type.
- Use `meta` as the content-type.
- Use `urn:csa:matter:av-metadata` as the uri.

Placement of the metadata SHALL be as follows:

- Use a single `trac` for time-synced data points as described in CMAF-Ingest Interface-1 Section 6.6 Requirements for Timed Metadata Tracks [https://dashif.org/Ingest/#interface-1-timed-metadata].
- Use a single Box for non-time synced data points.

11.7.7.11. ContainerOptionsStruct Type

This struct encodes the specific container type options struct

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**ContainerType**	ContainerFormatEnum	all				M
1	**CMAFContainerOptions**	CMAFContainerOptionsStruct					ContainerType == CMAF

11.7.7.11.1. ContainerType Field

This field SHALL indicate the container type chosen for this transport.

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11.7.7.11.2. CMAFContainerOptions Field

This field SHALL contain a CMAF Container Options if the ContainerType is set to CMAF, otherwise this field SHALL be omitted.

11.7.7.12. TransportZoneOptionsStruct Type

This struct encodes the options that configure the per Zone portion of a Trigger configuration.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**Zone**	ZoneID	all	X			M
1	**Sensitivity**	uint8	1 to 10				PERZONESENS

11.7.7.12.1. Zone Field

This field SHALL be a Motion ZoneID found in the Zone Management Cluster which SHALL cause the trigger to activate. If not null, motion in this zone will activate the trigger. If null, motion anywhere in the complement of the union of all the Zones defined in the Zone Management Cluster on this endpoint will activate the trigger.

11.7.7.12.2. Sensitivity Field

This field SHALL indicate how sensitive the trigger for the specified Zone is, and SHALL match the same implementation specifics as the Sensitivity attribute in the Zone Management Cluster. This field SHALL only be included when PerZoneSensitivity is supported, otherwise the value from MotionSensitivity is used.

11.7.7.13. TransportTriggerOptionsStruct Type

This struct encodes the conditions and options that configures the trigger for the push transport. The transport SHALL only start transmitting AV Streams when its associated trigger is activated.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**TriggerType**	TransportTriggerTypeEnum	all				M
1	**MotionZones**	list[TransportZoneOptionsStruct]		X			TriggerType == Motion
2	**MotionSensitivity**	uint8	1 to 10	X			TriggerType == Motion & !PERZONESENS

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
3	**MotionTimeControl**	TransportMotionTriggerTimeControlStruct					TriggerType == Motion
4	**MaxPreRollLen**	uint16	all				TriggerType == Command | TriggerType == Motion

11.7.7.13.1. TriggerType Field

This field SHALL indicate the type of the transport trigger.

11.7.7.13.2. MotionZones Field

This field SHALL be a list of TransportZoneOptionsStruct containing the Motion Zones to trigger on. If this list is null, empty, or if the Zone Management Cluster is not supported on this endpoint, then motion anywhere SHALL cause the trigger to activate. The maximum size of this list is MaxZones

11.7.7.13.3. MotionSensitivity Field

This field SHALL indicate how sensitive the trigger is to motion and SHALL match the same implementation specifics as Sensitivity. This field SHALL NOT be used if the PerZoneSensitivity feature is supported, as a Zone specific value is available in the TransportZoneOptionsStruct Sensitivity field. If this is null and the Zone Management Cluster is supported on this endpoint, then the value found in the Sensitivity attribute in the Zone Management Cluster SHALL be used. If this is null and the Zone Management Cluster is not supported on this endpoint, a value of 10 SHALL be used.

11.7.7.13.4. MotionTimeControl Field

This field SHALL control timing around repeated activation of the trigger (see TransportMotionTriggerTimeControlStruct). If TriggerType is Motion Value, this field SHALL be required.

11.7.7.13.5. MaxPreRollLen Field

This field SHALL indicate the maximum duration in milliseconds of pre-roll content that can be included, if the TriggerType is Motion Value or Command Value, when the trigger activates.

A value of 0 SHALL indicate that no extra segments beyond the one containing the trigger point will be sent.

When using a non 0 value, the value SHALL be greater than or equal to the value of the stream's KeyFrameInterval and it SHOULD be a multiple of that value if larger.

The actual amount transmitted will always be less than or equal to the per stream storage amount

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found in the MaxContentBufferSize.

Since a transmission caused by a trigger activation always begins on the Container Format's segment (or key-frame) boundary, if the trigger occurs mid segment, the entire segment still needs to be sent. This time delta between the actual trigger point and the start of the segment is counted as part of the pre-roll length. Thus, for more than the current segment to be sent as pre-roll, the full size of a segment must fit within the remainder of this length. For this reason, it is RECOMMENDED that a value of at least two times SegmentDuration be used so that a full segment is always included if available.

11.7.7.14. TransportMotionTriggerTimeControlStruct Type

This struct is used to encode a set of values for controlling the lifecycle of a motion triggered transport.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**InitialDuration**	uint16	min 1				M
1	**AugmentationDuration**	uint16	all				M
2	**MaxDuration**	elapsed-s	min InitialDuration				M
3	**BlindDuration**	uint16	all				M

When a Motion Trigger is activated, either by receiving a ManuallyTriggerTransport command, or when motion is initially detected which matches a configured motion trigger, the Node SHALL start the push transport configured with this trigger (see TransportOptionsStruct).

This places the Node in a Motion Detected state, at which point the Node SHALL internally track two values.

TimeSinceActivation

The time in seconds since the trigger was activated.

MotionDetectedDuration

Initially set to the InitialDuration value.

The transport SHALL remain active minimally for InitialDuration period before a PushTransportEnd event can occur.

However, if additional motion is detected during this MotionDetectedDuration period, the Node SHALL increase its value by the AugmentationDuration value. This process may occur repeatedly, but after the first increase of MotionDetectedDuration, the Node SHALL NOT increase the MotionDetectedDuration value unless the previous MotionDetectedDuration has been exceeded by the TimeSinceActivation.

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If the TimeSinceActivation value exceeds the MaxDuration or MotionDetectedDuration value, the Node SHALL generate a PushTransportEnd event and stop detecting motion for this trigger for the period of the BlindDuration value.

If SetTransportStatus is called anytime during this state and sets the TransportStatus to Inactive, the Motion Detected state is superseded.

Since multiple triggers (and corresponding push transports) may be activated by the same motion, the Node SHALL perform this process independently for each motion trigger activated.

11.7.7.14.1. InitialDuration Field

This field SHALL indicate the initial duration (in seconds) of the recording, following the initial trigger.

11.7.7.14.2. AugmentationDuration Field

This field SHALL indicate the duration (in seconds) that the MotionDetectedDuration value is to be extended by if motion is still detected during this period.

11.7.7.14.3. MaxDuration Field

This field SHALL indicate the maximum duration (in seconds) after initial motion detection that additional motion will be detected.

11.7.7.14.4. BlindDuration Field

This field SHALL indicate the duration (in seconds) after a transport finishes transmitting that the Node SHALL NOT activate the trigger again.

11.7.7.15. TransportOptionsStruct Type

This encodes the options and configuration of a transport.

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0	**StreamUsage**	StreamUsageEnum	all				M
0x1	**VideoStreamID**	VideoStreamID	all	X			O, D
0x2	**AudioStreamID**	AudioStreamID	all	X			O, D
0x3	**TLSEndpointID**	TLSEndpointID	0 to 65534				M
0x4	**URL**	string	13 to 2000				M

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x5	**TriggerOptions**	TransportTriggerOptionsStruct					M
0x6	**IngestMethod**	IngestMethodsEnum	all				M
0x7	**ContainerOptions**	ContainerOptionsStruct					M
0x8	**ExpiryTime**	epoch-s	all				O
0x9	**VideoStreams**	list[VideoStreamStruct]	1 to 16				O, [Rev >= v2].a+
0xA	**AudioStreams**	list[AudioStreamStruct]	1 to 16				O, [Rev >= v2].a+

11.7.7.15.1. StreamUsage Field

This field contains the StreamUsageEnum of this transport.

11.7.7.15.2. VideoStreamID Field

This field is deprecated and the VideoStreams field used instead.

If this field is encountered from clients implementing cluster revision 1, then the following SHALL be done:

- If not present, video isn't requested.
- If present and null, automatic video stream assignment is requested.
- If present and non-null, the specific video stream identified by the VideoStreamID SHALL be added as an entry to the VideoStreams field using the VideoStreamName of `video`.

11.7.7.15.3. AudioStreamID Field

This field is deprecated and the AudioStreams field used instead.

If this field is encountered from clients implementing cluster revision 1, then the following SHALL be done:

- If not present, audio isn't requested.
- If present and null, automatic audio stream assignment is requested.
- If present and non-null, the specific audio stream identified by the AudioStreamID SHALL be

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added as an entry to the AudioStreams field using the AudioStreamName of `audio`.

11.7.7.15.4. TLSEndpointID Field

This field SHALL be a TLSEndpointID representing a provisioned TLS Endpoint, which SHALL have valid TLSCAID and TLSCCDID values (see Chapter 14, Certificate Authority ID (CAID) Mapping and the ProvisionEndpoint command in the TLS Client Management Cluster sections in [MatterCore]).

11.7.7.15.5. URL Field

This field SHALL be a valid string in [RFC3986] format representing the upload location. The field SHALL use the `https` scheme which will be validated by the underlying TLSEndpointID.

When the IngestMethod is CMAFIngest, this SHALL be the CMAF publishing_point_URL [https://dashif.org/Ingest/#publishing_point_url] to transport the AV Stream to. The URL length does not need to include space for the full CMAF POST_URL fields which specify the session, track, and segment names as these will be internally appended. See Section 11.7.1.2, “Operation” for further restrictions on the characters allowed in the URL.

11.7.7.15.6. TriggerOptions Field

This field SHALL be of type TransportTriggerOptionsStruct and represents the Trigger Type and its sub options.

11.7.7.15.7. IngestMethod Field

This field SHALL be of type IngestMethodsEnum and represents the Ingest Method to be used.

11.7.7.15.8. ContainerOptions Field

This field SHALL be of type ContainerOptionsStruct and represents the type of Push AV Stream Container to be uploaded and any additional options relating to the Container Format used.

11.7.7.15.9. ExpiryTime Field

This field SHALL be an unsigned 32 bit integer representing the TTL in seconds of a transport allocation. If not present, the transport SHALL never expire.

11.7.7.15.10. VideoStreams Field

This field SHALL be a list of VideoStreamStruct which indicates the requested video streams and the stream names for this transport.

11.7.7.15.11. AudioStreams Field

This field SHALL be a list of AudioStreamStruct which indicates the requested audio streams and the stream names for this transport.

11.7.7.16. TransportConfigurationStruct Type

This encodes the current configuration of an allocated transport.

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Access Modifier: Fabric Scoped							
ID	**Name**	**Type**	**Constraint**	**Quality**	**Fallback**	**Access**	**Conformance**
0	**ConnectionID**	PushTransportConnectionID	0 to 65534				M
1	**TransportStatus**	TransportStatusEnum	all				M
2	**TransportOptions**	TransportOptionsStruct					O

11.7.7.16.1. ConnectionID Field

This field SHALL be a PushTransportConnectionID representing a unique transport.

11.7.7.16.2. TransportStatus Field

This field SHALL represent the Stream Transport Status of the transport.

11.7.7.16.3. TransportOptions Field

This field SHALL represent the Stream Transport Options of the transport.

11.7.7.17. PushTransportConnectionID Type

This data type is derived from uint16.

It represents an allocated transport. This value starts at 0 and monotonically increases by 1 with each new allocation provisioned by the Node. A value incremented past 65534 SHALL wrap to 0. The Node SHALL verify that the incremented ID does not match any other ID. If such a match is found, the ID SHALL be incremented until a unique ID is found.

11.7.8. Status Codes

11.7.8.1. StatusCodeEnum Type

This data type is derived from enum8.

Value	Name	Summary	Conformance
0x02	**InvalidTLSEndpoint**	The specified TLSEndpointID cannot be found.	M

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Value	Name	Summary	Conformance
0x03	**InvalidStream**	The specified VideoStreamID or AudioStreamID cannot be found.	M
0x04	**InvalidURL**	The specified URL is invalid.	M
0x05	**InvalidZone**	A specified ZoneID was invalid.	M
0x06	**InvalidCombination**	The specified combination of Ingestion method and Container format is not supported.	M
0x07	**InvalidTriggerType**	The trigger type is invalid for this command.	M
0x08	**InvalidTransportStatus**	The Stream Transport Status is invalid for this command.	M
0x09	**InvalidOptions**	The requested Container options are not supported with the streams indicated.	M
0x0A	**InvalidStreamUsage**	The requested StreamUsage is not allowed.	M
0x0B	**InvalidTime**	Time sync has not occurred yet.	M
0x0C	**InvalidPreRollLength**	The requested pre roll length is not compatible with the streams key frame interval.	Rev >= v2
0x0D	**DuplicateStreamValues**	The requested Streams info contained duplicate entries.	Rev >= v2

11.7.9. Attributes

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**SupportedFormats**	list[SupportedFormatStruct]	min 1	F		R V	M
0x0001	**CurrentConnections**	list[TransportConfigurationStruct]	desc	N		R V S	M

11.7.9.1. SupportedFormats Attribute

This attribute SHALL contain a list of Supported Container Format structs, which represents the combinations of Ingestion Method and Container Format that the Node supports. Nodes SHALL support at least the combination CMAFIngest,CMAF.

11.7.9.2. CurrentConnections Attribute

This attribute SHALL be a list of TransportConfigurationStruct which represents all the allocated connections added via AllocatePushTransport. When this attribute is read over a non Large Message (See Large Message Quality in the Data Model section of [MatterCore]) capable transport, the TransportOptions field SHALL NOT be included. To get the full details of the connections use the FindTransport command. The maximum size of this list is run-time dependent upon the resource constraints of the system as described in Resource Management and Stream Priorities and the currently used bandwidth of the total available specified by MaxNetworkBandwidth.

11.7.10. Commands

ID	Name	Direction	Response	Access	Quality	Conformance
0x00	**AllocatePushTransport**	client ⇒ server	AllocatePushTransportResponse	M F	L	M
0x01	**AllocatePushTransportResponse**	server ⇒ client	N		L	M
0x02	**DeallocatePushTransport**	client ⇒ server	Y	M F		M
0x03	**ModifyPushTransport**	client ⇒ server	Y	M F		M

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ID	Name	Direction	Response	Access	Quality	Conformance
0x04	**SetTransportStatus**	client ⇒ server	Y	M F		M
0x05	**ManuallyTriggerTransport**	client ⇒ server	Y	O F		M
0x06	**FindTransport**	client ⇒ server	FindTransportResponse	O F	L	M
0x07	**FindTransportResponse**	server ⇒ client	N		L	M

11.7.10.1. AllocatePushTransport Command

This command SHALL allocate a transport and return a PushTransportConnectionID.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**TransportOptions**	TransportOptionsStruct				M

11.7.10.1.1. TransportOptions Field

This field SHALL represent the configuration options of the transport to be allocated.

11.7.10.1.2. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If the UTCTime attribute of the Time Synchronization cluster is null:
  - Fail the command with the cluster-specific status code of InvalidTime, and end processing with no other side effects.
- If an entry for the TLSEndpointID field, of the passed in *TransportOptions*, is not found in the ProvisionedEndpoints list attribute of the TLS Client Management Cluster:
  - Fail the command with the cluster-specific status code of InvalidTLSEndpoint, and end processing with no other side effects.
- If the associated fabric of that entry does not match the accessing fabric:
  - Fail the command with the cluster-specific status code of InvalidTLSEndpoint, and end processing with no other side effects.
- If that entry does not contain a valid non-null CCDID:
  - Fail the command with the cluster-specific status code of InvalidTLSEndpoint, and end pro-

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cessing with no other side effects.

- If the requested combination of IngestMethod and ContainerType from the ContainerOptions fields, of the passed in *TransportOptions*, is not in SupportedFormats:
  - Fail the command with the cluster-specific status code of InvalidCombination, and end processing with no other side effects.
- If the URL is invalid:
  - Fail the command with the cluster-specific status code of InvalidURL, and end processing with no other side effects.
- If the StreamUsage field's value is not present in the StreamUsagePriorities list:
  - Fail the command with the cluster-specific status code of InvalidStreamUsage, and end processing with no other side effects.
- If the MotionZones field is present in the TriggerOptions field of TransportOptions:
  - If there are any entries with duplicate values for the Zone field in MotionZones:
    - Fail the command with the status code ALREADY_EXISTS, and end processing with no other side effects.
  - If the length of the MotionZones list is greater than MaxZones:
    - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
  - If the MotionZones list in the TriggerOptions field of TransportOptions contains an invalid ZoneID:
    - Fail the command with the cluster-specific status code of InvalidZone, and end processing with no other side effects.
- If this new transport would exceed the bandwidth limits:
  - Fail the command with RESOURCE_EXHAUSTED, and end processing with no other side effects.
- If the VideoStreams or AudioStreams fields and either the VideoStreamID or AudioStreamID fields are present in the passed in *TransportOptions*:
  - Fail the command with INVALID_COMMAND, and end processing with no other side effects.
- If the VideoStreamID field of the passed in *TransportOptions* is present:
  - If AllocatedVideoStreams is empty:
    - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
  - If that field is null:
    - Automatically select an existing video stream per the Resource Management and Stream Priorities.
  - Add the provided or selected VideoStreamID as a new entry in the passed in *TransportOptions* VideoStreams list using the VideoStreamName 'video'.
- If the AudioStreamID field of the passed in *TransportOptions* is present:

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    - If AllocatedAudioStreams is empty:
        - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
    - If that field is null:
        - Automatically select an existing audio stream per the Resource Management and Stream Priorities.
    - Add the provided or selected AudioStreamID as a new entry in the passed in *TransportOptions* AudioStreams list using the AudioStreamName 'audio'.
- If both the VideoStreams and AudioStreams fields of the passed in *TransportOptions* are not present or are empty:
    - Fail the command with the status code InvalidStream, and end processing with no other side effects.
- For each VideoStreamStruct entry in the VideoStreams field of the passed in *TransportOptions*:
    - If any fields present are a duplicate of any other entry in the list:
        - Fail the command with the cluster-specific status code of DuplicateStreamValues, and end processing with no other side effects.
    - If an entry for the VideoStreamID field is not found in the AllocatedVideoStreams list:
        - Fail the command with the cluster-specific status code of InvalidStream, and end processing with no other side effects.
    - If the ContainerOptions ContainerType is CMAF and the requested SegmentDuration is not a multiple of the KeyFrameInterval of that entry:
        - Fail the command with the cluster-specific status code of InvalidOptions, and end processing with no other side effects.
    - If the MaxPreRollLen field of the passed in *TransportOptions* is non-zero and is less than the KeyFrameInterval value of that entry:
        - Fail the command with the cluster-specific status code of InvalidPreRollLength, and end processing with no other side effects.
- For each AudioStreamStruct entry in the AudioStreams field of the passed in *TransportOptions*:
    - If any fields present are a duplicate of any other entry in the list:
        - Fail the command with the cluster-specific status code of DuplicateStreamValues, and end processing with no other side effects.
    - If the AudioStreamName field is a duplicate of any VideoStreamName entry in the VideoStreams list of the passed in *TransportOptions*:
        - Fail the command with the cluster-specific status code of DuplicateStreamValues, and end processing with no other side effects.
    - If an entry for the AudioStreamID field is not found in the AllocatedAudioStreams list:
        - Fail the command with the cluster-specific status code of InvalidStream, and end processing with no other side effects.
- Create a new PushTransportConnectionID

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- Using the TLSEndpointID field of the passed in *TransportOptions*, increment the ReferenceCount field of the matching EndpointID entry in the TLS Client Management Cluster's ProvisionedEndpoints list.
- Allocate a new TransportConfigurationStruct associated with the accessing fabric and set the ConnectionID and TransportOptions fields of that struct to the new values.
- Set the TransportStatus field of the new TransportConfigurationStruct to Inactive.
- Persistently store the resulting TransportConfigurationStruct.
- Update the contents of the CurrentConnections list attribute to include the new entry.
- If the TransportOptions field has a non null ExpiryTime field:
  - Keep track of the time in seconds since allocation for this transport, and once this exceeds the ExpiryTime the Node SHALL stop any transport for this transport and deallocate it.
- Return the resulting TransportConfigurationStruct of this allocation in the corresponding AllocatePushTransportResponse command.

On reboot the Node SHALL restart any transports which were transporting previously.

11.7.10.2. AllocatePushTransportResponse Command

This command SHALL be generated in response to a successful AllocatePushTransport command.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**TransportConfiguration**	TransportConfigurationStruct				M

11.7.10.2.1. TransportConfiguration Field

This field SHALL be a TransportConfigurationStruct representing the newly allocated transport.

For compatibility with clients implementing cluster revision 1, the VideoStreamID and AudioStreamID fields SHALL be populated with the first video and audio stream entries that exist in the VideoStreams and AudioStreams lists.

11.7.10.3. DeallocatePushTransport Command

This command SHALL be generated to request the Node deallocates the specified transport.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ConnectionID**	PushTransportConnectionID	0 to 65534			M

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11.7.10.3.1. ConnectionID Field

This field SHALL be a PushTransportConnectionID representing the allocated transport to deallocate.

11.7.10.3.2. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If no entry can be found for the passed in *ConnectionID* in the CurrentConnections list attribute:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- If the associated fabric of that entry is not the accessing fabric:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- If the transport is currently uploading:
  - Fail the command with the status code BUSY, and end processing with no other side effects.
- Remove that entry from the CurrentConnections list attribute.
- Using the TLSEndpointID field of that entry, decrement the ReferenceCount field of the matching EndpointID entry in the TLS Client Management Cluster's ProvisionedEndpoints list.
- Deallocate the transport using the passed in *ConnectionID*.

11.7.10.4. ModifyPushTransport Command

This command is used to request the Node modifies the configuration of the specified push transport.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ConnectionID**	PushTransportConnectionID	0 to 65534			M
1	**TransportOptions**	TransportOptionsStruct				M

11.7.10.4.1. ConnectionID Field

This field SHALL be a PushTransportConnectionID representing the transport to modify.

11.7.10.4.2. TransportOptions Field

This field SHALL represent the Transport Options to modify.

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11.7.10.4.3. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If the UTCTime attribute of the Time Synchronization cluster is null:
  - Fail the command with the cluster-specific status code of InvalidTime, and end processing with no other side effects.
- If no entry can be found for the passed in *ConnectionID* in the CurrentConnections list attribute:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- If the associated fabric of that entry is not the accessing fabric:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- If the transport is currently uploading:
  - Fail the command with the status code BUSY, and end processing with no other side effects.
- On that entry, update the TransportOptions field, of the TransportConfiguration field, to the passed in *TransportOptions*.

11.7.10.5. SetTransportStatus Command

This command SHALL be generated to request the Node modifies the Transport Status of a specified transport or all transports.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ConnectionID**	PushTransportConnectionID	all	X		M
1	**TransportStatus**	TransportStatusEnum	all			M

11.7.10.5.1. ConnectionID Field

This field SHALL be a PushTransportConnectionID representing the transport to modify. If null is passed, all transports belonging to the calling fabric will be modified.

11.7.10.5.2. TransportStatus Field

This field SHALL be a TransportStatusEnum and represent the new transport status to apply.

11.7.10.5.3. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If the passed in *TransportStatus* is set to Active, and the UTCTime attribute of the Time Synchronization cluster is null:

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    - Fail the command with the cluster-specific status code of InvalidTime, and end processing with no other side effects.
- Create a temporary list
- If the passed in *ConnectionID* field is null:
    - For each entry in CurrentConnections:
        - If the associated fabric of that entry is the accessing fabric:
            - Add the ConnectionID of that entry to the temporary list.
- Else:
    - If no entry can be found for the passed in *ConnectionID* in the CurrentConnections list:
        - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
    - If the associated fabric of that entry is not the accessing fabric:
        - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
    - Add *ConnectionID* to the temporary list

- For each PushTransportConnectionID in the temporary list:
    - If the passed in *TransportStatus* field is set to Inactive:
        - Disable transmissions on the transport identified by the id.
        - Remove any queued items for upload on the transport.
        - Terminate any media content uploads currently in progress by using the HTTP/2 stream cancel process, as defined in [RFC9113], Section 6.4.
        - Perform any Container specific session closing uploads.
        - Generate a PushTransportEnd event.
    - Else if the passed in *TransportStatus* field is set to Active:
        - Enable transmissions on the transport identified by the id.
        - If all Privacy Related attributes are False:
            - If the transport's trigger type is Continuous:
                - Begin transmission immediately.
                - Generate a PushTransportBegin event.
            - Else if the transport's trigger type is Command or Motion, and the underlying trigger is currently active and still within the Time Control Bounds:
                - Begin transmission immediately.
                - Generate a PushTransportBegin event.
    - Set TransportStatus of the transport identified by the id to the value in the TransportStatus field of this command.

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11.7.10.6. ManuallyTriggerTransport Command

This command SHALL be generated to request the Node to manually start the specified push transport.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ConnectionID**	PushTransportConnectionID	0 to 65534			M
1	**ActivationReason**	TriggerActivationReasonEnum	all			M
2	**TimeControl**	TransportMotionTriggerTimeControlStruct				O
3	**UserDefined**	octstr	max 256			[METADATA]

11.7.10.6.1. ConnectionID Field

This field SHALL be a PushTransportConnectionID representing the push transport to start or stop.

11.7.10.6.2. ActivationReason Field

This field SHALL provide information as to why the transport was started or stopped.

11.7.10.6.3. TimeControl Field

This field SHALL be a struct of type TransportMotionTriggerTimeControlStruct, but the BlindDuration field SHALL be ignored.

11.7.10.6.4. UserDefined Field

This field SHALL be an octet string representing arbitrary format user defined metadata that will be included in the recording via the UserDefined field of AVMetadataStruct. The format and meaning of this field is not defined in this specification and is up to the users, vendors, or ecosystems deploying it.

11.7.10.6.5. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If the UTCTime attribute of the Time Synchronization cluster is null:
  - Fail the command with the cluster-specific status code of InvalidTime, and end processing with no other side effects.
- If no mapping can be found for the ConnectionID:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side

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effects.

- If the associated fabric of the entry in the CurrentConnections attribute which is identified by ConnectionID is not the accessing fabric:
  - Fail the command with the status code NOT_FOUND, and end processing with no other side effects.
- If HardPrivacyModeOn is true:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
- If SoftLivestreamPrivacyModeEnabled is true and StreamUsage Field for the ConnectionID is LiveView:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
- If SoftRecordingPrivacyModeEnabled is true and StreamUsage Field for the ConnectionID is Recording or Analysis:
  - Fail the command with the status code INVALID_IN_STATE, and end processing with no other side effects.
- If the transport is already transmitting:
  - Fail the command with the status code BUSY, and end processing with no other side effects.
- If the TransportStatus of the push transport is Inactive:
  - Fail the command with a cluster-specific status code of InvalidTransportStatus, and end processing with no other side effects.
- If the trigger type for the specified transport is Continuous:
  - Fail the command with a cluster-specific status code of InvalidTriggerType, and end processing with no other side effects.
- If the trigger type for the specified transport is Command and the TimeControl field is not present:
  - Fail the command with the status code DYNAMIC_CONSTRAINT_ERROR, and end processing with no other side effects.
- If the TimeControl field is not present:
  - Look up the MotionTimeControl field in the in the allocated ConnectionID and use those values as the TimeControl.
- Begin transmission using the TimeControl values, but ignore the BlindDuration field.
- Generate a PushTransportBegin event with ConnectionID set to the ConnectionID field of this command, the ActivationReason field set to the ActivationReason field of this command and the TriggerType field set to Command.
- Asynchronously generate the PushTransportEnd event as normal when the TimeControl values indicate this transmission will end.

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11.7.10.7. FindTransport Command

This command SHALL return the Transport Configuration for the specified push transport or all allocated transports for the fabric if null.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ConnectionID**	PushTransportConnectionID	all	X		M

11.7.10.7.1. ConnectionID Field

This field SHALL be a PushTransportConnectionID or NULL representing the allocated push transport.

11.7.10.7.2. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If ConnectionID is null:
  - If the CurrentConnections list is empty:
    - Fail the command with the status code NOT_FOUND.
  - For each entry in CurrentConnections:
    - If the associated fabric of the entry matches the accessing fabric:
      - Add the full TransportConfigurationStruct to the resulting list.
- Else if ConnectionID is not null:
  - If an entry matching ConnectionID is not found in the CurrentConnections list:
    - Fail the command with the status code NOT_FOUND.
  - If the associated fabric of the matching list entry does not match the accessing fabric:
    - Fail the command with the status code NOT_FOUND.
  - Add a single entry to the resulting list containing the full Transport Configuration that matches ConnectionID.
- If the resulting list is empty:
  - Fail the command with the status code NOT_FOUND.
- Return the resulting list in the corresponding FindTransportResponse command.

11.7.10.8. FindTransportResponse Command

This command SHALL be generated in response to a successful FindTransport command.

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**TransportConfigurations**	list[TransportConfigurationStruct]				M

11.7.10.8.1. TransportConfigurations Field

This field SHALL be a list of Transport Configurations.

11.7.11. Events

ID	Name	Priority	Quality	Access	Conformance
0x00	**PushTransportBegin**	INFO		V	M
0x01	**PushTransportEnd**	INFO		V	M

11.7.11.1. PushTransportBegin Event

This event SHALL indicate a push transport transmission has begun.

With Motion and Continuous based triggers, this event SHALL be generated when each new recording starts uploading.

With Command based triggers, this event SHALL be generated when uploading starts via the ManuallyTriggerTransport command.

The data on this event SHALL contain the following information.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ConnectionID**	PushTransportConnectionID	0 to 65534			M
1	**TriggerType**	TransportTriggerTypeEnum	all			M
2	**ActivationReason**	TriggerActivationReasonEnum	all			TriggerType == Command
3	**ContainerType**	ContainerFormatEnum	all			Rev >= v2

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ID	Name	Type	Constraint	Quality	Fallback	Conformance
4	**CMAFSessionNumber**	uint64	all			ContainerType == CMAF

11.7.11.1.1. ConnectionID Field

This field SHALL be a PushTransportConnectionID representing the push transport which started transmitting.

11.7.11.1.2. TriggerType Field

This field SHALL represent the type of trigger which caused this event to be generated.

11.7.11.1.3. ActivationReason Field

This field SHALL only be present when the TriggerType is Command and provides the reason for the event.

11.7.11.1.4. ContainerType Field

This field SHALL indicate the container type chosen for this transport.

11.7.11.1.5. CMAFSessionNumber Field

This field SHALL represent the CMAF Session number of the recording session that was triggered.

11.7.11.2. PushTransportEnd Event

This event SHALL indicate a push transport upload of the indicated recording has completed.

With Motion, Continuous, and Command based triggers, this event SHALL be generated when each new recording finishes uploading.

The data on this event SHALL contain the following information.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ConnectionID**	PushTransportConnectionID	0 to 65534			M
1	**ContainerType**	ContainerFormatEnum	all			Rev >= v2
2	**CMAFSessionNumber**	uint64	all			ContainerType == CMAF

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11.7.11.2.1. ConnectionID Field

This field SHALL be a PushTransportConnectionID representing the push transport which stopped transmitting.

11.7.11.2.2. ContainerType field

This field SHALL indicate the container type chosen for this transport.

11.7.11.2.3. CMAFSessionNumber Field

This field SHALL represent the CMAF Session number of the recording session that ended.

11.8. Chime Cluster

This cluster provides facilities to configure and play Chime sounds, such as those used in a doorbell.

11.8.1. Revision History

The global ClusterRevision attribute value SHALL be the highest revision number in the table below.

Revision	Description
1	Initial release
2	Add ChimeStartedPlaying event, new optional param to PlayChimeSound to play a specific sound, and remove F quality from Installed-ChimeSounds

11.8.2. Classification

Hierarchy	Role	Scope	PICS Code
Base	Application	Endpoint	CHIME

11.8.3. Cluster ID

ID	Name
0x0556	Chime

11.8.4. Data Types

11.8.4.1. ChimeSoundStruct Type

This struct is used to encode information needed to define a Chime Sound.

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ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	**ChimeID**	uint8	all				M
1	**Name**	string	1 to 48				M

11.8.4.1.1. ChimeID Field

This field SHALL represent the unique ID for a Chime sound.

11.8.4.1.2. Name Field

This field SHALL represent the unique user friendly name of the Chime Sound.

11.8.5. Attributes

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0x0000	**InstalledChimeSounds**	list[ChimeSoundStruct]	1 to 255			R V	M
0x0001	**SelectedChime**	uint8	all	N		RW VO	M
0x0002	**Enabled**	bool	all	N		RW VO	M

11.8.5.1. InstalledChimeSounds Attribute

This attribute SHALL contain all installed chime sounds, represented by a list of Chime Sounds. Each entry in this list SHALL have a unique ChimeID value and a unique Name value.

11.8.5.2. SelectedChime Attribute

This attribute SHALL indicate the currently selected chime sound that will be played when PlayChimeSound is invoked and SHALL be the ChimeID value for the requested Chime Sound within InstalledChimeSounds.

This attribute MAY be written by the client to request a different chime sound. An attempt to write a value that is not contained within InstalledChimeSounds SHALL be failed with a NOT_FOUND response. Writes to this attribute while a chime is currently playing SHALL NOT affect the playback in progress and SHALL only apply starting at the next PlayChimeSound command invocation.

11.8.5.3. Enabled Attribute

This attribute SHALL indicate if chime sounds can currently be played or not, and MAY be written by the client to enable / disable playing of chime sounds.

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11.8.6. Commands

ID	Name	Direction	Response	Access	Quality	Conformance
0x00	**PlayChimeSound**	client ⇒ server	N	O		M

11.8.6.1. PlayChimeSound Command

This command will play the currently selected chime or the chime passed in. In either case the server SHALL generate the ChimeStartedPlaying event.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	**ChimeID**	uint8	all			[Rev >= v2]

11.8.6.1.1. ChimeID Field

This field SHALL represent the unique ID for a Chime sound to play if present, instead of the current value in SelectedChime.

11.8.6.1.2. Effect on Receipt

The following process SHALL be followed when the server receives this command:

- If Enabled is currently false:
  - The server SHALL immediately respond with a status code of SUCCESS, and end processing with no other side-effects.
- If a chime sound is currently being played:
  - The server SHALL immediately respond with a status code of SUCCESS, and end processing with no other side-effects.
- If the passed in *ChimeID* is present:
  - If an entry for the passed in *ChimeID* does not exist in the InstalledChimeSounds list:
    - The server SHALL immediately respond with a status code of NOT_FOUND, and end processing with no other side-effects.
  - Start playing the chime sound associated with *ChimeID*.
- Else
  - Start playing the chime sound associated with SelectedChime.
- Generate the ChimeStartedPlaying event.

11.8.7. Events

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ID	Name	Priority	Quality	Access	Conformance
0x00	**ChimeStarted-Playing**	INFO		V	Rev >= v2

11.8.7.1. ChimeStartedPlaying Event

This event SHALL indicate a Chime sound has just started playing.

The data on this event SHALL contain the following information.

ID	Name	Type	Constraint	Quality	Fallback	Confor-mance
0	**ChimeID**	uint8	all			M

11.8.7.1.1. ChimeID Field

This field SHALL represent the unique ID for the Chime sound that just started playing.

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